

TI 6365.2

INSTRUCTION BOOK

MODE SELECT BEACON SYSTEM (MODE-S) SENSOR

SYSTEM

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Operational Support
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TABLE OF CONTENTS

<i>Section</i>		<i>Page</i>
1.0	GENERAL INFORMATION AND REQUIREMENTS	1-1
1.1	INTRODUCTION	1-1
1.2	MODE SELECT BEACON SYSTEM (MODE S) INTERFACE DESCRIPTION	1-1
1.2.1	Mode S Interrogations and Replies	1-1
1.2.1.1	Mode S Surveillance Capabilities	1-3
1.2.1.2	Mode S Data Link	1-3
1.2.1.3	Mode S Sensor Functions	1-3
1.2.1.4	Mode S Transponder	1-6
1.2.2	Mode S and ATCRBS Surveillance Functions Comparison	1-6
1.2.2.1	Mode S and ATCRBS Surveillance Characteristics	1-7
1.2.2.2	Mode S Monopulse Direction Finding	1-7
1.2.2.3	Mode S Performance Standards	1-8
1.3	EQUIPMENT DESCRIPTION	1-9
1.3.1	Interrogator Equipment	1-9
1.3.1.1	Time of Year Clocks	1-23
1.3.1.2	Mode S Junction Box No. 1	1-23
1.3.1.3	Interrogator Cabinet	1-23
1.3.2	Data Processing System (DPS)	1-23
1.3.2.1	Processor Subsystem (PRCSS) Cabinet	1-24
1.3.2.2	Communications Junction Box Assembly	1-39
1.3.2.3	Sensor Control	1-39
1.3.3	Calibration Performance Monitoring Equipment (CPME)	1-40
1.3.3.1	CPME Description	1-42
1.3.4	Power Distribution Equipment	1-44
1.3.5	Azimuth Pulse Generator (APG) Modification	1-44
1.3.6	Mode S Beacon Video Reconstitutor (BVR) with ASR 7/8 Radars	1-44
1.4	EQUIPMENT SPECIFICATION DATA	1-44
1.5	EQUIPMENT AND ACCESSORIES REQUIRED (SUPPLIED)	1-45
1.6	EQUIPMENT REQUIRED (NOT SUPPLIED)	1-47
2.0	TECHNICAL DESCRIPTION	2-1
2.1	INTRODUCTION	2-1

TABLE OF CONTENTS (Continued)

<i>Section</i>		<i>Page</i>
	2.1.1	2-1
2.2	SIMPLIFIED THEORY OF OPERATION	2-1
	2.2.1 General Mode S System Operation	2-1
	2.2.1.1 Mode S Interrogations and Replies	2-3
	2.2.1.2 Mode S Interrogation and Reply Formats	2-6
	2.2.1.3 Acquisition and Lockout Protocols	2-17
	2.2.1.4 Basic Surveillance Protocols	2-19
	2.2.1.5 Standard-Length Uplink Message Protocol	2-20
	2.2.1.6 Standard-Length Downlink Message Protocol	2-22
	2.2.1.7 Extended Length Uplink Message Protocol	2-26
	2.2.1.8 Extended Length Downlink Message Protocol	2-29
	2.2.1.9 Message Protocol Independence	2-31
	2.2.1.10 Capability Reporting	2-32
	2.2.1.11 Aircraft Identification Protocol	2-32
	2.2.1.12 Utility Message Protocol	2-32
	2.2.2 Interrogator Functional Operation	2-32
	2.2.2.1	2-33
	2.2.2.2	2-33
	2.2.2.3	2-33
	2.2.3 Data Processing System (DPS) Functional Operation	2-34
	2.2.3.1	2-34
	2.2.3.2	2-34
	2.2.3.3	2-34
	2.2.3.4	2-34
	2.2.4 Calibration Performance Monitoring Equipment (CPME) Functional Operation	2-34
	2.2.4.1	2-34
	2.2.4.2	2-35
2.3	DETAILED THEORY OF OPERATION	2-35
	2.3.1 Mode S System Operation	2-35
	2.3.1.1 Channel Management	2-36
	2.3.1.2 Clock	2-40
	2.3.1.3 Modulator/Transmitter	2-40
	2.3.1.4 Antenna System	2-40
	2.3.1.5 Multichannel Receiver	2-43
	2.3.1.6 Mode S Reply Processing	2-43
	2.3.1.7 ATCRBS Reply Processing	2-45
	2.3.1.8 Surveillance Processing	2-47
	2.3.1.9 Data Link Processing	2-50

TABLE OF CONTENTS (Continued)

<i>Section</i>		<i>Page</i>
	2.3.1.10 Network Management	2-53
	2.3.1.11 Performance Monitoring	2-57
	2.3.1.12 Message Routing Management	2-58
	2.3.1.13 Data Extraction	2-59
	2.3.1.14 Site Adaptation	2-59
2.3.2	Interrogator Equipment Operation	2-60
	2.3.2.1 Transmitter Operation	2-60
	2.3.2.2 Receiver Operation	2-64
	2.3.2.3 Time of Year (TOY) Clock Operation	2-64
	2.3.2.4 Input/Output Buffering	2-64
	2.3.2.5 Power Distribution	2-64
	2.3.2.6 Environmental Controls Operation	2-65
	2.3.2.7 Performance Monitoring Circuit Operation	2-65
	2.3.2.8 Fault Detection and Fault Isolation Testing	2-66
2.3.3	Data Processing System (DPS) Equipment Operation	2-68
	2.3.3.1 Processor Subsystem (PRCSS)	2-68
	2.3.3.2 Sensor Control	2-81
	2.3.3.3 Common Digitizer (CD-2) Modification Kit	2-84
	2.3.3.4 Communication Junction Box (CJBX)	2-84
	2.3.3.5 Power Subsystem	2-90
2.3.4	Calibration Performance Monitoring Equipment (CPME) Operation	2-93
	2.3.4.1 Initial Acquisition	2-93
	2.3.4.2 Ground Initiated Comm-B Replay Messages	2-93
	2.3.4.3 Comm-A Message (Interrogation) — Comm-B Message (Reply)	2-94
	2.3.4.4 Comm-C Message (Interrogation) — Comm-D Message (Reply)	2-94
	2.3.4.5 CPME Status	2-94
	2.3.4.6 CPME Hardware Operation	2-94
2.3.5	Mode S Sensor-to-ATC Communications Link Messages	2-97
	2.3.5.1 Mode S Sensor/ATC Communicatons Link	2-97
2.3.6	Mode S Sensor-To-Non-ATC Communications Link Messages....	2-117
	2.3.6.1 Mode S Sensor/Non-ATC Communications Link	2-118
2.3.7	Mode S Sensor-To-ATC Enroute Surveillance Link Messages..	2-131
2.3.8	Mode S Sensor-To-ATC Terminal Surveillance Link Messages (for sites with ASR7 or ASR8)	2-157
2.3.9	Enroute Primary Radar Digitizer (CD-2) to Mode S Sensor Messages	2-157

TABLE OF CONTENTS (Continued)

<i>Section</i>		<i>Page</i>
	2.3.10 Terminal ASR-9 Primary Radar to/from Mode S Sensor Messages	2-157
	2.3.10.1 Health Check Message	2-158
	2.3.11 Mode S Interrogator/DPS Interface	2-158
3.0	OPERATION	3-1
3.1	INTRODUCTION	3-1
3.2	CONTROL AND INDICATORS	3-1
3.3	TURN-ON AND CHECKOUT	3-2
3.4	ON-SITE AND REMOTE MONITORING AND CONTROL	3-13
	3.4.1 On-Site Monitoring and Control	3-13
	3.4.2 Remote Maintenance Monitoring System and Control	3-13
	3.4.2.1 Continuous Poll	3-14
	3.4.2.2 Scheduled Poll	3-14
	3.4.2.3 Specific Poll	3-14
	3.4.3 Remote Terminal Control	3-14
	3.4.3.1	3-14
3.5	EQUIPMENT SHUTDOWN	3-17
4.0	STANDARDS AND TOLERANCES	4-1
4.1	INTRODUCTION	4-1
5.0	PERIODIC MAINTENANCE	5-1
5.1	INTRODUCTION	5-1
5.2	PERFORMANCE CHECKS	5-1
	5.2.1 Performance Checks at Mode S Site	5-1
	5.2.1.1	5-2
	5.2.2 Performance Checks by the RMMS/MPS	5-2
	5.2.2.1 Continuous Poll	5-2
	5.2.2.2 Scheduled Poll	5-2
	5.2.2.3 Specific Poll	5-2
	5.2.2.4 Logical Units	5-3
	5.2.2.5 Data Points	5-3
	5.2.2.6 RMS Master Logical Unit (LU20)	5-4
	5.2.2.7 LU20 Site Data Reports (SDRs)	5-6

TABLE OF CONTENTS (Continued)

<i>Section</i>		<i>Page</i>
5.3	OTHER MAINTENANCE TASKS	5-17
5.4	RECOMMENDED PERFORMANCE INTERVALS	5-17
6.0	MAINTENANCE PROCEDURES	6-1
6.1	INTRODUCTION	6-1
6.2	PERFORMANCE CHECK PROCEDURES	6-1
	6.2.1 Taking Monitor Reports	6-1
	6.2.1.1 RMS Master Logical Unit (LU20)	6-2
6.3	OTHER MAINTENANCE TASK PROCEDURES	6-3
6.4	SPECIAL MAINTENANCE PROCEDURES	6-3
7.0	CORRECTIVE MAINTENANCE	7-1
7.1	INTRODUCTION	7-1
	7.1.1 Maintenance Concept	7-1
	7.1.1.1 On-Site Maintenance	7-1
	7.1.1.2 Integrated Diagnostics	7-3
	7.1.1.3 Maintenance Control Center (MCC)	7-3
	7.1.1.4 Depot Level Maintenance	7-3
	7.1.1.5 Deferred Maintenance	7-3
	7.1.1.6 Preventive Maintenance	7-3
7.2	FAULT INDICATORS	7-4
	7.2.1 Interrogator Monitor Report Fault Messages	7-5
	7.2.2 DPS Monitor Report Fault Messages	7-9
	7.2.3 CPME Monitor Report Fault Messages	7-15
	7.2.3.1	7-15
	7.2.3.2	7-17
7.3	FAULT ANALYSIS PROCEDURES	7-22
7.4	REMOVAL AND INSTALLATION PROCEDURES	7-22
7.5	CABLE WIRING DATA	7-23
	7.5.1 Mode S Interrogator, RF Transfer Switch, and MSJB-1 System Cable List	7-23
	7.5.2 Mode S DPS System Cable List	7-24
7.6	PACKING INSTRUCTIONS	7-24
8.0	PARTS LIST	8-1
8.1	GENERAL	8-1

TABLE OF CONTENTS (Continued)

Section		Page
	8.1.1 Reference Designation	8-1
	8.1.2 Indent	8-1
	8.1.3 Name of Part/Description	8-1
	8.1.4 Mfgs. Code Number	8-1
	8.1.5 JAN/MIL Mfgs. Part No.	8-1
	8.1.6 Notes	8-1
9.0	INSTALLATION, INTEGRATION AND CHECKOUT	9-1
	9.1 INTRODUCTION	9-1
	9.2 INSTALLATION AND CHECKOUT	9-1
10.0	COMPUTER SOFTWARE	10-1
	10.1 INTRODUCTION	10-1
	10.2 PROGRAM HIERARCHY, DESCRIPTION AND LISTING INFORMATION	10-1
11.0	TROUBLESHOOTING SUPPORT AREA	11-1
	11.1 INTRODUCTION	11-1

LIST OF ILLUSTRATIONS

<i>Figure</i>		<i>Page</i>
1-1	Mode S Beacon System Elements	1-2
1-2	Mode S System — Enroute Site Configuration Interface to ATC Facilities, Block Diagram	1-4
1-3	Mode S System — ASR-9 Configuration (Terminal Site) Interface to ATC Facilities, Block Diagram	1-5
1-4	Comparison of ATCRBS and Mode S Aircraft Position Accuracy at Maximum Range	1-7
1-5	Mode S Sensor Monopulse Reply Degarbling	1-8
1-6	Mode S System Physical Relationship of Units	1-10
1-7	Mode S System Relationship of Units at a Terminal Site, Block Diagram	1-11
1-8	Mode S System Relationship of Units at an Enroute Site, Block Diagram	1-12
1-9	Interrogator Equipment Chart (5 Sheets)	1-18
1-10	DPS Equipment Chart (13 Sheets)	1-25
1-11	Calibration Performance Monitoring Equipment (CPME), Physical Relationship of Units	1-41
1-12	Calibration Performance Monitoring Equipment Chart	1-43
2-1	Mode S Sensor Functional Block Diagram	2-2
2-2	ATCRBS/Mode S All-Call Interrogation	2-4
2-3	ATCRBS Reply	2-4
2-4	Mode S Interrogation	2-5
2-5	Mode S Reply	2-6
2-6	Mode S Uplink Formats	2-7
2-7	Mode S Downlink Formats	2-8
2-8	ATCRBS/Mode S Time Sharing Diagram	2-37
2-9	Mode S Roll-Call Mode Scheduling Diagram	2-38
2-10	Mode S Interrogation and Reply Cycles	2-38
2-11	Channel Management Functional Block Diagram	2-41

LIST OF ILLUSTRATIONS (Continued)

<i>Figure</i>	<i>Page</i>
2-12 Modulator/Transmitter General Block Diagram	2-42
2-13 Sum and Difference Antenna Patterns	2-42
2-14 Mode S Joint Use Enroute Sensor General Block Diagram (Deleted)	2-44
2-15 Mode S Reply Processing General Block Diagram	2-44
2-16 ATCRBS Reply Processing General Block Diagram	2-46
2-17 Surveillance Processing General Block Diagram	2-48
2-18 Mode S Surveillance Processing General Block Diagram	2-48
2-19 ATCRBS Surveillance Processing General Block Diagram	2-50
2-20 Data Link Processing General Block Diagram	2-51
2-21 Input Message Processing General Block Diagram	2-52
2-22 Output Message Processing General Block Diagram	2-52
2-23 Network Management General Block Diagram	2-53
2-24 Sensor Coverage Map Grid Structure	2-54
2-25 Sensor Coverage Coordination General Block Diagram	2-56
2-26 Performance Monitoring General Block Diagram	2-58
2-27 Interrogator Block Diagram	2-61
2-28 Interrogator Functional Block Diagram (2 Sheets)	2-62
2-29 Data Processing System (DPS) Overview Diagram	2-68
2-30 Data Processing System (DPS) Block Diagram	2-69
2-31 Processor Subsystem Channel Block Diagram for a Basic Configuration	2-70
2-32 Processor Subsystem Channel Block Diagram for a Basic Configuration with Add-On Kits	2-71
2-33 CPU or IOP Block Diagram	2-73
2-34 Processor Subsystem (PRCSS) Channel Functional Block Diagram	2-74
2-35 PRCSS Channel Bus Interface Simplified Block Diagram	2-75
2-36 PRCSS Channel Interface with the Interrogator (Front End) Channels, Block Diagram	2-77

LIST OF ILLUSTRATIONS (Continued)

<i>Figure</i>	<i>Page</i>
2-37 Interrogator Interface (ITRI) General Block Diagram.....	2-78
2-38 Remote Mode S Sensor Control	2-83
2-39 ATC and Non-ATC Communications Interface Block Diagram.....	2-84
2-40 Communication Junction Box (CJBX6) With SDS Signal Designations (Front Panel View)	2-87
2-41 PRCSS Cabinets Interface with Communications Junction Box (CJBX6) Block Diagram.....	2-91
2-42 CJBX Interface with Common Digitizer (CD-2) Via Surveillance Data Selector (SDS) Control Block Diagram.....	2-92
3-1 Power Distribution Box.....	3-2
3-2 Mode S Remote Terminal (RT) Display Screen	3-15 ■
7-1 Mode S Maintenance Concept	7-2
7-1A Typical LU Data Report.....	7-5 ■
7-2 MSJB-1, RF Transfer Switch, and Interrogator System Cabling Diagram for a Terminal Site	7-25
7-3 Interrogator and DPS System Cabling Diagram for a Terminal Site	7-26
7-4 MSJB-1, RF Transfer Switches, and Interrogator System Cabling Diagram for an En route Site	7-27
7-5 Interrogator and DPS System Cabling Diagram for an En route Site	7-28
7-6 DPS System Cable Configuration Diagram (6 Sheets)	7-29
7-7 Communication Junction Box (Unit 6), Basic Configuration	7-35
7-8 Communication Junction Box (Unit 6) with SDS	7-35
7-9 Communication Expansion Junction Box (Unit 10)	7-36
7-10 Surveillance Expansion Junction Box (Unit 11)	7-36
7-11 Interrogator and RF Transfer Switch Cable Interconnection Diagram for Terminal Site and En route Site.....	7-37
7-12 Interrogator Cable Assemblies W64-W66, W76-W78, and W100-W102	7-38
7-13 Interrogators (Unit 2 and 3) RF Transfer Switch (Unit 9) Harness Assembly W67 for a Terminal Site	7-39
7-14 Interrogator (Units 2 and 8) RF Transfer Switches (Units 9 and 13) Harness Assembly W67 for an En route Site.....	7-40

LIST OF ILLUSTRATIONS (Continued)

<i>Figure</i>	<i>Page</i>
7-15 Cable Pinout for Interrogator Cable Assemblies W68, W69, W73, AND W74	7-41
7-16 Cable Pinout for Interrogator Cable Assemblies W70 AND W71	7-42
7-17 Cable Pinout for Interrogator Cable Assemblies W72 AND W75	7-43
7-18 Cable Pinout for Interrogator Cable Assemblies W79 AND W80	7-44
7-19 Type A-1 System Cable Assembly 3261 8423-XX Connections (4 Sheets)	7-45
7-20 Type B-1 System Cable Assembly 3262 0072 Connections	7-49
7-21 Type B-2 System Cable Assembly 3588 0459 Connections	7-50
7-22 Type C-1 System Cable Assembly 3262 0064 Connections	7-51
7-23 Type D-1 System Cable Assembly 3262 0080-01 Connections (3 Sheets)	7-52
7-24 Type E-1 System Cable Assembly 3262 0049-XX Connections	7-55
7-25 Type F-1 System Cable Assembly 3261 8472-XX Connections	7-56
7-26 Type G-1 System Cable Assembly 3262 0007-XX Connections	7-57
7-27 Type G-2 System Cable Assembly 3262 0015-XX Connections	7-58
7-28 Type H-1 System Cable Assembly 3262 0023-XX Connections	7-59
7-29 Type J-1 System Cable Assembly 3262 0171 Connections	7-60
7-30 Type J-2 System Cable Assembly 3262 0189 Connections	7-61
7-31 Type K-1 System Cable Assembly 3262 8399 Connections	7-62
11-1 Mode S System Typical Terminal Site	11-2
11-2 Mode S System Typical En route Site	11-3
11-3 Interrogator Cable Assemblies W64-W66, W76-W78, and W100-W102	11-4
11-4 Interrogator Cable Assembly W67	11-5
11-5 Interrogator Cable Assemblies W68-W75, W79, and W80	11-6
11-6 DPS Cable Assemblies, Processor to Junction Box (W1001-W1012, W1021-W1026, W1037-W1042 and W1045-W1170)	11-6
11-7 DPS Cable Assemblies, Processor to Processor (W1013-W1016)	11-6
11-8 DPS Cable Assemblies, Interrogator to Processor (W1029, W1030, and W1171-W1174)	11-7

LIST OF ILLUSTRATIONS (Continued)

<i>Figure</i>		<i>Page</i>
11-9	DPS Cable Assembly, Processor to Junction Box (W1019, W1020, W1027, W1028, W1033-W1036, and W1201-W1208)	11-7
11-10	DPS System Cables Wiring and Signal Distribution Diagram (14 Sheets)	11-8
11-11	Mode S Power Distribution Information Diagram (2 Sheets)	11-22

LIST OF TABLES

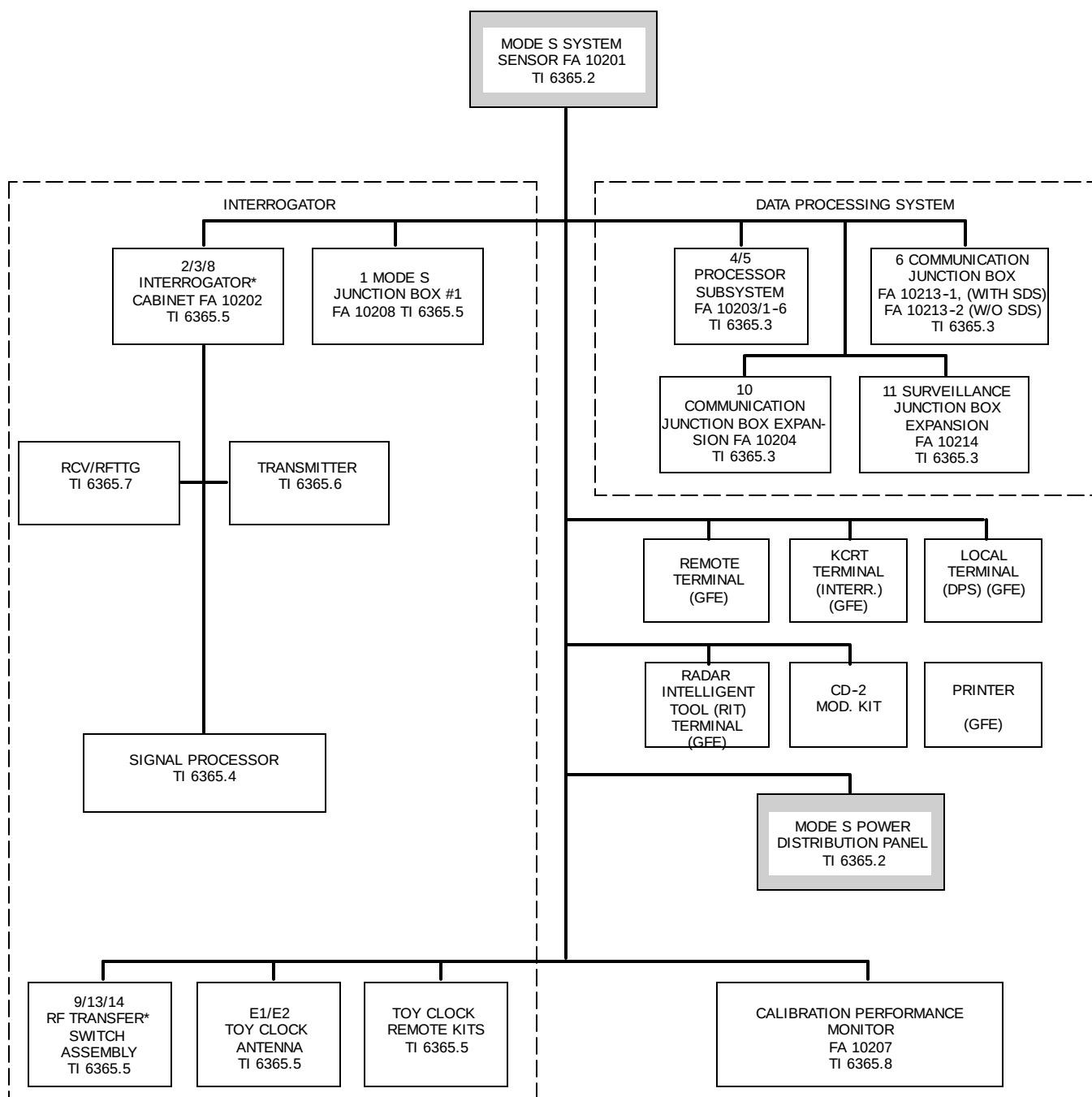
<i>Table</i>		<i>Page</i>
1-1	Mode S System Equipment	1-13
1-2	Mode S Equipment Specifications	1-44
1-3	Equipment Required (Supplied)	1-46
1-4	Equipment Required (Not Supplied)	1-47
2-1	Interrogations and Reply Field Index	2-10
2-2	Mode S-Only All-Call Interrogation Fields	2-12
2-3	All-Call Reply Fields	2-12
2-4	Surveillance and Comm-A Interrogation Fields	2-13
2-5	Surveillance and Comm-B Reply Fields	2-14
2-6	Comm-C (Extended Length Message-ELM) Interrogation Fields	2-15
2-7	Comm-D (ELM) Reply Fields	2-16
2-8	Air-to-Air Surveillance Interrogation Fields	2-16
2-9	Air-to-Air Surveillance Reply Fields	2-17
2-10	Example of Non-Selective Air-Initiated Comm-B Message Delivery	2-24
2-11	Example of Multisite Air-Initiated Comm-B Message Delivery	2-25
2-12	Example of Multisite Uplink ELM Delivery	2-28
2-13	Example of Multisite Downlink ELM Delivery	2-31
2-14	Information Field (I) Type Codes for Mode S Sensor/ATC Messages	2-102
2-15	ATC-to-Mode S Message Data Blocks	2-103
2-16	Mode S-to-ATC Message Data Blocks	2-109
2-17	Sensor-to-Sensor Message Data Blocks	2-116
2-18	Information Field (I) Type Codes for Mode S Sensor/Non-ATC Messages	2-119
2-19	Non-ATC-to-Mode S Message Data Blocks	2-122
2-20	Mode S-to-Non-ATC Message Data Blocks	2-128
2-21	Mode S Sensor to En route ATC Facility, Beacon Message	2-132
2-22	Mode S Sensor to En route ATC Facility, Beacon RTQC Message	2-138

LIST OF TABLES

<i>Table</i>		<i>Page</i>
2-23	Mode S Sensor to En route ATC Facility, Beacon Strobe Message	2-141
2-24	Mode S Sensor to En route ATC Facility, Search Message	2-143
2-25	Mode S Sensor to En route ATC Facility, Search RTQC Message	2-146
2-26	Mode S Sensor to En route ATC Facility, Search Strobe Message	2-148
2-27	Mode S Sensor to En route ATC Facility, Weather Map Message	2-150
2-28	Mode S Sensor to En route ATC Facility, CD-2/Mode S Status Message	2-153
2-29	Mode S Sensor to Terminal ATC Facility, Beacon Message or Beacon RTQC Message (for sites with ASR7 or ASR8 radar)	2-159
2-30	Mode S Sensor to Terminal ATC Facility, Search RTQC Message (for sites with ASR7 or ASR8 radar)	2-162
2-31	Mode S Sensor to Terminal ATC Facility, Mode S Status Message (for sites with ASR7 or ASR8 radar)	2-163
2-32	CD-2 to Mode S Sensor, Search Message	2-165
2-33	CD-2 to Mode S Sensor, Search RTQC Message	2-168
2-34	CD-2 to Mode S Sensor, Status Message	2-171
2-35	CD-2 to Mode S Sensor, Weather Map Message	2-175
2-36	CD-2 to Mode S Sensor, Search Strobe Message	2-178
2-37	ASR-9 to Mode S Sensor, Radar Target (RTGT) Report (Uplink)	2-181
2-38	ASR-9 to Mode S Sensor, Radar Search (RTQC) Report (Uplink)	2-188
2-39	ASR-9 to Mode S Sensor, System Status Message (Uplink)	2-196
2-40.	ASR-9 To/From Mode S Sensor, Health Check Message (Uplink and Downlink) ...	2-201
2-41	Mode S Sensor to ASR-9; Beacon, Beacon/Radar Merge, or Beacon RTQC Message Report (Downlink)	2-202
3-1	Mode S System Initial Turn-On and Checkout Procedures	3-3
3-2	Mode S Remote Terminal Control Functions	3-16
4-1	Standards and Tolerances	4-1
7-1	Interrogator Hardware Monitor Report Fault Messages	7-6
7-2	DPS Monitor Report Fault Messages	7-9

LIST OF TABLES

<i>Table</i>		<i>Page</i>
7-3	CPME Summary Conditions Monitor Report Status and Fault Messages for Logical Unit 2A	7-16
7-4	CPME Test Conditions Monitor Report Status and Fault Messages for Logical Units 2B Through 2F	7-18
9-1	Installation and Checkout Documentation	9-1
11-1	Support Data Figures	11-1



NOTE: AN * INDICATES THAT THE THIRD INTERROGATOR (UNIT 8) AND RF TRANSFER SWITCH (UNIT 14) ARE RESERVED FOR FUTURE EXPANSION.

MODE SELECT BEACON SYSTEM SENSOR FAMILY TREE CHART

1.0 GENERAL INFORMATION AND REQUIREMENTS

1.1 INTRODUCTION

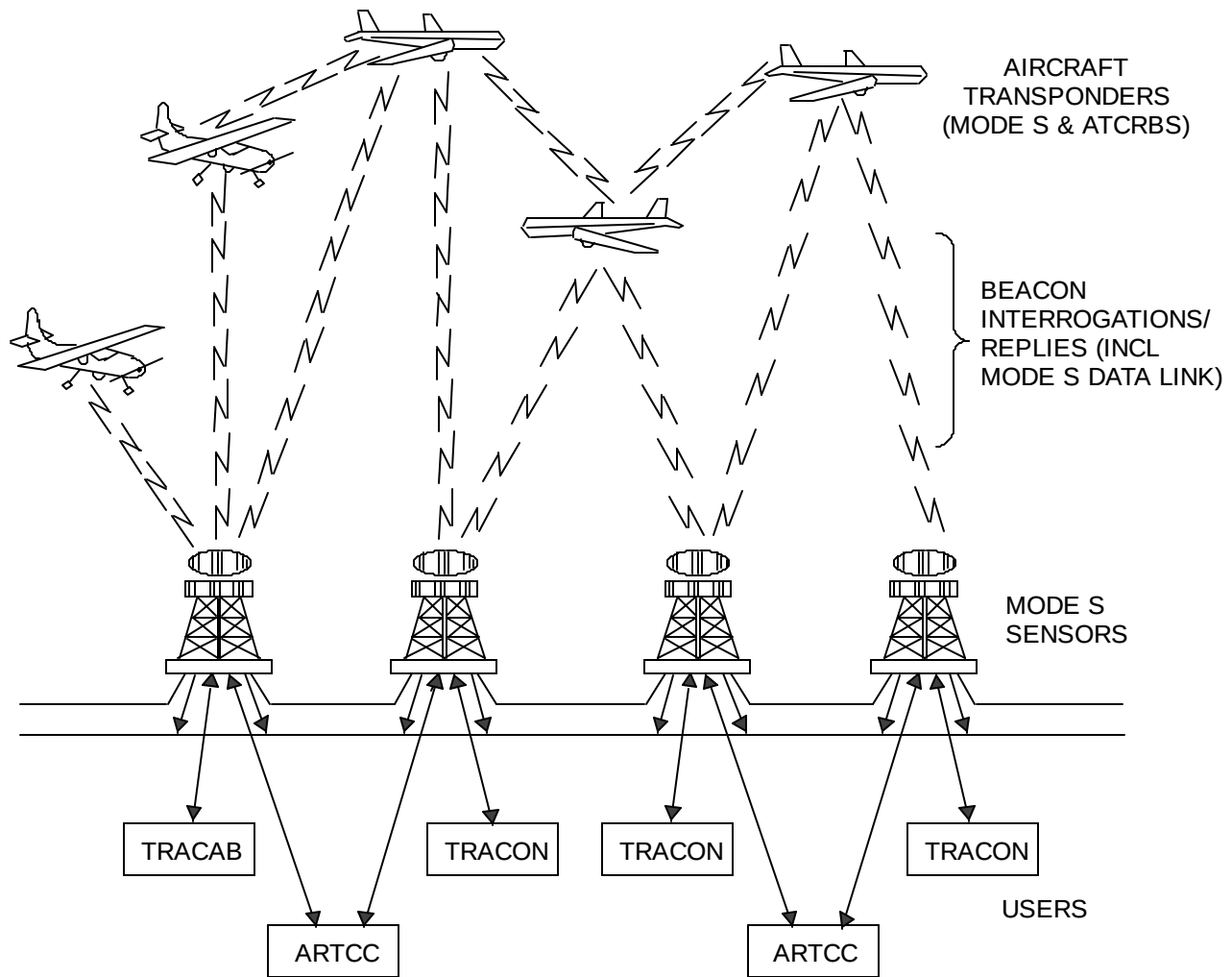
This system instruction manual is the on-site system maintenance manual for the Mode Select Beacon System (Mode S) Sensor. The Mode S sensor is comprised of the Interrogator (front end), Data Processing System (DPS), Calibration Performance Monitoring Equipment (CPME), and the power distribution equipment. The Mode S Sensor Family Tree Chart indicates the equipments that comprise the Mode S sensor. On the Family Tree Chart, each of the Mode S sensor equipments has a reference to the applicable document (TI 6365.X) that describes in detail that particular equipment. Data provided in this system manual and also in each equipment manual includes subjects such as: (1) general information and equipment requirements, (2) technical description, (3) operating instructions, (4) periodic maintenance instructions, (5) troubleshooting information, (6) removal and replacement procedures, (7) calibration/adjustment instructions, (8) checkout/re-test instructions, (9) part lists, and (10) references to installation, integration, checkout, and general software information. A detailed Mode S Sensor Family Tree Chart, which breaks down a particular equipment, is provided in the applicable equipment manual (TI 6365.X). The detailed Mode S Sensor Family Tree Chart in each equipment manual also provides information pertinent to equipment reference designations/unit numbers and additional equipment manuals.

1.2 MODE SELECT BEACON SYSTEM (MODE S) INTERFACE DESCRIPTION

The Mode Select Beacon System (which is commonly referred to in this manual as Mode S or Mode S System) is a combined secondary surveillance radar (beacon) and ground-air-ground data link system. The Mode S system is capable of providing the aircraft surveillance and communications necessary to support Air Traffic Control (ATC) automation in dense air traffic environments. Figure 1–1 shows the elements which comprise the Mode S System. To facilitate transition with existing beacon radar systems, Mode S is capable of common-channel interoperation with the current Air Traffic Control Radar Beacon System (ATCRBS). Mode S provides the accurate surveillance needed to support automated decision making, and the reliable communications needed to support data link services. In order to meet these requirements at enroute facilities, Mode S sensors may operate with back-to-back beacon antennas to provide twice the beacon data rate available from a standard antenna. When operating in conjunction with a terminal or enroute digitizer equipped ATC surveillance radar, a Mode S sensor uses the primary radar returns either to reinforce beacon tracks, or in cases of absence or failure of a transponder, to provide radar target reports.

1.2.1 Mode S Interrogations and Replies

Mode S uses the same interrogation and reply frequencies as ATCRBS, and the signal formats have been chosen to permit substantial commonality in hardware. This degree of compatibility permits smooth transition, in which: (1) Mode S interrogators provide surveillance of ATCRBS-equipped aircraft, and (2) Mode S transponders reply to ATCRBS interrogators. The fundamental difference between Mode S and ATCRBS is the manner of addressing aircraft, or selecting which aircraft will respond to an interrogation. In ATCRBS, all aircraft within the main beam of the interrogator are selected, interrogated, and then respond. As the beam sweeps around, all angles are interrogated, and all aircraft within line-of-sight of the antenna will respond. In Mode S, each aircraft is assigned a unique address code. Selection of which aircraft is to respond to an interrogation is accomplished by including the aircraft's address code in the interrogation. Each interrogation is directed to a particular aircraft and only that aircraft will respond to the interrogation.



ARTCC — AIR ROUTE TRAFFIC CONTROL CENTER
 TRACAB — TERMINAL RADAR APPROACH CONTROL FACILITY (TOWER CAB)
 TRACON — TERMINAL RADAR APPROACH CONTROL FACILITY

FIGURE 1-1. MODE S BEACON SYSTEM ELEMENTS

1.2.1.1 Mode S Surveillance Capabilities. Two major advantages are derived from the use of a discrete address for surveillance. First, an interrogator is now able to limit its interrogation to only those targets for which it has surveillance responsibility, rather than to continuously interrogate all targets within line-of-sight. This prevents surveillance system saturation caused by all transponders responding to all interrogators within line-of-sight. The second advantage concerns the appropriate timing of interrogations which ensures that the responses from aircraft do not overlap. This condition eliminates the mutual interference which results from the overlapping of replies from closely spaced aircraft (which is commonly referred to as synchronous garble). In addition to the improved surveillance capability of Mode S, the use of the discrete address in interrogations and replies permits the inclusion of messages to or from a particular aircraft, thereby providing the basis for a ground-air and air-ground digital data link. The Mode S system is comprised of sensors, transponders, and signal links which interface these elements. Mode S provides surveillance and ground-air-ground communication service to air traffic control facilities including the enroute Air Route Traffic Control Center (ARTCC), Terminal Radar Approach Control Facility (TRACON), and Terminal Radar Approach Control Facility Tower Cab (TRACAB) as shown in Figure 1-1. The Mode S sensor/ATC surveillance interface provides up to six data links from the Mode S sensor to enroute Air Traffic Control (ATC) facilities (refer to Figure 1-2). Each data link is comprised of up to six data line interfaces. Each data line interface supports one GFE modem. At a Mode S system enroute site configuration, the DPS receives source information for surveillance reports from two independent sources: (1) beacon data is provided by the Mode S sensor Interrogator (front end) and (2) primary radar data is provided by the Common Digitizer (CD-2) equipment via a Mode S CD-2 modification kit (refer to Figure 1-2). The DPS processes this surveillance information and then transmits the results to the enroute ATC facilities via the communication/surveillance junction box(es) and the optional surveillance data selector (SDS). Figure 1-3 shows a Mode S system terminal site with an ASR-9 radar and its interface (surveillance data lines and communication data lines) to an ATC facility. In this configuration, the Mode S surveillance data is sent back to the ASR-9 which provides the interface to the ATC facility. At sites without digitized primary radar, there is no primary radar interface with the Mode S system, which operates in a beacon-only configuration.

1.2.1.2 Mode S Data Link. The Mode S sensor provides a two-way digital data link for all Mode S equipped aircraft. Messages originating on the ground (from terminal facilities, enroute facilities, or non-ATC sources) will be sent to the aircraft. The appropriate acknowledgment which is received by the Mode S sensor from the aircraft will be relayed to the sender. The Mode S sensor also manages the data link so that any time an aircraft wishes to initiate an air-to-ground message, that message will be read out with a minimum of delay. The Mode S data link is used only for Mode S targets. The Mode S data link employs signal formats used for ATCRBS and adds to these formats the signal waveforms and message formats necessary to acquire Mode S equipped aircraft, (using discretely addressed surveillance and data link interrogations and replies).

1.2.1.3 Mode S Sensor Functions. To interrogate Mode S-equipped aircraft, the Mode S sensor maintains a file of the identity and approximate position of all aircraft within its defined area of coverage. Each Mode S sensor's operation is controlled by a prestored map defining its coverage volume, which may be different in normal operation and in the event of various system failures, for example: the failure of an adjacent sensor. In a netted configuration, each sensor may communicate with adjacent sensors via a common ATC facility to hand off targets as they pass from the region of one sensor's coverage to that of an adjacent sensor. In regions of overlapping coverage, this intersensor communication may be used to assist in the reacquisition of a lost target. Each Mode S sensor can provide surveillance and communication services to several ATC and/or non-ATC facilities, such as all those facilities whose areas of control responsibility overlap the coverage area of the Mode S sensor. The interface between the sensor and each control facility comprises a one-way circuit for the transmission of surveillance data (both radar and beacon) and a two-way circuit for the interchange of data link messages. The two-way circuit is also used to transmit various status and control messages between the sensor and the ATC facility.

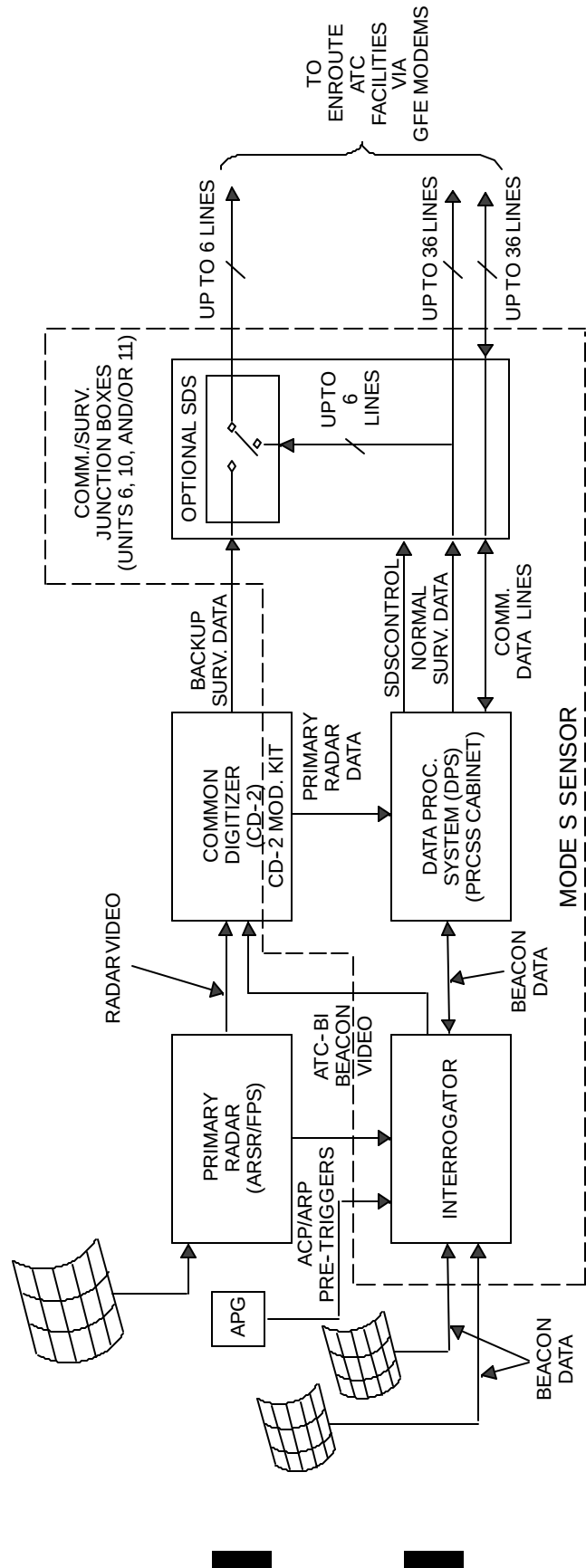
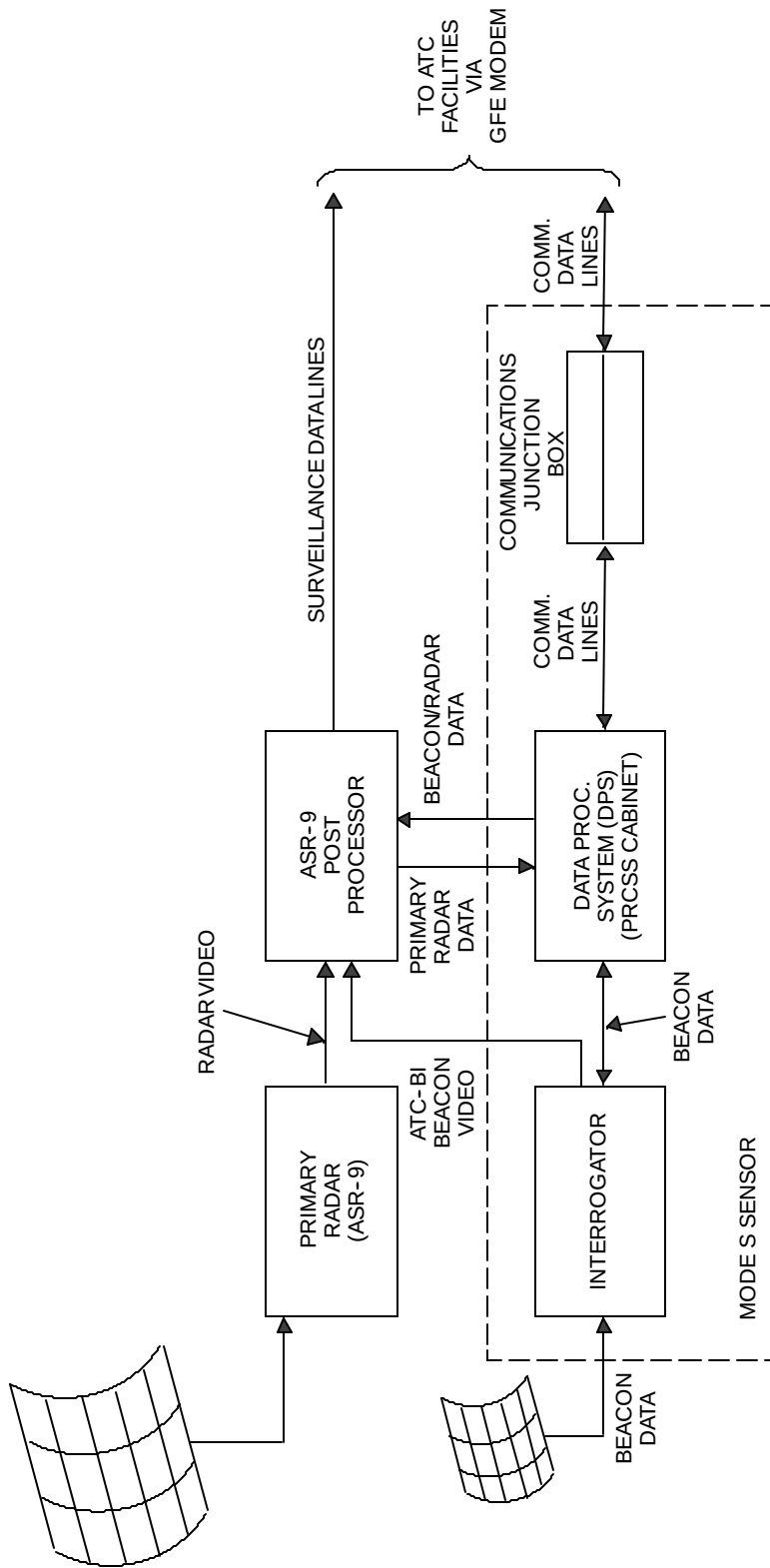


FIGURE 1-2. MODE S SYSTEM — ENROUTE SITE CONFIGURATION INTERFACE TO ATC FACILITIES, BLOCK DIAGRAM



NOTE:
 AT A TERMINAL SITE THAT USES AN ASR-7 OR ASR-8 RADAR, PRIMARY RADAR DATA IS NOT
 TRANSFERRED BETWEEN THE RADAR EQUIPMENT AND MODE S SENSOR. AT THESE SITES THE
 DPS SUPPLIES BEACON-ONLY DATA AND COMMUNICATIONS DATA TO ATC FACILITIES VIA COMM.
 JUNCTION BOX.

FIGURE 1-3. MODE S SYSTEM — ASR-9 CONFIGURATION (TERMINAL SITE) INTERFACE TO ATC FACILITIES,
 BLOCK DIAGRAM

1.2.1.3.1 When the Mode S sensor is in the Air Traffic Control Beacon Interrogator (ATC-BI) backup mode of operation, the Interrogator independently carries out beacon detection of aircraft. In ATC-BI backup mode, the Interrogator continues to monitor the overall status of the Interrogator, the transmitters power, the transmitters voltage standing wave ratio (VSWR), and status and values for front end (Interrogator and MSJB-1) cabinet temperatures. the aforementioned data may be displayed at the Interrogator maintenance terminal (KCRT) on user request. In addition, the Interrogator also makes this information available to the DPS. During ATC-BI backup mode, the DPS provides status and values for Interrogator transmitters power, the Interrogator transmitters VSWR, front end cabinet temperature data, an indication that the sensor is in ATC-BI backup mode, sensor status, and channel status. the DPS will provide the aforementioned information to the sensor control and status display point equipment (remote terminal, DPS local terminal, and the remote maintenance monitoring system-RMMS).

1.2.1.4 Mode S Transponder. The Mode S transponder includes all of the functions of an ATCRBS transponder, and also has the additional ability to decode Mode S interrogations and to format and transmit the appropriate replies. For data link, the transponder functions primarily as a modem. On receipt of a ground-to-air transmission, the transponder verifies the correctness of the received message by use of the error-detecting code. Once the message is verified, the transponder transfers the message contents to one or more external devices. For air-to-ground messages, the transponder accepts the message contents from an external input device, and formats and encodes the data for transmission as part of the reply to a subsequent interrogation.

1.2.2 Mode S and ATCRBS Surveillance Functions Comparison

The Mode S sensor provides surveillance of ATCRBS aircraft with quality enhancements better than that of ATCRBS surveillance equipment. Conventional ATCRBS processors determine azimuth through a technique known as a “sliding window detector”. This azimuth measurement technique (sliding window detector) uses a high pulse repetition frequency (PRF) of up to 450 per second. The high PRF of azimuth measurement provides the twelve or more replies needed for azimuth determination for a terminal sensor. A second characteristic of this azimuth measurement technique is that a failure to receive replies in the middle of the reply run can lead to the declaration of the presence of two targets, neither of which is at the azimuth of the true target. The phenomenon is known as an “azimuth split”. A comparison of ATCRBS and Mode S aircraft position accuracy at maximum range is presented in Figure 1-4.

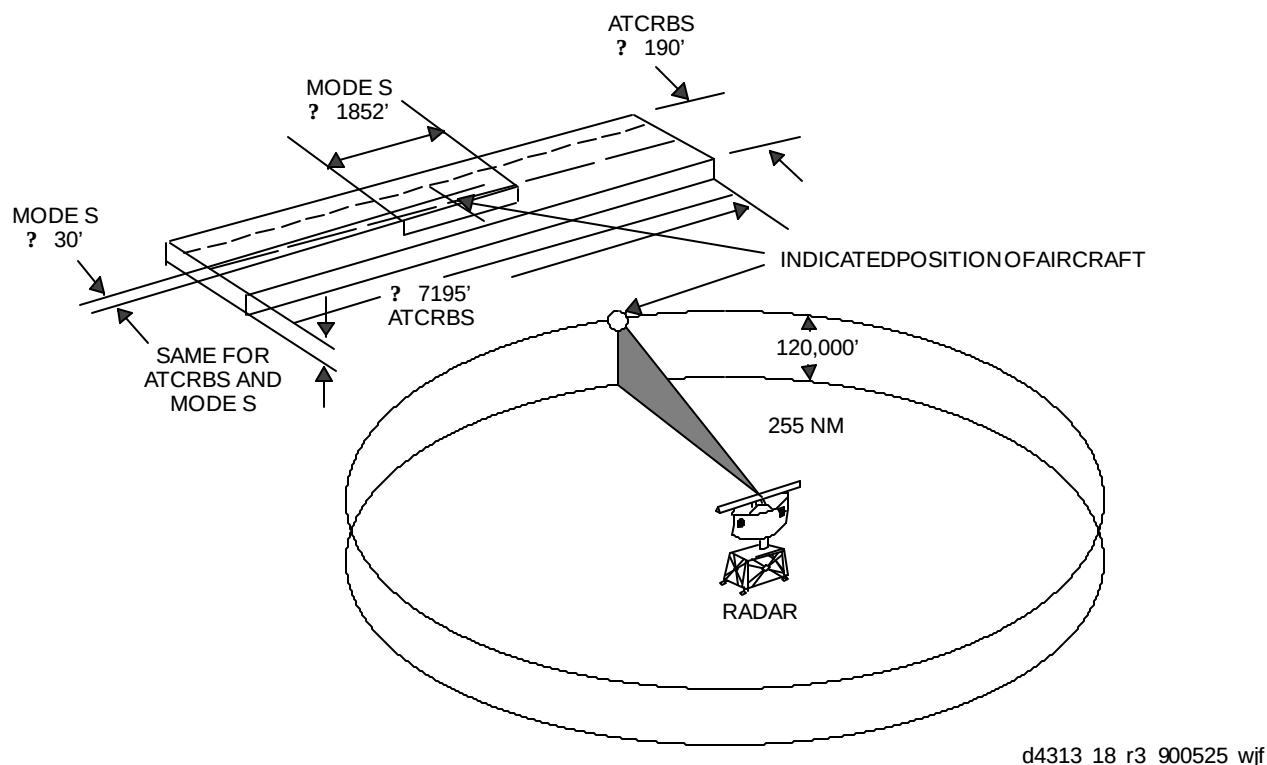


FIGURE 1-4. COMPARISON OF ATCRBS AND MODE S AIRCRAFT POSITION ACCURACY AT MAXIMUM RANGE

1.2.2.1 Mode S and ATCRBS Surveillance Characteristics. The principal characteristics of ATCRBS surveillance that are provided by Mode S are: (1) monopulse direction finding, (2) reduced interrogation rate, (3) improved reply degarbling, and (4) false target identification. In ATCRBS, two types of interrogations are accepted by an ATCRBS transponder: (1) a mode A interrogation, which has an 8 msec pulse spacing and elicits a 20.3 msec reply that contains one of 4,096 pilot entered identity codes, or (2) a mode C interrogation, which has a 21 msec pulse spacing and elicits a similar 20.3 msec reply that contains the aircraft's altitude. The use of Mode S monopulse direction finding on ATCRBS replies eliminates the disadvantages of the sliding window detector. Nominally the sensor could operate with only one reply per ATCRBS mode (mode A and mode C) for each target during one scan of the antenna. However, this would lead to frequent target report declarations due to the chance reception of the replies (commonly referred to as "fruit") elicited by adjacent sensors. To reduce this effect, the Mode S sensor is set to a PRF that produces four replies (two for each of the ATCRBS modes) within the antenna beamwidth. Typically, this is about one-third the PRF of a conventional sensor. This reduction in ATCRBS interrogation rate causes an immediate and significant reduction in the ATCRBS interference environment (for example: garbling, false target ID, and "fruiting") when an ATCRBS sensor is replaced by a Mode S sensor.

1.2.2.2 Mode S Monopulse Direction Finding. In addition to eliminating the disadvantages of the sliding window detector, another advantage of monopulse direction finding is its ability to aid in the decoding of synchronously garbled replies from aircraft near the same range and azimuth. An example of monopulse reply degarbling situation is shown in Figure 1-5. The monopulse estimates for each reply pulse readily identify the reply A or B to which each of the received pulses belong. In the example shown in Figure 1-5, the pulses are not overlapped so that pulse timing alone could have lead to the correct sorting of pulses into replies. Monopulse reply degarbling however continues to operate into regions of pulse overlap that could not be resolved

by pulse timing alone. Monopulse reply degarbling reduces the susceptibility of the ATCRBS mode to synchronous garble. A major problem in ATCRBS is the appearance of false targets due to reflection from large objects such as buildings or hillsides. The Mode S sensor is programmed to identify and flag such false targets using both target reply parameters and the pre-stored geometry of principal reflecting surfaces.

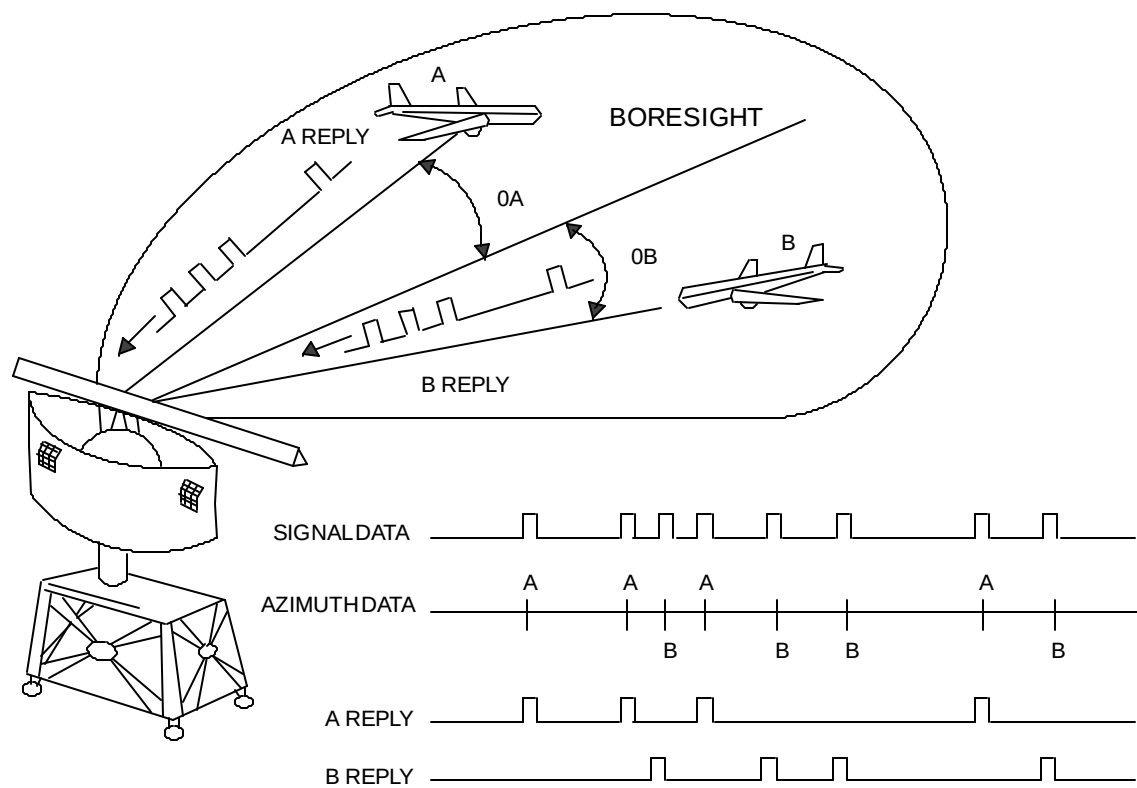


FIGURE 1-5. MODE S SENSOR MONOPULSE REPLY DEGARBLING

1.2.2.3 Mode S Performance Standards. The performance for the Mode S system are presented in the following list.

<u>Function</u>	<u>Description</u>
1. <u>Surveillance</u>	
a. Coverage	Up to 260 degrees azimuth (site dependent) Up to 255 nautical miles (site dependent)
b. Scan Rates:	Terminal: Approximately 12-13 rpm Enroute: Approximately 5-6 rpm
c. Range Accuracy:	? 30 feet, 25 feet rms jitter.

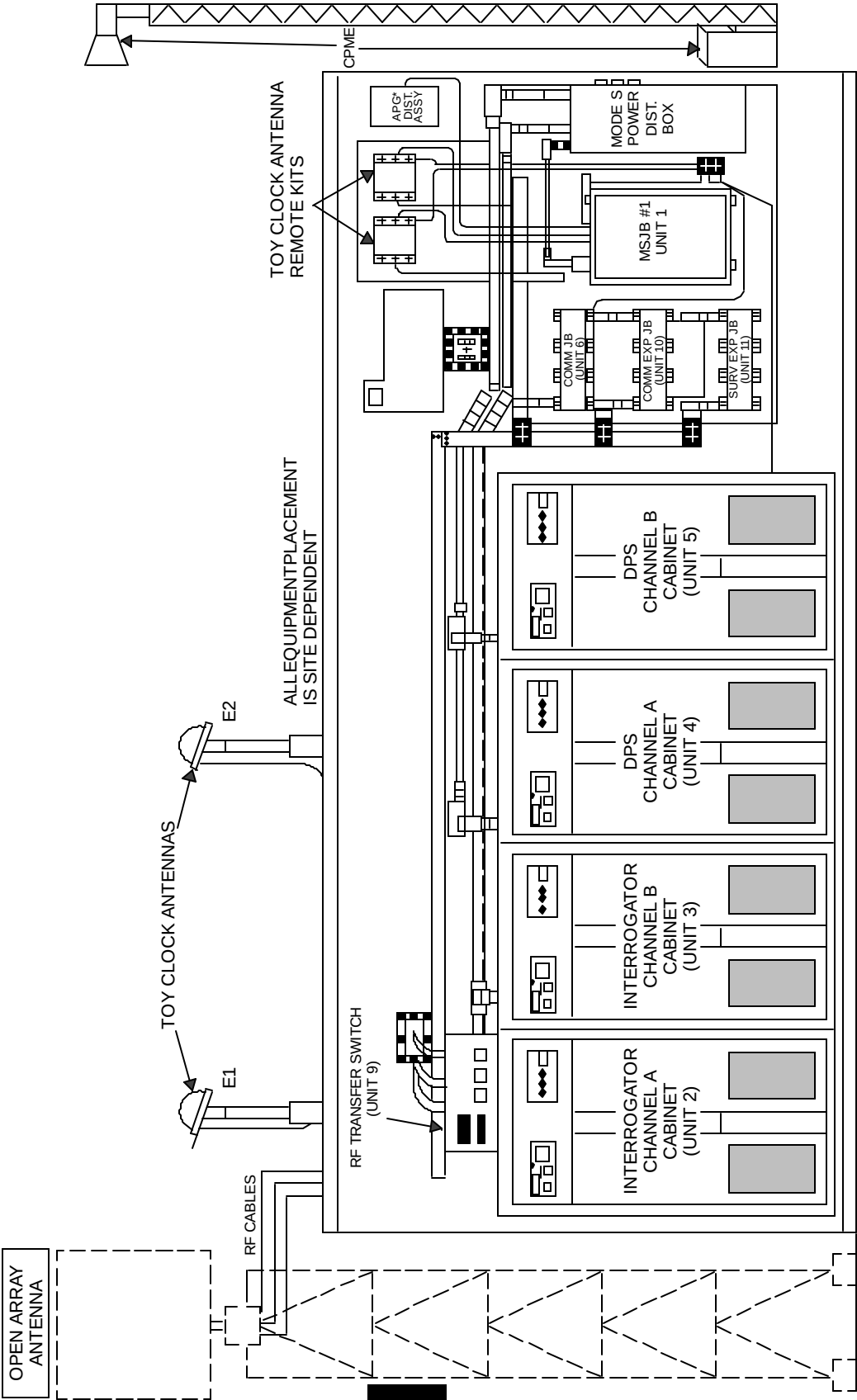
<u>Function</u>	<u>Description</u>
d. Azimuth Accuracy:	0.068 degrees, 1 sigma rms error.
e. Capacity:	Up to 700 aircraft per sensor (Certified 400 terminal/700 enroute tracks)
2. <u>Data Link</u>	
a. Capacity:	All identified ATC messages require a few per cent of available capacity.
b. Delivery Reliability:	Greater than 0.99 in one antenna scan interval for short tactical messages.
c. Undetected Error Rate:	Less than one error in 10^7 112-bit transmissions.
3. <u>System Reliability</u>	– Multiple coverage. – Automatic monitoring and network reconfiguration. – Automatic switching to standby sensor channel without loss of database in case of failure. – Remote maintenance monitor.

1.3 EQUIPMENT DESCRIPTION

Descriptions of the various equipments that comprise the Mode Select Beacon Sensor (Mode S) system are presented under the following headings: (1) Interrogator, (2) Data Processing System (DPS), (3) Calibration Performance Monitor Equipment (CPME), and (4) the power distribution equipment. Figures 1–6, 1–7 and 1–8 are illustrations that show the relationship of equipments (units) in the Mode S system. Table 1–1 provides a description of the units which comprise the Mode S equipments, the reference designator that applies to the unit, and the location of that unit in the Mode S system. If the unit is the lowest replaceable unit (LRU) for that equipment, an (LRU) designation is indicated in the Unit Nomenclature column of Table 1–1. For a more detailed breakdown of the various equipments (Interrogator, DPS, and CPME) to the LRU level, the user should refer to Section 7 of the applicable Mode S equipment instruction book (Interrogator-TI6365.5, DPS-TI6365.2, and CPME-TI6365.8).

1.3.1 Interrogator Equipment

The Interrogator equipment includes all lowest replaceable units (LRUs) and cabling within and between the time of year (TOY) clock antennas, TOY clock remote kits, Mode S junction box no. 1 (MSJB–1) (Unit 1), the two interrogator cabinets (Units 2 and 3), and the RF transfer switch (Unit 9) at terminal sites. Refer to Figure 1–7. At enroute sites, the interrogator equipment includes all LRUs and cabling within and between the TOY clock antennas, TOY clock remote kits, MSJB–1, the two Interrogator cabinets (Units 8 and 2), and the RF transfer switches (Units 13 and 9). Refer to Figure 1–8. The Interrogator equipment is functionally divided into a signal processor, a receiver/RF test target generator, a transmitter, power distribution circuits, and environmental monitoring and control circuits. The Interrogator interprets interrogation commands received from the data processing system (DPS) and sends RF transmit signals (via the transmitter) to the antenna. Interrogation replies and received from the antenna and processed in the receiver and signal processor. The Interrogator transmits the reply data to the DPS for further processing. All of the Interrogator units and circuits are tested by fault detection test software and fault isolated by fault isolation test (FIT) software. The tests reside in the signal processor and are run periodically as controlled by the signal processor, DPS, or remote maintenance monitoring system (RMMS). The FIT tests can also be controlled on-site using the KCRT terminal. The following paragraphs provide a physical and functional description of the TOY clock antennas, TOY clock remote kits, MSJB–1 cabinet and its units, and the Interrogator cabinet. Figure 1–9 is an equipment chart of the equipments that comprise the Interrogator.

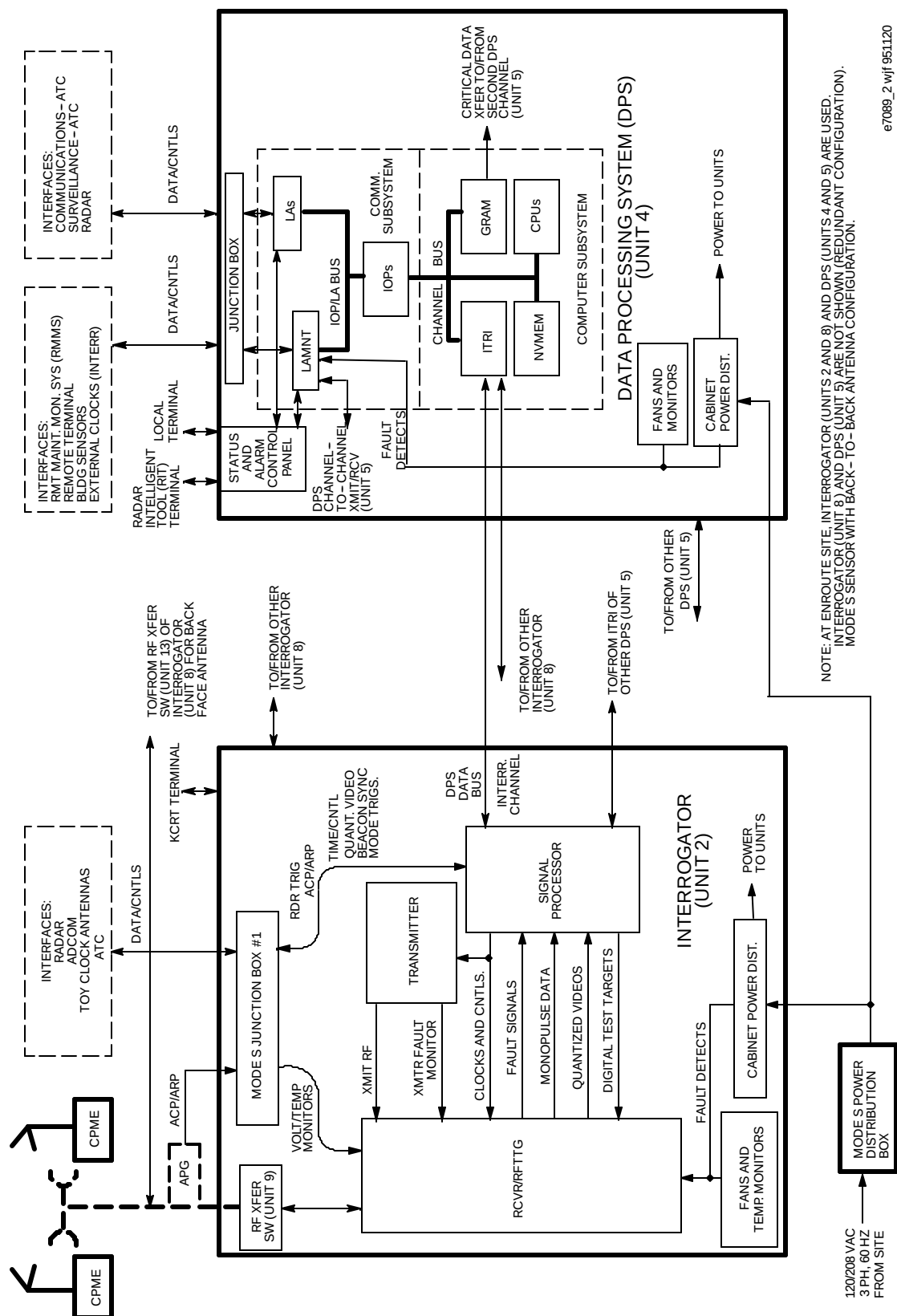


NOTE:
A THIRD INTERROGATOR CABINET IS RESERVED FOR FUTURE EXPANSION. FOR AN ENROUTE SITE. INTERROGATOR CHANNEL C CABINET (UNIT 8) AND RF TRANSFER SWITCH (UNIT 13) ARE LOCATED IN THE FIRST POSITION AND INTERROGATOR CHANNEL A CABINET (UNIT 2) AND RF TRANSFER SWITCH (UNIT 9) ARE LOCATED IN SECOND POSITION.
APG DISTRIBUTION ASSEMBLY ONLY LOCATED AT MODE-S SITES COLLOCATED WITH AN ASR-7, ASR-8, ARSR-1/2, FPS RADAR OR BEACON ONLY SITE.

FIGURE 1-6. MODE S SYSTEM PHYSICAL RELATIONSHIP OF UNITS



FIGURE 1-7. MODE S SYSTEM RELATIONSHIP OF UNITS AT A TERMINAL SITE, BLOCK DIAGRAM



e7089 2 wif 951120

TABLE 1–1. MODE S SYSTEM EQUIPMENT

Equipment	Reference Designator	Unit Nomenclature (LRU)*	Location
Interrogator	1A1	Time of Year (TOY) Clock Assembly No. 1 (LRU)	MSJB–1 cabinet
Interrogator	1A2	Time of Year (TOY) Clock Assembly No. 2 (LRU)	Refer to TI 6365.5
Interrogator	1A3	Video Trigger/Driver (LRU)	Refer to TI 6365.5
Interrogator	1PS1	Toy Clock Power Supply No. 1 (LRU)	Refer to TI 6365.5
Interrogator	1PS2	Toy Clock Power Supply No. 2 (LRU)	Refer to TI 6365.5
Interrogator	1PS3	+24V Power Supply (LRU)	Refer to TI 6365.5
Interrogator	1PS4	+5V Power Supply (LRU)	Refer to TI 6365.5
Interrogator	1RT1/1RT2	RT1 and RT2 Temperature Sensors (LRUs)	MSJB–1 cabinet
Interrogator	2/3/8A1	Control Panel	Interrogator cabinet
Interrogator	2/3/8A2	Connector Panel	Refer to TI 6365.5
Interrogator	2/3/8A17	Signal Monitor Panel	Interrogator cabinets
Interrogator	2/3/8A18	Blower Assembly	Units 2 and 3 are used at
Interrogator	2/3/8PS1	+15V Power Supply (LRU)	terminal sites.
Interrogator	2/3/8PS2	–15V Power Supply (LRU)	
Interrogator	2/3/8PS3	+5V Power Supply (LRU)	Interrogator cabinets
Interrogator	2/3/8PS4	+36V Power Supply (LRU)	Units 8 and 2 are used at
Interrogator	2/3/8PS5	+52V Power Supply (LRU)	enroute sites.
Interrogator	2/3/8PS6	Bulk Power Supply (LRU)	
Interrogator	2/3/8RT1 2/3/8RT8	RT1 through RT8 Temperature Sensors (LRU)	Interrogator cabinet
Interrogator	—	Receiver/RFTTG	Interrogator cabinet
	2/3/8A3	Video Processor/Monitor (LRU)	
	2/3/8A5	IF Monopulse Receiver (LRU)	Refer to TI 6365.7
	2/3/8A6	Omni Diplexer (LRU)	
	2/3/8A9	Local Oscillator (LRU)	
	2/3/8A19	Monopulse Diplexer (LRU)	
	2/3/8A20	RF Receiver (LRU)	
	2/3/8A14	RF Test Target Generator (LRU)	

TABLE 1-1. MODE S SYSTEM EQUIPMENT (Continued)

Equipment	Reference Designator	Unit Nomenclature (LRU)*	Location
Interrogator	2/3/84A	Signal Processor:	Interrogator cabinet
	2/3/8A4A103	ATCRBS Bracket Detector (LRU)	
	2/3/8A4A104/A105	ATCRBS Code Processor (LRU)	Refer to TI 6365.4
	2/3/8A4A106	ATCRBS Output FIFO (LRU)	
	2/3/8A4A107	Test Target Generator (LRU)	
	2/3/8A4A109	Mode S Interface (LRU)	
	2/3/8A4A110	Mode S Message Decoder (LRU)	
	2/3/8A4A111	Timing Expansion (LRU)	
	2/3/8A4A112	Timing Generator (LRU)	
	2/3/8A4A113	Modulation Control (LRU)	
	2/3/8A4A114	Independent ATCRBS (LRU)	
	2/3/8A4A115	Sensitivity Time Control (STC) Generator (LRU)	
	2/3/8A4A116	VSWR Monitor (LRU)	
	2/3/8A4A117	CCS I/O (LRU)	
	2/3/8A4A118	CPU Interface (LRU)	
	2/3/8A4A119	TOY/ARIES Interface (LRU)	
	2/3/8A4A122	CPU 68020 (LRU)	
Interrogator	———	Transmitter:	Interrogator cabinet
	2/3/8A7	Auxiliary Modulator/Driver (LRU)	
	2/3/8A8	Primary Modulator/Driver (LRU)	Refer to TI 6365.6
	2/3/8A10	Auxiliary Output Level Control (LRU)	
	2/3/8A11	Primary Output Level Control (LRU)	
	2/3/8A12	Auxiliary Capacitor Bank (LRU)	
	2/3/8A13	Auxiliary Power Amplifier (LRU)	
	2/3/8A15	Primary Capacitor Bank (LRU)	
Interrogator	2/3/8A16	Primary Power Amplifier (LRU)	
Interrogator	Unit 9	Transfer Switch Assembly	Top of Interrogator cabinet Unit 2
Interrogator	Unit 13	Transfer Switch Assembly	Top of Interrogator cabinet Unit 8
Interrogator	Unit 14	Transfer Switch Assembly	Reserved for future expansion capability

TABLE 1–1. MODE S SYSTEM EQUIPMENT (Continued)

Equipment	Reference Designator	Unit Nomenclature (LRU)*	Location
Interrogator	E1/E2	Time Of Year Clock Antennas (LRU)	Site
Interrogator		Time of Year Clock Remote Kits	Site building wall
Mode S Sensor		Power Distribution Panel	Site building wall
Interrogator		Power Distribution Box	Site building wall
Data Processing System (DPS)	4/5A1	I/O Connector Panel	Top rear area of PRCSS cabinet
DPS	4/5A2	Main Circuit Breaker Panel Assembly (LRU)	Top rear area of PRCSS cabinet
DPS	4/5A3	Control Panel Assembly, Status and Alarm (LRU)	Top rear area of PRCSS cabinet
DPS	4/5A4	Power Distribution Box Assembly	Bottom rear area of PRCSS cabinet
DPS	4/5A5	Temperature Sensor Assembly (LRU)	Top area of PRCSS cabinet
DPS	4/5A6	Temperature Sensor Assembly (LRU)	Top area of PRCSS cabinet
DPS	4/5B1	Backplane, Line Adapter (LRU)	Upper area of PRCSS cabinet
DPS	4/5B2	Backplane, Channel Bus (LRU)	Lower area of PRCSS cabinet
DPS	4/5B3	Blower Assembly, Centrifugal (LRU)	Lower area of PRCSS cabinet (front view)
DPS	4/5B4	Gate Circuit Breaker Panel Assembly	Upper area of PRCSS cabinet (front view)
DPS	4/5B5	Terminal Board Assembly (LRU)	Lower area of PRCSS cabinet
DPS	4/5B6	Power Supply Surveillance Data Selector Assembly (LRU)	Lower area of PRCSS cabinet
DPS	4/5B3/A1	Temperature Sensor Assembly (LRU)	Lower area of PRCSS cabinet
DPS	4/5B/PS1	+5 Volt Power Supply Assembly (LRU)	Lower area of PRCSS cabinet adjacent to B2 area

TABLE 1–1. MODE S SYSTEM EQUIPMENT (Continued)

Equipment	Reference Designator	Unit Nomenclature (LRU)*	Location
DPS	4/5B/PS2	+5 Volt Power Assembly Assembly (LRU)	Lower area of PRCSS cabinet below B/PS1
DPS	4/5B/PS3	+5 Volt Power Assembly (LRU)	Lower area of PRCSS cabinet opposite B/PS1
DPS	A/FL1	AC Line Filter (LRU)	Top area of PRCSS cabinet
DPS	A/VR1–VR4	Varistor Module (LRU)	Top area of PRCSS cabinet
DPS		Communication Junction Box Assembly (CJBX) with SDS Kit (2696 0161) (LRU)	Enroute site with CD–2 equipment
DPS		Communication Junction Box Assembly Kit (3594 1715) (LRU)	Terminal site
DPS		1. Communication Junction Box Assembly Kit (2696 0161) (LRU)	Technical Center Terminal Site No. 1
		2. Communication Expansion Junction Box Assembly Kit (3594 1574) (LRU)	
		3. Surveillance Expansion Junction Box Assembly Kit (3594 1582) (LRU)	
DPS		1. Communication Junction Box Assembly with SDS Kit (2696 0161) (LRU)	Technical Center Enroute Site No. 2
		2. Communication Expansion Junction Box Assembly Kit (3594 1574) (LRU)	
		3. Surveillance Expansion Junction Box Assembly Kit (3594 1582) (LRU)	
DPS		CD–2 Modification Kit (2695 8488) (Part of Mode S/DPS equipment installed in CD–2 Equipment)	Enroute Site with CD–2 Equipment

TABLE 1-1. MODE S SYSTEM EQUIPMENT (Continued)

Equipment	Reference Designator	Unit Nomenclature (LRU)*	Location
DPS		CD-2 Modification Kit (2695 8488) (Part of Mode S/DPS equipment installed in CD-2 Equipment	Technical Center Enroute Site No. 2 with SDS
Calibration Performance Monitor Equipment (CPME)	Unit 1	Battery Box Refer to Section 7 of TI 6365.8 for LRU identification.	Mounted along side of Electrical Equipment cabinet
CPME	Unit 2	Electrical Equipment Cabinet Refer to Section 7 of TI 6365.8 for LRU identification.	On concrete or asphalt pad inside building or on side of building
CPME	2A1	Receiver/Transmitter (LRU)	Top area of Electrical Equipment cabinet
CPME	2A2	Power Supply (LRU)	Middle area of Electrical Equipment cabinet
CPME	2A3	Hinged Panel Assembly	Bottom area of Electrical Equipment cabinet
CPME	Unit 3	CPME Antenna (LRU)	Mounted on a pole or mounted on side of building
CPME	W1	Antenna RF cable	Connector for equipment to antenna

Note: (LRU)* = Lowest Replaceable Unit.

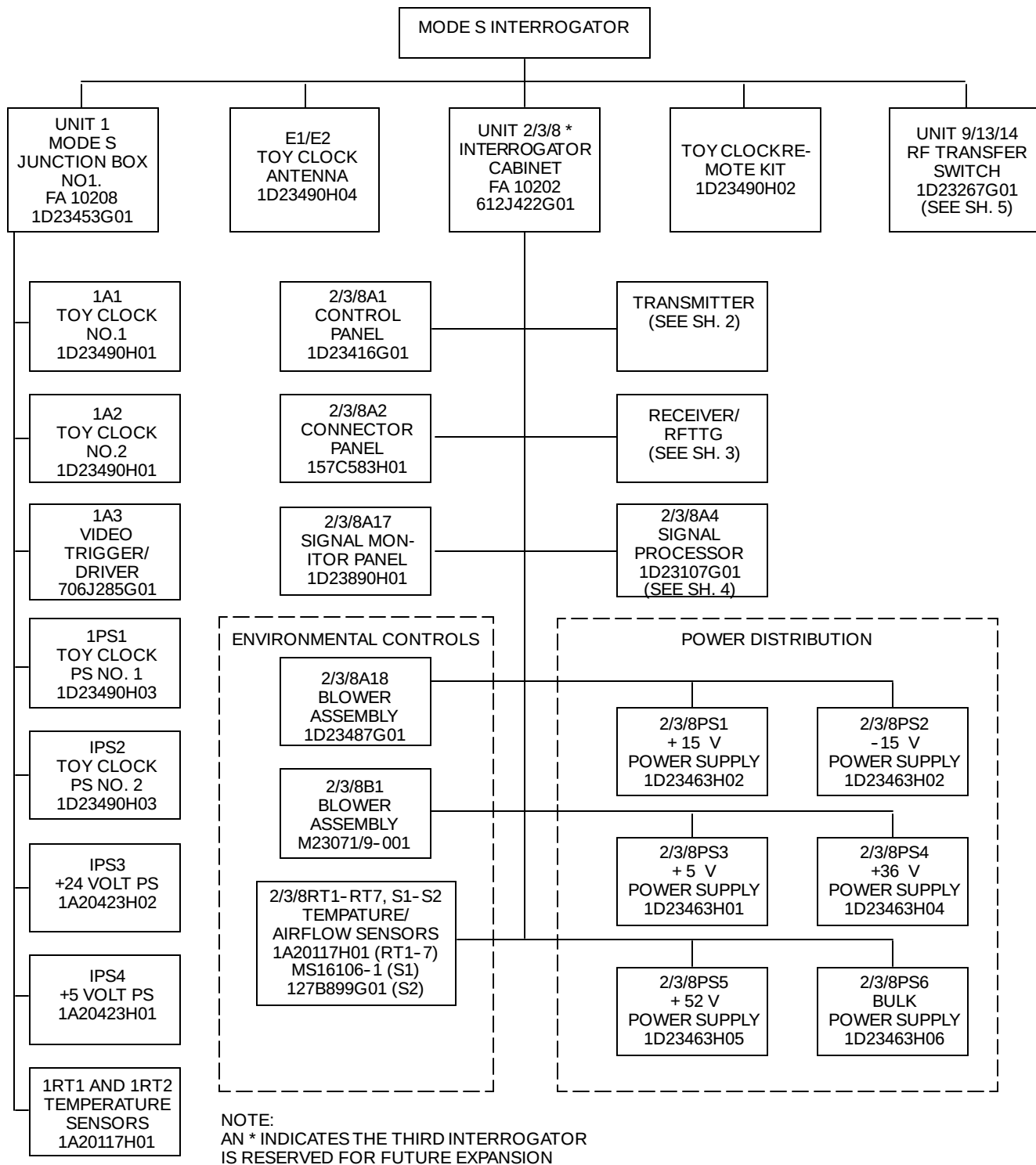


FIGURE 1-9. INTERROGATOR EQUIPMENT CHART (SHEET 1)

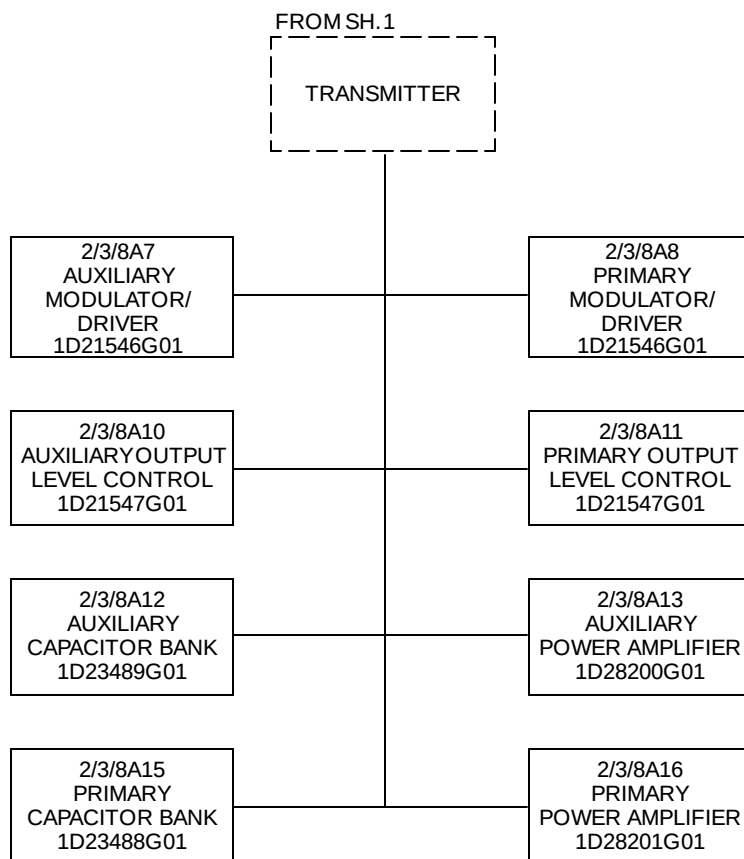


FIGURE 1-9. INTERROGATOR EQUIPMENT CHART (SHEET 2)

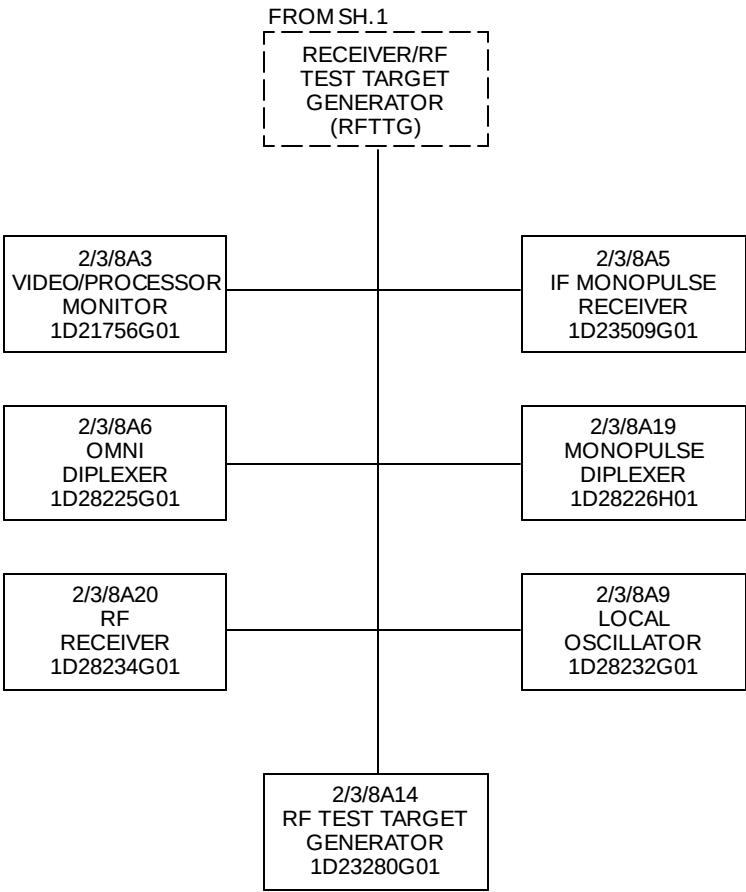


FIGURE 1–9. INTERROGATOR EQUIPMENT CHART (SHEET 3)

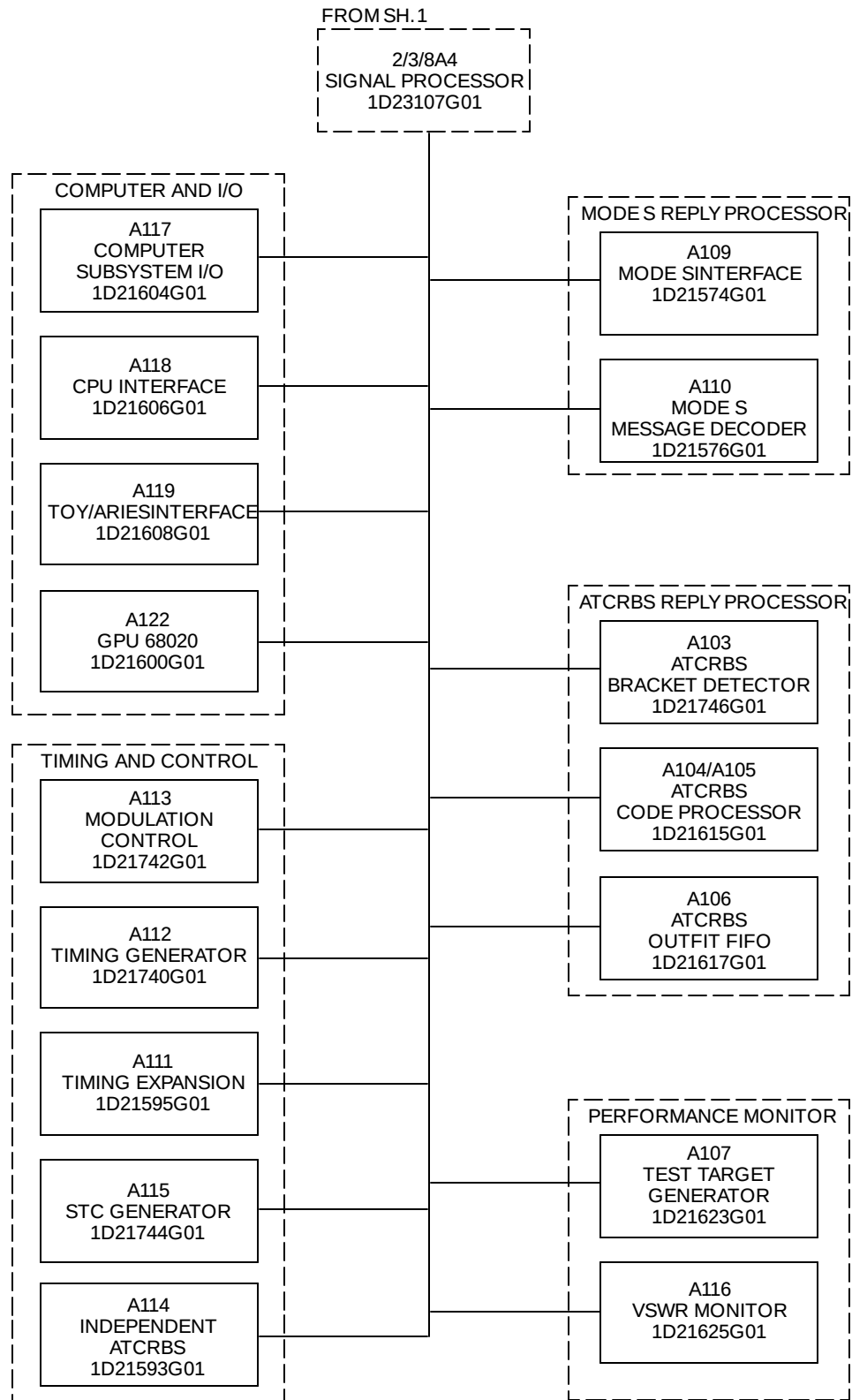
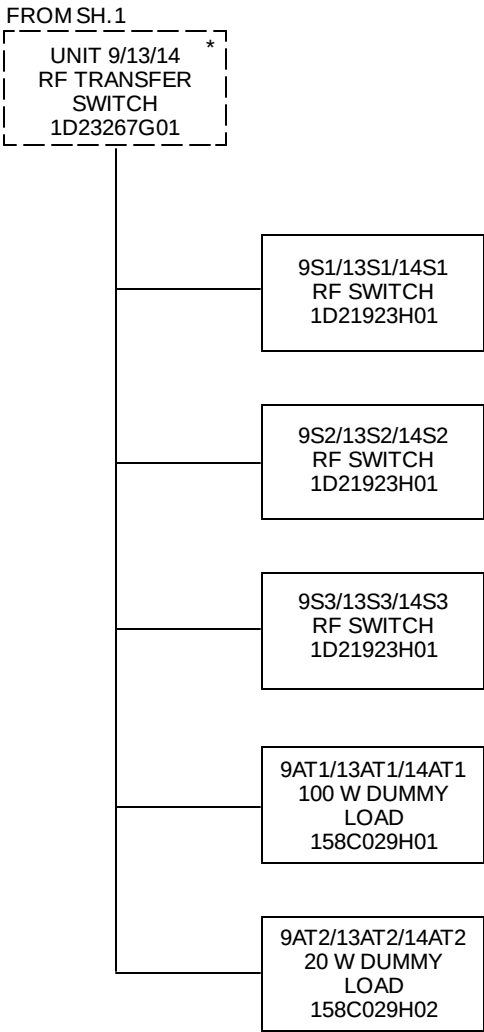


FIGURE 1-9. INTERROGATOR EQUIPMENT CHART (SHEET 4)



NOTE:
AN * INDICATES RF TRANSFER SWITCH (UNIT 14)
IS RESERVED FOR FUTURE EXPANSION

FIGURE 1–9. INTERROGATOR EQUIPMENT CHART (SHEET 5 — END)

1.3.1.1 Time of Year Clocks. The two time of year (TOY) clocks each consist of a TOY clock antenna, a remote kit, and a TOY clock assembly. Each TOY clock antenna receives time of year signals from a satellite and sends the signals to the associated remote kit. The remote kit contains a preamplifier that increases signal strength for the associated TOY clock assembly in the MSJB-1 cabinet. Power to and signals from each remote kit and antenna are sent through a coaxial cable. Refer to the Mode S Interrogator document TI 6365.5 for a detailed equipment description and additional information concerning the time of year (TOY) clocks.

1.3.1.2 Mode S Junction Box No. 1. The Mode S junction box no. 1 (MSJB-1) contains two TOY clocks, two TOY clock uninterruptible power supplies (UPS), a video trigger/driver, a +5 V power supply, a +24 V power supply, two temperature sensors, and power control and monitoring equipment. The TOY clocks provide a stable and accurate time of day for the signal processor. The video trigger/driver is an interface between the interrogator and equipment such as the azimuth pulse generator and radar. The temperature sensors send signals to the signal processor (via the receiver/RFTTG) indicating power and cabinet heat conditions. Power control/monitoring equipment includes a circuit breaker to protect against overloading conditions and an indicator light that lights whenever power from the circuit breaker is off. Each power supply, TOY clock, and temperature sensor sends fault detection signals to the signal processor through the receiver/RFTTG. The signal processor monitors the fault detect signals and declares a fault if any of the signals are out of tolerance. The MSJB-1 is an enclosure that hangs on the wall.

1.3.1.2.1 Refer to the Mode S Interrogator document TI 6365.5 for additional information concerning the Mode S junction box no. 1: (1) time of year (TOY) clocks, (2) video trigger/driver board, (3) TOY clock uninterruptible power supplies, (4) MSJB-1 +5-volt power supply, and (5) MSJB-1 +24-volt power supply.

1.3.1.3 Interrogator Cabinet. Each interrogator cabinet contains the transmitter, receiver/RFTTG, signal processor, environment monitoring, and power control functions for a single channel. The following discussion applies to each of the interrogator cabinets. Power and signal interfaces are made through connector panel A2 on top of the cabinet. The control and signal monitor panels are used during on-site maintenance to control/monitor cabinet power and temperature, monitor signals during adjustments and troubleshooting, and initiate tests. The power distribution LRUs are: the bulk supply, +5 volt, +15 volt, +36 volt, and +52 volt power supplies. The environment monitors are the temperature sensors RT1 through RT8. The airflow sensors are A18S1 and A18S2.

1.3.1.3.1 Refer to Mode S Interrogator document TI 6365.5 for additional information concerning the Interrogator cabinets, such as (1) the receiver/RF test target generator, (2) signal processor (3) transmitter, (4) control panel, (5) connector panel, (6) signal monitor panel, (7) blower assembly, (8) power supplies (+15V, -15V, +5V, +36V, +52V, and bulk), (9) environmental sensors, and (10) transfer switch assembly.

1.3.2 Data Processing System (DPS)

The various equipments that comprise the Data Processing System (DPS) are indicated in Table 1-1. An equipment chart for the DPS is presented in the 13 sheets of Figure 1-10. The following list describes the information contained on each page of the DPS Equipment Chart.

1. Mode S Site Configuration DPS Option List 7103934 (sheet 1 of figure 1-10).
2. Radar Kit 3594 1525-XX, Surveillance Kit 3594 1541-XX, and Communication Kit 3594 1558-XX (sheet 2 of figure 1-10).
3. Surveillance Data Selector (SDS) Kit 3594 1566-XX (sheet 3 of figure 1-10).
4. Communication Expansion Junction Box 3594 1574 (sheet 4 of figure 1-10).
5. Surveillance Expansion Junction Box Kit 3594 1582 (sheet 5 of figure 1-10).
6. CD-2 Modification Kit 2695 8488-XX (sheet 6 of figure 1-10).

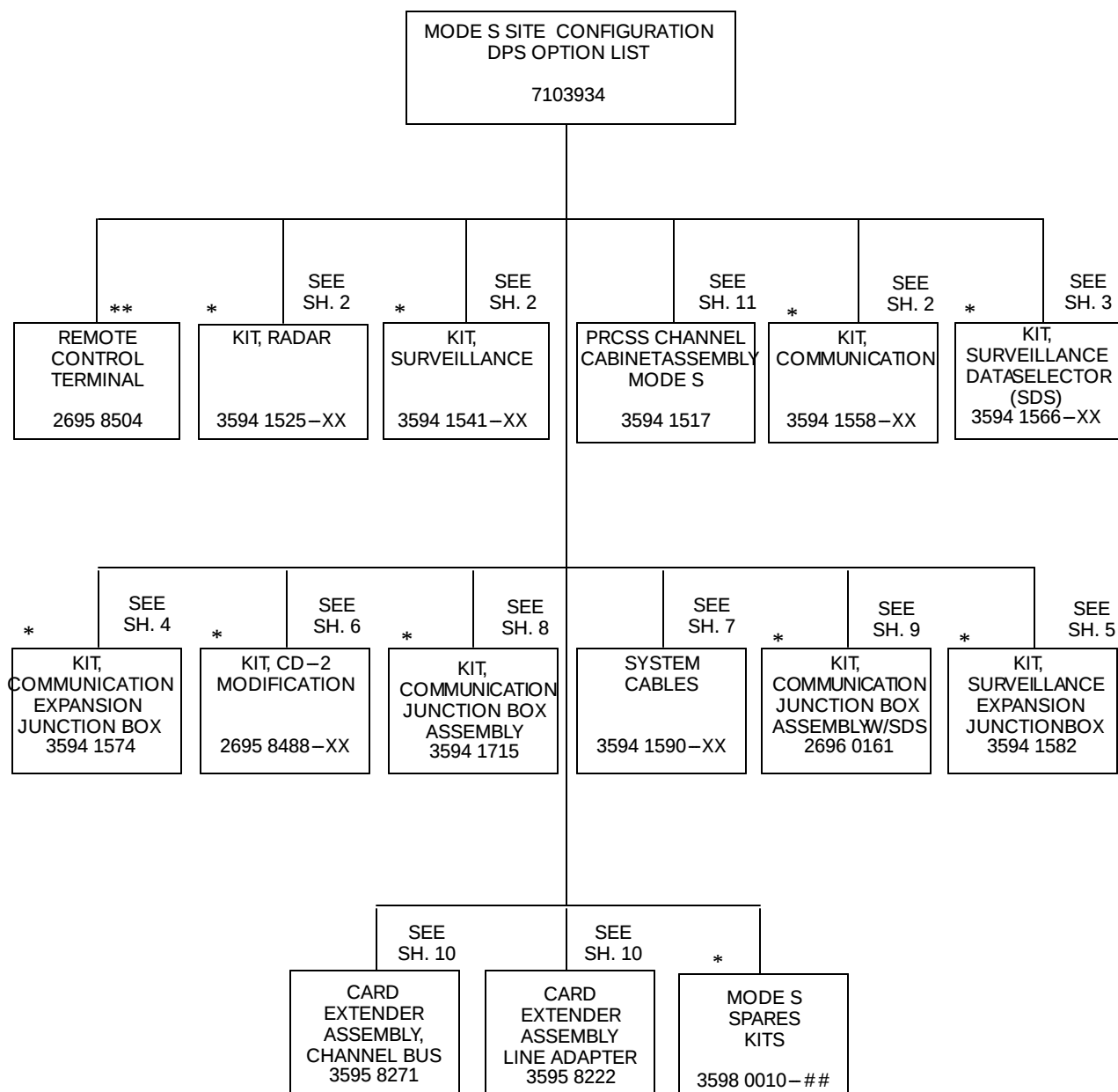
7. System Cables Basic 3594 1590–XX (sheet 7 of figure 1–10).
8. Communication Junction Box Assembly Kit 3594 1715 (sheet 8 of figure 1–10).
9. Communication Junction Box Assembly with SDS Kit 2696 0161 (sheet 9 of figure 1–10).
10. Card Extender Line Adapter (240 pin) 3595 8222 and Card Extender Channel Bus (320 pin) 3595 8271 (sheet 10 of figure 1–10).
11. PRCSS Channel Cabinet Assembly 3594 1517 (sheet 11 of figure 1–10).
12. Circuit Card Assembly (CCA Complement 3594 1608 (sheet 12 of figure 1–10).
13. Cable Card Assembly (CA CD ASSY) Complement 3594 1616 (sheet 13 of figure 1–10).

The DPS is comprised of two redundant Processor subsystem (PRCSS) cabinets, one for channel A and the other for channel B (one used as operational equipment and the other as backup equipment); a communication junction box (CJBX) or boxes depending on the DPS site configuration; a sensor control terminal, which is present at all site configurations; and the power subsystem.

1.3.2.1 Processor Subsystem (PRCSS) Cabinet. The Processor subsystem (PRCSS) cabinet, which is 80-inches high, 48-inches wide, and 34-inches deep, is comprised of the various units (equipments) listed in Table 1–1. Refer to section 1 of the DPS manual, TI 6365.3, for an area reference designation diagram of the various equipments that comprise the PRCSS cabinet.

1.3.2.1.1 The PRCSS consists of two redundant cabinets, (Units 4 and 5). These cabinets are identical in structure and function; one is intended to backup the other in the event of failure and shutdown of the operating PRCSS. Each cabinet is of heavy-duty construction, and is adequately shielded to guard against RFI/EMI leakage. The top of each cabinet houses the AC input openings, the air exhaust grille, and the I/O Connector Panel (A1) that provides connections to the comm junction box, to the other PRCSS, and to the Interrogator. The top of the cabinet also houses the Temperature Sensor assemblies (A5 and A6), used to activate indicators on status and alarm panel. The top front of each cabinet includes two control panels. The Status and Alarm Control Panel (A3) is on the right. It includes the power alarm enable switch, the lamp test and channel reset button, two temperature indicators (one for elevated temperature conditions and one for overheat), module power fault indicator, and connections for two external tape drives (TDIC CCA in RIT terminal emulates the tape drives and is used for loading System Image and for data extraction) and the local terminal. The Main Circuit Breaker Panel (A2), on the left, includes three ganged circuit breakers for system overload protection, and main power indicators.

1.3.2.1.2 Double doors on the front of the cabinet provide access to the card racks and power supplies inside. The card racks, power supplies, terminal boards etc. are mounted to a hinged gate that swings out of the cabinet for easy access to the electronics within. The PRCSS cabinet includes two card racks: the channel bus card module (B2) and the line adapter card module (B1). However, most installations populate only the channel bus card rack. In systems with expanded surveillance and communication functions, the line adapter card module supports all communications and surveillance line adapters, up to a maximum of six each. To the left of the line adapter card module is a panel of internal circuit breakers (B4) that protect the individual main power supplies, the blower assembly of the DPS, and the SDS power supply (if used).



- NOTES :
1. AN * DENOTES OPTIONAL KITS TO BE SELECTED.
 2. XX INDICATES WHERE DASH NUMBERS ARE USED FOR CABLE LENGTH OPTIONS.
 3. ## INDICATES WHERE DASH NUMBERS ARE USED FOR SPARE KIT TYPES.
 4. OTHER DASH NUMBERS ARE INTENTIONALLY OMITTED.
 5. ** NOT USED AS REMOTE TERMINAL, MAY BE USED AS KCRT TERMINAL

FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 1)

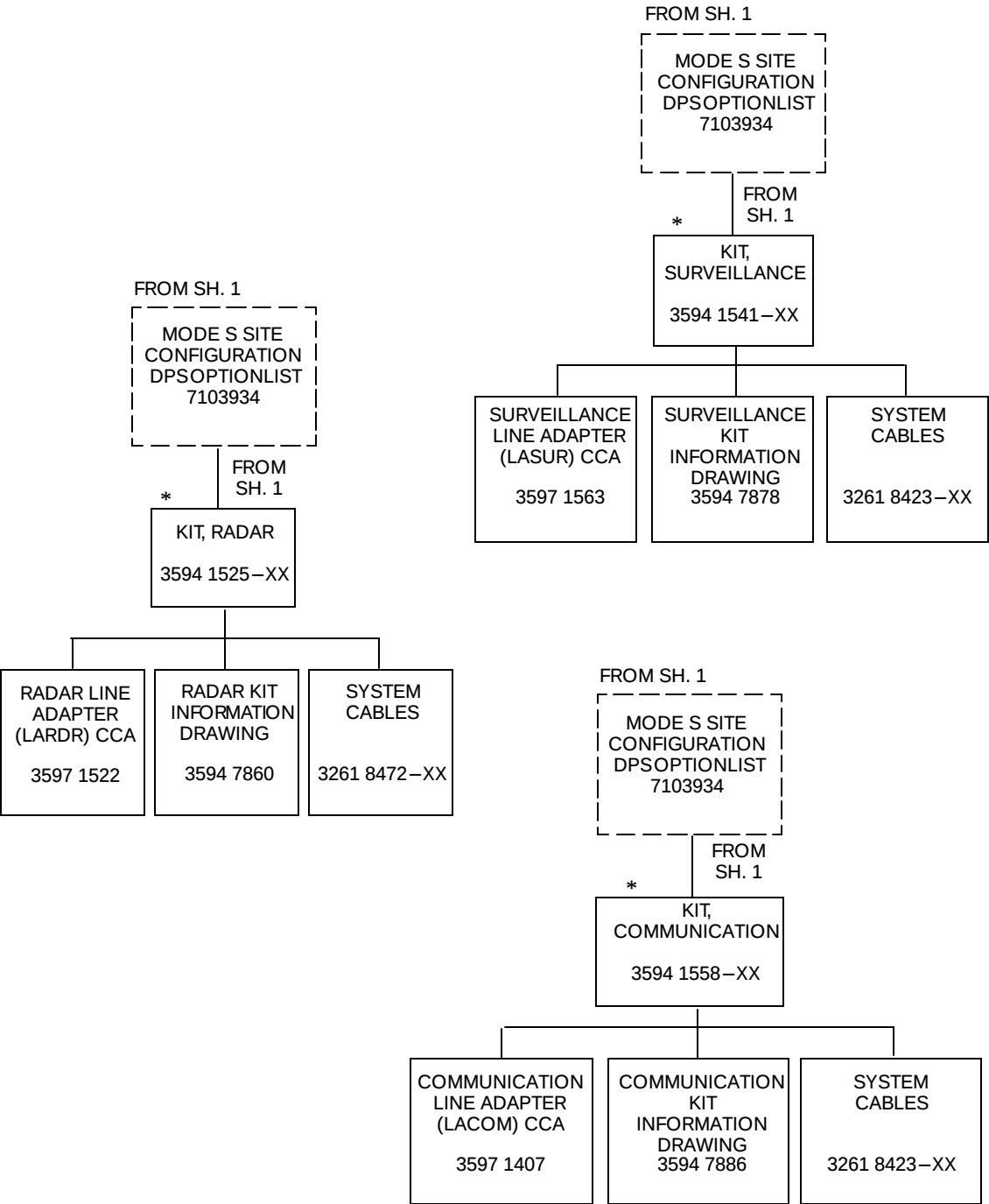
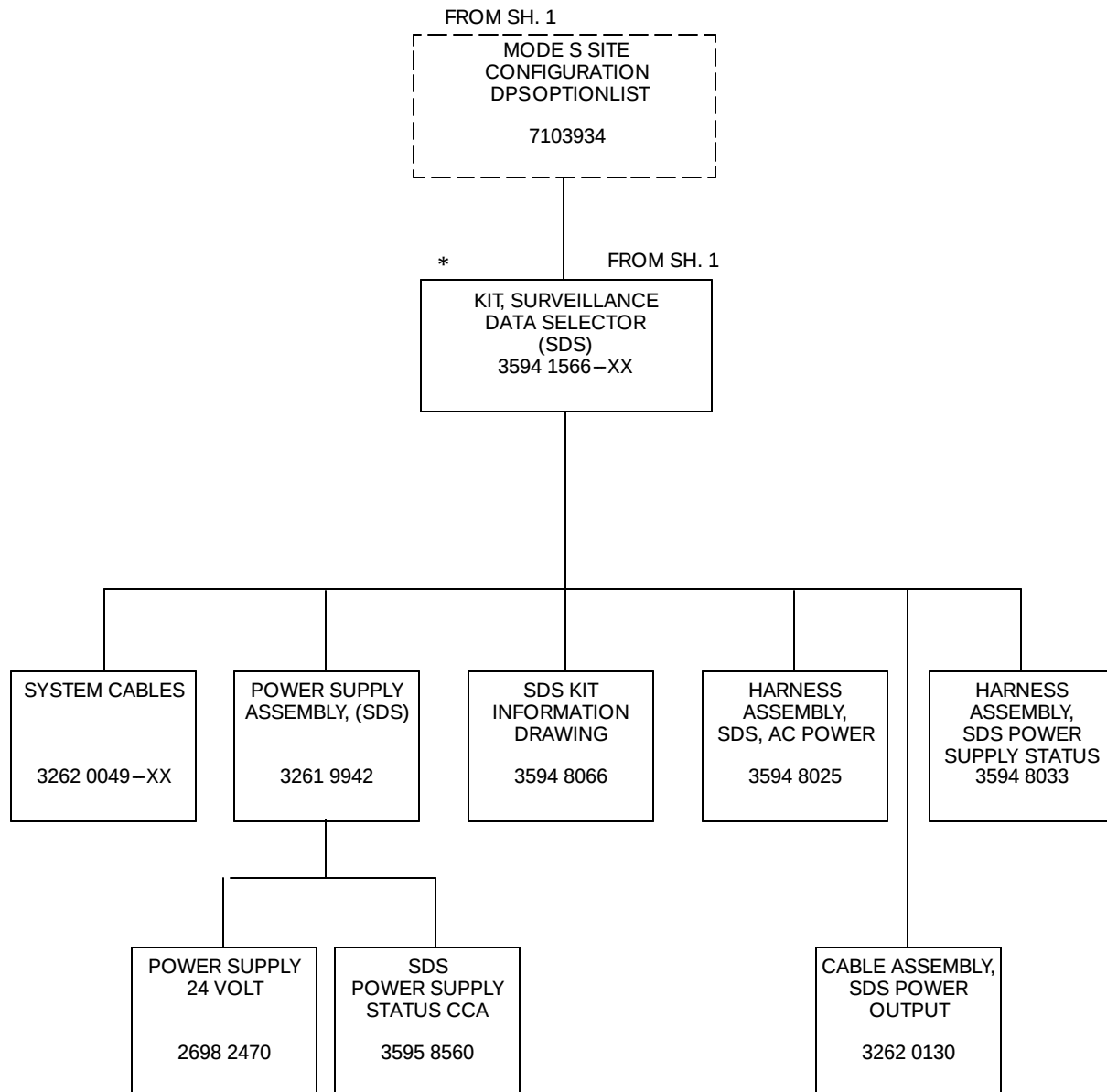


FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 2)



d4313_11_r4_920327_wjtf

FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 3)

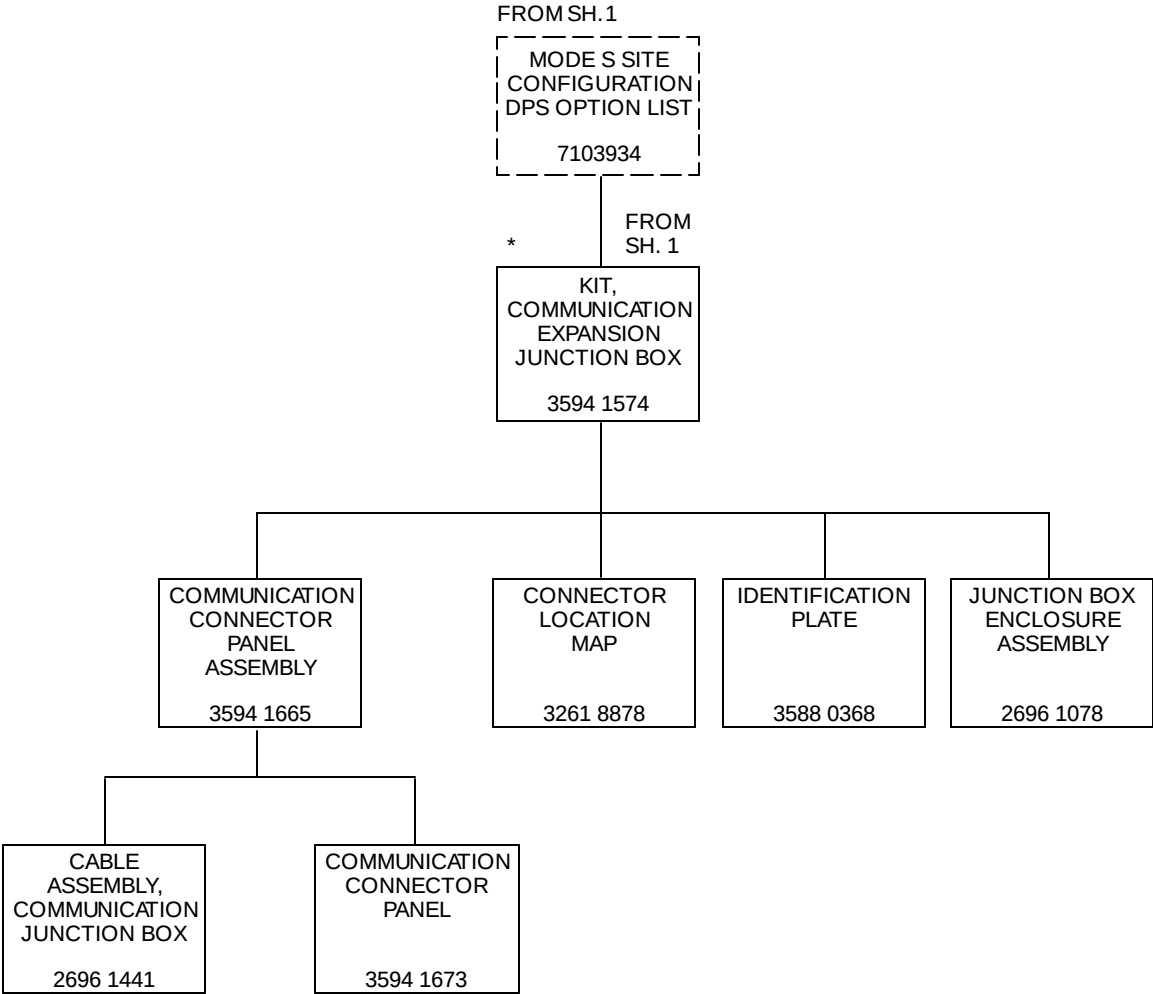


FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 4)

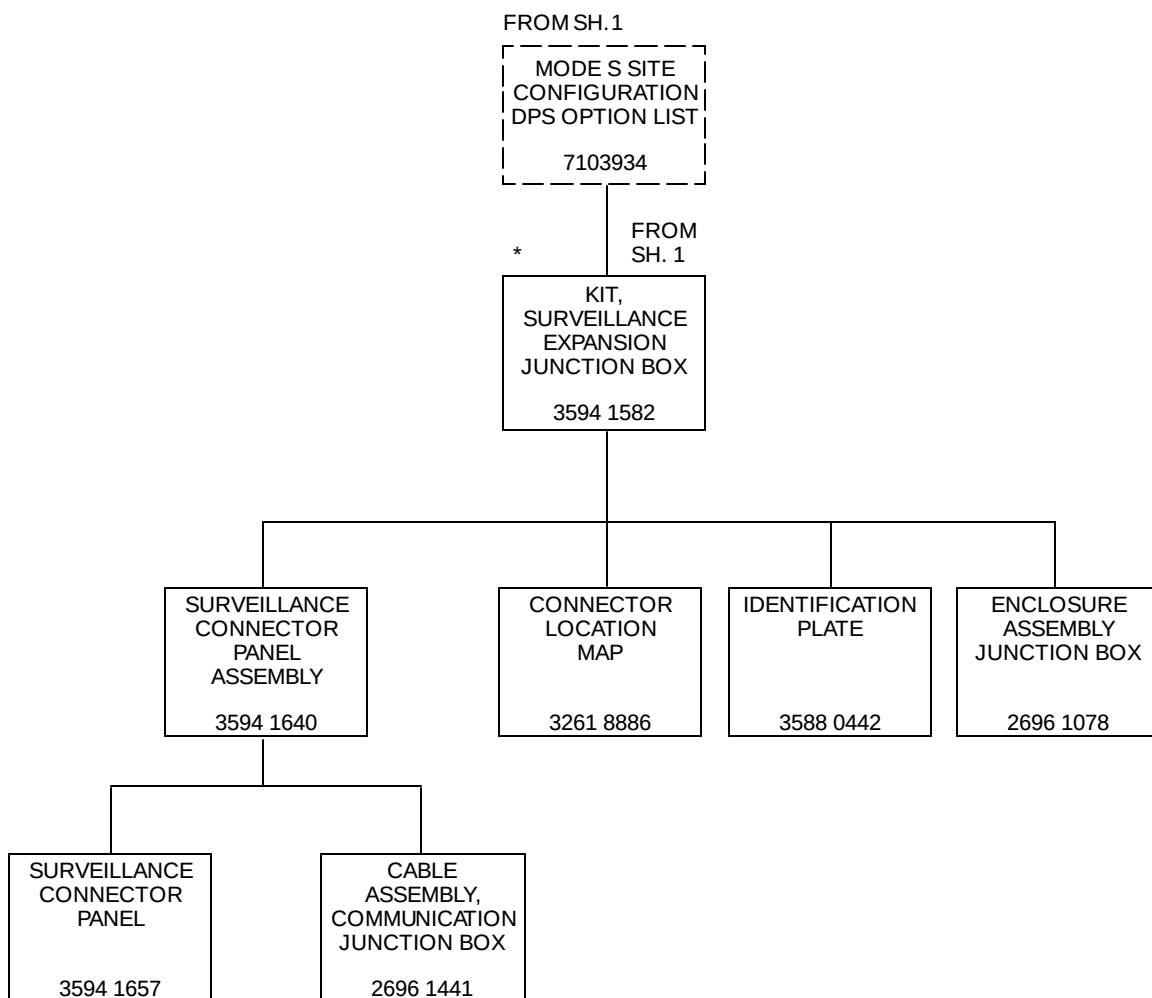


FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 5)

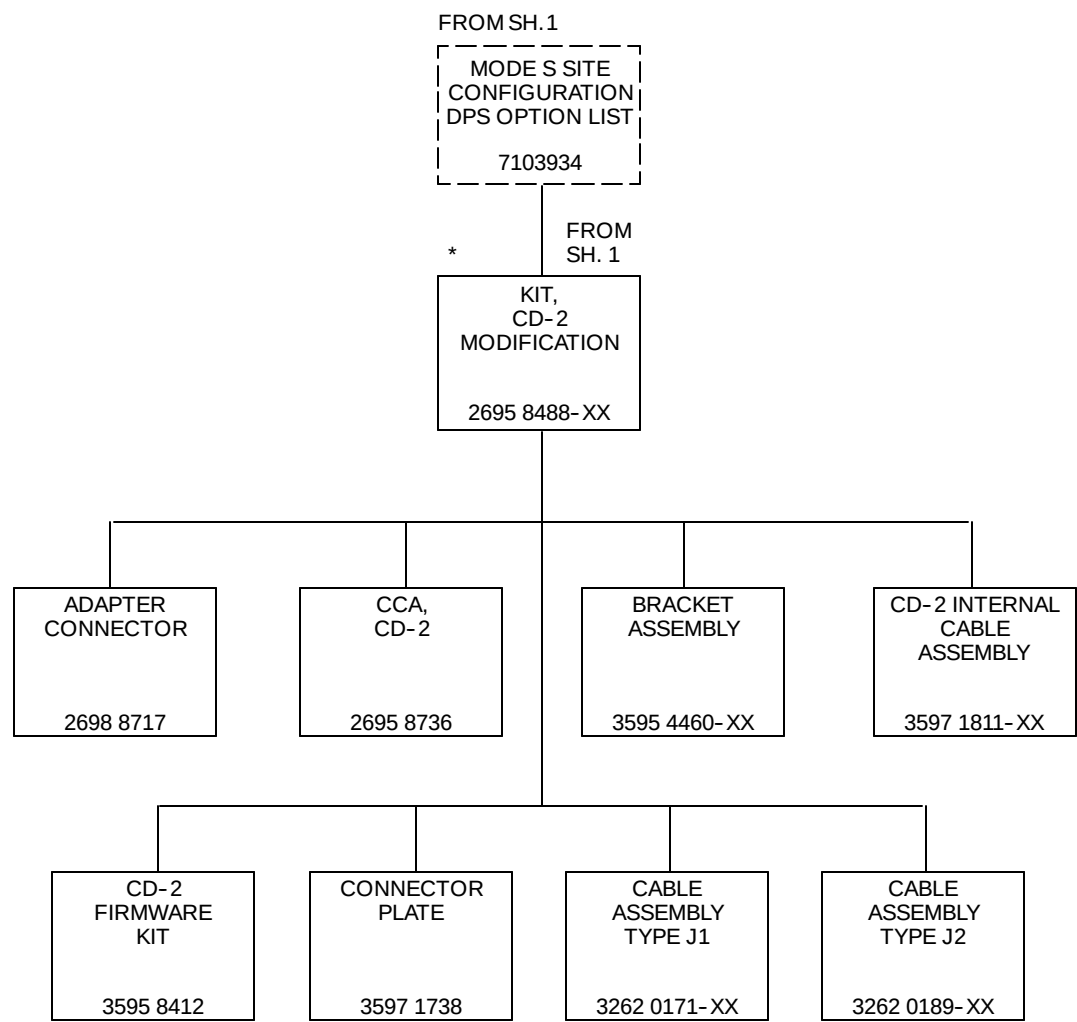


FIGURE 1–10. DPS EQUIPMENT CHART (SHEET 6)

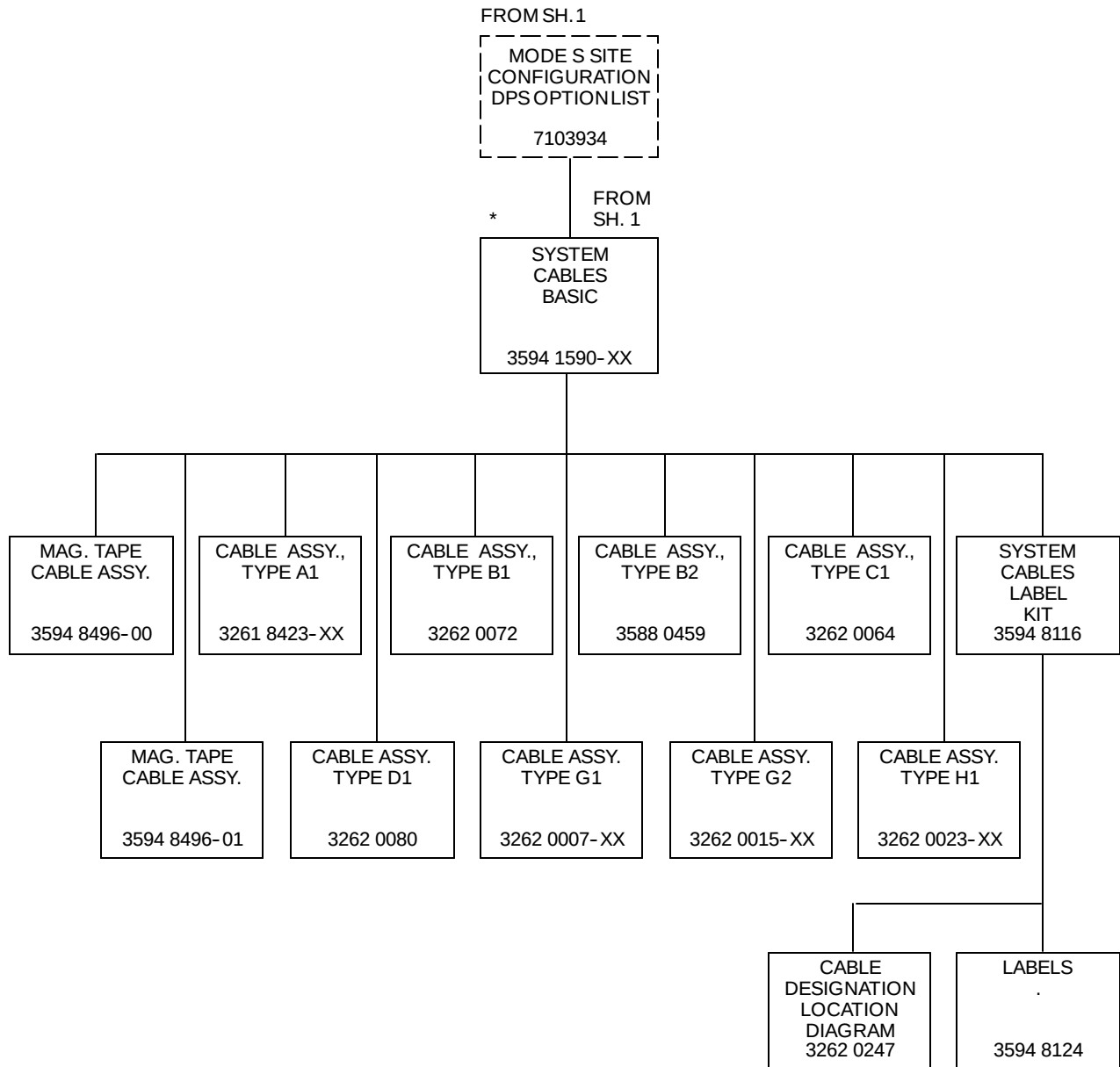
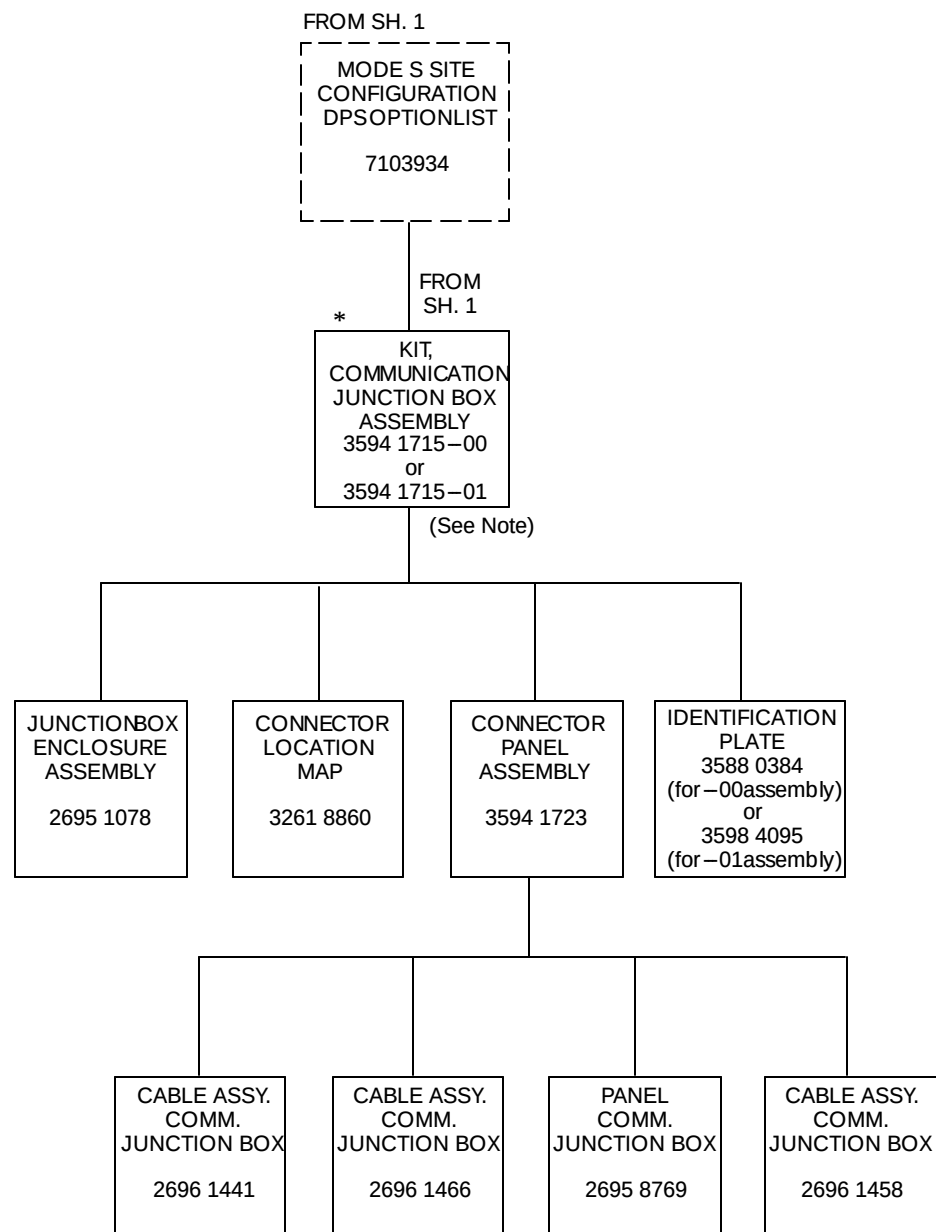
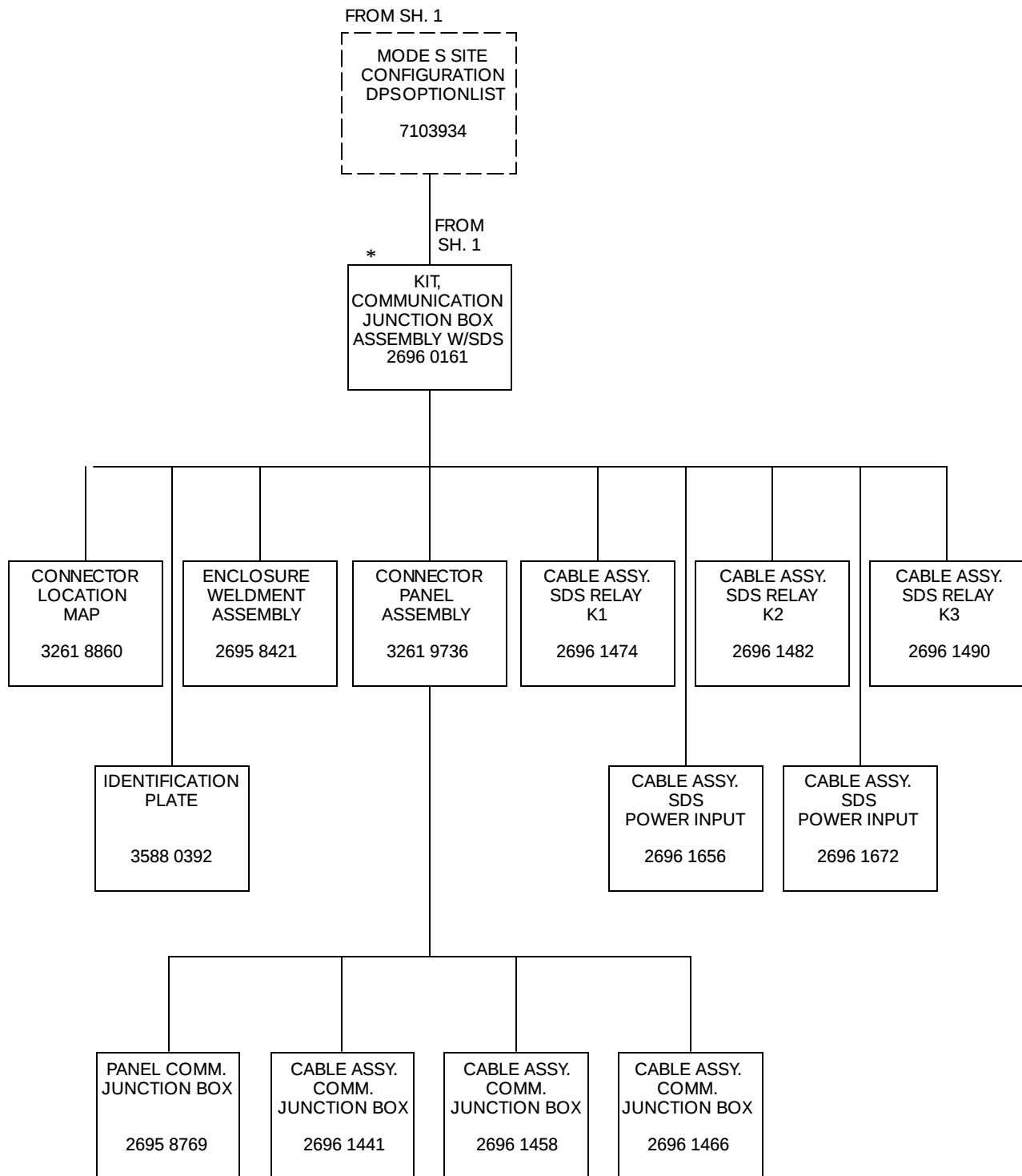


FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 7)



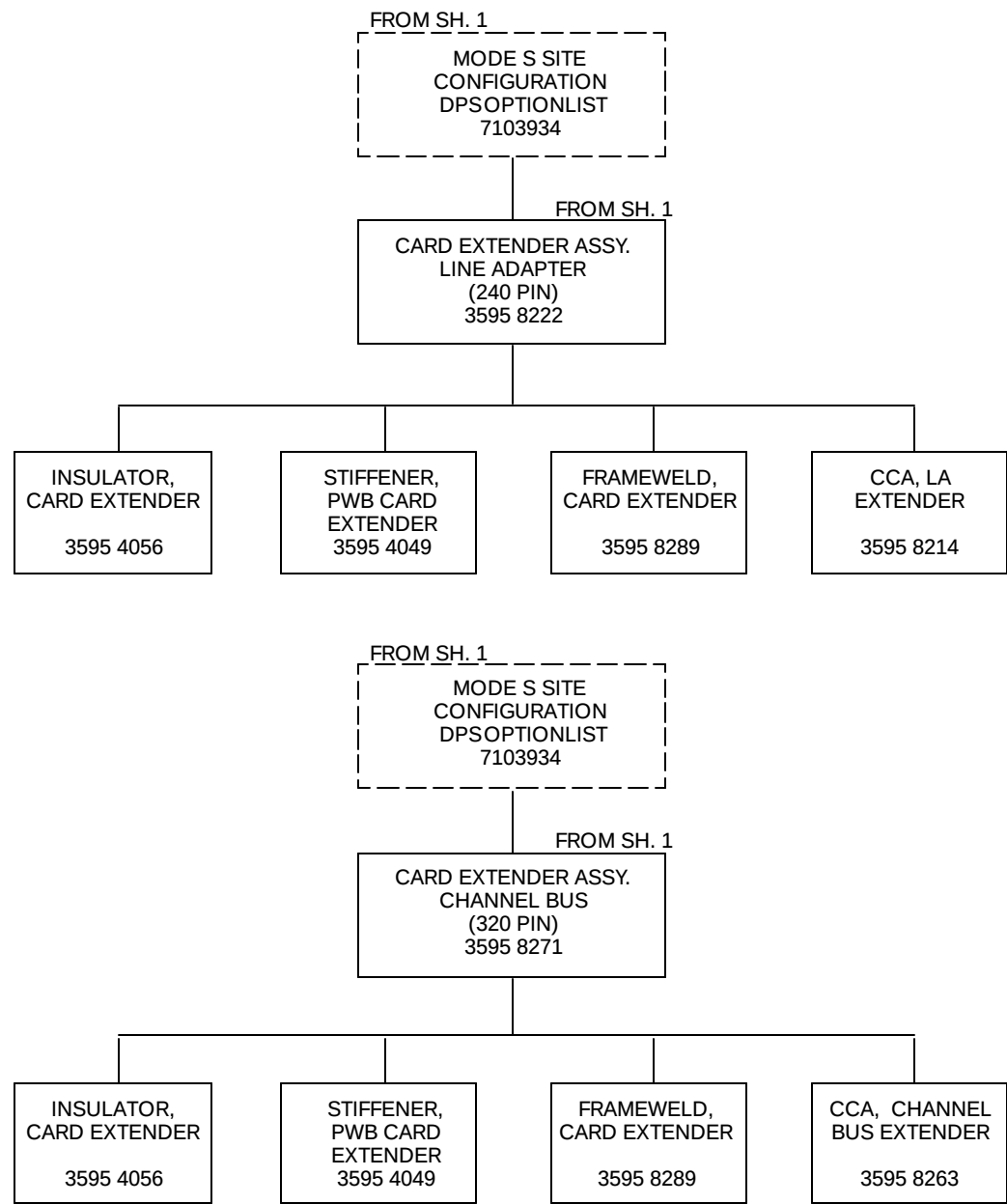
Note: Assembly 3594-1715-00 has main contract number on ID plate.
Assembly 3594-1715-01 has "Buy 10" contract number on ID plate.

FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 8)



d4313_9_r4_920327_WJF

FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 9)



d4313_12r6_920327_wjf

FIGURE 1–10. DPS EQUIPMENT CHART (SHEET 10)

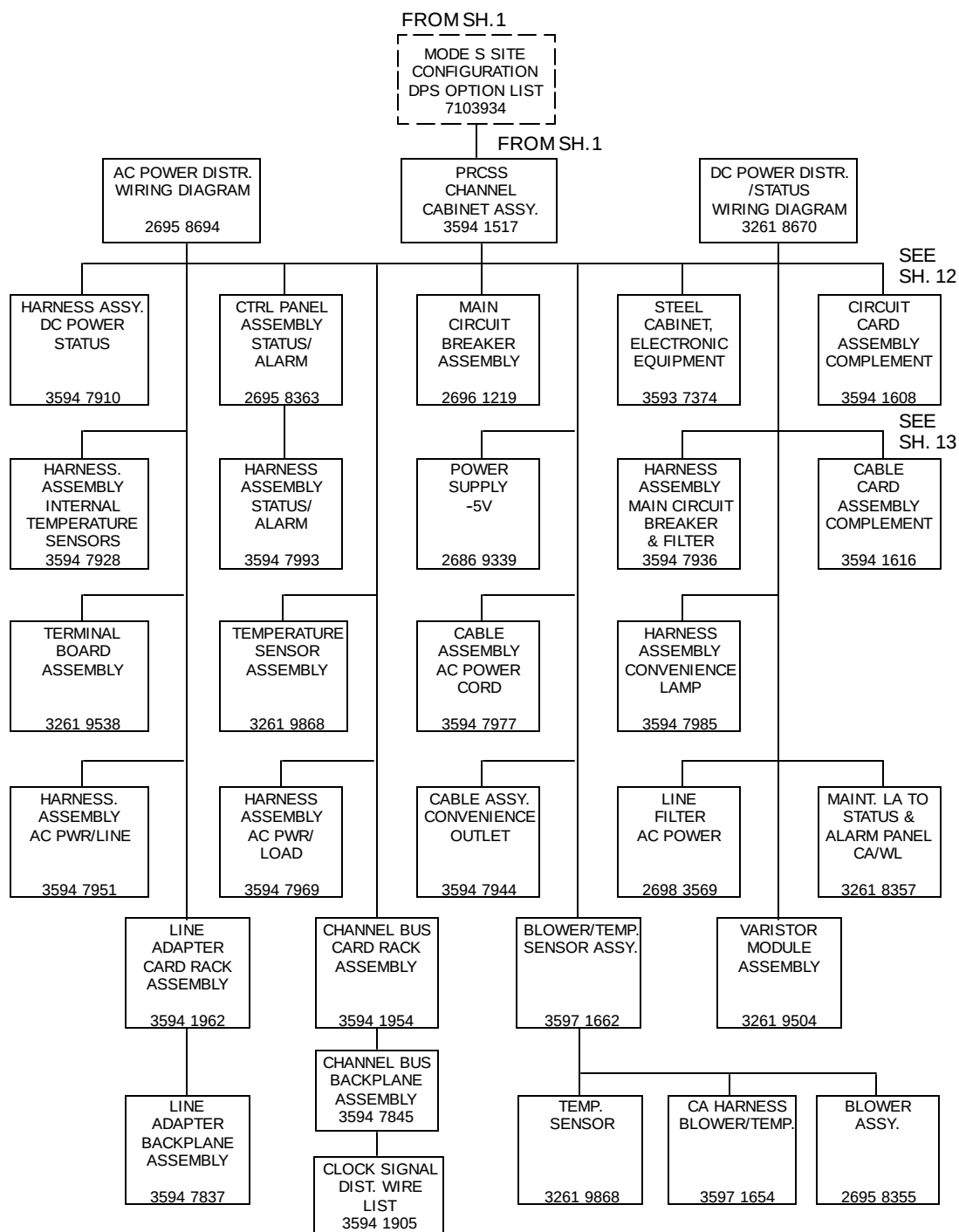


FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 11)

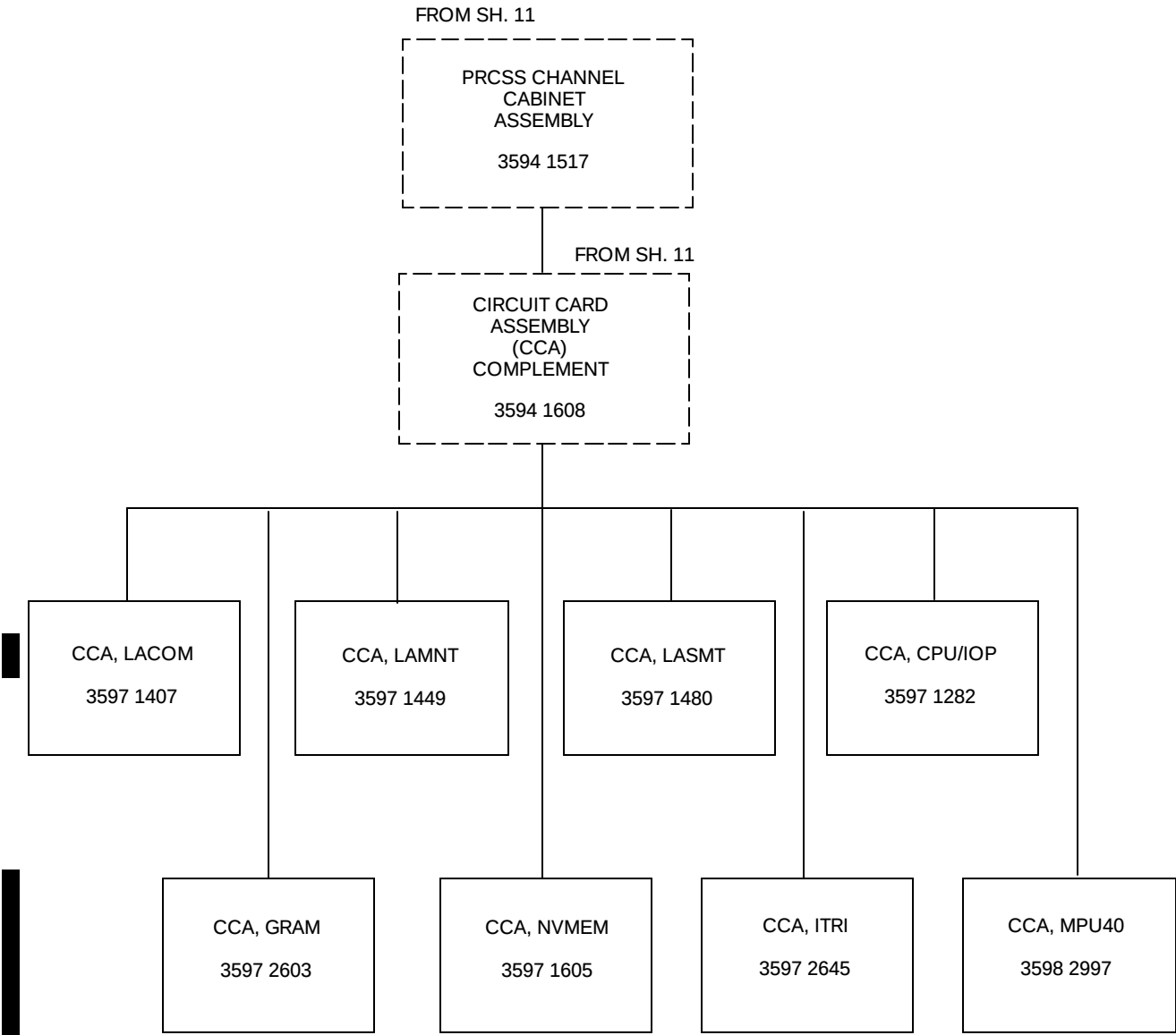
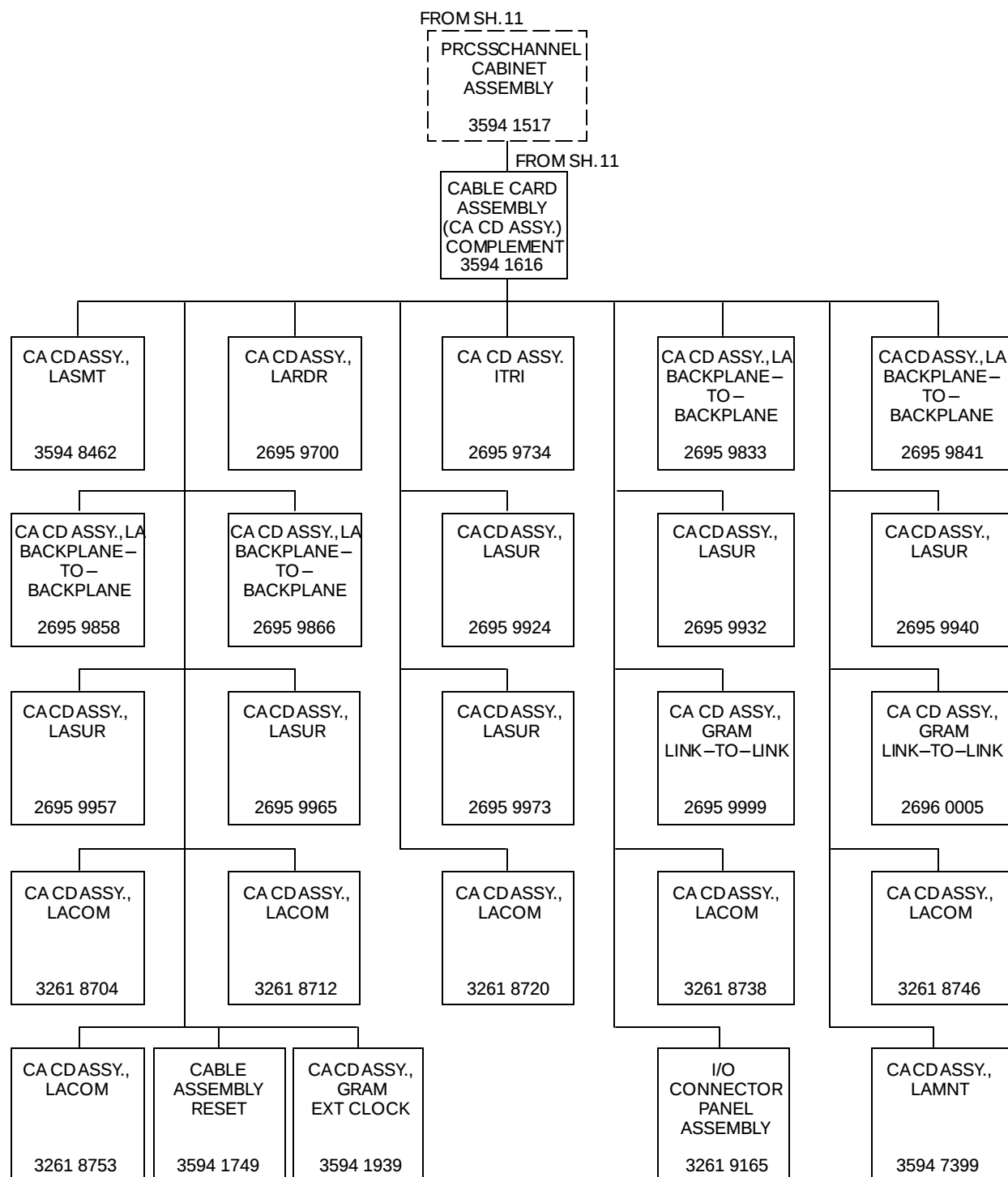


FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 12)



d4313_13_r4_901113_dbb

FIGURE 1-10. DPS EQUIPMENT CHART (SHEET 13 - END)

1.3.2.1.3 The main power supplies (+5V output) for the system are mounted on the gate to the left and right of the channel bus card rack. There may be as many as four supplies (B/PS1, B/PS2, B/PS3, and optionally, B/PS4). However, for most systems fielded, three supplies are used. These supplies operate in parallel. If one should fail, the remaining supplies increase output to compensate for the lost supply. The bottom of the cabinet contains the centrifugal blower assembly (B3) that cools the cabinet. The air intake for the blower package is located on the bottom front of the cabinet. Another temperature sensor (B3/A1) provides temperature monitoring for this area of the PRCSS cabinet. An optional +24V power supply (B6) is present in some PRCSS cabinets. This supply powers the Surveillance Data Switch (SDS). When present, the +24V power supply is located on the bottom left side of the gate next to the blower package.

1.3.2.1.4 The PRCSS is available in one of six standard configurations; these configurations are differentiated by their locations and interfacing equipment. The six standard configurations are as follows:

1. FAA Technical Center Terminal Site (FA 10203/1).
2. FAA Technical Center Enroute Site (FA 10203/2).
3. Enroute Site with CD-2 (FA 10203/3).
4. Enroute Site less CD-2 (FA 10203/4).
5. Terminal Site with ASR-9 (FA 10203/5).
6. Terminal Site less ASR-9 (FA 10203/6).

Some of the differences in the above systems are as follows:

- a. All enroute configurations include a Surveillance Data Selector (SDS) kit.
- b. All terminal configurations have no SDS kit.
- c. Configurations interfacing with a radar system (CD-2 or ASR-9) include a radar kit.

The following paragraphs describe the physical characteristics of the PRCSS subcomponents.

1.3.2.1.4.1 Channel Bus Card Rack Assembly. This card rack assembly holds the standard circuit cards of the DPS. This module can hold 26 cards, but only 17 positions are used in the standard configuration, 16 when a radar line adapter (LARDR) is not used. The module is positioned above the blower package in the PRCSS cabinet. The individual cards are interconnected by a backplane motherboard located on the rear of the card rack. On the front of the motherboard, three header connectors interface each card to the motherboard. From the back of the motherboard, signals are disseminated by a number of complex cable card assemblies to the I/O panel at the top of the cabinet, the connectors on the status and control panel, the line adapter card rack assembly and other parts of the PRCSS cabinet. Refer to section 1 of the DPS manual, TI 6365.3, for additional information concerning the various card assemblies that are used in the channel bus card rack assembly (area B2), P/N 3594 1954.

1.3.2.1.4.2 Line Adapter Card Rack Assembly. This card rack assembly is identical in dimensions to the channel bus card rack assembly, but is designed to hold 12 cards. This module houses surveillance and communications line adapters on expanded systems. Refer to section 1 of the DPS manual, TI 6365.3, for additional information concerning the various card assemblies that in the line adapter card rack assembly (area B1), P/N 3594 1962.

1.3.2.1.4.3 Power Supplies. The PRCSS cabinet contains up to four +5V power supplies and, in the enroute configurations a +24V SDS power supply. Three or four +5V power supplies provide the DC power for the electronics in the PRCSS cabinet. Each supply can provide up to 300A of the current. The

supplies are connected in parallel to a power bus, which distributes the +5V to the card rack backplanes. The supplies have load sharing capability, when one of the power supplies fails, voltage and current sensing circuits in each device sense the change and cause the remaining supplies to immediately compensate, avoiding the loss of system data or function due to power loss. Each supply includes a small rotary fan for positive airflow within the unit. The SDS power supply is a ferro-resonant device that supplies +24V used to operate the Surveillance Data Switch relay. The SDS relay is used to bypass the DPS during instances of system level failure, and send unprocessed radar data directly to the NAS or the enroute surveillance facility. This supply is only present in systems intended for enroute service.

1.3.2.1.4.4 Blower Assembly. The blower assembly provides a positive airflow of 859.4 cubic feet/minute minimum. It consists of two centrifugal-type blower elements powered by three permanent capacitor motors. The motors are powered by 3-phase source power. The assembly provides adequate cooling within the PRCSS cabinet, even with the loss of one motor.

1.3.2.2 Communications Junction Box Assembly. The Communication Junction Box (CJBX) is in a cabinet where the PRCSS A and B interface with external devices such as NAS telecommunication links, environmental sensors, and on-site radar equipment. The box is of heavy-duty construction, and is shielded to minimize EMI/RFI emissions.

The equivalent interfaces of two PRCSS cabinets are OR coupled in CJBX (Unit 6) and fed to a single connector. These interfaces include:

- a. Surveillance Line Adapter interface (1 to 6).
- b. Communication Line Adapter interface (1 to 6).
- c. Remote Terminal interface.
- d. Remote Maintenance Monitoring System (RMMS/MPS) interface.
- e. Program Support Facility (PSF) interface (not used).

Other interfaces are not ORed. These include SDS control lines, radar data lines, and building sensor lines (if used).

1.3.2.2.1 The surveillance data selector (SDS) switch is also located in the CJBX. This device is a set of relays used to bypass the DPS during system level failure conditions (failure of both PRCSS cabinets or a non-redundant system element). During these conditions, the relays disconnect the DPS surveillance line adapters from their telecommunication links, and immediately connect the links to the backup source for surveillance data: the CD-2. Power for the SDS is supplied by a +24V power supply external to the CJBX. The SDS is present in systems used at enroute-type facilities that include on-site radar.

1.3.2.2.2 Expanded systems require additional junction boxes (communication expansion and surveillance expansion) to handle the increased number of connections required. In this instance, as many as two more junction boxes are included in the DPS. The FAA Technical Centers (site nos. 1 and 2) use more than one junction box at each site. Refer to Table 1-1 and the equipment chart for the DPS equipment (figure 1-10).

1.3.2.3 Sensor Control. The sensor control function provides for control of major operating parameters of the Mode S sensor. The sensor control function is always assigned to one of several possible control points. The basic control point is the remote terminal. Two alternate control points are the local terminal and the Remote Maintenance Monitoring System (RMMS)/Maintenance Processing Subsystem (MPS). To transfer sensor control from the remote terminal to either the local terminal or the MPS, a control point request must be sent to the remote terminal from the MPS or the local terminal, and then control must be yielded at the remote terminal. Provision is made for transfer of control if the communication link is not connected.

1.3.2.3.1 The remote terminal interfaces with the sensor equipment through government furnished modems and cables that are input to the sensor at the Communications Junction Box (Unit 6). DPS system cables connect the terminal interface from the junction box to each PRCSS cabinet. The remote terminal function is provided by a Remote Terminal computer program that runs on a GFE IBM compatible PC. When the program is executing, it simulates a remote control/system status panel on the PC.

1.3.2.3.2 The local terminal is connected for use directly to the LTI connector J4 on the status and alarm panel of either DPS PRCSS cabinet for use. The terminal provides local control of the system to facilitate loading operational parameters and for use during sensor maintenance. (The KCRT terminal can also be connected to the Interrogator cabinets for use during Interrogator maintenance.) The local terminal is a government furnished personal computer. The baud rate at which communication between the sensor and the local terminal operates is selected by a site adaptable parameter. Two of the software packages that can be loaded into the local terminal are a communications package (GFE) that provides for data comm between the terminal and the MPS, and a DPS RMS (Remote Maintenance Subsystem) terminal package. The DPS RMS terminal software is used to obtain local control of the sensor. This software also provides for control of, or access to the following functions.

- a. Initial loading and start-up of the sensor.
- b. Modification of sensor site parameters.
- c. Initialization and control of diagnostic procedures.
- d. Reassignment of tasks within the computer subsystem.
- e. Switching of redundant elements from standby to operational status.
- f. Element reconfiguration and display of current channel configuration.
- g. Data extraction of the local sensor.
- h. Request and receipt of monitor (site data) and alarm reports.

1.3.2.3.3 The MPS interfaces with either PRCSS channel through the RMMS communications link interfaced to the sensor at the communication junction box (Unit 6). This communication link is always used for transfer of site data reports and sensor status information. The sensor information is sent to the MPS each time the site is polled by the MPS. To allow for direct control of sensor functions, the MPS can become the control point by sending appropriate message. At a remote site, the MPS function is emulated by the Mode S Interim Monitoring and Control (MSIMC) system.

1.3.3 Calibration Performance Monitoring Equipment (CPME)

The CPME is a remote Mode S test transponder, two or more of which are installed at fixed sites visible to a Mode S sensor and used for calibration and test of the Mode S sensor. Figure 1–11 shows the physical relationship of the units which comprise the CPME. The CPME provides full Mode S responses plus conventional Air Traffic Control Radar Beacon System (ATCRBS) Mode 3/A and 3/C, and military Mode 2 replies. The Mode S modes handled by the CPME include All-Call and surveillance uplink and downlink transmissions, single-segment uplink (Comm-A), and downlink (Comm-B) communication transmissions, and extended length uplink (Comm-C) and downlink Comm-D) communications transactions. The CPME also transmits altitude reports. The CPME provides built-in interface test capabilities that appear to the sensor to be functionally equivalent to those of an ordinary Mode S transponder interface.

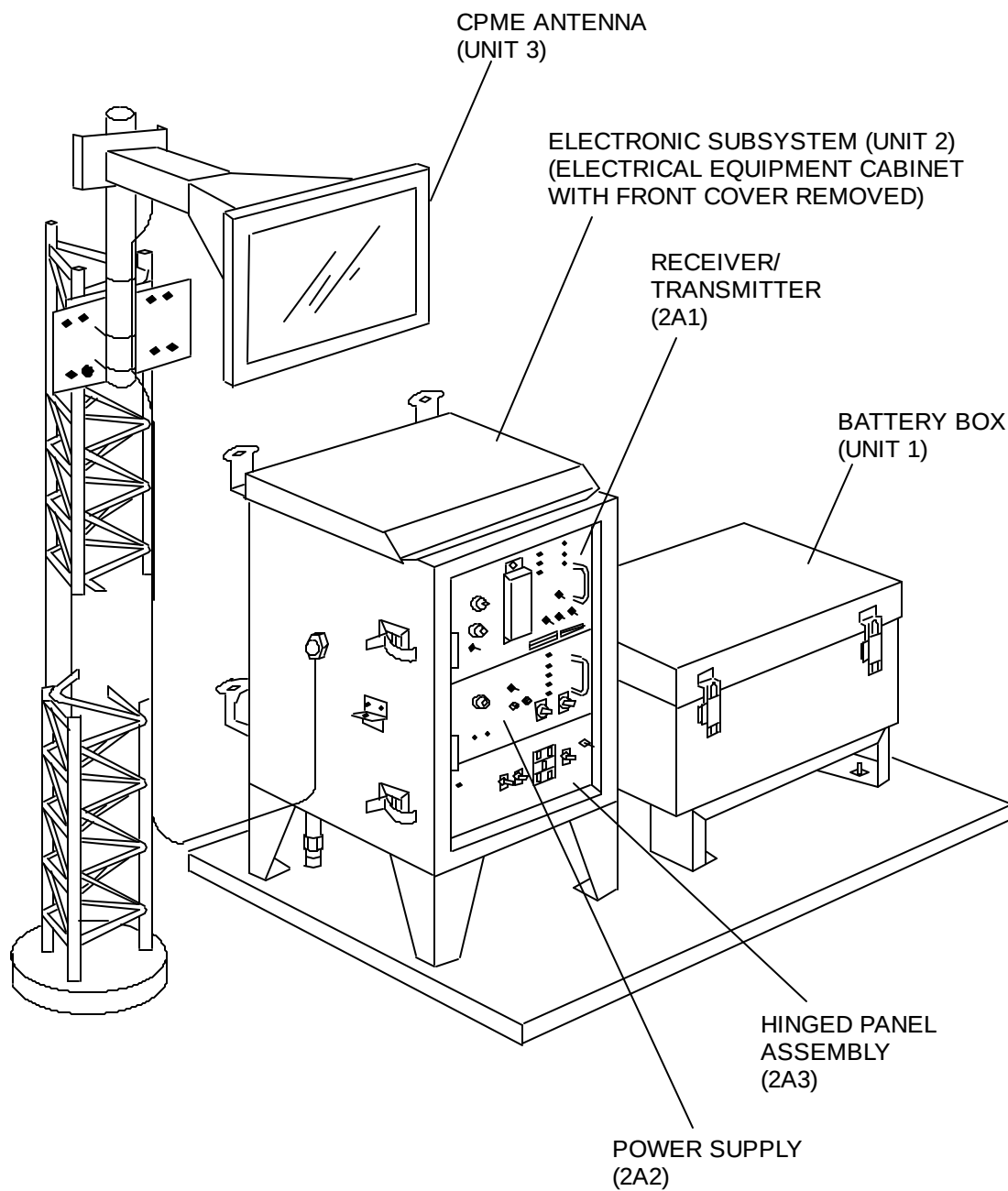


FIGURE 1-11. CALIBRATION PERFORMANCE MONITORING EQUIPMENT (CPME), PHYSICAL RELATIONSHIP OF UNITS

1.3.3.1 CPME Description. The CPME stands alone, remote from the sensor, and operates unattended. Two CPMEs are installed near each sensor. The CPME comprises a weatherproof lockable enclosure housing the receiver/transmitter, power supply, and hinged panel assembly. External to the enclosure is an antenna with radome and 150 feet of coaxial cable transmission line with connectors, and the battery box. The battery box contains four batteries, which provide the CPME with 48 volts emergency power. The CPME is designed to meet all performance specifications over the full environment range of conditions. The equipments that comprise the CPME are shown in the CPME equipment chart in Figure 1–12.

1.3.3.1.1 Battery Box Unit 1. The battery box is a white weatherproof metal box which houses four batteries. The batteries will provide 48-volts emergency power to the CPME. The battery box contains two heaters and two thermostats for environmental control. Refer to Mode S Calibration Performance Monitoring Equipment document TI 6365.8 for additional information concerning battery box unit 1.

1.3.3.1.2 Electrical Equipment Cabinet Unit 2. The electrical equipment cabinet houses the electronics of the CPME. The electrical equipment cabinet contains the receiver/transmitter and the power supply, as well as discrete components such as thermostats, circuit breakers, and a surge protector, are the LRUs of the electrical equipment cabinet. The cabinet, which is lockable, may be mounted on a concrete or asphalt pad, inside a building, or on the side of a building. The cabinet assembly is weatherproof and RF shielded. Refer to Mode S Calibration Performance Monitoring Equipment document TI 6365.8 for additional information concerning the electrical equipment cabinet unit 2.

1.3.3.1.3 Receiver/Transmitter (2A1). The receiver/transmitter (2A1) is a Lowest Replaceable Unit (LRU). The receiver/transmitter contains the electronics for the CPME. The receiver/transmitter contains 22 assemblies (A1 through A12, A14, A15, A17 through A20, AT1, AT2, FL1 and W1) as shown in Figure 1–12. Operation of all selectable functions is made via the front panel which also contains mounted lights for indicating replies to Mode S or ATCRBS interrogations. The front panel has a jack and selector switch for injection of an external Continuous Wave (CW) signal source which can be used to check the Mode S receiver bandwidth. The receiver/transmitter is rack mounted within the electrical equipment cabinet. Refer to Mode S Calibration Performance Monitoring Equipment document TI 6365.8 for additional information concerning the receiver/transmitter (2A1) equipment.

1.3.3.1.4 Power Supply (2A2). The power supply (2A2) is an LRU. The power supply contains six assemblies (A1 through A6), power transformer (T1), and wiring harness (W1) as shown in Figure 1–12. The power supply provides operating voltages for the receiver/transmitter and a charge voltage for the system backup batteries. The power supply also contains a switching circuit to apply battery power to the receiver/transmitter when ac input power is lost. The front panel contains indicator lights for monitoring the power supply status and a selector switch with test points to monitor power supply critical voltages and current. The power supply is rack mounted in the electrical equipment cabinet and is an LRU. Refer to Mode S Calibration Performance Monitoring Equipment document TI 6365.8 for additional information concerning power supply (2A2).

1.3.3.1.5 Hinged Panel Assembly (2A3). The hinged assembly (2A3) contains three circuit breakers, a ground fault interrupt ac outlet, a surge protector, and a terminal board. Refer to Mode S Calibration Performance Monitoring Equipment document TI 6365.8 for additional information concerning hinged panel assembly (2A3).

1.3.3.1.6 Calibration Performance Monitor Antenna Unit 3. Antenna unit 3 is a metal feedhorn with a Plexiglas cover that can be mounted on a pole or the side of a building. Refer to the Mode S Calibration Performance Monitoring Equipment document TI 6365.8 for additional information concerning calibration performance monitor unit 3, such as: (1) relationship of units diagram, (2) functional description, (3) equipment specifications, (4) dimensions, (5) technical description, (6) maintenance procedures, (7) corrective maintenance, and so forth.

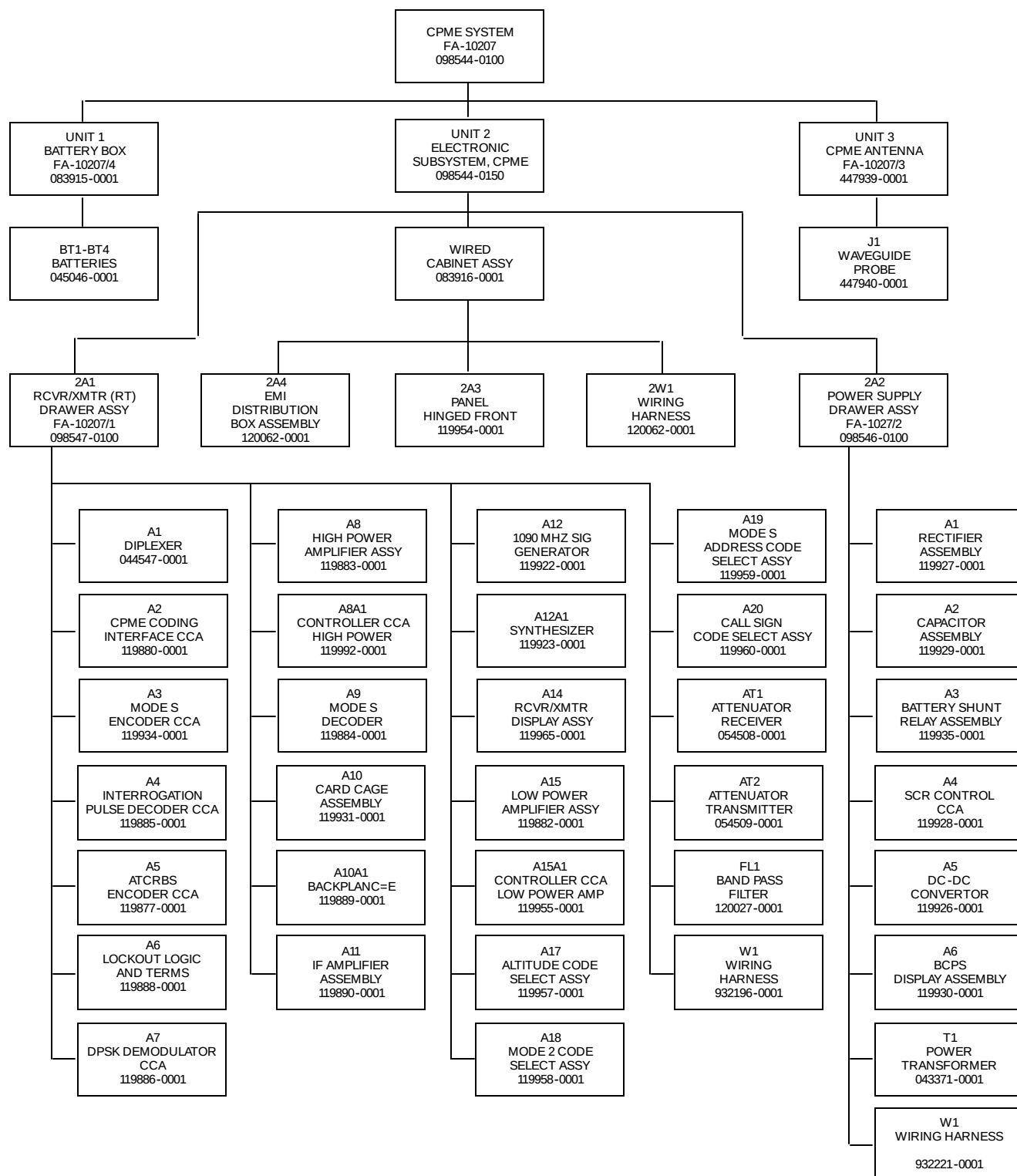


FIGURE 1-12. CALIBRATION PERFORMANCE MONITORING EQUIPMENT CHART

1.3.4 Power Distribution Equipment

The site power requirement for the Mode S sensor consists of a primary power source of 3-phase AC power, 177 VAC minimum, 239 VAC maximum, 57 to 63 Hz frequency, and 100 amperes maximum current. A site power distribution panel is required as shown in figures 3–1 and 11–11 with a 100A main circuit breaker, two 30A branch circuit breakers for the Interrogator units, two 40A branch circuit breakers for the DPS cabinets, and five 15A branch circuit breakers for the Mode S junction box MSJB1 and the four convenience outlets of the Interrogator and DPS cabinets. (The recommended typical power distribution panel should also include spare circuit breakers as shown in the figures.) Power distribution from the panel to the system cabinets is via conduit runs using color coded single conductor wiring as specified in figure 11–11.

1.3.5 Azimuth Pulse Generator (APG) Modification

The Mode S APG Modification applies to facilities configured either Mode-S/En route, Mode-S/ASR-7/8 or, Mode S En route beacon only. This implementation allows for lightning protection of the APG lines to the Mode-S encoders, improved DC voltage regulation to the encoders, distribution of Mode-S encoder data to other NAS equipment, and conditioning of the Azimuth Change Pulse (ACP)/Azimuth Reference Pulse (ARP) signals to alleviate excessive jitter.

1.3.6 Mode S Beacon Video Reconstitutor (BVR) with ASR 7/8 Radars.

The BVR is an independent self-contained unit located at the Air Traffic Control Automated Radar Terminal System (ARTSIIA/IIE) facility. The BVR is designed to operate as part of Mode S/ASR-7/8 configuration.

The primary purpose of the BVR is to provide regenerated ATCRBS beacon video for the ARTS IIA/IIE Radar Alphanumeric Display Subsystem (RADS) presentation. The reconstituted video is based on the information contained in the digital Mode S beacon surveillance messages.

1.4 EQUIPMENT SPECIFICATION DATA

Table 1–2 contains reference data for the equipment of the Mode S system equipment is listed in alphabetical order by unit nomenclature. The manufacturer/part number, FAA number, and the characteristics (electrical, power, environmental) of each unit are included. The equipment specifications for each of the Mode S system equipments are contained in the individual equipment instruction book (TI 6365.X) as indicated in the Equipment Specification Column of Table 1–2.

TABLE 1–2. MODE S EQUIPMENT SPECIFICATIONS

Nomenclature	Mfr/Part No.	FAA No.	Equipment Specifications
Interrogator	Northrop Grumman Corporation, P/N 612J422G01	FA 10202	For the electrical, power, and environmental characteristics of the Interrogator equipment, refer to the Interrogator Instruction Book TI 6365.5.
Data Processing System (DPS) for:	Lockheed Martin Corporation, P/N 3594 1509	FA 10203	For the electrical, power, and environmental characteristics of the Data Processing System (DPS) equipment, refer to the Data Processing System Instruction Book TI 6365.3.
1. FAA Technical Center Terminal Site No. 1	Plate ID 3588 0301	FA 10203/1	

TABLE 1–2. MODE S EQUIPMENT SPECIFICATIONS (Continued)

Nomenclature	Mfr/Part No.	FAA No.	Equipment Specifications
2. FAA Technical Center Enroute Site No. 2	Plate ID 3588 0319	FA 10203/2	
3. Enroute Site with CD–2	Plate ID 3588 0327	FA 10203/3	
4. Enroute Site less CD–2	Plate ID 3588 0335	FA 10203/4	
5. Terminal Site with ASR9	Plate ID 3588 0343	FA 10203/5	
6. Terminal Site less ASR9	Plate ID 3588 0350	FA 10203/6	
Local Terminal	GFE		Implemented in a GFE personal computer (PC). For characteristics of the Local Terminal, refer to the Local Terminal User's Manual TI 6365.11.
KCRT Terminal	GFE		For characteristics of the KCRT Terminal, refer to the Interrogator Instruction Book TI 6365.5.
Radar Intelligent Tool (RIT) Terminal	GFE		Implemented in a GFE personal computer (PC) with a Tape Drive Interface Card (TDIC) for magnetic tape emulation, an MX6–A card for data recording and a CD–ROM for viewing technical illustrations.
Remote Terminal	GFE		Implemented in a GFE personal computer (PC). For characteristics of the Remote Terminal, refer to Online Help.
Calibration Performance Monitoring Equipment (CPME) system	Wilcox Electric Inc. P/N 0985544–0100	FA 10207	For the electrical, power, and environmental characteristics of the CPME, refer to the Calibration Performance Monitoring Equipment (CPME) Instruction Book TI 6365.8.

1.5 EQUIPMENT AND ACCESSORIES REQUIRED (SUPPLIED)

Table 1–3 lists the equipment and accessories required for Mode S system operation that are supplied with each system. The equipment and accessories required for the Interrogator, Data Processing Subsystem (DPS), and Calibration Performance Monitoring Equipment (CPME) are defined in the respective equipment instruction books TI 6365.5, TI 6365.3, and TI 6365.8. The dimensions and weight and volume of the Interrogator, DPS, and CPME are also contained in the respective equipment instruction book.

TABLE 1–3. EQUIPMENT REQUIRED (SUPPLIED)

Quantity	Nomenclature	Part No.	FAA No.	Unit No.
Interrogator Configuration				
2	Interrogator Assembly	612J422G01	FA 10202	2/3/8
1	Mode S Junction Box No. 1	1D23453G01	FA 10208	1
2	TOY Clock Antenna (E1, E2)	1D23490H04		X
2	TOY Clock Remote Kit (A1, A2)	1D23490H02		X
1 (at Terminal and Enroute sites)	RF Transfer Switch Assembly	1D23267G01		9
1 (at Enroute site only)	RF Transfer Switch Assembly	1D23267G01		13
	Data Processing System (DPS) Configuration	3594 1509		
2	Processor Subsystem (PRCSS) Channel Cabinet Assembly for:	3594 1517		4/5
	FAATC Terminal Site		FA 10203/1	
	FAATC Enroute Site		FA 10203/2	
	Enroute Site With CD–2		FA 10203/3	
	Enroute Site Less CD–2		FA 10203/4	
	Terminal Site With ASR–9		FA 10203/5	
	Terminal Site Less ASR–9		FA 10203/6	
1 Max.*	Radar Kit	3594 1525–XX		
5 Max.*	Communication Kit	3594 1558–XX		
1*	Surveillance Data Selector (SDS) Kit	3594 1566–XX		
6 Max.*	Surveillance Kit	3594 1541–XX		
1*	Communication Junction Box Assembly	3594 1715	FA 10213/1	6
	or			
1*	Communication Junction Box Assembly with SDS	2696 0161	FA 10213/2	6
*	Communication Expansion Junction Box Assembly	3594 1574	FA 10204	10

TABLE 1–3. EQUIPMENT REQUIRED (SUPPLIED) (Continued)

Quantity	Nomenclature	Part No.	FAA No.	Unit No.
*	Surveillance Expansion Junction Box Assembly	3594 1582	FA 10214	11
*	CD–2 Modification Kit	2695 8488–XX		
1	System Cables	3594 1590–XX		
1	Card Extender, Channel Bus	3595 8271		
1	Card Extender, Line Adapter	3595 8222		
	Calibration Performance Monitor Equipment (CPME) Configuration	098544–0100	FA 10207	
1 (per CPME)	Battery Box (Unit 1)	083915–0001	FA 10207/4	
4 (per CPME)	Battery (BT1–BT4)	045046–0001		
1 (per CPME)	Electronic Subsystem, CPME (Unit 2)	098544–0150		
1 (per CPME)	Receiver/Transmitter Drawer Assembly (2A1)	098457–0100	FA 10207/1	
1 (per CPME)	Power Supply Drawer Assembly (2A2)	098546–0100	FA 10207/2	
1 (per CPME)	Hinged Front Panel (2A3)	119954–0001		
1 (per CPME)	CPME Antenna (Unit 3)	4477939–0001	FA 10207/3	
1 (per CPME)	CPME System Installation Kit			

Note: An * in Quantity column indicates a selected option.

1.6 EQUIPMENT REQUIRED (NOT SUPPLIED)

Table 1–4 lists the equipment and accessories required for Mode S system operation that are not supplied with the system. These items must be procured through FAA supply channels.

TABLE 1–4. EQUIPMENT REQUIRED (NOT SUPPLIED)

Qty	Nomenclature
1	Portable Terminal (Local Terminal-DPS)

TABLE 1-4. EQUIPMENT REQUIRED (NOT SUPPLIED) (Continued)

1	Printer for Local Terminal
1	Portable Terminal (KCRT Terminal Interrogator)
1	Portable Terminal (RIT Terminal-DPS) (Emulates Magnetic Tape Unit)
1	Portable Terminal (Remote Terminal)
Various	GFE cables

2.0 TECHNICAL DESCRIPTION

2.1 INTRODUCTION

This section contains the theory of operation for the Mode S system equipment in two levels: simplified and detailed. The simplified theory provides an overall description of the Mode S system, describes the interfaces with other equipment, and identifies the major functions performed by the Mode S system equipments. The Mode S system functional units are introduced with references made to the associated instruction books for further details. The simplified theory presents general Mode S operation. The detailed theory describes the functions of the various equipments that comprise the Mode S system: Interrogator, Data Processing System (DPS), and the Calibration Performance Monitoring Equipment (CPME).

2.1.1

For additional information and functional diagrams which concern the simplified and detailed theory of operation of the Mode S system equipments, the user must refer to the Interrogator instruction book TI 6365.5, the Data Processing System (DPS) instruction book TI 6365.3, and the Calibration Performance Monitoring Equipment (CPME) instruction book TI 6365.8. Additional Interrogator equipment instruction books are provided for the Transmitter – TI 6365.6, Receiver/RFTTG – TI 6365.7, and the Signal Processor TI 6365.4.

2.2 SIMPLIFIED THEORY OF OPERATION

The simplified theory of operation of the Mode S system is presented under the following headings:

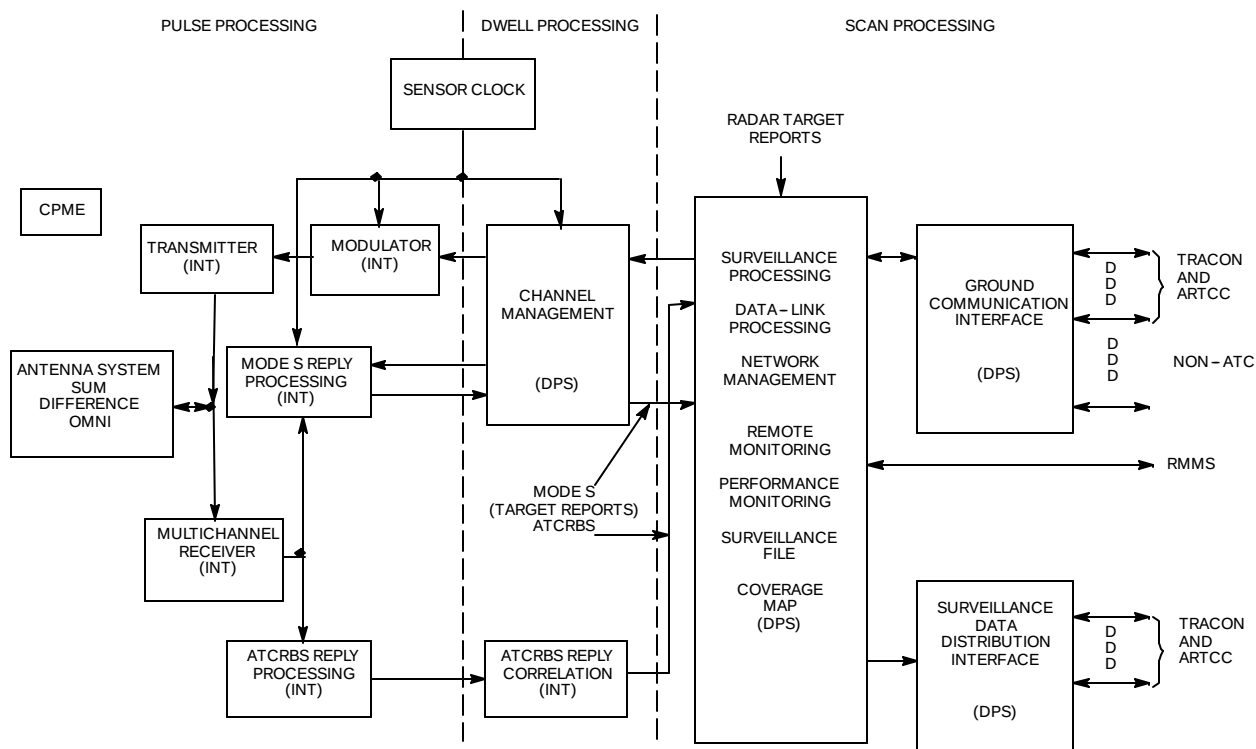
- a. General Mode S System Operation
- b. Interrogator Functional Operation
- c. Data Processing System (DPS) Functional Operation
- d. Calibration Performance Monitoring Equipment (CPME) Functional Operation

2.2.1 General Mode S System Operation

The Mode S sensor performs surveillance of Mode S and ATRBS equipped aircraft within its assigned area of coverage, and acts as a communication relay for data link messages between Mode S-equipped aircraft and Air Traffic Control (ATC) and non-ATC facilities. The Mode S sensor may accept digital target reports from a collocated radar, and merge these target reports into a common surveillance output stream, correlating the radar and beacon reports when both exist on the same target. The Mode S sensor interfaces with the airborne transponders via the RF link, and (when netted) also interfaces with adjacent sensors and ATC facilities via low-baud-rate digital communication circuits. A block diagram of the functional architecture of the Mode S sensor is presented in Figure 2–1. In Figure 2–1, most sensor functions are categorized according to the time scale on which they operate. Those pulse processing functions that operate on a microsecond time scale are: (1) modulator/transmitter, (2) multichannel receiver, (3) Mode S reply processor, and (4) ATRBS reply processor. Those dwell processing functions that involve channel transactions, and operate on a millisecond time scale proportionate with the dwell time of the Interrogator antenna on a target are: (1) channel management and (2) ATRBS reply correlation. Those scan processing functions that are paced by the antenna scan time are: (1) surveillance processing, (2) data link processing, (3) network management, (4) performance monitoring, and (5) coverage map. The two functional elements that are not categorized under pulse, dwell, or scan processing are the antenna system and clock.

Information concerning the general operation of the Mode S sensor is presented under the following headings:

- a. Mode S Interrogations and Replies
- b. Mode S Interrogations and Reply Formats
- c. Acquisition and Lockout Protocols
- d. Basic Surveillance Protocols
- e. Standard-Length Uplink Message Protocol
- f. Standard-Length Downlink Message Protocol
- g. Extended-Length Uplink Message Protocol
- h. Extended-Length Downlink Message Protocol
- i. Message Protocol Independence
- j. Capability Reporting
- k. Aircraft Identification Protocol
- l. Utility Message Protocol



NOTES:
 TRACON = TERMINAL RADAR APPROACH CONTROL FACILITY
 ARTCC = AIR ROUTE TRAFFIC CONTROL CENTER
 NON-ATC = NON-AIR TRAFFIC CONTROL LINES
 CPME = CALIBRATION PERFORMANCE MONITOR EQUIPMENT
 (INT) = INTERROGATOR
 (DPS) = DATA PROCESSING SYSTEM
 RMMS = REMOTE MAINTENANCE MONITORING SYSTEM

d4313_20_r5_920205_wjf

FIGURE 2-1. MODE S SENSOR FUNCTIONAL BLOCK DIAGRAM

2.2.1.1 Mode S Interrogations and Replies. There are six types of interrogations and replies used by Mode S for surveillance of ATCRBS and Mode S equipped aircraft, and for data link communication with Mode S transponder equipped aircraft. To maximize hardware compatibility between Mode S and ATCRBS, Mode S interrogations and replies use the same frequencies as those used for ATCRBS interrogations and replies. The types of interrogation and replies that are used by Mode S are as follows:

1. The ATCRBS/Mode S all-call interrogation, which is used for surveillance of ATCRBS equipped aircraft and also for acquisition of Mode S aircraft not already on the sensor's roll call.
2. The ATCRBS-only all-call interrogation, which is used for surveillance of ATCRBS equipped aircraft in conjunction with Mode S only all-call interrogation. This interrogation does not elicit a response from Mode S equipped aircraft.
3. The ATCRBS all-call interrogation, which is used to ensure detection of transponders that are incapable of interpreting the P4 pulse associated with the Mode S/3A/C all-call interrogations (item 1) or the Mode S/A/C only interrogations (item 2).
4. The Mode S interrogation, which is used for roll-call surveillance and data link communication to Mode S equipped aircraft. This type of interrogation is also used for the Mode S only all-call interrogation format that is required for: (1) the "stochastic acquisition" mode, (2) site addressed acquisition, and (3) lockout override functions.
5. The Mode S reply, which is used by Mode S transponders in response to: (1) Mode S interrogation, (2) ATCRBS/Mode S all-call interrogations, and (3) Mode S only all-call interrogations.

2.2.1.1.1 ATCRBS/Mode S and ATCRBS-Only All-Call Interrogations. The ATCRBS/Mode S and ATCRBS-only all-call interrogations are similar to the corresponding ATCRBS interrogations as defined in the U.S. National Aviation Standard for the Mark X Air Traffic Control Radar Beacon System (ATCRBS) Characteristics, but with an additional pulse P₄ following the P₃ pulse. Figure 2-2 presents a typical pulse relationship diagram of an ATCRBS/Mode S all-call interrogation. A P₄ pulsewidth of 1.6 msec defines the ATCRBS/Mode S all-call interrogation, while a P₄ pulse width of 0.8 msec defines the ATCRBS-only all-call interrogation. An ATCRBS transponder is unaffected by the presence of the P₄ pulse. The ATCRBS transponder will respond with a normal ATCRBS reply. A Mode S transponder will recognize the interrogation as a Mode S all-call or ATCRBS-only all-call and transmit a Mode S reply containing the aircraft's discrete address to the Mode S all-call interrogation (if it is not in a lockout state). The Mode S transponder will not respond to the ATCRBS-only all-call interrogation.

2.2.1.1.1.1 The ATCRBS all-call interrogation is similar to the previous two all-call interrogations but with the absence of the P₄ pulse to ensure detection by all aircraft transponders.

2.2.1.1.1.2 As in ATCRBS, Mode S interrogator sidelobe suppression (SLS or ISLS) is accomplished by the transmission of a control pulse P₅ on an SLS control pattern (usually omni-directional in azimuth). If this pulse is received by either an ATCRBS or Mode S transponder at an amplitude exceeding that of the P₁ pulse of the interrogation, the ATCRBS or Mode S transponder will not reply.

2.2.1.1.2 ATCRBS Reply. The ATCRBS reply signal characteristics are as defined in the U.S. National Aviation Standard for the Mark X Air Traffic Control Radar Beacon System (ATCRBS) Characteristics. The ATCRBS reply format is presented in Figure 2-3.

2.2.1.1.3 Mode S Interrogation. The Mode S interrogation is formed by three pulses, P₁, P₂ and P₆ as shown in Figure 2-4. Pulses P₁ and P₂ form the preamble. An ATCRBS transponder that receives this interrogation will interpret the pair of pulses as an ATCRBS sidelobe suppression command and will remain in suppression during the remainder of the Mode S interrogation. Without such sidelobe suppression, the subsequent Mode S P₆ pulse would, with high probability, trigger the ATCRBS transponder and cause a spurious reply.

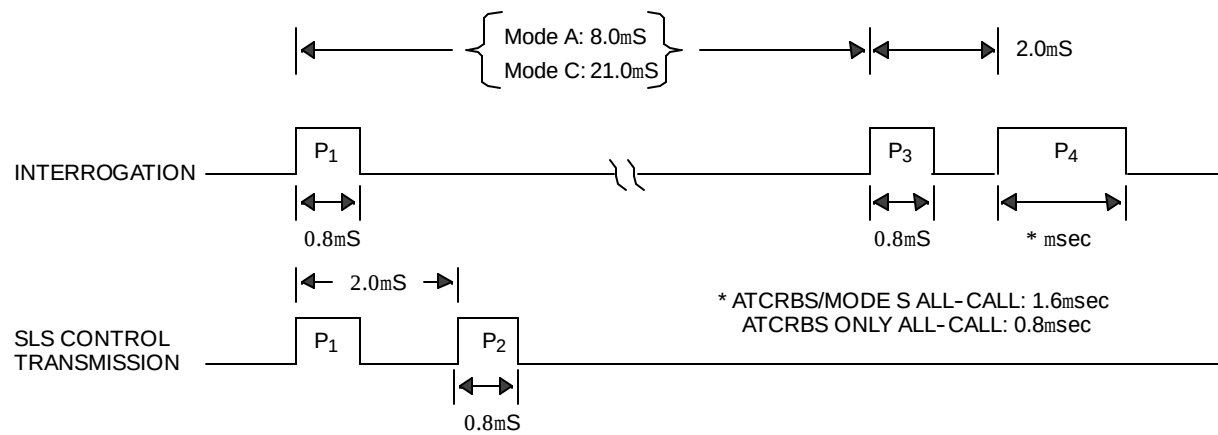


FIGURE 2-2. ATCRBS/MODE S ALL-CALL INTERROGATION

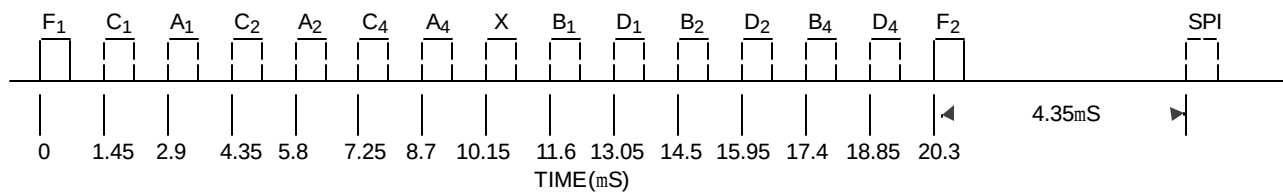


FIGURE 2-3. ATCRBS REPLY

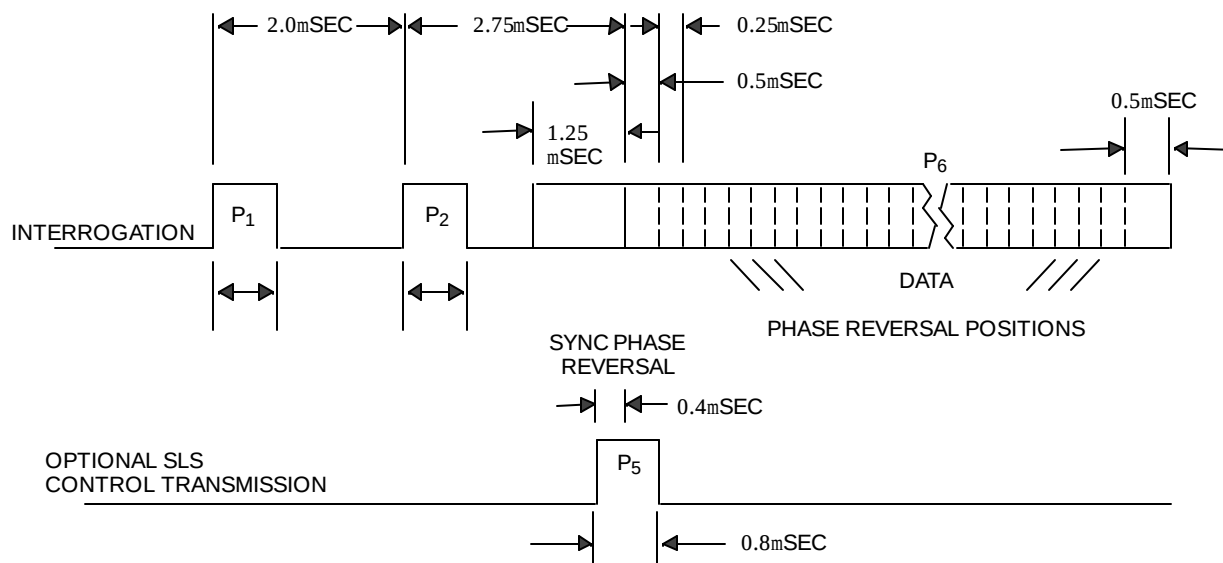


FIGURE 2-4. MODE S INTERROGATION

2.2.1.1.3.1 P₆ Pulse. The P₆ pulse of the Mode S interrogation contains the data in the form of Differential Phase Shift Keying (DPSK) modulation at a 4 Mbps rate. The first phase reversal within the P₆ pulse is the synch phase reversal used to start the DPSK decoding process. A phase reversal of the RF carrier at the beginning of a bit interval represents a binary ONE (1), while the absence of such a reversal denotes a binary ZERO (0). The 4 Mbps rate permits transmission of 112-bit interrogations within the nominal ATCRBS suppression interval. DPSK provides superior interference immunity, increased fade margin, and greater multipath immunity than pulse amplitude modulation (PAM).

2.2.1.1.3.2 Transmit Sidelobe Suppression. Transmit sidelobe suppression (SLS) is accomplished by the transmission of a control pulse (P₅) on an SLS control pattern. If the pulse amplitude received by the transponder exceeds the amplitude of the interrogation, the sync phase reversal will be obscured and the interrogation will be rejected. The P₅ pulse must be used with the Mode S-only all-call interrogation to prevent unwanted replies from aircraft in the sidelobes. With discrete address interrogations, transmit SLS is not required to prevent sidelobe replies. An aircraft will be interrogated only when in the mainbeam of the interrogator antenna. However, transmit SLS on discretely-addressed interrogations minimizes the probability of an aircraft erroneously accepting an interrogation directed to another aircraft; most such interrogations will be received through an interrogator antenna sidelobe, and thus will be rejected by the transponder without decoding the interrogation.

2.2.1.1.3.3 Receive Sidelobe Suppression (RSLs). RSLs is a receiver hardware modification being incorporated as an enhancement to Mode S operations in backup mode. It has a twofold purpose. The first is to reduce the long run lengths encountered in enroute operations. These excessive run lengths result from the wide beam width (low azimuth directivity) of the enroute antenna and result in split targets. To overcome this deficiency the RSLs modification utilizes the Delta fixed threshold to effectively reduce the beamwidth during reception of replies. It will not reduce the number of interrogations transmitted.

The second purpose of the RSLs is to reduce the number of sidelobe replies received. The ISLS backup mode function does not provide enough suppression to aircraft transmissions in the sidelobes. To overcome this deficiency, the RSLs modification utilizes the Omega fixed threshold to reduce the number of sidelobe replies during reception.

To achieve the reduction in run length and sidelobe replies, RSLS in Backup Mode utilizes the same comparator and thresholding circuits that are used in Mode S operation. The processing of the outputs of the threshold circuits is radically different between the two modes of operation. Whereas Mode S processing of these signals is done in the Signal Processor (either the ATRBS Bracket Detector Board or the Mode S Interface Board), RSLS for backup mode is done on the VPM board. A new circuit was added to take the outputs of the comparators, delay them to compensate for delays in the Aux quantizer, then AND them (they are both active low) with the Aux quantizer video.

2.2.1.1.4 Mode S Reply. A Mode S reply consists of a preamble and a data block containing 56 or 112 pulses. The signal format for a Mode S reply is presented in Figure 2–5. The preamble consists of a series of four pulses. The data block begins 8.0 ms after the leading edge of the first preamble pulse. Binary data are transmitted at a 1 Mbps data rate using pulse position modulation (PPM). As shown in Figure 2–5, during the 1.0 ms interval corresponding to each data bit, a 0.5 ms pulse is transmitted in the first half of the interval if the data bit is a 1, and in the second half of the interval if the data bit is a 0.

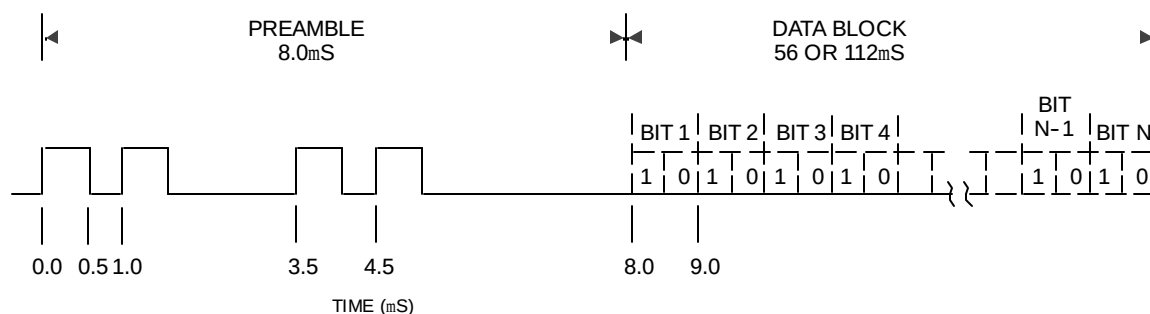


FIGURE 2–5. MODE S REPLY

2.2.1.1.4.1 The four-pulse preamble of a Mode S reply is designed to be easily distinguished from ATRBS replies. It can be reliably recognized and used as a source of reply timing in the presence of one overlapping ATRBS reply, while at the same time resulting in a low rate of false alarms arising from multiple ATRBS replies. The choice of PPM for the data modulation permits reliable bit detection in the presence of ATRBS interference. In addition, PPM results in a constant number of pulses in each reply, assuring sufficient energy for an accurate monopulse estimate. Operation at 1 Mbps data rate, in combination with the use of the 24-bit parity check coding, further enhances downlink reliability by permitting the correction of any error pattern which can result from a single ATRBS reply interfering with the desired Mode S reply.

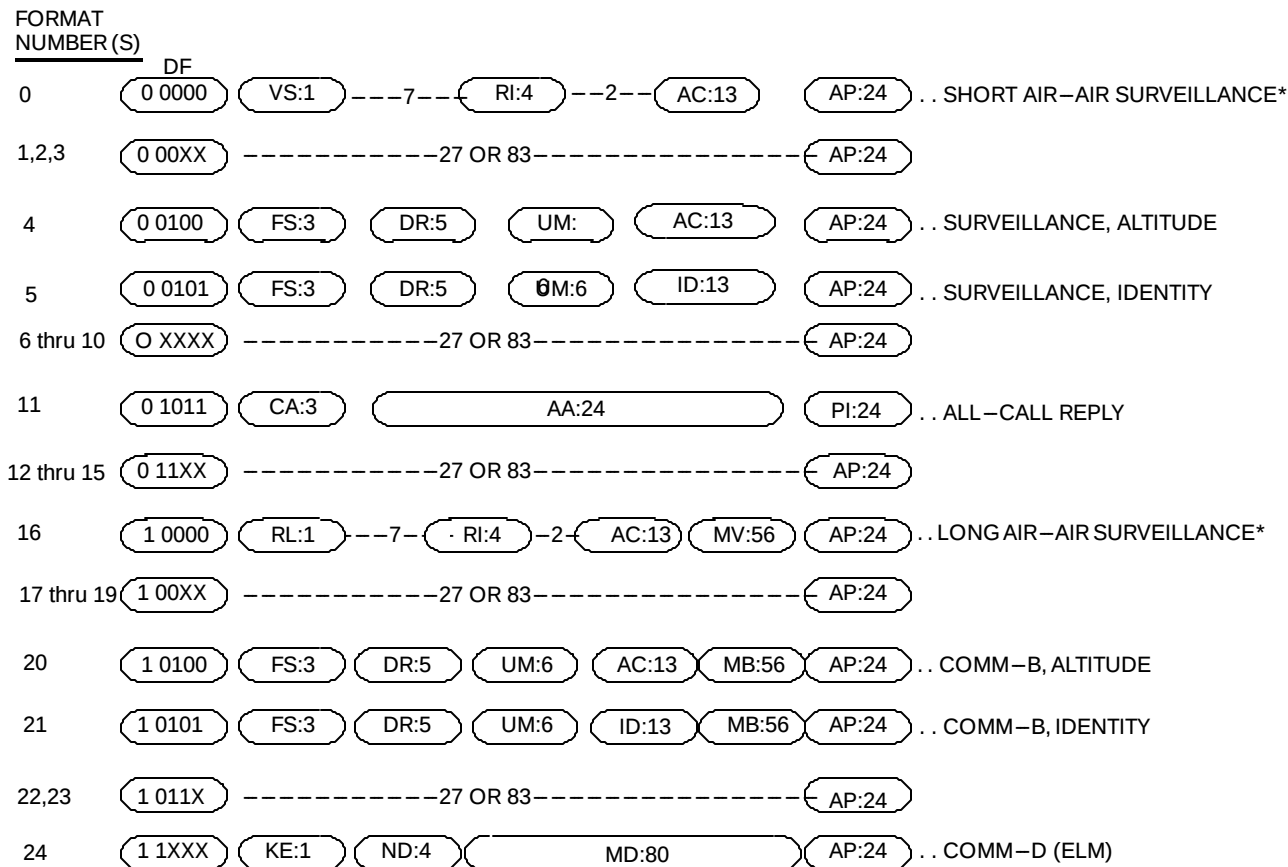
2.2.1.2 Mode S Interrogation and Reply Formats. The information transmitted in Mode S interrogations and replies is contained in data blocks that can carry either 56 or 112 bits of information. The interrogation data block is formed by the sequence of phase reversals within P₆, while the reply data block is represented by the pulse position modulation of the Mode S reply presented in Figure 2–5. Information within each data block is encoded in fields, each field existing for a dedicated purpose. All data blocks contain at least two essential fields, the format descriptor and the address/parity field. The format descriptor is transmitted at the beginning of each data block while the address/parity field is transmitted at the end. For different purposes and missions of the Mode S system, 25 different formats can be used; eight formats are presently defined as indicated in Figures 2–6 and 2–7. Figure 2–6 presents the Mode S uplink formats and Figure 2–7 presents the Mode S downlink formats. The following list provides information concerning Mode S field definitions that are used in the Mode S uplink and downlink formats presented in Figures 2–6 and 2–7.

FORMAT NUMBER(S)	UF														
0	0 0000	---	3	---	RL:1	---	4	---	AQ:1	---	18	---	AP:24	..	SHORT AIR-AIR SURVEILLANCE*
1,2,3	0 00XX	-----											AP:24		
4	0 0100	PC:3		RR:5		DI:3		SD:16		AP:24			..	SURVEILLANCE, ALTITUDE REQUEST	
5	0 0101	PC:3		RR:5		DI:3		SD:16		AP:24			..	SURVEILLANCE, IDENTITY REQUEST	
6 thru 10	0 XXXX	-----											AP:24		
11	0 1011	PR:4		II:4	-----								AP:24	..	MODE S ONLY ALL-CALL
12 thru 15	0 11XX	-----											AP:24		
16	1 0000	---	3	---	RL:1	---	4	---	AQ:1	---	18	---	MU:56	AP:24	.. LONG AIR-AIR SURVEILLANCE*
17 thru 19	1 00XX	-----											AP:24		
20	1 0101	PC:3		RR:5		DI:3		SD:16		MA:56		AP:24	..	COMM-A, ALTITUDE REQUEST	
21	1 0100	PC:3		RR:5		DI:3		SD:16		MA:56		AP:24	..	COMM-A, IDENTITY REQUEST	
22,23	1 011X	-----											AP:24		
24	1 1XXX	RC:2		NC:4				MC:80					AP:24	..	COMM-C (ELM)

NOTES;

1. AN "X" IN THE FIRST FIVE BITS INDICATES A VARIABLE CONDITION (1 OR 0) DEPENDING ON THE FORMAT NUMBER.
2. THE REMAINING FIELDS CONTAIN THE FIELD MNEMONIC AND NUMBER OF BITS (FOR EXAMP(E, RL)1).
3. AN --"N"-- DENOTES FREE CODING SPACE WITH "N" AVAILABLE BITS.
4. FOR UPLINK FORMATS (UF) 0 THROUGH 24, THE FORMAT NUMBER CORRESPONDS TO THE BINARY CODE IN THE FIVE BITS OF THE INTERROGATION.
5. THOSE FORMATS FOR WHICH NO APPLICATION IS DEFINED, REMAIN UNDEFINED IN LENGTH. DEPENDING ON FUTURE FORMAT ASSIGNMENTS, THEY MAY BE SHORT (56-BIT) OR LONG (112-BIT) FORMATS.
6. An "*" INDICATES NOT USED BY MODE S SENSOR

FIGURE 2-6. MODE S UPLINK FORMATS



NOTES;

1. AN "X" IN THE FIRST FIVE BITS INDICATES A VARIABLE CONDITION (1 OR 0) DEPENDING ON THE FORMAT NUMBER.
2. THE REMAINING FIELDS CONTAIN THE FIELD MNEMONIC AND NUMBER OF BITS (FOR EXAMPLE, VS:1).
3. AN --"N"-- DENOTES FREE CODING SPACE WITH "N" AVAILABLE BITS.
4. FOR DOWNLINK FORMATS (DF) 0 THROUGH 24, THE FORMAT NUMBER CORRESPONDS TO THE BINARY CODE IN THE FIVE BITS OF THE INTERROGATION.
5. THOSE FORMATS FOR WHICH NO APPLICATION IS DEFINED, REMAIN UNDEFINED IN LENGTH. DEPENDING ON FUTURE FORMAT ASSIGNMENTS, THEY MAY BE SHORT (56-BIT) OR LONG (112-BIT) FORMATS.
6. AN "*" INDICATES NOT USED BY MODE S SENSOR.

FIGURE 2-7. MODE S DOWNLINK FORMATS

Mode S Field Descriptions

Field Code	Field Name	Downlink (D)/Uplink (U) — Meaning	
AA	Address Announced	D	— Aircraft identification in all-call reply
AC	Altitude Code	D	— Aircraft altitude code
AP	Address/Parity	U/D	— Error detection field
AQ+	Acquisition	U	— Part of air-to-air protocol
CA	Capability	D	— Aircraft report of system capability
DF	Downlink Format	D	— Downlink descriptor
DI	Designator Identification	U	— Describes content of SD field
DR	Downlink Request	D	— Aircraft requests permission to send data
FS	Flight Status	D	— Aircraft's situation report
ID	Identification	D	— Equivalent to ATCRBS identity number
II	Interrogator Identification	U	— Site number of multisite features
KE	Control, ELM	D	— Part of Extended Length Message protocol
MA	Message, Comm-A	U	— Message to aircraft
MB	Message, Comm-B	D	— Message from aircraft
MC	Message, Comm-C	U	— Long message segment to aircraft
MD	Message, Comm-D	D	— Long message segment from aircraft
MU+	Message, Comm-U	U	— Air-to-air message to aircraft
MV+	Message, Comm-V	D	— Air-to-air message from aircraft
NC	Number, C-segment	U	— Part of ELM protocol
ND	Number, D-segment	D	— Part of ELM protocol
PC	Protocol	U	— Operating commands for the transponder
PI	Parity/Interr. Identity	D	— Reports source of interrogation
PR	Probability of Reply	U	— Used in stochastic acquisition mode
RC	Reply Control	U	— Part of ELM protocol
RI+	Reply Information	D	— Aircraft status information for TCAS*
RL+	Reply Length	U	— Commands air-to-air reply length
RR	Reply Request	U	— Commands details of reply
SD	Special Designator	U	— Control codes to transponder
UF	Uplink Format	U	— Format descriptor
UM	Utility Message	D	— Protocol message
VS	Vertical Status	D	— Aircraft status, airborne or on the ground

Note: *—TCAS = Traffic Alert and Collision Avoidance System

+ —Indicates not used by Mode S sensor

2.2.1.2.1 Table 2–1 presents information concerning the various interrogation and reply fields, subfields, and bit positions for uplink and downlink (UF and DF) message formats (refer to Figures 2–6 and 2–7). Tables 2–2 through 2–9 present significant field information applicable to the following interrogations and replies:

<u>Table No.</u>	<u>Title</u>
2–2	Mode S–Only All-Call Interrogation Fields
2–3	All-Call Reply Fields
2–4	Surveillance and Comm-A Interrogation Fields
2–5	Surveillance and Comm-B Reply Fields
2–6	Comm-C (Extended Length Message-ELM) Interrogation Fields
2–7	Comm-D (ELM) Reply Fields
2–8	Air-to-Air Surveillance Interrogation Fields
2–9	Air-to-Air Surveillance Reply Fields

TABLE 2–1. INTERROGATIONS AND REPLY FIELD INDEX

Ref.	Field	Sub Field	Name	Bits		Formats U = Up D = Down
				No.	Positions	
1	AA		Address Announced	24	9–32	D
2	AC		Altitude Code	13	20–32	D
3	AP		Address/Parity	24	33–56	U/D
				24	89–112	
4	AQ ⁺		Acquisition	1	14	U
5	CA		Capability	3	6–8	D
6	DF		Downlink Format	5	1–5	D
7	DI		Designator Identification	3	14–16	U
8	DR		Downlink Request	5	9–13	D
9	FS		Flight Status	3	6–8	D
10	ID		Identification	13	20–32	D
11	II		Interrogator Identification	4	10–13	U
12	KE		Control ELM	1	4	D
13	MA		Message, Comm-A	56	33–88	U
14	MB		Message, Comm-B	56	33–88	D
14.1	AIS		Aircraft Identification Subfield	48	41–88	D

TABLE 2–1. INTERROGATIONS AND REPLY FIELD INDEX (Continued)

Ref.	Field	Sub Field	Name	Bits		Formats U = Up D = Down
				No.	Positions	
15	MC		Message, Comm-C	80	9–88	U
15.1		SRS	Segment Request Subfield (RC=3)	16	9–24	U
16	MD		Message, Comm-D	80	9–88	D
16.1		TAS	Transmission Acknowledgement Subfield (KE=1)	16	17–32	D
17	MU ⁺		Message, Comm-U	56	33–88	U
18	MV ⁺		Message, Comm-V	56	33–88	D
19	NC		Number, C–Segment	4	5–8	U
20	ND		Number, D–Segment	4	5–8	D
21	PC		Protocol	3	6–8	U
22	PI		Parity/Interrogator Identity	24	33–56	D
23	PR		Probability of Reply	4	6–9	U
24	RC		Reply Control	2	3, 4	U
25	RI ⁺		Reply Information	4	14–17	D
26	RL ⁺		Reply Length	1	9	U
27	RR		Reply Request	5	9–13	U
28	SD		Special Designator	16	17–32	U
28.1		IIS	Interrogator Identification Subfield (DI=0, 1 or 7)	4	17–20	U
28.2		LOS	Lockout Subfield (DI=1 or 7)	1	26	U
28.3		MBS	Multisite Comm-B Subfield (DI=1)	2	21, 22	U
28.4		MES	Multisite ELM Subfield (DI=1)	3	23–25	U
28.5		RRS	Reply Request Subfield (DI=7)	4	21–24	U
28.6		RSS	Reservation Status Subfield (DI=1)	2	27, 28	U
28.7		TMS	Tactical Message Subfield (DI=1 or 7)	4	29–32	U
29	UF		Uplink Format	5	1–5	U
30	UM		Utility Message	6	14–19	D
30.1		IDS	Identifier Designator Subfield	2	18, 19	D
30.2		IIS	Interrogator Identification Subfield	4	14–17	D
31	VS		Vertical Status	1	6	D

Note: + – Indicates not used by Mode S sensor.

TABLE 2–2. MODE S–ONLY ALL–CALL INTERROGATION FIELDS

Field	Significance
Uplink Format (UF)	Identifies interrogation as a Mode S-only all-call format.
Probability of Reply (PR)	1. Specifies the probability to be used in generating a reply. 2. Used by the stochastic acquisition and lockout override modes.
Interrogator Identification (II)	1. Contains the site number of the interrogating site. 2. Used in conjunction with multisite lockout.
Address/Parity (AP)	Error detection coding combined with the “all ones” all-call address.

TABLE 2–3. ALL–CALL REPLY FIELDS

Field	Significance
Downlink Format (DF)	Identifies reply as an all-call reply.
Capability (CA)	1. Indicates communications capability of the transponder as: a. None (CA=0). b. Standard message uplink and downlink (CA=1). c. Standard message plus uplink ELM (CA=2). d. Standard message plus uplink and downlink ELM (CA=3). 2. Indicates presence of data link capability reporting (CA=1, 2, or 3).
Address Announced (AA)	Contains the aircraft’s 24-bit Mode S address.
Parity/Interrogator Identity (PI)	1. Combines error detection and the site number of the interrogator that elicited the reply. 2. Site number equals zero if non-selective interrogation is used.

TABLE 2–4. SURVEILLANCE AND COMM–A INTERROGATION FIELDS

Field	Significance
Uplink Format (UF)	1. Identifies interrogation as surveillance (UF=4, 5) or Comm-A (UF=20, 21) format. 2. Indicates whether reply should contain altitude (UF=4, 20) or ATCRBS code (UF=5, 21).
Protocol (PC)	1. Controls non-selective All-Call lockout. 2. Clears non-selective Comm-B and extended length message (ELM) transactions.
Reply Request (RR)	1. Indicates whether the reply will be short (RR<16) or long (RR=16). 2. Specifies Comm-B data selector 1 (BDS1) code of requested Comm-B message.
Designator ID (DI)	Indicates the structure of the special designator (SD) field.
Special Designator (SD)	Contains coding used for: a. Multisite lockout and communications protocols, b. Specification of the BDS2 code (if non-zero) of a requested Comm-B message, and c. Linking of Comm-A messages.
Comm-A Message (MA)	The 56-bit uplink message.
Address/Parity (AP)	1. Combined error detection and Mode S address coding. 2. All one's address in UF=20, 21 defines Comm-A broadcast.

TABLE 2–5. SURVEILLANCE AND COMM–B REPLY FIELDS

Field	Significance
Downlink Format (DF)	1. Identifies reply as surveillance (DF=4, 5) or Comm-B (DF=20, 21). 2. Indicates whether reply contains altitude (DF=4, 20) or ATCRBS code (DF=5, 21).
Flight Status (FS)	1. Indicates alert status of aircraft if: a. ATCRBS code is changed (alert condition will time out) b. ATCRBS code is an emergency code (alert condition will persist). 2. Indicates if pilot has activated the transponder SPI button. 3. Indicates whether the aircraft is airborne or on the ground.
Downlink Request (DR)	Indicates presence of waiting air-initiated Comm-B or downlink extended length message (ELM).
Utility Message (UM)	Identifies reserved site for multisite Comm-B, uplink ELM, or downlink ELM protocols.
Altitude Code (AC)	Altitude coded in 100-foot or 25-foot increments.
Identification (ID)	Aircraft 4096 ATCRBS code.
Comm-B Message (MB)	The 56-bit downlink message.
Address/Parity (AP)	Combined error detection and Mode S address coding.

TABLE 2–6. COMM–C (EXTENDED LENGTH MESSAGE–ELM) INTERROGATION FIELDS

Field	Significance
Uplink Format (UF)	Identifies interrogation as a Comm-C format.
Reply Control (RC)	1. Indicates initialization of an uplink extended length message (ELM) (RC=0). a. MC contains final segment and NC final segment number to indicate message length. b. Instructs transponder not to respond to this interrogation. 2. Indicates intermediate segment of uplink ELM (RC=1). a. MC contains intermediate segment. b. Instructs transponder not to reply to this interrogation. 3. Indicates segment delivery with technical acknowledgement request (RC=2). a. MC contains message segment. b. Instructs transponder to reply with Comm-D format with MD containing technical acknowledgement of uplink ELM delivery. 4. Indicates request for downlink ELM segments as indicated in segment request subfield of MC field (RC=3).
Number, C Segment (NC)	Identifies the segment number of the uplink ELM segment contained in the MC field.
Message, Comm-C (MC)	The 80-bit uplink Comm-C message.
Address/Parity (AP)	Combined error detection and Mode S address coding.

TABLE 2–7. COMM–D (ELM) REPLY FIELDS

Field	Significance
Downlink Format (DF)	Identifies reply as a Comm-D format.
Control, ELM (KE)	1. Indicates that ND and MD are part of a downlink extended length message — ELM (KE=0). 2. Indicates that MD contains Transmission Acknowledgement Subfield (TAS) in response to an uplink ELM (KE=1).
Number, D Segment (ND)	Identifies the segment number of the downlink ELM segment contained in the MD field.
Message, Comm-D (MD)	The 80-bit downlink Comm-D message.
Address/Parity (AP)	Combined error detection and Mode S address coding.

TABLE 2–8. AIR–TO–AIR SURVEILLANCE INTERROGATION FIELDS

Field	Significance
Uplink Format (UF)	Identifies interrogation as a short (UF=0) or long (UF=16) special surveillance format.
Reply Length (RL)	Indicates whether the transponder is to reply with a short (RL=0) or long (RL=1) special surveillance reply.
Acquisition (AQ)	1. Indicates if this is an air-to-air acquisition interrogation (AQ=1) or a non-acquisition interrogation (AQ=0). 2. Controls the contents of the reply information (RI) field of the elicited reply.
Message, Comm-U (MU)	The 56-bit message from the Traffic Alert and Collision Avoidance System (TCAS) if UF=16.
Address/Parity (AP)	Combined error detection and Mode S address coding.

TABLE 2–9. AIR–TO–AIR SURVEILLANCE REPLY FIELDS

Field	Significance
Downlink Format (DF)	Identifies the reply as a short (DF=0) or long (DF=16) special surveillance reply.
Vertical Status (VS)	Indicates whether the aircraft is airborne (VS=0) or on the ground (VS=1).
Reply Information (RI)	1. Contains maximum airspeed data for an acquisition reply (RI>7). 2. Contains Traffic Alert and Collision Avoidance System (TCAS) data for a non-acquisition reply (RI<7).
Altitude Code (AC)	Altitude coded in 100-foot or 25-foot increments.
Message, Comm-V (MV)	The 56-bit message from the replying aircraft to the TCAS aircraft (TCAS present if DF=16).
Address/Parity (AP)	Combined error detection and Mode S address coding.

2.2.1.3 Acquisition and Lockout Protocols. Acquisition of a Mode S aircraft initially occurs as the result of the receipt of a reply to an all-call interrogation. This reply, identified by DF = 11, contains the 3-bit capability (CA) field (used for capability reporting) and the 24-bit address announced (AA) field that contains the aircraft's Mode S address. The AA field is contained within the reply as data and is protected by the error coding of the parity/interrogator identity (PI) field. The PI field is encoded by the transponder using an address known by the sensor. The sensor can therefore perform the same error detection and correction on the all-call reply as the sensor does on the roll-call reply. Once acquired, an aircraft can be locked out from replying to subsequent all-call interrogations to minimize synchronous garble on the air-to-ground channel. The interrogation used by the sensor to elicit all-call replies depends upon the acquisition technique in effect at that site. The acquisition techniques are presented under the following headings:

- a. Non-selective acquisition and lockout.
- b. Site addressed acquisition and lockout.
- c. Stochastic acquisition.

2.2.1.3.1 Non-Selective Acquisition and Lockout. Non-selective acquisition uses the ATCRBS/Mode S all-call interrogation shown in Figure 2–2. After the address is acquired in the all-call reply, lockout is achieved by setting a code 1 in the protocol (PC) field of discrete surveillance interrogations, UF=4, 5, 20, or 21 (see Figure 2–6). Once the code 1 is set in the PC field, the non-selective lockout state remains in effect for a period of time. This function is refreshed every scan by the sensor to maintain the aircraft in the locked-out state.

2.2.1.3.2 Site Addressed Acquisition and Lockout. Site addressed acquisition uses the Mode S-only all-call interrogation, UF=11 (see Figure 2–6). The site address of the interrogating site is contained in the interrogator identification (II) field. Site address of 1 to 15 are valid (a site address of zero is interpreted as a non-selective interrogation). The transponder replies to this interrogation if it is not in a state of lockout to that specific site address. The transponder has a total of 16 independent lockout timers to maintain the lockout state requested by the ground sensors. For example, these are 15 site addressed lockout timers and one non-selective lockout timer.

2.2.1.3.2.1 The parity/interrogator identity (PI) field of an all-call reply, DF=11 (see Figure 2-7), elicited by a Mode S-only all-call interrogation is encoded using the site address received in the interrogator identification (II) field. The 4-bit site address is used as the lowest order four bits and then zero-filled to form a 24-bit "Mode S address". This address is used in the encoding of the PI field in exactly the same manner as the transponder Mode S address is used to generate the address/parity (AP) field. Sensors operating in the multisite mode decode all-call replies using their own interrogator identity code as the expected address. All-call "fruit" replies produced by adjacent interrogators will not be accepted by the local sensor since they would be encoded using a different site address. This filtering of all-call replies by site address eliminates the possibility of extraneous all-call tracks being formed by Mode S "fruit" replies.

2.2.1.3.2.2 Site addressed lockout is effected through coding in the special designator (SD) field of discrete surveillance interrogations, UF=4, 5, 20, or 21 (see Figure 2-6). The designator identification (DI) field defines the data content of the SD field. For DI equal to 0, 1, or 7, the first four bits of the SD field contain the site address in the interrogator identification subfield (IIS). Lockout to that address can be controlled by the sensor by setting the DI field equal to 1 or 7 and the lockout subfield (LOS) equal to 1. Once set, lockout for each site address remains in effect for approximately 18 seconds unless refreshed by the ground sensor.

2.2.1.3.2.3 The Mode S-only all-call interrogation elicits replies only from Mode S transponders. The Mode S-only all-call interrogation is therefore used in conjunction with the ATCRBS-only all-call interrogation (distinguished by a short P4 pulse as indicated in Figure 2-2). The ATCRBS-only all-call interrogator elicits replies only from ATCRBS transponders and therefore complements the Mode S-only all-call so that ATCRBS and Mode S transponders reply to one of the interrogations. This condition avoids the possibility of having the same aircraft under surveillance as both an ATCRBS and a Mode S aircraft.

2.2.1.3.3 Stochastic Acquisition. Stochastic acquisition uses the Mode S-only all-call interrogation, DF=11 (see Figure 2-7). The defined reply probability is contained in the probability of reply (PR) field and can be selected from the values 1, 1/2, 1/4, 1/8, or 1/16 for codes from 0 to 4, respectively. The transponder will not reply if a lockout condition applies. If a lockout condition does not apply, the transponder executes a random process and replies only if it is consistent with the defined reply probability. For example, if the transponder generated a random number between zero and one, it would only reply to a PR code 2 if: (1) a lockout condition did not apply, and (2) the generated random number was less than or equal to 0.250. This random reply process permits the reception of ungarbled all-call replies at the expense of additional delay (and additional interrogations) required for acquisition. The ability to assign a range of probabilities is intended to permit relatively rapid automatic reacquisition of worst case traffic loads. Information concerning the implementation of stochastic acquisition is provided under the following headings: (1) initial acquisition mode and (2) adaptive acquisition mode.

2.2.1.3.3.1 Initial Acquisition Mode. This mode is executed after a period of Mode S sensor inactivity. Initial acquisition mode consists of periodic Mode S-only all-call interrogations (4 to 6 per beam dwell) followed by a listening interval out to the range of interest. These interrogations are interspersed with scheduled Mode S intervals to permit discrete interrogation and lockout of Mode S address acquired on the previous scan. The minimum probability assignment used for this purpose is a site adaptation parameter chosen to match the Mode S traffic load handled by this site. The program for reacquisition begins at this lowest probability level and then moves to higher probability levels after several scans in order to reduce the overall acquisition time. Aircraft not acquire initially, as well as all-call garble situations which occur during normal operation, are handled as described under the adaptive acquisition mode heading.

2.2.1.3.3.2 Adaptive Acquisition Mode. Undecodable Mode S replies received during the all-call listening interval are tested for range correlation. Reports formed of two or more replies per dwell are interpreted as evidence of an all-call synchronous garble occurrence. A trial Mode S track is initiated at the approximately range and azimuth of the correlated replies. On the next scan, the sensor interrogates by use of a Mode S-only all-call with a specified reply probability. With high probability, an ungarbled reply will be received from one of the two transponders in the garble situation during the four or more interrogation opportunities, thereby permitting discrete interrogation and lockout on the following scan.

If acquisition is not successful, the trial track will be dropped since the continued garble situation will lead to the initiation of a fresh trial track. Residual garbling caused by more than two aircraft in the initial garble set will also result in a new trial track. The last aircraft in the garble set will be acquired by the normal ATCRBS/Mode S all-call interrogation process in use at that site.

2.2.1.4 Basic Surveillance Protocols. Information concerning the basic surveillance protocols is presented the following three headings:

1. Interrogation reply protocol.
2. Alert protocol.
3. Special designator (SD) protocol.

2.2.1.4.1 Interrogation Reply Record. Basic roll-call surveillance of Mode S aircraft is performed using surveillance or Comm-A interrogations. A surveillance or Comm-A interrogation (UF=4 or 20, see Figure 2–6) will elicit a short surveillance reply with altitude (DF=4, see Figure 2–7) if the value of the reply request (RR) field of the interrogation is less than 16. If RR is greater than or equal to 16, a UF=4 or 20 interrogation will elicit a Comm-B reply with altitude (DF=20). A similar rule applies to the readout of Mode A code information. The interrogation/reply protocol for altitude and identity requests is presented in the following list.

<u>Surveillance Interrogation/Reply Protocol</u>		
<u>Interrogation</u>	<u>Reply</u>	
<u>Uplink Format</u>	<u>Downlink Format RR < 16</u>	<u>Downlink Format RR > or = 16</u>
UF4	DF4	DF20
UF5	DF5	DF21
UF20	DF4	DF20
UF21	DF5	DF21

2.2.1.4.2 Alert Protocol. Knowledge of the current value of the transponder Mode A code is important to the ATC system because: (1) the code is used to signal emergency conditions and (2) the code is needed to identify an aircraft in handoff to non-Mode S facilities. Provision is made to notify the ground any time that the Mode A code is changed. Notice of change is detected on the ground by receipt of a surveillance or Comm B reply (DF=4, 20 or 21, see Figure 2–7) with the flight status (FS) field containing code value 2, 3, or 4. This condition defines an “alert” situation and results in the sensor scheduling an additional interrogation. This additional interrogation (UF equals 5 or 21) causes a readout of the current Mode A value in the elicited reply. A normal code change will result in the reporting of an alert condition for a period of approximately 18 seconds. This reporting period allows sufficient time for ground interrogators to detect the alert condition and to read the Mode A code. If an emergency Mode A code is used, the alert state will not time out, but will remain on for the duration of the declared emergency. In the absence of an alert condition, the surveillance protocol will routinely elicit altitude and not identity, except on the first scan of a roll-call interrogation, when both are requested.

2.2.1.4.3 Special Designator (SD) Protocol. The special designator (SD) field can be used for special communications or surveillance protocols. Provision is made for the SD field to be multiplexed for different uses. The significance of the code in the SD field is indicated by the designator identification (DI) field. A DI code value of zero means that the SD field is not used. DI=1 indicates that the SD field contains multisite and linked Comm-A information.

DI=7 indicates that the SD field contains multisite lockout data and a Comm-B extended data readout request.

2.2.1.5 Standard-Length Uplink Message Protocol. Comm-A interrogations (UF=20 or 21, see Figure 2–6) are used for the ground-to-air transmission of short “tactical” ATC messages. It must be assumed that, before any such transfer is initiated, the ground system has knowledge (from a previous capability — CA report) of the Comm-A capability of the addressed aircraft and also of the set of permissible message types. Information concerning standard-length uplink message protocol is presented under the following headings:

- a. Single Segment Comm-A Protocol.
- b. Linked Comm-A Message Protocol.
- c. Linked Comm-A Message Delivery.

2.2.1.5.1 Single Segment Comm-A Protocol. The standard uplink message is transmitted by a Comm-A interrogation (UF=20 or 21, see Figure 2–6) with the appropriate message header and message text in the message, Comm-A (MA) field. The reply will ordinarily be a surveillance reply with altitude unless some transaction involving an air-to-ground data link transmission is being carried on simultaneously. An acceptable reply received by the interrogator constitutes an implicit transponder technical acknowledgement of the ground-to-air message. If no reply is received, the message is scheduled for one or more repeat transmissions.

2.2.1.5.2 Linked Comm-A Message Protocol. Provision is made for the linking of up to four Comm-A message components (received on different interrogation/reply cycles) into a single message. For short messages, the linked Comm-A message protocol is more efficient than the extended length message (ELM) protocol, because: (1) Comm-A delivery can take place during a surveillance measurement and (2) no coordination with adjacent sites is required before message delivery. Therefore, the linked Comm-A message will be used for messages that are too long for a single Comm-A segment. The linked Comm-A message will also be used for message delivery requirements that can not be satisfied by the ELM. Coding for the linking of Comm-A message components is contained in the special designator (SD) field and is transparent to the transponder. Detection and assembly of the linked Comm-A message components will be performed by a data link device interfaced to the transponder. The data link device will wait until all message components have been received before releasing the message for further processing.

2.2.1.5.2.1 The Comm-A protocol for single component messages has been implemented so that no coordination with adjacent sites is required before delivery to the aircraft. Each linked Comm-A component must include the site address in the interrogation identification subfield (IIS) of the special designator (SD) field. The condition permits the correct association of interleaved components that may occur when more than one sensor at a time is delivering a linked Comm-A message to the same aircraft. The following protocol for linked Comm-A messages must be observed:

- a. Only one linked Comm-A can be in process for a given site/aircraft pair. This condition prevents ambiguities that could lead to garbled and lost messages.
- b. Component one is delivered first. This condition announces the initiation of a linked Comm-A message from a particular site number. The avionics can then wait until the final component is received before displaying the message. The first component can also serve as a message cancel for partially delivered message since the receipt of an initial component should clear any previous linked Comm-A message for that particular site.
- c. The first component must be successfully delivered before delivering the next component. This condition eliminates the need for any additional technical acknowledgement other than the reply to the interrogation that delivered the component. This condition also eliminates the need for a special message start protocol.

- d. Components two and three (if used) may be delivered in either order but must be successfully delivered before delivering the last component. This distinction in the coding of these components (rather than a single code for “intermediate component”) is necessary. This component numbering scheme prevents errors if a particular component is delivered more than once due to the occurrence of a downlink failure, which prevented the sensor from receiving the technical acknowledgement to a correctly delivered component.
- e. The final component is the last component transmitted. This announces the end of a linked Comm-A message transfer. If this condition is used in combination with the other codes, this permits the transfer of linked Comm-A messages with two, three, or four components.
- f. The final component must be successfully delivered before beginning a new linked Comm-A message to the same aircraft. This condition eliminates the need for a special message clear protocol.

2.2.1.5.2.2 The resulting message capacity (in bits) for a linked Comm-A message is presented in the following list:

<u>Number of Components</u>	<u>Number of Message Bits</u>
2	112
3	168
4	224

2.2.1.5.3 Linked Comm-A Message Delivery. The following text concerns linked Comm-A message delivery. An example of a Comm-A message, which has three components is presented in the following steps. The sensor will repeat delivery of each component until technical acknowledgement is received. Refer to Figures 2–6 and 2–7 for applicable message formats (UF and DF).

1. Interrogation: Comm-A message.
Relevant fields:
 - a. Tactical Message Subfield (TMS) — Code for first component of linked Comm-A message.
 - b. MA — Comm-A message, component 1.
2. Reply: Surveillance or Comm-B message — First component technical acknowledgement.
3. Interrogation: Comm-A message.
Relevant fields:
 - a. TMS — Code for second component of linked Comm-A messages.
 - b. MA — Comm-A message, component 2.
4. Reply: Surveillance or Comm-B message — Second component technical acknowledgement.

5. Interrogation: Comm-A message.
Relevant fields:
 - a. TMS — Code for final (third) component of linked Comm-A message.
 - b. MA — Comm-A message, component 3.
6. Reply: Surveillance or Comm-B message — Final component technical acknowledgement.

2.2.1.6 Standard-Length Downlink Message Protocol. The Comm-B message air-to-ground data link carries two kinds of message traffic: (1) ground-initiated, for readout of on-board data sources and (2) air-initiated. Information concerning standard-length message protocol is presented under the following headings:

- a. Ground-Initiated Transfer of Comm-B Messages.
- b. Air-Initiated Transfer of Comm-B Messages.
- c. Linked Comm-B Messages.
- d. Comm-B Broadcast Messages.

2.2.1.6.1 Ground-Initiated Transfer of Comm-B Messages. A ground initiated Comm-B message transfer is triggered by a surveillance or Comm-A interrogation with an reply request (RR) code greater than 16 (see Figure 2–6). This interrogation contains an 8-bit message request code called the Comm-B Data Selector (BDS). For convenience of description, the high order four bits are referred to as the BDS1 code and the low order four bits as the BDS2 code. The RR code value minus 16 equals the BDS1 value of the message requested by the interrogation. Therefore, an RR value of 17 is equivalent to a BDS1 value of 1 or “0001” in binary form. The BDS2 value of the requested message is assumed to be zero (0000) unless it is otherwise specified in the special designator (SD) field of the requesting interrogation. When the designator identification (DI) field code equals 7, the SD field contains an extended data readout request containing the reply request subfield (RRS) that specifies the BDS2 code. Therefore, an interrogation with RR=19, DI=7, and RRS=5 specifies a message with a BDS1 value of 3 and a BDS2 of 5, (0011 and 0101).

2.2.1.6.1.1 A ground-initiated Comm-B request triggers an immediate Comm-B reply with the Message, Comm-B (MB) field containing the requested data. If the aircraft is not equipped to supply the requested Comm-B Data Selector (BDS) code data, the aircraft responds with an MB field of 56 zeros. No acknowledgements are involved in the case of ground-initiated Comm-B transactions, failure to receive the data correctly on the ground simply causes a repeat of the request. An example of a Comm-B message delivery is presented in the following steps. Refer to Figures 2–6 and 2–7 for applicable message formats (UF and DF)

1. Interrogation: Surveillance or Comm-A message.
Relevant fields:
 - a. Reply request (RR=19) — Readout of Comm-B message with BDS1=3.
 - b. Designator identification (DI=7) — Extended data readout in special designator (SD) field.
 - c. Reply request subfield (RRS=5) — BDS2 of requested Comm-B message.
2. Reply: Comm-B message.
Relevant fields: Message, Comm—B (MB) — Ground-initiated Comm-B message.

2.2.1.6.2 Air-Initiated Transfer of Comm-B Messages. The presence of an air-initiated Comm-B message causes the downlink request (DR) code of every surveillance or Comm-B reply (DF=4, 5, 20, or 21) to be set to an appropriate value that indicates an air-initiated Comm-B message is waiting. The protocol used by the sensor to transfer this message depends upon the operating environment of that sensor. Information concerning air-initiated transfer of Comm-B messages is presented in the following text: (1) non-selective transfer of air-initiated Comm-B messages, (2) multisite transfer of air-initiated Comm-B messages, and (3) air-directed multisite Comm-B message delivery.

- a. Non-selective Transfer of Air-Initiated Comm-B Message. The non-selective protocol is used if the sensor does not have overlapping coverage with any other Mode S sensor or if the sensor communicates with its neighboring sensors to ensure that only one sensor at a time is responsible for Comm-B message transfer. In this case, the sensor transmits a surveillance or Comm-A interrogation with RR=16 (read air-initiated Comm-B) and either DI=7 (no extended data readout) or an RRS=0 if DI=7. This specifies a BDS1 and a BDS2 value of zero, which uniquely defines a request to read an air-initiated Comm-B message. The transponder responds with the waiting Comm-B message.

Because of the asymmetry between sensor and transponder (with the sensor not routinely replying to every air-to-ground transmission), an explicit ground technical acknowledgement is required. When the air-to-ground message is successfully received by the sensor, the sensor sets a code value of 4 into the protocol (PC) field of the next surveillance or Comm-A interrogation. When PC=4 (closeout Comm-B) is received by the transponder and the Comm-B message has been read at least once, the message is cleared. The transaction is complete when the ground receives the elicited reply, indicating successful receipt of the closeout Comm-B instruction. This protocol can result in the receipt of more than one closeout Comm-B instruction for a particular message if the interrogation is received by the transponder, but the ground does not receive the reply. The absence of the expected reply will cause the ground to repeat the interrogation at the next opportunity.

If the transponder has a second air-initiated Comm-B message waiting to be transferred, the transponder indicates its presence to the ground by sending the appropriate downlink request (DR) code in its reply to the message that delivered the closeout Comm-B acknowledgement to the previous message. It must be noted that the avionics equipment must not permit a message to be closed out until after the message has been read at least once. This condition prevents the multiple closeouts that can be received due to a downlink failure from closing out a second waiting message that has not as yet been transferred to the ground. The possibility of downlink failure also prohibits the sensor from both reading and closing out a Comm-B message in the same interrogation, because multiple delivery of such an interrogation would certainly result in the loss of an undelivered Comm-B message. An example a non-selective air-initiated Comm-B message delivery is presented in Table 2–10. Refer to Figure 2–6 and 2–7 for applicable message format (UF and DF) information.

- b. Multisite Transfer of Air-Initiated Comm-B Messages. If it is possible for more than one sensor to service an air-initiated Comm-B message request, the multisite protocol must be used to avoid duplicate message delivery and possible loss of messages. This loss can occur when: (1) a transponder has more than one air-initiated message ready for delivery and (2) the interleaved read and closeout instructions from the multiple sites results in the following sequence. Sensor A has closed out message 1 and has had a downlink failure in reading message 2. This is followed by a closeout from sensor B (intending to closeout message 1) that in fact closes out message 2, which is lost since it has not been successfully read by either sensor. The multisite protocol handles this problem by allowing any sensor to read the air-initiated message, but uses only one reserved site to perform the closeout.

TABLE 2-10. EXAMPLE OF NON-SELECTIVE AIR-INITIATED COMM-B MESSAGE DELIVERY

(1) INTERR.	(1) REPLY	RELEVANT FIELDS	MEANING
S or A	—	—	—
—	S or B	DR=1	Air-initiated Comm-B message waiting.
S or A	—	RR=16	Requested BDS1 code equals zero.
		DI? 7	Requested BDS2 code equals zero.
		or DI=7 and RRS=0	Requested BDS2 code equals zero.
—	B	DR=1 ⁽²⁾ MB	Air-initiated Comm-B message waiting. Air-initiated Comm-B message.
S or A	—	PC=4	Non-selective Comm-B message closeout.
—	S or B	DR=0 ⁽³⁾	Comm-B message closed out.
Notes: 1. S = Surveillance, A = Comm-A, B = Comm-B.			
2. The message has not yet been closed out.			
3. If DR=1 in the reply to an air-initiated Comm-B cancellation, this condition indicates the presence of another air-initiated Comm-B message.			

When a sensor using the multisite protocol receives a surveillance or Comm-B reply with a downlink request (DR) code that indicates an air-initiated Comm-B is waiting, the sensor responds with a surveillance or Comm-A interrogation with:

RR code = 16 (Read air-initiated Comm-B message)
 DI code = 1 (Multisite SD field)
 MBS code = 1 (Comm-B reservation)
 RSS code = 1 (Report Comm-B reservation status in UM)
 IIS code = sensor site address

The transponder stores the address of the first site to make such a reservation and includes this “reserved” site number in the interrogator-identification subfield (IIS) of the utility message (UM) field of its reply. A site that sees its own address in the reply knows that it has the responsibility for handling the message. A second site that reads the Comm-B message before it is cleared (by the first site) will know that this is a duplicate delivery since the reserved site number is different from its own site number. The reserved site clears a delivered air-initiated Comm-B message with a surveillance or Comm-A interrogation with:

DI code = 1 (Multisite SD field)
 MBS code = 2 (Comm-B message close out)
 IIS code = sensor site address

The transponder does not clear the message (set DR=0) until a closeout message is received from the reserved site and the message has been read at least once. If the message is not closed out by the received site within 18 seconds, the reservation times out and can be reserved by any other sensor that is in contact with the aircraft. An example of a multisite air-initiated Comm-B message delivery is presented in Table 2–11. In this example, the local site interrogator address equals 4. Refer to Figures 2–6 and 2–7 for applicable message format (UF and DF) information.

TABLE 2–11. EXAMPLE OF MULTISITE AIR-INITIATED COMM-B MESSAGE DELIVERY

(1) INTERR.	(1) REPLY	RELEVANT FIELDS	MEANING
S or A	—	—	—
—	S or B	DR=1	Air-initiated Comm-B message waiting.
S or A	—	RR=16 DI=1 MBS=1 RSS=1 IIS=4	Readout of air-initiated Comm-B message. Multisite special designator (SD) field. Comm-B message reservation. Request for Comm-B message reservation status in utility message (UM). Local site's interrogator identification.
—	B	DR=1 IDS=1 IIS=4 ⁽²⁾ MB	Air-initiated Comm-B message waiting. Comm-B message reservation status is in IIS Site 4 (the local site) is the reserved site. Contains the air-initiated message.
S or A	—	DI=1 MBS=2 IIS=4	Multisite SD field. Multisite Comm-B message closeout. Local site's interrogator identification.
—	S or B	DR=0 ⁽³⁾	Comm-B message closed out (IIS field is also cleared in the transponder).

- Notes: 1. S = Surveillance, A = Comm-A, B = Comm-B.
2. If IIS= 4, the local site is not the reserved site and no further action is taken on the message.
3. If DR=1 in the reply to a Comm-B message closeout, this condition indicates the presence of another air-initiated Comm-B message.

- c. Air-Directed Multisite Comm-B Message Delivery. The multisite Comm-B message protocol makes it possible for an aircraft to route an air-initiated message to a particular sensor. This capability is important in a multisite environment since it makes it possible for a pilot acknowledgement message to be routed to the sensor that delivered the message that is being acknowledged. When a Comm-A message is delivered, the data link avionics equipment will receive the special designator (SD) field along with the Message, Comm-A (MA) field, and will thus be able to associate the sending site address (in the IIS subfield) with the Comm-A message. When a pilot acknowledgement is generated, the data link avionics equipment will cause the transponder to reserve the message for the desired site address at the same time that the message is made available for delivery.

Therefore, even the initial reply which indicates a message waiting (in the downlink request — DR field) will show that a reservation already exists for the desired site (in the utility message — UM field). This allows the desired sensor to initiate readout of the message and then issue the closeout.

2.2.1.6.3 Linked Comm-B Messages. The Mode S link protocols accommodate linked Comm-B messages. As for the uplink case, the transponder is transparent to the linking protocol. A block of three consecutive Comm-B data selector (BDS) codes is reserved for linked Comm-B messages, one each for components 2 through 4 of the linked Comm-B message. The data link avionics equipment initiates a linked Comm-B transfer by sending the first message component as an air-initiated message. The sensor detects the linking code in the received Comm-B message component and initiates a ground-initiated Comm-B transfer to read out the other message components. When the sensor has received all of the Comm-B message components, the sensor closes out the air-initiated message that began the sequence. Receipt of this closeout by the transponder serves as a technical acknowledgement for the entire linked Comm-B message. This protocol works with either the non-selective or multisite delivery protocols. In the latter case, the presence of a reservation during the entire delivery process ensures that all message components are read by the same site.

2.2.1.6.4 Comm-B Broadcast Messages. The air-initiated Comm-B protocol described in previous text is designed to deliver a Comm-B message initiated on board an aircraft to a single sensor. Certain types of messages may require delivery to more than one sensor. In this case, the broadcast Comm-B protocol is used. The broadcast Comm-B protocol uses the downlink request (DR) field to signal the presence of a broadcast Comm-B message that is available for delivery. Any sensor in contact with the transponder may read the message using the same code as for an air-initiated Comm-B message. The key difference is that the sensor cannot clear a broadcast message; the message clears automatically after 18 seconds. During this period when the message is active, the message can be read by every sensor in contact with the transponder.

2.2.1.6.4.1 Provision is made in the coding of the downlink request (DR) field to distinguish between two different broadcast messages (broadcast message 1 and broadcast message 2). The transponder will send successive broadcast messages using alternate message codes. This permits the sensor to detect a change in a broadcast message and eliminates the need for the sensor to read the message every scan. Therefore, a sensor that has read broadcast message 1 from a particular transponder on one scan can safely avoid reading the same message on subsequent scans (until it times out). The presence of a new broadcast message would be indicated by DF=5 (broadcast message 2 is available). Provision is made for an air-initiated Comm-B message to interrupt the delivery of a broadcast Comm-B message. In this case, the broadcast message delivery is resumed for the full 18-second period following the delivery of the air-initiated message.

2.2.1.7 Extended Length Uplink Message Protocol. The extended length message (ELM) protocol provides for more efficient transmission of long data link messages by permitting the grouping of up to 16 message segments into a single entity that can be acknowledged by a single reply. Each segment is included in a single Comm-C transmission (UF=24, see Figure 2–6). The limit of 16 segments refers solely to the manner in which the message is transferred over the link. Longer messages can be accommodated through the use of a linking indicator within the message header in the first segment of an ELM. Information concerning extended-length message (ELM) protocol is presented under the following headings:

- a. Basic Ground-to-Air Extended Length Message Transfer.
- b. Non-Selective Uplink ELM Protocol.
- c. Multisite Uplink ELM Protocol.

2.2.1.7.1 Basic Ground-to-Air Extended Length Message Transfer. Ground-to-Air Extended Length Messages are transmitted using the Comm-C format with three different reply control codes (RC=0,

1, or 2). The three reply control codes designate an initializing segment, intermediate segments, and a final segment. It must be noted that the minimum length of a ground-to-air ELM is two segments since the protocol requires at least an initializing and a final segment. The transfer of all segments may take place without any intervening air-to-ground replies, as described in the following text. In this way, channel loading is minimized. Message segments (one per Comm-C interrogation) may be transmitted at a rate up to one per 50 msec. This minimum spacing is required to permit the resuppression of ATRBS transponders. Delivery of the message may take place during a single scan or over a few scans depending on the length of the message, the channel interference level, and the sensor loading. Normally, sufficient time will be available within one scan to permit complete delivery of the message. Information concerning ELM segment transfer is presented in the following text.

- a. **Initializing Segment Transfer.** The extended length message transaction for an N-segment message (segment numbers 0 through N-1) is initiated by a Comm-C interrogation with RC code = 0. The transponder does not reply. Receipt of this interrogation (in effect a "dial up") causes the ELM interface within the transponder to initialize its message storage and bookkeeping registers in preparation for a new ELM transfer. Also delivered in the initial interrogation is the text of the final message segment in the Message, Comm-C (MC) field, and its segment number (N-1) in the number, C segment (NC) field. This "last segment first" protocol is used to inform the transponder of the length of the message. If an initializing segment is received before the completion of an earlier ground-to-air ELM transfer, the effect is to abort the earlier message and replace it with the newer message. If the ELM processor fails to receive an initializing segment, it will not store any received segments.
- b. **Intermediate Segment Transfers.** Message delivery proceeds with the transmission of intermediate segments (any sequence of N-2 segments chosen from segments N-2 through 0) via Comm-C interrogations with RC code = 1 (intermediate). The transponder does not reply. Each message segment is identified with its segment number in the number, C-segment (NC) field. The ELM processor stores each segment in the appropriate storage location based on this number. In this way, the message processor reassembles the message, and its bookkeeping function keeps track of which segments have been received. Intermediate segments may be delivered in any order, once the ELM processor has been initialized with segment N-1. If the entire message consists of only two segments, there will be no intermediate transfers.
- c. **Final Segment Transfer.** The interrogator transmits the final segment of a Comm-C interrogation with RC code = 2 (final). The Comm-C interrogation segment number is in the number, C-segment (NC) field and the text is in the Message, Comm-C (MC) field. This reply control (RC) code elicits a Comm-D format (DF code = 24) with control, ELM (KE) code = 1 (acknowledgement in MD) and a cumulative transponder technical acknowledgement in the MD field. The cumulative transponder technical acknowledgement subfield (TAS) consists of a bit string (maximum length of 16 bits) that indicates which segments of the ELM have been received. The first bit represents the state of the first (N=0) segment (with the states defined as: 1 = segment received, and 0 = segment not received). Therefore, at all times this field represents the current status of segment delivery from the time of ELM initiation. If no reply to the final segment is received, all segments which have not been acknowledged as delivered are also retransmitted. If one or more segments of the ELM were not received by the transponder, this fact is indicated by zeros in the corresponding bit positions in transmission acknowledgement subfield (TAS). The interrogator retransmits the missing segments with RC code = 1, except for the final segment which has RC code = 2 to request an updated TAS. This process continues until the ground receives a cumulative TAS which indicates that all segments have been delivered. When all segments have been received by the transponder, the interrogator knows that its last transfer was final and closes out the transaction. The ELM processor in the transponder transfers the message to the appropriate output device as soon as the ELM processor senses the presence of all segments.

2.2.1.7.2 **Non-Selective Uplink ELM Protocol.** The non-selective protocol may be used when only one sensor at a time has the responsibility for delivering an uplink ELM message. By use of this protocol,

a sensor delivers an uplink ELM as described in previous text. The sensor then terminates the delivery by sending a PC code = 5 (closeout uplink ELM) in a surveillance or Comm-A interrogation.

2.2.1.7.3 Multisite Uplink ELM Protocol. The multisite uplink ELM protocol is used to coordinate the activities of multiple sensors so that only one sensor at a time is reserved for uplink ELM activity. Before beginning an uplink ELM, a sensor uses the multisite protocol to obtain a reservation by transmitting a surveillance or Comm-A interrogation with the following relevant fields and subfields:

DI code = 1 (Multisite special designator — SD)
 MES code = 1 (Uplink ELM reservation)
 RSS code = 2 (Report uplink ELM reservation status in utility message — UM)
 IIS code = sensor site address

2.2.1.7.3.1 The transponder reply to this interrogation includes the site address (in the interrogator identification subfield (IIS) of the UM field) and the type of reservation mode by the interrogator (in the identification designator subfield, IDS=2) of the sensor currently reserved for uplink activity. IDS=2 indicates uplink ELM (Comm-C) reservation active. A sensor recognizing its own site address as the reserved site will proceed with the uplink ELM. If a different site is designated as the reserved site, uplink ELM activity is not initiated this scan and a reservation request may be attempted again on the following scan. When the uplink ELM delivery is complete, the reserved sensor closes out the delivery by sending a surveillance or Comm-A interrogation with: (1) DI code = 1 (multisite SD), (2) MES code = 2 (uplink ELM closeout), and (3) IIS code = the sensor site address. The transponder only accepts the closeout if the IIS field in the interrogation matches the reserved uplink ELM IIS code stored in the transponder.

2.2.1.7.3.2 A reservation in effect for 18 seconds from the time of last delivery activity (the reservation itself, any segment delivery, or TAS request). If delivery activity ceases, it is assumed that the sensor is no longer in contact and the reservation is cancelled by the transponder timing out. The transponder is then able to grant a reservation to any other sensor that is currently in contact. An example of a multisite uplink ELM delivery is presented in Table 2–12. This example presents multisite delivery of 3-segment uplink ELM by a local site with an interrogator address equal to 6.

TABLE 2–12. EXAMPLE OF MULTISITE UPLINK ELM DELIVERY

INTERR. (1)	REPLY (1)	RELEVANT FIELDS	MEANING
S or A	—	DI=1 MES=1 RSS=2 IIS=6	Multisite special designator (SD) field. Uplink ELM reservation. Request for uplink ELM reservation status in utility message (UM). Local site's interrogator identification.
—	S or B	IDS=2 IIS=6(2)	Uplink ELM reservation status is in IIS. Site 6 (the local site) is the reserved site.
C	—	RC=0 NC=2 MC	Initializing segment delivery. Announces a three segment ELM. Segment 3 of uplink ELM.
—	None	—	Reply not elicited by RC=0.

TABLE 2–12. EXAMPLE OF MULTISITE UPLINK ELM DELIVERY (Continued)

INTERR. (1)	REPLY (1)	RELEVANT FIELDS	MEANING
C	—	RC=1 NC=1 MC	Intermediate segment delivery. Indicates MC contains segment 2. Segment 2 of uplink ELM.
—	None	—	Reply not elicited by RC=1.
C	—	RC=2 NC=0 MC	Segment delivery and request trans- mission acknowledgement. Indicates MC contains segment 1. Segment 1 of uplink ELM.
—	D	KE=1 TAS (3)	MD contains transmission acknowledge- ment of uplink ELM in TAS. Transmission acknowledgement of seg- ments 1 to 3.
S or A	—	DI=1 MES=2 IIS=6	Multisite SD field. Multisite uplink ELM closeout. Local site's interrogator identification.
—	S or B	—	Transmission acknowledgement of closeout command.
Notes: 1. S = Surveillance, A = Comm-A, B = Comm-B, C = Comm-C, D = Comm-D			
2. If IIS? 6, the local site is not the reserved site and no further action is taken on this scan. The IIS subfield is in the UM field.			
3. The sensor will resend any segment not acknowledged by TAS.			

2.2.1.8 Extended Length Downlink Message Protocol. The transfer of an air-to-ground ELM is similar to the ground-to-air process. Differences between the two protocols result primarily from the fact that: (1) all channel activity is ground initiated and (2) the transponder can reply with longer communications formats only when given a specific permission by the ground. As with the uplink ELM, provision is made for the linking of downlink ELMs. Information concerning extended length downlink message protocol is presented under the following headings:

- a. Basic Air-to-Ground Extended Length Message Transfer.
- b. Non-Selective Downlink ELM Protocol.
- c. Multisite Downlink ELM Protocol.

2.2.1.8.1 Basic Air-to-Ground Extended Length Message Transfer. Air-to-ground extended length messages are transmitted (under ground control) using the Comm-D reply (DF=24, see Figure 2–7). Information concerning initialization of air-to-ground ELM transfer and message delivery is presented in the following text.

- a. Initialization. An N-segment air-to-ground ELM transfer is initiated by a surveillance or Comm-B reply containing DR code > 16. The number of message segments to be transferred is equal to the DR code number minus 15. Therefore, a DR code of 21 indicates the presence of a 6-segment downlink ELM ready for delivery.

- b. **Message Delivery.** The interrogator requests the air-to-ground transmission of ELM segments using a single Comm-C transmission with RC code = 3 (request Comm-D). In this format, the first 16-bits of the MC field form a special 16-bit Segment Request Subfield (SRS), in which the successive bit positions correspond to segment numbers 0 through 15. The designated response is a series of Comm-D replies containing those message segments for which the corresponding SRS bit is set to one. (The transponder is not told which segments have been successfully received, but those which are to be transmitted.) The successive Comm-D replies of the response are transmitted with a nominal spacing of 136 msec between preambles (16 msec between the end of a reply and the succeeding preamble). Although the precise spacing of segments in the response is known to the interrogator, each segment is transmitted as a full Comm-D reply with a preamble in order to resynchronize the reply decoder in case the preceding segment is not correctly received.

After the complete response to the Comm-C interrogation has been received, another Comm-C interrogation with an updated SRS subfield is transmitted to request segments not yet received (either because they were not requested in the first response, or because they were received in error). The transponder replies again with the requested segments. The cycle is repeated until all segments have been received. As with uplink ELMs this process may take place within a scan or over several scans. When all segments have been received, the complete message is transferred to the designated recipient.

2.2.1.8.2 Non-Selective Downlink ELM Protocol. When the non-selective protocol is used, a sensor delivers a downlink ELM as described in previous text. The sensor then terminates the delivery by sending a PC code = 6 (closeout downlink ELM) in a surveillance or Comm-A interrogation. As with the air-initiated Comm-B message, the transponder will not clear a downlink ELM unless it has been read out at least once. This condition is necessary because multiple closeouts can be received due to downlink failure. Without this readout requirement, a second waiting message that has not as yet been transferred to the ground could be inadvertently cleared.

2.2.1.8.3 Multisite Downlink ELM Protocol. When the multisite downlink ELM protocol is used, a sensor obtains a reservation before beginning the readout of a downlink ELM message. This is done by transmitting a surveillance or Comm-A interrogation with the following relevant fields and subfields:

DI code	=	1 (Multisite special designator — SD)
MES code	=	3 (Downlink ELM reservation)
RSS code	=	3 (Report downlink ELM reservation status in utility message — UM)
IIS code	=	Sensor site address

2.2.1.8.3.1 The transponder reply to this interrogation includes the site address (in the interrogator identification subfield (IIS) of the UM field) and the type of reservation made by the interrogator (in the identification designator subfield, IDS=3) of the sensor currently reserved for uplink activity. IDS=3 indicates downlink ELM (Comm-D) reservation active. A sensor recognizing its own site address as the reserved site will proceed with the downlink ELM transfer. If a different site is designated as the reserved site, downlink ELM activity is not initiated on this scan. A new reservation request may be attempted during the next scan if the downlink request (DR) code indicates that a downlink ELM message is still pending. When the downlink ELM delivery is complete, the reserved sensor closes out the delivery by sending a surveillance or Comm-A interrogation with: (1) DI code = 1(multisite SD), (2) MES code = 4 (downlink ELM closeout), and (3) IIS code = the sensor site address. The transponder only accepts the closeout if the interrogator identification subfield (IIS) in the interrogation matches the reserved downlink ELM IIS code stored in the transponder. A reservation will timeout if 18 seconds pass with no delivery activity. The transponder is then able to grant a reservation to another site for transfer of the downlink ELM. An example of a multisite downlink ELM delivery is presented in Table 2–13. Information concerning multisite delivery of a 2-segment downlink ELM to a local site with an interrogator address equal to 2 is presented in Table 2–13.

TABLE 2-13. EXAMPLE OF MULTISITE DOWNLINK ELM DELIVERY

INTERR. (1)	REPLY (1)	RELEVANT FIELDS	MEANING
S or A	—	—	—
—	S or B	DR=17	Announces presence of 2-segment downlink ELM.
S or A	—	DI=1 MES=3 RSS=3 IIS=2	Multisite special designator (SD). Downlink ELM reservation. Request for Comm-D reservation status in UM. Local site's interrogator identification.
—	S or B	IDS=3 IIS=2 ⁽²⁾	Downlink ELM reservation status is in IIS. Site 2 (the local site) is the reserved Site.
C	—	RC=3 SRS	MC contains request for downlink ELM delivery in SRS Indicates delivery request for segments 1 and 2.
—	D	KE=0 ⁽³⁾ NC=1 MD	Downlink ELM in MD. Indicates MD contains segment 1. Segment 1 of downlink ELM.
—	D	KE=0 ⁽³⁾ NC=1 MD	Downlink ELM in MD. Indicates MD contains segment 2. Segment 2 of downlink ELM.
S or A	—	DI=1 MES=4 IIS=2	Multisite SD field. Multisite uplink ELM closeout. Local site's interrogator identity.
—	S or B	DR=0 ⁽⁴⁾	Downlink ELM closed out.
Notes: 1. S = Surveillance, A = Comm-A, B = Comm-B, C = Comm-C, D = Comm-D			
2. If IIS? 2, the local site is not the reserved site and no further action is taken on this message			
3. If all segments are not successfully received, the sensor will send another SRS requesting redelivery of missing segment(s).			
4. If DR? 0 in reply to a downlink ELM closeout, this condition indicates the presence of another air-initiated message (either B or downlink ELM depending on the value of DR).			

2.2.1.9 Message Protocol Independence. The Mode S communication protocols are defined in such a way that the Comm-A, Comm-B, uplink ELM, and downlink ELM protocols are completely independent. This means that delivery activity for the different message types can be freely interleaved without restriction. If the multisite protocol is used, it is possible for Comm-B, uplink ELM, and downlink ELM reservations to be granted to three different sites.

2.2.1.10 Capability Reporting. The ground system must know what capability an aircraft has to accept and provide data link information. This is accomplished through the reporting of transponder capability. Information concerning capability reporting is presented under the following headings: (1) Capability Report and (2) Data Link Capability Report.

2.2.1.10.1 Capability Report. Basic transponder capability is reported in the CA field of the Mode S all-call reply (DF=11). The CA field informs the ground system of the capability. If CA=0, this indicates no data link capability. If CA=1, this indicates Comm-A/B capability. If CA=2, this indicates Comm-A/B and uplink ELM capability. If CA=3, this indicates Comm-A/B, uplink ELM, and downlink ELM capability. For each category of capability, the transponder can handle both the non-selective and the multisite modes. Additional details of the specific transponder capability for codes 1 through 3 are obtained by reading the transponder data link capability report via a ground-initiated Comm-B message. A transponder reporting CA code = 2 can handle uplink ELMs. This includes the ability to transmit the single Comm-D format required for technical acknowledgement of the uplink ELM transfer. A transponder reporting CA code = 3 has the communication capability of the uplink ELM transponder, and also has the registers and logic to accept and process downlink extended length messages. This transponder also has the higher duty cycle transmitter needed to send bursts of long Mode S replies.

2.2.1.10.2 Data Link Capability Report. A sensor obtains the data link capability report of a transponder so equipped by sending a ground-initiated Comm-B request to read out Comm-B data selector 1 (DBS1) code = 1 and DBS2 code = 0. The data link capability report contains details that inform the ground of the specific data link capability that currently exists onboard the aircraft. This information is given insufficient detail to define the services that the avionics equipment can handle. The capability is updated onboard the transponder at least every four seconds to ensure that the reported capability reflects the current status of the avionics equipment. The data link capability will normally be read by a sensor immediately after the aircraft is put on the sensor's roll-call. In addition, any time that the transponder detects a change in aircraft capability, the transponder causes a transfer of the data link capability report to the ground as a Comm-B broadcast message.

2.2.1.11 Aircraft Identification Protocol. All aircraft use an aircraft identification code for air traffic control purposes. Some aircraft (usually general aviation) use the aircraft registration number that is painted on the aircraft. Other aircraft (principally those used for air carrier and military flights) use variable aircraft identification codes based on the commercial flight number or the military call sign. Mode S transponders with Comm-B capability could automatically report this aircraft identification data. General aviation aircraft could permanently report the registration number. Other aircraft could use a pilot input device to allow manual selection of the identification code for each flight. A sensor obtains the aircraft identification report by sending a ground-initiated Comm-B request to read out Comm-B data selector 1 (BDS1) code = 2 and BDS2 code = 0. If the aircraft identification code is changed in flight, the ground is notified by delivery of the aircraft identification message using the Comm-B broadcast protocol.

2.2.1.12 Utility Message Protocol. The function of the utility message (UM) field has been previously described in connection with the multisite communications protocols. If the content of the UM field in a reply is not specified by the interrogation, the UM field contains the site number reserved for multisite Comm-B or downlink ELM activity (if any). If both message protocols are in use, the reporting of the Comm-B reserved site takes precedence over the reporting of the downlink ELM reserved site. The specific activity (Comm-B or downlink ELM) being reported is defined by the identification designator subfield (IDS) of the UM field. This is followed by the interrogator identification subfield (IIS) that gives the site number of the reserved site. The voluntary reporting of reserved B and downlink ELM sites eliminates unnecessary attempts by other sensors to transfer a message if a reservation has already been granted.

2.2.2 Interrogator Functional Operation

Each Mode S system contains an Interrogator that consists of two Interrogator cabinets (channel A-unit 2 and channel B-unit 3), one Mode S junction box number 1 (unit 1), two time of year (TOY) clocks, and an RF transfer switch (unit 9) at a terminal site. At an enroute site, the Interrogator equipments

are comprised of two Interrogator cabinets (channel C-unit 8 and channel A-unit 2), one Mode S junction box number 1 (unit 1), two time of year (TOY) clocks, and RF transfer switch (13 and 9). Refer to section 1 for further physical description details. The Mode S Interrogator has two basic functions: (1) locate and identify aircraft and (2) fault detection testing. The units used by the interrogator to perform these functions are: transmitter, receiver/RFTTG, signal processor, time of year (TOY) clocks, power distribution, environmental control circuits, and input/output buffering.

The functions performed within an interrogator are as follows:

- S** Amplifies and modulates RF signals for transmission through the antenna.
- S** Processes Mode S and ATCRBS RF replies into digital monopulse data and quantized video signals.
- S** Interfaces with the radar, RF antenna, and Data Processing System (DPS).
- S** Processes control information received from the DPS, and provides target reports, timing information, and status reports to the DPS.
- S** Provides power distribution circuits for the interrogator and MSJB-1 cabinets.
- S** Performs environmental control and monitoring to maintain temperature, and to detect elevated and overheat temperature conditions.
- S** Performs fault detection functions.
- S** Generates test targets for testing.
- S** Provides a stable and accurate time of year.

2.2.2.1 The Interrogator interrogates and processes replies from Mode S and ATCRBS equipped aircraft. To provide surveillance of both ATCRBS and Mode S equipped aircraft with minimal mutual interference, the RF Channel is time-shared between ATCRBS (and Mode S all-call) modes, and the roll-call mode. Following each ATCRBS interrogation, the replies are processed for an interval corresponding to the maximum desired coverage range at the current antenna azimuth. Channel time is allocated to each Mode S interrogation and reply based upon a prediction of aircraft range. The intervals of time devoted to processing roll-call and all-call activity are called periods (ALL-CALL-PERIOD). During a Mode S roll-call period, one or more roll-call schedules are produced. A schedule (ROLL-CALL SCHEDULE) is a set of interrogation and reply times (TRANSACTIONS) that allows the sensor to carry out one transaction per aircraft to some or all of the aircraft on the active target list. Each transaction consists of interrogation and reply periods. Each interrogation consists of INTERROGATION and SIDE LOBE SUPPRESSION CONTROL transmissions that correspond to primary and auxiliary transmissions for the transmitter. ATCRBS replies and Mode S replies are processed by the receiver/RFTTG and signal processor.

2.2.2.2 Functionally, not only is an interrogator required to locate and identify aircraft, but also to insure that the interrogator is performing accurately and up to specifications. To do this, fault detection tests are performed during every scan. Fault detection testing is performed only when there are no targets to be processed, that is, when there is a NULL period. Fault detection testing includes power/VSWR tests (measuring transmit go/no/ go status, etc.). Specified power/VSWR tests have to be completed every scan while performance test have to be completed over a period of eight scans. Thus, power/VSWR tests have priority over performance tests. If a fault is detected during fault detection testing, the suspect channel is switched off line and the other channel is switched on line by the DPS and fault isolation test (FIT) is initiated. FIT performs a series of tests to isolate the fault and identify the LRU for replacement to correct the fault.

2.2.2.3 Interrogator operations to accomplish the above functions can be grouped into transmit, receive, power distribution, environmental control, TOY clock, input/output buffering, performance monitoring circuit,

and fault detection and fault isolation operations. The various Interrogator equipment operations are discussed in detail under the detailed theory of operation header, which appears later in this section. For additional information and diagrams concerning Interrogator operations, the user must also refer to the Interrogator instruction book TI 6365.INTERR, the transmitter instruction book TI 6365.5, the Receiver/RF Test Target Generator (RFTTG) instruction book TI 6365.7, and the Signal Processor instruction book TI 6365.4.

2.2.3 Data Processing System (DPS) Functional Operation

The Data Processing System (DPS) is comprised of the Processor subsystem (PRCSS cabinets — units 4 and 5), Sensor Control, Common Digitizer-2 (CD-2) modification kit, the communication/surveillance junction boxes (units 6, 10, and 11), and the power subsystem.

2.2.3.1 The DPS external interfaces provide connections to units outside of the sensor, such as: (1) communications link to the Air Traffic Control (ATC) facility which handle uplink and downlink messages to/from the aircraft; (2) communications link to non-ATC facilities that provide uplink and downlink messages to/from the aircraft; (3) surveillance interfaces which handle surveillance and status data to FAA ATC facilities; (4) digitized primary radar interfaces which handle primary radar data from the ASR-9 radar equipment, surveillance data to the ASR-9 radar equipment, and primary radar data from Common Digitizer (CD-2) equipped primary radar; (5) Remote Maintenance Monitor System (RMMS) interface which handles automatic and remote control of the periodic sensor tasks; and (6) remote terminal sensor control; and (7) remote program support facilities interface which handles data extraction and down-line load of new programs.

2.2.3.2 The DPS external interfaces provide connections to equipment internal to the sensor but external to the DPS cabinets, such as the interrogator units and the local terminal used for maintenance and sensor control. The DPS internal interface to the interrogator provides for modulation control, performance monitor control, and site adaptable parameters. The DPS internal interface from the interrogator provides for ATRBS target reports, Mode S target reports, performance monitor status, data extraction information, station time, time of year (TOY), azimuth information, and north mark data.

2.2.3.3 The DPS internal interfaces to the sensor control provide for the selection of the sensor channel, selection of the primary radar channel input, selection of backup surveillance data, and an indication of the selected channels. The DPS also interfaces with a magnetic tape unit and the Common Digitizer (CD-2) modification kit. The DPS interfaces with the local terminal (portable terminal) to provide for local terminal start up functions, data extraction, and diagnostic programs. The DPS interface with the CD-2 modification kit is used to extract primary radar data from the CD-2 equipment that provides primary radar target information and primary radar weather reports.

2.2.3.4 The operation of the various DPS equipments are discussed in detail under the detailed theory of operation header, which appears later in this section. For additional information and diagrams concerning Data Processing System (DPS) operations, the user must also refer to the Data Processing System (DPS) instruction book TI 6365.3.

2.2.4 Calibration Performance Monitoring Equipment (CPME) Functional Operation

The CPME is a remote Mode S test transponder, two or more of which are installed at fixed sites visible to a Mode S sensor and used for calibration and test of the Mode S sensor. The CPME provides full Mode S responses plus conventional Air Traffic Control Radar Beacon System (ATCRBS) Mode 3/A and 3/C, and military Mode 2. The Mode S modes handled by the CPME include All-Call and surveillance uplink and downlink transmissions, single-segment uplink (Comm-A) and downlink (Comm-B) communication transmissions, and extended length uplink (Comm-C) and downlink (Comm-D) communications transactions. The CPME also transmits altitude reports. The CPME provides built-in interface test capabilities that appear to the sensor to be functionally equivalent to those of an ordinary Mode S transponder interface.

2.2.4.1 The CPME stands alone, remote from the sensor. The CPME operates unattended. Two CPMEs are installed near each sensor. The CPME comprises a weatherproof lockable enclosure housing the receiver/

transmitter, power supply, and hinged panel assembly. External to the enclosure is an antenna with random and 150 feet of coaxial cable transmission line with connectors, and the battery box. The battery box contains four batteries, which provide the CPME with 48 volts emergency power.

2.2.4.2 The functional operation of the various CPME units are discussed in detail under the detailed theory of operation header, which appears later in this section. For additional information and diagrams concerning CPME operations, the user must refer to the Calibration Performance Monitoring Equipment (CPME) instruction book TI 6365.8.

2.3 DETAILED THEORY OF OPERATION

The detailed theory of operation of the Mode S system and the Mode S equipment is presented under the following headings:

- a. Mode S System Operation
- b. Interrogation Equipment Operation
- c. Data Processing System (DPS) Equipment Operation
- d. Calibration Performance Monitoring Equipment (CPME) Operation
- e. Mode S Sensor-To-ATC Communications Link Messages
- f. Mode S Sensor-To-Non-ATC Communications Link Messages

2.3.1 Mode S System Operation

The Mode S beacon system sensor performs surveillance of Mode S-and ATCRBS-equipped aircraft within its assigned area of coverage. The Mode S sensor operates a store and forward communications relay for data link communication between aircraft at ATC facilities. The Mode S sensor can accept digitized radar reports from a collocated radar and can combine these radar reports with beacon reports into a composite surveillance output stream. The Mode S sensor correlates these radar and beacon reports when both exist on the same target. The functional structure of the Mode S sensor is presented in Figure 2–1. Most Mode S sensor functions are classified according to the time scale in which the functions operate. The pulse processing functions that involve the generation and processing of signals and operate on a microsecond time scale are: (1) the modulator/transmitter, (2) multichannel receiver, (3) Mode S reply processor, and (4) the ATCRBS reply processor. The dwell processing functions that involve channel transactions and operate on a millisecond time scale are: (1) channel management and (2) ATCRBS reply correlation. The scan processing functions that are paced by the antenna scan time and operate on a time scale on the order of a second are: (1) surveillance processing, (2) data link processing, (3) network management, and (4) performance monitoring. The following text describes these functions and various other functions within the Mode S sensor. Because of the importance to overall Mode S sensor operation, channel management is described first, followed by the functions responsible for the generation and processing of channel signals (pulse processing). The functions involved in surveillance, data link message distribution, and multisensor network coordination are also described in the following text. The Mode S sensor functional operation is presented under the following headings:

- a. Channel Management
- b. Clock
- c. Modulator/Transmitter
- d. Antenna System
- e. Multichannel Receiver

- f. Mode S Reply Processing
- g. ATCRBS Reply Processing
- h. Surveillance Processing
- i. Data Link Processing
- j. Network Management
- k. Performance Monitoring
- l. Message Routing Management
- m. Data Extraction
- n. Site Adaptation

2.3.1.1 Channel Management. Channel management regulates all activity on the RF channel through control of the modulator/transmitter and the ATCRBS and Mode S reply processors. Channel management's principal function is the scheduling of ATCRBS and Mode S interrogations. When configured for back-to-back antenna operation, channel management separately schedules interrogations for each antenna face, and controls switching between the faces. To provide surveillance of both ATCRBS and Mode S equipped aircraft with minimal mutual interference, the RF channel is time-shared between the all-call mode and the roll-call mode as illustrated in Figure 2–8. Information concerning channel management is presented, as follows: (1) ATCRBS/Mode S and all-call scheduling, (2) Mode S roll-call scheduling, and (3) channel management organization.

2.3.1.1.1 ATCRBS/Mode S and All-Call Scheduling. ATCRBS interrogations are scheduled at the beginning of an ATCRBS period as determined by a site adaptation input that defines the sensor time line. To reduce the incidence of “fruit” and second-time-around target reports, these interrogations can be delayed from the nominal interrogation time. In any of the aforementioned modes, a range of ATCRBS interrogation rates can be accommodated. In order to provide a maximum of channel time for roll-call activity, the sensor has been designed for a nominal ATCRBS rate of two interrogations per mode. Following each ATCRBS interrogation, the sensor processes ATCRBS replies for an interval corresponding to the maximum desired coverage range at the current antenna azimuth. When the desired coverage range is short, initiation of the subsequent Mode S interval may be delayed to allow replies from longer-range ATCRBS targets to “ring out” so that they do not interfere with roll-call replies in the following roll-call interval.

2.3.1.1.1.1 In order to conserve channel time, Mode S all-call acquisitions also performed during the ATCRBS listening period. Replies from unacquired Mode S and ATCRBS aircraft may be elicited one of two ways: (1) through use of the ATCRBS/Mode S all-call interrogation, and (2) by using a Mode S-only all-call interrogation followed by an ATCRBS-only all-call interrogation. The latter approach must be used with the site addressed acquisition (stochastic acquisition) and lockout override functions because the coding for these functions is contained in format UF=11 (the Mode S-only All-Call).

2.3.1.1.1.2. A channel management frame utilizing Overlapped All Call Interrogations is used, at times, in enroute configurations. By using a frame with overlapped all call interrogations, the time needed for all call interrogations to Mode S targets may be overlapped with the time needed for all call interrogations to ATCRBS targets. This is more helpful in enroute configurations since the additional sensor coverage range requires that more time is spent for the interrogations and replies to traverse the distance between the sensor and the limits of the coverage area.

2.3.1.1.2 Mode S Roll-Call Scheduling. Scheduling of Mode S roll-call interrogations and replies occurs under the following ground rules:

- a. Mode S interrogations are addressed only to aircraft within the antenna beam.
- b. Channel time is allocated to each Mode S interrogation and reply based upon a prediction of aircraft range.
- c. Mode S surveillance and data link procedures may require more than one interrogation to each aircraft. Therefore, the sensor is able to reinterrogate an aircraft while it remains in the beam.

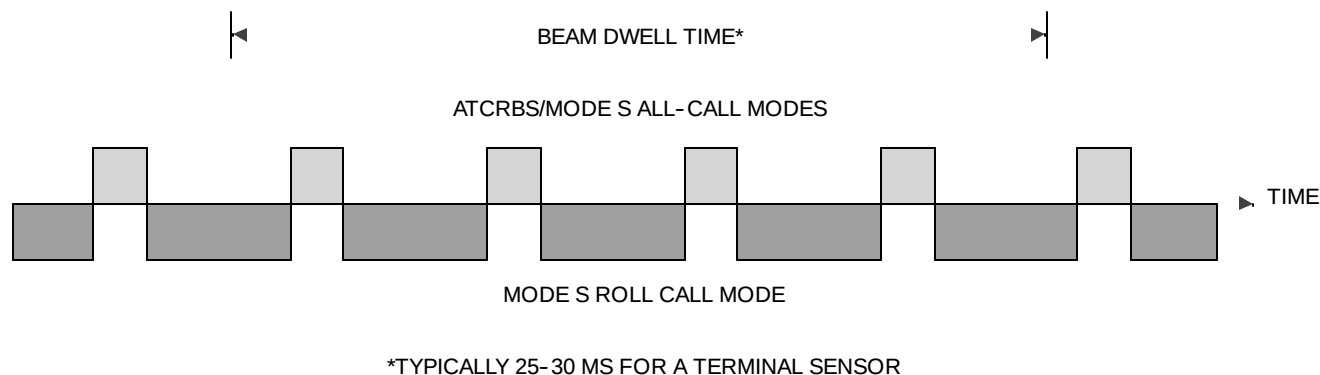
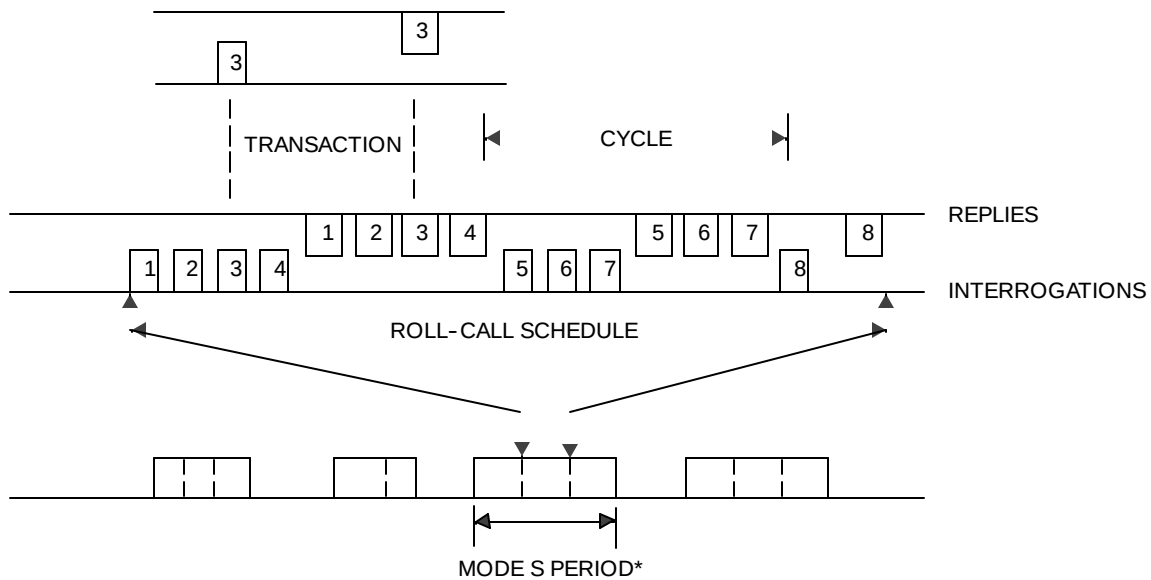


FIGURE 2-8. ATCRBS/MODE S TIME SHARING DIAGRAM

2.3.1.1.2.1 The sensor maintains an active target list, which is comprised of those Mode S aircraft that are within the antenna beam. The sensor makes repeated passes through this list and schedules discretely-addressed Mode S interrogations and replies on a nonconflicting basis. A single aircraft may appear on one or more of the resulting schedules of interrogations and replies, so that multiple surveillance and communication tasks can be accomplished. In the case of a failure to receive a reply, the capability for repeated scheduling of interrogations to an aircraft provides a high overall surveillance/communication reliability. The principal elements of Mode S roll-call scheduling are illustrated in Figure 2-9. The intervals of time devoted to Mode S roll-call activity are called Mode S periods. During a Mode S period, one or more roll-call schedules are produced. A schedule is a set of interrogations and reply times that allows the sensor to carry out one interrogation-reply pair per aircraft to some or all of the aircraft on the active target list without providing a second transaction to any aircraft. The interrogations are timed so that nonoverlapping blocks of channel time are assigned to each individual interrogation and reply. If sufficient time is available to schedule all aircraft on the list, the time is allocated to aircraft according to a preassigned transaction priority.

2.3.1.1.2.2 Roll-call scheduling commences with the first (longest range) aircraft on the list by scheduling an interrogation at the assigned start time of the schedule. The expected reply arrival time is then computed and a suitable listening period provided. Subsequent aircraft are scheduled by placing their reply listening periods in sequence and computing the corresponding interrogation times. A cycle is completed when the next interrogation, if so scheduled, would overlap the first reply. This interrogation is deferred to start a new cycle. Several types of transactions must be efficiently combined in forming a Mode S schedule. Since the aircraft on the active target list are in various stages of completion, with respect to Mode S activity, each may require a different kind of transaction. Figure 2-10(A) illustrates a typical cycle comprised of long and short interrogations, coupled with long and short replies. The cycles shown in Figure 2-10(B) and 2-10(C) illustrate the inclusion of downlink and uplink extended length message (ELM) transactions.



*THIS MODE S PERIOD COMPRISES THREE SCHEDULES. THE SECOND INCLUDES EIGHT TRANSACTIONS, GROUPED IN THREE CYCLES OF 4, 3, AND 1 TRANSACTIONS, RESPECTIVELY.

FIGURE 2-9. MODE S ROLL-CALL MODE SCHEDULING DIAGRAM

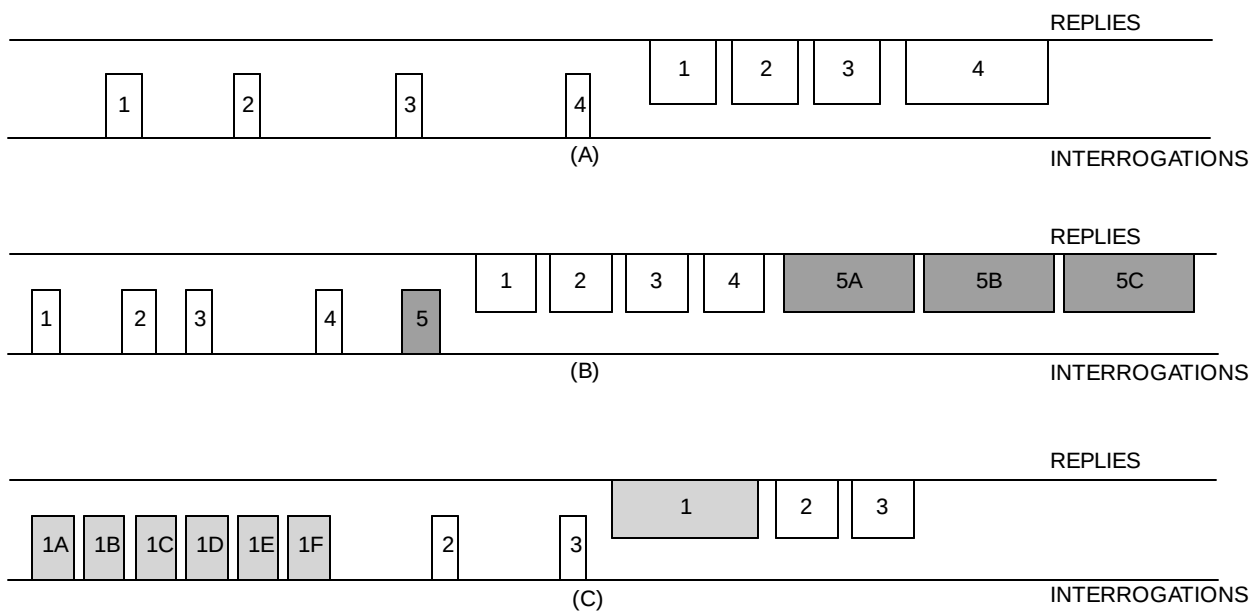


FIGURE 2-10. MODE S INTERROGATION AND REPLY CYCLES

2.3.1.1.3 Channel Management Organization. The three subfunctions which comprise channel management are: (1) target control, (2) transaction block creation, and the (3) ITRI scheduler. The data flow between these subfunctions, and their interfaces with other sensor functions, are illustrated in Figure 2–11. Channel management receives inputs from surveillance processing, data-link processing, and network management. Surveillance processing provides channel management with predicted position (azimuth and range) of Mode S aircraft. Data-link processing provides organized lists of pending uplink messages for each Mode S aircraft. Network management controls the track state to define the kinds of service (both surveillance and communication) to be afforded each aircraft. Channel management also receives inputs from the performance monitor functions of both the Interrogator and the DPS. The Interrogator performance monitor function periodically provides TTG control, blocks for subsequent scheduling by channel management. The DPS performance monitor provides commands to channel management which control scheduling during sensor calibration. Channel management has control over the modulator/transmitter unit and the Mode S and ATCRBS reply processors. Channel management communicates with these units by generating interrogation and reply control commands, and by receiving Mode S reply data blocks. ATCRBS reply data blocks are sent directly to surveillance processing. When aircraft leave the antenna beam, a record of channel activity and downlink message content is placed in a released target list, and is then passed on to the data-link processing, surveillance processing, and network management functions.

1. Target Control: Target Control provides the interface between the other two subfunctions (transaction block creation and ITRI scheduler) to provide the first Mode S roll-call schedule for each roll-call period. Target control controls the functioning of the transaction block creation task by requesting entering target lists for the specified azimuth limits of an upcoming roll-call period. Target control merges this entering target list into the active target list. In order to facilitate the computation of a non-conflicting schedule of interrogations and replies, an active target list is arranged in order of decreasing target range. This order is maintained each time the list is modified. In the back-to-back antenna configuration, two active target lists are maintained. Targets which are leaving the beam or are completely serviced are moved from the active target list into a released target list. These targets are then forwarded to data-link processing, surveillance processing, and/or network management for further processing.

Target control, operating on the contents of the active target list, prepares the first roll-call schedule of a roll-call period. Subsequent schedules, if any, are prepared by the ITRI scheduler. If insufficient time remains for a complete schedule, the available time is allocated based on transaction priority and type, as follows: (1) Level 1 — interrogations for altitude, ATCRBS ID, and high priority Comm A/Comm-B messages, (2) Level 2 — high priority uplink extended length messages (ELMs), (3) Level 3 — additional (normal priority) Comm-A/Comm-B messages, (4) Level 4 — normal priority uplink uplink ELMs and (5) downlink ELMs. The output of the first roll-call schedule processing consists of the interrogation and reply control blocks to be transmitted by the ITRI scheduler, along with information to support the subsequent reply processing. Target control then passes this first schedule to the ITRI scheduler. Target control also receives TTG control blocks from the Interrogator. The requested TTG interrogations and replies are built and are passed on to the ITRI for scheduling during a TTG period or during available time at the end of a roll-call period.

2. Transaction Block Creation: At regular intervals, target control directs transaction block creation (TBC) to provide a list of aircraft about to enter the antenna beam. TBC obtains this list from the surveillance file via the data manager. TBC consults the surveillance file to obtain the current status of the target, and prepares one or more surveillance transactions. TBC also checks the active message list for the existence of Comm-A, Comm-B, or ELM requests. If uplink messages and/or downlink message requests are pending for an aircraft entering the beam, TBC determines the number and type of transactions required to accomplish these tasks. The output of TBC is a list of data blocks called the entering target list (one block for each new aircraft), which contains a complete specification of the required set of transactions needed to accomplish all pending surveillance and communication tasks. For the back-to-back antenna configuration, the list consists of two sublists which correspond to the two beams.

3. ITRI Scheduler: The channel management software which resides on the ITRI is referred to as the ITRI scheduler. The ITRI scheduler controls the Interrogators by transmitting commands for the modulator/transmitters and reply processors. These commands assure that all ATCRBS and Mode S activities take place at the proper time and in the proper sequence. The ITRI scheduler is driven by the absolute frame structure. During a roll-call period, the ITRI scheduler is activated by the arrival of the first schedule provided by target control. During periods following a roll-call period, the absolute frame structure dictates whether the ITRI scheduler will schedule all-call interrogations, TTG control blocks, or be idle. The ITRI scheduler receives the first schedule of a roll-call period from target control, and then transmits it to Interrogator. If sufficient time remains in the roll-call period, the ITRI schedule determines which transactions (if any) are eligible for subsequent schedules according to the same priority scheme described in target control. When the ITRI scheduler receives Mode S replies, the ITRI scheduler examines the contents of each reply. If the transaction was successful, the ITRI scheduler modifies the target's data block and then schedules and transmits the next pending transaction in the subsequent schedule. If the transaction was unsuccessful, the transaction is repeated in the next schedule, and the next pending transaction is, delayed to a later schedule. The ITRI scheduler also creates new transactions (if needed). Finally, the ITRI scheduler marks the completion of targets for which no further transactions are pending.

2.3.1.2 Clock. Each Mode S sensor is equipped with a "time-of-day" clock that permits precisely-timed coordination of activities at different sensors. Site-to-site synchronization of + 2.5 milliseconds is required for utilization of multi-site surveillance data. System time is continuously available to the modulator/transmitter, Mode S and ATCRBS reply processors, and channel management. In addition, the Mode S sensor clock provides timing references for the generation of both Mode S and ATCRBS interrogation waveforms, and the demodulation of replies.

2.3.1.3 Modulator/Transmitter. The modulator/transmitter accepts digital control inputs from channel management and generates the requisite RF interrogation signals. For each interrogation to be transmitted, the control inputs specify the mode, the transmission time (with 1/15 ms resolution), and for a Mode S interrogation, the contents of the data block prior to encoding. The modulator generates the sequence of parity check bits and combines them with the specified discrete address. Figure 2-12 is a general block diagram of the modulator/transmitter. Mode S interrogations may be transmitted at a low or high power under control of channel management. ATCRBS transmit power is site- adaptable with nominal values of 300 watts for a terminal sensor and 800 watts for an enroute sensor. An auxiliary transmitter is required for the generation of the sidelobe suppression (SLS) control pulses, since the Mode S SLS pulse is transmitted simultaneously with the Mode S interrogation.

2.3.1.4 Antenna System. The Mode S sensor employs a rotating fan-beam antenna having a sum (S) pattern and a difference (D) pattern. The interrogation is transmitted and the reply received on the sum pattern (refer to Figure 2-13a). The reply is also received on the difference pattern, and the ratio of the signals received on the difference and sum patterns is used to estimate the off-boresight angle of the target (refer to Figure 2-13b). In addition to the sum and difference patterns, the antenna system includes an SLS control pattern. This pattern is used for the transmission of the P₂ and P₅ sidelobe suppression pulses (refer to Figures 2-2 and 2-4). The SLS control pattern is also used on receive, where comparison of the amplitudes of the same signal received on the sum pattern and the control pattern permits rejection of signals received via a sidelobe of the sum pattern. To perform its function, the gain of the SLS control pattern exceeds that of the sum pattern in all directions except those corresponding to the main beam of the sum pattern. Information concerning the antenna system is presented under the following headings: (1) antenna configuration, (2) azimuth pattern characteristics, and (3) elevation pattern characteristics.

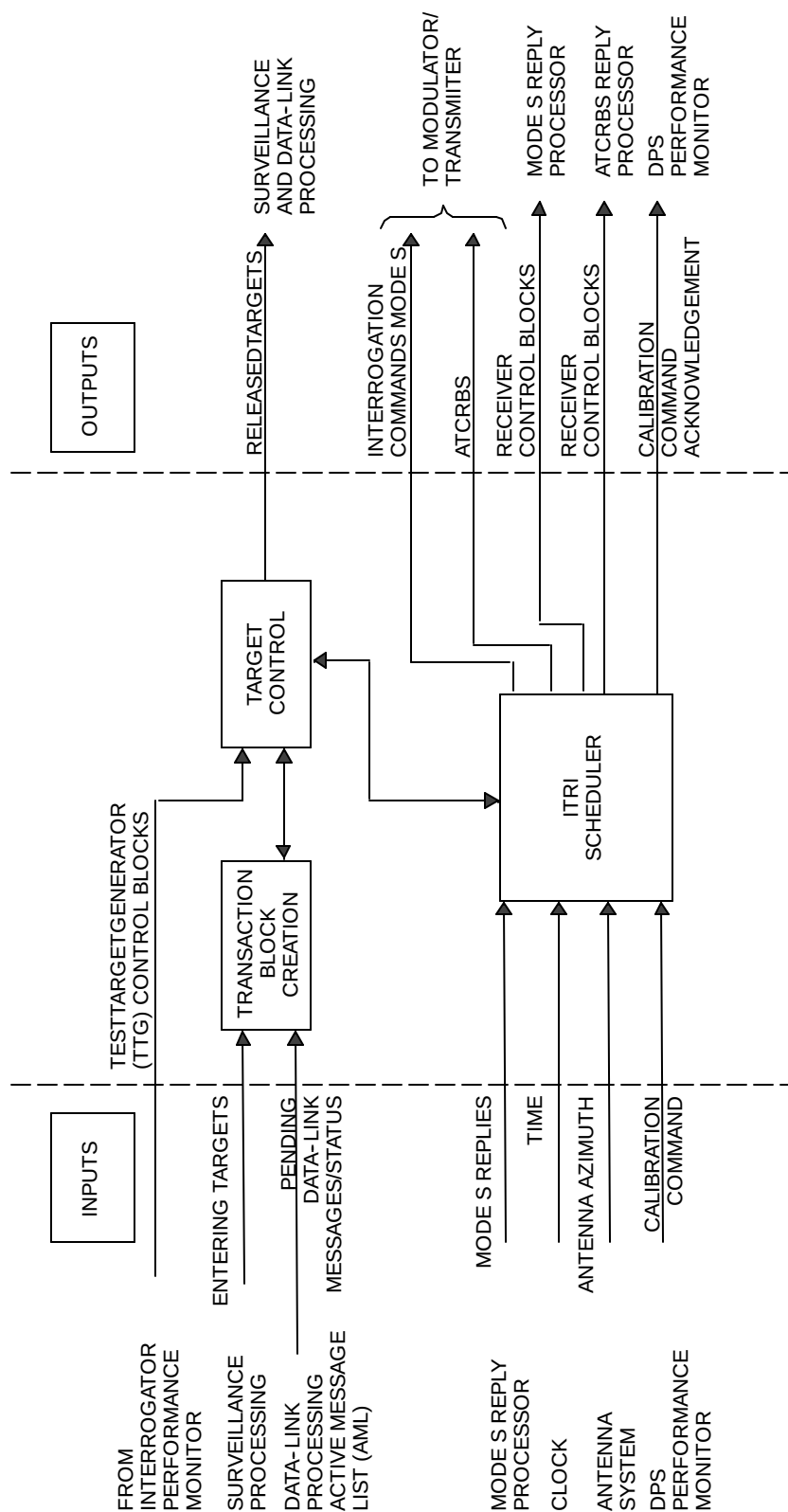


FIGURE 2-11. CHANNEL MANAGEMENT FUNCTIONAL BLOCK DIAGRAM

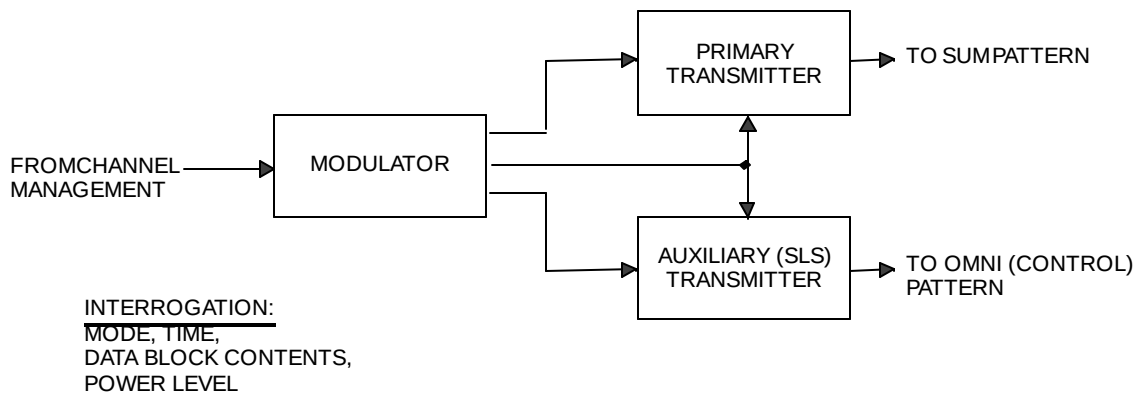


FIGURE 2-12. MODULATOR/TRANSMITTER GENERAL BLOCK DIAGRAM

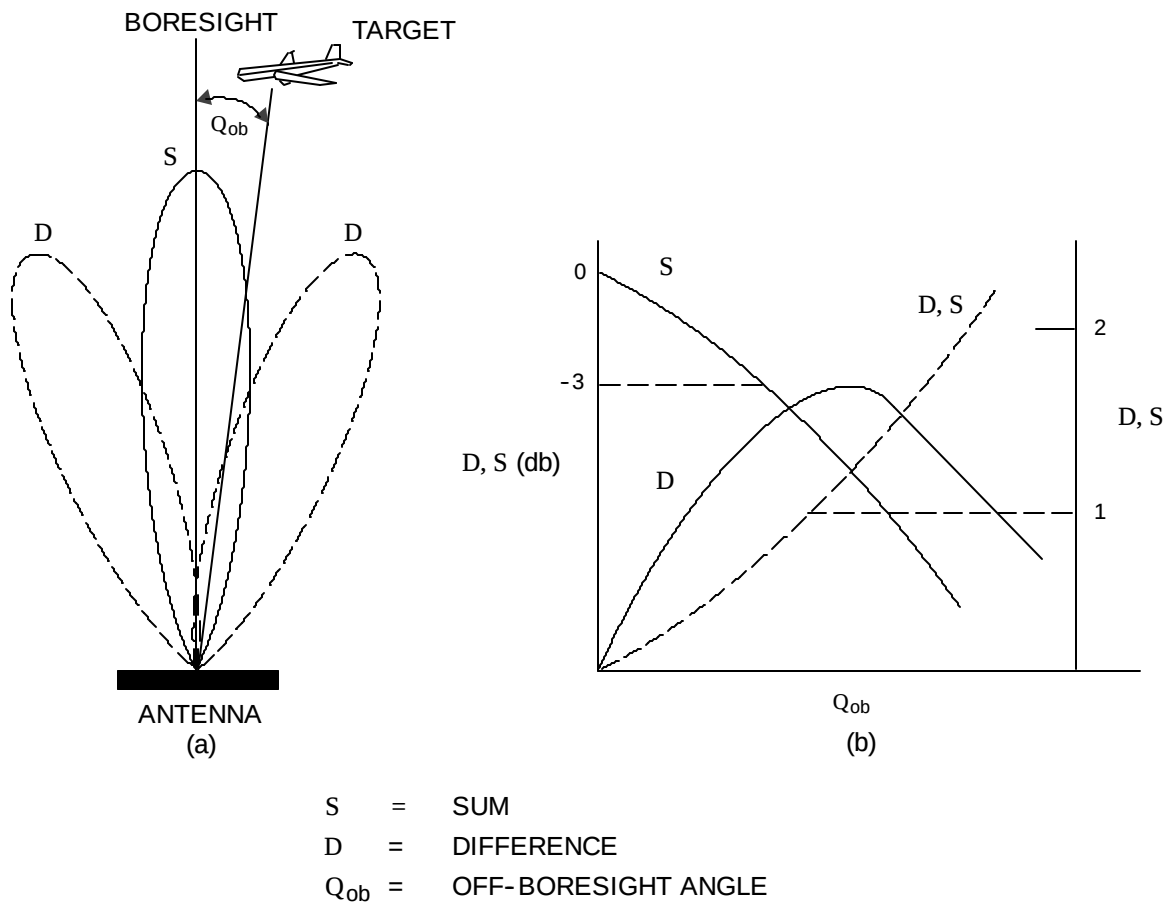


FIGURE 2-13. SUM AND DIFFERENCE ANTENNA PATTERNS

2.3.1.4.1 Antenna Configuration. The Mode S antenna may either stand alone, mounted on its own pedestal, or it may share a pedestal with a radar sensor. The latter will normally be the case when a Mode S antenna system and radar are collocated. When sharing a pedestal with a radar antenna, the Mode S antenna may employ a completely separate radiating structure, mounted, for example, above the radar antenna. On slowly rotating radar antennas (long-range air route surveillance radars which rotate at 5 to 6 rpm), back-to-back Mode S antennas may be used to realize a data update rate twice that of the host radar. The back-to-back antennas may be mounted atop the collocated radar antenna if space permits. If space is restricted, such as in a radome, the forward antenna may be chin-mounted on the radar feed support boom with the rearward antenna mounted at approximately the same height on the backside of the radar antenna.

2.3.1.4.2 Azimuth Pattern Characteristics. The choice of azimuth beamwidth is constrained by two conflicting requirements. A broad beam results in higher target capacity (the sensor can provide discrete-address surveillance and data-link service to a larger number of aircraft). A narrow beam provides higher azimuth measurement accuracy. The most generally useful range of 3-db azimuth beamwidth for the Mode S sensor appears to be 2.4 to 4 degrees. Both sum and difference patterns must have low amplitude azimuth sidelobes to minimize the effects of sidelobes interference on reply detection and monopulse azimuth estimation.

2.3.1.4.3 Elevation Pattern Characteristics. A rapid cutoff of the antenna patterns at the horizon is desirable to minimize the energy incident on, and reflected by, the ground, and thereby to: (1) decrease the magnitude of lobing and (2) reduce the error introduced in the azimuth measurement due to off-axis multipath signals. While the specific requirements on lower edge cutoff are highly site-dependent, a cutoff rate of approximately 2 dB per degree appears adequate for most typical site environments. Above the horizon, the Mode S sensor must provide coverage to elevation angles of 30 to 40 degrees. Current beacon sensors use antennas having relatively constant gain with the elevation angle (termed sector beams); used with sensitivity-time control (STC), this minimizes the dynamic range requirements of the receiver. A more tapered pattern (intermediate between the sector beam and the cosecant-squared pattern frequently used in radar applications) is preferred for Mode S. The tapered pattern has greater gain near the horizon, which provides for a higher fade margin for long-range, low-elevation angle targets. The tapered patterns' lower gain at high elevation angles is balanced by the reduced range of aircraft at those elevations.

2.3.1.5 Multichannel Receiver. The multichannel receiver accepts the three RF signals from the sum, difference, and omni (control) patterns of the antenna system, and produces three outputs for use by the ATCRBS and Mode S reply processors. These outputs are: (1) log S — the log amplitude of the sum pattern signal, (2) log W — the log amplitude of the omni pattern, and (3) $f(D/S)$ • a bipolar video signal proportional to the ratio of the difference and sum pattern signals. The log S signal is the principal receiver output, used for the detection of reply pulses. Comparison of the amplitudes of the log S and log W signals indicates whether a pulse detected in the log S output was received via the main lobe ($\log S > T + \log W$) or via a side lobe ($\log S < T + \log W$) of the sum pattern (where T = a preset threshold). The digitized value of the $f(D/S)$ signal from each received pulse is compared with a prestored antenna/receiver calibration curve to provide an estimate of the angle of arrival of the signal relative to the antenna boresight.

2.3.1.6 Mode S Reply Processing. Mode S reply processing operates on the multichannel receiver outputs to detect and decode Mode S all-call and roll-call replies, and to generate an estimate of aircraft range and azimuth from each detected reply. The principal steps in Mode S reply processing are shown in Figure 2-15. Information concerning Mode S reply processing is presented under the following headings: (1) preamble detection, (2) message bit processing, and (3) message decoding and error correction.

FIGURE 2-14. MODE S JOINT USE ENROUTE SENSOR GENERAL BLOCK DIAGRAM
(DELETED)

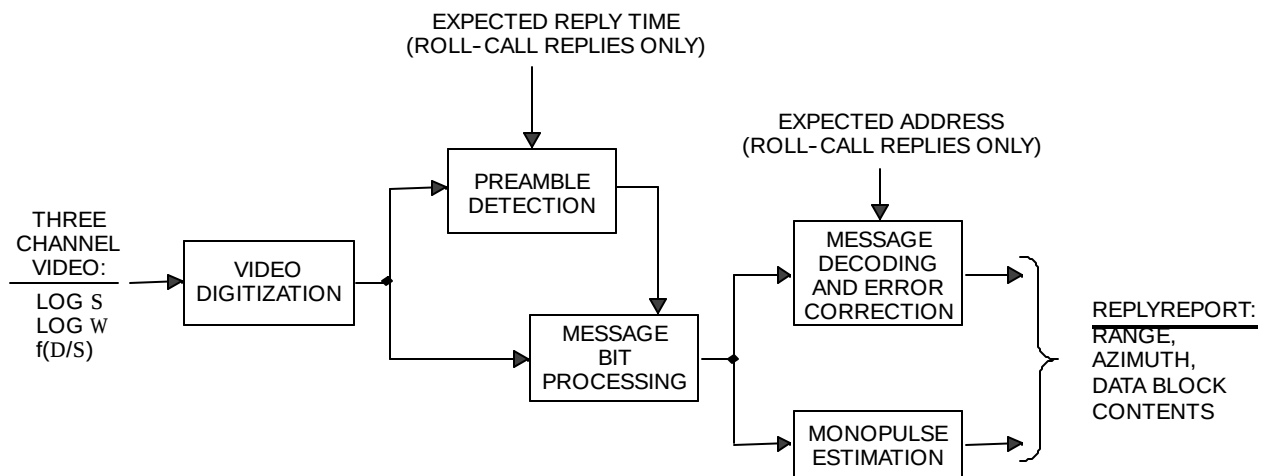


FIGURE 2-15. MODE S REPLY PROCESSING GENERAL BLOCK DIAGRAM

2.3.1.6.1 Preamble Detection. Mode S replies are detected on the basis of the four-pulse preamble waveform preceding the reply data block (refer to Figure 2–5). The preamble detector provides accurate time-of-arrival estimation for aircraft ranging, synchronization of message bit processing, and reply decoding. In the case of replies to roll-call interrogations, channel management provides to the preamble detector an estimate of expected reply time and an uncertainty window. A reply is accepted only if its preamble is detected within this window. Since the reply processor cannot start decoding a new reply when it is still decoding an earlier one, the use of this window minimizes the probability that the reply decoder will miss the desired reply due to a Mode S fruit condition.

2.3.1.6.2 Message Bit Processing. Message bit processing operates on sampled video waveforms to produce a sequence of demodulated message bits, and a confidence bit (high or low) for each demodulated message bit. Since a message bit is transmitted as a pulse in one of two possible positions, depending on whether the bit value is a 0 or a 1, bit decisions are based primarily on the relative amplitudes of the signals received in these two pulse positions. A sidelobe flag is used to help resolve ambiguous situations in which a signal is received in both pulse positions. Bit decisions are indicated as high confidence only when a mainbeam signal appears in one pulse position, and either no signal or a sidelobe signal appears in the other.

2.3.1.6.3 Message Decoding and Error Correction. Message decoding uses the parity check code to detect errors in the demodulated message. Since the parity check bits for roll-call replies are combined with the transponder address, the decoder must know the expected address (supplied by channel management) in order to perform error detection. Whenever a decoded reply contains errors, error correction is attempted if the total number of low-confidence bits in the reply does not exceed a preset threshold. The sum of this threshold minimizes the possibility of erroneously correcting a reply that contains a very large number of errors. Error correction will be successful only if the following conditions are present: (1) all errors are confined within a span of 24 contiguous bits and (2) all errors occur in bits flagged as low confidence. A garbled condition by a single strong ATCRBS reply, which can result in bit decision errors spanning no more than 24 bits, usually results in a correctable error pattern. Therefore, the Mode S data block will be correctly decoded unless the data block is garbled by more than one strong ATCRBS reply.

2.3.1.7 ATCRBS Reply Processing. The ATCRBS processing subsystem has been designed to produce accurate, reliable target reports at low interrogation rates in order to maximize the channel time available for the roll-call mode of operation. Monopulse information is used to determine target azimuth and also to assist in the decoding of overlapped replies. Sidelobe fruit replies are detected and rejected by comparison of the signals received on the sum and omni patterns. When ATCRBS Mode A and C replies are processed, the extracted codes are then included in the target report. Figure 2–16 is a general block diagram of ATCRBS reply processing. ATCRBS reply processing is performed in two consecutive steps as follows: (1) reply decoding, and (2) reply correlation.

2.3.1.7.1 Reply Decoding. Reply decoding operates on the three video outputs of the multichannel receiver to detect ATCRBS replies. For each detected reply, reply decoding provides an estimate of target range, azimuth, code, and code confidence for each mode (A or C) in which the target responded. A general block diagram of the major elements of ATCRBS reply decoding is shown in Figure 2–16. Information concerning ATCRBS reply decoding is presented under the following headings: (1) video digitization, (2) bracket detection, (3) garble sensing and Phantom elimination, (4) code extraction, and (5) monopulse estimation.

2.3.1.7.1.1 Video Digitization. Video digitization provides a digital representation of pulses whose widths are within an acceptable range. This digital representation includes leading edge location, a monopulse sample, and a sidelobe flag. Overlay long pulses are assumed to have resulted from an overlapping of pulses from two replies, and pseudo-leading edges are inserted based on the observed trailing edge position. A monopulse sample is produced for each pulse of acceptable width, except for pulses which are overlapped enough to result in garbled monopulse samples.

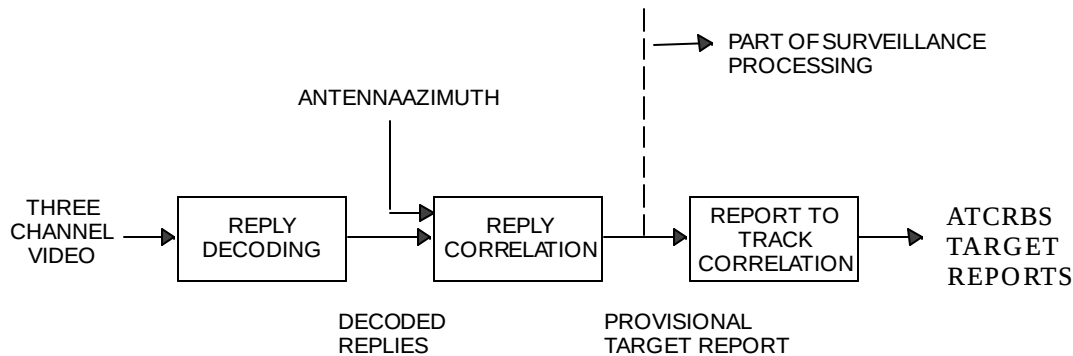


FIGURE 2–16. ATCRBS REPLY PROCESSING GENERAL BLOCK DIAGRAM

2.3.1.7.1.2 Bracket Detection. Bracket Detection is based on the detection of two pulses in the leading edge data stream that have the appropriate spacing characteristics. Bracket pulse pairs for which both pulses are flagged as sidelobe pulses are not declared, and thus sidelobe replies are not decoded.

2.3.1.7.1.3 Garble Sensing and Phantom Elimination. Garble sensing is based on the time separation of two detected bracket pairs that would result in overlapping pulse decode regions for the two replies. The incorrect declaration of a bracket pair made up of pulses from two garbling replies is termed a phantom. Phantoms produced by two garbling replies will be correctly eliminated if no additional garbling reply occurs for 20.3 ms.

2.3.1.7.1.4 Code Extraction. Code extraction is initiated by the declaration of a bracket pair that defines the possible information pulse locations. Information pulses are associated with a reply (bracket pair) on the basis of their leading edge location relative to the bracket pair, and their monopulse sample values relative to a monopulse reference. The initial monopulse reference is the monopulse sample of the first framing pulse, except when that pulse is in a possible garble region of an earlier reply or is labelled as sidelobe. If the first framing pulse is not suitable, the monopulse sample of the second framing pulse is used as the monopulse reference. If that pulse is also not suitable, the reply is rejected. Each information pulse decision is accompanied by a confidence bit (high/low confidence). A pulse is decoded as high confidence “1” if it falls within one of the acceptable pulse decoding regions of the reply being decoded, and its monopulse sample correlates with the monopulse reference of that reply and not with any garbling reply (if any). A pulse position is decoded as high confidence “0” if no pulse is detected in its decode region, and no pulse leading edge is detected in the sample position just ahead of this decode region.

2.3.1.7.1.5 Monopulse Estimation. Each pulse which lies within a valid information pulse position of a detected reply, and which correlates with the monopulse reference of that reply, is used to update the reference by averaging the reference value and the sample value. The monopulse estimate for a decoded ATCRBS reply is the final value of the monopulse reference for the reply at the time the F₂ pulse has been processed. In order to avoid an erroneous monopulse estimate because of a bad reference sample at the start of the pulse decoding, an ATCRBS monopulse estimate is not declared unless it includes at least two monopulse samples.

2.3.1.7.2 Reply Correlation. The function of reply correlation is to combine all replies received from a transponder in one scan into a single target report. Each reply received from reply processing

is correlated (compared) in range, azimuth, and high-confidence code pulse with existing target report files. If a reply does not correlate with any existing target report file, a new file is started with the reply. Code correlation is done by comparing only high-confidence code positions of a reply with the high-confidence code positions of the code estimate contained in the target report file. This code estimate (Mode A or C) is updated by forming a new estimate, which consists of the composite of the current file code estimate and the high-confidence code positions of the correlating reply. Likewise, the composite confidence bit sequence for each reply mode is updated by adding to the high confidence positions in the target report file any new high confidence positions in the correlating reply. The azimuth estimate provided in a target report consists of the azimuth estimate of the reply closest to boresight when replies are received only on one side of boresight, or the average of the azimuths of the two replies which straddle boresight.

2.3.1.8 Surveillance Processing. Surveillance processing maintains target files on all Mode S and ATCRBS aircraft within the sensor's coverage volume. The data flow between the main elements of surveillance processing are shown in Figure 2–17. The main functions of surveillance processing are:

- a. To predict next-scan position of Mode S aircraft for interrogation scheduling;
- b. To edit and correct ATCRBS target reports based upon data from previous scans;
- c. To perform radar/beacon correlation of target reports from a collocated radar;
- d. To disseminate composite ATCRBS/radar and Mode S/radar surveillance data to ATC users.

2.3.1.8.1 Mode S Surveillance Processing. Mode S surveillance processing operates on the set of Mode S replies received from an aircraft during a scan to produce a target report on that aircraft and a prediction of its position on the next scan. Figure 2–18 is a general block diagram of Mode S surveillance processing. A single Mode S report for a given address is selected to represent the surveillance data for each track. Many replies with the same address may be present, either because of multiple roll-calls for data-link activity or because this aircraft is just being acquired and has answered an all-call. The report selection process includes finding the reply that is near beam center to maximize azimuthal accuracy, and also finding the reply at shortest range to exclude false (reflected) targets.

2.3.1.8.1.1 The Mode S all-call reports whose addresses do not match an existing track are examined in the track initiation process to determine if an address match can be made with uncorrelated reports held over from the previous scan. When a match is found, a test for false track due to reflection is made. If the test is negative, a newly-initiated track is entered into the surveillance file. The still-unmatched reports are saved for one scan and are also used in radar correlation. The occurrence of synchronous garble of Mode S all-call replies is signified by the receipt of two or more undecodable all-call replies at a consistent range during a beam dwell. When this condition is detected, an acquisition track is initiated at the range and azimuth of the detected reply. Channel management will then schedule stochastic Mode S-only all-call. If a successful reply is received, the aircraft will then be discretely interrogated and locked out to remove it from the garbling set. Acquisition tracks are only retained for one scan since continued garbling will result in generation of a fresh acquisition track each scan.

2.3.1.8.1.2 All Mode S reports that match with tracks and radar reports that correlate with coasted tracks are used to update and project the Mode S track ahead one scan. This new projection is used by channel management determine when to schedule the next roll-call interrogation to the aircraft. It must be noted, that all references to the antenna scan time for a back-to-back antenna sensor configuration should be understood to mean one-half the scan time of a single-face antenna rotating at the same period rate of speed.

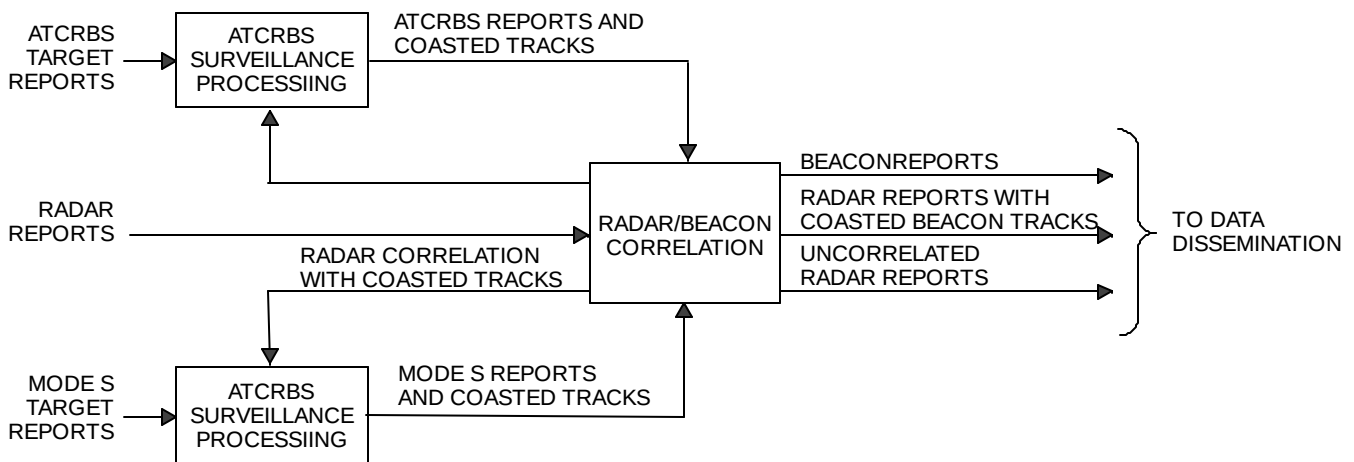


FIGURE 2-17. SURVEILLANCE PROCESSING GENERAL BLOCK DIAGRAM

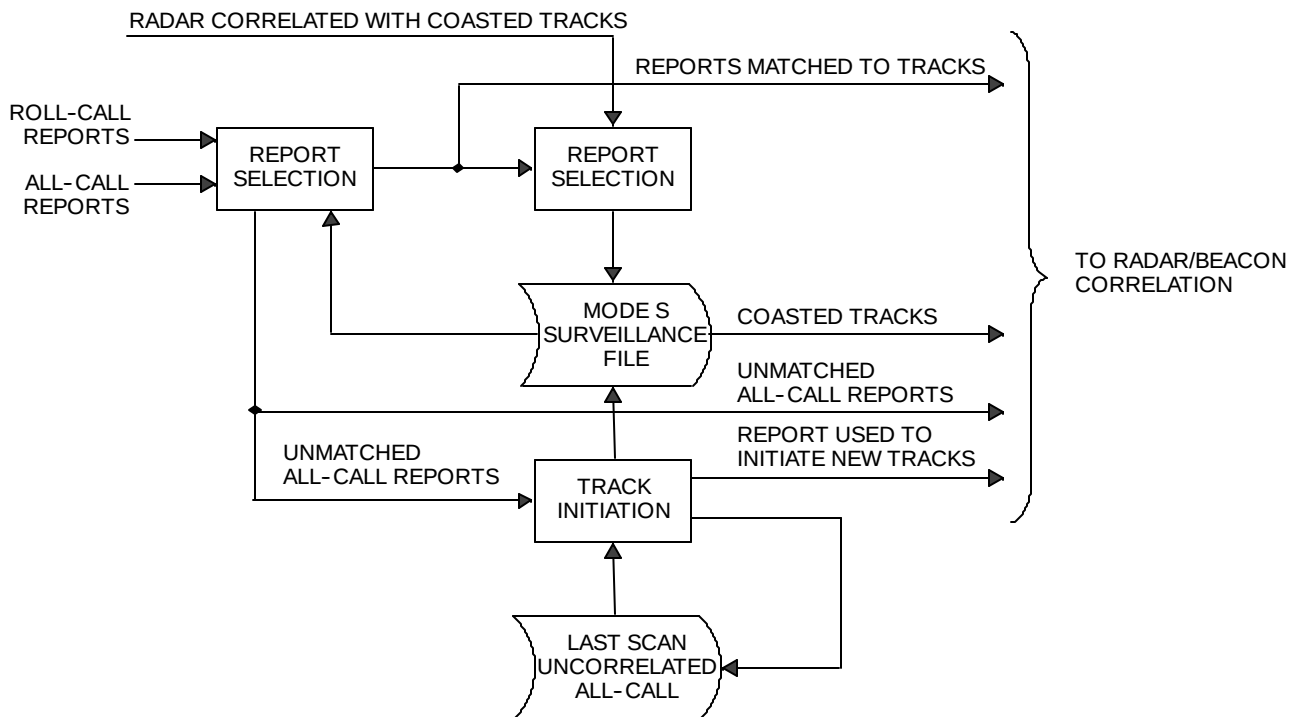


FIGURE 2-18. MODE S SURVEILLANCE PROCESSING GENERAL BLOCK DIAGRAM

2.3.1.8.2 **ATCRBS Surveillance Processing.** ATCRBS surveillance processing edits and corrects ATCRBS target reports by comparing reports received on the current scan with target tracks derived from reports received on previous scans. Figure 2–19 presents a general block diagram of ATCRBS surveillance processing. Editing includes eliminating residual fruit replies which may have been passed by the ATCRBS reply correlator, and flagging reports suspected of being due to reflections (false targets). When target reports correlate with an existing track, missing or low-confidence code bits may be corrected by insertion of the corresponding bits from the track file.

2.3.1.8.2.1 The lack of a unique identity code as part of each ATCRBS reply (as opposed to Mode S replies) makes necessary a report-to-track correlation function as the first step in ATCRBS surveillance processing. Report-to-track correlation is based upon the several attributes of the report. A report is said to be associated with a track if it falls within a specified range, azimuth, and altitude association interval. If only one ATCRBS report associates with a single track, that report/track pair is declared correlated. For those cases where more than one report associates with a track, or more than one track associates with a report, or if both of these conditions occur, report and track correlations are based upon the following factors: (1) code match, (2) number of replies in the report, (3) altitude match, (4) track maturity, and (5) distance parameter.

2.3.1.8.2.2 During track update, a special code situation is sensed. If the Mode A code of the report does not match the code of the track, it is assumed that a code change has been made in the aircraft and a transition situation is noted. After three scans of consistent new code, the track file code is updated. An ATCRBS fruit reply is rejected only if it is not radar reinforced, the target did not correlate with a track, and one of the following conditions are present: (1) the report was generated on the basis of two replies (one Mode A and one Mode C), (2) the report was generated on the basis of Mode C only reply, (3) the report was one of a pair of possible code swap reports as indicated by the code swap field of the ATCRBS target report data block, and (4) the report occurred without a monopulse azimuth estimate as indicated by the boresight field of ATCRBS target report data block.

2.3.1.8.2.3 The test for determining whether a report is false is initiated by comparing the ATCRBS report azimuth with a stored table of reflections zones. If the report is in one of the reflection zones, then an image location is calculated and compared against known tracks. If a known track with matching altitude and Mode A code could have caused the report, the report and any track started from it will be tagged as potentially false. Each scan thereafter, the false track is examined to see if it continues to satisfy the “false track” criteria.

2.3.1.8.3 **Radar Beacon Correlation.** The last step in beacon surveillance processing prior to data dissemination is the comparison of radar reports from a collocated primary radar with the result of Mode S and ATCRBS report processing. First, the incoming primary radar reports must be corrected for the offset between the radar antenna boresight and the beacon antenna boresight. This collimation correction is applied by adding a constant to the range of the radar report and another constant to the azimuth of the radar report. The corrected radar report range is then compared to the current listening window of the sensor, and reports beyond the listening window are discarded. The corrected radar reports that remain then form the input to the radar beacon correlation function. A radar report may be identified as associated with a beacon report if the two reports are sufficiently close in both range and azimuth. If no beacon report associates with a radar report, the radar report will be disseminated as a normal radar report. If at least one radar report is associated with a beacon report, the beacon report is said to be radar reinforced, and will be disseminated as a radar-reinforced beacon report. If more than one radar report associates with a beacon report, the closest radar report will be dropped, and the remaining radar report(s) will be disseminated as normal radar reports. When a radar report is disseminated as normal radar report, it can also be used in the process of radar substitution. In this process, beacon tracks which have not received any correlating beacon reports on this scan are compared to radar reports which have not associated with beacon reports. If the beacon track and the radar report are sufficiently close in range and azimuth, the report and the track is said to be associate. If one and only one radar report associates with a track, and that radar report associates with no other track, then the radar report is used by the

track update function as a substitute for a beacon position measurement for the track on this scan. In addition, the range and azimuth differences between radar reports and beacon reports with which they correlate are accumulated, and used periodically to update the collimation corrections used in radar report correction.

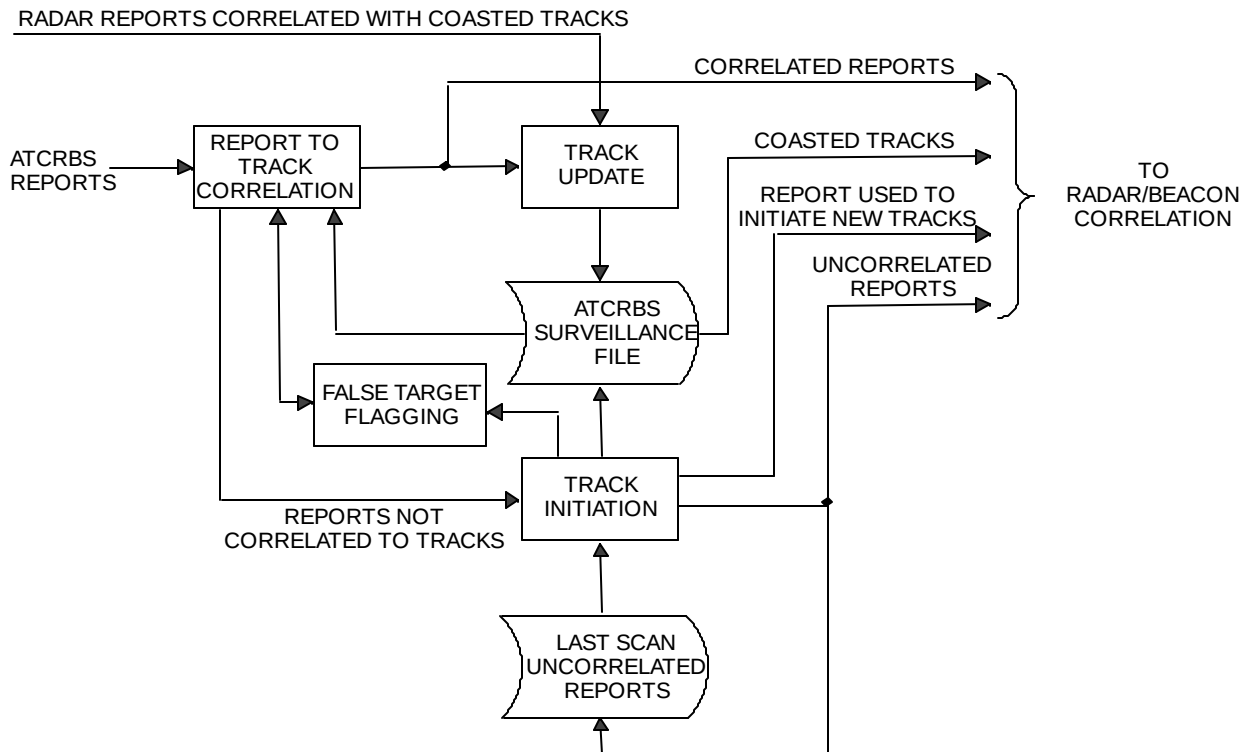


FIGURE 2–19. ATCRBS SURVEILLANCE PROCESSING GENERAL BLOCK DIAGRAM

2.3.1.8.4 Surveillance Data Output. At the completion of surveillance processing there are several classes of reports to be disseminated, such as: (1) beacon, radar reinforced, (2) beacon, not radar reinforced, (3) radar correlated with coasted beacon tracks and (4) uncorrelated radar reports are also disseminated. Reports are selected for dissemination to connect ATC facilities through the use of a site adaptable dissemination map. This dissemination map defines the areas of interest for each ATC facility in terms of minimum and maximum range and altitude for each 11–¼ degree azimuth sector.

2.3.1.9 Data Link Processing. Data link processing regulates the flow of messages on the air-ground link. Data link processing is accomplished through the maintenance of a file, called the active message list, which contains a record of all of the pending communications activities. Entries in this file are organized by Mode S address and are used by channel management to determine the number and type of interrogations and replies to schedule for an aircraft when it is available in the antenna main-beam. As shown in Figure 2–20, the two major subfunctions of data link processing operate to update the active message list. Input processing handles messages received from ground users of Mode S and is involved with additions of ground-to-air messages to the active message list. Output processing examines the transaction record prepared by channel

management. The transaction record together with and reply message contents indicate which communication activities are complete and which, if any, transponders are requesting an air-to-ground message transfer. Information concerning data link processing is present under the following headings: (1) input message processing and (2) output message processing.

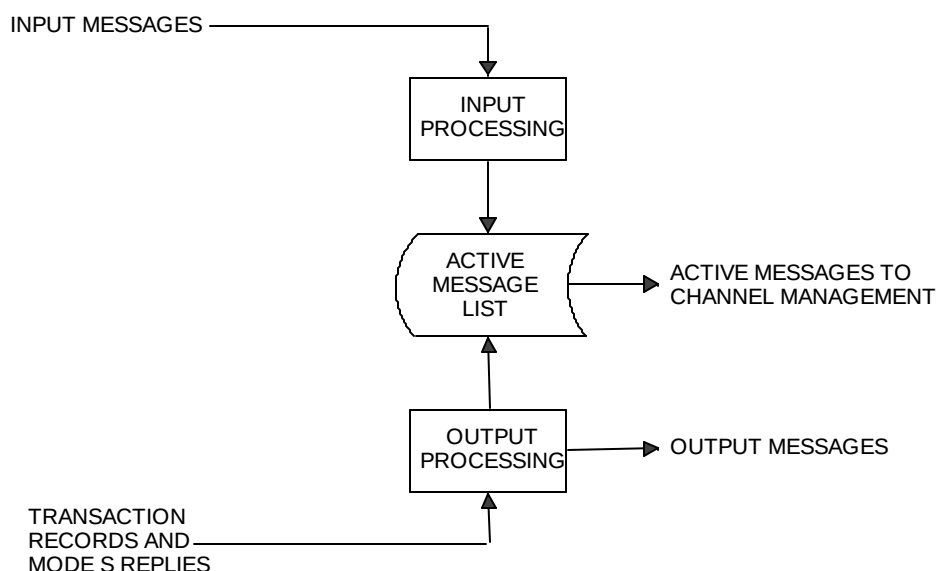


FIGURE 2-20. DATA LINK PROCESSING GENERAL BLOCK DIAGRAM

2.3.1.9.1 Input Message Processing. Communication messages are received from the ATC interface. Figure 2-21 presents a general block diagram of data link input message processing. In order to be considered for delivery during the current scan, a message must be received by the sensor at least two sectors ($22\frac{1}{2}$ degrees) ahead of azimuth at which the addressed aircraft will enter the antenna beam. A test is made to determine the status of the addressed Mode S aircraft in the local surveillance file. If the Mode S aircraft is not contained in the file, an immediate reject notice is generated and sent to the message originator. If the Mode S aircraft is in the file but has not replied to interrogations for one or more scans, requests for uplink delivery will be accepted but the sender will be issued a message delay notice. The message type is detected and the active message list is updated. The active message list update takes place according to an implicit and a user-defined priority. The implicit structure of data link processing and channel management places downlink extended length messages (ELMs) at a lower priority than the shorter, standard message or uplink ELMs. Further priority assignment can be made by the ATC authority. If a message is flagged as priority, it enters the active message list for the addressed aircraft ahead of any pending low priority message of the same category.

2.3.1.9.2 Output Message Processing. Mode S replies from the current scan are checked for an indication of successful uplink message delivery and for requests for new air-to-ground message transfers. Downlink messages included in the replies are detected (flagged as complete) and routed to intended recipients as indicated by information stored in active message list (for ground-initiated messages), or to all appropriate ground facilities (for pilot-initiated messages). In the latter case, the dissemination to ATC and non-ATC users depends only on aircraft position. Output message processing also issues ATCRBS ID messages and data link capability messages. The final task of output message processing involves updating the list to reflect additions and completions for this scan, and the generation of delivery notices. Figure 2-22 presents a general block diagram of data link output message processing.

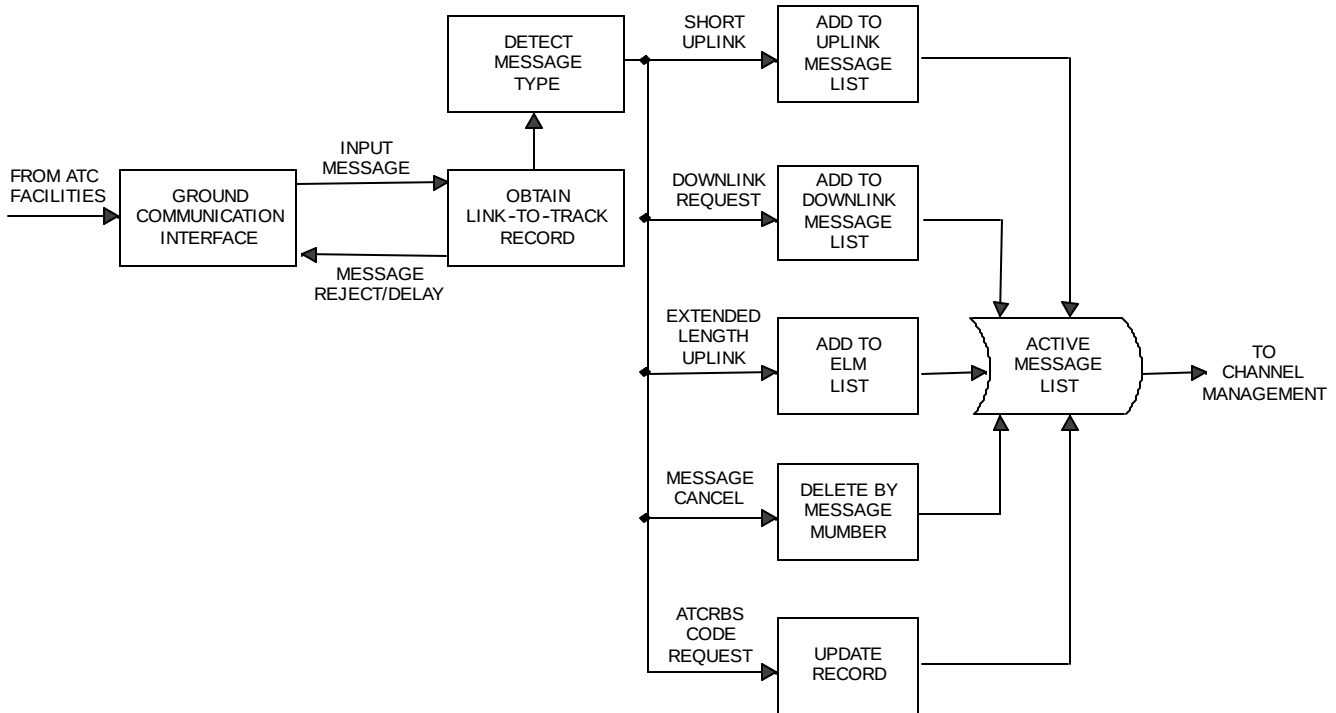


FIGURE 2-21. INPUT MESSAGE PROCESSING GENERAL BLOCK DIAGRAM

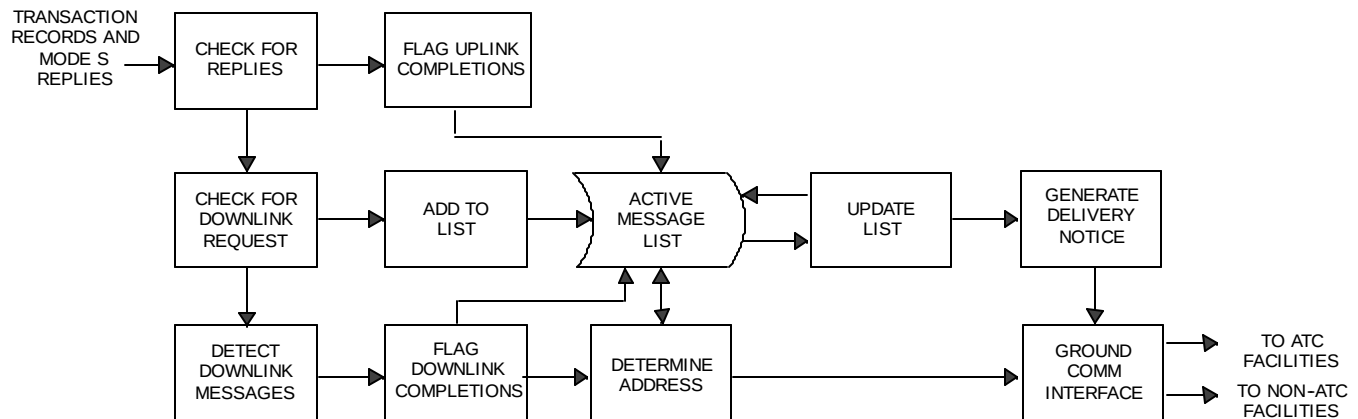


FIGURE 2-22. OUTPUT MESSAGE PROCESSING GENERAL BLOCK DIAGRAM

2.3.1.10 Network Management. Network management provides for continuity of surveillance and data link services in situations where adjacent sensors have overlapping coverage. When netted, Mode S sensors exchange surveillance data to hand off targets between sensors and to maintain surveillance continuity and rapid target reacquisition in the event of a temporary link interruption. This multisensor coordination is directed by the network management function, operating under the control of the sensor coverage map. An overall block diagram of the network management function is presented in Figure 2–23. Once each scan, each Mode S track is processed by coverage coordination to determine if a boundary crossing has occurred. For non-netted sensors, this event will cause the coverage coordination function to manage Mode S lockout status in order to permit acquisition by means of all-call interrogations. In the case of netted sensors, a check is made for a boundary crossing or a track state change. Either of these events will cause control message handling to initiate a sequence of messages. A boundary crossing results in a flow of track data to the sensor whose boundary was crossed. A track state change results in a request or cancellation of a previous request for surveillance data to fill in during a link interruption. Control message handling also processes network control messages from adjacent sensors and ATC facilities, and retrieves or stores data in the surveillance file as required. Information concerning network management is presented under the following headings: (1) sensor coverage map, (2) sensor coverage coordination, (3) netted sensor configuration, and (4) non-netted sensor configuration.

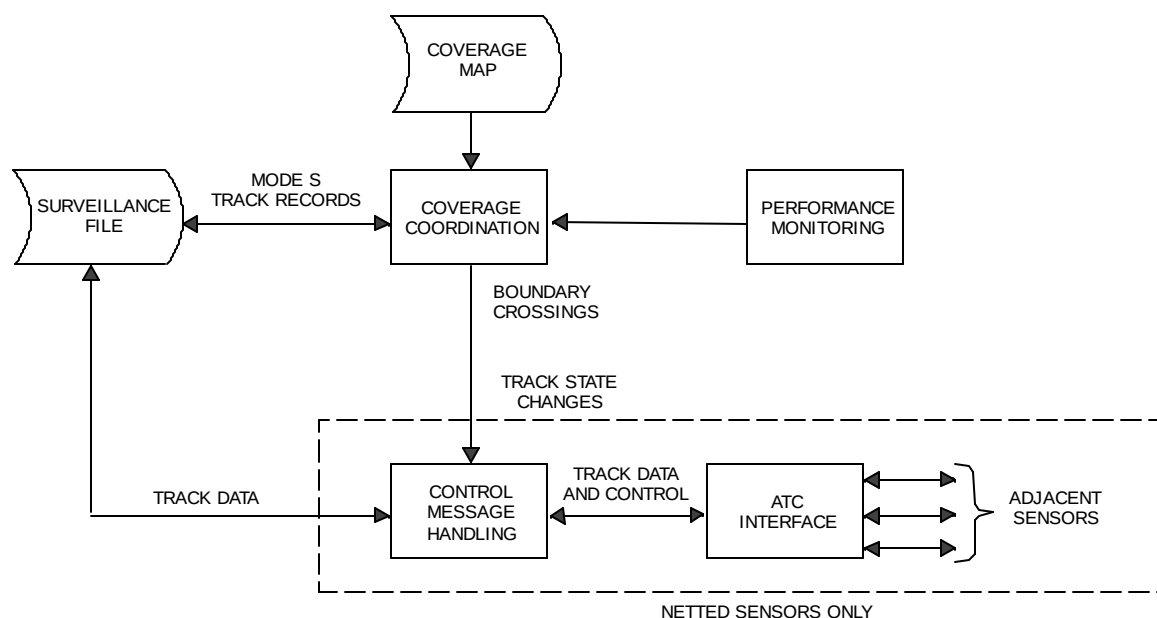
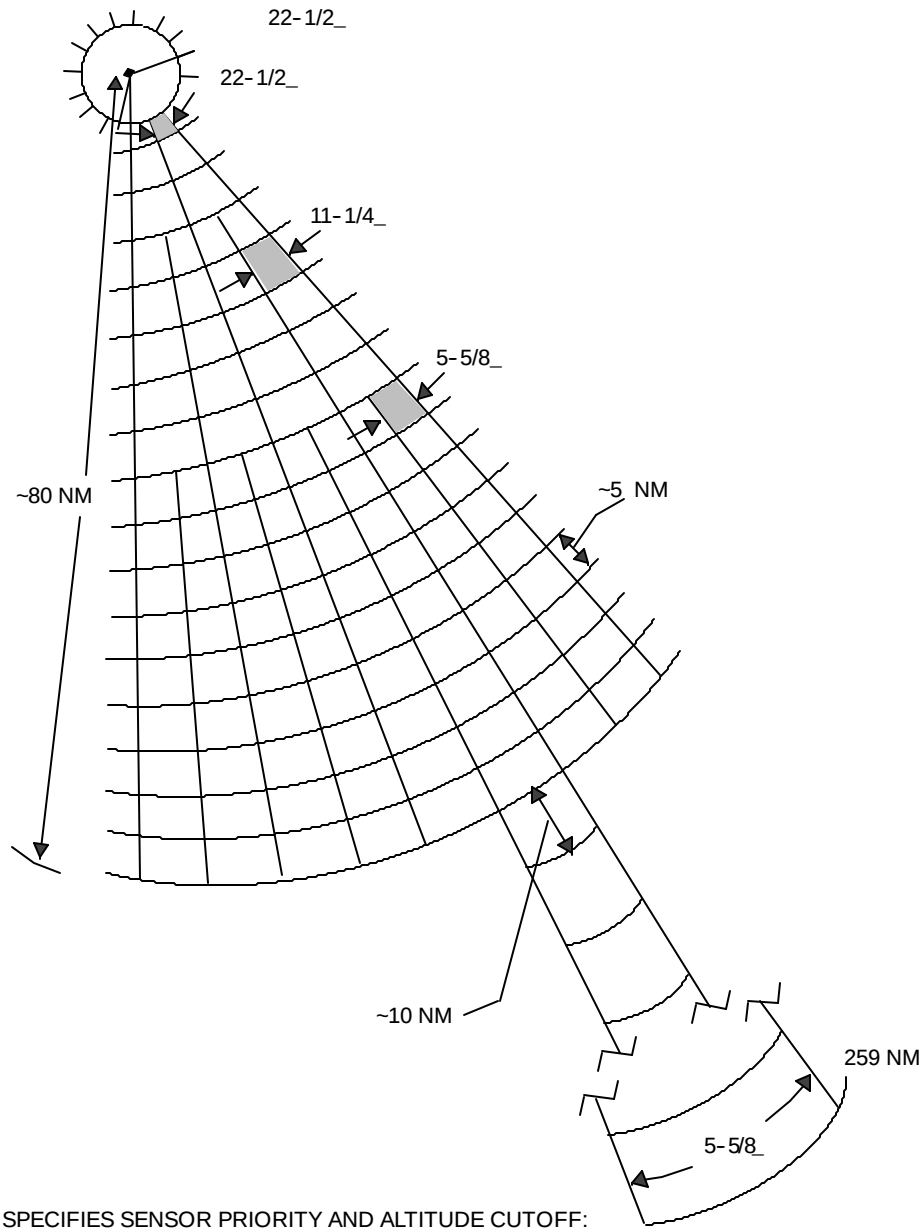


FIGURE 2–23. NETWORK MANAGEMENT GENERAL BLOCK DIAGRAM

2.3.1.10.1 Sensor Coverage Map. The extent and type of coverage to be provided by each sensor is controlled by a data file known as the sensor coverage map. Two major boundaries are defined by the sensor coverage map: (1) the maximum range at which the sensor is to provide surveillance coverage, and (2) the area where the sensor is assigned primary responsibility (for uncontrolled aircraft only). The coverage map is implemented in a grid structure as shown in Figure 2–24. For each element of the grid (termed a cell) a sensor priority list is specified, with a lower altitude cutoff defined for each sensor. The position of a sensor in the priority list specifies its surveillance function in that cell: (1) primary, (2) secondary, or (3) backup. Sensor coverage boundaries are thus defined by a change in sensor ordering between adjacent cells. For each controlled aircraft, a message from an ATC facility assigning primary or secondary status has the effect of overriding the map designation.



NOTES:

1. EACH CELL SPECIFIES SENSOR PRIORITY AND ALTITUDE CUTOFF:
 S_1, A_1 ← PRIMARY SENSOR
 S_2, A_2 ← NORMAL SURVEILLANCE
 S_3, A_3 ← BACKUP COVERAGE
2. NM = NAUTICAL MILES.

FIGURE 2-24. SENSOR COVERAGE MAP GRID STRUCTURE

2.3.1.10.1.1 The sensor coverage area is now determined in SAPs by the ATCRBS Radar Range Mask, or ARRM. This table, added to the Surveillance Processing SAPs, allows the user to input the listening window which corresponds to the maximum range for each of 32 sectors of 11.25°

2.3.1.10.1.2 In the event of a sensor failure, that sensor is flagged as inactive and the sensor is deleted from the sensor priority list. In this way, the primary and secondary coverage areas of active sensors are automatically enlarged to take over the area formerly serviced by the failed sensor. In the event of failure of sensor-to-sensor communications for a netted sensor, the sensor affected is temporarily treated as non-connected. That is, aircraft are unlocked to Mode S all-call interrogations at the surveillance boundaries of this sensor in order to permit acquisition by its normal all-call interrogations.

2.3.1.10.2 Sensor Coverage Coordination. The elements of sensor coverage coordination are presented in Figure 2–25. The current position of the track is used to determine the present cell index. If the cell index has not changed since the last scan and no failure/recovery of an adjacent sensor has occurred, no further action on coverage assignment is performed. Otherwise, the sensor priority list for this cell is retrieved from the coverage map, deleting those sensors that the performance monitoring function has declared to be in a failed state. The resulting sensor assignment is compared to the previous assignment (stored as part of the track record) to detect the occurrence of a boundary crossing. If boundary crossing is detected, the action depends upon whether or not the sensor is netted to adjacent sensors.

2.3.1.10.3 Netted Sensor Configuration. In configurations where the local and adjacent sensors communicate via a common ATC facility, an indication is passed to control message handling to initiate the appropriate sequence of messages. The present track state is then compared to the previous state to detect the beginning or end of a link interruption. Once more, an indication is passed to control message handling to begin or terminate the flow of adjacent sensor data. An additional check is made to determine if the track was just initiated as a result of the local sensor's all-call interrogations. If the aforementioned track was just initiated, surveillance data on this track is disseminated to the other assigned sensor(s) in the cell by control message handling, which uses the same message sequence as for a surveillance boundary crossing. (Note: The Network Management function for netted sensor messages is not currently implemented in the Mode S system.)

2.3.1.10.3.1 When the new cell data indicates a primary/secondary status different from the one in effect for an uncontrolled aircraft, a primary coordination message exchange is initiated. A typical primary coordination exchange involves a local sensor initiating a request for primary assignment to an adjacent sensor when an aircraft first enters the local sensor's primary coverage area. If the adjacent sensor has not as yet acquired the track, the adjacent sensor will respond with a "primary approved" message. If the adjacent sensor has the aircraft in track and assigned primary (due to the overlapping primary assignment needed for continuous primary coverage), the adjacent sensor will send a "primary disapproved" message. The coordination is completed when the adjacent sensor's map no longer indicates primary status for this aircraft. The adjacent sensor sends a request for secondary assignment to the local sensor and assigns itself secondary when an "accept primary" response is received.

2.3.1.10.3.2 Certain events cause a sequence of messages to be exchanged between Mode S sensors. These events and the resulting messages are indicated in the following list. Message on ATCRBS targets are limited to an exchange of track data on request for Mode-C equipped aircraft. The sequence of ATCRBS control messages used to establish this exchange is similar to that indicated in the following list for the Mode S track state change event.

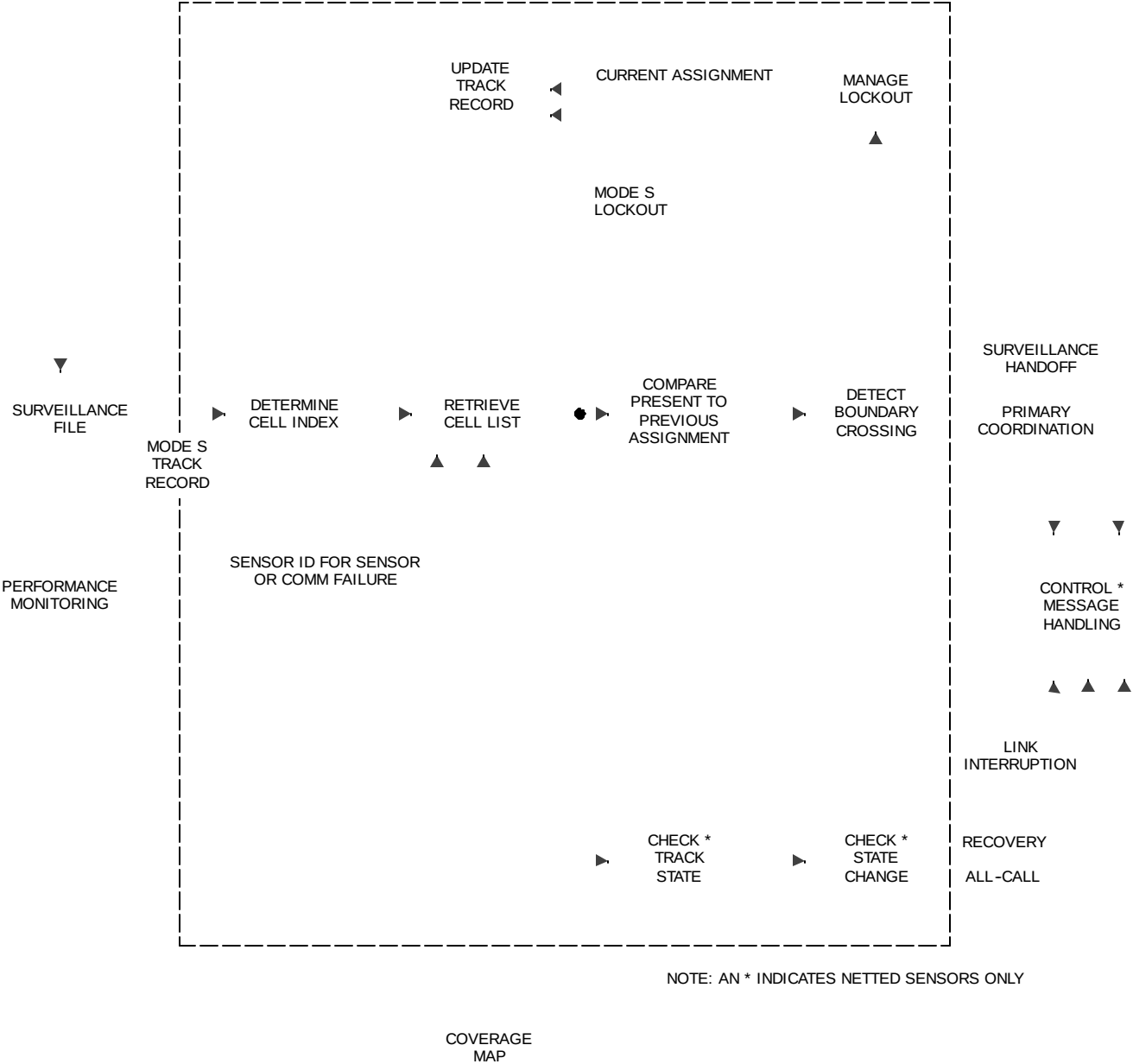


FIGURE 2–25. SENSOR COVERAGE COORDINATION GENERAL BLOCK DIAGRAM

Control Message Handling List

Event	Message Sequence	To/From
Surveillance Handoff	Send Data Start to Send Track Data to Receive Cancel Request from Send Data Stop to	The New Active Sensor in the Cell
Track State Change	Send Data Request to Receive Data Start from Receive Track Data from Send Cancel Request to Receive Data Stop to	The Other Active Sensor(s) in the Cell
All-Call Acquisition	Send Data Start to Send Track Data to Receive Cancel Request from Send Data Stop to	The Other Active Sensor(s) in the Cell
Primary/Secondary Coordination (for uncontrolled aircraft)	Send Request for Primary/ Secondary Assignment to Receive Primary Approved to Disapproved/Accepted from	The Other Active Sensor(s) in the Cell

2.3.1.10.4 Non-Netted Sensor Configuration. In configurations in which an adjacent assigned sensor does not communicate with the local sensor (either permanently or temporarily because of a communications failure), it is necessary to provide another means of surveillance handoff. This can be accomplished by using site-addressed all-call interrogations or, alternately, by intermittently unlocking each Mode S aircraft to all-call interrogations for a few scan. The parameters for a cycle of unlock and lock periods for the latter approach are chosen so as to minimize possible interference while providing adequate opportunities for acquisition. Communication coordination for either configuration is handled by the transponder multisite communications features.

2.3.1.11 Performance Monitoring. The ability of the sensor to perform its surveillance and communication tasks is continuously checked by the performance monitoring function. Figure 2–26 presents a general block diagram of the performance monitoring functions. Three categories of performance monitoring checks are performed:

1. Overall checks performed by the calibration performance monitoring equipment (CPME) for proper surveillance and data-link operation.
2. Internal checks on the status of the sensor hardware and software.
3. External checks on the status of adjacent sensors (when netted) and their ability to provide the local sensor with correct surveillance data.

2.3.1.11.1 Results of Performance Checks. The results of these performance checks are evaluated once per scan to determine the status of the sensor. The possible status conditions are: (1) normal operation (condition green) — no abnormal indications, (2) marginal operation (condition yellow) — operational but becoming marginal, and (3) failed state (condition red) — sensor operation ceases. Declared sensor status is reported once per scan to the ATC interface along with condition codes that define the reason(s) for the yellow or red condition.

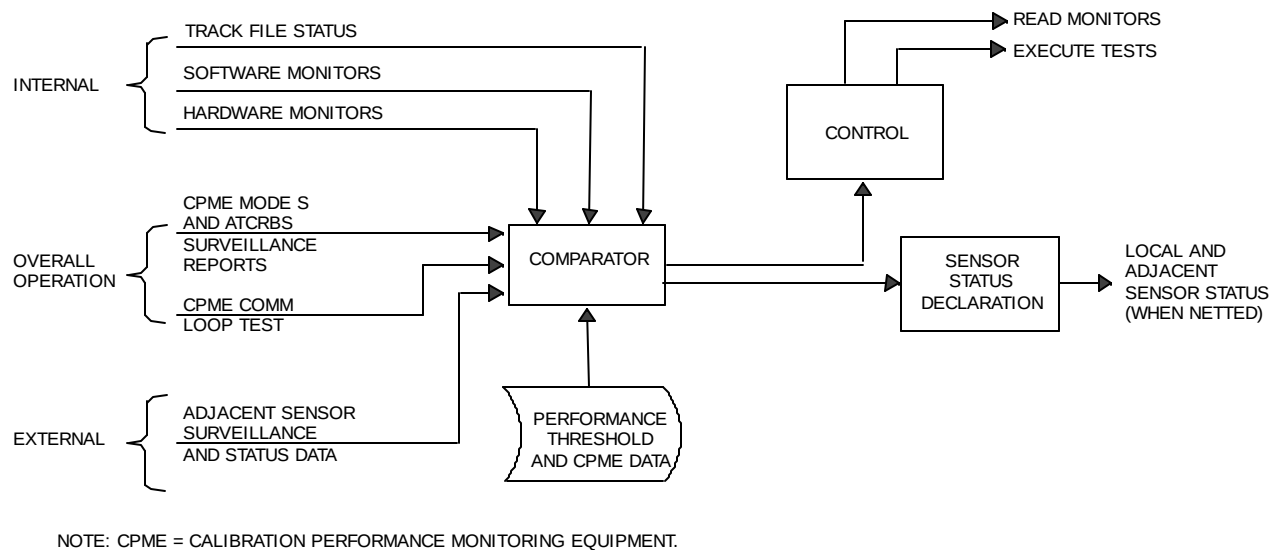


FIGURE 2-26. PERFORMANCE MONITORING GENERAL BLOCK DIAGRAM

2.3.1.11.2 Performance Monitoring Function Interfaces. The sensor performance monitoring function interfaces with the FAA's remote maintenance monitoring system (RMMS). The RMMS also receives the sensor status message and, in addition, has the capability of remotely controlling sensor functions, such as: (1) initial loading and start up, (2) parameter modification, and (3) the switching of redundant elements.

2.3.1.11.3 Calibration Performance Monitoring Equipment (CPME). The calibration performance monitoring equipment (CPME) is a transponder-like device, several of which are deployed in close proximity to each sensor. The CPME serves as the basis for the overall operational surveillance checks by providing replies from "aircraft" with known identification and position (range, azimuth, and altitude). Overall operational communication checks are performed by loop tests with the CPME. In these tests, an uplink test message delivered to the CPME causes it to initiate a downlink message with the same text as contained in the uplink message.

2.3.1.12 Message Routing Management. Message handling between sensor functions and the external interfaces is performed by the sensor message routing management function. Input messages are routed from the external interfaces (ATC, non-ATC users, and adjacent sensors) to the local network management, performance monitoring, and data link processing functions. Data link capability requests from the ATC or non-ATC user are forwarded to message routing management. Non-ATC users can also request aircraft information, such as (1) a request for aircraft state and (2) a request for aircraft position. Additional tasks are performed on outgoing messages, particularly to support requirements of data link users not connected to ATC facilities. (Note: The Message Routing Management function for netted sensor messages is not currently implemented in the Mode S system.)

2.3.1.13 Data Extraction. Data Extraction software resides on both the Local Terminal and the DPS. This software allows the user to gather operational data from a Mode S Sensor. Data provided by important Sensor functions may be recorded onto disk of the Radar Intelligent Tool (RIT) terminal via the Tape Drive Interface Card (TDIC) input to the RIT terminal. This data may later be used in conjunction with the Sun System Data Reduction program, or the FAA RBAT system, to provide specialized reports that describe in detail how the Sensor is functioning. The Data Extraction function allows the user to formulate requests from the Local Terminal for the data to be extracted. These requests, in turn, are sent from the Local Terminal to the DPS. Once the request has been received by the DPS, the user can start and stop Data Extraction at any time. Data Extraction is invoked by the user selecting the Data Extraction control option from the Local Terminal Main Menu. The user presses Function key F5 to obtain the Data Extraction main screen.

2.3.1.13.1 Data is extracted from an operating Mode S Sensor upon the receipt of the user request. This request is constructed by combining data types which specify what information is to be extracted by category. The category is then “initiated”; that is installed on the Sensor. In addition, optional filtering can be specified so that only values matching the filters are recorded. When categories are initiated on the Sensor, these categories are inspected for the data types that they contain. These data types are copied into a data structure known as the Interface DE Request Table. A copy of this table exists on each processor in the DPS. When Data Extraction is started, the various Sensor functions consult the Interface DE Request Table and determine what data is to be extracted. If the RIT terminal is not available for receiving the data, the user is notified immediately. If the RIT terminal becomes unavailable while the data is being extracted, the Data Extraction process is halted, and the user is notified. Data Extraction can then be restarted once the problem with the RIT terminal has been corrected.

2.3.1.13.2 When the user starts Data Extraction (DE), the Site Adaptation Table, the Current Antenna Azimuth, the Channel Configuration Table, the most recent Northmark data, the Current Station Time, and the raw calibration data are immediately extracted and written to the RIT terminal. The Sensor functions then read the Interface DE Request Table, and if the Table indicates a Data Extraction request for that function, the data is extracted and sent to the selected output device. The Data Extraction process will continue until it is halted by the user. Additionally, the user can specify as a filter value a number of antenna scans during which data is extracted. The Data Extraction halts automatically when the number of antenna scans specified in the filter is reached. For additional information concerning Data Extraction functions on the Local Terminal, the user should refer to the Local Terminal information contained in the Mode S Local Terminal Users Manual, software instruction book TI 6365.11.

2.3.1.14 Site Adaptation. Site adaptation is a function that is provided for establishing or changing a Mode S Sensor’s operating parameters. These operating parameters are also known as Site Adaptable Parameters (SAPs). Site Adaptable Parameters are used to customize the software for each Mode S Sensor. This customized Sensor software is known as the System Image. Once the System Image software has been installed on a Mode S Sensor, the SAPs may be modified by means of the Local terminal. The modified SAPs can then temporarily replace the existing SAPs. The original SAPs can be restored or the modified SAPs can permanently replace the existing SAPs at the Mode S Sensor.

2.3.1.14.1 Two versions of the Site Adaptation program exist: (1) a version that runs on the PSF computer and (2) a version that runs on the Local Terminal. The PSF version is used solely to establish and modify SAPs for a System Image that is to be initially installed on a Mode S Sensor. The Local Terminal version is used to modify (and possibly replace) SAPs after System Image has been installed on the Mode S Sensor.

2.3.1.14.2 Site Adaptation is performed at the Mode S site upon request from the user by selecting function F4 (SAPs) from the Local Terminal (LT) Main Menu. The Baseline SAP Main Menu is displayed on the Local Terminal screen. The first time the user selects the SAPs, BASELINE.SAP will be loaded from the hard disk. Any changes in the SAP data (reading another file, modifying parameters, or down loading from the Sensor) remain in memory. The next time SAPs are selected from the Main Menu, the previously modified SAPs will be displayed. The user can display SAPs from the Sensor by selecting the F3 key from the SAP Main Menu (request the Sensor download). SAP downloads may be requested from both channels and are displayed on the same screen, in two column format (one column for each channel).

For additional information concerning Site Adaptable Parameter (SAP) functions, the user should refer to the Local Terminal information contained in the Mode S Local Terminal Users Manual, software instruction book TI 6365.11. SAPs displayed on the local terminal can be uploaded to the sensor for temporary (F4 on SAP main menu) or permanent (F5 on SAP main menu) installation. Instructions for loading SAPs to the sensor are provided in the Local Terminal Users Manual.

2.3.2 Interrogator Equipment Operation

Interrogator equipment operation is presented under the following headings:

- a. Transmitter Operation
- b. Receiver Operation
- c. Time of Year Clock Operation
- d. Input/Output Buffering
- e. Power Distribution
- f. Environmental Controls Operation
- g. Performance Monitoring Circuit Operation
- h. Fault Detection and Fault Isolation Testing

Figure 2–27 is a block diagram of the Interrogator equipments. Figure 2–28 is a functional block diagram of the units contained in the Interrogator cabinet and MSJB–1. For additional information and functional diagrams concerning the Interrogator equipments, the user must refer to the Interrogator instruction book TI 6365.5, the Transmitter instruction book TI 6365.6, the Receiver/RFTTG instruction book TI 6365.7, and the Signal Processor instruction book TI 6365.4.

2.3.2.1 Transmitter Operation. During each interrogation period, two beams are transmitted. The main directional beam [sum pattern] is used to radiate Mode S and ATCRBS interrogation pulses generated by the primary transmitter, while the omni-directional beam [omni pattern] is used to radiate side-lobe suppression pulse generated by the auxiliary transmitter. The transmitter channels receive signal LOW POWER RF (at 1030 MHz) from the local oscillator. Modulation control signals (MODULATION) from the signal processor modulate the RF signals to form the desired transmitted waveform. The modulated RF is amplified into high power RF signals and adjusted in amplitude for each site. The AMPLITUDE ADJUST control signals from the signal processor (A4) also adjust the final output power depending upon the power level to be transmitted: Mode S low power, Mode S high power, or ATCRBS power.

2.3.2.1.1 The primary and auxiliary transmitters send the transmit signals (PRI XMIT RF and AUX XMIT RF) to the monopulse and omni diplexers (A19 and A6), respectively. The diplexers are filters that direct the 1030–MHz transmit signals from the transmitter to the antenna and the direct the 1090–MHz replies from the antennas to the receiver. For terminal sites, an RF transfer switch (unit 9) is used to connect one Interrogator channel output to the antenna and the other to the dummy loads. For enroute sites, two RF transfer switches (units 9 and 13) are used to connect one Interrogator channel output to the front antenna and the other to the back antenna. The signal processor in each Interrogator cabinet controls and monitors the RF transfer switch settings. The transmitter also monitors forward power amplitudes within each channel. Signals that represent detected faults (XMTR FAULTS) are sent to the receiver/RFTTG video processor/monitor (A3) for processing into digital fault signals (part of RECEIVER MONITOR BUS) that are sent to the signal processor (A4). refer to TI 6365.6, Transmitter Equipment Instruction Book, for more details of transmitter operation.

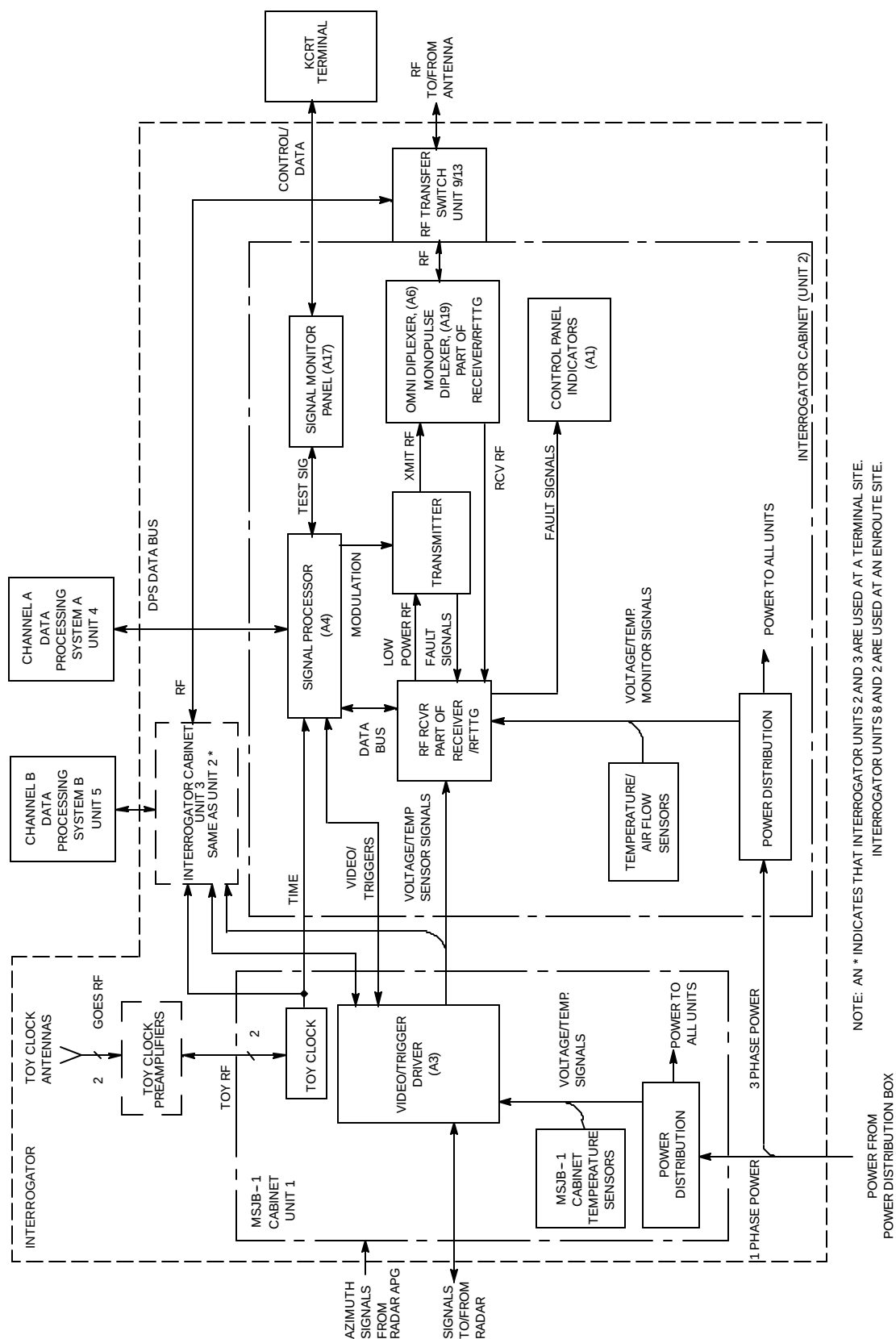
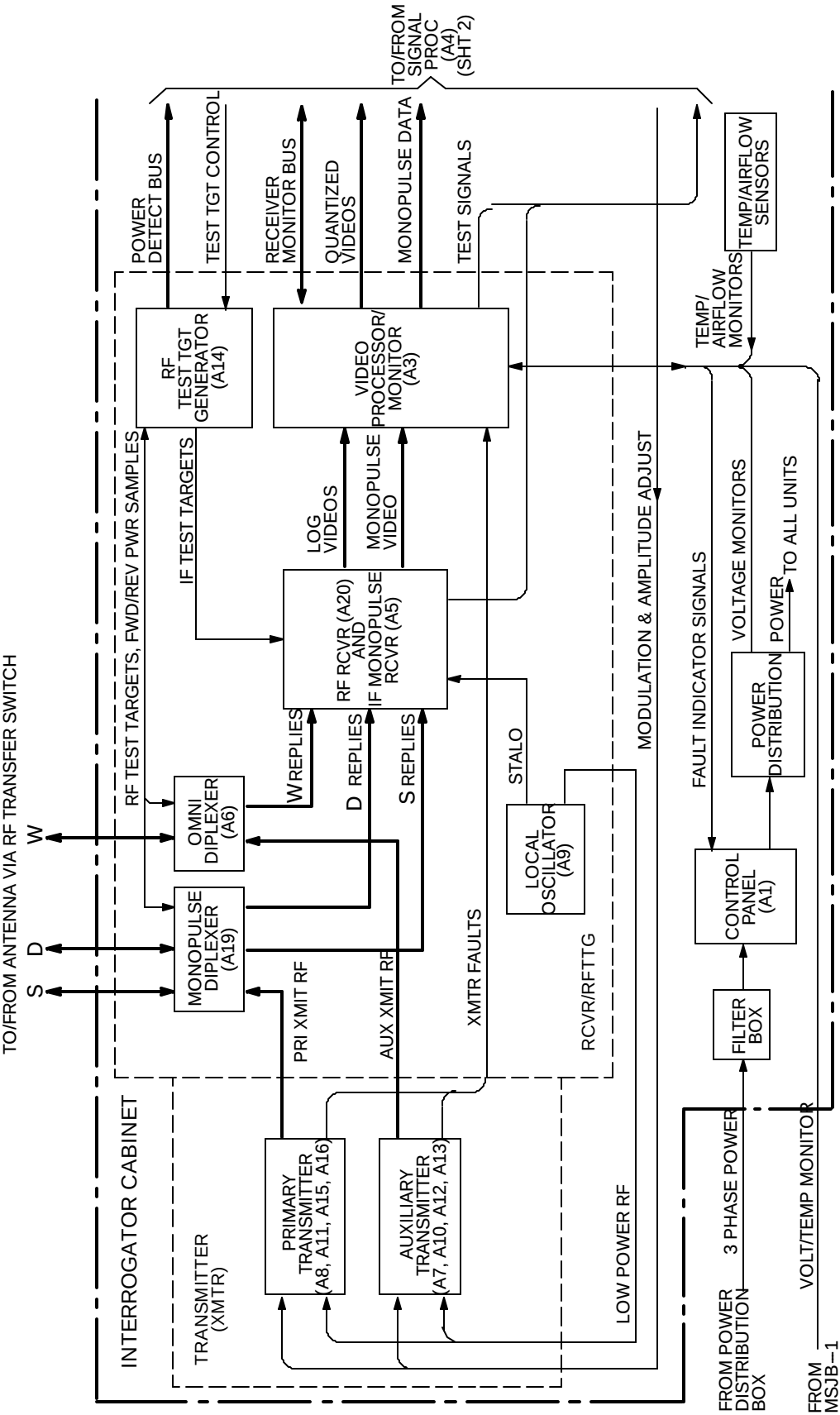
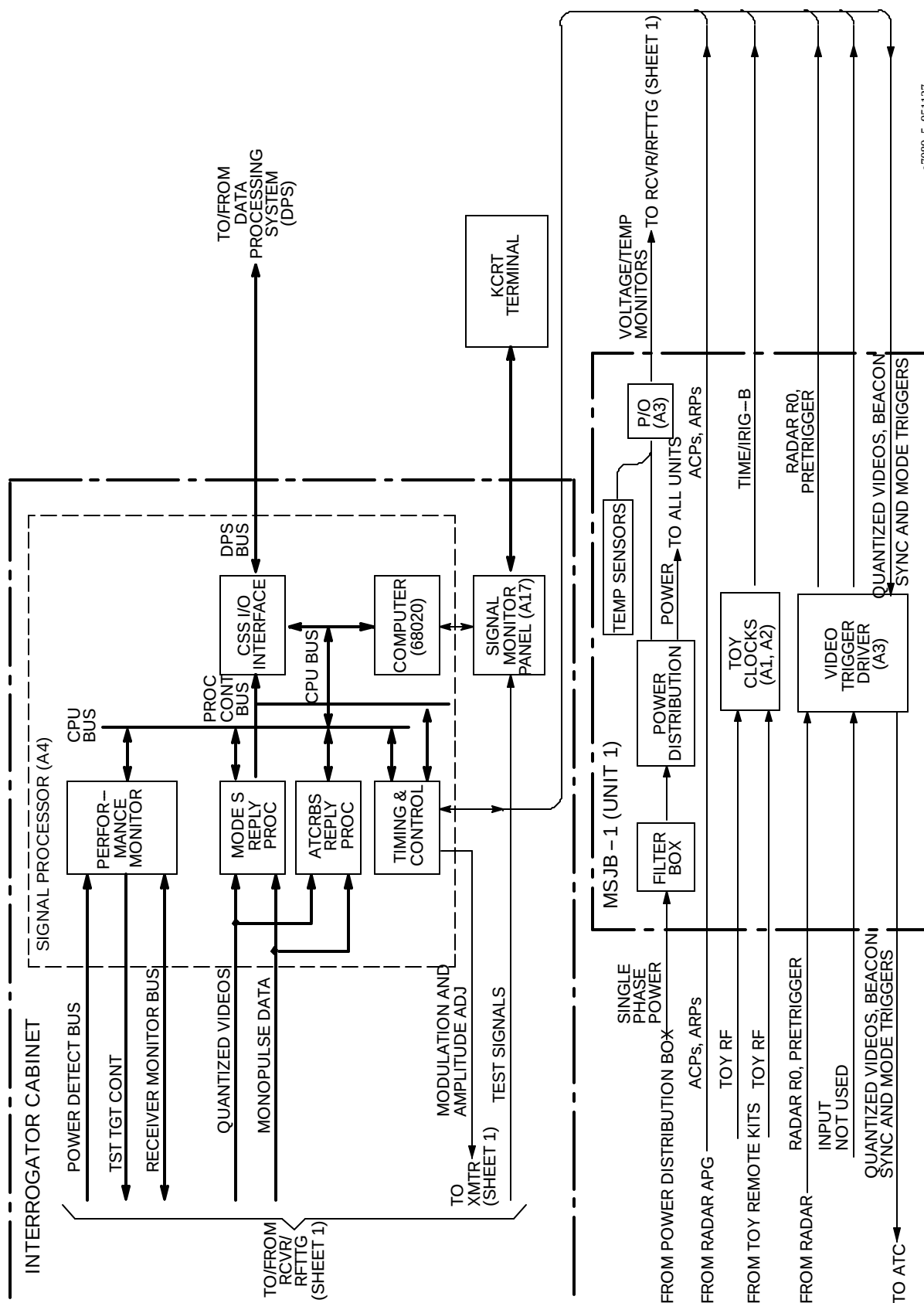


FIGURE 2-27. INTERROGATOR BLOCK DIAGRAM



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FIGURE 2-28. INTERROGATOR FUNCTIONAL BLOCK DIAGRAM (SHEET 1)



e7089_5_951127

FIGURE 2-28. INTERROGATOR FUNCTIONAL BLOCK DIAGRAM (SHEET 2 — END)

2.3.2.2 Receiver Operation. RF replies are received at the antenna, routed through the RF transfer switch and diplexers, and processed in the three-channel receiver. The receiver processes the Mode S and ATCRBS RF replies into digital monopulse data and quantized video signals and sends them to the signal processor (A4). The signal processor decodes and processes transponder replies further and sends reply data to the DPS via the DPS Bus.

2.3.2.2.1 Replies received from the sum (S), delta (D), and omni (W) receive beams are sent through the RF transfer switch to the omni and monopulse diplexers. The diplexers direct the 1090-MHz replies (S, D, and W REPLIES) to the receiver circuits. The receiver circuits are located in three units: RF receiver (A20), IF monopulse receiver (A5), and video processor/monitor (A3). RF receiver (A20) mixes the received RF replies with the 1030-MHz STALO signal (from the local oscillator A9) to convert the replies to IF signals. Four log detectors within IF monopulse receiver (A5) then detect the IF inputs to recover the modulation signals (LOG VIDEOS). The IF monopulse receiver also processes the S and D replies through a monopulse detector circuit to generate monopulse signal (MONOPULSE VIDEO) that is proportional in amplitude to the off-boresite processor/monitor (A3). The video processor/monitor processes the LOG VIDEOS into eight quantized videos and converts the MONOPULSE VIDEO from an analog level to a digital value. The QUANTIZED VIDEOS and MONOPULSE DATA are sent to the signal processor (A4) where they are used for reply processing.

2.3.2.2.2 The signal processor (A4) processes the reply signals in the Mode S and ATCRBS reply processors, a computer subsystem input/output (CSS I/O) interface, and a computer [central processing unit (CPU 68020)]. The QUANTIZED VIDEO signals are received in the Mode S or ATCRBS reply processors. The Mode S reply monopulse estimates are extracted from the QUANTIZED VIDEO and MONOPULSE DATA inputs and are compiled into a Mode S target report. The ATCRBS reply processor processes replies received from ATCRBS interrogations. Range, azimuth and monopulse estimates of the target are detected and sent to the internal computer for reply-to-reply processing. The Mode S and ATCRBS target reports are sent to both channels of the DPS on the DPS BUS.

2.3.2.3 Time of Year (TOY) Clock Operation. Interrogator processing includes time of year (TOY) data in replies sent to the DPS. To ensure the correct time of year is used throughout the Mode S System network, a common source for standard time of year is used. The time is obtained by two TOY clocks from the geostationary orbiting environmental satellite (GOES) signals. Each TOY clock configuration consists of an antenna, a remote kit, a TOY clock, and an uninterruptible power supply. Signals that contain the desired time (GOES RF) are amplified by the TOY clock remote kits (TOY RF) and sent to the TOY clocks. The TOY clocks (A1 and A2 in MJSB-1) extract the time of year from the TOY RF signal and send the digital result (signal TIME) to the signal processor (A4) in the interrogation cabinet. A timer, internal to each TOY clock, provides local station time in the event communication with the selected GOES fails. The Inter-Range Instrumentation Group-B (IRIG-B) output of each TOY clock contains time accuracy bits. Time inaccuracies of 1, 5, 50, and 500 milliseconds can be detected and reported in the output. The signal processor (A4) runs fault detection tests to check for any accuracy bit that is set. A TOY clock fault is declared if any bit is set. A VSP was added to select whether or not to report TOY clock red codes. When selected, a TOY red code will not be reported, and the Mode S will not reconfigure.

2.3.2.4 Input/Output Buffering. The Interrogator provides input/output buffering for certain signals entering or exiting from/to equipment outside the Mode S equipment. The input/output buffering circuits are on video/trigger driver unit (A3) in the MSJB-1 cabinet. The driver circuits receive triggers from the radar (RADAR R0, PRETRIGGER) and route these signals to the signal processor (A4) in each Interrogator cabinet. The driver circuits also receive QUANTIZED VIDEOS, and BEACON SYNC AND MODE TRIGGERS from the signal processor (A4) in each Interrogator cabinet. The signal processor signals are sent through a multiplexer internal to the video trigger driver (A3 in MSJB-1), which selects which Interrogators inputs are to be sent to the destination outside the Mode S system. Also, azimuth change pulses (ACPs) and azimuth reference pulses (ARPs) from the azimuth pulse generator (APG) are routed directly through the MSJB-1 cabinet to the signal processor (A4) in the Interrogator cabinet.

2.3.2.5 Power Distribution. Power in the Interrogator is received at each cabinet from the power distribution box as three-phase power with neutral and earth ground (3 PHASE POWER). In each Interrogator

cabinet, there is a filter box that filters out unwanted RF and provides lightning protection to internal circuits. The filtered AC is applied to power supplies that convert the AC to DC. Each power supply output is distributed to LRUs, and each output is monitored for undervoltage conditions by performance monitoring circuits.

2.3.2.6 Environmental Controls Operation. The environmental of the MSJB-1 is cooled by unaided airflow only. The MSJB-1 cabinet temperature is monitored by a temperature sensor next to each power supply. The Interrogator cabinet temperature is mainly controlled by blower assembly (A18) and primary amplifier fan (B1). The primary amplifier fan prevents the amplifier from overheating. The RF receiver contains a heater that keeps the temperature within the RF receiver relatively constant. The temperature of each power supply, cabinet air temperature, and RF receiver temperature are monitored by temperature sensors. Cabinet airflow is monitored by two sensors in blower assembly (A18). The output of each sensor is sent via the video processor/monitor (A3) to the signal processor (A4).

2.3.2.7 Performance Monitoring Circuit Operation. The signal processor (A4) continuously monitors the performance of the Interrogator during on-line (and off-line) operation. Video processor/monitor (A3) acts as an interface between the signal processor (A4) and the rest of the Interrogator.

2.3.2.7.1 Power/VSWR Monitoring Circuits. The signal processor (A4) measures transmitter power amplitudes, performs VSWR measurements, and measures receiver signal and noise amplitudes, using the video processor/monitor (A3) and RF test target generator (A14). To measure forward and reverse power, the RF signals from the transmitter are sampled in the omni and monopulse duplexers (FWD/REV PWR SAMPLES). These samples are processed by the RF test target generator (A14) into digital samples that are sent to the signal processor (A4) on the POWER DET BUS and analyzed by the signal processor. VSWR is measured by comparing forward and reverse RF samples obtained in the same manner. Receiver noise amplitudes are measured by the signal processor (A4) through the video processor/monitor (A3) and the RF test target generator (A14) with no RF inputs into the receiver.

2.3.2.7.2 Test Target Generator Circuits. The signal processor (A4) and the receiver/RFTTG contain the test target generator circuits to test the performance of the various processing circuits. A test target generator in the signal processor (A4) performance monitor generates test vectors (digital) that simulate the quantized outputs of the receiver/RFTTG. The vectors are injected into the inputs of the selected reply processor. The output of the selected reply processor is then compared with the expected result to determine correct/faulty operation. The RF test target generator (A14) in the receiver generates RF and IF TEST TARGETS under control of the signal processor (TEST TGT CONT). The RF TEST TARGETS are injected into the duplexers. The IF test targets are sent to the IF monopulse receiver (A5). The test targets are routed through the receive paths to test the performance of the receiver and are sent to the video processor/monitor. The video processor/monitor (A3) sends the resulting QUANTIZED VIDEOS and MONOPULSE DATA to the signal processor (A4). The signal processor then compares the output of the receiver with the expected result to determine correct or faulty operation.

2.3.2.7.3 Environmental/Power Monitoring Circuits. Each temperature sensor, airflow sensor, and power supply output signal (VOLTAGE/TEMP/AIRFLOW MONITORS) is sampled by the video processor/monitor (A3) and converted to a digital signal that is accessed by the signal processor (A4) performance monitor over the RECEIVER MONITOR BUS. MSJB-1 cabinet temperature and (VOLTAGE/TEMP MONITORS) are sent through video/trigger driver unit (A3 of MSJB-1) to the video processor/monitor (A3) of the RCVR/RFTTG unit. These VOLTAGE/TEMP/MONITOR signals are sampled in the same manner as the Interrogator temperature/voltage signals. Faults detected in the power distribution and environmental control operation can cause an indicator on Interrogator control panel (A1) to be turned on. If the signal processor (A4) detects a fault that should turn on an indicator, the signal processor sends a command via the RECEIVER MONITOR BUS to the video processor/monitor (A3) for storage. The video processor/processor sends indicator turnon commands (FAULT INDICATOR SIGNALS) to the control panel (A1) to turn on the selected indicator.

2.3.2.7.4 Accessible Performance Monitoring Test Signals. Certain signals generated during fault detection testing and fault isolation testing are also sent to test connectors on signal monitor panel (A17)

as TEST SIGS. These signals can be viewed with test equipment using the data from the appropriate instruction book. Signals available at the signal monitor panel (A17) are described in each equipment instruction book. The signal monitor panel (A17) are described in each equipment instruction book. The signal monitor panel (A17) is also used to connect the KCRT terminal to the Interrogator cabinet. The KCRT terminal is used during maintenance to control the Interrogator and access information generated by the Interrogator.

2.3.2.7.4.1 Manual testing and control of the Interrogator can be performed by the use of the KCRT terminal. The KCRT terminal is connected to the KCRT connector (J4) located on the signal monitor panel (A17) of the Interrogator. An Interrogator man-machine interface (MMI) is provided for use on the KCRT terminal for testing and control purposes. The user can manually test the Interrogator (via the KCRT terminal) for the proper functioning of the Interrogator's hardware components, and can also perform monitoring, calibration, and alignment of the Interrogator. The user can check the various lowest replaceable units (LRUs) of the Interrogator and associated equipments by monitoring (via the KCRT terminal) the data points for the Interrogator A, B, C, hardware logical units (LU26, LU27, LU28) in the RMS Monitor Report (selection number 1 of the RMS menu displayed on the KCRT terminal). The LRUs are monitored by means of status reporting, where an "A" in the Monitor Report indicates a normal condition, a "C" indicates a red alarm condition, or a "D" indicates a yellow alarm condition. Some of the Interrogator LRUs that are checked are: the transmitter, receiver, signal processor, omni diplexer, monopulse diplexer, RF test target generator (RFTTG), power supplies, cabinet temperature sensors, and blowers. Other LRUs that are checked include the azimuth pulse generators (APG1 and APG2), time of year clocks (TOY1 and TOY2), the video trigger driver of MSJB-1, MSJB-1 temperature sensors, and RF transfer switches. General software, hardware, and software configuration errors are also checked by LU26 through LU28.

2.3.2.7.4.2 Various Interrogator operational conditions can be checked by monitoring the data points for the Interrogator A, B, and C conditions logical units (LU30, LU32, and LU34) in the site RMS Monitor Report. These various conditions are monitored by means of status reporting where an "A" in the Monitor Report for the data form indicates a normal condition, a "C" indicates a red alarm condition, a "D" indicates a yellow alarm condition, and an "E" indicates a pre-alarm condition. Some of the conditions that are checked are: (1) counting the occurrences of beacon strobes, (2) counting the occurrences of CW interference, (3) measuring the differences for both on-boresight and off-boresight test targets, (4) counting the occurrences of read status, read data, or write data errors, (5) checking the sum, delta, or omni channel gain and noise, (6) checking the Mode 2, Mode 3/A, or Mode C counts, (7) checking the VSWR for the main or auxiliary transmitter, and (8) measuring the cabinet temperatures for the Interrogator and MSJB-1.

2.3.2.7.4.3 Various Interrogator (front end) status conditions can be checked by monitoring the data points for the Interrogator A, B, and C FE status conditions logical units (LU31, LU32, and LU33) in the Monitor Report. The logical units (LU31, LU32, and LU33) use the same data point ID's 20 through 9B (which indicate an LRU in the Interrogator or associated equipments) that are used for logical units LU26, LU27, and LU28 in the Monitor Report. The status field for a data point can contain an "A" for a normal condition, a "C" for a red alarm, a "D" for a yellow alarm, or an "E" for a pre-alarm condition. The value field for a data point identifies the failed test ID (if any), a 0 indicates no failures. For additional information concerning logical units, data points, and message fields; the user should refer to the Mode S to Remote Maintenance System document, TM-PA-0018/073/XX and the Local Terminal document, TM-PA-0018/077/XX.

2.3.2.8 Fault Detection and Fault Isolation Testing. During the Interrogator startup process and after the startup process, each Interrogator regularly conducts a series of built-in tests (BITs), such as: download, of EEPROM into RAM tests, CPU self test, and various performance tests. Built-in tests are performed on the Interrogator's transmission functions, the ACTRBS and Mode S reply processing functions, timing functions, and control functions. The Interrogator also conducts performance monitoring tests, on a regular basis, to ensure such items as: transmitter power is within tolerance, the receiver's gain and noise criteria are within an acceptable range, and that the Interrogator cabinet temperature sensors are not indicating an elevated cabinet temperature. The outputs from the performance monitoring circuits in the signal processor (A4) are

monitored by fault detection tests during each scan. If a fault is detected, fault isolation tests (FITs) are run to isolate the fault to the defective LRU(s). The following paragraphs describe fault detection and FIT processing for the Interrogator. The following text describes testing to the functional unit only. Detailed test descriptions are contained in the associated instruction book for the transmitter, receiver/RFTTG, and signal processor. Test descriptions for the power distribution and environmental controls are contained in the detailed theory section of the Interrogator instruction book.

2.3.2.8.1 Fault Detection Testing. Faults in Interrogator performance are detected by tests run during startup and during each scan of an eight-scan cycle. During startup, the signal processor (A4) is started (bootstrapped), tested, initialized, configured, and loaded with site parameters. The following list indicates the periods of the startup process with their purposes. Refer to the signal processor instruction book for details of test descriptions and operation.

<u>Period</u>	<u>Purpose</u>
Bootstrap	Initializes CPU and runs Interrogator diagnostics to verify equipment operation. Verifies random access memory (RAM) and electrically erasable programmable read only memory (EEPROM) operation.
CPU Self-test	Tests CPU ability to write and read the EEPROM and test target generator (TTG).
Sig Proc Initialize	Initializes the site parameters, error flags, STC curve, and TTG RAM.
APG and TOY Select	Selects the TOY clock, RF transfer switch position, and APG for on-line operation.

2.3.2.8.1.1 After startup, Interrogator performance during operation is tested during an eight-scan cycle. Both the on-line channel and the standby channel are tested. If a fault is detected, FIT is run to isolate the fault to an LRU. Each scan has a fault detection testing period that begins when a null period is available. The testing is divided into two parts: power/VSWR testing and performance testing. Within a scan, the power/VSWR testing is completed; then the performance tests are started. The performance tests runs to completion unless a target needs to be processed (no null). If a target must be processed, the performance testing is interrupted until the next scan, after power/VSWR testing is completed. At that time, the performance tests are resumed where testing was interrupted during the previous scan. All performance tests must be completed once during each fault detection testing cycle (eight scans). Each test is listed with a hexadecimal test number. Information concerning the test number and test descriptions is provided in the associated equipment instruction book.

2.3.2.8.2 Fault Isolation Testing. Fault Isolation Testing (FIT) is initiated whenever fault detection testing detects a fault. The FIT is a series of tests run in sequence, which test specific functions of the Interrogator until all tests are run with no failures detected or until a fault is detected. In general, the signal processor (A4) is tested first, then the receiver/RFTTG, the transmitter, the power supplies, and the temperature/airflow status, in that order. Each FIT configures the Interrogator for the test, operates the Interrogator for as long as the test requires, and then analyzes the results for fault indications. Any failed FIT causes further testing in the area of the fault until an LRU (or up to three candidate LRUs) is identified as the cause. A message identifying the suspected faulty LRU(s) is sent to the DPS. If more than one LRU is suspected, up to three LRUs can be identified in priority order: primary, secondary, and tertiary. The primary LRU should be replaced and FIT performed to check the repair. If FIT provides the same message, the secondary LRU should be replaced and FIT performed. This should be repeated if the same message is again repeated. If replacing the three LRUs does not correct the fault, manual fault isolation is required.

2.3.3 Data Processing System (DPS) Equipment Operation

DPS equipment operation is presented under the following headings:

- a. Processor Subsystem (PRCSS)
- b. Sensor Control
- c. Common Digitizer (CD-2) Modification Kit
- d. Communication Junction Box (CJBX)
- e. Power Subsystem

Refer to Figures 2-29 and 2-30 which present an overview diagram of how the DPS fits into the Mode Select Beacon Sensor (Mode S) system and a block diagram of the DPS. For additional information and functional diagrams concerning the Data Processing System (DPS) equipments, the user must refer to the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1 Processor Subsystem (PRCSS). The processor subsystem (PRCSS) channel block diagram for a basic configuration is presented in Figure 2-31. Figure 2-32 presents the PRCSS channel block diagram for a basic configuration with add-on-kits. The Processor subsystem (PRCSS) is comprised of the computer subsystem and communication subsystem. The computer subsystem of the PRCSS is comprised of the following elements: (1) central processing units (CPU) circuits, (2) a PRCSS channel bus, (3) global random access memory (GRAM) circuit card assemblies, (4) non-volatile memory (NVMEM) circuit card assembly, and (5) Interrogator interface (ITRI) circuit card assemblies. The communications subsystem is comprised of the following elements: (1) and input/output processor (IOP) circuits, (2) the PRCSS channel bus, (3) communication line adapter (LACOM) circuit card assembly, (4) radar line adapter (LARDR) circuit card assembly, (5) surveillance line adapter (LASUR) circuit card assemblies, (6) magnetic tape line adapter (LASMT) circuit card assembly and (7) a maintenance line adapter (LAMNT) circuit card assembly.

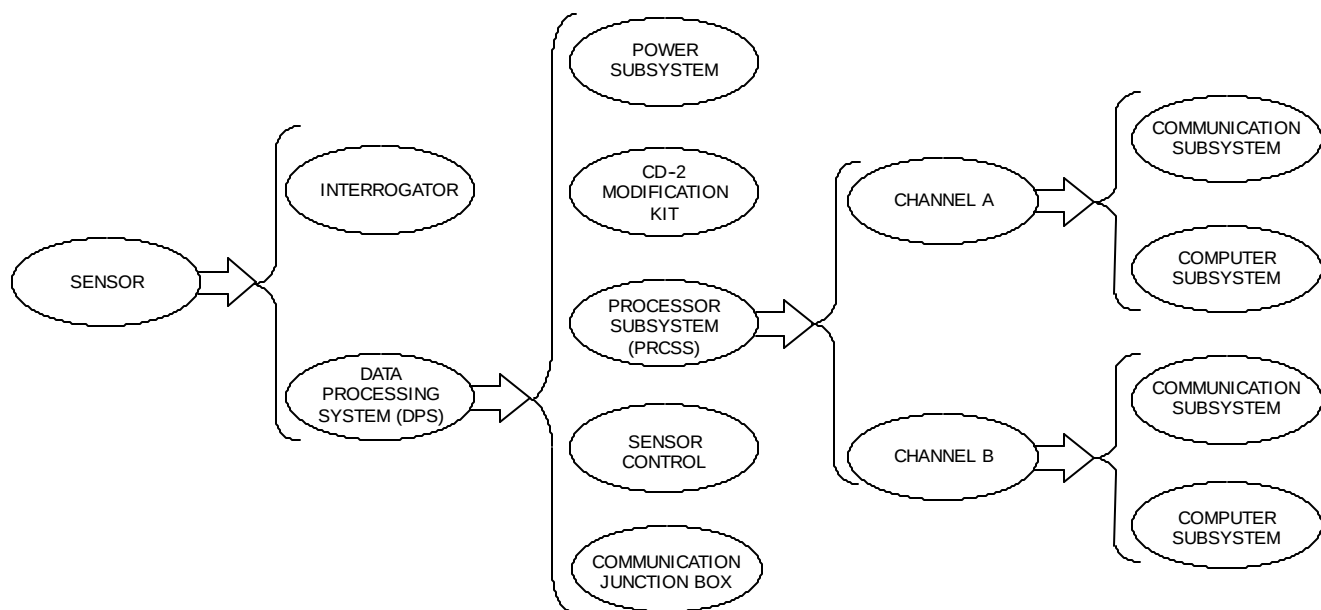
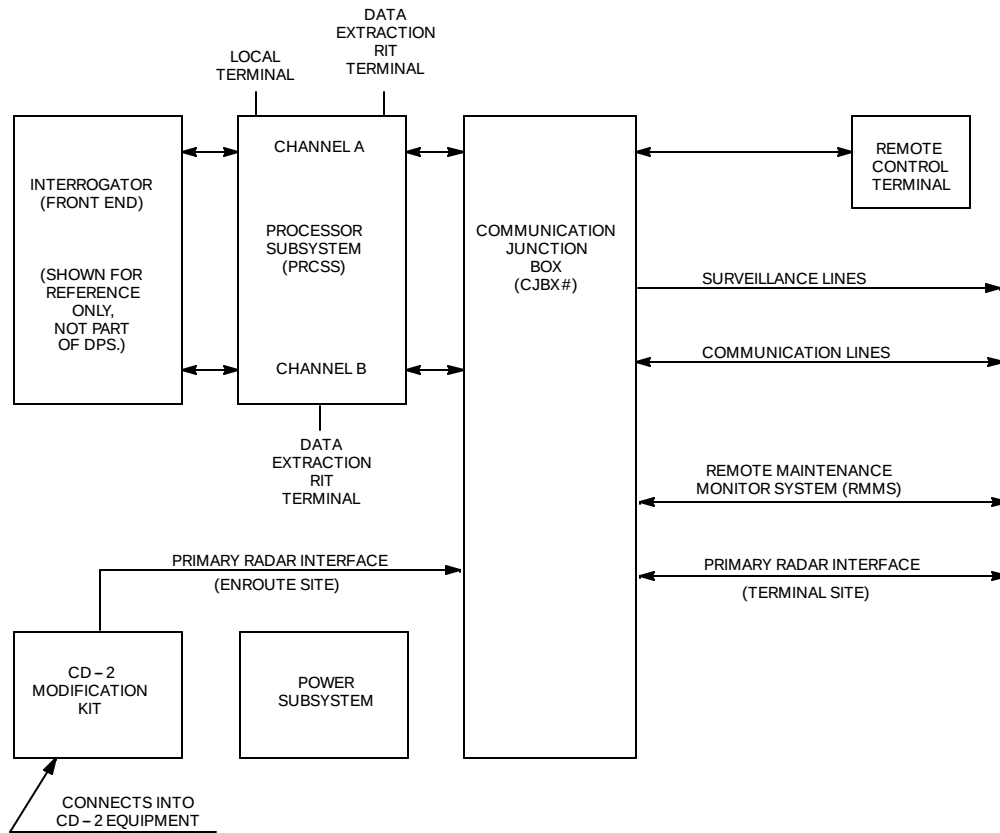


FIGURE 2-29. DATA PROCESSING SYSTEM (DPS) OVERVIEW DIAGRAM



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FIGURE 2-30. DATA PROCESSING SYSTEM (DPS) BLOCK DIAGRAM

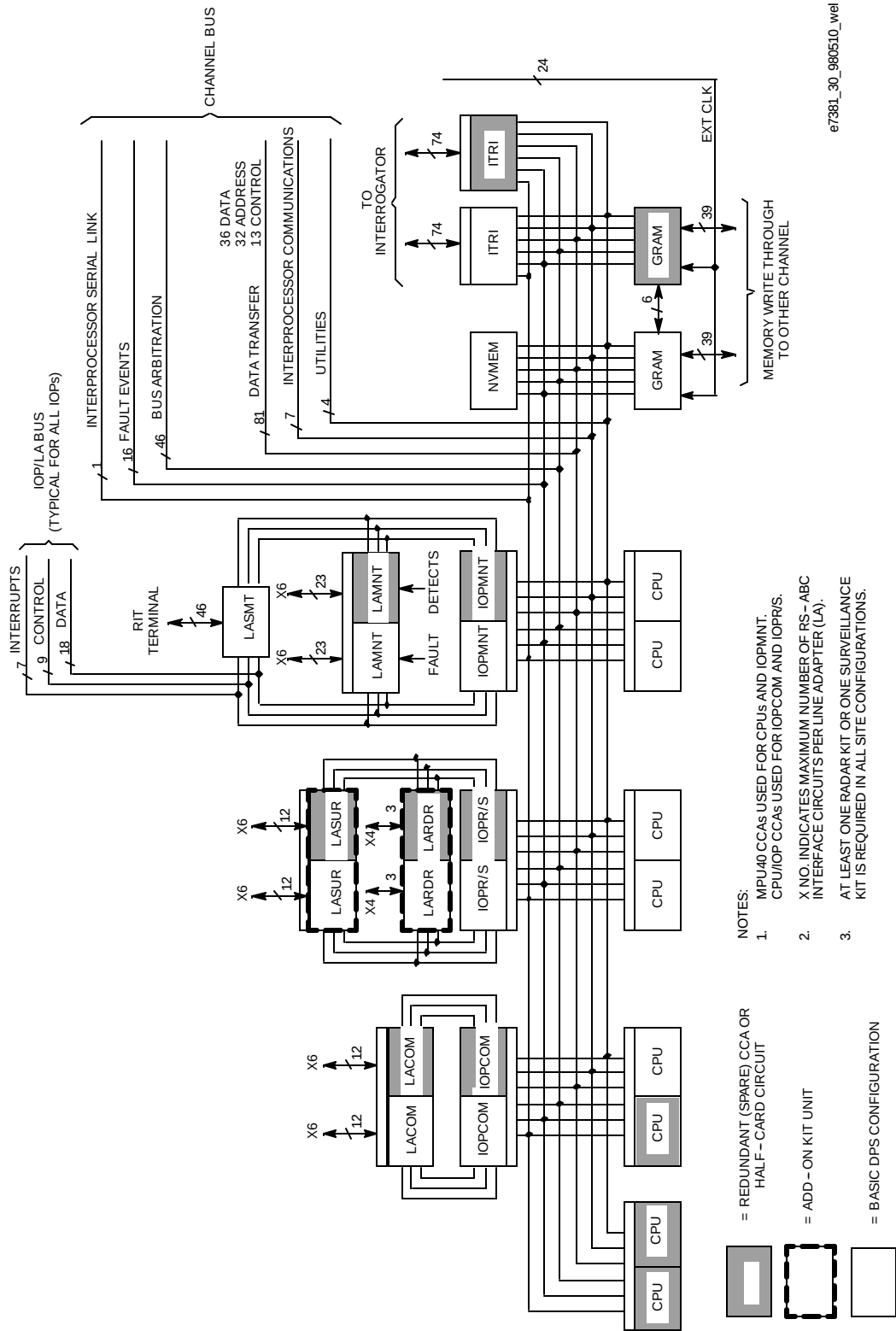


FIGURE 2-31. PROCESSOR SUBSYSTEM CHANNEL BLOCK DIAGRAM FOR A BASIC CONFIGURATION

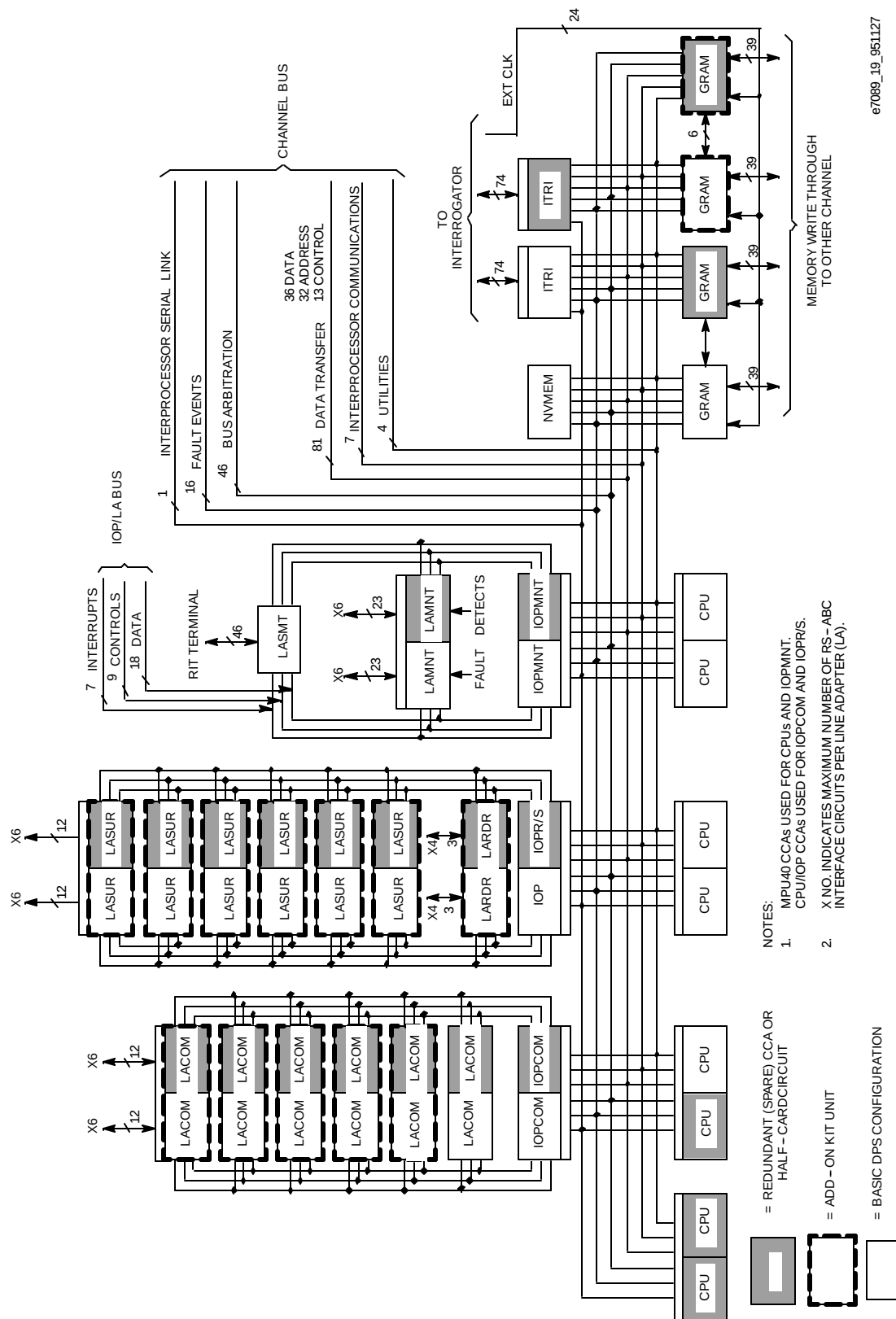


FIGURE 2-32. PROCESSOR SUBSYSTEM CHANNEL BLOCK DIAGRAM FOR A BASIC CONFIGURATION WITH ADD-ON KITS

2.3.3.1.1 PRCSS Computer Subsystem. The PRCSS computer subsystem is comprised of the following elements: (1) CPU/IOP or MPU40 CCAs, (2) PRCSS channel bus, (3) GRAM CCA, (4) NVMEM CCA, and (5) an ITRI CCA.

2.3.3.1.1.1 CPU and IOP. The CPU circuits of the computer subsystem and the IOP circuits of the communication subsystem are comprised of the same type of logic circuits and are implemented with the same type of circuit card assembly. The IOP circuit has a line adapter interface which is present but not used on the CPU circuit. The CPU and IOP circuits, which are interchangeable for maintenance purposes, are implemented with either 68020-based CPU/IOP circuit card assemblies or 68040-based MPU40 circuit card assemblies. The two types of circuit card assemblies (CPU/IOP and MPU40) are functionally the same but are not directly interchangeable because the software build must be constructed to match the configuration of CCAs used. The functional description of the CPU is the same as the IOP unless otherwise specified. The CPU (or IOP) circuit card assembly is comprised of two physical (redundant) CPU (or IOP) circuits on each card assembly. Each CPU (or IOP) circuit is a stand-alone processor unit with a primary microprocessor, a shadow microprocessor, a local dynamic random access memory (DRAM) and controls, a programmable read-only memory (PROM) and controls, a local bus, address decoding circuits, error detection and reporting circuits, a status port, interrupt logic circuits, an access protection unit (APU), timing logic circuits, a channel bus interface, and a line adapter interface (not used when the circuit card assembly is in a CPU card slot). For a CPU (or IOP) logical processor detailed block diagram refer to Figure 2–33.

2.3.3.1.1.2 PRCSS Channel Bus. The PRCSS channel bus provides bus arbitration which is used to gain access to the data transfer bus. The data transfer bus is used to move information (data) between devices. Figures 2–31 and 2–32 present various configurations of the PRCSS channel and the type of interface provided by the PRCSS channel bus, such as: (1) interprocessor serial link, (2) fault event, (3) bus arbitration, (4) data transfer, (5) interprocessor communications, and (6) utilities. Figure 2–34 presents a functional block diagram of the PRCSS channel. Figure 2–35 presents a simplified block diagram of the PRCSS channel bus interface.

2.3.3.1.1.2.1 The fault event bus is used to signal the existence of a fault from the CPU, IOP, ITRI, GRAM, and/or NVMEM circuits. The bus arbitration bus interfaces with the CPU, IOP, ITRI, GRAM and NVMEM circuits and is used to gain access to the data transfer bus. The data transfer bus interfaces with all of the aforementioned circuit card assemblies and is used to move information (data) between devices. The interprocessor communications bus interfaces with the CPU, IOP, ITRI, and GRAM circuits and is used to send interrupts to the destination processor.

2.3.3.1.1.2.2 The PRCSS channel bus has the following characteristics: (1) multi-mastership capability, (2) a centralized arbiter which accepts bus requests and issues an encoded grant identification (grant ID), (3) priority which rotates after the grant ID is used and inhibits further requests when a memory bank is busy, (4) an interleaved memory configuration — the basic system configuration uses two-way memory interleave operation, and the expanded system configuration uses a four-way memory interleave operation, (5) individual locking of up to four memory banks for indivisible operations — the PRCSS channel bus and all unlocked memory banks remain available for use, (6) memory with input and output queues, (7) a non-multiplexed address bus and a non-multiplexed data bus, (8) address, data, grant ID, response ID, and processor ID lines which are protected by byte parity, (9) read operation by means of separate bus transactions for high throughput capability, (10) separate interprocessor communications for high throughput capability, (11) synchronous operation— all operations and circuit card assemblies are in sync with central channel clock, (12) a unique encoded ID of five bits for each card slot, and (13) all signal lines terminated at each end.

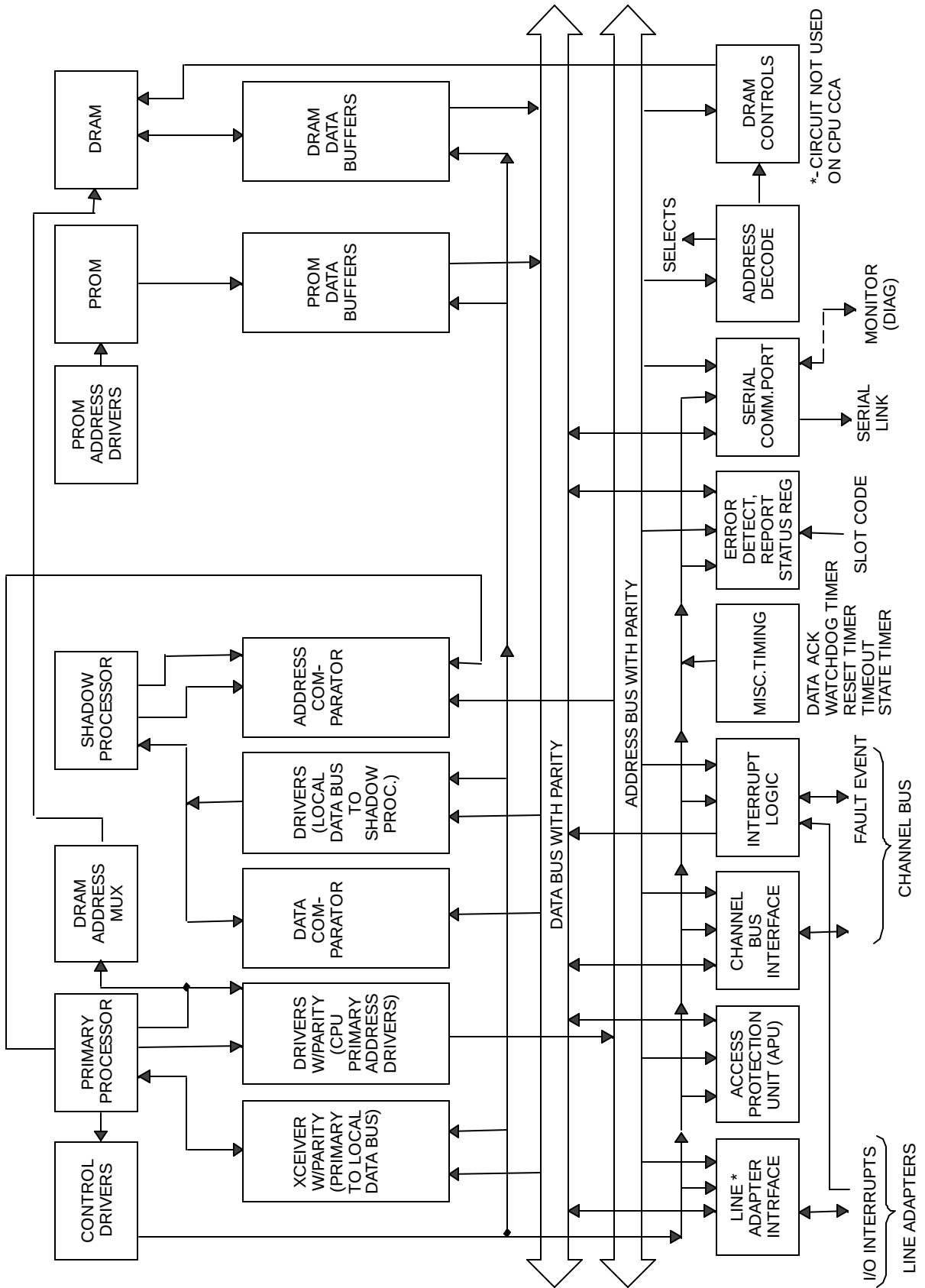
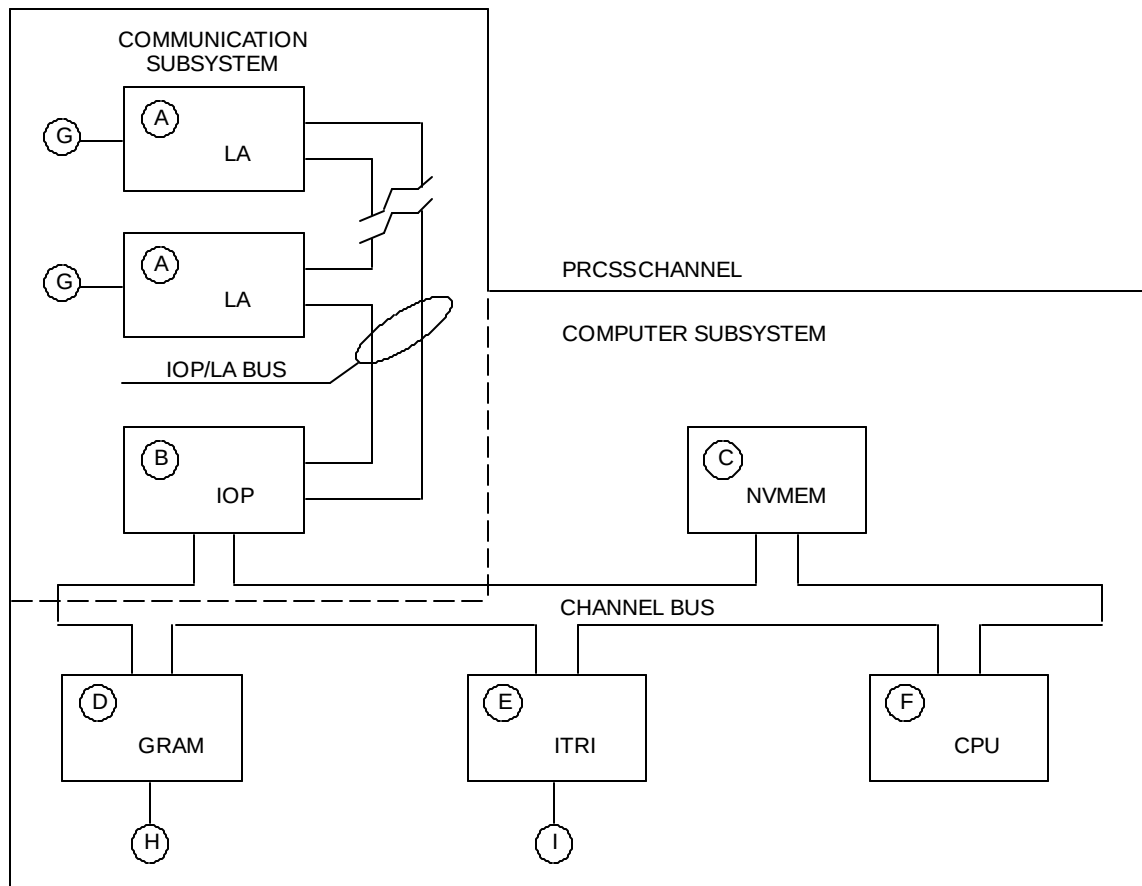


FIGURE 2-33. CPU OR IOP BLOCK DIAGRAM



A. LA = LINE ADAPTER. THERE ARE REDUNDANT LAs.

B. IOP = INPUT/OUTPUT PROCESSOR.

- THERE ARE MULTIPLE IOPs.
- THERE ARE REDUNDANT IOPs.

C. NVMEM= NON-VOLATILE RANDOM ACCESS MEMORY.

D. GRAM = GLOBAL RANDOM ACCESS MEMORY.

- THE GRAM IS REDUNDANT

E. ITRI = INTERROGATOR INTERFACE.

- THERE ARE REDUNDANT ITRIs.

F. CPU = CENTRAL PROCESSING UNIT.

- THERE ARE MULTIPLE CPUs.
- THERE ARE REDUNDANT CPUs.

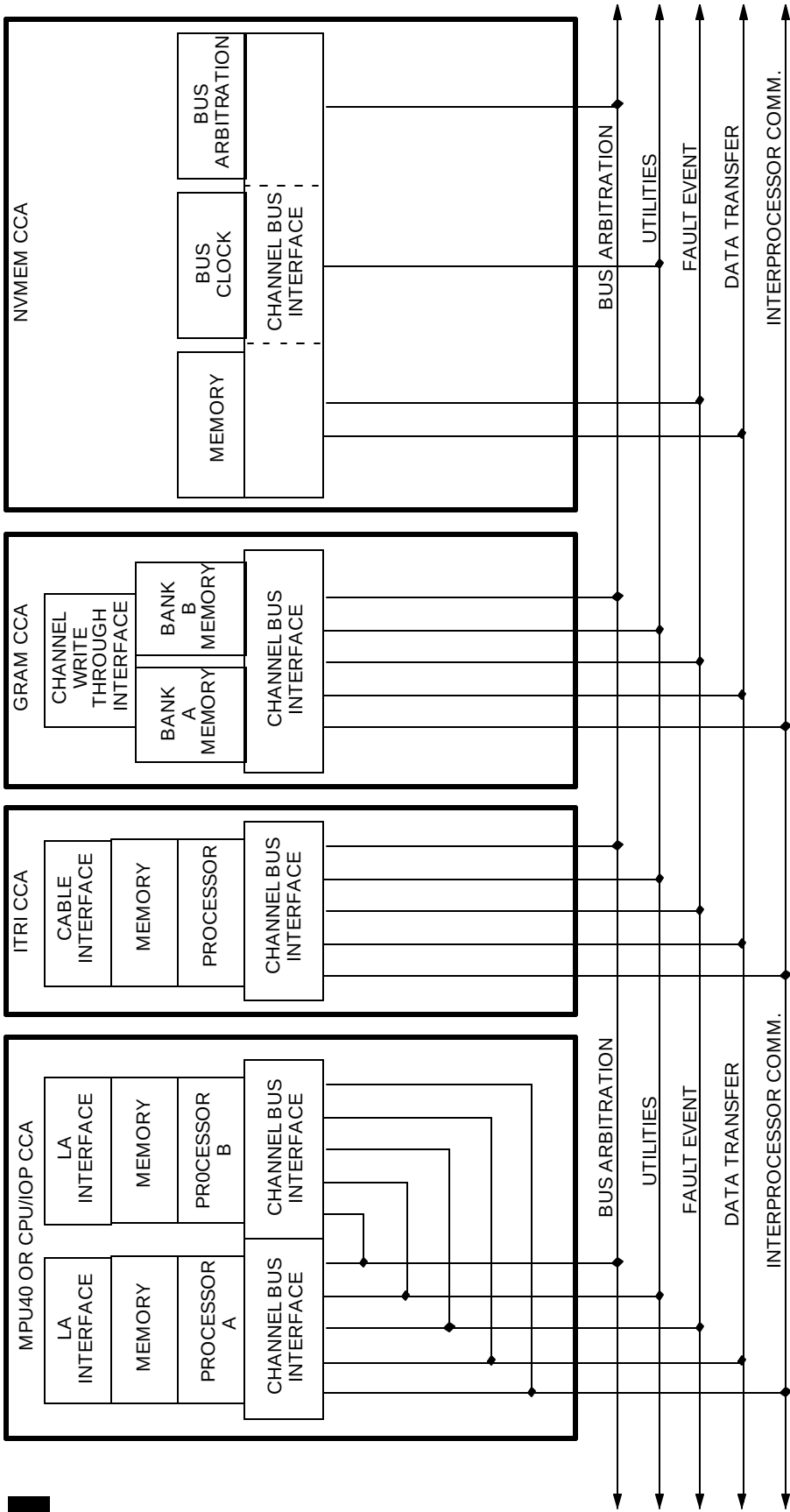
G. OUTPUT TO EXTERNAL FAA ELEMENTS AND/OR OTHER SENSOR ELEMENTS.

H. OUTPUT TO OTHER PRCSS CHANNEL GRAM (CRITICAL DATA TRANSFER).

I. OUTPUT TO INTERROGATOR.

THE INDICATOR BLOCKS ON THIS DIAGRAM ONLY INDICATE THE RELATIONSHIP OF THE SHOWN FUNCTIONAL UNITS. THEY DO NOT INDICATE QUANTITY.

FIGURE 2-34. PROCESSOR SUBSYSTEM (PRCSS) CHANNEL FUNCTIONAL BLOCK DIAGRAM



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FIGURE 2-35. PRCSS CHANNEL BUS INTERFACE SIMPLIFIED BLOCK DIAGRAM

2.3.3.1.1.3 Global Random Access Memory (GRAM). The global random access memory (GRAM) CCA provides for global memory accessibility by all CPU, IOP, and ITRI circuit card assemblies. Each of the users of GRAM access memory via the PRCSS channel bus, which is controlled by a centralized arbiter located on the NVMEM CCA (refer to Figure 2–34). Each access to GRAM memory can fetch or store eight, sixteen, and thirty-two bits of information. Per byte parity is maintained on all GRAM memory. Each GRAM CCA has two banks (0 and 1) of memory in which each bank provides two megabytes of storage. Global RAM can be expanded by the addition of a second set of two GRAM CCA's, one expansion CCA active and the other expansion CCA standby, making a total of four GRAM CCA's two active and two standby. The I/O register and interprocessor communication logic circuits are located on the memory bank zero side. During the initial power up state or reset state, the diagnostics are run. If no error is detected in that GRAM CCA, the memory is fully updated and that GRAM circuit card assembly is assigned to the active or standby mode after the system has been configured. Data written to the active GRAM is copied to the GRAM circuits in the other PRCSS channel by use of the memory write-through link. A copy of the write data is also made in the redundant (standby) GRAM card in each PRCSS.

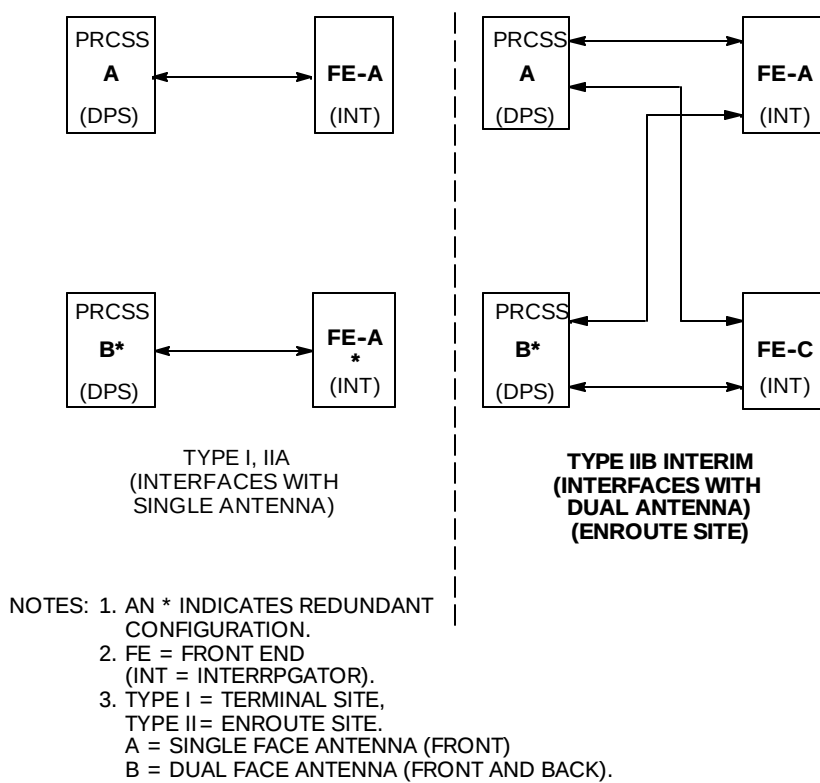
2.3.3.1.1.3.1 For the functional and detailed theory of operation of the global random access memory (GRAM) circuit card assembly and its functional interface with the CPU, IOP, ITRI, and NVMEM of the DPS; refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.1.4 Non-Volatile Memory (NVMEM). The PRCSS non-volatile memory (NVMEM) circuit card assembly is comprised of the following logic circuits: (1) a channel clock circuit, (2) a channel bus arbiter logic circuit, (3) a non-volatile storage and storage control logic circuits, and (4) an error detection logic circuit which detects an error in arbitration, a bus parity error, an invalid address, or an invalid operation. The NVMEM circuit card assembly is accessible by any CPU, IOP, or ITRI circuit (refer to Figure 2–34). The NVMEM circuit card assembly channel bus arbiter logic circuit performs the following functions: (1) resolves bus contention for requestors, (2) increments grant increment priority (GIP) rule, (3) performs circuit card assembly arbitration for 125 nanoseconds (2 clocks), (4) returns the encoded bus grant ID to the requestors, and (5) performs error detection of invalid codes, no bus requests, and prolonged bus mastership.

2.3.3.1.1.4.1 The NVMEM circuit card assembly PRCSS channel bus interface logic circuits perform the following functions: (1) support channel bus protocol, (2) decodes for operation selection, (3) sequences EE-PROM or I/O responses, and (5) performs error detection on parity errors. The NVMEM circuit card assembly electrically erasable PROM (EEPROM) logic circuit performs the following functions: (1) memory storage of two megabytes, (2) long word data operations, (3) semaphore access protection, (4) power cycling write inhibit operations, and (5) error detection of double accesses, misaligned address, and denied accesses.

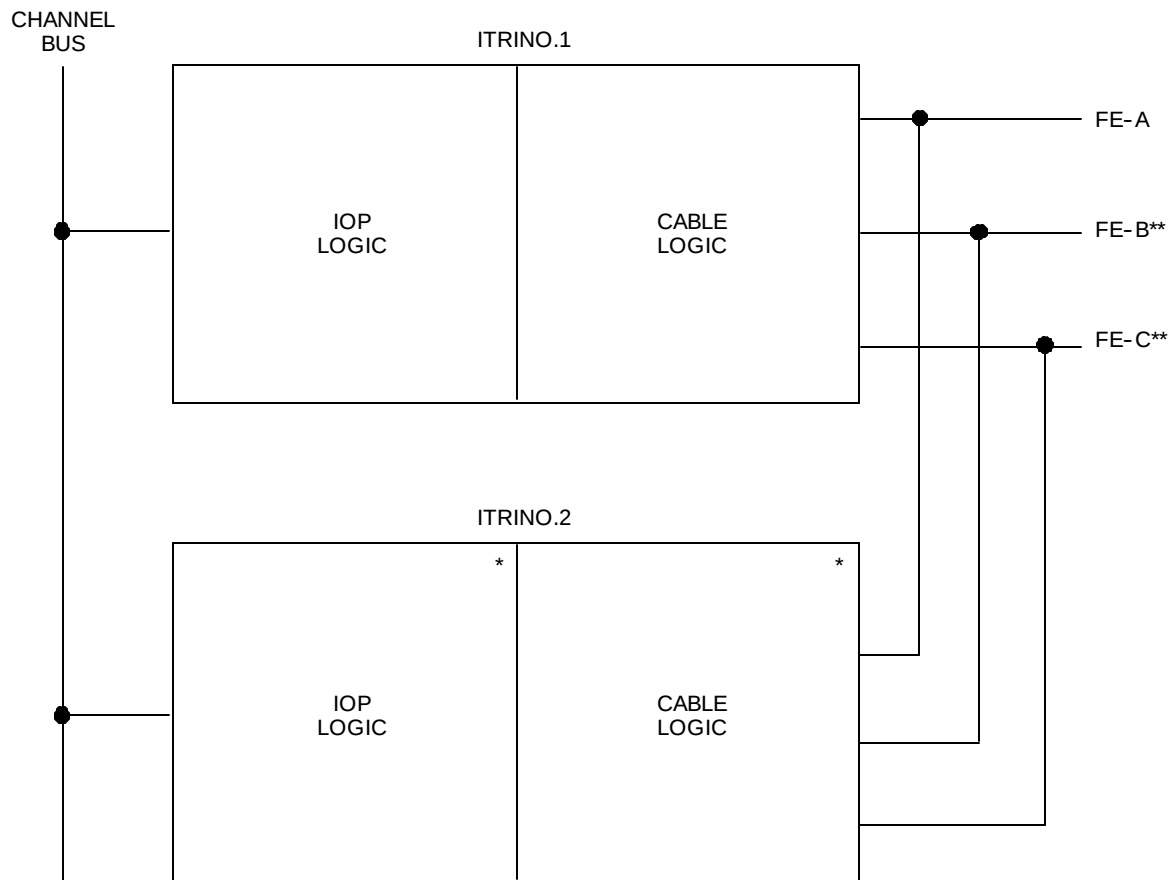
2.3.3.1.1.4.2 For the functional and detailed theory of operation of the non-volatile memory (NVMEM) circuit card assembly and its functional interface with the CPU, IOP and ITRI; the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3

2.3.3.1.1.5 Interrogator Interface (ITRI). Each ITRI circuit contains the interface and control logic for transfer of data between a PRCSS channel and up to three interrogator units. Each PRCSS channel contains two ITRI circuit cards that operate as an active/backup pair. Each ITRI Circuit Card Assembly (CCA) contains one ITRI circuit. All communications between the interrogator and the DPS are carried out through the ITRI circuits. Figure 2–36 presents a block diagram of the PRCSS (A or B) interface with the interrogator (front end–FE) unit configurations. The ITRI CCA provides bidirectional parallel message oriented communications with up to three front-end interrogators (current configurations use only two interrogators). The ITRI circuit performs the channel management scheduling function that directs interrogator operations. For each message sent to an interrogator, the ITRI fetches the message from global memory, formats the data as required, and transfers the message to the appropriate interrogator channel. For each message received from an interrogator, the ITRI identifies and checks the message, then stores the message in global memory for subsequent CPU processing. Each ITRI circuit includes channel bus interface logic (similar to an IOP circuit) and a cable logic interface. The ITRI circuit is, in effect, an IOP circuit together with a specialized line adapter. The cable logic controls the routing of the message to or from the addressed interrogator. The cable logic interface of the ITRI to the interrogator accommodates three separate cable assemblies with a worst case transfer rate of 571K words/second across the cable interface (refer to Figure 2–37). The primary and backup ITRI CCA's provide for a wire-ORed configuration for the cable interface.



e7089_8_951127_WJF

FIGURE 2–36. PRCSS CHANNEL INTERFACE WITH THE INTERROGATOR (FRONT END) CHANNELS, BLOCK DIAGRAM



NOTES:

1. AN * INDICATES REDUNDANT LOGIC.
2. THE ** INDICATE THAT THE FE-B AND FE-C OUTPUTS ARE NOT USED AT A TERMINAL SITE. AT AN ENROUTE SITE, THE FE-C AND FE-A OUTPUTS ARE USED AND THE FE-B OUTPUT IS NOT USED.
3. FE=FRONT END (INTERROGATOR A, B, OR C).

FIGURE 2-37. INTERROGATOR INTERFACE (ITRI) GENERAL BLOCK DIAGRAM

2.3.3.1.1.5.1 The IOP logic circuit of the ITRI CCA performs the following functions: (1) interfaces with the PRCSS channel bus, (2) directs message traffic between the PRCSS channel bus and the cable logic interface, (3) provides a local memory with parity, (4) interfaces with the cable logic circuit on the ITRI card assembly, and (5) performs various channel management functions on the ITRI CCA. The input/output processor (IOP) logic circuits are the standard CPU/IOP logic which uses the 68020 processor with a 68020 shadow processor configuration, transceiver and driver circuits which interface with local bus, a local memory configuration (256K bytes of RAM and 32K or PROM), address decode circuits, access protection unit (APU), interrupt logic circuits, channel bus interface, miscellaneous timing circuits, error detection/reporting and status logic circuits, and cable interface via the local bus. For the functional and detailed theory of operation of the various logic circuits used by the IOP portion of the ITRI card assembly, the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.1.5.2 The cable logic circuits of the ITRI circuit card assembly performs the following functions: (1) transfers block oriented messages between the ITRI circuit card assembly and one of three Interrogator channels, (2) resolves priority in case of concurrent need for service, and (3) provides error handling on the occurrence of error during data transfers. The cable logic circuits in automatic mode provide the following functions:

- a. Transfers messages from the ITRI output FIFO memory to the dedicated Interrogator channel receive FIFO buffer.
- b. Transfers messages from the Interrogator channel read FIFO buffer to the ITRI input FIFO memory.
- c. Prioritizes read/write operations as to destination and message types.
- d. Provides message/word level error detection.
- e. Acts as the master in the data transfer process.

2.3.3.1.1.5.3 For the functional and detailed theory of the various logic circuits used by the cable logic circuits of the ITRI card assembly, the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.2 PRCSS Communication Subsystem. The PRCSS communication subsystem is comprised of Input/Output Processor (IOP) circuit card assemblies (CPU/IOP or MPU40 CCAs) and various line adapter (LA) circuit card assemblies. The CPU/IOP or MPU40 circuit card assembly can be used as either a CPU or IOP because of identical logic circuits as described in paragraph 2.3.3.1.1.1. The line adapter (LA) CCAs that comprise the PRCSS communication subsystem are:

- a. Communications line adapter (LACOM),
- b. Primary radar line adapter (LARDR),
- c. Surveillance line adapter (LASUR),
- d. Magnetic tape line adapter (LASMT), and
- e. Maintenance line adapter (LAMNT).

The line adapter logic circuits are divided into two areas: (1) the common logic which is peculiar to all line adapter CCAs and (2) the device dependent port logic circuits which are unique to that particular line adapter. For the functional and detailed theory of operation of the various line adapter (LA) card assemblies and their functional interface with the IOP of the PRCSS communication subsystem and the device interface, the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.2.1 Communications Line Adapter (LACOM). The communications line adapter (LACOM) device dependent logic circuits provide six multi-protocol communication controllers (MPCC-68560) which implement zero-bit insertion, frame flags, address fields, and control fields. The MPCC also provides direct memory access request generation for each data byte or word that is transmitted or received by the LACOM. The MPCC interrupts the processor (MPU-68000) at the end of the frame and also for abnormal occurrences. The LACOM also contains three direct memory access (DMA) controllers (68440). Each four-channel DMA is dedicated to two MPCCs. The DMA controller provides high-speed transfer of data from the local static random access memory (SRAM) to the MPCC. Local bus arbitration is provided via a local bus arbiter module (BAM) which generated the bus request and bus grant acknowledge signals sent to the external I/O interface circuit of the LACOM common logic. The DMA controller also interrupts the processor (MPU) for abnormal conditions or data block move completions. Six tri-state RS-422/423 driver and receiver circuits with wire-ORed connections to the secondary LACOM circuit. The LACOM is configured as a DTE.

2.3.3.1.2.1.1 For the functional and detailed theory of operation of the communications line adapter (LACOM) and its interface with the IOP of the PRCSS communication subsystem and the communication device interface, the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.2.2 Radar Line Adapter (LARDR). The radar line adapter (LARDR) circuit card assembly provides the interface between the primary radar's serial links and the radar IOP (IOPR/S) circuit card assembly for the communications subsystem. The primary radar interface is either a Westinghouse electric Company (WEC) ASR-9 or the Mode S common digitizer CD-2 modification kit. The Mode S CD-2 modification kit is the interface to the Eaton CD-2 radar system and must conform to WEC's ASR-9 interface circuit design. When the CD-2 is connected to the Mode S system, CD-2 is only used as an uplink for data to the Mode S system. The ASR-9 connection is a full-duplex link with a greater message loading capacity than the CD-2 radar system.

2.3.3.1.2.2.1 The radar line adapter (LARDR) is a microcomputer based on an MC68000, 16-bit microprocessor (MPU). The microprocessor operates at 10 megahertz and supports the following LARDR logic circuits: (1) local memory, (2) MPU peripheral and control, (3) an IOP interface, and (4) a radar interface. The LARDR circuit card assembly, as is the case with most other line adapters, has a primary and secondary line adapter logic circuit on the same circuit card assembly. The primary and secondary logic circuits are identical in function, design, and layout. The serial radar links are wire-ORed at the card edge to provide a redundant system of operation for a single interface. For the IOP connection, the primary LARDR circuit is dedicated to the primary IOP logic circuit and the secondary LARDR circuit is dedicated to the secondary IOP logic circuit. For the radar interface, the serial links are not wire-ORed in the communication junction box (CJBX) as for the other line adapters, but have separate lines of service.

2.3.3.1.2.2.2 For the functional and detailed theory of operation of the radar line adapter (LARDR) and its functional interface with the IOP of the PRCSS communication subsystem and the radar device interface, the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.2.3 Surveillance Line Adapter (LASUR). The surveillance line adapter (LASUR) circuit card assembly is used to interface the input/output processor (IOP) circuit card assembly and the outside world. The surveillance line adapter (LASUR) communicates with the common digitizer (CD-2) interface at an enroute site. The LASUR interface is a transmit only configuration from the Mode S sensor link. A data loop back circuit, which is multiplexed between six serial circuits, is provided for on the LASUR circuit card assembly. The surveillance line adapter (LASUR) device dependent logic circuits are comprised of: (1) a port control circuit which includes the surveillance transmission circuit and (2) a surveillance loopback multiplexing circuit.

2.3.3.1.2.3.1 The LASUR port control circuit is comprised of six transmit only circuits with an RS-ABC interface. The LASUR line adapter is configured as a DTE. The six transmit only circuit use a tri-state RS-422/423 interface and are wire-ORed to the secondary LASUR circuit on the card assembly. The surveillance port control register (SPCR) controls transmission parameters such as: (1) surveillance word size, (2) internal or external (INT/EXT) baud rate selection, (3) 13-bit parity enable control, and (4) controls for loopback control decode circuit which provides for local or remote loopback of data. The baud rate clock can be selected from an internal source (multi-protocol communications controller-MPCC) or an external source (from the modem clock). A readback register is used to check the FIFO status for surveillance transmission circuits, and test mode inputs (TM6-TM1) from the ports.

2.3.3.1.2.3.2 The surveillance loopback multiplexing logic circuit provides the following features: (1) any one of six surveillance circuits can be placed in local or remote data loopback mode, while the remaining five surveillance circuits continue to perform in normal operation mode, (2) the receipt of loopback test data is controlled by one multi-protocol communications controller, (3) the transmission of loopback data is handled by a multiplexer circuit, and (4) data loopback circuit selection bits from the surveillance port control register (SPCR) via the loopback control decode circuits which control the multiplexer circuits.

2.3.3.1.2.3.3 For the functional and detailed theory of operation of the surveillance line adapter (LASUR) and its functional interface with the IOP of the PRCSS communications subsystem and the surveillance

device interface, the user must refer to Section 2 of the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.2.4 Magnetic Tape Line Adapter (LASMT). The magnetic tape line adapter (LASMT) coordinates the transfer of information between the maintenance IOP (IOPMNT), which interfaces with the LASMT circuit card assembly and the Radar Intelligent Tool (RIT) terminal via the Tape Drive Interface Card (TDIC) of the RIT terminal. The Tape Drive Interface Card (TDIC), which is installed in the RIT terminal, is used as a replacement for (emulates) a magnetic tape drive to upload/download data to/from the Mode S Sensor. The LASMT circuit card assembly interconnects with the RIT terminal via the TAPE1 and TAPE2 connectors (J1 and J2), on the Status and Alarm panel (A3) of the DPS. The LASMT is able to transmit and receive data blocks at a rate up to 320K bytes/second. The LASMT circuit card assembly is designed around a 68000 (10 MHz) based microprocessor (MPU), a 68450 direct memory access controller (DMAC), a local memory, an interrupt circuit, programmable interval timers (PITS), error detection circuits, IOP/LA interface circuit, and a formatter interface circuit. The LASMT formatter interface circuits have unidirectional signal lines with separate read data and write data paths from and to the tape drive formatter circuits. Formatter internal and external loopback circuits are positioned on the write and read data lines for test purposes. The LASMT logic circuit is not redundant. Because the LASMT must interface with a primary and secondary IOP, the interface logic circuit merges the signals from the primary and secondary IOP circuits.

2.3.3.1.2.4.1 There is only one line adapter circuit on the LASMT circuit card assembly. All other line adapters in this system have redundant line adapter circuits. The microprocessor unit (MPU) performs read/write operations from/to the formatter interface control logic. For the functional and detailed theory of operation of the magnetic tape line adapter (LASMT) and its functional interface with the IOP of the PRCSS communication subsystem and the RIT terminal interface, the user must refer to Section 2 of the Mode S Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.1.2.5 Maintenance Line Adapter (LAMNT). The maintenance line adapter (LAMNT) CCA provides the logic circuits that are required to interface the maintenance input/output (IOPMNT) of the Mode S data processing system to the various monitoring and control devices, both local and remote. Each LAMNT circuit card assembly shall contain two independent LAMNT logic circuits which are designated primary and secondary. The interfaces to the various control devices of the two LAMNT logic circuits are wire-ORed with the exception of the channel-to-channel interfaces which are totally independent of each other.

2.3.3.1.2.5.1 The maintenance line adapter (LAMNT) has a redundant circuit on each card assembly (two complete LAMNT logic circuits on one card). The LAMNT device dependent logic circuit is comprised of six multi-protocol communication controllers (MPCC-68560) which interface with the data bus, address bus, and control lines of the LAMNT common logic circuits inputs. The MPCC output circuits interface with the receiver/driver circuits of the six interfaces. The port control circuit controls the enable pulse to the receiver/driver circuits.

2.3.3.1.2.5.2 The maintenance line adapter (LAMNT) also interfaces with the status and alarm panel (A3) and auxiliary status (ASTAT) circuit. The auxiliary status (ASTAT) and panel control logic circuits receive inputs from: (1) the remote contacts that indicate the status of devices outside of the DPS cabinet but internal to the Mode S system environment, (2) the three temperature sensors, (3) the power supply and blower fault lines, (4) channel reset push button, and (5) the status and alarm panel. The ASTAT and panel control logic circuits are non-redundant on the LAMNT circuit card assembly except for the power fail detect circuit which is non-redundant. For the functional and detailed theory operation of the maintenance line adapter (LAMNT) and its functional interface with the IOP of the PRCSS communications subsystem and the device interface, the user must refer to the Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.2 Sensor Control. The local and remote sensor control function, performed using a GFE standard personal computer (PC) terminal, provides the capability to override automatic system control and manually select the sensor configuration, which includes control over the following system elements: (1) the sensor

channel, and (2) backup mode of operation. The sensor control also provides an indication of the current system status. Standard computer terminals can serve as local or remote control devices. The use of a standard terminal also provides an efficient man-machine interface, flexibility, and also the advantage of an “off-the-shelf” product. The Local Terminal at the sensor site interfaces with the maintenance line adapter (LAMNT) of the Processor subsystem (PRCSS) via the Local Terminal Interface (LTI) connector, located on the status and alarm control panel (A3) in the top front area (A3) of the PRCSS cabinet. The KCRT terminal at a sensor site interfaces with the Interrogator via the KCRT input connector (J4), located on the signal monitor panel (2/3/8A17) on the top front area of the Interrogator cabinet. For additional information concerning GFE software package, startup, and description of the Interrogator menus displayed on the KCRT terminal, the user must refer to Section 3 of the Interrogator instruction book TI 6365.5. The remote terminal interfaces with the maintenance line adapter (LAMNT) of the Processor subsystem (PRCSS) via a GFE modem and communications junction box (CJBX, unit 6).

2.3.3.2.1 Local Terminal. The local terminal provides the local control point for the Mode S sensor. The local terminal consists of a GFE supplied AT class, IBM compatible PC with a color VGA display. Each Processor subsystem (PRCSS) of a DPS has its own separate port for connection of the Local Terminal. This arrangement permits the connection of the Local Terminal directly to both channels. The configuration allows maintenance of either channel in the event that the channel-to-channel path is not operational. When the channel-to-channel path is operational, commands are issued to either channel from a Local Terminal connected to only one of the channels. Commands are directed to the desired channel via the channel identification (ID). The software package is the local terminal program. The local terminal program provides the following functions: (1) initial loading and start-up of the Mode S sensor, (2) modification of Mode S sensor site parameters, (3) initiation and control of diagnostic procedures, (4) reassignment of processor images within the computer subsystem, (5) switching of redundant elements from standby to operational status, (6) element reconfiguration, and (7) data extraction at the local Mode S sensor. The following additional functions may also be controlled by the local terminal program: (1) reading the status of various data points and (2) calibration of the Interrogator.

2.3.3.2.1.1 With the local terminal program running, the Local Terminal uses asynchronous, character-oriented communications. The baud rate is 9600. After the Local Terminal is connected to the LT1 jack on the DPS status and alarm control panel assembly, communication with the Local Terminal is established by the DPS RMS. The DPS maintenance line adapter (LAMNT) card assembly senses the presence of the Local Terminal via an active request to send (RS) signal. The LAMNT sets its baud rate to the site adaptable parameter (SAP) value and then issues three signals (data mode-DM, receiver ready-RR, and clear-to-send-CS) that indicate to the Local Terminal that the line is being opened. The LAMNT then waits for the Local Terminal to transmit data. The Local Terminal, with the local terminal program running, then transmits an initialization message. The initialization message will be the baud rate request message. The message is transmitted repeatedly at increasing baud rates, beginning at 4800 and then 9600 baud. As soon as the SAP data rate is reached, the LAMNT recognizes the baud rate request message and responds by returning a SAP baud rate message to the Local Terminal. The LAMNT then switches to the DPS mode protocol as does the Local Terminal.

2.3.3.2.2 Remote Terminal. The remote terminal (RT) serves as the remote control point for the Mode S sensor. The remote terminal equipment consists of a GFE supplied AT class, IBM compatible personal computer (PC) with a color VGA display and remote terminal software. The remote terminal is located in the TRACON or ARTCC and is used for interaction with the Mode S Sensor. The remote terminal is connected to the Mode S Sensor via GFE modems and interfaces with the DPS Processor Subsystem (PRCSS) cabinet via the remote control output cable connector (RMT CONT OUT) on the communications junction box (CJBX, unit 6). The remote control A or B (RMT CONT A or RMT CONT B) cable connector on the CJBX, unit 6 interfaces with the LAMNT circuit card assembly via the J2 cable connector on the I/O connector panel assembly of the respective DPS A (unit 4) or DPS B (unit 5). Refer to Figure 2-38 for a block diagram of Mode S sensor control for the remote terminal. At the Mode S sensor site, the two sensor channels (A and B) are wire-ORed within the CJBX, unit 6 to the collocated modem. The remote terminal communicates with each sensor channel via the DPS remote monitoring subsystem (RMS) software residing in each channel. The Mode S remote terminal display

screen is organized into four display areas: (1) Sensor Status, (2) Channel Select/Status, (3) Sensor Control Select/Status, and (4) Communication and Command Status. Seven function keys positioned across the top of the remote terminal keyboard (F2 through F8) are used to implement Mode S Sensor actions when the remote terminal has control.

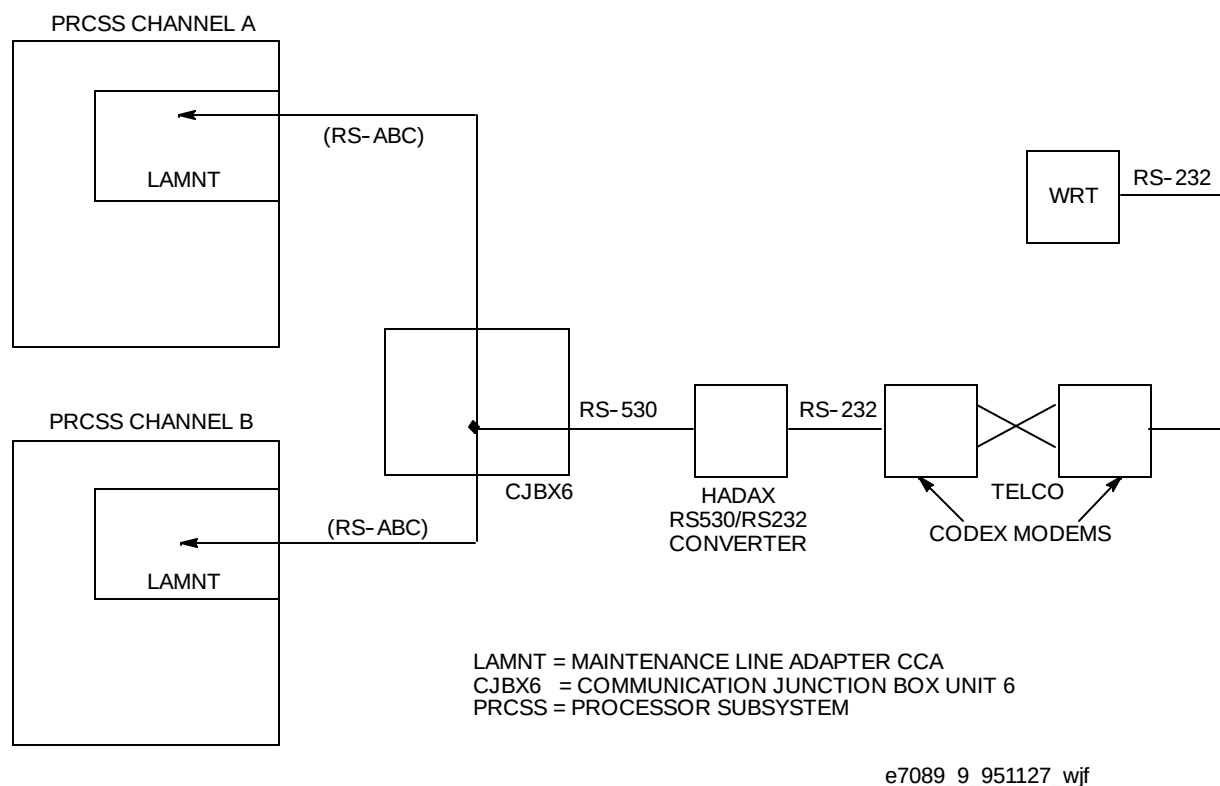


FIGURE 2-38. REMOTE MODE S SENSOR CONTROL

2.3.3.2.2.1 The Sensor Status area displays the sensor's active operational mode (Mode S or ATC-BI). The active mode appears in either a 'green' or 'yellow' display. A 'red' indication means a hard alarm condition exists. The Channel Select/Status area displays the Mode S and ATCBI channel statuses. This control area allows the user to select the active channel when the "Panel Control" indicator is 'green'. The "Online" and "Standby" channel statuses are displayed in either 'green' or 'yellow' and the "Offline" channel statuses are displayed in 'red'. The Sensor Control Select/Status area allows the user to request ATC control (CTRL-F6) or to release control to a maintenance (Maint) terminal (CTRL-F7). The remote terminal has control when the "Panel Control" indicator is 'green'. The user can only issue sensor commands when the remote terminal has control. The remote terminal does not have control when the "Panel Control" indicator is 'red'.

2.3.3.2.2.2 The Communication and Command Status area is comprised of the Comm Stat and Cmnd Stat indicators. When the Comm Stat indicator is green, it indicates that communications are fine ("Comm Ok"). If there is a communications break, the Comm Stat indicator will change to 'red' to indicate a break ("Comm Bk"). The Cmnd Stat indicator displays the status of commands sent from the remote terminal. When a sensor command is sent and acknowledged, this indicator displays "Sent" in 'green'. If the command is accepted by the sensor, the indicator displays "Accept". However, if a command is sent and acknowledged but it is not accepted, the Cmnd Stat indicator will display "Reject" or Cmnd Err".

2.3.3.3 Common Digitizer (CD-2) Modification Kit. The common digitizer (CD-2) modification kit is the device through which the Mode S Data Processing System (DPS) obtains radar and weather information via serial communications ports. The Mode S CD-2 interface CCA (MCDI), which is the mod. kit, will tap into the CD-2 system internal data bus via the control and interface module (CIM) backplane. Each Mode S CD-2 interface CCA (channel A and channel B) is a double width VME wire-wrapped card assembly, which is commonly referred to as a “large CD-2 module circuit card assembly”. The CD-2 interface board contains two 96-pin connectors, which plug into the CD-2 interface module (CIM) backplane. One connector (P1) interfaces to the general purpose interface board (GPIB) bus, the CD-2 interface, and clock. The second 96-pin connector (P2) interfaces with the radar line adapter (LARDR) of the Processor Subsystem (PRCSS) through the communications junction box (CJBX, unit 6). The MCDI CCA is based around a 68000 microprocessor unit (MPU), local memory, and various devices to service the MPU and handle the special functions associated with the MCDI CCA. The MCDI CCA handles the reformatting of messages and transmission of five message types to the Processor subsystem (PRCSS). These five message types are: (1) the CD-2 status, (2) search real time quality control (RTQC) information, (3) search strobe, (4) search target data, and (5) weather information. For the detailed and functional theory of operation of the CD-2 modification kit, the user must refer to the CD-2 Modification Kit Instructions, SSM-CD2-001.

2.3.3.4 Communication Junction Box (CJBX). The communication junction box (CJBX) is usually a wall mounted unit that forms the demarcation between Processor Subsystem (PRCSS) channels A and B and external FAA equipment (refer to Figure 2-39). Remote control, surveillance, communications, remote maintenance monitoring system (RMMS), remote programming support facility, and digitized primary radar will be connected via the communications junction box (CJBX). There are three types of communication junction boxes (units 6, 10, and 11). The primary communication junction box (unit 6) is used for communication/surveillance interface at each site configuration. If communication expansion is necessary at a site configuration, an additional communication junction box (unit 10) is required. If surveillance expansion is necessary at a site configuration, an additional surveillance junction box (unit 11) is required. Each CJBX is comprised of an interconnection section (cable connections) and, at some site configurations, an optional surveillance data selector (SDS) relay control. The interconnection section is a passive connection of PRCSS channels A and B cable connectors and a corresponding output of cable connectors, which provide for cable connections to ATC facilities. The surveillance data selector (SDS) control is a 24-pole, form C electromechanical relay used to select either the primary or Mode-S (PRCSS channel A or channel B) as the source of surveillance data. Each CJBX is capable of containing up to three SDS relays with each SDS capable of handling three surveillance links at an enroute site. The SDS is powered and controlled by a module contained in channel A and B PRCSS equipment racks.

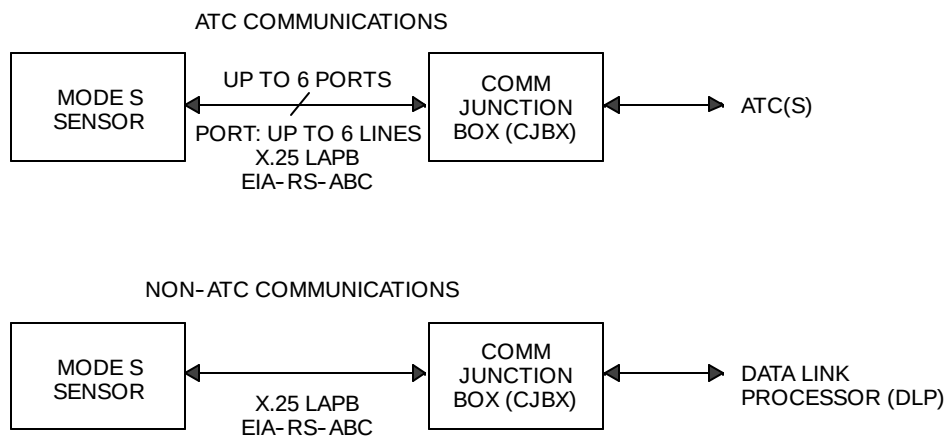


FIGURE 2-39. ATC AND NON-ATC COMMUNICATIONS INTERFACE BLOCK DIAGRAM

2.3.3.4.1 CJBX Connections. The communication junction box (CJBX) contains a series of 25-pin D connectors mounted on the top plate. The CJBX contains two types of interface circuits, those where signals from Mode S channels A and B are combined in the junction box (wire-ORed) and those where signals from each channel remain independent. The interfaces which are wire-ORed require three 25-pin connectors on the CJBX, two inputs with suffixes A and B, and an output with suffix OUT. The following list indicates the wire-OR connections and separate connections of the communications junction box (CJBX). Additional information concerning the circuit connection, the type of interface, number of connector pins, and the number of connections for PRCSS channel A, PRCSS channel B, and output is provided in this list.

Communication Junction Box (CJBX) Connections
(Wired-ORed Connections)

Circuit Connection	Interface	No. of Pins	PRCSS CH-A	PRCSS CH-B	OUT
Local Control	—	25	(not connected in this system configuration)		
Surveillance (DTE)	RS-ABC	25	6*	6*	6*
Communications (DTE)	RS-ABC	25	6*	6*	6*
Programming Support Facility (DTE)	—	25	(not connected in this system configuration)		
Remote Control (CJBX6 to Modem)	RS-ABC (multipoint interface)	25	1	1	1
Remote Maintenance Monitoring System (RMMS)	RS-ABC (multipoint interface)	25	1	1	1

NOTE

An * indicates there are six communication connectors and six surveillance connectors on CJBX, unit 6. On the communications expansion junction box (CJBX, unit 10) there can be up to 30 connectors (if required). On the surveillance expansion junction box (CJBX, unit 11) there can be up to 30 connectors (if required). Therefore, a maximum of 36 communications connectors can be configured with CJBX-6 and CJBX-10. Likewise, a maximum of 36 surveillance connectors can be configured with CJBX-6 and CJBX-11.

Communication Junction Box (CJBX) Connections
(Separate Connections)

Circuit Connection	Interface	No. of Pins	PRCSS CH –A	PRCSS CH –B	OUT
Digitized Primary Radar A (DCE)	RS –ABC	25	2	0	2
Digitized Primary Radar B (DCE)	RS –ABC	25	0	2	2
Surveillance Data Selector Connectors:					
SDS Control A	Power		1	0	0
SDS Control B	Power		0	1	0
SDS SR# (From CD–2)	RS –ABC	25	Up to 6 connections (if used)		
SDS OUT# (To ATC)	RS –ABC	25	Up to 6 connections (if used)		

2.3.3.4.1.1 The 25-pin connectors used are specified in the EIA–RS –ABC document. The type of connector (male or female) used on the CJBX is determined by its status as either the Data Circuit Terminating Equipment (DCE) or Data Terminal Equipment (DTE). The DCE connectors make use of female contacts in a male shell. The DTE connectors make use of male contacts in a female shell. The basic junction box (unit 6) supports up to six communication lines. If Mode S is required to support more than six lines, an additional communication expansion junction box must be added. The expansion box (unit 10) can support an additional 30 communication lines. Figure 2–40 shows the communications junction box (CJBX6) signal designations. The connectors used for communication lines are designated as follows:

COMM An Communication input channel A
 COMM Bn Communication input channel B
 COMM OUTn Communication output

NOTE

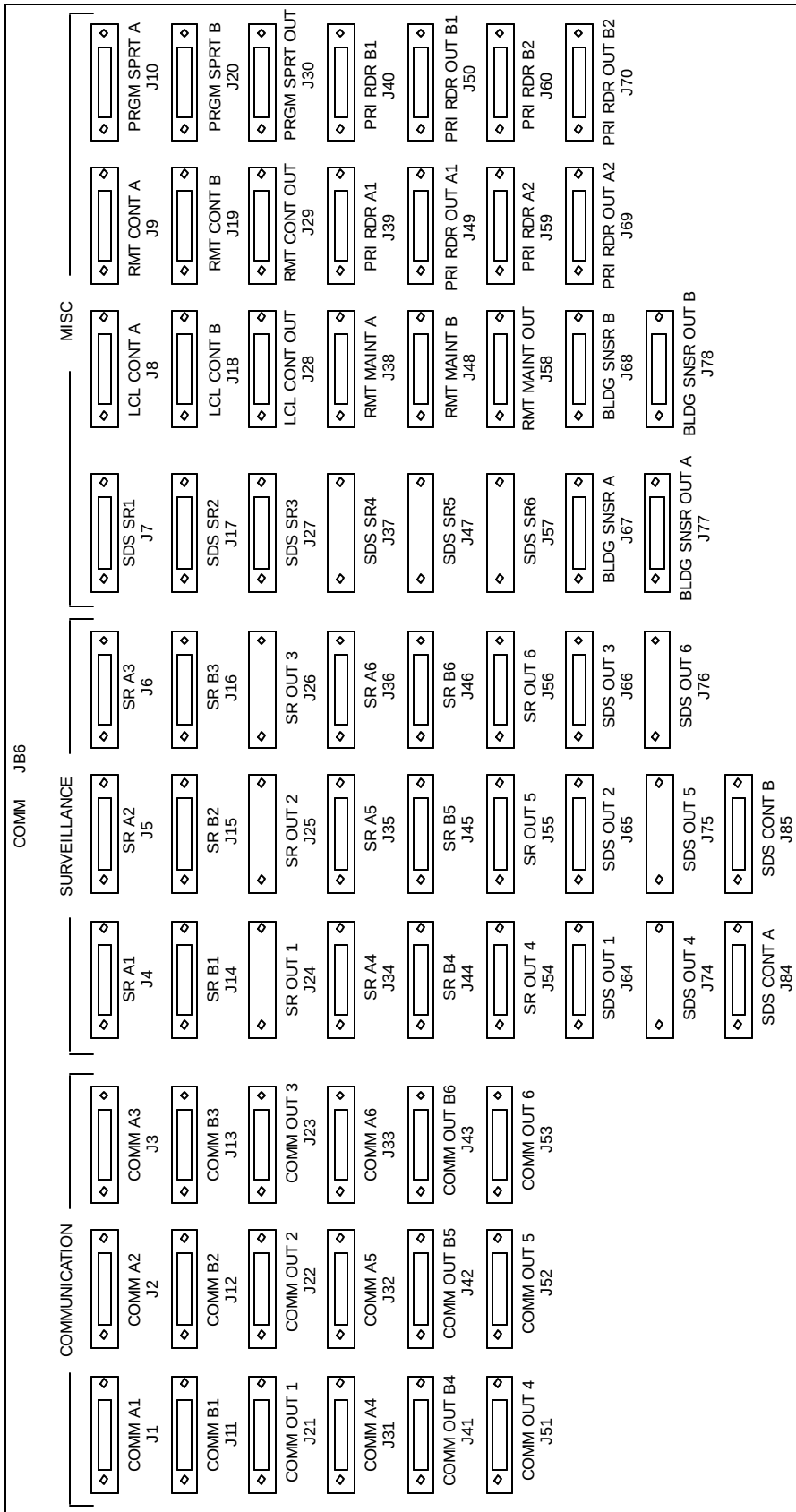
Where n = 1 to 6 in the primary CJBX (unit 6), and n = 7 to 36 in the expansion CJBX (unit 10).

The connectors used for surveillance lines are designated as follows:

SR An Surveillance input from channel A
 SR Bn Surveillance input from channel B
 SR OUTn Surveillance output to ATC

NOTE

Where n = 1 to 6 in the primary CJBX (unit 6), and n = 7 to 36 in the surveillance expansion (CJBX11 unit 11).



NOTE: J24, J25, J26, J37, J47, J57, J74, J75, AND J76
HAVE BLANKING PLATES IN THESE LOCATIONS

FIGURE 2-40. COMMUNICATION JUNCTION BOX (CJBX6) WITH SDS SIGNAL DESIGNATIONS (FRONT PANEL VIEW)

The connectors used for surveillance lines with SDS feature (at an en route site) are designated as follows (refer to Figures 2-40 and 2-42):

SDS SRn	Backup surveillance input from CD-2
SDS OUTn	Surveillance output to ATC
SDS CONT A	SDS power input channel A
SDS CONT B	SDS power input channel B

NOTE

Where n = 1 to 6 (if used). Blanking plates are used for SDS SR4 to SDS SR6, SDS OUT 4 to SDS OUT 6, and SR OUT 1 to SR OUT 3 in Figure 2-40.

The remaining miscellaneous connectors are designated as follows:

BLDG SNSR A BLDG SNSR B BLDG SNSR OUT A BLDG SNSR OUT B	Building sensor connectors are not used in this system configuration.
LCL CONT A LCL CONT B LCL CONT OUT	Local control connectors are not used in this system configuration.
RMT CONT A RMT CONT B RMT CONT OUT	Remote control input sent to channel A or channel B. Remote control input from remote terminal via modem to CJBX6.
PRGM SPRT A PRGM SPRT B PRGM SPRT OUT	Programming support facility connectors are not used in this system configuration.
RMT MAINT A RMT MAINT B RMT MAINT OUT	Remote Maintenance Monitoring System input sent to channel A or channel B. Remote Maintenance Monitoring System (RMMS) interface connector.
PRI RDR An PRI RDR Bn	Primary radar input sent to channel A. Primary radar input sent to channel B.
PRI RDR OUT An PRI RDR OUT Bn	Primary radar interface connector for channel A (CD-2 at en route site, ASR-# at terminal site). Primary radar interface connector for channel B (CD-2 at en route site, ASR-# at terminal site).

NOTE

Where n = 1 or 2.

2.3.3.4.1.2 The drive capability is sufficient for up to 200 feet of cable from the CJBX. The cable shall be 24 gauge shielded twisted pair with a characteristic impedance of 100 ohms + 10%. This guarantees the above electrical characteristics and drive capability. Cables from the CJBX to GFE modems are furnished

by the FAA. The following list provides the cable connector pin number, the circuit name, the RS–ABC interface mnemonic, and also the RS–232C mnemonic.

Pin No.	Circuit	RS–ABC Mnemonic	Pin No.	Circuit	RS–232C Mnemonic
1	Shield	—	1	Protective Ground	AA
2	Send Data	SD	2	Transmitted Data	BA
3	Receive Data	RD	3	Received Data	BB
4	Request to Send	RS	4	Request to Send	CA
5	Clear to Send	SC	5	Clear to Send	CB
6	Data Mode	DM	6	Data Set Ready	CC
7	Signal Ground	SG	7	Signal Ground (Common Return)	AB
8	Receiver Ready	RR	8	Received Line Signal Detector	CF
9	Receive Timing	RR	9	(Reserved for Data Set Testing)	—
10	Receiver Ready	RR	10	(Reserved for Data Set Testing)	—
11	Terminal Timing	TT	11	Unassigned	
12	Send Timing	ST	12	Sec. Received Line Sig. Detector	SCF
13	Clear to Send	CS	13	Sec. Clear to Send	SCB
14	Send Data	SD	14	Secondary Transmitted Data	SBA
15	Send Timing	ST	15	Transmission Signal Element Timing (DCE Source)	DB
16	Receive Data	RD	16	Secondary Received Data	SBB
17	Receive Timing	RT	17	Receiver Signal Element Timing (DCE Source)	DD
18	Local Loopback	LL	18	Unassigned	—
19	Request to Send	RS	19	Secondary Request to Send	SCA
20	—	—	20	Data Terminal Ready	CD
21	Remote Loopback	RL	21	Signal Quality Detector	CG
22	Data Mode	DM	22	Ring Indicator	CE
23	—	—	23	Data Signal Rate Selector (DTE/DCE Source)	CH/CI
24	Terminal Timing	TT	24	Transmit Signal Element Timing (DTE Source)	DA
25	Test Mode	TM	25	Unassigned	—

2.3.3.4.2 CJBX Input/Output Interface. Figure 2–41 provides a block diagram which shows the connections between the Processor subsystem (PRCSS) A and B cabinets and the communications junction box (CJBX – unit 6). Six connections are provided for each PRCSS A and B cabinet to the CJBX for the surveillance interface (SRA1 through SRA6 inputs, SRB1 through SRB6 inputs) and SROUT1 through SROUT6 surveillance outputs to the ATC. Six connections are provided for each PRCSS A and B cabinet to the CJBX for the communications interface (COMM A1 through COMM A6 inputs, COMM B1 through COMM B6 inputs) and COMM OUT1 through COMM OUT6 outputs to the ATC/non–ATC. The remaining connections are provided for each PRCSS A and B cabinet to the CJBX for the remote control interface (RMT CONT A, RMT CONT B) and RMT CONT OUT to the remote (IRCP) terminal; program support facility interface (PRGM SPRT A input, PRGM SPRT B input) and PRGM SPRT OUT are not used; remote maintenance monitoring system interface (RMT MAINT A input, RMT MAINT B input) and RMT MAINT OUT to the Maintenance Processor Subsystem (MPS); and the primary radar interface (PRI RDR A1 and A2 inputs, PRI RDR B1 and B2 inputs) and PRI RDR OUT A1 and A2 and PRI RDR OUT B1 and B2 to the respective radar interface equipments (i.e. CD–2/ASR–9). Connections for the PRCSS A and PRCSS B cabinets are also provided for the surveillance data selector (SDS) control (SDS CONT A and SDS CONT B), SDS backup surveillance lines SDS SR1 to SDS SR3 from the CD–2 at an en route site, and SDS OUT1 to SDS OUT3 lines to the ATC.

2.3.3.4.2.1 A block diagram of the primary communication junction box (CJBX6, unit 6) with the surveillance data selector (SDS) relay switch (en route site configuration) is shown in Figure 2–42. The surveillance data selector (SDS) switch allows surveillance data to be selected from either the surveillance circuits of the Processor subsystem (PRCSS A or PRCSS B), via the SR A1 through SR A6 or SR B1 through SR B6 surveillance data input connectors on CJBX6 or from the Common Digitizer (CD–2) backup surveillance lines, via the SDS SR1 through SDS SR3 input connectors on CJBX6. If SDS backup surveillance data is selected, the SDS surveillance data is sent to the Air Traffic Control (ATC) facility via the SDS OUT1 through SDS OUT3 connectors on CJBX6. Refer to communication junction box (CJBX6) with SDS connector diagram in Figure 2–40 for the SDS input and output connectors that are used. The input and output connectors that are not used on CJBX6 with SDS have blanking plates installed in those locations (SDS SR4 through SDS SR6, SDS OUT4 through SDS OUT6, and SR OUT1 through SR OUT 3). The SDS input and output connector locations with blanking plates are reserved for future expansion capability if SDS relays K4 through K6 are installed. Power (+24 VDC) to the SDS relays K1 through K3 is received from PRCSS cabinet A or B via the SDS CONT A or SDS CONT B connector (J84 or J85). For additional information concerning the surveillance data selector (SDS) control feature, the user must refer to Section 2 of the Mode S Data Processing System (DPS) instruction book TI 6365.3.

2.3.3.5 Power Subsystem. The DPS power subsystem has three phase input power supplied to the Processor subsystem (PRCSS) AC power control via an input power wall circuit breaker (power distribution panel). The three phase input power circuits to the PRCSS cabinet are protected from line transients and lightning surges by means of a surge suppressor circuit. Three phase input power is then sent to the AC power control cabinet which is located in the top front portion of the PRCSS cabinet next to the PRCSS status and alarm panel. This AC Power control circuit is comprised of an input power terminal board, a varistor circuit, primary power input circuit breakers, power on indicators, and an electromagnetic interference (EMI) line filter circuit.

2.3.3.5.1 This AC input power control enclosure is in the upper portion of the PRCSS cabinet. The three phase filtered output power from the AC input power control circuit is then sent to the terminal board in the PRCSS cabinet. The filtered three phase power is then applied to the four primary power supplies via its respective circuit breaker. Three phase power from the terminal board is also applied to the PRCSS cabinet blower circuit via a circuit breaker. If any of the power supply or blower circuit breakers are open, a CIRCUIT BREAKER OPEN indicator will light and the audible POWER ALARM will sound if the ENABLE/DISABLE toggle switch is set to position ENABLE. The POWER ALARM, CIRCUIT BREAKER OPEN indicator, and the ENABLE/DISABLE toggle switch are located on the PRCSS status and alarm panel, which is located in the top front portion (area A3) of the PRCSS cabinet next to the PRCSS Circuit breaker panel.

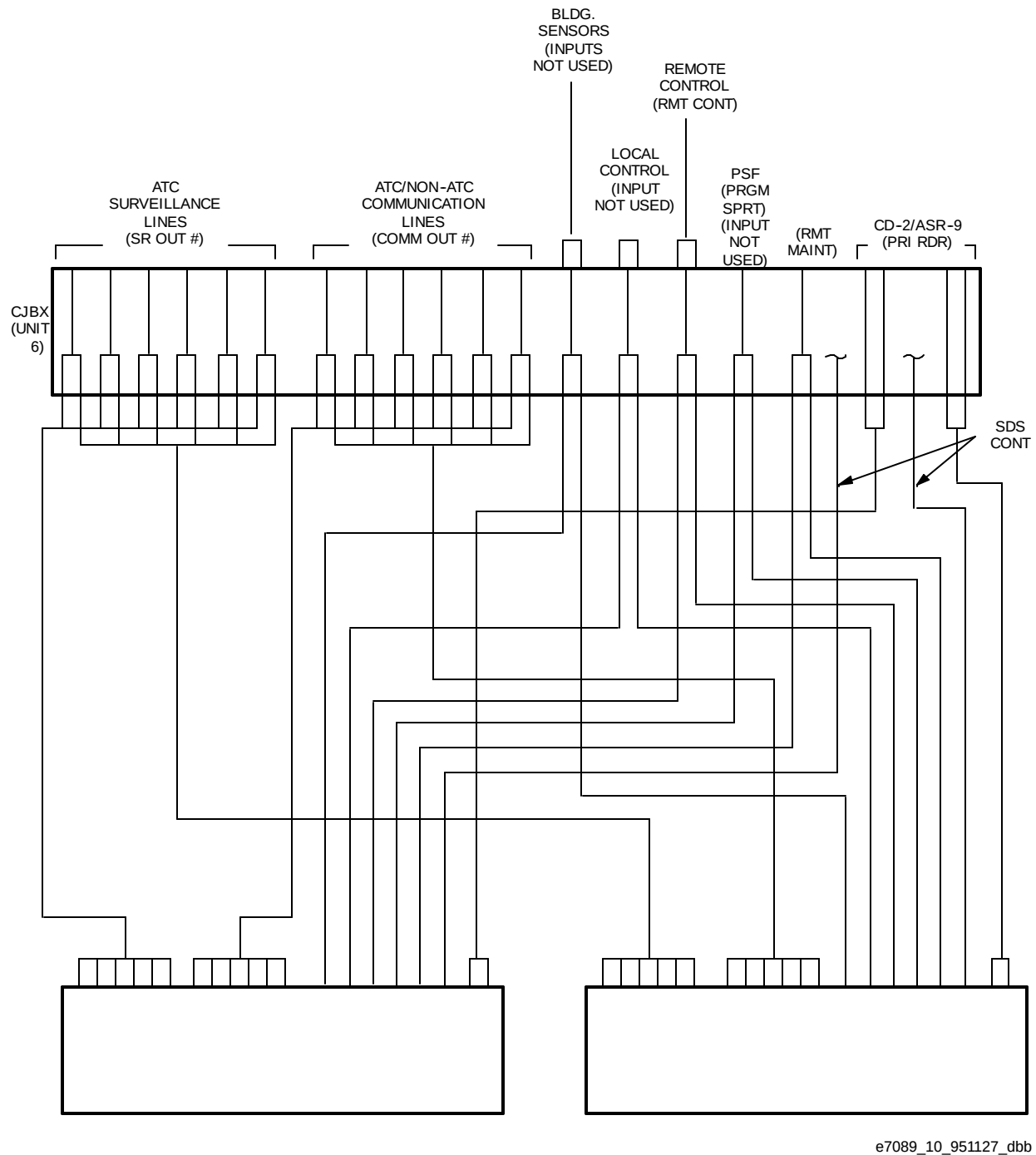
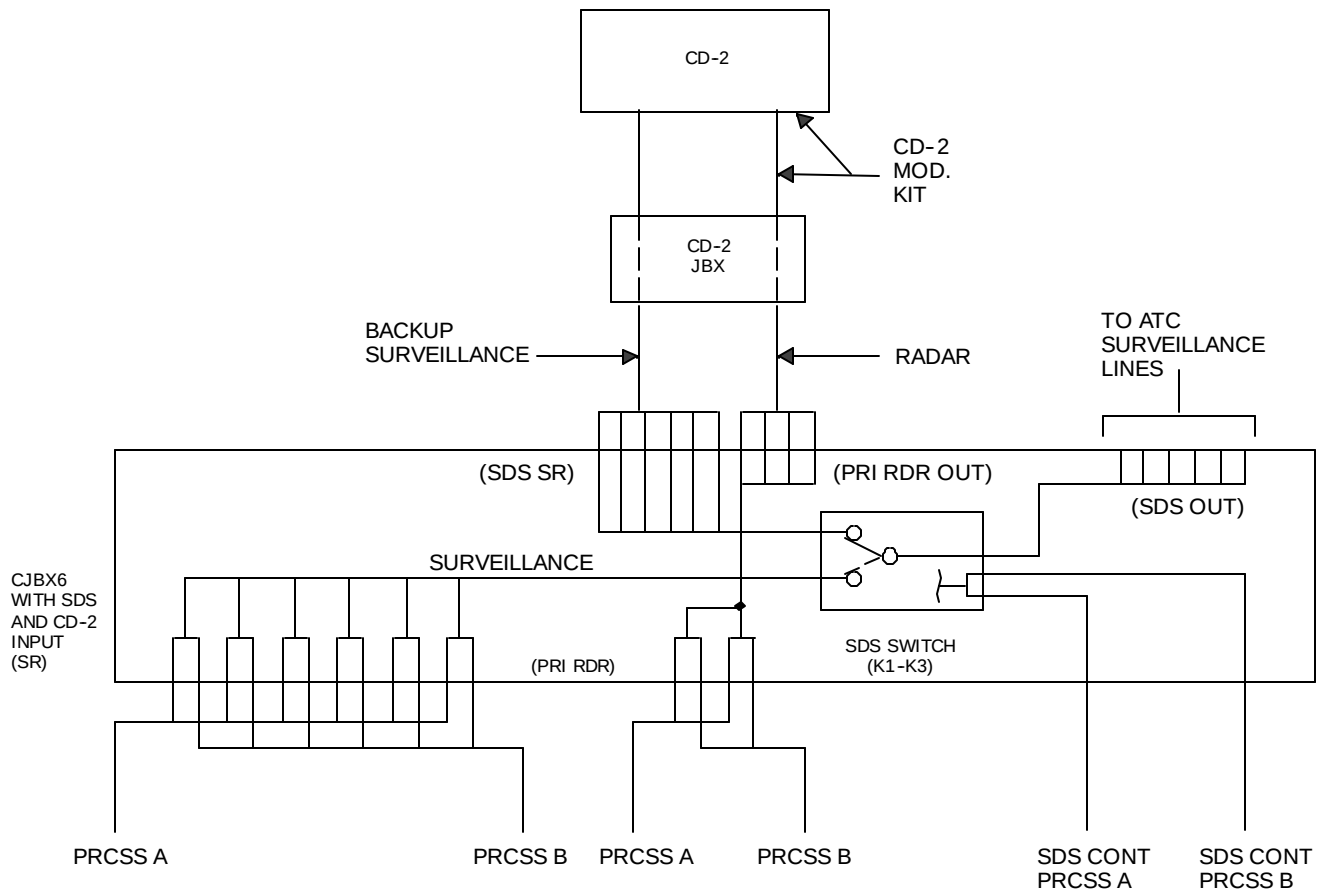


FIGURE 2-41. PRCS CABINETS INTERFACE WITH COMMUNICATIONS JUNCTION BOX (CJBX6) BLOCK DIAGRAM



NOTE:
THIS CONFIGURATION IS FOR AN ENROUTE SITE WITH COMM JUNCTION BOX 6 WITH SDS, AN SDS KIT, AND A CD-2 MOD KIT INSTALLED.

FIGURE 2-42. CJBX INTERFACE WITH COMMON DIGITIZER (CD-2) VIA SURVEILLANCE DATA SELECTOR (SDS) CONTROL BLOCK DIAGRAM

2.3.4 Calibration Performance Monitoring Equipment (CPME) Operation

The CPME is used as a test transponder for the Mode S system. The CPME is also used to generate the monopulse table within the Mode S system. The CPME provides a means for calibration and performance monitoring of various functions of the Mode S sensors with which the CPME is associated. The position of the CPME is recorded in its associated Mode S sensor and in adjacent sensors. Sensor interrogations elicit replies from the CPME. A variable reply delay in the CPME allows the CPME replay to be positioned in an area of little air traffic. Calibrated variable transmitter power provides sufficient signal for the Mode S sensor without the CPME radiating excessive levels. The CPME capability of being driven from an external signal source permits measurement of the Mode S monopulse operation over transponder reply frequencies. The CPME provides conventional ATRBS Mode 3/A and Mode C capabilities, military Mode 2 capability, and full Mode S capability. The Mode S modes of operation handled by the CPME include All-Call and surveillance uplink and downlink transmissions, single-segment message uplink (Comm-A) and downlink (Comm-B) communications transmissions, and extended length message uplink (Comm-C) and downlink (Comm-D) communications transactions. The CPME includes a simulated standard message interface, a simulated extended length message (ELM) interface, and a simulated altimeter input. These simulated interfaces provide Comm-B and Comm-D downlink transmissions in accordance with the protocols established by the U.S. National Aviation Standard for Mode S. Information concerning the operation of the CPME is presented under the following headings:

- a. Initial Acquisition
- b. Ground Initiated Comm-B Reply Messages
- c. Comm-A Message (Interrogation) — Comm-B Message (Reply)
- d. Comm-C Message (Interrogation) — Comm-D Message (Reply)
- e. CPME Status

2.3.4.1 Initial Acquisition. An initial acquisition check is performed at the beginning of Mode S sensor operation (startup condition) or following the interruption of CPME operation. For either of the aforementioned conditions, the CPME is unlocked to Mode S and, therefore, must be acquired by the Mode S sensor and placed on the roll call. For this operation, the CPME has a unique address for the particular site.

2.3.4.2 Ground Initiated Comm-B Replay Messages. Periodically, the Mode S sensor will elicit Comm-B replay messages from each CPME. The interrogation (Comm-A, identity request) will contain a reply request (RR) of 17 or 18. A value of RR = 18 requests the CPME to send its 48-bit call sign field in its replay (RR = 18, reply length = long, MB content = aircraft identification). The call sign occupies the 48-bit aircraft (flight) identification subfield (AIS/FIS) in bits 41 through 88 of the Comm-B message (MB) field in the Comm-B reply (refer to Figure 2-6 and 2-7 and Table 2-1). The CPME call sign is programmed by jumper wires on a plug-in assembly on the front panel of the receiver/transmitter unit.

2.3.4.2.1 The CPME has a series of fault indicators which are ORed together. These indicators determine the coded value put in the flight status (FS) field of all CPME surveillance and Comm-B replies (refer to Figure 2-7 and Table 2-1). If no error condition exists, the FS code is 1. If one or more error conditions (faults) exist, the FS field contains a value of 5 in bits 6 through 8. FS=5 indicates a special pulse identification (SPI) condition. If an FS=5 is received, an interrogation message with an RR=17 will be generated to elicit the CPME status. An interrogation with RR=17 requests that an extended capability report be included in the Comm-B message reply from the CPME. The CPME status report contains seven status information bits which are inserted into the Comm-A capability subfield of the Comm-B Message (MB field) in bits 45 through 51, as follows: (1) bit 45 – oscillator fault, (2) bit 46 – AC line power fault, (3) bit 47 – inside of CPME cabinet under temperature, (4) bit 48 – inside of CPME cabinet over temperature, (5)

bit 49 – transmitter RF source level degradation greater than 3 dB, (6) bit 50 – protocol (PC)=7 state, and (7) bit 51 – Mode S all-call lockout status. If any one of the aforementioned conditions is true, the respective bit is transmitted as a logic “1” in the CPME Comm-B message reply.

2.3.4.3 Comm-A Message (Interrogation) — Comm-B Message (Reply). The CPME, in response to a Comm-A message interrogation from the sensor, will participate in a loop performance check. The Comm-A message will be stored in the CPME’s standard message interface and, in response to the next sensor interrogation, the CPME will indicate that the stored message is waiting to be delivered. The CPME will deliver this identical message to the sensor as a Comm-B reply following a surveillance or Comm-A message authorization. The sensor will check to determine that the message received is the same as the message that was transmitted (loop performance check).

2.3.4.4 Comm-C Message (Interrogation) — Comm-D Message (Reply). The Comm-C message (interrogation) and Comm-D message (reply) are extended length messages (ELMs) which contain 112 bits of information in their message fields (refer to Figures 2–6 and 2–7). The Comm-C message from the sensor will be stored in the CPME’s extended length message (ELM) interface. The CPME in response to the next interrogation from the sensor will indicate that this message is waiting to be delivered. The CPME will then transmit this identical message to the sensor as a Comm-D message reply following a Comm-C message authorization. The sensor will check to determine that the Comm-D message that was received from the CPME is the same as the Comm-C message that was transmitted to the CPME. The resetting of the downlink request (DR) field will also be checked at this time. For additional information concerning an example of downlink ELM delivery and resetting of the downlink request (DR field) in a Comm-B reply, the user should refer to the example of a downlink ELM delivery in Table 2–13, last field DR=0 (downlink ELM closed out).

2.3.4.5 CPME Status. The CPME, in response to properly addressed interrogations from the sensor with a reply request field RR=17 in the Comm-A message, will respond with a Comm-B message reply (in the MB field, bits 33 through 88) formatted as an extended data link capability report. This format is determined by dividing the MB field into six subfields, as follows: (1) Comm-B data selector 1 (BDS1), (2) Comm-B data selector 2 (BDS2), (3) continuation subfield (CFS), (4) Comm-A capability subfield (ACS), (5) Comm-B capability subfield (BCS), and (6) Extended capability subfield (ECS).

<u>Subfield Mnemonic</u>	<u>MB Field Bit No.</u>	<u>Hex Code</u>
BDS1	33 ? 36	1
BDS2	37 ? 40	0
CFS	41 ? 44	0
ACS	45 ? 64	*
BCS	65 ? 80	8000
ECS	81 ? 88	FF

NOTE

An * indicates that the first seven bits of the ACS (bits 45 through 51) subfield are determined by the states of the CPME status. The remaining bits (52 through 64) are transmitted as logic “0”.

2.3.4.6 CPME Hardware Operation. The CPME is comprised of the following units: (1) battery box unit 1, (2) electrical equipment cabinet, (3) receiver/transmitter drawer assembly (2A1), (4) power supply drawer assembly (2A2), (5) hinged panel assembly (2A3), (6) EMI distribution box assembly (2A4), and (7) CPME antenna (unit 3). Refer to Figure 1–11 for the CPME relationship of units. For additional information and diagrams concerning the detailed theory of operation of the aforementioned CPME units, the user must refer to Section 2 of the Calibration Performance Monitoring Equipment (CPME) instruction book, TI 6365.8.

2.3.4.6.1 Battery Box Unit 1. The battery box (unit 1) is a white weatherproof metal box which houses four batteries. The batteries will provide 48 volts emergency power to the CPME for up to 2 hours; shutdown is automatic after the batteries have discharged by 50 percent of their ampere-hour rating. The battery box contains two heaters and two thermostats for environmental control.

2.3.4.6.2 Electrical Equipment Cabinet. The electrical equipment cabinet shown in Figure 1–11 houses the electronic components of the CPME. The electrical equipment cabinet contains the receiver/transmitter, power supply, EMI distribution box assembly, and a hinged panel assembly. The receiver/transmitter, power supply, EMI distribution box, and discrete components such as thermostats and circuit breakers are the lowest replaceable units of the electrical equipment cabinet. The electrical equipment cabinet is lockable and may be mounted on a concrete or asphalt pad, inside a building, or on the side of a building. The cabinet assembly is weatherproof and radio frequency (RF) shielded.

2.3.4.6.3 Receiver/Transmitter (2A1). Receiver/transmitter (2A1) shown in Figure 1–11 is a lowest replaceable unit (LRU). The receiver/transmitter contains the electronics for the CPME. The receiver/transmitter contains various assemblies as shown in Figure 1–12. Operation of all selectable functions is made via the front panel which also contains mounted lamps for indicated replies to Mode S or ATCRBS interrogations. The front panel RF SOURCE SELECT switch is used when an external continuous wave (CW) signal source is applied. The CW signal can be used to measure Mode S monopulse operation over transponder reply frequencies. The receiver/transmitter is rack mounted (on slides) in the electrical equipment cabinet. Information concerning Mode S operation and ATCRBS operation of the receiver/transmitter (2A1) is presented in the following text.

2.3.4.6.3.1 Mode S Operation. During Mode S operation, the CPME antenna (unit 3) receives the Mode S interrogation and the interrogation is sent to diplexer (A1) of the receiver/transmitter assembly. Diplexer (A1) separates the receive signal from the transmit signal (refer to Figure 1–12 for receiver/transmitter assemblies). The RCV PATH ATTEN (AT1) on the front panel assembly is adjusted so that a normal received signal does not saturate the IF amplifier assembly (A11). The IF amplifier assembly (A11) mixes the 1030 MHz receiver signal with the 1090 MHz local oscillator signal to produce a 60 MHz IF signal. The IF signal is amplified and passed to a detector/amplifier. One output of the detector/amplifier is a 60 MHz differential phase-shift keying (DPSK) signal that is applied to DPSK demodulator assembly (A7). The other output of the detector/amplifier is the detected IF pulse train. The detected IF pulse train is amplified and then processed by an amplitude comparator. The amplitude comparator output is applied to interrogation pulse decoder assembly (A4). Interrogation pulse decoder assembly (A4) generates the Mode S reply triggers. The Mode S reply triggers can be delayed by the setting of front panel EX REPLY DLY SEL switch (S1). The delayed Mode S reply triggers are applied to lockout logic and timers assembly (A6).

2.3.4.6.3.1.1 DPSK demodulator assembly (A7) detects the DPSK signal and passes the received Mode S data (DPSK data) to the Mode S decoder assembly (A9). The DPSK data is loaded into the uplink shift register. The output of the uplink shift register will be loaded into the comm A/B shift register or the extended length message (ELM) memory/assembler, depending on the type of interrogation. The comm A/B shift register output is applied to coding interface assembly (A2). The ELM memory/assembler output is applied to the Mode S encoder assembly (A3). The CPME call sign code is selected by call sign code select assembly (A20) and applied to the call sign shift register in coding interface assembly (A2). The CPME address code is selected by Mode S address code select assembly (A19) and is applied to the address shift register in coding interface assembly (A2). The CPME altitude code (AC) is selected by altitude code select assembly (A17) and is applied to the altitude shift register in coding interface assembly (A2). The CPME Mode A code is selected by the setting of front panel Mode A CODE SEL switch (S2) and is applied to the identity shift register in coding interface assembly (A2). The long/short message multiplexer combines the required data to develop a long or short message and applies the data to the altitude/identity multiplexer. The altitude/identity multiplexer inserts the altitude code or identity code into the message as applicable. The long/short message is applied to Mode S encoder assembly (A3) as the AC/ID/ADRS FIELD signal.

2.3.4.6.3.1.2 Mode S encoder assembly (A3) combines the message header information with the ELM data and develops the Mode S reply (MESSAGE OUT) signal. The MESSAGE OUT signal is applied to

lockout logic and timers assembly (A6). The MESSAGE OUT signal will be passed through the pulse position modulator when triggered by the Mode S reply triggers. The Mode S REPLY signal is applied to low-power amplifier assembly (A15) as modulation pulses. Low-power amplifier assembly (A15) has two outputs (low power and high power). The selection of low-power or high-power operation is determined by the particular CPME installation site requirements. The high-power or low-power mode is selected by the receiver/transmitter front panel POWER SELECT HIGH/LOW switch (S4). In the low-power mode, the output of low-power amplifier assembly (A15) is applied to band-pass filter (FL1). In the high-power mode, low-power amplifier (A15) output is applied to high-power amplifier assembly (A8). The output of high-power amplifier assembly (A8) is applied to band-pass filter (FL1). The RF is passed through band-pass filter (FL1) and applied to attenuators (AT3, AT4, and/or AT5) and transmit variable attenuator (AT2). XMT PATH ATTEN (AT2) is located on the receiver/transmitter front panel assembly. The attenuators attenuate the output of high-power amplifier assembly (A8) to the desired site level. The output of the attenuator(s) is fed to diplexer (A1). The output of diplexer (A1) is radiated by CPME antenna (unit 3).

2.3.4.6.3.2 ATCRBS Operation. During ATCRBS operation, the CPME antenna (unit 3) receives the ATCRBS interrogation which is sent to diplexer (A1) of the receiver/transmitter assembly. Diplexer (A1) separates the receive signal from the transmit signal. The RCV PATH ATTEN (AT1) on the front panel assembly is adjusted so that a normal received signal does not saturate the IF amplifier assembly (A11). The IF amplifier assembly (A11) demodulates the received ATCRBS interrogations and passes detected pulses to interrogation pulse decoder assembly (A4). The pulse train signal is a replica of the demodulation pulses. The ATCRBS trigger enable (ATCRBS TRIG ENBL) signal is applied to the trigger generator and enables the ATCRBS trigger generator. If the received interrogation rate is greater than the limit determined by the ATCRBS reply rate limit setting, the ATCRBS trigger generator will not be enabled 100 percent. The ATCRBS trigger generator, of the interrogation pulse decoder assembly (A4), will develop ATCRBS triggers at the pulse train rate when it is enabled by the ATCRBS trigger enable. The ATCRBS triggers are applied to the three mode latches (Mode 3/A, Mode C, and Mode 2). The spacing on the ATCRBS triggers determines which mode latch will be enabled. The ATCRBS trigger and the delayed gate time must be the same for the mode latch to be enabled.

2.3.4.6.3.2.1 The Mode 3/A reply code is determined by the receiver/transmitter front panel assembly Mode A CODE SEL switch (S2). The Mode 3/A code is applied to ATCRBS encoder assembly (A5) Mode 3/A shift register. The Mode 3/A reply code out of the shift register will be passed through the mode select gate if the Mode 3/A gate is enabled. The Mode C code is determined by jumpers on altitude code select assembly (A17). The Mode C reply code is applied to ATCRBS encoder assembly (A5) Mode C shift register. The Mode C reply code out of the shift register will be passed through the mode select gate if the Mode C gate is enabled. The Mode 2 code is determined by jumpers on Mode 2 code select assembly (A18). The Mode 2 reply code is applied to ATCRBS encoder assembly (A5) Mode 2 shift register. The Mode 2 reply code out of the shift register will be passed through the mode select gate if the Mode 2 gate is enabled. The selected reply code is passed through the mode select OR gate and applied to the 0.45-microsecond pulse generator of the ATCRBS encoder assembly (A5).

2.3.4.6.3.2.2 The pulse generator, of the ATCRBS encoder assembly (A4), will generate properly spaced modulation pulses. The pulse generator also drives the ATCRBS reply indicator driver, which lights the front panel RECEIVER TRANSMITTER STATUS ATCRBS REPLY indicator (DS1) when the CPME replies to an ATCRBS interrogation. The output of the pulse generator is applied to low-power amplifier assembly (A15) as modulation pulses. Low-power amplifier assembly (A15) has two outputs: low power and high power. The selection of low-power or high-power operation is determined by the particular CPME installation site requirements. The high-power or low-power mode is selected by the receiver/transmitter front panel POWER SELECT HIGH/LOW switch (S4). In the low-power mode, the output of low-power amplifier assembly (A15) is applied to band-pass filter (FL1). In the high-power mode, low-power assembly (A15) output is applied to high-power amplifier assembly (A8). The output of the high-power amplifier assembly (A8) is applied to band-pass filter (FL1). The radio frequency (RF) is passed through band-pass filter (FL1) and applied to attenuators (AT3, AT4, and/or AT5) and variable XMT PATH ATTEN (AT2) on the receiver/transmitter front panel. The attenuators attenuate the output of high-power amplifier assembly (A8) to the desired site level. The output of the attenuator(s) is fed to diplexer (A1). The output of the diplexer (A1) is radiated by CPME antenna (unit 3).

2.3.4.6.4 Power Supply (2A2). Power supply (2A2) shown in Figure 1–11 is a lowest replaceable unit (LRU). The power supply contains six assemblies (A1 through A6). The power supply provides operating voltages for the receiver/transmitter (2A1) and a charge voltage for the system backup batteries. The power supply also supplies operating voltages to the receiver/transmitter when AC input power is lost and the batteries are connected. The front panel contains indicator lamps (DS1 through DS7) for monitoring the power supply status and a TEST SELECTOR switch (S2) with test points (TP0 and TP1) for monitoring power supply critical voltages and current. The BCPS DC OUT circuit breaker (CB1), on the front panel of the power supply, protects the power supply circuits from an overload. CB1 opens at approximately 12 amperes. The DC–DC CONV circuit breaker (CB2), when in the off position, disconnects the DC–DC converter and external equipment from the DC line. A BATTERY ENABLE switch (S1), when in the one position, enables the battery shunt relay to connect the battery to the DC line. A 5 AMP fuse (F2) fuses battery current. The 1/2 AMP SLOW BLOW fuse (F1) fuses the 28 VAC output from transformer T1.

2.3.4.6.5 Hinged Panel Assembly (2A3). The hinged panel assembly shown in Figure 1–11 provides overcurrent protection from the CPME AC input lines. The AC input to the CPME power supply may be switched on or off (BCPS circuit breaker CB2) at the hinged panel assembly. The hinged panel assembly provides overcurrent protection and on/off switching of AC for the battery box heaters with a BATTERY HEATER circuit breaker (CB1). The hinged panel assembly has a CONVENIENCE OUTLET circuit breaker (CB3) for turning AC power on or off to the ground-fault interrupt AC CONVENIENCE OUTLET (J1) that is used for test equipment.

2.3.4.6.6 EMI Distribution Box Assembly (2A4). Filter (FL1) provides electromagnetic interference (EMI) filtering for the battery +48 volt line. Filter FL2 provides EMI filtering for the battery return line. Filter FL3 provides EMI filtering for the AC input power hot line. Filter FL4 provides EMI filtering for the AC neutral input power line. Filter FL5 provides EMI filtering for the battery heater AC power line. Resistors R1, R2, and R3, capacitor C1, and inductor L1 remove the power supply semiconductor-controlled rectifier (SCR) noise from the AC input power lines. Transient suppressor RV1 provides transient protection for the AC input power lines.

2.3.4.6.7 CPME Antenna (Unit 3). The CPME antenna shown in Figure 1–11 is a metal feed horn with an epoxy-fiberglass radome and hardware for mounting on a pole or the side of a building. The antenna measures 55.88 centimeters (22 inches) high, 72.39 centimeters (28.5 inches) wide, and 81.28 centimeters (32 inches) deep. The CPME antenna receives interrogations from the Mode S equipment and transmits replies to the Mode S equipment. The antenna is a horn type antenna with waveguide probe (J1) excitation.

2.3.5 Mode S Sensor-to-ATC Communications Link Messages

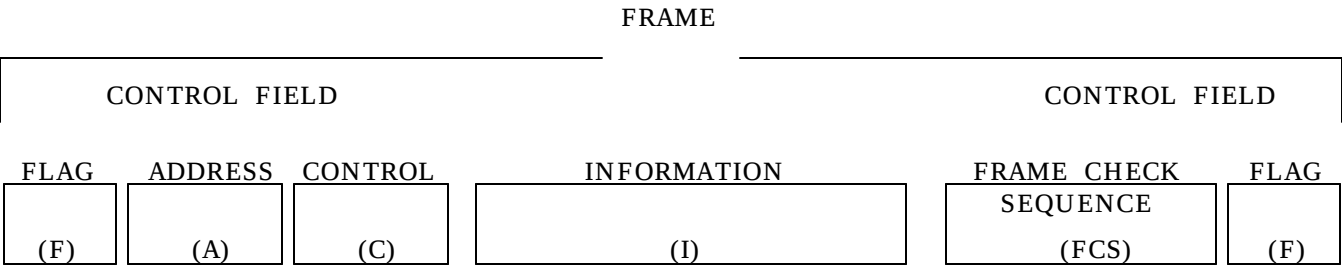
The Mode S Sensor-to-Air Traffic Control (ATC) communications link provides for the interchange of data link, status, and control messages between the Mode S Sensor and ATC facilities. The ATC facilities can send data to particular Mode S equipped aircraft by means of the Standard Uplink Message or ELM Uplink Message. The ATC facilities can request data from or about particular Mode S equipped aircraft by means of the request for Downlink Data Message, ATCRBS ID Request Message or Data Link Capability Request Message. Standard Uplink and ELM Uplink message types can be cancelled (Message Cancellation Request). The ATC facilities also can define the control state of Mode S equipped aircraft (Mode S Aircraft Control State Message), issue test messages, and signal failure or recovery of the ATC facility itself or of other sensors. The Mode S sensor can send data to the ATC facilities to indicate the outcome of received message (message rejection/delay notice, Uplink Delivery Notice, and/or Test Response) and provide data on Mode S equipped aircraft (Standard Downlink, ELM Downlink, Data Link Capability, and/or ATCRBS ID Code messages). The Mode S sensor can also send data to the ATC facilities which indicate signal conditions in the Mode S sensor (such as, the Track Alert Message or the Status Message).

2.3.5.1 Mode S Sensor/ATC Communications Link. This communications link also carries messages from one Mode S sensor to another Mode S sensor via an ATC facility. These messages provide for exchange of tracking data for both Mode S and Air Traffic Control Radar Beacon System (ATCRBS) equipped aircraft.

Messages of this type include Data Start, Data Stop, Data Request, Track Data, Cancel Request, ATCRBS Data Start, ATCRBS Data Stop, ATCRBS Data Request, ATCRBS Track Data, and ATCRBS Cancel Request. Messages are provided for handoff of Mode S equipped aircraft (Primary Coordination Message), and as a warning of duplicate Mode S IDs (Track Alert message). Status messages provided include Status Message, Adjacent Sensor Status Request, and Adjacent Sensor Status Response. The Mode S sensor to ATC communications interface provides up to six full duplex data links from the Mode S sensor to ATC facilities. Each data link is comprised of up to six full duplex data line EIA–STD–RS–ABC interfaces. Each data line interface supports one Government Furnished Equipment (GFE) modem at up to 9600 bits per second data rate. All six data lines from one data link operate at the same data rate, if in use. Output data messages will be equally distributed among the modems that are in use. Information concerning Mode S Sensor/ATC communication link messages is presented in the following order:

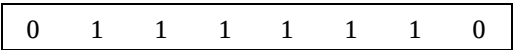
- a. Mode S Sensor-to-ATC Message Protocol
- b. Message Types

2.3.5.1.1 Mode S Sensor-to-ATC Message Protocol. Communication messages transmitted between the Mode S sensor and ATC facilities conform to CCITT X.25–1984, level 2, Link Access Procedure Balanced (LAPB), for a balanced, point-to-point, two-way interchange. All transmissions in LAPB are composed of frames; the basic frame structure is as follows:



Messages are comprised of two separate levels, an outer Frame Format level and an inner Information Field level. The Frame Format is the outer-level message structure needed to carry out transfer procedures on the link: the transmission of a message to a receiving station; the testing of that message by the receiving interface to determine its acceptability; the issuance of an acceptance or rejection response by the recipient; and further action (in case of a rejection or no response) by the transmitting station. Acceptance and rejection criteria are based on the Frame Format requirements which are tested by the interface of the receiving station, and not on an inner-level interpretation of message contents. Each frame carries several fixed-length fields of control data relating to interface functions. These include the address of the receiving station, a control field, a frame check sequence (error detection code), and flags to denote the beginning and end of the frame. The variable-length Information Field (I) becomes the inner-level message when the interface of a receiving status has accepted a message and stripped off the control fields.

2.3.5.1.1.1 The basic unit of transmission is the frame, which is a bit sequence containing at least 32 bits between flags (eight address, eight control, and 16 frame check sequence bits). Each message between a Mode S Sensor and an ATC facility can be accommodated within one frame. Transmissions conform to the structure: F, A, C, I, FCS, and F (except for a data link control sequence where there is no information field). The Flag character (F) at the beginning and end of every message frame is the standard eight-bit sequence as follows:



The address field (A) consists of one octet (eight bits). Coding for the address field is defined in CCITT X.25. The basic control field (C) length consists of one octet (eight bits). Control field coding is defined in CCITT X.25. The information field consists of an integer number of octets. Information field formats are defined in the following text. All frames include a 16-bit frame check sequence (FCS) just prior to the closing flag for error detection purposes. The frame check sequence is defined in CCITT X.25.

2.3.5.1.2 Message Types. The following Mode S Sensor/ATC messages cross the communications interfaces. These message types are grouped accordingly to the sender and the receiver of the message and according to the contents of the Information Field (I). These message types are grouped as follows: (1) ATC-to-Mode S messages, (2) Mode S-to-ATC messages, and (3) Sensor-to-Sensor messages. Refer to Table 2–14 for information field type codes for Mode S Sensor/ATC messages.

2.3.5.1.2.1 ATC-To-Mode S Messages. The ATC-to-Mode S communication link messages are divided into two categories: (1) ATC-to-Mode S Uplink Messages and (2) ATC-to-Mode S Status/Control Messages. Information concerning ATC-to-Mode S message data blocks is provided in Table 2–15.

a. ATC-to-Mode S Uplink Messages

1. Standard Uplink Message (to a Mode S aircraft).
2. ELM Uplink Message (to a Mode S aircraft).
3. Request for Downlink Data (from a Mode S aircraft).
4. ATCRBS ID Request (to a Mode S aircraft).
5. Message Cancellation Request (of a previous Uplink Message).

NOTE

The term “Standard Uplink” refers to a message sent to an aircraft by one or more Comm-A transmissions.

STANDARD UPLINK MESSAGE

0010	0001	MODE S ADDRESS	MSG NO.	P	EXP	PT	SC	SP	MA (REPEATED)
1	8	32	40	44	47	48	50	56	56 + 56(SC + 1)

ELM UPLINK MESSAGE

0010	0010	MODE S ADDRESS	MSG NO.	P	EXP	SP	ELM LENGTH	ELM TEXT (MAX 5120)
1	8	32	40	44	47	50	56	56 + 80 (ELM LENGTH + 1)

REQUEST FOR DOWNLINK DATA MESSAGE

0010	0011	MODE S ADDRESS	MSG NO.	P	EXP	SP	BDS1	BDS2
1	8	32	40	44	47	48	52	56

ATCRBS ID REQUEST MESSAGE

0010	0100	MODE S ADDRESS	MSG NO.
1	8	32	40

MESSAGE CANCELLATION REQUEST MESSAGE

0010	0101	MODE S ADDRESS	MSG NO.	REF MSG NO.	REF TYPE CODE
1	8	32	40	48	56

b. ATC-to-Mode S Status/Control Messages

1. Test Message (to a Mode S sensor).
2. ATC Failure/Recovery Message (to a Mode sensor).
3. Data Link Capability Request.
4. Sensor/Failure Recovery Message.
5. Mode S Aircraft Control State Message.

TEST MESSAGE

0110	0001	MSG NO.	TEST DATA
1	8	16	64

ATC FAILURE/RECOVERY MESSAGE

1001	1001	MSG NO.	STATE	SP
1	8	16	18	24

DATA LINK CAPABILITY REQUEST MESSAGE

0000	0010	MODE S ADDRESS	MSG NO.
1	8	32	40

SENSOR FAILURE/RECOVERY MESSAGE

0110	0101	MSG NO.	SID	SP	SSTAT
1	8	16	26	30	32

MODE S AIRCRAFT CONTROL STATE MESSAGE

1001	1010	MSG NO.	IP	IS	IU	MODE S ADDRESS (REPEATED)
1	8	16	19	22	24	24+24 (IP + IS + IU)

TABLE 2–14. INFORMATION FIELD (I) TYPE CODES FOR MODE S SENSOR/ATC MESSAGES

From/To	Message Type	Prefix	Suffix	Description
ATC-to-Mode S	Uplink Message	0010	0001	Standard Uplink
		0010	0010	ELM Uplink
		0010	0011	Request for Downlink Data
		0010	0100	ATCRBS ID Request
		0010	0101	Message Cancellation Request
	Status/Control Messages	0110	0001	Test
		1001	1001	ATC Failure/Recovery
		0000	0010	Data Link Capability Request
		0110	0101	Sensor Failure/Recovery Message
		1001	1010	Mode S Aircraft Control State Message
Mode S-to-ATC	Sensor Response Messages	0011	0001	Message Rejection/Delay Notice
		0011	0010	Uplink Delivery Notice
	Downlink Messages	0100	0001	Pilot Downlink
		0100	0100	Ground Initiated Downlink
		0100	0000	Broadcast Downlink
		0100	0010	ELM Downlink
		0011	0100	Data Link Capability
		0100	0101	ATCRBS ID Code
	Sensor Performance/Status Messages	0110	0010	Test Response
		0110	0100	Status Message
		0110	1111	Sensor Recovery Notice Message
		1001	1100	Track Alert Message
Sensor-to-Sensor	Mode S Network Control Messages	1001	0001	Data Start
		1001	0010	Data Stop
		1001	0011	Data Request
		1001	0100	Track Data
		1001	0101	Cancel Request
		1001	1101	Primary Coordination
		1001	1110	Track Alert Message
	ATCRBS Network Control Messages	1101	0001	ATCRBS Data Start
		1101	0010	ATCRBS Data Stop
		1101	0011	ATCRBS Data Request
		1101	0100	ATCRBS Track Data
		1101	0101	ATCRBS Cancel Request

TABLE 2-15. ATC-TO-MODE S MESSAGE DATA BLOCKS

Data Block	Description																		
Type Code	The Type Code block is an eight-bit block (bits 1 through 8) that begins every Information Field (I).																		
MSG No.	Message number is an eight-bit integer which numbers all types of messages concerning a particular Mode S target sent by a particular facility. This field is also kept as an unambiguous number in messages not concerning any one aircraft (like type codes 0110 0010 and 1001 1100). ATC facilities will only assign message numbers zero to 127. Zero is excluded so that 127 messages about each aircraft and 127 messages not about any one aircraft from ATC, can be distinguished at a time. When a message that has this field is used as a response to a request, the message number assigned in the request is placed in this field. When the message is not used as a response, a message number is assigned by the sender.																		
MODE S ADDRESS	The Mode S address is the unique 24-bit identification code of a Mode S equipped aircraft.																		
P	Priority (P) is a four-bit data block which provides a user-supplied priority tag (for Uplink Messages only). This block contains a binary integer, with 0 being the lowest and 15 the highest priority. Delivery of every Uplink Message is scheduled as promptly as possible, regardless of the priority value; only if a queue of messages exists for a particular aircraft is a message given precedent over an earlier message with lower priority.																		
EXP	Expiration (EXP) is a three-bit data block encoding "Time to Expiration" of a message (for Uplink Messages only). This block contains a binary integer, with the values 0 through 7 representing the number of scans for which delivery should be attempted, encoded as shown in the following list.																		
	<table> <tr> <th><u>EXP Value</u></th><th><u>Time To Expiration (In Scan Periods)</u></th></tr> <tr> <td>0</td><td>Default Value</td></tr> <tr> <td>1</td><td>1</td></tr> <tr> <td>2</td><td>2</td></tr> <tr> <td>3</td><td>4</td></tr> <tr> <td>4</td><td>8</td></tr> <tr> <td>5</td><td>16</td></tr> <tr> <td>6</td><td>32</td></tr> <tr> <td>7</td><td>No Expiration</td></tr> </table>	<u>EXP Value</u>	<u>Time To Expiration (In Scan Periods)</u>	0	Default Value	1	1	2	2	3	4	4	8	5	16	6	32	7	No Expiration
<u>EXP Value</u>	<u>Time To Expiration (In Scan Periods)</u>																		
0	Default Value																		
1	1																		
2	2																		
3	4																		
4	8																		
5	16																		
6	32																		
7	No Expiration																		
PT	Packet Type (PT) is a one-bit block present only in a Standard Uplink Message. This PT block serves to characterize the message for communication system purpose.																		
SC	Segment Count (SC) is a two-bit integer block present in a Pilot Uplink or a Standard Downlink Message. The SC value is one less than the number of message text segments comprising the complete message. Values of SC greater than zero correspond to "linked Comm-A or Comm-B messages".																		

TABLE 2–15. ATC–TO–MODE S MESSAGE DATA BLOCKS (Continued)

Data Block	Description
SP	The Square (SP) field represents the needed spacing required in some messages in order to minimize shifting in the Mode S sensor. This data block will be set equal to zero.
MA	Comm-A Message Text (MA) is a 56-bit block present in Standard Uplink Messages. A message may contain up to four MA blocks, as specified by the value of SC.
ELM LENGTH or LENGTH	ELM Length or Length is a six-bit block present only in ELM messages. These data blocks contain binary integers specifying the number of 80-bit segments comprising the ELM Text block. This data block value is one less than the number of segments. The values of ELM length are 1 (2 segments) to 63 (64 segments) in the ELM Uplink message. The values of length are 0 (1 segment) to 15 (16 segments) in the ELM Downlink Message.
NOTE	
An ELM Length value greater than 15 in an ELM Uplink message defines a set of linked ELMs. Processing of such a message to support the linking protocol is a task of the Data Link Processing function of the Mode S sensor.	
ELM TEXT	ELM Text is a variable-length ELM data field. For an ELM Uplink message, the length is a multiple of 80 bits with a maximum of 5120 bits, corresponding to a sequence of four linked ELMs, each comprised of 16 consecutive Comm-C ground-to-air transmissions. For an ELM Downlink message, the length is a multiple of 80 bits with a maximum of 1280 bits, corresponding to a sequence of 16 consecutive Comm-D air-to-ground transmissions. Each transmission carries an 80-bit field of text.
BDS1 and BDS2	Comm B data selectors (BDS1 and BDS2) are two 4-bit data blocks present only in a Request for Downlink Data. Together BDS1 and BDS2 identify the particular 56-bit data register within the Mode S transponder that is to be read out and the data contents downlinked using a ground-initiated Comm-B message. BDS1=0001 and BDS2=0000 or 0001 signifies a request for an extended capability report from an airborne Mode S installation.
REF MSG NO.	Referenced Message Number is an eight-bit binary integer present only in messages that are sent in response to requests or that are in reference to other messages. In a message cancellation request, the REF MSG NO. represents the message number of the message whose cancellation is being requested.
REF TYPE CODE	Referenced Type Code is the eight-bit Type Code block corresponding to a message whose cancellation is being requested.
TEST DATA	Test Data is a 48-bit block present in a Test Message. The contents may be defined by the originator. Each message may be different.

TABLE 2-15. ATC-TO-MODE S MESSAGE DATA BLOCKS (Continued)

Data Block	Description
STATE	STATE is a two-bit data block occurring only in an ATC Failure/Recovery Message. This STATE data block is used to inform a Mode S sensor of a change in the operational status of an ATC facility: (1) a “00” indicates not used, (2) a “01” indicates a failure condition, (3) a “10” indicates a recovery from a failure condition, and (4) a “11” indicates a recovery from a failure condition and a loss of data base.
SID	The Sensor ID field is a 10-bit field that uniquely identifies the Mode S sensor whose status is being reported.
SSTAT	The two-bit Sensor Status (SSTAT) as reported by the ATC facility is interpreted as follows: (1) “00” indicates not used, (2) a “01” indicates Sensor not failed, (3) a “10” indicates Sensor failed, and (4) a “11” indicates Sensor communications failed
IP	Primary Index is a three-bit binary integer present in a Mode S Aircraft Control State Message. The Primary Index (IP) value is the number of Mode S addresses in the message representing controlled aircraft for which the Sensor is being assigned primary. If IP is not zero, the corresponding Mode S addresses occupy bit 25 through bits 24+24 (IP) of the message.
IS	Secondary Index is a three-bit binary integer present in a Mode S Aircraft Control State Message. The IS value is the number of Mode S addresses in the message representing controlled aircraft for which the sensor is being assigned secondary. If IS is not zero, the corresponding Mode S addresses occupy bit 25+24 (IP) through bits 24+24 (IP+IS) of the message.
IU	Uncontrolled Index is a two-bit binary integer present in a Mode S Aircraft Control State Message. The IU value is the number of Mode S addresses in the message representing uncontrolled aircraft. If IU is not zero, the corresponding Mode S addresses occupy bit 25+24 (IP+IS) through bits 24+24 (IP+IS+IU) of the message.

NOTE

The maximum length of the Mode S Aircraft Control State Message, corresponding to IP=7, IS=7, and IU=3 is 432 bits.

For example:	Bits 1 through 24	=	24 bits
	+ IP = 7 X 24	=	168 bits
	+ IS = 7 X 24	=	168 bits
	+ IU = 3 X 24	=	72 bits
	Total	=	432 bits

2.3.5.1.2.2 Mode S-to-ATC Messages. The Mode S-to-ATC communication link messages are divided into three categories: (1) Sensor Response Messages, (2) Downlink Messages, and (3) Sensor Performance/Status Messages. Information concerning Mode S-to-ATC message data blocks is provided in Table 2–16.

1. Mode S-to-ATC Sensor Response Messages
 - a. Message Rejection/Delay Notice (with respect to a Mode S aircraft).
 - b. Uplink Delivery Notice (with respect to a previous Uplink Message).

MESSAGE REJECTION/DELAY NOTICE MESSAGE

0011	0001	MODE S ADDRESS	REF MSG NO.	QUAL	SP
1	8	32	40	43	48

UPLINK DELIVERY NOTICE MESSAGE

0011	0010	MODE S ADDRESS	REF MSG NO.	DI	SP
1	8	32	40	41	48

2. Mode S-To-ATC Downlink Messages (Each Message From A Mode S Aircraft)
 - a. Pilot Downlink Message (see following note).
 - b. Ground Initiated Downlink Message.
 - c. Broadcast Downlink Message (see following note).
 - d. ELM Downlink Message.
 - e. Data Link Capability Message.
 - f. ATCRBS ID Code Message.

NOTE

The terms “Pilot Downlink” and “Broadcast Downlink” refer to a message received from an aircraft by one or more Comm-B transmissions.

PILOT DOWNLINK MESSAGE

0100	0001	MODE S ADDRESS	MSG NO.	SC	SP	MB (REPEATED)
1	8	32	40	42	48	104 + 56 (SC)

GROUND-INITIATED DOWNLINK MESSAGE

0100	0100	MODE S ADDRESS	MSG NO.	BDS1	BDS2	MB
1	8	32	40	44	48	104

BROADCAST DOWNLINK MESSAGE

0100	0000	MODE S ADDRESS	MSG NO.	MB
1	8	32	40	96

ELM DOWNLINK MESSAGE

0100	0010	MODE S ADDRESS	MSG NO.	SP	LENGTH	ELM TEXT (MAX 1280)
1	8	32	40	42	48	48 + 80 (LENGTH + 1)

DATA LINK CAPABILITY MESSAGE

0011	0100	MODE S ADDRESS	MSG NO.	SP	CA	ECA
1	8	32	40	41	44	96

ATCRBS ID CODE MESSAGE

0100	0101	MODE S ADDRESS	MSG NO.	SP	ATCRBS ID
1	8	32	40	44	56

3. Mode S-to-ATC Sensor Performance/Status Messages
 - a. Test Response Message (from a Mode S sensor).
 - b. Status Message.
 - c. Sensor Recovery Notice Message.
 - d. Track Alert Message (from a Mode S sensor on a given target).

TEST RESPONSE MESSAGE

0110	0010	MSG NO.	TEST RESPONSE
1	8	16	64

STATUS MESSAGE

0110	0100	NO. OF MODE S TRACKS	NO. OF ATCRBS TRACKS	NO. OF RADAR-ONLY TRACKS	SP	NEW CODE FLAG (N)	CODE OVERFLOW FLAG (V)	NORTHMARK TIME
1	8	18	28	38	42	43	44	58

STATUS MESSAGE (CONT.)

SENSOR ID	TARGET OVERLOAD	SENSOR STATUS	CHANNEL IN USE (CH)	NO. OF BCN STROBE WORDS = 2(i)	NO. OF CW INTERFERENCE WORDS = 2(m)	NO.OF RED CODES = k	NO.OF YELLOW CODES = j
68	69	71	72	80	88	96	104

STATUS MESSAGE (CONT.)

BORESIGHT AZ OF BEGINNING OF STROBE OVERLOAD = $B_S(1)$	BORESIGHT AZ OF END OF STROBE OVERLOAD = $B_E(1)$	$B_S(2)$	$B_E(2)$	...	$B_S(i)$	$B_E(i)$
120	136					
104 + 32(i)						

STATUS MESSAGE (CONT.)

BORESIGHT AZ OF BEGINNING OF CW INTERFERENCE = $C_S(1)$	BORESIGHT AZ OF END OF CW INTERFERENCE = $C_E(1)$	$C_S(2)$	$C_E(2)$	...	$C_S(m)$	$C_E(m)$
104 + 32(i + m)						

STATUS MESSAGE (CONT.)

YELLOW CONDITION CODE Y(1)	Y(2)	...	Y(i)	RED CONDITION CODE R(1)	R(2)	...	R(k)
104 + 16(2i + 2m + j)				104 + 16(2i + 2m + j + k)			

SENSOR RECOVERY NOTICE MESSAGE

0110	1111	SENSOR ID	RECOVERY TYPE
1	8	18	24

TRACK ALERT MESSAGE

1001	1100	MSG NO.	MODE S ADDRESS	RANGE 1	SP	AZIMUTH 1	RANGE 2	SP	AZIMUTH 1
1	8	16	40	56	58	72	88	90	104

TABLE 2–16. MODE S–TO–ATC MESSAGE DATA BLOCKS

Data Block	Description
	NOTE Information concerning Mode S-to-ATC message data blocks, such as: Type Code, Mode S Address, Reference Message Number, Segment Count (SC), Length, ELM Text, Comm B Data Selectors (BDS1 and BDS2), Sensor ID (SID), and Spare are defined in previous text for ATC-To-Mode S Message Data Blocks.
MSG No.	Output messages for the ATC shall contain 8-bit message number fields. These fields shall be set as follows: 1. For message which are responses to uplink messages (rejection and delivery notices), the field contains a Referenced Message Number. The field shall be set equal to the value in the message number field in the corresponding Uplink Message. The message number field shall also be set in this manner for the following types of messages: tactical downlink and tactical downlink with position (when generated in response to a request for downlink data), test response, aircraft state, aircraft position, ATCRBS ID Code (when generated in response to an ATCRBS ID request), and data link capability (when generated in response to a capability request). 2. For other messages referring to a Mode S aircraft (except Track Alert), the message number shall be set to 128. 3. For all other messages containing a message number field (i.e., Track Alert and messages without a Mode S address field) the message number shall be set to 128.
QUAL	The Qualifier is a three-bit block present in a Message Rejection/Delay Notice only, with the following coding: 1. A “000” indicates target not on file (rejection). 2. A “001” indicates target not in roll-call mode (delay). 3. A “010” indicates Active Message List (AML) sublist overflow (rejection). 4. A “011” indicates Sensor not primary (rejection of ELM uplink). 5. A “100” indicates targets lack ELM capability (rejection).

NOTE

Other code values are not assigned.

TABLE 2–16. MODE S–TO–ATC MESSAGE DATA BLOCKS (Continued)

Data Block	Description
DI	The Delivery Indicator (DI) is a one-bit block present in a Message Delivery Notice only. The DI datablock reports on the success of a referenced Uplink Message, with the following coding: (1) a “0” indicates that the message was successfully delivered and (2) a “1” indicates that the message expired and was not delivered.
MB	The Comm-B Message Text (MB) field is a 56-bit block used for Downlink Message text similar to the Uplink MA block. A Pilot Downlink Message may contain up to four MB blocks as specified by the value of SC. Broadcast and Ground Initiated Comm-B messages each contain only a single MB block. Coding of MB is not constrained by the Mode S sensor but must conform to the draft Minimum Operational Performance Standard for Mode S Data Link Processor. The MB for Pilot Downlink Messages begins with a two-bit Linked Comm-B Subfield (LBS) that is used to support linking of MB message components. In a linked Comm-B message, the LBS is present only in the first component of the linked message.
CA and ECA	Capability is a 55-bit block that indicates the input/output capability of the aircraft. The Capability function is divided into two subfields defined as follows: 1. CA (Basic Capability) — This field is the three-bit CA field contained in Mode S All-Call replies. 2. ECA (Extended Capability) — This 52-bit field contains the data link capability when the CA field so indicates. The ECA field coding is supported by the data link avionics and the ground ATC and non-ATC users, and is transparent to the Mode S sensor.
NOTE	
The leading four-bits of the MB are not assigned and the final 52-bits form the ECA.	
ATCRBS ID	ATCRBS ID is a 12-bit block which contains the 4096-code value from a Mode S transponder. The coding is the standard Mode 3/A format used in Surveillance Reports on ATCRBS targets.
TEST RESPONSE	Test Response data is a 48-bit block present in a Test Response Message. The data must be an exact duplicate of the test data contained in the corresponding test message.
RANGE AND AZIMUTH	The Range and Azimuth fields, respectively 16- and 14-bits long, in the Track Alert Message, are the measured ranges and azimuths of the two targets involved in the duplicate address alert situation. The MSB of the range is 128 nautical miles, and the MSB of the azimuth is 180 degrees.

TABLE 2–16. MODE S–TO–ATC MESSAGE DATA BLOCKS (Continued)

Data Block	Description
NO. OF MODE S TRACKS	Number of Mode S tracks in the sensor Track File. This number may be equal to or greater than the number of ATRBS targets reported to a particular ATC facility.
NO. OF ATRBS TRACKS	Number of ATRBS tracks in the sensor Track File. This number may be equal to or greater than the number of ATRBS targets reported to a particular ATC facility.
NO. OF RADAR–ONLY TRACKS	This field will be set to zero.
NEW CODE FLAG (N)	The new Code Flag (N) is set when a change (addition or deletion) is made to Beacon Strobe, Yellow, or Red code list contained in the previous message.
CODE OVERFLOW FLAG (V)	The Code Overflow Flag (V) is set when the number of either Red or Yellow sensor conditions exceeds the number that can be reported in this message. Codes are placed in the list in increasing numerical order. The red codes have priority and will be included up to the maximum of “k”. When “k” is exceeded, the higher numbered codes are truncated and are not sent. When the yellow code limit of “j” is exceeded or the total of “k” + “j” = 487, the higher numbered yellow codes are truncated and are not sent.
NORTHMARK TIME	Northmark Time is the time that the antenna boresight passed through true north. Northmark Time is indicated by means of the seconds portion of the Mode sensor realtime clock, with a maximum decimal value of 59.992 seconds and a minimum decimal value of 0 seconds. The Northmark Time segment has the following binary coded (BCD) format: <pre> graph TD B1[45 47] --- L1[THREE-BIT BCD REPRESENTATION OF SECONDS X 10] B2[48 51] --- L2[FOUR-BIT BCD REPRESENTATION OF SECS] B3[52 58] --- L3[SEVEN-BIT BINARY FRACTION (BIT 58 = 1/128 SEC)] </pre>
TARGET OVERLOAD	Target Overload is a one-bit block in the Status Message. When a “0” is present, this indicates a no overload condition. When a “1” is present, this indicates a target overload condition.
SENSOR STATUS	Sensor Status is a two-bit block in the Status Message. When a “00” is present, this indicates a green condition. A yellow condition (01) is no longer reported. If a “10 or “11” is present, this indicates a red condition.
CHANNEL IN USE (CH)	Channel in Use is a one-bit block in the Status Message. When a “0” is present, this indicates that Channel A is in use. When a “1” is present, this indicates that Channel B is in use.

TABLE 2-16. MODE S-TO-ATC MESSAGE DATA BLOCKS (Continued)

Data Block	Description										
NO. OF BCN STROBE WORDS = 2 (i)	Number of Beacon Strobe Words is an eight-bit block in the Status Message. The number of strobe words is “i”, with a maximum value of i = 10. Therefore, the maximum number of strobe words is $2i = 20$.										
NO. OF CW INTER-FERENCE WORDS = 2 (m)	Number of CW interference section = m. Maximum value of m = 10. The maximum number of CW interference words (=2m) therefore is 20.										
NO. OF RED CODES = k	Number of Red Codes is an eight-bit block in the Status Message. The maximum value of k = 255, and k + j is less than (<) 466.										
NO. OF YELLOW CODES = j	Number of Yellow Codes is an eight-bit block in the Status Message. The maximum value of j = 255, and k + j is 466.										
BORESIGHT AZ OF BEGINNING OF STROBE OVERLOAD = B _S (i) and BORE-SIGHT AZ OF END OF STROBE OVERLOAD = B _E (i)	Boresight Azimuth of Beginning of Strobe Overload = B _S (i) and End of Strobe Overload = B _E (i) always occur in pairs. In the event that B _E (i) has not been found at the end of the report, the Hex value FFFF is substituted in the corresponding 16 bits reserved for the corresponding B _E report. The completed pair, B _S and B _E , will occur as the first report in the next message. The units of B _S and B _E are Azimuth Units (AUs).										
BORESIGHT AZ OF BEGINNING OF CW INTERFERENCE = C _S (m) and BORE-SIGHT AZ OF END OF CW INTERFERENCE = C _E (m)	C _S (m) and C _E (m) always occur in pairs. In the event that the C _E (m) has not been found at the end of the report, the Hex value FFFF is substituted in the corresponding 16 bits reserved for the corresponding C _E report. The completed pairs, C _S and C _E will occur as the first report in the next message. The units of C _S and C _E are Azimuth Units.										
YELLOW CON-DITION CODE Y(j)	Yellow condition codes as detected by the Mode S sensor performance monitor are transmitted in ascending numerical order.										
RED CONDITION CODE R(k)	Red condition codes as detected by the Mode S sensor performance monitor are transmitted in ascending numerical order.										
RECOVERY TYPE	Recovery Type is a six-bit binary integer block present in a Sensor Recovery Notice Message. The code values are defined as listed below:										
	<table> <tr> <th>Recovery Type Value</th><th>Definition</th></tr> <tr> <td>0</td><td>Not used</td></tr> <tr> <td>1</td><td>Loss of full data base</td></tr> <tr> <td>2</td><td>Loss of message data base</td></tr> <tr> <td>3 to 63</td><td>Not used</td></tr> </table>	Recovery Type Value	Definition	0	Not used	1	Loss of full data base	2	Loss of message data base	3 to 63	Not used
Recovery Type Value	Definition										
0	Not used										
1	Loss of full data base										
2	Loss of message data base										
3 to 63	Not used										

2.3.5.1.2.3 Sensor-To-Sensor Messages. Sensor-to-Sensor messages are generated to support netted Mode S operations. For sites at which netted operation is implemented, the netting will be accomplished using Mode S/ATC data links rather than dedicated links. For this reason, these messages are part of the Mode S/ATC interface and must satisfy all of the format and protocol requirements of that interface. As shown in Table 2–14, there are two groups of such messages: Mode S Network Control and ATCRBS Network Control. For each message type within these groups, Table 2–14 gives the name of the messages and its Type Code. The Information Field of each message contains the following two data blocks: Sending Sensor ID (SSID) and Receiving Sensor ID (RSID). Each of these blocks is a 10-bit block encoded like the SID block. The ATC link shall provide the appropriate sensor IDs for these fields as indicated by internal sensor message route addressing. The Sensor-to-Sensor messages are as follows: (1) Mode S Network Control Messages and (2) ATCRBS Network Control Messages. Information concerning Sensor-to-Sensor message data blocks is provided in Table 2–17.

NOTE

In handling a Sensor-to-Sensor Message, the only actions to be performed by an ATC facility are as follows: (1) determine that the message falls into a Sensor-to-Sensor category based on its Type Code, (2) determine the routing of the message by using the RSID data block, and (3) retransmit the message to the receiving sensor with its Information Field (I) unchanged. For the most part, the prefix of the Type Code of a Sensor-to-Sensor messages identifies the message. However, one Mode S-to-ATC message type (Track Alert) uses the same prefix as the Mode S Network Control Messages.

- a. Sensor-to-Sensor Mode S Network Control Messages (Routed Via ATC Facility)
 1. Data Start.
 2. Data Stop.
 3. Data Request.
 4. Track Data.
 5. Cancel Request.
 6. Primary Coordination.
 7. Track Alert Message.

DATA START MESSAGE

1001	0001	MODE S ID	MEAS RANGE (MR)	SP	MEAS AZ (MA)
1	8	32	48	50	64

DATA START MESSAGE (CONT.)

RANGE RATE (RR)	AZ RATE (AR)	ALT	SP	TIME REF
72	80	92	96	104

DATA START MESSAGE (CONT.)

BASIC CAPABILITY ALT (CA)	RSID	SSID	SP
107	117	127	128

DATA STOP MESSAGE

1001	0010	MSG S ID	RSID	SSID	SP
1	8	32	42	52	56

DATA REQUEST MESSAGE

1001	0011	MODE S ID	RSID	SSID	SP
1	8	32	42	52	56

TRACK DATA MESSAGE

1001	0100	MODE S ID	MEAS. RANGE (MR)	SP	MEAS. AZ (MA)
1	8	32	48	50	64

TRACK DATA MESSAGE (CONT.)

RANGE RATE (RR)	AZ RATE (AR)	ALT	SP	TIME REF
72	80	92	96	104

TRACK DATA MESSAGE (CONT.)

BASIC CAPABILITY ALT (CA)	RSID	SSID	SP
107	117	127	128

CANCEL REQUEST MESSAGE

1001	0101	MODE S ID	RSID	SSID	SP
1	8	32	42	52	56

PRIMARY COORDINATION MESSAGE

1001	1101	MODE S ID	VALUE (VAL)	RSID	SSID	SP
1	8	32	35	45	55	65

TRACK ALERT MESSAGE

1001	1110	MODE S ID	RSID	SSID	SP
1	8	32	42	52	56

b. Sensor-to-Sensor ATCRBS Network Control Messages (Routed Via ATC Facility)

1. ATCRBS Data Start.
2. ATCRBS Data Stop.
3. ATCRBS Data Request.
4. ATCRBS Track Data.
5. ATCRBS Cancel Request.

ATCRBS DATA START MESSAGE

1101	0001	REQUESTING SENSOR TRACK NUMBER	ATCRBS CODE	CODE CONFIDENCE	MEAS. RANGE (MR)	MEAS. AZ (MA)
1	8	20	32	44	60	74

ATCRBS DATA START MESSAGE (CONT.)

RANGE RATE (RR)	AZ RATE (AR)	ALT	ALT CONFIDENCE	SP	TIME REF	RSID	SSID	SP
82	90	102	114	120	128	138	148	152

ATCRBS DATA STOP MESSAGE

1101	0010	REQUESTING SENSOR TRACK NUMBER	ATCRBS CODE	RSID	SSID	SP
1	8	20	32	42	52	56

ATCRBS DATA REQUEST MESSAGE

1101	0011	REQUESTING SENSOR TRACK NUMBER	ATCRBS CODE	RANGE	SP	AZ	ALT	RSID	SSID
1	8	20	32	48	50	64	76	86	96

ATCRBS TRACK DATA MESSAGE

1101	0100	REQUESTING SENSOR TRACK NUMBER	ATCRBS CODE	CODE CONFIDENCE	MEAS. RANGE (MR)	MEAS. AZ (MA)
1	8	20	32	44	60	74

ATCRBS TRACK DATA MESSAGE (CONT.)

RANGE RATE (RR)	AZ RATE (AR)	ALT	ALT CONFIDENCE	SP	TIME REF	RSID	SSID	SP
82	90	102	114	120	128	138	148	152

ATCRBS CANCEL REQUEST MESSAGE

1101	0101	REQUESTING SENSOR TRACK NUMBER	ATCRBS CODE	RSID	SSID	SP
1	8	20	32	42	52	56

TABLE 2-17. SENSOR-TO-SENSOR MESSAGE DATA BLOCKS

Data Block	Description
NOTE	
Information Concerning Sensor-to-Sensor message data blocks, such as: Type Code, Spare (SP), etc. are defined in previous text.	
SSID	The Sending Sensor Identification (SSID) is a 10-bit block that uniquely identifies the Mode S sensor that is sending the message.
RSID	The Receiving Sensor Identification (RSID) is a 10-bit block that uniquely identifies the Mode S sensor that is to receive the message.
MODE S ID	Mode S ID (Mode S Address) is the unique 24-bit code of a Mode S-equipped aircraft.
MEAS. RANGE (MR)	Measured Range (MR) is a 16-bit binary integer defining measured aircraft range in range units, RUs (MSB = 32768, LSB = 1).
MEAS. AZ (MA)	Measured Azimuth (MA) is a 14-bit binary integer defining measured aircraft azimuth in azimuth units, AUs (MSB = 8192, LSB = 1).
RANGE RATE (RR)	Range Rate (RR) is an eight-bit block, which contains the predicted one-way range rate of the aircraft coded as a signed binary integer, in units of range units/sec. (MSB = 128, LSB = 1).
AZ RATE (AR)	Azimuth Rate (AR) is an eight-bit block containing the predicted azimuth rate of the aircraft coded as a signed binary integer, in units of azimuth units/sec. (MSB = 128, LSB = 2).
ALT	Altitude (ALT) is a 12-bit binary integer containing the reported Mode C altitude of the aircraft in feet (MSB = 204800, LSB = 100).
TIME REF	Time Reference is an eight-bit binary integer defining the time reference in seconds. (MSB = 16, LSB = .0125).

TABLE 2–17. SENSOR–TO–SENSOR MESSAGE DATA BLOCKS (Continued)

Data Block	Description																		
BASIC CAPABILITY (CA)	Basic Capability (CA) is a three-bit subfield of the 55-bit block that indicates the input/output capability of the aircraft. CA coding is as follows: <table> <tr> <th><u>Code</u></th><th><u>Meaning</u></th></tr> <tr> <td>000</td><td>No extended capability report is available.</td></tr> <tr> <td>001</td><td>Comm A/B extended capability report is available.</td></tr> <tr> <td>010</td><td>Comm A/B/C and extended capability report is available.</td></tr> <tr> <td>011</td><td>Comm A/B/C/D are extended capability report is available.</td></tr> <tr> <td>100</td><td>Not assigned.</td></tr> <tr> <td>101</td><td>Not assigned.</td></tr> <tr> <td>110</td><td>Not assigned.</td></tr> <tr> <td>111</td><td>Not assigned.</td></tr> </table>	<u>Code</u>	<u>Meaning</u>	000	No extended capability report is available.	001	Comm A/B extended capability report is available.	010	Comm A/B/C and extended capability report is available.	011	Comm A/B/C/D are extended capability report is available.	100	Not assigned.	101	Not assigned.	110	Not assigned.	111	Not assigned.
<u>Code</u>	<u>Meaning</u>																		
000	No extended capability report is available.																		
001	Comm A/B extended capability report is available.																		
010	Comm A/B/C and extended capability report is available.																		
011	Comm A/B/C/D are extended capability report is available.																		
100	Not assigned.																		
101	Not assigned.																		
110	Not assigned.																		
111	Not assigned.																		
VALUE (VAL)	Value (VAL) is a three-bit field contained in a Primary Coordination Message. The VAL field is coded as follows: <table> <tr> <th><u>Code</u></th><th><u>Meaning</u></th></tr> <tr> <td>000</td><td>Not used.</td></tr> <tr> <td>001</td><td>Request for primary.</td></tr> <tr> <td>010</td><td>Primary Assignment is approved.</td></tr> <tr> <td>011</td><td>Primary Assignment is disapproved.</td></tr> <tr> <td>100</td><td>Primary Assignment is accepted.</td></tr> <tr> <td>101</td><td>Request for Secondary.</td></tr> <tr> <td>110</td><td>Controlled Primary Notice.</td></tr> <tr> <td>111</td><td>Not used.</td></tr> </table>	<u>Code</u>	<u>Meaning</u>	000	Not used.	001	Request for primary.	010	Primary Assignment is approved.	011	Primary Assignment is disapproved.	100	Primary Assignment is accepted.	101	Request for Secondary.	110	Controlled Primary Notice.	111	Not used.
<u>Code</u>	<u>Meaning</u>																		
000	Not used.																		
001	Request for primary.																		
010	Primary Assignment is approved.																		
011	Primary Assignment is disapproved.																		
100	Primary Assignment is accepted.																		
101	Request for Secondary.																		
110	Controlled Primary Notice.																		
111	Not used.																		
ATCRBS CODE	ATCRBS Code is a 12-bit block that contains the ATCRBS 4096-code value from a Mode S transponder.																		
TRACK NUMBER	Track Number is a 12-bit block identifying the track file number.																		

2.3.6 Mode S Sensor-To-Non-ATC Communications Link Messages

The Mode S Sensor-to-non-ATC communications link provides for the interchange of data link, status, and control messages between the Mode S Sensor and non-ATC facilities. Non-ATC communication links provide the medium for two way communication between a Mode S Sensor and a non-ATC facility. Messages sent from the non-ATC facility to the Mode S Sensor include:

- Standard Uplink
- ELM Uplink
- Request for Downlink Data
- Message Cancellation Request
- Data Link Capability Request

- f. Request for Aircraft State
- g. Request for Aircraft Position
- h. Test Message

Messages sent from the Mode S Sensor to the non-ATC facility include:

- a. Message Rejection/Delay Notice
- b. Message Rejection/Delay Notice with Sensor IDs
- c. Uplink Delivery Notice
- d. Pilot Downlink
- e. Pilot Downlink with Position
- f. Ground-Initiated Downlink
- g. Ground-Initiated Downlink with Position
- h. Broadcast Downlink
- i. Broadcast Downlink with Position
- j. ELM Downlink
- k. ELM Downlink with Position
- l. Data Link Capability
- m. ATRBS ID Code
- n. Aircraft State
- o. Aircraft Position
- p. Track Drop
- q. Test Response Message
- r. Track Alert Message
- s. Sensor Recovery Notice Message

2.3.6.1 Mode S Sensor/Non-ATC Communications Link. The Mode S Sensor to non-ATC communications interface provides up to six full duplex data links from the Mode S Sensor to non-ATC facilities. Each data link is comprised of up to six full duplex data line EIA-STD-RS-ABC interfaces. Each data line interface supports one GFE odem at up to 9600 bits per second data rate. All six data lines from one data link operate at the same data rate if in use. Output data messages will be equally distributed among the modems that are in use. Each link can support internal or external clocking. When external clocking is provided, the data line interfaces can simultaneously transmit and/or receive data at rates up to 56,000 bits per second. The total simultaneous transmit and receive data rate of a communication link will not exceed 56,000 bits per second, in either direction per port. Information concerning Mode S Sensor/Non-ATC communication link messages is presented in the following order:

- a. Mode S Sensor-To-Non-ATC Message Protocol
- b. Message Types

2.3.6.1.1 Mode S Sensor-To-Non-ATC Message Protocol. For information concerning the message frame structure and frame formats, the user should refer to the information contained in previous text (Mode S Sensor-To-ATC Message Protocol).

2.3.6.1.2 Message Types. The following non-ATC messages cross the communication interfaces. These messages types are grouped according to the sender and the receiver of the message, and according to the contents of the Information Field (I). These message types are grouped as follows: (1) Non-ATC-to-Mode S Messages and (2) Mode S-to-Non-ATC Messages. Refer to Table 2–18 for information concerning Information Field Type Codes for Mode S Sensor/Non-ATC Messages.

TABLE 2–18. INFORMATION FIELD (I) TYPE CODES FOR MODE S SENSOR/NON-ATC MESSAGES

From/To	Message Type	Prefix	Suffix	Description
Non-ATC to-Mode S	Uplink Messages	0010 0010 0010 0010 0010	1001 1010 1011 0100 1101	Standard Uplink ELM Uplink Request for Downlink Data ATCRBS ID Request Message Cancellation Request
	Status Request	0000 0000 0000 0110	1010 1000 1001 0001	Data Link Capability Request Request for Aircraft State Request for Aircraft Position Test Message
Mode S-to- Non-ATC	Sensor Response Message	0011 0011	0001 1001	Message Rejection/Delay Notice Message Rejection/Delay Notice with Sensor IDs
		0011	0010	Uplink Delivery Notice
	Downlink Messages	0100 0100 0100 0100	0001 1001 0100 1100	Pilot Downlink Pilot Downlink with Position Ground-Initiated Downlink Ground-Initiated Downlink with Position
		0100 0100 0100 0100 0011 0100	0000 1000 0010 1010 0100 0101	Broadcast Downlink Broadcast Downlink with Position ELM Downlink ELM Downlink with Position Data Link Capability ATCRBS ID Code
	Status Messages	0100 0100 0100 0110 0110 1001	1110 1111 1011 1111 0010 1100	Aircraft State Aircraft Position Track Drop Sensor Recovery Notice Message Test Response Message Track Alert Message

2.3.6.1.2.1 Non-ATC-To-Mode S Messages. The non-ATC-to-Mode S communication link messages are divided into two categories: (1) Uplink Messages and (2) Status Request Messages. Information concerning non-ATC-to-Mode S message data blocks is provided in Table 2–19.

1. Non-ATC-to-Mode S Uplink Messages

- a. Standard Uplink.
- b. ELM Uplink.
- c. Request for Downlink Data.
- d. ATCRBS ID Request.
- e. Message Cancellation Request.

NOTE

The term “Standard Uplink” refers to a message sent to an aircraft by one or more Comm-A transmissions.

STANDARD UPLINK MESSAGE

0010	1001	MODE S ADDRESS	MSG NO.	P	EXP	PT	SC
1	8	32	40	44	47	48	50

STANDARD UPLINK MESSAGE (CONT.)

SP	MA (REPEATED)
56	56 + 56 (SC + 1)

ELM UPLINK MESSAGE

0010	1010	MODE S ADDRESS	MSG NO.	P	EXP	SP	ELM LENGTH
1	8	32	40	44	47	50	56

ELM UPLINK MESSAGE (CONT.)

ELM TEXT (MAX 5120)
56 + 80 (LENGTH + 1)

REQUEST FOR DOWNLINK DATA MESSAGE

0010	1011	MODE S ADDRESS	MSG NO.	P	EXP	SP	BDS1	BDS2
1	8	32	40	44	47	48	52	56

ATCRBS ID REQUEST MESSAGE

0010	0100	MODE S ADDRESS	MSG NO.
1	8	32	40

MESSAGE CANCELLATION REQUEST MESSAGE

0010	1101	MODE S ADDRESS	MSG NO.	REF MSG NO.	REF TYPE CODE
1	8	32	40	48	56

2. Non-ATC-To-Mode S Status Request Messages

- a. Data Link Capability Request.
- b. Request for Aircraft State.
- c. Request for Aircraft Position.
- d. Test Message (to a Mode S Sensor).

DATA LINK CAPABILITY REQUEST MESSAGE

0000	1010	MODE S ADDRESS	MSG NO.
1	8	32	40

REQUEST FOR AIRCRAFT STATE MESSAGE

0000	1000	MODE S ADDRESS	MSG NO.
1	8	32	40

REQUEST FOR AIRCRAFT POSITION MESSAGE

0000	1001	MODE S ADDRESS	MSG NO.
1	8	32	40

TEST MESSAGE

0110	0001	MSG NO.	TEST DATA
1	8	16	64

TABLE 2–19. NON–ATC–TO–MODE S MESSAGE DATA BLOCKS

Data Block	Description																		
TYPE CODE	The Type Code Block is an eight-bit control block that begins every Information Field. The code values are shown in Table 2–18.																		
MODE S ADDRESS	Mode S Address is the unique 24-bit identification code of a Mode S-equipped aircraft.																		
MSG NO.	Message Number is an eight-bit binary integer which numbers all types of messages concerning a particular Mode S target sent by a particular facility. This field is also kept as an unambiguous number in messages not concerning any one aircraft (like Type Codes 0110 0010 and 1001 1100). Non-ATC facilities will only assign message numbers zero to 127. Zero is excluded so that 127 messages about each aircraft and 127 messages not about any one aircraft from Non-ATC, can be distinguished at a time. When a message that has this field is used as a response to a request, the message number assigned in the request is placed in this field. When the message is not used as a response, a message number is assigned by the sender.																		
P	Priority (P) is a four-bit data block which provides a user supplied priority tag (for Uplink Messages only). This data block contains a binary integer, with 0 being the lowest and 15 the highest priority. Delivery of every Uplink Message is scheduled as promptly as possible, regardless of the priority value; only if a queue of messages exists for a particular aircraft is a message given precedence over an earlier one with lower priority.																		
EXP	Expiration (EXP) is a three-bit data block encoding “Time To Expiration” of a message (for Uplink Messages only). This data block is a binary integer, with the values 0 through 7 representing the number of scans for which delivery should be attempted, encoded as follows: <table><tr><th>EXP Value</th><th>Time To Expiration (in Scan Periods)</th></tr><tr><td>0</td><td>Default Value</td></tr><tr><td>1</td><td>1</td></tr><tr><td>2</td><td>2</td></tr><tr><td>3</td><td>4</td></tr><tr><td>4</td><td>8</td></tr><tr><td>5</td><td>16</td></tr><tr><td>6</td><td>32</td></tr><tr><td>7</td><td>No Expiration</td></tr></table>	EXP Value	Time To Expiration (in Scan Periods)	0	Default Value	1	1	2	2	3	4	4	8	5	16	6	32	7	No Expiration
EXP Value	Time To Expiration (in Scan Periods)																		
0	Default Value																		
1	1																		
2	2																		
3	4																		
4	8																		
5	16																		
6	32																		
7	No Expiration																		
PT	Packet Type (PT) is a one-bit block present only in a Standard Uplink Message. This PT block serves to characterize the message for communication system purposes.																		
SC	Segment Count (SC) is a two-bit binary integer block present in a Pilot Uplink or a Standard Downlink Message. The SC value is one less than the number of messages text segments comprising the complete message. Values of SC greater than zero correspond to “linked Comm-A or Comm-B” messages.																		

TABLE 2-19. NON-ATC-TO-MODE S MESSAGE DATA BLOCKS (Continued)

Data Block	Description
SP	The Spare (SP) field represents the needed spacing required in some messages to meet the integral number of octets of CCITT X.25. The SP field is set to zero.
MA	Comm-A message text (MA) is a 56-bit block present in Standard Uplink Messages. A message may contain up to four MA blocks, as specified by the value of SC.
ELM LENGTH	ELM Length is a six-block present only in ELM messages. This data block contains binary integers specifying the number of 80-bit segments comprising the ELM Text block. This data block value is one less than the number of segments. The values of ELM length are 1 (2 segments) to 63 (64 segments) in the ELM Uplink Message.
NOTE	
An ELM length value greater than 15 in an ELM Uplink message defines a set of linked ELMs. Processing of such a message to support the linking protocol is a task of the Data Link Processing function of the Mode S sensor.	
ELM TEXT	ELM Text is a variable-length ELM data field. For an ELM Uplink message, the length is a multiple of 80 bits with a maximum of 5120 bits, corresponding to a sequence of four linked ELMs, each comprised of 16 consecutive Comm-C ground-to-air transmissions. For an ELM Downlink message, the length is a multiple of 80 bits with a maximum of 1280 bits corresponding to a sequence of 16 consecutive Comm-D air-to-ground transmissions. Each transmission carries an 80-bit field of text.
BDS1 and BDS2	BDS1 and BDS2 are two 4-bit blocks present only in a Request for Downlink Data. Together BDS1 and BDS2 identify the particular 56-bit data register, within the Mode S transponder, that is to be read out and the data contents downlinked using a ground-initiated Comm-B message. BDS1=0001 and BDS2=0000 or 0001 signify a request for an extended capability report from an airborne Mode S installation.
REF MSG NO.	Referenced Message Number is an eight-bit binary integer present only in messages that are sent in response to requests or that are in reference to other messages. In a message cancellation request, the REF MSG NO. represents the message number of the message whose cancellation is being requested.
REF TYPE CODE	Referenced Type Code is the eight-bit Type Code block corresponding to a message whose cancellation is being requested.
TEST DATA	Test Data is a 48-bit block present in a Test Message. The contents may be defined by the originator. Each message may be different.

2.3.6.1.2.2 Mode S-to-Non-ATC Messages. The Mode S-to-non-ATC communication link messages are divided into three categories: (1) Sensor Response Messages, (2) Downlink Messages, and (3) Status Messages. Information concerning Mode S-to-non-ATC message data blocks is provided in Table 2–20.

1. Mode S-To-Non-ATC Sensor Response Messages
 - a. Message Rejection/Delay Notice.
 - b. Message Rejection/Delay Notice with Sensor IDs.
 - c. Uplink Delivery Notice.

MESSAGE REJECTION/DELAY NOTICE MESSAGE

0011	0001	MODE S ADDRESS	MSG NO.	QUAL	SP	SID
1	8	32	40	43	46	56

MESSAGE REJECTION NOTICE WITH SENSOR IDs MESSAGE

0011	1001	MODE S ADDRESS	MSG NO.	QUAL	SP	SID1
1	8	32	40	43	46	56

MESSAGE REJECTION NOTICE WITH SENSOR ID's MESSAGE (CONT.)

SP	SID2	SP	SID
62	72	78	88

UPLINK DELIVERY NOTICE MESSAGE

0011	0010	MODE S ADDRESS	MSG NO.	D1	SP	SID
1	8	32	40	41	46	56

2. Mode S-To-Non-ATC Downlink Messages
 - a. Pilot Downlink.
 - b. Pilot Downlink With Position.
 - c. Ground Initiated Downlink Message.
 - d. Ground Initiated Downlink Message With Position.
 - e. Broadcast Downlink Message.
 - f. Broadcast Downlink Message With Position.
 - g. ELM Downlink.

- h. ELM Downlink With Position.
- i. Data Link Capability.
- j. ATCRBS ID Code.

NOTE

The terms “Pilot Downlink” and “Pilot Downlink With Position” refer to messages received from an aircraft by one or more Comm-B transmissions.

PILOT DOWNLINK MESSAGE

0100	0001	MODE S ADDRESS	MSG NO.	SC	SP	MB (REPEATED)	SP	SID
1	8	32	40	42	48	$N = 104 + 56(SC)$	$N + 6$	$N + 16$

PILOT DOWNLINK WITH POSITION MESSAGE

0100	1001	MODE S ADDRESS	MSG NO.	SC	SP	MB (REPEATED)	SP
1	8	32	40	42	48	$N = 104 + 56(SC)$	$N + 2$

PILOT DOWNLINK WITH POSITION MESSAGE (CONT.)

A/C RANGE	A/C AZIMUTH	SP	RANGE RATE	AZIMUTH RATE	A/C STATE
$N + 12$	$N + 24$	$N + 26$	$N + 32$	$N + 40$	$N + 44$

PILOT DOWNLINK WITH POSITION MESSAGE (CONT.)

ALTITUDE	SP	SID
$N + 56$	$N + 62$	$N + 72$

GROUND INITIATED DOWNLINK MESSAGE

0100	0100	MODE S ADDRESS	MSG NO.	BDS1	BDS2	MB	SP	
1	8	32	40	44	48	104	110	120 SID

GROUND INITIATED DOWNLINK WITH POSITION MESSAGE

0100	1100	MODE S ADDRESS	MSG NO.	BDS1	BDS2	MB	SP	A/C RANGE
1	8	32	40	44	48	104	106	116

GROUND INITIATED DOWNLINK WITH POSITION MESSAGE (CONT.)

A/C AZIMUTH	SP	RANGE RATE	AZIMUTH RATE	A/C STATE	ALTITUDE	SP	SID
128	130	136	144	148	160	166	176

BROADCAST DOWNLINK MESSAGE

0100	0000	MODE S ADDRESS	MSG NO.	MB	SP	SID
1	8	32	40	96	102	112

BROADCAST DOWNLINK WITH POSITION MESSAGE

0100	1000	MODE S ADDRESS	MSG NO.	MB	SP	A/C RANGE
1	8	32	40	96	102	108

BROADCAST DOWNLINK WITH POSITION MESSAGE (CONT.)

A/C AZIMUTH	SP	RANGE RATE	AZIMUTH RATE	A/C STATE	ALTITUDE	SP	SID
120	122	128	136	140	152	158	168

ELM DOWNLINK MESSAGE

0100	0010	MODE S ADDRESS	MSG NO.	SP	LENGTH	ELM TEXT (MAX 1280)	SP	SID
1	8	32	40	42	48	$N = 48 + 80 (\text{LENGTH} + 1)$	$N + 6$	$N + 16$

ELM DOWNLINK WITH POSITION MESSAGE

0100	1010	MODE S ADDRESS	MSG NO.	SP	ELM LENGTH	ELM TEXT (MAX 1280)	SP
1	8	32	40	42	48	$M = 48 + 80 (\text{LENGTH} + 1)$	$M + 2$

ELM DOWNLINK WITH POSITION MESSAGE (CONT.)

A/C RANGE	A/C AZIMUTH	SP	RANGE RATE	AZIMUTH RATE	A/C STATE
$M + 12$	$M + 24$	$M + 26$	$M + 32$	$M = 40$	$M + 44$

ELM DOWNLINK WITH POSITION MESSAGE (CONT.)

ALTITUDE	SP	SID
$M + 56$	$M + 62$	$M + 72$

DATA LINK CAPABILITY MESSAGE

0011	0100	MODE S ADDRESS	MSG NO.	SP	CA	ECA	SP	SID
1	8	32	40	41	44	96	102	112

ATCRBS ID CODE MESSAGE

0100	0101	MODE S ADDRESS	MSG NO.	SP	ATCRBS ID	SP	SID
1	8	32	40	44	56	70	80

3. Mode S-To-Non-ATC Status Messages

- a. Aircraft State.
- b. Aircraft Position.
- c. Track Drop.
- d. Test Response Message (from a Mode S Sensor).
- e. Track Alert Message.
- f. Sensor Recovery Notice Message.

AIRCRAFT STATE MESSAGE

0100	1110	MODE S ADDRESS	MSG NO.	A/C STATE	SP	SID
1	8	32	40	44	46	56

AIRCRAFT POSITION MESSAGE

0100	1111	MODE S ADDRESS	MSG NO.	SP	A/C RANGE	A/C AZIMUTH	SP
1	8	32	40	42	52	64	66

AIRCRAFT POSITION MESSAGE (CONT)

RANGE RATE	AZIMUTH RATE	A/C STATE	ALTITUDE	SP	SID
72	80	84	96	102	112

TRACK DROP MESSAGE

0100	1011	MODE S ADDRESS	MSG NO.	SP	SID1	SP	SID2	SP	SID
1	8	32	40	46	56	62	72	78	88

TEST RESPONSE MESSAGE

0110	0010	MSG NO.	TEST RESPONSE	SP	SID
1	8	16	64	70	80

TRACK ALERT MESSAGE

1001	1100	MSG NO.	MODE S ADDRESS	RANGE 1	SP	AZIMUTH 1	RANGE 2	SP
1	8	16	40	56	58	72	88	90

TRACK ALERT MESSAGE (CONT)

ALTITUDE 2	SP	SID
104	118	128

SENSOR RECOVERY NOTICE MESSAGE

0110	1111	SID	RECOVERY TYPE
1	8	18	24

TABLE 2–20. MODE S–TO–NON–ATC MESSAGE DATA BLOCKS

Data Block	Description												
NOTE													
Information concerning Mode S-to-non-ATC message data blocks, such as: Type Code, Mode S Address, Segment Count (SC), ELM Length, ELM Text, and Spare (SP) are defined in previous text for non-ATC-to-Mode S message data blocks.													
MSG NO.	Output messages for Mode S-to-non-ATC facilities shall contain eight-bit Message Number fields. These MSG No. fields shall be set as follows: 1. For messages which are responses to Uplink Messages (Rejection and Delivery Notices), the field contains a referenced message number. The field shall be set equal to the value in the message number field in the corresponding Uplink Message. The message number field shall also be set in this manner for the following types of message: Tactical Downlink and Tactical Downlink With Position (when generated in response to a request for downlink data), Aircraft State, Aircraft Position, and Data Link Capability (when generated in response to a capability request). 2. For other messages referring to a Mode S aircraft (except Track Alert), the message number shall be set to 128. 3. For all other messages containing a message number field (i.e., Track Alert and messages without a Mode S Address field), the MSG No. shall be set to 128.												
QUAL	The Qualifier (QUAL) is a three-bit block present in a Message Rejection/Delay Notice only, with the following coding: <table><tr><th><u>Code</u></th><th><u>Meaning</u></th></tr><tr><td>000</td><td>Target not on file (rejection).</td></tr><tr><td>001</td><td>Target not in roll-call mode (delay).</td></tr><tr><td>010</td><td>AML sublist overflow (rejection).</td></tr><tr><td>011</td><td>Sensor not primary (rejection of ELM uplink).</td></tr><tr><td>100</td><td>Targets lack ELM capability (rejection).</td></tr></table>	<u>Code</u>	<u>Meaning</u>	000	Target not on file (rejection).	001	Target not in roll-call mode (delay).	010	AML sublist overflow (rejection).	011	Sensor not primary (rejection of ELM uplink).	100	Targets lack ELM capability (rejection).
<u>Code</u>	<u>Meaning</u>												
000	Target not on file (rejection).												
001	Target not in roll-call mode (delay).												
010	AML sublist overflow (rejection).												
011	Sensor not primary (rejection of ELM uplink).												
100	Targets lack ELM capability (rejection).												

NOTE

Code values 101 through 111 are not assigned.

TABLE 2–20. MODE S–TO–NON–ATC MESSAGE DATA BLOCKS (Continued)

Data Block	Description						
SID, SID1, and SID2	Mode S Sensor ID (SID) is a 10-bit field containing the unique identity code of the Mode S Sensor sending the message. The SID1 and SID2 blocks are 10-bit fields, each of which contain the unique identity code of an adjacent Mode S Sensor. SID1 and SID2 are used by the reporting Sensor when it is not primary for an aircraft to which an Uplink ELM is being sent, or when the Sensor is in the process of dropping a track. SID1 and SID2 identify the adjacent Sensors that are assigned to the aircraft, with the highest-priority Sensor given in SID1 and the second-highest priority Sensor in SID2. If fewer than two adjacent Sensors are assigned, either SID2 or both SID1 and SID2 will contain zeroes.						
DI	The Delivery Indicator (DI) is a one-bit block present in an Uplink Delivery Notice only. DI reports on the success of a referenced Uplink Message with the following coding: <table> <tr> <th><u>Code</u></th><th><u>Meaning</u></th></tr> <tr> <td>0</td><td>Message successfully delivered.</td></tr> <tr> <td>1</td><td>Message expired, undelivered.</td></tr> </table>	<u>Code</u>	<u>Meaning</u>	0	Message successfully delivered.	1	Message expired, undelivered.
<u>Code</u>	<u>Meaning</u>						
0	Message successfully delivered.						
1	Message expired, undelivered.						
MB	The Comm-B Message Text (MB) field is a 56-bit block used for Downlink Message text similar to the Uplink MA block. A Pilot Downlink Message may contain up to four MB blocks as specified by the value of Segment Count (SC). Broadcast and Ground Initiated Comm-B Messages each contain only a single MB block. The MB for Pilot Downlink Messages begins with a two-bit Linked Comm-B Subfield (LBS) that is used to support linking of MB messages. The LSB is present only in the first component of the linked message.						
A/C RANGE	Aircraft Range (A/C Range) is a 10-bit block containing the measured one-way range to the aircraft coded as a binary integer, in units of nautical miles (MSB=128 and LSB=0.25).						
A/C AZIMUTH	Aircraft Azimuth (A/C Azimuth) is a 12-bit block containing the measured azimuth of the aircraft coded as a binary integer in units of degrees (MSB=180 and LSB=0.088).						
RANGE RATE (RR)	Range Rate (RR) is a six-bit block containing the predicted one-way range rate of the aircraft coded as a signed binary integer, in units of nautical miles/second (MSB=0.2 and LSB=0.0125).						
AZIMUTH RATE	Azimuth Rate (AR) is an eight-bit block containing the predicted azimuth rate of the aircraft coded as a signed binary integer, in units of degrees/second (MSB=2 and LSB=0.031).						

TABLE 2–20. MODE S–TO–NON–ATC MESSAGE DATA BLOCKS (Continued)

Data Block	Description														
A/C STATE	Aircraft State (A/C State) is a four-bit block containing the track state of the aircraft, its control state (controlled or uncontrolled), and the priority assignment of the Sensor with respect to the aircraft (primary or secondary). The coding is as follows: <table><tr><th><u>Code</u></th><th><u>Meaning</u></th></tr><tr><td>000</td><td>Aircraft not on file.</td></tr><tr><td>001</td><td>Aircraft on file but not on full track (i.e., track is not currently being updated by roll-call replies).</td></tr><tr><td>010</td><td>Aircraft in full track, controlled primary.</td></tr><tr><td>011</td><td>Aircraft in full track, controlled secondary.</td></tr><tr><td>100</td><td>Aircraft in full track, uncontrolled primary.</td></tr><tr><td>101</td><td>Aircraft in full track, uncontrolled secondary.</td></tr></table>	<u>Code</u>	<u>Meaning</u>	000	Aircraft not on file.	001	Aircraft on file but not on full track (i.e., track is not currently being updated by roll-call replies).	010	Aircraft in full track, controlled primary.	011	Aircraft in full track, controlled secondary.	100	Aircraft in full track, uncontrolled primary.	101	Aircraft in full track, uncontrolled secondary.
<u>Code</u>	<u>Meaning</u>														
000	Aircraft not on file.														
001	Aircraft on file but not on full track (i.e., track is not currently being updated by roll-call replies).														
010	Aircraft in full track, controlled primary.														
011	Aircraft in full track, controlled secondary.														
100	Aircraft in full track, uncontrolled primary.														
101	Aircraft in full track, uncontrolled secondary.														
ALTITUDE	Altitude is a 12-bit field containing the reported Mode C altitude of the target (in feet) if valid, otherwise, this field contains zeros. The Altitude field is encoded as a sign bit (set to “0” for positive values), followed by an 11-bit binary integer with MSB=102,400 and LSB=100. Negative Mode C altitudes are reported in two’s complement with the sign bit set.														
CA AND ECA	Capability is a 55-bit block that indicates the input/output capability of the aircraft. The Capability field is divided into two subfields defined as follows: 1. CA (Basic Capability) — This field is the tree-bit CA field contained in Mode S All-Call replies. 2. ECA (Extended Capability) — This 52-bit field contains the data link capability when the CA field so indicates. The ECA field’s coding is supported by the data link avionics and the ground ATC and non-ATC users, and is transparent to the Mode S Sensor.														

NOTE

The remaining code values are not assigned.

NOTE

The leading four bits of the MB are not assigned and the final 52 bits form the ECA.

TABLE 2–20. MODE S–TO–NON–ATC MESSAGE DATA BLOCKS (Continued)

Data Block	Description										
RECOVERY TYPE	Recovery Type is a six-bit binary integer block present in a Sensor Recovery Notice Message. The code values are defined as listed below: <table> <tr> <th><u>Code</u></th><th><u>Meaning</u></th></tr> <tr> <td>0</td><td>Not used.</td></tr> <tr> <td>1</td><td>Loss of full data base.</td></tr> <tr> <td>2</td><td>Loss of message data base.</td></tr> <tr> <td>3 to 63</td><td>Not used.</td></tr> </table>	<u>Code</u>	<u>Meaning</u>	0	Not used.	1	Loss of full data base.	2	Loss of message data base.	3 to 63	Not used.
<u>Code</u>	<u>Meaning</u>										
0	Not used.										
1	Loss of full data base.										
2	Loss of message data base.										
3 to 63	Not used.										
ATCRBS ID	ATCRBS ID is a 12-bit block which provides the 4096-code value from a Mode S transponder. The coding is the standard Mode 3/A format used in Surveillance Reports on ATCRBS targets.										
TEST RESPONSE	Test Response data is a 48-bit block present only in a Test Response Message. The data must be an exact duplicate of the Test data contained in the corresponding Test Message.										
RANGE 1 and RANGE 2	The Range 1 and 2 fields are each 16-bits long, in the Track Alert Message, and are the measured ranges of the two targets that are involved in the duplicate address alert situation. The MSB of the range is 128 nautical miles.										
AZIMUTH 1 and AZIMUTH 2	The Azimuth 1 and 2 fields are each 14-bits long, in the Track Alert Message, and are the measured azimuth of the two targets that are involved in the duplicate address alert situation. The MSB of the azimuth is 180 degrees.										

2.3.7 Mode S Sensor–To–ATC Enroute Surveillance Link Messages

The Mode S Sensor–to–ATC surveillance link provides for transmission of surveillance messages between the Mode S Sensor and air traffic control (ATC) en route facilities (refer to Figure 1–2). This surveillance link allows for the transfer of the following message types: (1) beacon messages, (2) beacon real time quality control (RTQC) messages, (3) beacon strobe messages, (4) search messages (5) search RTQC messages, (6) search strobe messages, (7) weather map messages, and (8) Common Digitizer (CD–2)/Mode S status messages. For information concerning Mode S Sensor-to-en route ATC facility surveillance link message formats, message labels, and message descriptions; refer to Tables 2–21 through 2–28.

TABLE 2–21. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON MESSAGE

Word	Bit(s)	ID	Bit Description
1	12 thru 10	Message Label	NOTE Beacon targets are both ATCRBS and Mode S, and the reports from these targets are used to generate Beacon Messages. A 011 in Message Label bits 12 through 10 indicates a Beacon Message. Bit 12 is the first message bit.
		MODE 2	This field is set to 1 for Mode 2 reports with all confidence bits high. This field is set to 0 for all other Mode 2 reports.
		MODE 3/A	This field indicates the validity of the Mode 3/A code data. For ATCRBS targets: If the Sensor indicates Mode 3/A code confidence as all high confidence bits, then the data is valid, and this field is set to 1. If the code confidence does not have all high confidence bits or if the Mode 3/A code data is not present; then the data is not valid and this field is set to 0. The Sensor may replace the Mode 3/A code of an ATCRBS report with the Mode 3/A code from a track. When this condition occurs, the Mode 3/A confidence bits from the track are used (instead of the report code confidence bits) to validate the report code according to the aforementioned ATCRBS target rules. For Mode S targets: The ATCRBS 3/A code is acquired by the Sensor in a different way than for ATCRBS targets. As long as there is a 3/A code available for a Mode S target, this field is set to 1. If the 3/A code is not available, this field is set to 0.
	7	MODE C	This field indicates the validity of the Mode S altitude data. For ATCRBS reports: If the Sensor has successfully decoded the Mode C data for a report; then the data is valid, and this field is set to 1. If the sensor has not decoded the Mode C data, or if the Mode C data is not present; then the data is not valid, and this field is set to 0. For Mode S targets: If the surveillance reply has altitude data present i.e., the Mode C data is present and valid, then this field is set to 1. If the altitude data is not present in any reply, then this field is set to 0.

TABLE 2–21. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	6	SPI	Special Identity (SPI) field. This field is the Special Identity (SPI) bit in the Mode S Sensor. For ATCRBS targets: This field is set to 1 when the SPI bit in the ATCRBS reply is 1 with high confidence. For Mode S targets: This field is set to 1 when the Flight Status (FS) field in the Mode S reply indicates the Special Identity (SPI) function is set in the transponder; otherwise, this field is set to 0.
	5	R/R	Radar Reinforced (R/R) bit. When this field is set to 1, this indicates that a radar report reinforced the beacon data (no radar data is reported).
	4	7700	When this field is set to 1, this indicates a Mode 3/A aircraft emergency.
	3	7600	When this field is set to 1, this indicates a Mode 3/A aircraft radio failure.
	2	FAA	When this field is set to 1, this indicates FAA as a user.
	1	AF	When this field is set to 1, this indicates a false target.
	0	Parity	Odd parity bit (message bit 13).
2	12	RANGE	128 NM
	11		64 NM
	10		32 NM
	9		16 NM
	8		8 NM
	7		4 NM
	6		2 NM
	5		1 NM
	4		1/2 NM
	3		1/4 NM
	2		1/8 NM
	1	SPEC TARGET	Special Target bit. This field is set to 0.
	0	Parity	Odd parity bit (message bit 26).

TABLE 2–21. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North.
	11		1024	ACPs	
	10		512	ACPs	
	9		256	ACPs	
	8		128	ACPs	
	7		64	ACPs	
	6		32	ACPs	
	5		16	ACPs	
	4		8	ACPs	
	3		4	ACPs	
	2		2	ACPs	
	1		1	ACP	
	0	AZIMUTH Parity	Odd parity bit (message bit 39).		
4 When R/L=0 (bit 2)	12	AIMS PRES	AIMS Present bit. This field is set to 0.		
	11	MSB/AIMS	This field is set to 0.		
	10	LSB/AIMS	This field is set to 0.		
	9	DISC	Discrete field. When this field is set to 1, this indicates that the report is based on replies with discrete Mode 3/A codes.		
	8	MODE 2 ‘X’	When this field is set to 1, this indicates valid Mode 2 ‘X’ responses.		
	7	MODE 3/A ‘X’	When this field is set to 1, this indicates valid Mode 3/A ‘X’ responses.		
	6	TIME –IN – STORAGE	4	SEC.	Four-bit field contains the elapsed time in seconds which represents the time between the reported azimuth of data (time of measurement), and insertion of the message into the channel for delivery. Encoded as a binary integer with MSB=4 SEC., LSB=0.5 SEC.
	5		2	SEC.	
	4		1	SEC.	
	3		0.5	SEC.	

TABLE 2–21. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	2	R/L	Run–Length (R/L) bit. This field indicates which of the two formats is used for the fourth word of the Beacon Message. When this field is set to 1, the fourth word contains Beacon run-length. When this field is set to 0, the fourth word contains discrete Beacon indicator information. The R/L setting is site adaptable. The default value is 0.
	1	MTI	Moving Target Indicator (MTI) bit. When this field is set to 1, this indicates that the radar reinforced bit is set and the associated Search target was reported with its MTI flag set.
	0	Parity	Odd parity bit (message bit 52).
4 When R/L = 1 (bit 2)	12	BEACON RUN– LENGTH	64 ACPs ATCRBS Target Reports:
	11		32 ACPs a. All high confidence bits that match all confidence bits in the track file code, transmitted with run-length = 24.
	10		16 ACPs
	9		8 ACPs b. All high confidence bits that have not been correlated with an existing track, transmitted with run-length = 20.
	8		4 ACPs
	7		2 ACPs c. Some low confidence bits that match the high confidence bits in the track file code, transmitted with run-length = 22.
			d. The code in transition bit set, transmitted with run-length = 10.
			e. All other ATCRBS reports with run-length = 8.
			Mode S Target Reports:
			a. Mode S only target reports, run-length = 26.
			b. Mode S, ATCRBS Reinforced, run-length = 28.
			c. Mode S, Illegal Mode S ID, run-length = 30.
	6	TIME–IN– STORAGE	4 SEC. Four-bit field contains the elapsed time in seconds which represents the time between the reported azimuth of data (time of measurement), and insertion of the message into the channel for delivery. Encoded as a binary integer with MSB = 4 SEC., LSB = 0.5 SEC.
	5		2 SEC.
	4		1 SEC.
	3	TIME–IN– STORAGE	0.5 SEC.

TABLE 2–21. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	2	R/L	Run–Length (R/L) bit. This field indicates which of the two formats is used for the fourth Word of the Beacon Message. When this field is set to 0, the fourth Word contains discrete Beacon indicator information. The R/L setting is site adaptable. The default value is 0.
	1	MTI	Moving Target Indicator (MTI) bit. When this field is set to 1, this indicates that the radar reinforced bit is set and the associated Search target was reported with its MTI flag set.
	0	Parity	Odd parity bit (message bit 52).
5	12	MODE 3/A	Bits 12 through 1 contain the Mode 3/A code of the target if the Mode 3/A Valid indicator (Word 1, bit 8) is set to 1; otherwise, these bits contain 0s.
	11	BEACON CODE	
	10		
	9		
	8		
	7		
	6		
	5		
	4		
	3		
	2		
	1		
	0	Parity	Odd parity bit (message bit 65).
6	12	MODE 2	Bits 12 through 1 contain the Mode 2 code of the target if the Mode 2 Valid indicator (Word 1, bit 9) is set to 1; otherwise, these bits contain 0s.
	11	BEACON CODE	
	10		
	9		

TABLE 2–21. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description	
	8		A2	
	7		C4	
	6		A4	
	5		B1	
	4		D1	
	3		B2	
	2		D2	
	1		B4	
	0	Parity	Odd parity bit (message bit 78).	
7	12	MODE C	SIGN	Contains the reported Mode C altitude of the target (in feet) if the Mode C Valid indicator (Word 1, bit 7) is set to 1; otherwise, these bits contain all 0s. This field is encoded as a two's complement number with: MSB = 102, 400 Ft, LSB = 100 Ft
	11	ALTITUDE	102,400 Ft	
	10		51,200 Ft	
	9		25,600 Ft	
	8		12,800 Ft	
	7		6,400 Ft	
	6		3,200 Ft	
	5		1,600 Ft	
	4		800 Ft	
	3		400 Ft	
	2		200 Ft	
	1		100 Ft	
	0	Parity	Odd parity bit (message bit 91).	

TABLE 2–22. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON RTQC MESSAGE

Word	Bit(s)	ID	Bit Description
1	12 thru 10	Message Label	NOTE Mode S reports the Beacon Real Time Quality Control (RTQC) target using the same message format as used for a Beacon target. The Beacon RTQC target is not a real target or a test target, it exists only as a surveillance message. The data for the Beacon RTQC target is site adaptable.
			A 111 in Message Label bits 12 through 10 indicates a Beacon Real Time Quality Control (RTQC) Message is present. Bit 12 is the first message bit.
			MODE 2 This field is set to 1 for Mode 2 reports with all confidence bits high. This field is set to 0 for all other Mode 2 reports.
			MODE 3/A This field indicates the validity of the Mode 3/A code data. (Refer to Beacon Message bit description for Word 1, bit 8.)
			MODE C This field indicates the validity of the Mode S altitude data. (Refer to Beacon Message bit description for Word 1, bit 7.)
			SPI Special Identity (SPI) field. This field is the Special Identity (SPI) bit in the Mode S Sensor. (Refer to Beacon Message bit description for Word 1, bit 6.)
			– In the Beacon RTQC Message, this bit is set to 0.
			– In the Beacon RTQC Message, this bit is set to 0.
			– In the Beacon RTQC Message, this bit is set to 0.
			FAA When this field is set to 1, this indicates FAA as a user.
			AF When this field is set to 1, this indicates a joint-use site.
			Parity Odd parity bit (message bit 13).

TABLE 2–22. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description			
2	12	RANGE	128	NM	Eleven-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.125 NM	
	11		64	NM		
	10		32	NM		
	9		16	NM		
	8		8	NM		
	7		4	NM		
	6		2	NM		
	5		1	NM		
	4		1/2	NM		
	3		1/4	NM		
	2		1/8	NM		
	1	–	In the Beacon RTQC Message, this bit is set to 0.			
0	Parity	Odd parity bit (message bit 26).				
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180°) LSB = 1 ACP (approx. 0.088°)	
	11		1024	ACPs		
	10		512	ACPs		
	9		256	ACPs		
	8		128	ACPs		
	7		64	ACPs		
	6		32	ACPs		
	5		16	ACPs		
	4		8	ACPs		
	3		4	ACPs		
	2		2	ACPs		
	1		AZIMUTH	1		ACP
	0		Parity	Odd parity bit (message bit 39).		

TABLE 2-22. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description			
4	12	BEACON RUN – LENGTH	64	ACPs	For bit description, refer to Beacon Message (Word 4, Bits 12 through 7, R/L bit 2 is set to 1). MSB = 64 ACPs, LSB = 2 ACPs	
	11		32	ACPs		
	10		16	ACPs		
	9		8	ACPs		
	8		4	ACPs		
	7	BEACON RUN – LENGTH	2	ACPs		
	6		TIME –IN – STORAGE	4	SEC.	For bit description, refer to Beacon Message (Word 4, Bits 6 through 3, R/L bit 2 is set to 1). MSB = 4 SEC., LSB = 0.5 SEC.
	5			2	SEC.	
	4			1	SEC.	
	3			0.5	SEC.	
	2	R/L		Run –Length (R/L) bit. In the Beacon RTQC Message, this bit is set to 1.		
	1	–	In the Beacon RTQC Message, this bit is set to 0.			
0	Parity	Odd parity bit (message bit 52).				
5	12	MODE 3/A	A4	Contains the Mode 3/A code of the target if the Mode 3/A Valid indicator (Word 1, bit 8) is set to 1; otherwise, these bits contain 0s.		
	11	BEACON CODE	A2			
	10		A1			
	9		B4			
	8		B2			
	7		B1			
	6		C4	<u>Mode 3/A Beacon Code</u>		
	5	C2				
	4	C1				

TABLE 2–22. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	3		D4
	2		D2
	1		D1
	0	Parity	Odd parity bit (message bit 65).
6	12 thru 1	–	In the Beacon RTQC Message, these bits are set to 0.
7	12 thru 1	–	In the Beacon RTQC Message, these bits are set to 0.
	0	Parity	Odd parity bit (message bit 78).

TABLE 2–23. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON STROBE MESSAGE

Word	Bit(s)	ID	Bit Description
			NOTE
			Beacon Strobe Message generation produces a four word message which is comprised of a message label, user identification, range, azimuth, run-length, and time-in-storage data.
1	12	TEST	In the Beacon Strobe Message, this bit is set to 0. Bit 12 is the first message bit.
	11 thru 3	Message Label	A 001100010 in Message Label bits 11 through 3 indicates a Beacon Strobe Message.
	2	FAA	When this field is set to 1, this indicates FAA as a user.
	1	AF	When this field is set to 1, this indicates a joint-use site.
	0	Parity	Odd parity bit (message bit 13).

TABLE 2–23. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description			
2	12	RANGE	128	NM	Eleven-bit field contains pseudo-range with value in Nautical Miles (NM) coded as a binary integer. MSB = 128 NM, LSB = 0.125 NM	
	11		64	NM		
	10		32	NM		
	9		16	NM		
	8		8	NM		
	7		4	NM		
	6		2	NM		
	5		1	NM		
	4		0.5	NM		
	3		0.250	NM		
	2		0.125	NM		
	1	Not Used	In the Beacon Strobe Message, this bit is set to 0.			
	0	Parity	Odd parity bit (message bit 39).			
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180°) LSB = 1 ACP (approx. 0.088°)	
	11		1024	ACPs		
	10		512	ACPs		
	9		256	ACPs		
	8		128	ACPs		
	7		64	ACPs		
	6		32	ACPs		
	5		16	ACPs		
	4		8	ACPs		
	3		4	ACPs		
	2		2	ACPs		
	1	AZIMUTH	1	ACP		

TABLE 2-23. MODE S SENSOR TO EN ROUTE ATC FACILITY, BEACON STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description	
	0	Parity	Odd parity bit (message bit 39).	
4	12	RUN – LENGTH	256 ACPs	Seven-bit binary integer which defines the azimuth extent of the detected strobe in Azimuth Change Pulses (ACPs). MSB = 256 ACPs, LSB = 4 ACPs
	11		128 ACPs	
	10		64 ACPs	
	9		32 ACPs	
	8		16 ACPs	
	7		8 ACPs	
	6		4 ACPs	
		RUN – LENGTH		
	5	TIME –IN – STORAGE	2 SEC.	Five-bit binary integer which defines the elapsed time in seconds representing the time between the reported azimuth of the data (time of measurement), and the insertion of the message into the channel for delivery. MSB = 2 SEC., LSB = 0.125 SEC.
	4		1 SEC.	
	3		1/2 SEC.	
	2		1/4 SEC.	
	1	TIME –IN – STORAGE	1/8 SEC.	
	0		Parity	Odd parity bit (message bit 52).

TABLE 2-24. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH MESSAGE

Word	Bit(s)	ID	Bit Description	
			NOTE	
			The Search Message contains Search reports that are not merged with Beacon reports.	
1	12	TEST	When this field is set to 1, this indicates the presence of a test report. Bit 12 is the first message bit.	
	11 thru 3	Message Label	A 001101100 in Message Label bits 11 through 3 indicates a Search Message.	

TABLE 2–24. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	2	FAA	When this field is set to 1, this indicates FAA as a user.
	1	AF	When this field is set to 1, this indicates a joint-use site.
	0	Parity	Odd parity bit (message bit 13).
2	12	RANGE	128 NM Eleven-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.125 NM
	11		64 NM
	10		32 NM
	9		16 NM
	8		8 NM
	7		4 NM
	6		2 NM
	5		1 NM
	4		1/2 NM
	3		1/4 NM
	2	RANGE	1/8 NM
	1	Not Used	In the Search Message, this field is set to 0.
	0	Parity	Odd parity bit (message bit 26).

TABLE 2–24. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180°) LSB = 1 ACP (approx. 0.088°)
	11		1024	ACPs	
	10		512	ACPs	
	9		256	ACPs	
	8		128	ACPs	
	7		64	ACPs	
	6		32	ACPs	
	5		16	ACPs	
	4		8	ACPs	
	3		4	ACPs	
	2		2	ACPs	
	1		1	ACP	
	0	Parity	Odd parity bit (message bit 39).		
4	12	AIMS PRES	AIMS Present bit. When This field is set to 1, this indicates that ATCRBS Improved Mark IV System is present.		
	11,10	Not Used	In the Search Message, these fields are set to 0.		
	9	RUN – LENGTH	16	ACPs	Run –Length bits in Azimuth Change Pulses (ACPs). MSB = 16 ACPs, LSB = 4 ACPs
	8		8	ACPs	
	7		4	ACPs	
	6	TIME – IN – STORAGE	4	SEC.	Time – in – Storage bits in seconds.
	5		2	SEC.	MSB = 4 SEC., LSB = 0.25 SEC.
	4		1	SEC.	
	3		1/2	SEC.	
	2		1/4	SEC.	
	1	MTI	Moving Target Indicator (MTI) bit. This field is set to 1 when a Search target is reported with the MTI flag set.		
	0	Parity	Odd parity bit (message bit 52).		

TABLE 2–25. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH RTQC MESSAGE

Word	Bit(s)	ID	Bit Description
1			NOTE
			The Search Real Time Quality Control (RTQC) Message is used to send Search RTQC reports (one message per scan). The Search RTQC Message is transmitted (after receipt from the CD–2) when the current antenna azimuth equals the azimuth in the message plus a site adaptable bias whose default is 1536 Azimuth units (Au).
	12	Not Used	In the Search RTQC Message, this field is set to 1. Bit 12 is the first message bit.
	11 thru 3	Message Label	A 001001000 in the Message Label bits 11 through 3 indicates a Search RTQC Message is present.
	2	FAA	When this field is set to 1, this indicates the FAA as a user.
	1	AF	When this field is set to 1, this indicates a joint-use site.
	0	Parity	Odd parity bit (message bit 13).
2	12	RANGE	128 NM
	11		64 NM
	10		32 NM
	9		16 NM
	8		8 NM
	7		4 NM
	6		2 NM
	5		1 NM
	4		1/2 NM
	3		1/4 NM
	2		1/8 NM
	1	Not Used	This field is not used and is set to 0.
	0	Parity	Odd parity bit (message bit 26).

TABLE 2–25. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180°) LSB = 1 ACP (approx. 0.088°)
	11		1024	ACPs	
	10		512	ACPs	
	9		256	ACPs	
	8		128	ACPs	
	7		64	ACPs	
	6		32	ACPs	
	5		16	ACPs	
	4		8	ACPs	
	3		4	ACPs	
	2		2	ACPs	
	1	AZIMUTH	1	ACP	
	0	Parity	Odd parity bit (message bit 39).		
4	12	RUN – LENGTH	256	ACPs	Run –Length bits in Azimuth Change Pulses (ACPs).
	11		128	ACPs	MSB = 256 ACPs, LSB = 4 ACPs
	10		64	ACPs	
	9		32	ACPs	
	8		16	ACPs	
	7		8	ACPs	
	6	RUN – LENGTH	4	ACPs	

TABLE 2–25. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	5	TIME–IN–STORAGE	2 SEC. Time–In–Storage bits in seconds.
	4		1 SEC. MSB = 2 SEC., LSB = 0.125 SEC.
	3		1/2 SEC.
	2		1/4 SEC.
	1		1/8 SEC.
	0	Parity	Odd parity bit (message bit 52).

TABLE 2–26. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH STROBE MESSAGE

Word	Bit(s)	ID	Bit Description
			NOTE
			The Search Strobe Message data fields are set by the CD–2.
1	12	TEST	When this field is set to 1, this indicates a test report. Bit 12 is the first message bit.
	11 thru 3	Message Label	A 001100001 in Message Label bits 11 through 3 indicates a Search Strobe Message.
	2	FAA	When the field is set to 1, this indicates FAA as a user.
	1	AF	When this field is set to 1, this indicates a joint-use site.
	0	Parity	Odd parity bit (message bit 13).

TABLE 2–26. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
2	12	RANGE	128	NM	Eleven-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.125 NM
	11		64	NM	
	10		32	NM	
	9		16	NM	
	8		8	NM	
	7		4	NM	
	6		2	NM	
	5		1	NM	
	4		1/2	NM	
	3		1/4	NM	
	2		1/8	NM	
	1	Not Used	This field is not used and is set to 0.		
	0	Parity	Odd parity bit (message bit 26).		
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180°) LSB = 1 ACP (approx. 0.088°)
	11		1024	ACPs	
	10		512	ACPs	
	9		256	ACPs	
	8		128	ACPs	
	7		64	ACPs	
	6		32	ACPs	
	5		16	ACPs	
	4		8	ACPs	
	3		4	ACPs	
	2		2	ACPs	
	1		AZIMUTH	1	

TABLE 2–26. MODE S SENSOR TO EN ROUTE ATC FACILITY, SEARCH STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	0	Parity	Odd parity bit (message bit 39).
4	12	RUN – LENGTH	256 ACPs Run–Length bits in Azimuth Change Pulses (ACPs).
	11		128 ACPs MSB = 256 ACPs, LSB = 4 ACPs
	10		64 ACPs
	9		32 ACPs
	8		16 ACPs
	7		8 ACPs
	6	RUN – LENGTH	4 ACPs
	5	TIME–IN– STORAGE	2 SEC. Time–In–Storage bits in seconds.
	4		1 SEC. MSB = 2 SEC., LSB = 0.125 SEC.
	3		1/2 SEC.
	2		1/4 SEC.
	1	TIME–IN– STORAGE	1/8 SEC.
	0	Parity	Odd parity bit (message bit 52).

TABLE 2–27. MODE S SENSOR TO EN ROUTE ATC FACILITY, WEATHER MAP MESSAGE

Word	Bit(s)	ID	Bit Description
1			NOTE
	12	TEST	The Weather Map Message data fields are set by the CD–2. When this field is set to 1, this indicates a test report. Bit 12 is the first message bit.
	11 thru 6	Message Label	A 000000 in Message Label bits 11 through 6 indicates a Weather Map Message.

TABLE 2-27. MODE S SENSOR TO EN ROUTE ATC FACILITY, WEATHER MAP MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	5	WEATHER MAP	A 101 in bits 5 through 3 indicates heavy weather.
	4		A 110 in bits 5 through 3 indicates light weather.
	3		
	2	FAA	When this field is set to 1, this indicates FAA as a user.
	1	AF	When this field is set to 1, this indicates a joint-use site.
	0	Parity	Odd parity bit (message bit 13).
2	12	RANGE (START)	128 NM
	11		64 NM
	10		32 NM
	9		16 NM
	8		8 NM
	7		4 NM
	6		2 NM
	5		1 NM
	4		1/2 NM
	3		1/4 NM
	2,1	Not Used	These fields are not used and are set to 0.
	0	Parity	Odd parity bit (message bit 26).

TABLE 2–27. MODE S SENSOR TO EN ROUTE ATC FACILITY, WEATHER MAP MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
3	12	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180_) LSB = 1 ACP (approx. 0.088_)
	11		1024	ACPs	
	10		512	ACPs	
	9		256	ACPs	
	8		128	ACPs	
	7		64	ACPs	
	6		32	ACPs	
	5		16	ACPs	
	4		8	ACPs	
	3		4	ACPs	
	2		2	ACPs	
	1		1	ACPs	
	0	Parity	Odd parity bit (message bit 39).		
4	12	RANGE (STOP)	128	NM	Ten-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.25 NM
	11		64	NM	
	10		32	NM	
	9		16	NM	
	8		8	NM	
	7		4	NM	
	6		2	NM	
	5		1	NM	
	4		1/2	NM	
	3		1/4	NM	
	2,1	Not Used	These fields are not used and are set to 0.		
	0	Parity	Odd parity bit (message bit 52).		

TABLE 2–28. MODE S SENSOR TO EN ROUTE ATC FACILITY, CD–2/MODE S STATUS MESSAGE

Word	Bit(s)	ID	Bit Description
1			NOTE The CD–2/Mode S Status Message is used to send the status of Mode S and the search (primary) radar.
	12	TEST	When this field is set to 1, this indicates a test report. Bit 12 is the first message bit.
	11 thru 3	Message Label	A 000110000 in Message Label bits 11 through 3 indicates a CD–2/Mode S Status Message.
	2	FAA	When this field is set to 1, this indicates FAA as a user.
	1	AF	When this field is set to 1, this indicates a joint-use site.
	0	Parity	Odd parity bit (message bit 13).
2	12	RADAR CHAN	Radar Channel bit. When this field is set to 1, this indicates that channel “A” is selected.
	11	BEACON CHAN	Beacon Channel bit. When this field is set to 1, this indicates that channel “A” is selected (Mode S generated).
	10	CD–2 ALARM	When this field is set to 1, this indicates a CD–2 processing alarm condition.
	9	BEACON ALARM O/L	On-line Beacon Alarm bit. When this field is set to 1, this indicates that the active Mode S channel is in a “yellow” condition.
	8	BEACON OFFSET	This field is site adaptable, the default condition is 0. When this field is set to 1, this condition causes the special formatting of Search Messages and Beacon Messages to be invoked. The requirements for this format processing are discussed in the following text. This format processing is not intended to be used during capacity loading of the Sensor. Therefore, additional Sensor load that results when this processing is active, shall not be considered as part of the capacity of the Sensor. The following conditions will be performed when the Beacon Offset field bit 8 is set to 1:

TABLE 2–28. MODE S SENSOR TO EN ROUTE ATC FACILITY, CD–2/MODE S STATUS MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
			a. Formatting will add 0.5 NM to the range of all output Beacon Report Messages. b. Formatting will provide messages for all Search targets that are received by the Sensor. This condition requires that the Search target reports, which are normally lost during radar beacon correlation, be retained for formatting purposes. Search Messages that are produced under these conditions will not be released any sooner than or later than any other Search Messages that were produced under normal processing conditions. The Beacon Offset process allows the current ATC facilities to use their current data extraction and data reduction programs. These programs are used during facility checkout.
	7	AIMS ALARM	ATCRBS Identification Friend or Foe, <u>Mark XII System</u> (AIMS Alarm). This field is set to 0.
	6	POLAR	Antenna Polarization bit. When this field is set to 1, this indicates Circular Polarization (CP).
	5	BEACON ALARM S/B	Standby (S/B) Beacon Alarm bit. This field is set to 1 when the standby Mode S channel is in a “red” condition.
	4	RBPM ALARM O/L	On-line Radar Beacon Performance Monitor (RBPM) Alarm bit. When this field is set to 1, this indicates an alarm condition on a status line from the beacon radar.
	3	SERV ALARM O/P	CD–2 CIM’s Output Service Alarm bit. When this field is set to 1, this indicates an alarm condition from the CD–2 Control Interface Module (CIM).
	2	HT ALARM	CD–2 Military Interface Group (MIG) Request Line Alarm bit. When this field is set to 1, this indicates a CD–2 MIG request line parity or request line failure alarm condition.
	1	OH ALARM	CD–2 Overheat Alarm bit. When this field is set to 1, this indicates an overheated CD–2 module or CD–2 cabinet alarm condition.
	0	Parity	Odd parity bit (message bit 26).

TABLE 2–28. MODE S SENSOR TO EN ROUTE ATC FACILITY, CD–2/MODE S STATUS MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
3	12	MIL TIM ALARM	Military Timing Alarm bit. When this field is set to 1, this indicates a Military output service alarm condition or a Military modem alarm condition.
	11	MIG ALARM	Military Interface Group (MIG) Alarm bit. When this field is set to 1, this indicates that the Sensor is reporting a “red” condition. This bit set to 1 indicates an AIMS azimuth, range, or pretrigger alarm condition; a MIG buffer overload alarm condition; or a MIG half-scan inhibit alarm condition.
	10	BEACON RTQC ALARM	Beacon Real Time Quality Control (RTQC) Alarm bit. When this field is set to 1, this indicates the Sensor is reporting a “red” condition. This bit set to 1 indicates any out-of-tolerance condition for the Beacon RTQC target.
	9	SEARCH RTQC ALARM	Search Real Time Quality Control (RTQC) Alarm bit. When this field is set to 1, this indicates any out-of-tolerance condition for the Search RTQC target.
	8	RANGE ALARM	When this field is set to 1, this indicates that a CD–2 Beacon Reply Converter (BRC) range word generator or a Search Target Extractor (STE) range word generator alarm condition is present.
	7,6	Not Used	These bits are not used and are set to 0.
	5	SEARCH MAX RLD ON	Search Maximum Run–Length Discrimination (RLD) ON/OFF status bit. When this bit is set ON, this enables the RLD feature in the CD–2 Search Target Extractor (STE) circuit.
	4	SEARCH MIN RLD ON	Search Minimum Run–Length Discrimination (RLD) ON/OFF status bit. When this bit is set ON, this enables the RLD feature in the CD–2 Search Target Extractor (STE) circuit.
	3	AZ/SYNC ALARM	Azimuth/Synchronization Alarm bit. When this bit is set to 1, this indicates the presence of one of the following alarm conditions: (a) Beacon Reply Converter (BRC) azimuth word generator alarm, (b) Search Target Extractor (STE) azimuth word generator alarm, (c) external azimuth data source alarm, or (d) internal Azimuth Data Converter (ADC) alarm (if the ADC is the source of azimuth data).
	2	USAF MAP ON	When this field is set to 1, this indicates enabling of one or more USAF map sectors in the CD–2 Military Interface Group (MIG).
	1	Not Used	This bit is not used and is set to 0.
	0	Parity	Odd parity bit (message bit 39).

TABLE 2–28. MODE S SENSOR TO EN ROUTE ATC FACILITY, CD–2/MODE S STATUS MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
4	12	WEA ON	Weather On bit. When set “on”, this bit enables the weather contouring function in the CD–2. The weather contouring function is reset whenever weather reporting is inhibited by the CD–2 CIM’s overload alarm condition.
	11	WEA ALARM	When this field is set to 1, this condition indicates that a weather contouring function alarm exists when the weather on function is enabled (refer to Word 4, bit 12).
	10	Not Used	This field is not used and is set to 0.
	9	FAA MO-DEM ALARM	FAA Modem alarm bit. When this field is set to 1, this indicates failure of a modem clock signal for any operational CD–2 CIM output data channel.
	8	HS INHIB ALARM	CD–2 Half–Scan (HS) Inhibit Alarm bit. When this field is set to 1, this indicates that there is “old data” in the CD–2 CIM (or “old data” in the MIG for the USAF status).
	7	BUFF OL ALARM	CD–2 Buffer Overload (OL) Alarm bit. When this field is set to 1, this indicates a CD–2 throttling condition on the surveillance lines.
	6,5,4	Not Used	These bits are not used and are set to 0 in the Status Message.
	3	FAA CHAN ON	When this field is set to 1, this indicates the enabling of FAA data channel 3.
	2		When this field is set to 1, this indicates the enabling of FAA data channel 2.
	1		When this field is set to 1, this indicates the enabling of FAA data channel 1.
	0	Parity	Odd parity bit (message bit 52).

2.3.8 Mode S Sensor-To-ATC Terminal Surveillance Link Messages (for sites with ASR7 or ASR8)

For terminal sites equipped with ASR7 or ASR8 primary radar, the Mode S Sensor sends Beacon (only) surveillance messages to the ATC terminal (ARTS-2A) facility. This surveillance link is used for transmitting the following message types: (1) beacon message, (2) beacon RTQC message, (3) search RTQC message, and (4) Mode S Status message. Surveillance message words contain 13 bits each, with 12 data bits (bits 12 through 1) and an odd parity bit (bits 0). The data contents of each of these four message types is given in Tables 2-29 through 2-31. The beacon targets are both ATCRBS and Mode S, and these target reports are used to generate Beacon messages (Table 2-29). The same format is used for Beacon RTQC messages. The beacon RTQC target (for terminal sites with ASR7 or ASR8 radar) exists only as a surveillance message. The data for the beacon RTQC message is site adaptable.

2.3.9 Enroute Primary Radar Digitizer (CD-2) to Mode S Sensor Messages

At a Mode S Sensor en route system (refer to Figure 1-2), the primary radar common digitizer (CD-2) sends the following radar data message types to Mode S for correlation with beacon data: (1) search messages, (2) search real time quality control (RTQC) messages, (3) status messages, (4) weather map messages, and (5) search strobe messages. For information concerning primary radar common digitizer (CD-2)-to-Mode S Sensor message formats, message labels, and message descriptions; refer to Table 2-32 through 2-36.

2.3.10 Terminal ASR-9 Primary Radar to/from Mode S Sensor Messages

At a Mode S Sensor terminal system (refer to Figure 1-3), the ASR-9 post processor-to-Mode S link allows for the transfer of uplink and downlink messages. The ASR-9 sends (uplink) primary radar target reports to Mode S. Mode S correlates the radar reports with the Mode S Beacon reports. The merged results are returned to the ASR-9 (down link) for subsequent transmittal to ATC facilities. The messages sent from the ASR-9 post processor to the Mode S Sensor are as follows: (1) radar target reports (RTGT), (2) search real time quality control (RTQC) messages, (3) ASR-9 system status messages and (4) the health check message. The use of the health check is described in paragraph 2.3.10.1. For information concerning ASR-9-to-Mode S Sensor (uplink) message formats, message labels, and message descriptions; refer to Tables 2-37 through 2-40. The messages sent from the Mode S Sensor to the ASR-9 post processor are as follows: (1) radar target reports that did not merge with a Mode S beacon report, (2) search RTQC messages, (3) beacon only messages, (4) radar/beacon merged messages, (5) beacon RTQC messages, (6) system status messages and (7) the health check message return. For information concerning Mode S sensor-to-ASR-9 (down link) message formats, message labels, and message descriptions, refer to Table 2-37 for radar target report information (unchanged message format for uplink and downlink messages). Refer to Table 2-38 for search RTQC message information (unchanged message format for uplink and downlink messages). Refer to Table 2-40 for information concerning the message formats for the uplink and downlink health check messages. Table 2-41 provides information concerning downlink message formats message labels, and message descriptions for beacon only messages, radar/beacon merged messages, and beacon RTQC messages. Refer to Table 2-39 for information concerning ASR-9-to-Mode S Sensor (uplink) system status messages which are the same as system status downlink messages except for Word 4 as indicated in the following list:

Word	Bit	Description
4	49	Mode S Channel A Unavailable
4	64	Mode S Channel B Unavailable
4	63	Mode S Channel A Front End Interrogator Summary Alarm
4	62	Mode S Channel B Front End Interrogator Summary Alarm
4	61	Mode S Channel A Comm Subsystem Summary Alarm
4	60	Mode S Channel B Comm Subsystem Summary Alarm
4	59	Mode S Channel A Computer Subsystem Summary Alarm
4	58	Mode S Channel B Computer Subsystem Summary Alarm
4	57	Mode S Channel A or B Maintenance Alarm

2.3.10.1 Health Check Message. The Health Check message is used to facilitate prompt corrective action for certain interface failures on the ASR-9 and Mode S link. At odd-sector intervals, the health check message is sent from the ASR-9 to the DPS where it is looped back to the ASR-9. The on-line MIP in the ASR-9 detects the return of each health check. If three consecutive health checks are not returned to the ASR-9, corrective action is initiated. To restore the link to operational status, the ASR-9 will either switch its post processors or force the Mode S to reconfigure itself. First, the ASR-9 switches its post processors if a switch can be made. If the link failure persists, the ASR-9 triggers Mode S reconfiguration by holding both uplink discrete lines low (an action referred to as an ASR-9 “kick” command). These two discrete lines form the ASR-9 on line/no-fault logic. When both lines are in the same state for more than 150 milliseconds, the Mode S detects an alarm condition and reconfigures the Mode S/ASR-9 link by performing either an IOP switch or, if necessary, a DPS channel switch. (When a “kick” occurs, one of the Mode S Maintenance IOPs should report at least one 1155 trap with line status 05.) If a working path is found, receipt of three consecutive health check messages (a return-to-normal condition) will halt the reconfiguration actions. If no working path is found, successive reconfigurations will result in Mode S going into backup mode.

2.3.11 Mode S Interrogator/DPS Interface

The interface between the DPS and the Interrogator is designed to support the most complex sensor configuration (two PRCSS cabinets and three Interrogator channels). Therefore, each PRCSS is designed with three separate sets of drivers and receivers, one set for each possible front end. Also, each Interrogator has two separate receiver/driver sets to interface two DPS Channels.

TABLE 2–29. MODE S SENSOR TO TERMINAL ATC FACILITY, BEACON MESSAGE OR BEACON RTQC MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR)

Word	Bit(s)	ID	Bit Description
1	12 thru 10	Message Label	0 1 1 indicates a Beacon Message. 1 1 1 indicates a Beacon RTQC message
	9	Mode 2	Set to 1 for Mode 2 reports with all confidence bits high: set to 0 for all other Mode 2 reports, and non Mode 2 reports.
	8 and 7	Mode 3/A Validation	This field indicates the validity of the Mode 3/A code data. For ATCRBS targets, if the sensor indicates Mode 3/A code confidence as all high confidence, then the data is valid, and this field is set to all 1's (11). If the code confidence is not all high confidence, or if the Mode 3/A code data is not present, then the data is not valid, and this field is set to all 0's (00). The sensor may replace the Mode 3/A code of an ATCRBS report with the Mode 3/A code from a track. When this happens, the Mode 3/A confidence bits from the track are used, instead of the report code confidence bits, to validate the report code according to the above ATCRBS target rules. For Mode S targets, the ATCRBS 3/A code is acquired by the sensor in a different way than for ATCRBS targets. As long as there is a 3/A code in the surveillance file for a Mode S target, that code is present and valid, and this field is set to all 1's. If the code is not present, this field is set to all 0's.
	6	SPI	This field is the SPI bit in the Mode S sensor. For ATCRBS targets, this is set to 1 when the SPI bit is 1 with high confidence. For Mode S targets, this is set to 1 when the FS field in Mode S reply indicates the identify (SPI) function is set in the transponder. Otherwise this field is set to 0.
	5	R/R (Radar Reinforced)	When set to 1, indicates that a radar report reinforced the beacon data (no radar data is reported). (Not applicable at these sites.)
	4	7700	When set to 1, indicates a Mode 3/A aircraft emergency.
	3	7600	When set to 1, indicates a Mode 3/A aircraft radio failure.
	2	FAA	Used to report Mode 3/A "x" bit.
	1	AF	When set to 1 in a Beacon RTQC Message, indicates an RTQC sector synchronization.

TABLE 2–29. MODE S SENSOR TO TERMINAL ATC FACILITY, BEACON MESSAGE OR BEACON RTQC MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR) (Continued)

Word	Bit(s)	ID	Bit Description
2	12 thru 1	Terminal Range	Twelve-bit binary integer that defines range in Nautical Miles (NM). MSB = 32NM LSB = 1/64NM
3	12 thru 1	Azimuth	Twelve-bit binary integer defining azimuth in Azimuth Change Pulses (ACPs), measured from true north. MSB = 2048 (180_) LSB = 1 (approximately .088_) If message is an RTQC sector synchronization message, azimuth setting represents the antenna position. The values are every 11.25_ increment (0_, 11.25_, 22.5_, ...) from north (zero) to 348.75_. The MSB of this field is 180_.
4	12,11, and 10	ARTS –IIIA Quality	<u>Condition</u> <u>Setting</u> Mode 3/A Validation 111 = all 1's
			Mode 3/A Validation 011 = zero
	9	DISC (Discrete)	When set to 1, indicates report is based on replies with discrete Mode 3/A codes.
	8	Mode 2 “X”	When set to 1, indicates valid Mode 2 “X” responses.
	7 thru 3	Beacon Hit Count	This field is an unsigned binary integer, the settings are given in decimal. The settings of this field and related conditions are listed below. <u>ATCRBS Target Reports:</u> a. The target report has all high confidence bits that match all confidence bits in the track file code, transmitted with beacon hit count = 20. b. All high confidence bits that have not been correlated with an existing track, transmitted with beacon hit count = 16. c. Some low confidence bits, but all high confidence bits match the high confidence bits in the track file code, transmitted with beacon hit count = 18. d. The code in transition bit set, transmitted with beacon hit count = 8. e. All other ATCRBS reports transmitted with beacon hit count = 7.

TABLE 2–29. MODE S SENSOR TO TERMINAL ATC FACILITY, BEACON MESSAGE OR BEACON RTQC MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR) (Continued)

Word	Bit(s)	ID	Bit Description
	2 and 1	Mode C Validation	<u>Mode S Target Reports:</u> a. Mode S only target report, beacon hit count = 21. b. Mode S, ATCRBS reinforced, beacon hit count = 28. c. Mode S, Illegal Mode S ID, beacon hit count = 30. This field indicates the validity of the Mode S altitude data. For ATCRBS reports, if the sensor has successfully decoded the Mode C data for a report, then the data is valid, and this field is set to 11. If the sensor has not successfully decoded the Mode C data, or if the Mode C data is not present, then the data is not valid, and this field is set to 00.
5	12 thru 1	Mode 3/A Beacon Code	Contains the Mode 3/A code of the target if the Mode 3/A Valid indicator is set to 11, otherwise it contains 0's.
6	12 thru 1	Mode 2 Beacon Code	Contains the Mode 2 code of the target if the Mode 2 indicator is set to 1, otherwise it contains 0's.
7	12 thru 1	Mode C Altitude	Contains the reported Mode C altitude of the target (in feet) if the Mode C Valid indicator is set to 11; otherwise it is set to all 0's. It is encoded as a two's complement number with: MSB = 102,400 and LSB = 100. Bit 12 is the sign bit where 0 = + and 1 = –.

TABLE 2–30. MODE S SENSOR TO TERMINAL ATC FACILITY, SEARCH RTQC MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR)

Word	Bit(s)	ID	Bit Description
1	12	Not Used	Set to 1.
	11 thru 3	Message Label	0 0 1 0 0 1 0 0 0 indicates a Search RTQC Message.
	2	FAA	Set to 0.
	1	AF	Set to 1, indicating a sector synchronization message.
2	12 thru 1	Terminal Range	Set to 0, Terminal range represents the range of the search target in nautical miles. LSB = 1/64 nm.
3	12 thru 1	Azimuth	Twelve-bit binary integer defining azimuth in Azimuth Change Pulses (ACPs), measured from true north. MSB = 2048 (180_) and LSB = 1 (approx. 0.088_)
4	12 and 11	Radar Quality	Set to 0
	10 thru 8	Radar Confidence	Set to 0
	7 and 6	Tracking Eligibility	Set to 0
	5 thru 3	ARTS–IIIA Radar Quality	Set to 0
	2	Spare	Set to 0
	1	Scan to Scan Correlated	Set to 0

TABLE 2–31. MODE S SENSOR TO TERMINAL ATC FACILITY, MODE S STATUS MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR)

Word	Bit(s)	ID	Bit Description
NOTE The Mode S Status Message is used to send the status of Mode S. The setting of the beacon status is discussed below. Since there is no primary radar to Mode S connectivity in this configuration, there will be no radar status reports, the search status fields are set to indicate failure of the radar as shown below.			
1	12	TEST	When set to 1, indicates a test report.
	11 thru 3	Message Label	0 0 0 1 1 0 0 0 0 indicates a Status Message.
	2	Not Used	Set to 0.
	1	Not Used	Set to 0.
2	12	Channel On–Line	Set to 0.
	11	Channel A Alarm	Set to 0.
	10	Channel A Unavailable	Set to 0.
	9	Channel B Alarm	Set to 0.
	8	Channel B Unavailable	Set to 0.
	7	Post Proc. Online	Set to 0.
	6	Post Proc. A Alarm	Set to 1.
	5	Post Proc. A Unavailable	Set to 1.
	4	Post Proc. B Alarm	Set to 1.
	3	Post Proc. B Unavailable	Set to 1.

TABLE 2–31. MODE S SENSOR TO TERMINAL ATC FACILITY, MODE S STATUS MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR) (Continued)

Word	Bit(s)	ID	Bit Description
	2	WX Proc. On Line	Set to 0.
	1	WX Channel Alarm	Set to 0.
3	12	High Voltage On	Set to 0.
	11	Antenna Polarization	Set to 0.
	10	Antenna Rotation	Set to 0.
	9	SCIP On–Line	Set to 0.
	8	SCIP Chan A Alarm	Set to 0.
	7	SCIP Chan B Alarm	Set to 0.
	6	Maintenance Alarm	Set to 0.
	5	Comm Link A Alarm	Set to 0.
	9	Comm Link B Alarm	Set to 0.
	3	RMS Alarm	Set to 0.
	2 and 1	Spare	Set to 0.
4	12	Beacon Channel On–Line	When set to 1 indicates channel A is active, when set to 0 indicates channel B is active.
	11	Beacon Channel A Alarm	Set to 1 when channel A is in red condition.

TABLE 2–31. MODE S SENSOR TO TERMINAL ATC FACILITY, MODE S STATUS MESSAGE (FOR SITES WITH ASR7 OR ASR8 RADAR) (Continued)

Word	Bit(s)	ID	Bit Description
	10	Beacon Channel B Alarm	Set to 1 when channel B is in red condition
	9 thru 1	Reserved for Mode S	9 Mode S Channel A unavailable 8 Mode S Channel B unavailable 7 Mode S Ch. A Interrogator Summary Alarm 6 Mode S Ch. B Interrogator Summary Alarm 5 Mode S Ch. A Comm Subsystem Summary Alarm 4 Mode S Ch. B Comm Subsystem Summary Alarm 3 Mode S Ch. A Computer Subsystem Summary Alarm 2 Mode S Ch. B Computer Subsystem Summary Alarm 1 Mode S Ch. A or B Maintenance Alarm

TABLE 2–32. CD–2 TO MODE S SENSOR, SEARCH MESSAGE

Word	Bit(s)	ID	Bit Description
1	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	TEST	When set to 1, indicates a test report.
	10 thru 2	Message Label	A 001101100 in Message Label bits 10 through 2 indicates a Search Message.
	1	FAA	When set to 1, indicates FAA as a user.
	0	AF	When set to 1, indicates a joint-use site.

TABLE 2–32. CD–2 TO MODE S SENSOR, SEARCH MESSAGE (CONT.)

Word	Bit(s)	ID	Bit Description
2	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	RANGE	128 NM Thirteen-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.03125 NM
	11		
	10		
	9		
	8		
	7		
	6		
	5		
	4		
	3		
	2		
	1		
	0		
		RANGE	1/32 NM
3	15	X	Don't Care Condition
	14	X	Don't Care Condition

TABLE 2-32. CD-2 TO MODE S SENSOR, SEARCH MESSAGE (CONT.)

Word	Bit(s)	ID	Bit Description		
	13	AZIMUTH	8192	ACPs	Fourteen-bit binary integer defining azimuth in Improved Azimuth Change Pulses (IACPs), measured from true North. MSB = 8192 ACPs (180_), LSB = 1 ACP (approx. 0.022_)
	12		4096	ACPs	
	11		2048	ACPs	
	10		1024	ACPs	
	9		512	ACPs	
	8		256	ACPs	
	7		128	ACPs	
	6		64	ACPs	
	5		32	ACPs	
	4		16	ACPs	
	3		8	ACPs	
	2		4	ACPs	
	1		2	ACPs	
	0	AZIMUTH	1	ACP	
4	15	X	Don't Care Condition		
	14	X	Don't Care Condition		
	13	X	Don't Care Condition		
	12	X	Don't Care Condition		
	11	Not Used	Set to 0.		
	10	Not Used	Set to 0.		
	9	Not Used	Set to 0.		
	8	RUN – LENGTH (ACP)	16	ACPs	Three-bit binary integer which defines run-length in Azimuth Change Pulses (ACPs). MSB = 16 ACPs, LSB = 4 ACPs
	7	RUN – LENGTH	8	ACPs	

TABLE 2–32. CD–2 TO MODE S SENSOR, SEARCH MESSAGE (CONT.)

Word	Bit(s)	ID	Bit Description
	6	RUN LENGTH	4 ACPs
	5	Not Used	Set to 0.
	4	Not Used	Set to 0.
	3	Not Used	Set to 0.
	2	Not Used	Set to 0.
	1	Not Used	Set to 0.
	0	MTI Flag	Set to 1, when Search target reported with MTI flag set.

TABLE 2–33. CD–2 TO MODE S SENSOR, SEARCH RTQC MESSAGE

Word	Bit(s)	ID	Bit Description
1	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	Not Used	
	10 thru 2	Message Label	A 001001000 in Message Label bits 10 through 2 indicates a Search RTQC Message.
	1	FAA	When set to 1, indicates FAA as a user.
	0	AF	When set to 1, indicates a joint-use site.
2	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition

TABLE 2-33. CD-2 TO MODE S SENSOR, SEARCH RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description	
	11	RANGE	128 NM	Eleven-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.125 NM
	10		64 NM	
	9		32 NM	
	8		16 NM	
	7		8 NM	
	6		4 NM	
	5		2 NM	
	4		1 NM	
	3		1/2 NM	
	2		1/4 NM	
	1	RANGE	1/8 NM	
	0	Not Used	Set to 0.	
3	15	X	Don't Care Condition	
	14	X	Don't Care Condition	
	13	X	Don't Care Condition	
	12	X	Don't Care Condition	

TABLE 2–33. CD–2 TO MODE S SENSOR, SEARCH RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
	11	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs, LSB = 1 ACP
	10		1024	ACPs	
	9		512	ACPs	
	8		256	ACPs	
	7		128	ACPs	
	6		64	ACPs	
	5		32	ACPs	
	4		16	ACPs	
	3		8	ACPs	
	2		4	ACPs	
	1		2	ACPs	
	0	AZIMUTH	1	ACP	
4	15	X	Don't Care Condition		
	14	X	Don't Care Condition		
	13	X	Don't Care Condition		
	12	X	Don't Care Condition		
	11	RUN – LENGTH	256	ACPs	Seven-bit binary integer which defines run-length in Azimuth Change Pulses (ACPs). MSB = 256 ACPs, LSB = 4 ACPs
	10		128	ACPs	
	9		64	ACPs	
	8		32	ACPs	
	7		16	ACPs	
	6		8	ACPs	
	5		4	ACPs	
		RUN – LENGTH			
	4	Not Used	Set to 0.		
	3	Not Used	Set to 0.		

TABLE 2–33. CD–2 TO MODE S SENSOR, SEARCH RTQC MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	2	Not Used	Set to 0.
	1	Not Used	Set to 0.
	0	Not Used	Set to 0.

TABLE 2–34. CD–2 TO MODE S SENSOR, STATUS MESSAGE

Word	Bit(s)	ID	Bit Description
1	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	TEST	When set to 1, indicates a test report.
	10 thru 2	Message Label	A 000110000 in Message Label bits 10 through 2 indicates a Status Message.
	1	FAA	When set to 1, indicates FAA as a user.
	0	AF	When set to 1, indicates a joint-use site.
2	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	RADAR CHAN	When set to 1, indicates channel A is selected.
	10	BEACON CHAN	When set to 1, indicates channel A is selected (Mode S generated).
	9	CD–2 ALARM	When set to 1, indicates an alarm condition.
	8	BEACON ALARM	On-line Beacon Alarm. This field is set to 1 when the active Mode S channel is in a “yellow” condition.

TABLE 2–34. CD–2 TO MODE S SENSOR, STATUS MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	7	BEACON OFFSET	This field is site adaptable, the default condition is set to 0. When this field is set to 1, this condition causes the special formatting of Search Messages and Beacon Messages to be invoked. The requirements for this processing are discussed in the following text. Beacon Offset processing is not intended to be used during capacity loading of the Sensor. Therefore, additional Sensor load which results when this processing is active will not be considered as part of the capacity of the Sensor. The following events will occur when the Beacon Offset field is set to 1: a. Formatting will add 0.5 NM to the range of all output Beacon Report Messages. b. Formatting will provide messages for all Search targets that are received by the Sensor. This condition requires that the Search target reports, which are normally lost during radar beacon correlation, be retained for formatting purposes. Search Messages that are produced under these conditions will not be released any sooner than or later than any other Search Messages that were produced under normal processing conditions. The Beacon Offset process allows the current ATC facilities to use their current data extraction and data reduction programs. These programs are used during facility checkout.
	6	AIMS ALARM	<u>A</u> ir Traffic Control Radar Beacon System, <u>I</u> dentification Friend or <u>F</u> oe, <u>M</u> ark XII <u>S</u> ystem (AIMS) Alarm bit is set to 0.
	5	POLAR	Antenna Polarization bit. When this field is set to 1, this indicates Circular Polarization (CP).
	4	BEACON ALARM S/B	Standby (S/B) Beacon Alarm bit. This field is set to 1 when the standby Mode S channel is in a “red” condition.
	3	RBPM ALARM O/L	CD–2 On-line Radar Beacon Performance Monitor (RBPM) Alarm bit. When this field is set to 1, this indicates an alarm condition.
	2	SERV ALARM O/P	CD–2 CIM’s Output Service Alarm bit. When this field is set to 1, this indicates an alarm condition.
	1	HT ALARM	CD–2 Military Interface Group (MIG) Request Line Alarm bit. When this field is set to 1, this indicates an alarm condition.
	0	OH ALARM	CD–2 Overheat Alarm bit. When this field is set to 1, this indicates an alarm condition.

TABLE 2–34. CD–2 TO MODE S SENSOR, STATUS MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
3	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	MIL TIM ALARM	Military Timing Alarm bit. When this field is set to 1, this indicates an alarm condition.
	10	MIG ALARM	Military Interface Group (MIG) Alarm bit. When this field is set to 1, this indicates an alarm condition.
	9	BEACON RTQC ALARM	Beacon RTQC Message Alarm bit. When this field is set to 1, this indicates that the Sensor is reporting a “red” condition.
	8	SEARCH RTQC ALARM	Search RTQC Message Alarm bit. When this field is set to 1, this indicates an alarm condition.
	7	RANGE ALARM	When this field is set to 1, this indicates a range alarm condition.
	6	Not Used	Set to 0.
	5	Not Used	Set to 0.
	4	SEARCH MAX RLD	Search Maximum Run–Length Discrimination (RLD) ON/OFF status.
	3	SEARCH MIN RLD ON	Search Minimum Run–Length Discrimination (RLD) ON/OFF status.
	2	AZ/SYNC ALARM	Azimuth/Synchronization Alarm bit. When this field is set to 1, this indicates an alarm condition.
	1	USAF MAP ON	When this field is set to 1, this indicates the enabling of one or more USAF map sectors.
	0	Not Used	Set to 0.

TABLE 2–34. CD–2 TO MODE S SENSOR, STATUS MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
4	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	WEA ON	Weather On bit. When this field is set to 1, this indicates an “on” condition.
	10	WEA ALARM	Weather Alarm bit. When this field is set to 1, this indicates an alarm condition.
	9	Not Used	Set to 0.
	8	FAA MO-DEM ALARM	When this field is set to 1, this indicates an alarm condition.
	7	HS INHIB ALARM	Half–Scan–Inhibit Alarm bit. When this field is set to 1, this indicates an alarm condition.
	6	BUF OL ALARM	Buffer Overload Alarm bit. When this field is set to 1, this indicates an alarm condition.
	5	Not Used	Set to 0.
	4	Not Used	Set to 0.
	3	Not Used	Set to 0.
	2	FAA CHAN ON	FAA Modem Channel On bit. When this field is set to 1, this indicates Channel 3 is ON.
	1	FAA CHAN ON	FAA Modem Channel On bit. When this field is set to 1, this indicates Channel 2 is ON.
	0	FAA CHAN ON	FAA Modem Channel On bit. When this field is set to 1, this indicates Channel 1 is ON.

TABLE 2–35. CD–2 TO MODE S SENSOR, WEATHER MAP MESSAGE

Word	Bit(s)	ID	Bit Description
1	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	TEST	When this field is set to 1, this indicates a test report.
	10 thru 5	Message Label	A 000000 in Message Label bits 10 through 5 indicates a Weather Map Message.
	4	WEATHER MAP WEATHER MAP	101 = Heavy Weather
	3		110 = Light Weather
	2		
	1	FAA	When this field is set to 1, this indicates FAA as a user.
	0	AF	When this field is set to 1, this indicates a joint-use site.
2	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition

TABLE 2–35. CD–2 TO MODE S SENSOR, WEATHER MAP
MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
	11	RANGE (START)	128	NM	Ten-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.25 NM
	10		64	NM	
	9		32	NM	
	8		16	NM	
	7		8	NM	
	6		4	NM	
	5		2	NM	
	4		1	NM	
	3		1/2	NM	
	2		1/4	NM	
	1	Not Used	Set to 0.		
	0	Not Used	Set to 0.		
3	15	X	Don't Care Condition		
	14	X	Don't Care Condition		
	13	X	Don't Care Condition		
	12	X	Don't Care Condition		

TABLE 2–35. CD–2 TO MODE S SENSOR, WEATHER MAP
MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
	11	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180_) LSB = 1 ACP (approx. 0.088_)
	10		1024	ACPs	
	9		512	ACPs	
	8		256	ACPs	
	7		128	ACPs	
	6		64	ACPs	
	5		32	ACPs	
	4		16	ACPs	
	3		8	ACPs	
	2		4	ACPs	
	1		2	ACPs	
	0	AZIMUTH	1	ACP	
4	15	X	Don't Care Condition		
	14	X	Don't Care Condition		
	13	X	Don't Care Condition		
	12	X	Don't Care Condition		

TABLE 2–35. CD–2 TO MODE S SENSOR, WEATHER MAP MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
	11	RANGE (STOP)	128	NM	Ten-bit binary integer which defines range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.25 NM
	10		64	NM	
	9		32	NM	
	8		16	NM	
	7		8	NM	
	6		4	NM	
	5		2	NM	
	4		1	NM	
	3		1/2	NM	
	2	RANGE (STOP)	1/4	NM	
	1	Not Used	Set to 0.		
	0	Not Used	Set to 0.		

TABLE 2–36. CD–2 TO MODE S SENSOR, SEARCH STROBE MESSAGE

Word	Bit(s)	ID	Bit Description
1	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition
	11	TEST	When this field is set to 1, this indicates a test report.
	10 thru 3	Message Label	A 001100001 in Message Label bits 10 through 3 indicates a Search Strobe Message.
	1	FAA	When this field is set to 1, this indicates FAA as a user.

TABLE 2–36. CD–2 TO MODE S SENSOR, SEARCH STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	0	AF	When this field is set to 1, this indicates a joint-use site.
2	11	RANGE	128 NM Eleven-bit binary integer defining range in Nautical Miles (NM). MSB = 128 NM, LSB = 0.125 NM
	10		
	9		
	8		
	7		
	6		
	5		
	4		
	3		
	2		
	1		
	0	Not Used	Set to 0.
3	15	X	Don't Care Condition
	14	X	Don't Care Condition
	13	X	Don't Care Condition
	12	X	Don't Care Condition

TABLE 2–36. CD–2 TO MODE S SENSOR, SEARCH STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description		
	11	AZIMUTH	2048	ACPs	Twelve-bit binary integer which defines azimuth in Azimuth Change Pulses (ACPs), measured from true North. MSB = 2048 ACPs (180_) LSB = 1 ACP (approx. 0.088_)
	10		1024	ACPs	
	9		512	ACPs	
	8		256	ACPs	
	7		128	ACPs	
	6		64	ACPs	
	5		32	ACPs	
	4		16	ACPs	
	3		8	ACPs	
	2		4	ACPs	
	1		2	ACPs	
	0	AZIMUTH	1	ACP	
4	15	X	Don't Care Condition		
	14	X	Don't Care Condition		
	13	X	Don't Care Condition		
	12	X	Don't Care Condition		
	11	RUN – LENGTH	256	ACPs	Seven-bit binary integer which defines run-length in Azimuth Change Pulses (ACPs). MSB = 256 ACPs LSB = 4 ACPs
	10		128	ACPs	
	9		64	ACPs	
	8		32	ACPs	
	7		16	ACPs	
	6		8	ACPs	
	5		4	ACPs	
	4	Not Used	Set to 0.		
	3	Not Used	Set to 0.		

TABLE 2–36. CD–2 TO MODE S SENSOR, SEARCH STROBE MESSAGE (Continued)

Word	Bit(s)	ID	Bit Description
	2	Not Used	Set to 0.
	1	Not Used	Set to 0.
	0	Not Used	Set to 0.

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK)

Word	Bit(s)	ID	Bit Description
			NOTE
			The Radar Target (RTGT) Report consists of seven Words of 16–bits for each word.
1	8	X	Don't Care Condition
	7	X	Don't Care Condition
	6	X	Don't Care Condition
	5	X	Don't Care Condition
	4	TEST	Radar Test Target
	3	0	<u>Target Report</u>
	2	0	<u>Message Label</u>
	1	1	
	16	1	
	15	0	
	14	1	
	13	1	
	12	0	
	11	0	
	10	REFL	1 = Moving Target Indicator (MTI) Reflector
	9	0	This bit is always set to Zero (0).

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
2	24	X	Don't Care Condition
	23	X	Don't Care Condition
	22	X	Don't Care Condition
	21	X	Don't Care Condition
	20	RANGE	32 Nautical Miles
	19		16 Nautical Miles
	18		8 Nautical Miles
	17		4 Nautical Miles
	32		2 Nautical Miles
	31		1 Nautical Mile
	30		1/2 Nautical Mile
	29		1/4 Nautical Mile
	28		1/8 Nautical Mile
	27		1/16 Nautical Mile
	26		1/32 Nautical Mile
	25		1/64 Nautical Mile
			<u>Range</u>
			<u>of the</u>
			<u>C&I Report</u>
3	40	X	Don't Care Condition
	39	X	Don't Care Condition
	38	X	Don't Care Condition
	37	X	Don't Care Condition

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	36	AZIMUTH	2048 ACPs
	35		1024 ACPs
	34		512 ACPs
	33		256 ACPs
	48		128 ACPs
	47		64 ACPs
	46		32 ACPs
	45		16 ACPs
	44		8 ACPs
	43		4 ACPs
	42		2 ACPs
	41	AZIMUTH	1 ACP
4	56	X	Don't Care Condition
	55	X	Don't Care Condition
	54	X	Don't Care Condition
	53	X	Don't Care Condition
			<u>Quality and Confidence</u>
	52	QUAL	00 = One CPI Report
	51	QUAL	01 = Two CPI Reports of Different Types (both PRFs)
			10 = Two or more CPI Reports, Same PRF
			11 = Three or more CPI Reports and both PRFs

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	50	CONF	000 = Targets in Road Traffic Map Areas
	49	CONF	001 = Targets in Heavy Clutter
	64	CONF	010 = Interference
			011 = Thermal, False Targets and Aircraft < “R”
			100 = Max. Doppler and Zero Filter
			101 = Thermal, False Targets and Aircraft > “R”
			<u>Tracking Eligibility</u>
	63	TRACK	00 = The target report shall not be allowed to initiate new track reports or update existing track reports.
	62	TRACK	01 = The target report shall be allowed to correlate with existing established tracks (i.e., tracks in state 3 through state 6) and to promote a tentative track (i.e., a track in state S2 or S7) to an established track (state S3).
			10 = The target report shall be allowed to correlate with existing tracks and initiate new tracks.
			11 = The target report shall be allowed to correlate with existing established tracks (i.e., a track in state S3 through S6) but shall not be allowed to initiate new tracks or to promote non-established tracks (i.e., tracks in state S0, S1, S2, and S7).
			<u>ARTS–IIIA QUALITY</u>
	61	QUAL	Determined from Target Confidence and Quality:
	60	QUAL	Bits 61, 60, 59 are set using the target quality and confidence according to the following rules. The ARTS–IIIA Quality bits are set to closely resemble the Quality currently presented to the ARTS–IIIA by the Sensor Receiver and Processor (SRAP).
	59	QUAL	

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description		
			Bits 61, 60, 59	Confidence	Quality
			000	0	0
			000	0	1
			000	0	2
			000	0	3
			001	1	0
			001	2	0
			010	1	1
			011	3	0
			011	1	2
			011	4	0
			100	1	3
			100	5	0
			100	4	1
			101	3	1
			101	4	2
			101	5	1
			110	3	2
			110	4	3
			110	5	2
			111	3	3
			111	5	3
	58	0	This bit is always set to Zero (0).		
	57	CORRL	0 = Uncorrelated 1 = Correlated (Tracked)		
5	72	X	Don't Care Condition		
	71	X	Don't Care Condition		
	70	X	Don't Care Condition		
	69	X	Don't Care Condition		
	68		Not Used		
	67		Not Used		

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	66		48 DB
	65		24 DB
	80		12 DB
	79		6 DB
	78		3 DB
	77		1.5 DB
	76		3/4 DB
	75		3/8 DB
	74		3/16 DB
	73		3/32 DB
			<u>C&I Report Strength</u>
6	88	X	Don't Care Condition
	87	X	Don't Care Condition
	86	X	Don't Care Condition
	85	X	Don't Care Condition
			<u>Max. Filter PRF #1 (High PRF)</u>
			Bits 81–84 specify which filter, associated with PRF #1, contains the largest amplitude. Bits 81–84 are set as follows:
	84	Max Filter PRF #1	<u>Bits 84–81</u> <u>ASR–9 Filter Number</u>
	83		0001 –3
			0010 –2
	82		0011 –1
			0100 –0
	81		0101 +0
			0110 +1
			0111 +2
			1000 +3
			1001 +4
			1010 –4
	96	Quality	<u>Quality</u>
	95	Quality	Copy of word 4, bits 52 and 51.

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			<u>Interpolated Doppler Velocity PRF #1 (High PRF)</u>
	94		32
	93		16
	92		8
	91		4
	90		2
	89		1
7	104	X	Don't Care Condition
	103	X	Don't Care Condition
	102	X	Don't Care Condition
	101	X	Don't Care Condition
			<u>Max. Filter PRF #2 (Low PRF)</u>
			Bits 97–100 specify which filter, associated with PRF #2, contains the largest amplitude. Bits 97–100 are set as follows:
	100	Max Filter PRF #2	<u>Bits 100–97</u> <u>ASR–9 Filter Number</u>
	99		0001 –3
			0010 –2
	98		0011 –1
			0100 –0
	97		0101 +0
			0110 +1
			0111 +2
			1000 +3
	112	Refl	1 = MT1 Reflector (copy of word 1, bit 10)
	111	Accy	Degraded Accuracy Flag

TABLE 2–37. ASR–9 TO MODE S SENSOR, RADAR TARGET (RTGT) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			<u>Interpolated Doppler Velocity PRF #2 (Low PRF)</u>
	110		32
	109		16
	108		8
	107		4
	106		2
	105		1

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK)

Word	Bit(s)	ID	Bit Description
			NOTE
			The Search Real Time Quality Control (RTQC) Report consists of seven words, with 16 bits in each word. One Search RTQC Report is sent every scan.
1	8	X	Don't Care Condition
	7	X	Don't Care Condition
	6	X	Don't Care Condition
	5	X	Don't Care Condition

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	4	1	<u>Search</u>
	3	0	<u>RTQC</u>
	2	0	<u>Message Label</u>
	1	1	
	16	0	
	15	0	
	14	1	
	13	0	
	12	0	
	11	0	
	10	0	This bit is always set to Zero (0).
	9	0	This bit is always set to Zero (0).
2	24	X	Don't Care Condition
	23	X	Don't Care Condition
	22	X	Don't Care Condition
	21	X	Don't Care Condition

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	20	RANGE	32 Nautical Miles
	19		16 Nautical Miles
	18		8 Nautical Miles
	17		4 Nautical Miles
	32		2 Nautical Miles
	31		1 Nautical Mile
	30		1/2 Nautical Mile
	29		1/4 Nautical Mile
	28		1/8 Nautical Mile
	27		1/16 Nautical Mile
	26		1/32 Nautical Mile
	25		1/64 Nautical Mile
		RANGE	
3	40	X	Don't Care Condition
	39	X	Don't Care Condition
	38	X	Don't Care Condition
	37	X	Don't Care Condition

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	36	AZIMUTH	2048 ACPs
	35		1024 ACPs
	34		512 ACPs
	33		256 ACPs
	48		128 ACPs
	47		64 ACPs
	46		32 ACPs
	45		16 ACPs
	44		8 ACPs
	43		4 ACPs
	42		2 ACPs
	41	AZIMUTH	1 ACP
4	56	X	Don't Care Condition
	55	X	Don't Care Condition
	54	X	Don't Care Condition
	53	X	Don't Care Condition
			<u>Quality and Confidence</u>
	52	QUAL	00 = One CPI Report
	51	QUAL	01 = Two CPI Reports of different types (both PRFs)
			10 = Two or more CPI Reports, Same PRF
			11 = Three or more CPI reports and both PRFs

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	50	CONF	000 = Targets in Road Traffic Map Areas
	49	CONF	001 = Targets in Heavy Clutter
	64	CONF	010 = Interference 011 = Thermal, False Targets and Aircraft < “R” 100 = Max. Doppler and Zero Filter 101 = Thermal, False Targets and Aircraft > “R” <u>Tracking Eligibility</u>
	63	TRACK	00 = The target report shall not be allowed to initiate new track reports or update existing track reports.
	62	TRACK	01 = The target report shall be allowed to correlate with existing established tracks (i.e., tracks in state 3 through state 6) and to promote a tentative track (i.e., a track in state S2 or S7) to an established track (state S3). 10 = The target report shall be allowed to correlate with existing tracks and initiate new tracks. 11 = The target report shall be allowed to correlate with existing established tracks (i.e., a track in state S3 through S6) but shall not be allowed to initiate new tracks or to promote non-established tracks (i.e., tracks in state S0, S1, S2, and S7). <u>ARTS–IIIA QUALITY</u>
	61	QUAL	Determined from Target Confidence and Quality:
	60	QUAL	

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description		
	59	QUAL	Bits 61, 60, 59	Confidence	Quality
			000	0	0
			000	0	1
			000	0	2
			000	0	3
			001	1	0
			001	2	0
			010	1	1
			011	3	0
			011	1	2
			011	4	0
			100	1	3
			100	5	0
			100	4	1
			101	3	1
			101	4	2
			101	5	1
			110	3	2
			110	4	3
			110	5	2
			111	3	3
			111	5	3
	58	0	This bit is always set to Zero (0).		
	57	CORRL	0 = Uncorrelated 1 = Correlated (Tracked)		
5	72	X	Don't Care Condition		
	71	X	Don't Care Condition		
	70	X	Don't Care Condition		
	69	X	Don't Care Condition		
	68		Not Used		
	67		Not Used		

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	66		48 DB <u>Search</u>
	65		24 DB <u>RTQC</u>
	80		12 DB <u>Strength</u>
	79		6 DB
	78		3 DB
	77		1.5 DB
	76		3/4 DB
	75		3/8 DB
	74		3/16 DB
	73		3/32 DB
6	88	X	Don't Care Condition
	87	X	Don't Care Condition
	86	X	Don't Care Condition
	85	X	Don't Care Condition
	<u>Max. Filter PRF #1</u>		
	Bits 81–84 specify which filter, associated with PRF # 1, contains the largest amplitude. Bits 81–84 are set as follows:		
	84	Max. Filter PRF #1	<u>Bits 84–81</u> <u>ASR–9 Filter Number</u>
	83		0001 –3
			0010 –2
	82		0011 –1
			0100 –0
	81		0101 +0
			0110 +1
			0111 +2
			1000 +3
			1001 +4
			1010 –4
	96	Quality	Copy of word 4, bit 52
	95	Quality	Copy of word 4, bit 51

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			<u>Interpolated Doppler Velocity PRF #1</u>
	94		32
	93		16
	92		8
	91		4
	90		2
	89		1
7	104	X	Don't Care Condition
	103	X	Don't Care Condition
	102	X	Don't Care Condition
	101	X	Don't Care Condition
			<u>Max. Filter PRF #2</u>
			Bits 97–100 specify which filter, associated with PRF #2, contains the largest amplitude. Bits 97–100 are set as follows:
	100	Max. Filter PRF #2	<u>Bits 100–97</u> <u>ASR–9 Filter Number</u>
	99		0001 –3
			0010 –2
	98		0011 –1
			0100 –0
	97		0101 +0
			0110 +1
			0111 +2
			1000 +3
	112	Refl	1 = MTI Reflector (copy of word 1, bit 10)
	111	Accy	Degraded Accuracy Flag

TABLE 2–38. ASR–9 TO MODE S SENSOR, RADAR SEARCH (RTQC) REPORT (UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			<u>Interpolated Doppler Velocity PRF #2</u>
	110		32
	109		16
	108		8
	107		4
	106		2
	105		1

TABLE 2–39. ASR–9 TO MODE S SENSOR, SYSTEM STATUS MESSAGE (UPLINK)

Word	Bit(s)	ID	Bit Description
			NOTE
			The ASR–9 System Status Messages occur every antenna scan and whenever a change in the system status occurs. The ASR–9 System Status Message consist of four Words of 16–bits each. Unless otherwise stated, bits are active high.
1	8	X	Don't Care Condition
	7	X	Don't Care Condition
	6	X	Don't Care Condition
	5	X	Don't Care Condition
	4	TEST	Status Test Message when set to 1.

TABLE 2–39. ASR–9 TO MODE S SENSOR, SYSTEM STATUS MESSAGE
(UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	3	0	<u>Status</u>
	2	0	<u>Message</u>
	1	0	<u>Label</u>
	16	1	
	15	1	
	14	0	
	13	0	
	12	0	
	11	0	
	10	0	
	9	0	
2	24	X	Don't Care Condition
	23	X	Don't Care Condition
	22	X	Don't Care Condition
	21	X	Don't Care Condition

TABLE 2–39. ASR–9 TO MODE S SENSOR, SYSTEM STATUS MESSAGE
(UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
3	20	STATUS	Channel On-line (1 = Channel A)
	19	ALARM	Channel A Alarm
	18		Channel A Unavailable
	17		Channel B Alarm
	32		Channel B Unavailable
	31		Post Processor On-line (1 = Channel A)
	30		Post Processor A Alarm
	29		Post Processor A Unavailable
	28		Post Processor B Alarm
	27		Post Processor B Unavailable
	26	STATUS	Weather (Wx) Processor On-line (1 = 6 Level On-line)
	25	ALARM	Weather (Wx) Channel Alarm
	40	X	Don't Care Condition
	39	X	Don't Care Condition
	38	X	Don't Care Condition
	37	X	Don't Care Condition

TABLE 2–39. ASR–9 TO MODE S SENSOR, SYSTEM STATUS MESSAGE
(UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
3 (Cont.)	36	STATUS ALARM	High Voltage On
	35		Antenna Polarization 0 = Linear Polarization (LP), 1 = Circular Polarization (CP)
	34		Antenna Rotation (1 = Antenna is Rotating)
	33		Surveillance and Communication Interface Processor (SCIP) On-line
	48		SCIP Channel A Alarm
	47		SCIP Channel B Alarm
	46		Maintenance Alarm
	45		Comm Link A Alarm
	44		Comm Link B Alarm
	43		RMS Alarm
	42		Spare (Set to Zero).
	41	STATUS ALARM	Spare (Set to Zero).

TABLE 2–39. ASR–9 TO MODE S SENSOR, SYSTEM STATUS MESSAGE
(UPLINK) (Continued)

Word	Bit(s)	ID	Bit Description
4	56	X	Don't Care Condition
	55	X	Don't Care Condition
	54	X	Don't Care Condition
	53	X	Don't Care Condition
	52	STATUS	Beacon Channel (ATCBI/Mode S) On-line, 1 = Channel A
	51	ALARM	Beacon Channel (ATCBI/Mode S) A Alarm
	50		Beacon Channel (ATCBI/Mode S) B Alarm
	49		Bits 49 and 64 through 57 contain 0's in uplink message. In downlink message these bits contain status information entered by Mode S. (See paragraph 2.3.10 for status entries made by Mode S to create downlink status message returned to ASR–9 from Mode S.)
	64		
	63		
	62		
	61		
	60		
	59		
	58	STATUS	
	57	ALARM	

TABLE 2–40. ASR–9 TO/FROM MODE S SENSOR, HEALTH CHECK MESSAGE
(UPLINK AND DOWNLINK)

Word	Bit(s)	ID	Bit Description
1	15 thru 12	–	NOTE For each word, bits 7 through 0 are transmitted first, followed by bits 15 through 8. Unused bits, set to 0.
	11	Test Message	Defined for consistency only, bit set to 0.
	10 thru 7	Message Identification	The time message has a value of 1001 in Message Identification field.
	6 thru 0	–	Unused bits, set to 0.
2	15 thru 5	–	Unused, bits set to 0
	4 thru 0	Hours	Hours (0–23).
3	15 thru 6	–	Unused bits, set to 0.
	5 thru 0	Minutes	Minutes (0–59).
4	15 thru 6	–	Unused bits, set to 0.
	5 thru 0	Seconds	Seconds (0–59).
5	15 thru 5	–	Unused bits, set to 0.

TABLE 2–40. ASR–9 TO/FROM MODE S SENSOR, HEALTH CHECK MESSAGE
(UPLINK AND DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	4 thru 0	Months	Months (1–12).
6	15 thru 6 5 thru 0	– Days	Unused bits, set to 0. Days (1–31).
7	15 thru 12 11 thru 0	– Year	Unused bits, set to 0. Year (1978–...)
8	15 thru 12 11 thru 0	– IBI/Mode S Status	Unused bits, set to 0. Used to send status of the DPS/Interrogator to the ASR9 while in IBI mode.

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR
BEACON RTQC MESSAGE REPORT (DOWNLINK)

Word	Bit(s)	ID	Bit Description
1	8	X	Don't Care Condition
	7	X	Don't Care Condition
	6	X	Don't Care Condition
	5	X	Don't Care Condition
	4	BCN	0 = Beacon Target Message or 1 = Beacon RTQC Message
	3	1	Beacon Message Identifier
	2	1	Beacon Message Identifier

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			NOTE
			When bits 2 and 3 are both set to One (1), this indicates that this message is a Beacon Message.
			<u>Mode 2 Validation</u>
	1	M2VAL	This field indicates the validity of the Mode 2 code data. If the Sensor indicates Mode 2 code confidence as all high confidence, then the data is valid, and this field is set to One (1). If the code confidence is not all high confidence, or if the Mode 2 code data is not present; then the data is not valid, and this field is set to Zero (0).
			<u>Mode 3/A Validation</u>
	16	M3VAL	This field indicates the validity of the Mode 3/A code data.
	15	M3VAL	For ATCRBS targets: If the Sensor indicates Mode 3/A code confidence as all high confidence, then the data is valid, and this field is set to all 1s. If the code confidence is not all high confidence, or if the Mode 3/A code data is not present; then the data is not valid, and this field is set to all 0s. The Sensor may replace the Mode 3/A code of an ATCRBS report with the Mode 3/A code from a track. When this condition occurs, the Mode 3/A confidence bits from the track are used (instead of the report code confidence bits) to validate the report code according to the aforementioned ATCRBS rules. For Mode S targets: The ATCRBS Mode 3/A code is acquired by the Sensor in a different way than for ATCRBS targets. As long as there is a Mode 3/A code available for the Mode S target, and that code is present and valid, this field is set to all 1s. If the code is not available, this field is set to all 0s.
			<u>Identification</u>
	14	IDENT	This field is the Special Identity (SPI) bit in the Mode S Sensor. For ATCRBS targets, this bit is set to 1 when the SPI bit is one with high confidence. For Mode S targets, this bit is set to 1 when the FS field in the Mode S reply indicates the SPI function is set in the transponder; otherwise, this field is set to 0.
	13	R/B	1 = Radar Merged With Beacon

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			NOTE
			Beacon/Radar Merge Message is the same format as a Beacon Message, except that bit 13 is set to 1 and C&I target strength and filter information is inserted into Words 8, 9, and 10.
	12	7700	1 = 7700 Civil or Military Emergency
	11	7600	1 = 7600 Civil or Military Radio Failure
	10	3XVAL	1 = Validated Mode 3/A “X” responses
	9	False Target	1 = False Target flag, normally set to 0.
2	24	X	Don’t Care Condition
	23	X	Don’t Care Condition
	22	X	Don’t Care Condition
	21	X	Don’t Care Condition
	20	RANGE	32 Nautical Miles <u>Range</u>
	19		16 Nautical Miles
	18		8 Nautical Miles
	17		4 Nautical Miles
	32		2 Nautical Miles
	31		1 Nautical Mile
	30		1/2 Nautical Mile
	29		1/4 Nautical Mile
	28		1/8 Nautical Mile
	27		1/16 Nautical Mile
	26		1/32 Nautical Mile
	25	RANGE	1/64 Nautical Mile

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
3	40	X	Don't Care Condition
	39	X	Don't Care Condition
	38	X	Don't Care Condition
	37	X	Don't Care Condition
	36	AZIMUTH	2048 ACPs
	35		1024 ACPs
	34		512 ACPs
	33		256 ACPs
	48		128 ACPs
	47		64 ACPs
	46		32 ACPs
	45		16 ACPs
	44		8 ACPs
	43		4 ACPs
	42		2 ACPs
	41		1 ACP
			<u>Azimuth</u>
4	56	X	Don't Care Condition
	55	X	Don't Care Condition
	54	X	Don't Care Condition
	53	X	Don't Care Condition

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description						
			<u>ARTS –IIIA Quality</u>						
	52	QUAL	When this Beacon report is radar reinforced, the ARTS –IIIA quality field (of the reinforcing radar report) is used to fill this field. When the Beacon report is not radar reinforced, this field represents the quality of the Beacon report. The settings, in binary, for Beacon quality and related conditions are listed below: <table><tr><td><u>Condition</u></td><td><u>Setting</u></td></tr><tr><td>Mode 3/A validation = all 1s</td><td>111</td></tr><tr><td>Mode 3/A validation = 0.</td><td>011</td></tr></table>	<u>Condition</u>	<u>Setting</u>	Mode 3/A validation = all 1s	111	Mode 3/A validation = 0.	011
<u>Condition</u>	<u>Setting</u>								
Mode 3/A validation = all 1s	111								
Mode 3/A validation = 0.	011								
	51	QUAL							
	50	QUAL							
	49	DSCRE	1 = Discrete Mode 3/A Code. Information is in C and D Fields.						
	64	2XVAL	1 = Validated Mode 2 “X” Responses.						
	63	BCN HIT COUNT	16 BCN Hits						
	62		8 BCN Hits						
	61		4 BCN Hits						
	60		2 BCN Hits						
	59		1 BCN Hits						
		BCN HIT COUNT	(BCN = Beacon)						
			<u>ATCRBS Target Reports</u>						
			a. The target report has all high confidence bits that match all confidence bits in the track file code, transmitted with beacon hit count = 20.						
			b. All high confidence bits that have not been correlated with an existing track, transmitted with beacon hit count =16.						
			c. Some low confidence bits, but all high confidence bits match the high confidence bits in the track file code, transmitted with beacon hit count = 18.						
			d. The code in transition bit set, transmitted with beacon hit count = 8.						
			e. All other ATCRBS reports transmitted with beacon hit count = 7						

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			<u>Mode S Target Reports:</u> a. Mode S only target report, BCN hit count = 21. b. Mode S, ATCRBS Reinforced, BCN hit count = 28. c. Mode S, Illegal Mode S ID, BCN hit count = 30. <u>Mode C Validation</u>
	58	MCVAL	This field indicates the validity of the Mode C altitude data.
	57	MCVAL	For ATCRBS reports: If the Sensor has successfully decoded the Mode C data for a report, then the data is valid, and this field is set to all 1s. If the Sensor has not successfully decoded the Mode C data, or if the Mode C data is not present; then the data is not valid, and this field is set to all 0s. For Mode S targets: If there is a surveillance reply with altitude data (not just the altitude field), the Mode C data is present and valid, and this field is set to all 1s. If altitude data is not present in any reply, this field is set to all 0s.
5	72	X	Don't Care Condition
	71	X	Don't Care Condition
	70	X	Don't Care Condition
	69	X	Don't Care Condition

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	68	A4	<u>Mode 3/A Code (Octal)</u>
	67	A2	
	66	A1	
	65	B4	
	80	B2	
	79	B1	
	78	C4	
	77	C2	
	76	C1	
	75	D4	
	74	D2	
	73	D1	

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
6	88	X	Don't Care Condition
	87	X	Don't Care Condition
	86	X	Don't Care Condition
	85	X	Don't Care Condition
	84	D4	<u>Mode 2 Code (Octal)</u>
	83	C1	
	82	A1	
	81	C2	
	96	A2	
	95	C4	
	94	A4	
	93	B1	
	92	D1	
	91	B2	
	90	D2	
	89	B4	
7	104	X	Don't Care Condition
	103	X	Don't Care Condition
	102	X	Don't Care Condition
	101	X	Don't Care Condition
	100	SIGN	1 = Negative altitude in two's (2's) complement form.

TABLE 2-41. MODE S SENSOR TO ASR-9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description	
	99	ALTITUDE 100's of Ft	102,400	Ft
	98		51,200	Ft
	97		25,600	Ft
	112		12,800	Ft <u>Mode C Altitude</u>
	111		6,400	Ft
	110		3,200	Ft
	109		1,600	Ft
	108		800	Ft <u>Identifies Altitude of Target Report</u>
	107		400	Ft
	106		200	Ft
	105	ALTITUDE 100's of Ft	100	Ft
8	120	X	Don't Care Condition	
	119	X	Don't Care Condition	
	118	X	Don't Care Condition	
	117	X	Don't Care Condition	
	116		Not Used	
	115		Not Used	

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
	114		48 DB
	113		24 DB
	128		12 DB
	127		6 DB
	126		3 DB
	125		1.5 DB
	124		3/4 DB
	123		3/8 DB
	122		3/16 DB
	121		3/32 DB
9	136	X	Don't Care Condition
	135	X	Don't Care Condition
	134	X	Don't Care Condition
	133	X	Don't Care Condition
	132		<u>Max. Filter PRF #1 (High PRF)</u>
	131		Bits 132–129 specify which filter, associated with PRF #1, contains the largest amplitude. Bits 132–129 are set as follows:
	130		
	129		

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			Bits 132–129 ASR–9 Filter Number 0001–3 0010–2 0011–1 0100–0 0101+0 0110+1 0111+2 1000+3 1001+4 1010–4
			Quality Copy of word 4, bit 52 from merged C & I Report
	144	Quality	Copy of word 4, bit 52 from merged C & I Report
	143	Quality	Copy of word 4, bit 51 from merged C & I Report
	142		32 <u>Interpolated Doppler Velocity PRF #1</u>
	141		16
	140		8
	139		4
	138		2
	137		1
10	152	X	Don't Care Condition
	151	X	Don't Care Condition
	150	X	Don't Care Condition
	149	X	Don't Care Condition
	148		<u>Max. Filter PRF #2 (Low PRF)</u>
	147		Bits 148–145 specify which filter, associated with PRF #2, contains the largest amplitude. Bits, 148–145 are set as follows:
	146		
	145		

TABLE 2–41. MODE S SENSOR TO ASR–9; BEACON, BEACON/RADAR MERGE, OR BEACON RTQC MESSAGE REPORT (DOWNLINK) (Continued)

Word	Bit(s)	ID	Bit Description
			Bits 148–145 ASR–9 Filter Number 0001–3 0010–2 0011–1 0100–0 0101+0 0110+1 0111+2 1000+3
	160	Refl	1 = MT1 Reflector (copy of word 1, bit 10 from merged C&I report)
	159	Accy	Degraded Accuracy Flag
	158		32
	157		16
	156		8
	155		4
	154		2
	153		1
			Interpolated Doppler Velocity PRF #2

3.0 OPERATION

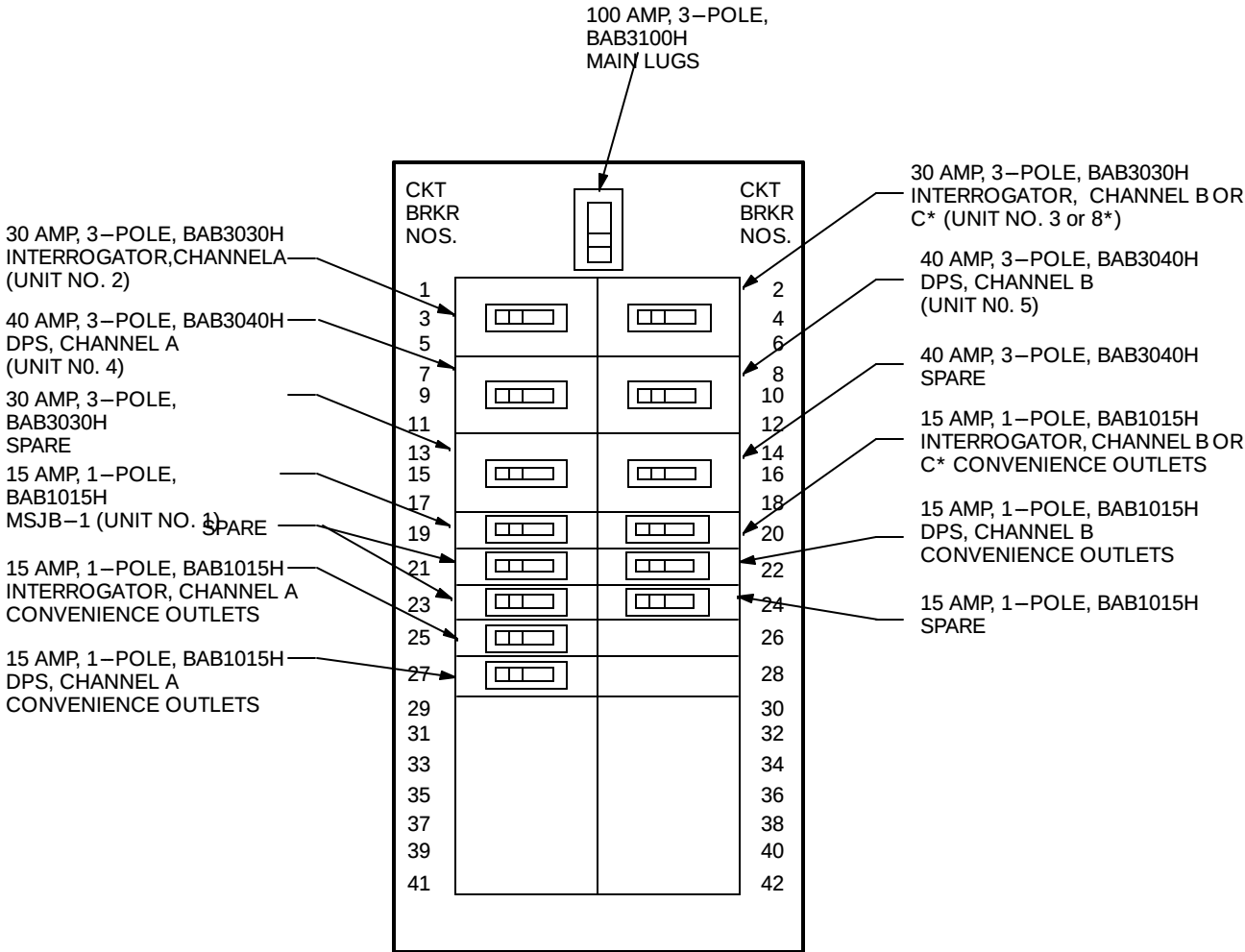
3.1 INTRODUCTION

This section contains an overview of in-service operation, including local and remote monitoring. This section also provides the operation and checkout procedures for the Mode S system. The section is divided into controls and indicators, initial system turn-on and checkout, remote monitoring and control, and system equipment shutdown.

3.2 CONTROL AND INDICATORS

The Mode S System Power Distribution Panel controls and functional information is presented in Figure 3–1. Additional information concerning the power control circuit breaker functions and control indications on the various Mode S System equipments (Interrogators MSJB–1 and DPS) is presented in the Indications-Remarks column of Table 3–1. Information concerning the controls and indicators pertinent to various Mode S equipment can be found in the applicable equipment instruction book as indicated in the following list.

<u>Equipment</u>	<u>Instruction Book</u>
1. Interrogator	a. Interrogator Instruction Book TI 6365.5 b. RCVR/RFTTG Instruction Book TI 6365.7 c. Transmitter Instruction Book TI 6365.6 d. Signal Processor Instruction Book TI 6365.4
2. Mode S Junction Box 1 (MSJB–1)	Interrogator Instruction Book TI 6365.5
3. Data Processing System (DPS)	Data Processing System (DPS) Instruction Book TI 6265.2
4. Calibration Performance Monitor Equipment (CPME)	Calibration Performance Monitor Equipment (CPME) Instruction Book TI 6365.8
5. Radar Intelligent Tool (RIT) Terminal	Government Furnished Equipment (GFE) Instruction Manual; SPB–RIT–001
6. Local Terminal-DPS	Local Terminal User's Manual TI 6365.11
7. KCRT Terminal-Interrogator)	Interrogator Instruction Book TI 6365.5
8. Remote Terminal	Online Help



NOTE: AN * INDICATES THAT INTERROGATOR CHANNEL C IS NOT USED AT A TERMINAL SITE. AT AN EN ROUTE SITE, INTERROGATOR CHANNELS A AND C ARE USED. THE THIRD INTERROGATOR CONFIGURATION IS RESERVED FOR FUTURE EXPANSION.

d4313_104_r3_910131_dbb

FIGURE 3-1. POWER DISTRIBUTION BOX

3.3 TURN-ON AND CHECKOUT

The Mode S System initial turn-on and checkout procedures are presented in Table 3-1. The Location column defines the equipment and various subassemblies of the applicable equipment. The Item column defines the applicable control or indicator for that equipment or subassembly. The Action column describes the procedure to be taken. The Indication-Remarks column provides the indication or information concerning the function of the applicable control or indicator. Additional information concerning the individual equipment turn-on and checkout procedures for the various Mode S System equipments is presented in the Interrogation instruction book TI 6365.5 Data Processing System (DPS) instruction book TI 6365.3, and Calibration Performance Monitor Equipment (CPME) instruction book TI 6365.8. Information concerning the local terminal power up procedure, hook-up procedure, and bootload procedure is provided in Section 3 of the DPS instruction book TI 6365.3.

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES

Location	Item	Action	Indication — Remarks
1. Calibration Performance Monitor Equipment (CPME)			
a. Hinged Panel Assembly	BCPS circuit breaker	Set to ON.	Power supply AC ON indicator lights. Connects AC line voltage to power supply.
b. Power Supply	BCPS DC OUT circuit breaker	Set to ON.	V Bus indicator lights. Connects V BUS to battery shunt relay assembly A3.
c. Power Supply	DC DC CONV circuit breaker	Set to ON.	Connects V Bus to DC–DC converter and out of power supply (when enabled by receiver/transmitter DC POWER switch).
d. Power Supply	BATTERY ENABLE switch	Set to on (up).	BAT ENABLE indicator lights. Enables battery back-up in case AC power fails.
e. Receiver/Transmitter	DC POWER switch	Set to ON.	RECEIVER TRANSMITTER STATUS XMT RF SOURCE, LOW PA SHUTDOWN, and HIGH PA SHUTDOWN indicators lights. Power supply +5V OUT, +12V OUT, and –12V OUT indicators light. Connects 48 volts to low- and high-power amplifiers and enables power supply DC–DC converter.
f. Receiver/Transmitter	TRANSMITTER switch	Set to NORMAL.	LOW PA SHUTDOWN and HIGH PA SHUTDOWN indicators go out. Energizes receiver/transmitter.
g. Hinged Panel Assembly	BATTERY HEATER circuit breaker	Set to ON.	Connect AC line voltage to battery heaters via thermostats.
h. Hinged Panel Assembly	CONVENIENCE OUTLET circuit breaker	Set to ON.	Connects AC line voltage to convenience outlet.

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
i. Receiver/Transmitter	ANTENNA TEST switch	Set to OFF.	Deactivates antenna test circuit (momentary on switch). In Test position, transmitter develops 0.5–microsecond pulses at a 2–Khz rate.
j. Receiver/Transmitter	RF SOURCE SELECT switch	Set to INTL.	Selects internal transmitter source frequency.
k. Receiver/Transmitter	Four coded cards	Per recorded site data.	Determines address, altitude, call sign, and Mode 2 codes.
l. Receiver/Transmitter	POWER SELECT switch	Per recorded site data.	Selects high- or low-power operation in conjunction with RF cable routing.
m. Receiver/Transmitter	MODE A CODE SELECT switch	Per recorded site data.	Selects Mode A code.
n. Receiver/Transmitter	EX REPLY DLY SEL	Per recorded site data.	Adds additional delay to CPME reply.
o. Receiver/Transmitter	RCV PATH ATTEN control	Per recorded site data.	Adjusted to provide interrogation within the dynamic range of the CPME receiver.
p. Receiver/Transmitter	XMT PATH ATTEN control	Per recorded site data.	Adjusted to provide CPME reply within the dynamic range of the Mode S sensor receiver.
2. Site Power Distribution Panel (PDP)	Main Lugs circuit breaker (top area)	Set to ON.	
	Interrogation Channel A circuit breaker	Set to ON.	MAIN POWER A, B, and C indicators on Interrogator A light.
	Interrogation Channel B circuit breaker	Set to ON.	MAIN POWER A, B, and C indicators on Interrogator B light.
	Interrogation Channel C circuit breaker	Set to ON.	MAIN POWER A, B, and C indicators on Interrogator C light (if applicable to system configuration).

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
3. MSJB–1 Cabinet	DPS Channel A circuit breaker	Set to ON.	AC MAIN POWER L1, L2, and L3 indicators light on applicable DPS–PRCSS cabinet.
	DPS Channel B circuit breaker	Set to ON.	AC MAIN POWER L1, L2, and L3 indicators light on applicable DPS–PRCSS cabinet.
	MSJB–1 circuit breaker	Set to ON.	Supplies single-phase input power to MSJB–1 cabinet.
	Interrogator Channel A convenience outlets circuit breaker	Set to ON.	Supplies power to Interrogator Channel A cabinet convenience outlets.
	Interrogator Channel B convenience outlets circuit breaker	Set to ON.	Supplies power to Interrogator Channel B cabinet convenience outlets.
	Interrogator Channel C convenience outlets circuit breaker	Set to ON.	If applicable to system configuration. Supplies power to Interrogator Channel C cabinet convenience outlets.
	DPS Channel A convenience outlets circuit breaker	Set to ON.	CONV PWR indicator lights on applicable DPS–PRCSS cabinet.
	DPS Channel B convenience outlets circuit breaker	Set to ON.	CONV PWR indicator lights on applicable DPS–PRCSS cabinet.
	POWER ON circuit breaker	Set to ON.	LINE INDICATOR DS1, XDS1 lights. If indicator is not lit, refer to Section 7 — Corrective Maintenance of Interrogator instruction book, TI 6365.5.
	Cabinet door	Open door.	

TABLE 3-1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
4. All Interrogator Cabinets	Power on switch on N.B.S. TIME monitors (TOY clocks) (2)	Set to ON.	N.B.S. TIME display on each monitor (TOY clock) indicates time for east or west. Applicable EAST or WEST indicator is lit.
	Uninterruptible D.C. power supplies (2)	Observe indicators.	Green BATTERY CONDITION indicator should light. (GREEN=OK, RED + LOW condition.) RATE OF CHARGE indicator should blink while charging and remain lit when charged.
	a. CONTROL PANEL Input MAIN POWER A, B, and C indicators	Observe that indicators are lit.	If any of the input MAIN POWER A, B, or C indicators are not lit, check applicable circuit breaker on site Power Distribution Panel. If applicable circuit breaker is ON check main power input to applicable Interrogator Cabinet as described in Section 7 — Corrective Maintenance of Interrogator instruction book, TI 6365.5.
	b. CONTROL PANEL Cabinet MAIN POWER breaker	Set to ON.	
	c. CONTROL PANEL BRK ALARM circuit breaker	Set to ON.	Alarm should remain silent. If alarm sounds, check MAIN POWER circuit breaker. If circuit breaker is not tripped, refer to Section 7 — Corrective Maintenance of Interrogator instruction book, TI 6365.5.

TABLE 3-1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
d. SIGNAL MONITOR PANEL	Seven FAULT MONITORS: 1. AIR FLOW (red) 2. INTERROGATOR TEMP ELEVATED (amber) 3. INTERROGATOR OVERTEMP (red) 4. INTERROGATOR PWR SPLY (red) 5. MSJB-1 TEMP ELEVATED (amber) 6. MSJB-1 OVERTEMP (red) 7. MSJB-1 PWR SPLY (red)	Observe that all seven indicators are “not” lit.	If any of the FAULT MONITORS are lit, refer to Section 7 — Corrective Maintenance of Interrogator instruction book, TI 6365.5.
e. CONTROL PANEL	LAMP TEST pushbutton	Press and observe that FAULT MONITORS light.	If any of FAULT MONITORS do not light, replace applicable LED. If fault condition persists, refer to Section 7 — Corrective Maintenance of Interrogator instruction book, TI 6365.5.
5. All DPS-PRCSS Cabinets			
a. Main circuit breaker panel	CONV PWR indicator	Observe that indicator is lit.	If indicator is not lit, check applicable DPS convenience outlet circuit breaker onsite in Power Distribution Panel. If circuit breaker is not tripped, refer to Section 7 — Corrective Maintenance of DPS instruction book, TI 6365.3.

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
b. Main circuit breaker panel	AC LINE MAIN POWER L1, L2, and L3 indicators	Observe that indicators are lit.	If input AC LINE MAIN POWER L1, L2, or L3 indicator is not lit, check applicable DPS channel circuit breaker in Power Distribution Panel. If circuit breaker is not tripped, refer to Section 7 — Corrective Maintenance of DPS instruction book, TI 6365.3.
c. Status and alarm control panel	TAPE1 and TAPE2 connectors	Observe that the tape emulation cables are plugged into the TAPE1 and TAPE2 connectors on the appropriate PRCSS.	Connects Radar Intelligent Tool (RIT) terminal to system.
d. Status and alarm control panel	LT1 jack	If local terminal is to be used, make certain that terminal connector is plugged into LT1 jack via an RS232 to an EIA Converter.	Make certain that local terminal is operational (powered up and boot loaded). Refer to Section 3 — Operation of DPS instruction book, TI 6365.3 to make portable terminal operational.
e. Cabinet	Doors	Open doors.	
f. Status and alarm control panel	POWER FAULT AUDIBLE ALARM switch	Set to ENABLE.	
g. Gate circuit breaker panel	BLOWER circuit breaker	Set to on (up).	

TABLE 3-1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
h. Gate circuit breaker panel	MAIN POWER SUPPLIES PS1, PS2, and PS3 circuit breakers	Set to on (up).	
i. Gate circuit breaker panel	SDS PS7 circuit breaker (if applicable)	Set to on (up).	
j. Main circuit breaker panel	Cabinet MAIN POWER SWITCH	Set to on (up).	Blower and power supply fans are operation.
k. Status and alarm control panel	CHANNEL RESET push button	Press push button.	Resets DPS channel.
6. Radar Intelligent Tool (RIT) Terminal replaces magnetic tape drive unit)	POWER ON/OFF switch (on power director)	Set to on position.	POWER on indicator lights (red).
	COMPUTER power switch (on power director)	Set to on position.	COMPUTER power on indicator lights (red).
	MONITOR power switch (on power director)	Set to on position.	MONITOR power on indicator lights (red).
	PRINTER power switch (on power director)	Set to on position. (if applicable).	PRINTER power on indicator lights (red).
	AUX 1 power switch (on power director)	Set to on position. (if applicable).	AUX 1 power on indicator lights (red).
	AUX 2 power switch (on power director)	Set to on position. (if applicable).	AUX 2 power on indicator lights (red).
	Power on/off switch (on computer)	Set to position 1 (on).	High-speed microprocessor indicator (00) lights (amber).

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
	DISK indicator (on computer) RESET push button (on computer)		NOTE If low-speed microprocessor has been selected, low-speed microprocessor indicator lights when power is applied to computer. DISK indicator lights when hard disk is accessed. If RESET push button is pressed, this enables restart of computer without powering off/on. The Power–On Self Test is also performed at this time. NOTE When RIT terminal powers on correctly, a software proprietary notice appears on display screen. The user must press ENTER on keyboard to get to the C:\> prompt. At the prompt, type “dir” and then press ENTER. Screen displays Directory of C:\.

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
			The Mode S system image file resides in the TDIC directory. To upload the System Image file from the RIT terminal to the Mode S sensor, the user must type “cd/tdic” and then press ENTER. To run the Uploading Executable program (ULOAD.EXE), the user must type “upload” at the prompt C:\tdic> and then press ENTER. When the ULOAD.EXE program is executed, the TDIC IMAGE UPLOADING screen appears on the RIT display monitor. The image file name, image file size, and number of bytes sent to the sensor appears on the display screen. The file name is the name of the system image file which will be uploaded to the sensor. For example, 11192.img is the Mode S image file (for release 1.1.19.2) shipped with the RIT terminal and resides in the TDIC directory. The RIT terminal, at this point, simulates a tape drive and is controlled by the local terminal or RESET push button (on the status and alarm control panel of the DPS). Press RESET push button on DPS to complete upload of system image to Mode S sensor.

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
7. Local Terminal	Display screen	When Local Terminal is powered on, bootloaded, and made operational; observe that Local Terminal Main Status Screen is displayed.	Mode S System becomes operational after powering up system, System Image program has been loaded into DPS memories (NVMEM and GRAM) from RIT terminal, and no faults or error conditions are present in the Mode S System equipment.
8. All DPS–PRCSS Cabinets	NOTE After initialization has been completed and the Mode S System is fully operational, check the following controls		
a. Status and alarm control panel	POWER FAULT CIRCUIT BREAKER OPEN indicators	Observe that indicator is “not” lit.	If POWER FAULT CIRCUIT BREAKER OPEN indicator is lit, refer to Section 7 — Corrective Maintenance of DPS instruction book, TI 6365.3.
b. Status and alarm control panel	TEMPERATURE ELEVATED and OVERHEAT indicators	Observe that indicator is “not” lit.	If a TEMPERATURE fault indicator is lit, refer to Section 7 — Corrective Maintenance of DPS instruction book, TI 6365.3.

TABLE 3–1. MODE S SYSTEM INITIAL TURN-ON AND CHECKOUT PROCEDURES
(Continued)

Location	Item	Action	Indication — Remarks
c. Status and alarm control panel	MODULE POWER FAULT indicator	Observe that indicator is “not” lit.	If POWER FAULT indicator is lit, refer to Section 7 — Corrective Maintenance of DPS instruction book, TI 6365.3.
d. Status and alarm control panel	LAMP TEST push button	Press and observe that four fault indicators light and power fault audible alarm sounds.	If any of the fault indicators do not light, replace bulb. If fault condition persists, refer to Section 7 — Corrective Maintenance of DPS instruction book, TI 6365.3.

3.4 ON–SITE AND REMOTE MONITORING AND CONTROL

In the Mode S System, on-site and remote monitoring and control is provided by the DPS Remote Monitoring Subsystem (RMS) software functions. The DPS RMS at the Mode S sensor site consists of DPS software and hardware working jointly to monitor Mode S sensor performance and to provide communications capability to the Mode S on-site local terminal (for DPS), the Remote Terminal, and also to the Remote Maintenance Monitoring System (RMMS/MPS). The DPS RMS provides for such items as monitor reports, control point requests, restart and sensor control, reconfiguration, System Image updating, calibration, and diagnostic testing of the DPS and Interrogator by use the local terminal, which is connected to the local terminal interface (LT1) jack on the status and alarm control panel of the DPS. Off-line testing of the Interrogator is available by use of the KCRT terminal. Information concerning the local terminal (for DPS) and KCRT terminal (for Interrogator) is presented in the following On-Site Monitoring and Control text. Information concerning the RMMS/MPS is presented in the following Remote Maintenance and Monitoring System and Control text. The remote terminal information is presented in the following Remote Terminal Control text.

3.4.1 On-Site Monitoring and Control

The local terminal main status screen, menus, and submenus are self-explanatory. Information concerning use of the local terminal main status screen and procedures for its use is provided in Section 3 of the DPS Instruction Book TI 6365.3. Detailed information describing the local terminal menus and submenus is provided in the Local Terminal User’s Manual, TI 6365.11. KCRT information is provided in the Interrogator Instruction Book TI 6365.5.

3.4.2 Remote Maintenance Monitoring System and Control

The Mode S sensor also interfaces with the Remote Maintenance Monitor System (RMMS), which provides the capability to automatically and remotely control the periodic tasks of Mode S sensor equipment performance monitoring, and also control the recording of site data. The Mode S sensor-to-RMMS/MPS interface also provides control of the Mode S sensor sites and supports fault isolation and testing via Mode S sensor resident diagnostics. The DPS Remote Monitoring Subsystem (RMS) at the Mode S sensor site consists

of the DPS software and hardware working together to monitor Mode S sensor performance and provide communication capability with the RMMS/MPS. The DPS RMS communications port is capable of transmitting and/or receiving synchronous data simultaneously at selectable baud rates over an RS-ABC type interface. The MPS terminal is currently implemented by use of the Mode Select Beacon System Interim Maintenance and Control (MSIMC) terminal. The MSIMC terminal is a PC configured with an I/O interface board (Emulex DCP-286i) that supports the synchronous communications two way alternate and two way simultaneous modes. The MPS simulator software is loaded onto the PC hard drive and executed under MS DOS. Use of the MSIMC is described in the MSIMC User's Manual, TI 6365.12. Message transfer between the DPS RMS and the RMMS/MPS is accomplished by polling procedures initiated by the RMMS/MPS, as follows: (1) continuous poll, (2) scheduled poll, or (3) specific poll.

3.4.2.1 Continuous Poll. The continuous poll is a supervisory command and response procedure used by the RMMS/MPS to retrieve messages from the sensor. The RMMS/MPS initiates this procedure at regular intervals to allow the transfer of messages to the RMMS/MPS.

3.4.2.2 Scheduled Poll. The scheduled poll is a supervisory command and response procedure used by the RMMS/MPS to retrieve all monitored data from the sensor. The RMMS/MPS initiates this procedure upon operator request or at regular intervals to cause the transfer of site data. When the DPS RMS receives a valid scheduled poll, the DPS RMS initiates a message transfer of all continuously monitored performance data via site data reports (SDRs/Monitor Reports). Alarm threshold values, including pre-alarm threshold filter values, are grouped by logical units (LUs) for site data reporting. A logical unit (LU) is defined as hardware or conditions that have functional and/or physical commonality. Alarm threshold value logical units are transmitted during scheduled polls only if nondefault values are present. In addition, the DPS RMS may interleave pending priority messages with the aforementioned responses.

3.4.2.3 Specific Poll. The specific poll is a supervisory command and response procedure used by the RMMS/MPS to retrieve current performance data from a single, specific logical unit (LU) of the DPS RMS. The RMMS/MPS initiates this procedure when an operator request (command) is invoked to provide the requested data. When the DPS RMS receives a valid specific poll, the DPS RMS initiates a message transfer of the continuously monitored performance data via a site data report (SDR/Monitor Report) for the applicable addressed logic unit. In addition, the DPS RMS interleaves pending priority messages with the SDR/Monitor Report responses.

3.4.3 Remote Terminal Control

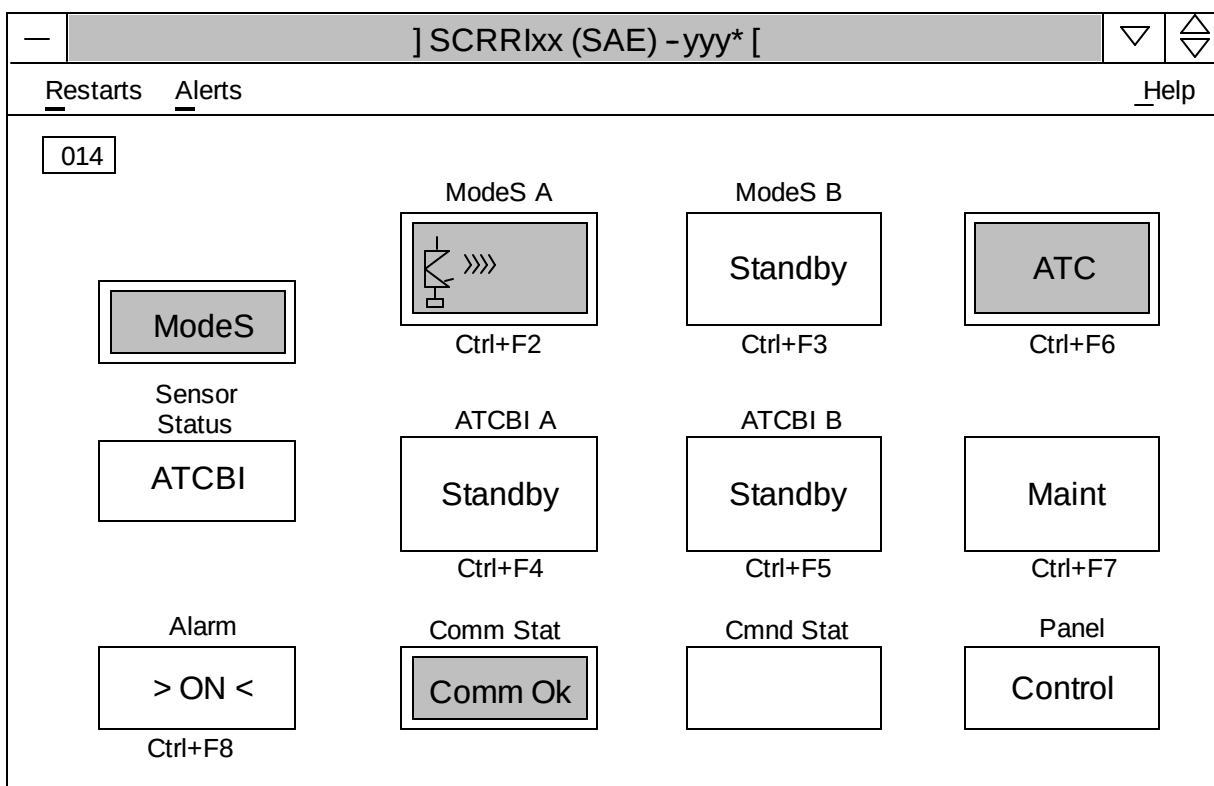
The GFE remote terminal serves as the remote control point for the Mode S Sensor. The remote terminal software runs on an AT class IBM compatible PC with a color VGA display. The remote terminal is located in the air traffic control (ATC) tower and is used for interaction with the Mode S sensor. The remote terminal is connected to the Mode S sensor via GFE modems and the communications junction box, unit 6 (CJBX-6). At the Mode S Sensor site, the two sensor channels are wire-ORed together within the communications junction box (CJBX6) and connected to the collocated modem. The remote terminal uses the ADCCP protocol modified for asynchronous communication. The RMT CONT OUT (J29) cable connector on CJBX6 is used to connect with the GFE modem. The RMT CONT A (J9) and RMT CONT B (J19) cable connectors on CJBX6 are used to connect with the I/O Connector Panel (A1) on top of the respective DPS A or DPS B cabinet. The remote terminal communicates with each Mode S Sensor channel via the DPS Remote Monitoring Subsystem (RMS) software residing in each channel. The remote terminal is connected to both channels. Detailed information concerning use of the remote terminal software and operations is available in the Remote IRCP.

3.4.3.1 The control functions of the seven available remote terminal function keys (CTRL+F2 through CTRL+F8) cause a Mode S sensor action to occur when the remote terminal has control. The Sensor Status and Channel Select/Status displayed on the RT display screen use control functions CTRL+F2 through CTRL+F5 to send commands that change the sensor configuration and control which channel is active. Each control function displays Online, Standby, or Offline status. The Online and Standby statuses can be displayed either in 'green' or 'yellow'. A Green status indicates that the sensor passed all internal diagnostic

tests. A Yellow status indicates that the sensor either did not pass all diagnostic tests or has a soft alarm but is fully operational. A Standby channel displayed in 'yellow' or 'green' can be selected as Online for either Mode S or ATCBI mode of operation. The control functions are described in Table 3–2.

NOTE

Online status in the Mode S channel indicators (A or B) is shown as a beacon antenna icon (see figure 3–2).



*xx = software version identifier; yyy = 3 or 4 letter site identifier.

FIGURE 3–2. Mode S Remote Terminal (RT) Display Screen

TABLE 3–2. MODE S REMOTE TERMINAL CONTROL FUNCTIONS

Control Function	Description
CTRL + F2	CTRL + F2 is the Mode S mode Channel A Online request. Control function CTRL + F2 displays Mode S Channel Status and allows the user to toggle the Standby channel to Online when the sensor is in the Mode S mode of operation, if the Standby channel is 'green' or 'yellow'. Control function CTRL + F2 also allows the user to switch from ATCBI mode to Mode S mode if the Mode S mode is available. When switching channels in Mode S mode, the channel status becomes 'yellow', CTRL + F2 will flash, and the Comm Stat indicator will indicate Sent. The Cmnd Stat indicator should indicate that the command is Accept(ed). CTRL –F2 will stop flashing and the Standby channel becomes 'green'. The user may also enter Mode S mode from ATCBI mode using the CTRL + F2 command. If the system is in ATCBI mode, the user can enter Mode S mode by pressing CTRL + F2. The Mode S mode channels will remain 'yellow' for approximately 90 seconds after the system enters Mode S mode in order to go through acquisition. If the system is in ATCBI mode with Channel B Online when the user opts to return to Mode S mode, the user may select CTRL + F2. Regardless of which channel is requested, the system will return with Mode S mode Channel B Online.
CTRL + F3	CTRL + F3 is the Mode S mode Channel B Online request. This function allows the user to request that Channel B be placed Online in the Mode S mode. Channel B must be in Standby and in either 'green' or 'yellow' status. The CTRL + F3 function is similar to the CTRL + F2 function and follows the same procedure when selected. This command can also be used to enter the Mode S mode from the ATCBI mode if the sensor is in ATCBI mode and Mode S Channel B is in Standby with a 'green' or 'yellow' status.
CTRL + F4	CTRL + F4 is the ATCBI mode Channel A Online request. Control function CTRL + F4 displays ATCBI mode Channel status and allows the user to toggle the Standby 'yellow' or 'green' Channel to Online when the sensor is in ATCBI mode. CTRL + F4 also allows the user to switch from Mode S mode to ATCBI mode. CTRL + F4 allows the user to request that Channel A be placed Online in ATCBI mode, when available. ATCBI Channel A must be in Standby with the display indicating the channel as either a 'green' or 'yellow' status. Upon executing the CTRL + F4 function, the system switches from Mode S mode into ATCBI mode, and the Mode S channels become Offline with a 'red' status. Initially ATCBI Channel A will be Online yellow, CTRL + F4 will be flashing, and the Cmnd Stat indicator will display the commands of Sent and Accept. The Standby ATCBI channel will switch from a 'yellow' to 'green' status. The user can also switch ATCBI channels by using the CTRL + F4 command. The channel being switched must be in Standby with a 'green' or 'yellow' status. Selection of this channel causes it to become Online with a 'yellow' status while the Mode S sensor performs diagnostics. Upon completion of these diagnostics, the channel goes Online with a 'green' status.

TABLE 3–2. MODE S REMOTE TERMINAL CONTROL FUNCTIONS (Continued)

Control Function	Description
CTRL + F5	CTRL + F5 is the ATCBI mode channel B Online request. CTRL + F5 allows the user to request that Channel B be placed Online in the ATCBI mode. The channel must be in Standby with a 'green' or 'yellow' status. This command can also be used to enter the ATCBI mode from the Mode S mode or to switch ATCBI channels.
CTRL + F6	CTRL + F6 is a request for ATC control of the sensor. This command sends a message to the Mode S sensor requesting the priority control point terminal to relinquish control to the remote terminal at the ATC.
CTRL + F7	CTRL + F7 is a release of control to a maintenance (Maint) terminal. This command sends a message to the Mode S sensor releasing control from the remote terminal (ATC) to the requesting control point terminal (Maint).
CTRL + F6 & CTRL + F7	Control terminals include the terminal, remote MPS terminal, and the remote terminals used by Air Traffic Control (ATC) personnel. The use of CTRL + F6 and CTRL + F7 depends on the control state of the remote terminal. When the remote terminal has control, the Panel Control indicator is displayed in 'green' and the Maint indicator is displayed in 'gray'. When the remote terminal does not have control, the Maint indicator is displayed in 'red' and the Panel Control indicator is displayed in 'gray'. When the remote terminal has control and another terminal requests control, the Maint indicator will flash on and off in 'red'. Pressing the CTRL + F7 keys relinquishes control to the requesting terminal. When the remote terminal does not have control, pressing the CTRL + F6 keys will request control from the current control terminal. This action causes the ATC indicator to flash on and off in 'green' until either the control point terminal releases control or a time-out occurs. If a release occurs, the ATC indicator will display 'green' and the Maint indicator will display 'gray'. If the terminal that is in control denies the request, the ATC indicator will continue to flash 'green' until the request times-out after 90 seconds.
CTRL + F8	The Remote Terminal has an audible alarm via the PC's speaker. It sounds following any RED condition or Communication Break. Once an alarm occurs it can be silenced by (CTRL + F8). The audible alarm will be re-enabled if the sensor status returns to normal (no red condition).

3.5 EQUIPMENT SHUTDOWN

The Mode S System equipment shutdown is basically the reverse of the Mode S System initial turn-on and checkout procedures defined in Table 3–1. The Mode S System equipment should be turned-off in the following order: (1) all DPS–PRCSS cabinets, (2) all Interrogator cabinets, (3) Mode S Junction Box–1 (MSJB–1), (4) Power Distribution Panel, and (5) Calibration Performance Monitor Equipment (CPME), and (6) all attached terminals (if desired). Additional information concerning individual equipment shutdown (for maintenance or other purposes) of the various Mode S System equipment is presented in Section 3, paragraph 3.6 of the Interrogator instruction book TI 6365.5, and Data Processing System (DPS) instruction book TI 6365.3, and paragraph 3.5 of the Calibration Performance Monitor Equipment (CPME) instruction book TI 6365.8.

4.0 STANDARDS AND TOLERANCES

4.1 INTRODUCTION

Table 4–1 lists the standards and tolerances for the power input to the Mode S system. Refer to the Procedure Reference Paragraph column to find the paragraph in section 6 where the parameter value was established. The values in the Standard column are optimum values based on original design. The maximum deviation permitted at power start-up, above and below the standard value, is listed in the Tolerance Limit column.

NOTE

The standards and tolerances information for the various Mode S system equipment are presented in Section 4 of the following instruction books: (1) Interrogator Instruction Book, TI 6365.5, (2) Data Processing System (DPS) Instruction Book, TI 6365.3, and (3) Calibration Performance Monitor Equipment (CPME) Instruction Book, TI 6365.8.

TABLE 4–1. STANDARDS AND TOLERANCES

Parameter	Procedure Reference Paragraph	Standard	Tolerance Limit
Mode S System Power:			
Mode S Power Distribution Box Main Circuit Breaker	6.2	208 VAC, 3–Phase, 100 Ampere	208 VAC ? 31V
Mode S Power Distribution Box to Interrogator Channel A, B, and/or C (Units 2/3/8)	6.2	208 VAC, 3–Phase, 30 Ampere	208 VAC ? 31V (Phase A–B, B–C and C–A)
Mode S Power Distribution Box to DPS Channel A and B (Units 4 and 5)	6.2	208 VAC, 3–Phase, 40 Ampere	208 VAC ? 31V (Phase A–B, B–C and C–A)
Mode S Power Distribution Box to MSJB–1 (Unit 1)	6.2	120 VAC, 1–Phase, 15 Ampere	120 VAC ? 18V

TABLE 4–1. STANDARDS AND TOLERANCES (Continued)

Parameter	Procedure Reference Paragraph	Standard	Tolerance Limit
Mode S Power Distribution Box to Interrogator Convenience Re- ceptacles	6.2	120 VAC, 1–Phase, 15 Ampere	120 VAC ? 18V
Mode S Power Distribution Box to DPS Convenience Receptacles	6.2	120 VAC, 1–Phase, 15 Ampere	120 VAC ? 18V

5.0 PERIODIC MAINTENANCE

5.1 INTRODUCTION

This section provides information concerning the performance checks and other preventive maintenance tasks that are performed periodically on the Mode S equipment to ensure proper performance. The data is presented in tabular format (whenever applicable) for onsite maintenance checks. Refer to the applicable Mode S equipment instruction book for performance checks, preventive maintenance tasks, and recommended performance check intervals for each equipment:

- a. Interrogator instruction book, TI 6365.5
 1. Transmitter instruction book, TI 6365.6
 2. Receiver/RFTTG instruction book, TI 6365.7
 3. Signal Processor instruction book, TI 6365.4
- b. Data Processing System (DPS) instruction book, TI 6365.3
- c. Calibration Performance Monitor Equipment (CPME) instruction book, TI 6365.8

Information concerning performance checks, preventive maintenance tasks, and the recommended performance check intervals for the Mode S junction box 1 (MSJB-1) is also contained in the Interrogator instruction book, TI 6365.5.

5.2 PERFORMANCE CHECKS

The performance checks required for the various Mode S equipment are presented in the applicable Mode S equipment instruction book as indicated in paragraph 5.1. These various performance checks tests, and other control functions can be performed at a Mode S sensor site via a local terminal (DPS) or KCRT terminal (Interrogator), or remotely by the Remote Maintenance Monitoring System (RMMS/MPS). The MPS terminal may be implemented by use of the Mode S Interim Monitoring and Control (MSIMC) terminal.

5.2.1 Performance Checks at Mode S Site

At a Mode S sensor site, a local terminal (which interfaces with the DPS) or a KCRT terminal (which interfaces with the Interrogator) allows the user to implement performance checks and various tasks via menus displayed on the terminal's screen. (Refer to paragraph 3.4 for information concerning on-site monitoring and control and also remote maintenance monitoring and control.) The local terminal (DPS) or the KCRT terminal (Interrogator) at a Mode S site or the RMMS/MPS at the ATC facility can request various maintenance/diagnostic functions to be performed. The local/KCRT terminal at the Mode S site implements these functions when selected by directly interfacing with the Mode S sensor, but the remotely located RMMS/MPS must poll the Mode S sensor via one of its regular polling procedures. At a Mode S site, the local/KCRT terminal provides the user with the capability of performing checks and test functions, monitoring of sensor operation, data extraction, and site adaptation, via the various menus and submenus displayed on the terminal's screen. A detailed discussion on how to use the local terminal (DPS) or KCRT terminal (Interrogator), the menu selections, and the various functions that are checked, tested, or monitored is provided in Section 3 of the DPS instruction book TI 6365.3 and the Interrogator instruction book TI 6365.5.

5.2.1.1 A detailed description of the maintenance data that can be obtained by use of the local terminal is available in the Mode S Local Terminal User's Manual, TI 6365.11.

5.2.2 Performance Checks by the RMMS/MPS

The Mode S sensor and the Remote Maintenance Monitor System (RMMS/MPS) at the ATC facility provide the capability to automate and remotely control periodic tasks of the Mode S equipment such as, performance monitoring checks and recording of site data. The Mode S sensor and the RMMS/MPS also provide for fault isolation by means of the Remote Monitoring Subsystem (RMS) software and hardware which is resident in the DPS. This is done by remotely invoking the sensor diagnostics. An alarm message is generated by the sensor and sent to the RMMS/MPS whenever the DPS RMS determines that a monitored data point, status, or condition is sensed as transitioning to an alarm state. Alarm messages have highest priority and are transmitted to the RMMS/MPS on the next polling request and are transmitted on a first-in, first-out basis as they are detected. If monitored parameter errors are detected but the number of occurrences has not reached the 'hard alarm' declaration level, the errors are reported as soft alarms (pre-alarms). The communication interface between the DPS RMS and the RMMS/MPS is via a maintenance line adapter (LAMNT) circuit card assembly in the DPS. Message transfer is accomplished by polling procedures (continuous poll, scheduled poll, and specific poll) initiated by the RMMS/MPS.

5.2.2.1 Continuous Poll. A continuous poll message is defined as the supervisory command and response procedure used by the RMMS/MPS to retrieve messages from the sensor. The RMMS/MPS initiates this procedure at regular intervals to allow the transfer sensor of messages to the RMMS/MPS. On receipt of a valid continuous poll, the addressed DPS RMS, initiates a message transfer if it has a priority message to send.

5.2.2.2 Scheduled Poll. A scheduled poll is defined as the supervisory command and response procedure used by the RMMS/MPS to retrieve all monitored data from the sensor. The RMMS/MPS initiates this procedure upon operator request or at regular intervals to cause the transfer of site data. On receipt of a valid scheduled poll, the addressed DPS RMS initiates a message transfer of all continuously monitored performance data via Site Data Reports (SDRs). Alarm threshold values, including pre-alarm threshold filter values, are grouped by logical units (a logical unit is defined as hardware, functions, or conditions that have functional and/or physical commonality) for site data reporting. Alarm threshold value logical units are transmitted during scheduled polls only if nondefault values are present. In addition, the DPS RMS may interleave pending priority message with the above responses. The specific responses by a DPS RMS to a valid scheduled poll are as follows:

- a. If a logical unit (LU) is unable to respond, that logical unit Site Data Report (SDR) is transmitted with a message function code of "g". No other fields are transmitted for that logical unit.
- b. If any priority messages are pending, the DPS RMS responds by sending the priority messages. If alarm conditions are detected or a terminal message has been entered between message block transmissions (prior to termination of the scheduled polling procedure), the DPS RMS queues and transmits the new priority message(s) during the poll.
- c. When no priority messages are waiting for transfer, the DPS RMS responds by sending the SDR messages, as applicable, for all DPS RMS logical units. The DPS RMS terminates the scheduled polling sequence only after all SDRs have been forwarded.
- d. Site Data Reports (SDRs) in response to a scheduled poll will be transmitted as messages of maximum size with priority messages (if any) between these messages.

5.2.2.3 Specific Poll. A specific poll is defined as the supervisory command and response procedure used by the RMMS/MPS to retrieve current sensor performance data from a single, specific logical unit of a DPS RMS. The RMMS/MPS initiates this procedure on operator command either on a rapid, repetitive basis or a one-time basis to provide the requested data. On receipt of a valid specific poll, the addressed

DPS RMS initiates a message transfer of the continuously monitored performance data by means of an SDR for the addressed logical unit only. In addition, the DPS RMS interleaves pending priority messages with the SDR responses. The specific response(s) by a DPS RMS to a valid specific poll are as follows:

- a. If the LU is unable to respond, that logical SDR is transmitted with a message function code of “g”. No other fields are transmitted for that logical unit.
- b. If any priority messages are pending, the DPS RMS responds by sending the priority messages. If alarm conditions are detected between message block transmissions (prior to termination of the scheduled polling procedure), the DPS RMS queues and transmits the new priority message(s) during this poll.
- c. When no priority messages are waiting for transfer, the DPS RMS responds by sending only the SDR for the addressed logical unit. The DPS RMS terminates the specific polling sequence only after the logical unit SDR or appropriate response has been forwarded.

5.2.2.4 Logical Units. The addressing scheme used between the RMMS/MPS and the DPS RMS is designed to identify any specific source or destination of information in the link. Within a given DPS RMS, common or related information is grouped into logical units (LUs) for ease of processing these conditions/parameters. The lowest logical unit (LU) is defined as a destination address. Any point below the logical unit (LU) is identified as a data point (DP). Addresses are identified by single 8-bit characters within the range Hex 20 through Hex FF. Hexadecimal values are shown in single quotes (for example: ‘20’ or ‘FF’). ASCII characters are enclosed in double quotes (for example: “G” or “g”). Binary Coded integers are shown without delimiters.

5.2.2.5 Data Points. The Mode S facility has multiple groups of related data points, each of which has been partitioned as a separate logical unit. Each Mode S channel has its own DPS RMS address (site adaptable). Both the continuous poll and scheduled poll messages are directed to all Mode S logical units (LUs), but the specific poll message is directed to a specific Mode S logical unit (LU). The logic unit (LU) name, logic unit identification, and reference to Section 7 of the Mode S equipment instruction book in which the logic unit condition/parameter information can be found is presented in the following list. Additional information concerning the various logical units (LUs), data points, parameters, and message contents is provided in the Mode S Local Terminal User’s Manual, TI 6365.11. Information concerning Logical Unit 20 is presented in paragraph 5.2.2.6.

Logical Unit Name	LU ID (Hex)		Mode S Equipment Instruction Book No.
	Chan A	Chan B	
RMS Master	20	A0	TI 6365.2
Local Terminal	21	A1	TI 6365.3
Environmental	22	A2	TI 6365.3
RMS Communications	23	A3	TI 6365.3
PRCSS Communications	24	A4	TI 6365.3
DPS Hardware	25	A5	TI 6365.3
Interrogators A, B, and C Hardware	26–28	A6–A8	TI 6365.5
PRCSS Software Conditions	29	A9	TI 6365.3
CPME Summary Conditions	2A	AA	TI 6365.2

Logical Unit Name	LU ID (Hex)		Mode S Equipment Instruction Book No.
	Chan A	Chan B	
CPME 1 thru 5 Test Conditions	2B–2F	AB–AF	TI 6365.2
Interrogators A, B, and C Conditions	30, 32, 34	B0, B2, B4	TI 6365.5
Interrogators A, B, and C FE Status Conditions	31, 33, 35	B1, B3, B5	TI 6365.5
RMS Parameters	36	B6	TI 6365.11
PRCSS Comm Link Parameters	37	B7	TI 6365.11
CPME Summary Conditions Parameters	38	B8	TI 6365.11
CPME Test Conditions Parameters	39	B9	TI 6365.11
PRCSS Software Conditions Parameters	3A	BA	TI 6365.11
Restart Report	3B	BB	TI 6365.11
Configuration Report	3C	BC	TI 6365.11
Interrogator Diagnostic Test Report	3D	BD	TI 6365.11
CPME Status Report	3E	BE	TI 6365.11
Interrogators A, B, and C Conditions Parameters	41–43	C1–C3	TI 6365.11
PCRSS Interrogator Conditions Parameters	44	C4	TI 6365.11
Interrogators A, B, and C Calibration Status Parameters	45–48	C5–C8	TI 6365.11
ATCBI Measured Data Report	49	C9	TI 6365.11

NOTES

1. Channel B LUs A0–C9 are applicable for DPS in system with dual-face antenna operation (two simultaneously active channel, A and B).
2. For system parameters and status reports (LU36–49), refer to Mode S Local Terminal User's Manual TI 6365.11.

5.2.2.6 RMS Master Logical Unit (LU20). Mode S equipment performance checks, data points, and conditions either controlled or monitored are grouped in the RMS Master Logical Unit (LU20). The following data points are included in Logical Unit 20.

LU 20 Data Point	Function
20	Minutes and Seconds of last Northmark
21	Sensor Status
22	Remote Terminal Control Priority
23	Remote Terminal Line Status
24	Remote Terminal Control State
25	Local Terminal Control Priority
26	Local Terminal Line Status
27	Local Terminal Control State
28	MPS Control Priority
29	MPS Line Status
2A	MPS Control State
2B	Control Point Request
2C	Sensor Type
2D	Channel A_A Sensor Channel Status
2E	Channel B_A Sensor Channel Status
2F	Unused (used for sensor types BI and BF)
30	Unused (used for sensor types BI and BF)
31	Unused (used for sensor types BI and BF)
32	Unused (used for sensor types BI and BF)
33	Channel A_A Sensor Channel State
34	Channel B_A Sensor Channel State
35	Unused (used for sensor types BI and BF)
36	Unused (used for sensor types BI and BF)
37	Unused (used for sensor types BI and BF)
38	Unused (used for sensor types BI and BF)
39	Degraded Sensor Channel
3A	Number of CPMEs
3B	CPME 1 Status
3C	CPME 2 Status
3D	CPME 3 Status
3E	CPME 4 Status
3F	CPME 5 Status
40	Status of Automatic Sensor Channel Switching
41	Status of Automatic Switching of Radar

LU 20 Data Point	Function
42	Status of Automatic Selection of ASR9 Backup Mode
43	Status of Automatic SDS Switching
44	Status of Automatic Selection of a New Primary CPME
45	Status of Automatic Entrance Into ATCBI Backup Mode
46	ASR9 Backup Status
47	ATCBI Backup Status
48	Position of SDS Switch
49	Current Radar Channel
4A	On/Off Status of Radar Channel A
4B	On/Off Status of Radar Channel B
4C	ATCBI Sensor Status
4D	ATCBI Sensor Channel A_A Status
4E	ATCBI Sensor Channel A_C Status
4F	ATCBI Sensor Channel B_A Status
50	ATCBI Sensor Channel B_C Status
51	Interrogator A ATCBI Primary Transmitter Power Status
52	Interrogator A ATCBI Auxiliary Transmitter Power Status
53	Interrogator A ATCBI Primary Transmitter VSWR Status
54	Interrogator A ATCBI Auxiliary Transmitter VSWR Status
55	Interrogator A ATCBI Overall Temperature Status
56	Interrogator B ATCBI Primary Transmitter Power Status
57	Interrogator B ATCBI Auxiliary Transmitter Power Status
58	Interrogator B ATCBI Primary Transmitter VSWR Status
59	Interrogator B ATCBI Auxiliary Transmitter VSWR Status
5A	Interrogator B ATCBI Overall Temperature Status
5B	Channel A Priority Message Reporting Status
5C	Channel B Priority Message Reporting Status

5.2.2.7 LU20 Site Data Reports (SDRs). The following information represents the site data reports (SDRs) for the RMS Master Logical Unit (LU20) message fields. The status and value are supplied for all data points (20 through 2D) in numerical order of the data point (DP) identification, starting with data point 20. The message field information is self-explanatory in that the various message fields are identified, the number of bits in that particular message field are indicated, the range of values that exist for that particular function are shown, and the additional information is indicated for each value. It must be noted, that all equipment, data points, or conditions that are being either controlled or monitored, which pertain to the Mode S facility, are grouped in the RMS Master Logical Unit (LU20).

LU	0	MSG FUNCTION	STATUS	VALUE	STATUS	VALUE	STATUS	VALUE
STATUS	VALUE	-----	-----	-----	STATUS	VALUE	STATUS	VALUE

This message specifies:

- The status of each sensor channel.
- The status of each CPME
- The enable/disabled state, online status of, and current priority of each control point.
- The enable/disabled status and mode of operation selected for each sensor control mode.

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
— —	Logical Unit	8	‘20’	RMS Master
— —	Message Function	8	“G”, “g”	G = Site Data Report g = LU Nonresponsive or Offline
20	Status	8	“A”	A = Normal
	Value (Time Stamp)*	16	0–5959	Minutes and seconds of last North mark in BCD code
21	Status	8	“A,C,E”	A = Normal C = Hard Alarm E = Soft Alarm
	Value** (Sensor Status)	16	0–2	0 = Normal 1 = Soft Alarm 2 = Hard Alarm
22	Status	8	“A”	A = Normal
	Value (Rmt Trm Ctrl Pri)	16	1,2	Remote Terminal Control Priority: 1 = Priority #1 2 = Priority #2
23	Status	8	“A”	A = Normal
	Value (Rmt Trm Status)	16	0,1	Remote Terminal Line Status: 0 = Offline 1 = Online

* This Status/Value is not displayed on MPS consoles.

** Indicates overall sensor status. Soft Alarm = any soft alarm condition within sensor.
Hard Alarm = Total loss of sensor’s mission capability.

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
24	Status	8	"A"	A = Normal
	Value (Rmt Trm Status)	16	2,6	Remote Terminal Control State: 2 = No response (from operator) 6 = Enabled
25	Status	8	"A"	A = Normal
	Value (Lcl Trm Ctrl Pri)	16	1,2	Local Terminal Control Priority: 1 = Priority #1 2 = Priority #2
26	Status	8	"A"	A = Normal
	Value (Lcl Trm Status)	16	0,1	Local Terminal Line Status: 0 = Offline 1 = Online
27	Status	8	"A"	A = Normal
	Value (Lcl Trm Status)	16	2,3,6	Local Terminal Control State: 2 = No response (from operator) 3 = Disabled 6 = Enabled
28	Status	8	"A"	A = Normal
	Value (MPS Control Pri)	16	1,2	MPS Control Priority: 1 = Priority #1 2 = Priority #2
29	Status	8	"A"	A = Normal
	Value (MPS Trm Status)	16	0,1	MPS Line Status: 0 = Offline 1 = Online
2A	Status	8	"A"	A = Normal
	Value (MPS Trm Status)	16	2,3,6	MPS Control State: 2 = No response (from operator) 3 = Disabled 6 = Enabled
2B	Status	8	"A"	A = Normal
	Value (Ctrl Point Reqst)	16	0,2,3,5	Control Point Request for #1 Priority: 0 = No request 2 = Local Terminal 3 = MPS 5 = Remote Terminal

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
2C	Status	8	"A"	A = Normal
	Value (Sensor Type)	16	0,1,2	Sensor Type: 0 = Type A 1 = Type B Interim (BI) 2 = Type B Final (BF)
2D–32	Status	8	"A,B,C,E"	A = Normal B = Not Monitored C = Hard Alarm E = Soft alarm
	Value (SC n Stat)	16	0–4	Sensor Channel Status: 0 = Normal 1 = Soft alarm 2 = Hard Alarm 3 = Status Unavailable (Channel-to-channel link not available) 4 = Not Configurable

These value fields are used to represent the status of all sensor channels found at the sensor from which the status report is being received. The relationship between the status field (SC n Stat) and the number and identity (n) of sensor channels for which status is being reported depends upon the sensor channel type. The identity is decoded as follows:

DP	FIELD	SENSOR CHANNEL TYPE		
		TYPE A	TYPE BI	TYPE BF
2D	SC 1 Stat	Channel A_A	Channel A_AC	Channel A_AB
2E	SC 2 Stat	Channel B_A	Channel B_AC	Channel B_AC
2F	SC 3 Stat	None	None	Channel A_CB
30	SC 4 Stat	None	None	Channel B_AB
31	SC 5 Stat	None	None	Channel B_AC
32	SC 6 Stat	None	None	Channel B_CB

NOTE

For Sensor Channel Types A and BI, status fields SC3 through SC6 report a status of "4" indicating not configurable.

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
33–38	Status	8	“A”	A = Normal
	Value (SC n Stat On/Off)	16	0–2,5	Sensor Channel State: 0 = Not Configurable 1 = Online (Active) 2 = Standby 3 = Offline

These value fields are used to represent the on/off state of all sensor channels found at the sensor from which the status report is being received. The relationship between the status field (SC n ON/OFF) and the number and identity (n) of sensor channels for which status is being reported depends upon the sensor channel type. The identity is decoded as follows:

DP	FIELD	SENSOR CHANNEL TYPE		
		TYPE A	TYPE BI	TYPE BF
33	SC 1 On/Off	Channel A_A	Channel A_AC	Channel A_AB
34	SC 2 On/Off	Channel B_A	Channel B_AC	Channel B_AB
35	SC 3 On/Off	None	None	Channel A_CB
36	SC 4 On/Off	None	None	Channel B_CB
37	SC 5 On/Off	None	None	Channel B_AC
38	SC 6 On/Off	None	None	Channel B_CB

NOTE

For Sensor Channel Types A and BI, status fields SC3 through SC6 report a status of “4” indicating not configurable.

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
39	Status	8	“A”	A = Normal
	Value (Degraded Snsr Ch)	16	0,A–F	Degraded Sensor Channel (for Type B configurations only. The selected channel is degraded to the configuration shown). 0 = Not Degraded A = Degraded to A_A B = Degraded to B_A C = Degraded to A_B D = Degraded to B_B E = Degraded to A_C F = Degraded to B_C
3A	Status	8	“A”	A = Normal
	Value (Num CPMEs)	16	0–5	Number of CPMEs at this site

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
3B–3F	Status	8	“A,C,E”	A = Normal
	Value (CPME n Status) where n = 1 to 5	16	0–2,4	Status of CPME n: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 4 = Not Configurable
40	Status	8	“A”	A = Normal
	Value (Auto SC Switching)	16	0,1	Status of Automatic Sensor Channel Switching: 0 = Disabled 1 = Enabled
41	Status	8	“A”	A = Normal
	Value (Auto R–Ch Switchng)	16	0,1	Status of Automatic Switching of Radar Channels: 0 = Disabled 1 = Enabled
42	Status	8	“A”	A = Normal
	Value (ASR9 ED)	16	0,1	Status of Automatic Selection of ASR–9 Backup Mode: 0 = Disabled 1 = Enabled
43	Status	8	“A”	A = Normal
	Value (Auto SDS Switching)	16	0,1	Status of Automatic SDS Switching: 0 = Disabled 1 = Enabled
44	Status	8	“A”	“A” = Normal
	Value (Auto CPME PriSel)	16	0,1	Status of Automatic Selection of a New Primary CPME Upon Primary CPME Failure: 0 = Disabled 1 = Enabled
45	Status	8	“A”	A = Normal
	Value (Auto ATCBI Bk Up)	16	0,1	Status of Automatic entrance into ATCBI Backup Mode: 0 = Disabled 1 = Enabled

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
46	Status	8	"A,C,E"	A = Normal
	Value (Auto ASR9 Bk Up)	16	0,1	ASR 9 Backup status: 0 = Running ASR 9 1 = Not in ASR 9 Backup
47	Status	8	"A"	A = Normal
	Value (ATCBI Bk Up Status)	16	0–4	ATCBI Backup status: 0 = Not in ATCBI Backup 1 = Running ATCBI Channel A_A 2 = Running ATCBI Channel A_C 3 = Running ATCBI Channel B_A 4 = Running ATCBI Channel B_C
48	Status	8	"A"	A = Normal
	Value (SDS Switch Posit)	16	0–2	Position of SDS switch: 0 = No SDS switch 1 = Radar 2 = MODE S
49	Status	8	"A"	A = Normal
	Value (Radar ID)	16	0–2	Current radar channel: 0 = No channel 1 = Channel A 2 = Channel B
4A	Status	8	"A"	A = Normal
	Value (Value A Status)	16	0,1	On/Off Status of Radar Channel A: 0 = Off 1 = On
4B	Status	8	"A"	A = Normal
	Value (Radar B Status)	16	0,1	On/Off Status of Radar Channel B: 0 = Off 1 = On
4C	Status	8	"A,B,C,E"	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
	Value (ATCBI Snsr Status)	16	0–4	ATCBI Sensor Status: 0 = Normal 1 = Soft alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
4D	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AA Stat)	16	0–4	ATCBI sensor channel A_A status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable (channel–to–channel link not available) 4 = Not Configurable
4E	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AC Stat)	16	0–4	ATCBI sensor channel A_C status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable (channel–to–channel link not available) 4 = Not Configurable
4F	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI BA Stat)	16	0–4	ATCBI sensor channel B_A status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable (channel–to–channel link not available) 4 = Not Configurable
50	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
	Value (ATCBI BC Stat)	16	0–4	ATCBI sensor channel B_C status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable (channel–to–channel link not available) 4 = Not Configurable
51	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AA PriTxPwr)	16	0–4	Interrogator A ATCBI primary transmitter power status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
52	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AA AuxTxPwr)	16	0–4	Interrogator A ATCBI auxiliary transmitter power status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status unavailable 4 = Not Configurable
53	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AA PriTxVSWR)	16	0–4	Interrogator A ATCBI primary transmitter VSWR status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
54	Status	8	"A,B,C,E"	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AA AuxTxVSWR)	16	0–4	Interrogator A ATCBI auxiliary VSWR status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
55	Status	8	"A,B,C,E"	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI AA Temp)	16	0–4	Interrogator A ATCBI overall temperature status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
56	Status	8	"A,B,C,E"	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI BA PriTxPwr)	16	0–4	Interrogator B ATCBI primary transmitter power status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
57	Status	8	"A,B,C,E"	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
	Value (ATCBI BA AuxTxPwr)	16	0–4	Interrogator A ATCBI auxiliary transmitter power status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
58	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI BA PriTxVSWR)	16	0–4	Interrogator B ATCBI primary transmitter VSWR status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
59	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI BA AuxTxVSWR)	16	0–4	Interrogator B ATCBI auxiliary transmitter VSWR status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable
5A	Status	8	“A,B,C,E”	A = Normal B = Not Monitored C = Hard Alarm E = Soft Alarm
	Value (ATCBI BA Temp)	16	0–4	Interrogator B ATCBI overall tempature status: 0 = Normal 1 = Soft Alarm 2 = Hard Alarm 3 = Status Unavailable 4 = Not Configurable

<u>DP</u>	<u>Field</u>	<u>No. of Bits</u>	<u>Range of Values</u>	<u>Additional Information</u>
5B	Status	8	“A”	A = Normal
	Value (Ch A Msg Reprtn)	16	0–1	Channel A priority message reporting status: 0 = Rprt/Allowed 1 = Inhbt/Lcl Cntrl
5C	Status	8	“A”	A = Normal
	Value (Ch B Msg Reprtn)	16	0–1	Channel B priority message reporting status: 0 = Rprt/Allowed 1 = Inhbt/Lcl Cntrl

NOTE

Sensor channel configurations are identified above by type notation such as “Channel A_AB”.

1. The first letter identifies the DPS channel (Channel A in this example).
2. The second letter (the letter after the underscore character) identifies the front face interrogator selected for this DPS channel (Interrogator A in this example).
3. It should be noted that for a Terminal Sensor configuration, only a single (front face) antenna is available. Here each Interrogator is dedicated to its respective DPS channel, and only the antenna can be switched to one DPS Channel/Interrogator pair or the other.
4. The second letter (if present) after the underscore character identifies the backface antenna selected for the DPS channel (Interrogator B in this example).
5. Configurations A_A and B_A apply to non back-to-back Terminal Sensors, A_AC and B_AC apply to back-to-back B–Interim Sensors, and A_AB, A_AC, A_CB, B_AB, B_AC, and B_CB apply to B-Final Sensors. Configurations A_B and B_B are not used.

5.3 OTHER MAINTENANCE TASKS

Other maintenance tasks include cleaning, lubrication, adjustments, calibration, etc., that do not require equipment removal from the site. Information concerning these other maintenance tasks is provided for in the applicable Mode S equipment instruction book. For a list of the applicable Mode S equipment instruction books, refer to paragraph 5.1.

5.4 RECOMMENDED PERFORMANCE INTERVALS

The performance checks that are to be performed periodically and the recommended intervals at which these performance checks are to be accomplished are presented in the applicable Mode S equipment instruction book. For a list of the applicable Mode S equipment instruction books, refer to paragraph 5.1.

6.0 MAINTENANCE PROCEDURES

6.1 INTRODUCTION

This section provides information concerning the maintenance procedures required by scheduled and unscheduled maintenance activities for the various Mode S equipment. This section is divided into performance check procedures, other maintenance procedures, and special maintenance procedures. The user must refer to the applicable Mode S equipment instruction book for the performance check procedures, other maintenance procedures, and special maintenance procedures information which is contained in the following manuals.

1. Interrogator instruction book TI 6365.5
 - a. Transmitter instruction book TI 6365.6
 - b. Receiver/RFTTG instruction book TI 6365.7
 - c. Signal Processor instruction book TI 6365.4
2. Data Processing System (DPS) instruction book TI 6365.3
3. Calibration Performance Monitor Equipment (CPMD) instruction book TI 6365.8

6.2 PERFORMANCE CHECK PROCEDURES

Information concerning the various Mode S system equipment performance check procedures for the performance checks (described in section 5) is presented in section 6 of the applicable Mode S equipment instruction book. To check that the Interrogator and MSJB-1 main power and convenience outlet power (as defined in Table 4-1) are applied to the Interrogator and MSJB-1 cabinets, the user must refer to the Interrogator Turn On and Checkout procedure in Table 3-2 of the Interrogator equipment instruction book TI 6365.5. If an error condition exists in the Interrogator and MSJB-1 turn on and performance checks, refer to Table 7-1 (Interrogator Fault Messages) of TI 6365.5 for the corrective maintenance action to be taken. To check that the DPS main power and convenience outlet power (as defined in Table 4-1) are applied to the DPS PRCSS cabinets, the user must refer to Table 3-1 (Function of Controls on PRCSS Main Circuit Breaker Panel A2) of the DPS equipment instruction book, TI 6365.3, and Table 3-1 (Mode S System Initial Turn-on and Checkout Procedures) of this manual. If an error condition exists in the DPS PRCSS cabinet power turn-on and checkout, the user must refer to Table 7-1 (DPS Fault Messages) of TI 6365.3 for the corrective maintenance action to be taken. The performance check procedure for the taking of monitor reports in order to ascertain the status of the Mode S system equipment is presented in paragraph 6.2.1. The bottom line on the local terminal is used to display status, error messages, and alarm messages to the user. Detailed information concerning the DPS RMS description, the RMS man machine interface, status and alarm displays, the local terminal Main Menu, and the taking of monitor reports is presented in Section 3 of the Mode S Local Terminal Users Manual TI 6365.11 and Sections 3, 6, and 7 of the Mode S DPS instruction book TI 6365.3.

6.2.1 Taking Monitor Reports

In order to obtain a report about the status of the Mode S system at a site, the user can monitor, via the local terminal, the Remote Monitoring Subsystem (RMS) Master Logical Unit (LU20) as presented in procedural step format in paragraph 6.2.1.1. In order to obtain a complete report about the status of Interrogator equipment, Data Processing System (DPS) equipment, and the Calibration Performance

Monitor Equipment (CPME); refer to Section 7 of the applicable Mode S equipment instruction book (refer to 6.1) for information which concerns monitoring the various Logical Units (LUs) as indicated in the following list.

LU Name and LU Identification	Equipment Instruction Book
RMS Master Logical Unit (LU20/A0)	TI 6365.2
Local Terminal Logical Unit (LU21/A1)	TI 6365.3
Environmental Logical Unit (LU22/A2)	TI 6365.3
RMS Communications Logical Unit (LU23/A3)	TI 6365.3
PRCSS Communications Logical Unit (LU24/A4)	TI 6365.3
DPS Hardware Logical Unit (LU25/A5)	TI 6365.3
Interrogators A, B and C Hardware Logical Units (LU26–28/A6–A8)	TI 6365.5
PRCSS Software Conditions Logical Unit (LU29/A9)	TI 6365.3
CPME Summary Conditions Logical Unit (LU2A/AA)	TI 6365.2
CPME 1 Thru 5 Test Conditions Logical Units (LU2B–2F/AB–BF)	TI 6365.2
Interrogators A, B and C Conditions Logical Units (LU30, 32, and 34/B0, B2, B4)	TI 6365.5
Interrogators A, B and C FE Status Conditions Logical Units (LU31, 33, and 35/B1, B3, B5)	TI 6365.5

NOTE

1. Second set of LU numbers (following slash) will only occur during dual-face antenna operation.
2. For system parameters and status report functions (LU36–49/B6–C9), refer to Mode S Local Terminal Users Manual TI 6365.11.

6.2.1.1 RMS Master Logical Unit (LU20). The user must perform this procedure at the local terminal to obtain a monitor report (site data report) about the Mode S system.

1. Display the main menu on the local terminal screen. (Refer to the Local Terminal Users Manual TI 6365.11 Section 3 for information concerning the main menu.)
2. At the local terminal main menu, press function key F3, the logical unit status and data report selections are then displayed (LU Data menu).
3. Select PRCSS Channel and Port for display. (If necessary press Control and F1 keys and/or Control and F2 keys to select desired Chan and Port as displayed on top status line.
4. At the LU Data menu, select logical unit 20, RMS Master, and press Enter.

5. A monitor report screen for the RMS Master Logical Unit (LU20) will appear on your screen. There are 61 data points (20 through 5C) identified for this Logical Unit. Each data point has a status and value shown. The status is represented by letter "A" which indicates a normal status. The status will be normal (A) for all data points in Logic Unit 20. Proceed to next step.

NOTE

For additional information concerning Logical Unit (LU20) data points, range of values, and definitions; the user should refer to paragraph 5.2.2.6.

6. When all the data points (20 through 5C) for Remote Monitoring Subsystem Logical Unit 20 have been checked, press the Escape (Esc) key twice to return to the local terminal main menu.

NOTE

To obtain a complete monitor report about the status of the Interrogator, Data Processing System (DPS), and the Calibration Performance Monitor Equipment (CPME), the user must refer to the applicable Mode S equipment instruction book as indicated in paragraph 6.2.1. Information concerning performance check procedures for monitoring of the Logical Units (LU##), Data Points (DP##), and also maintenance procedures (if any) are presented in the applicable equipment instruction book.

6.3 OTHER MAINTENANCE TASK PROCEDURES

Information concerning the various Mode S system equipment other maintenance task procedures (listed in Section 5) is presented in Section 6 of the applicable Mode S equipment instruction book as indicated in paragraph 6.1. General cleaning and inspection procedures are also included as part of other maintenance task procedures in the applicable Mode S equipment instruction book.

6.4 SPECIAL MAINTENANCE PROCEDURES

Information concerning the various Mode S system equipment special maintenance procedures (for tasks that are not listed in Section 5) is presented in the applicable Mode S equipment instruction book as indicated in paragraph 6.1. Special maintenance procedures include such topics as special adjustments and/or special calibration procedures and, whenever applicable, are contained in the Mode S equipment instruction book.

7.0 CORRECTIVE MAINTENANCE

7.1 INTRODUCTION

This section provides information concerning the corrective maintenance instructions for on site analysis, fault isolation, and repair of faults in the various Mode S system equipment such as the Interrogator, Data Processing System (DPS), and Calibration Performance Monitor Equipment (CPME). Maintenance information is provided in this section concerning the lowest replaceable unit (LRU) for a particular logical unit (LU)/data point (DP), the LRU reference designation, and a reference to the applicable Mode S system equipment instruction book. This information is presented under the Fault Indicators text (paragraph 7.2) in tabular format. The applicable equipment instruction book contains corrective maintenance information and instructions, fault indications, fault analysis procedures, removal and installation procedures, wiring data, and packing instructions (when applicable). The maintenance concept for the Mode S system is presented in paragraph 7.1.1

7.1.1 Maintenance Concept

The Mode S maintenance concept is a three-level concept as illustrated in Figure 7-1. On-site (organizational level) maintenance includes Built-In-Test (BIT)/Fault Isolation Test (FIT) performance monitoring circuitry and software that automatically detects and isolates failures to a faulty line replaceable unit (LRU). System status and/or fault messages will be sent via the DPS Remote Maintenance Monitoring Subsystem (RMMS) to the Maintenance Processing System (MPS) at the Maintenance Control Center (MCC). Note that the MPS terminal is currently implemented by the Mode S Interim Monitoring and Control (MSIMC) terminal. The current sensor and channel status (red, yellow, or green) is also displayed on the Remote Terminal located at the ATC facility. The RMS messages are also available on site by the use of the local terminal. The Remote Maintenance Monitoring System (RMMS/MPS) requests status and diagnostic information from the DPS Remote Monitoring Subsystem (RMS) via the RMMS/MPS-DPS RMS data link. The Mode S can be queried from the RMMS/MPS for additional diagnostic information when required. Integrated diagnostics are incorporated in the Mode S system to provide complete fault coverage by means of a combination of BIT/FIT performance monitoring and off-line diagnostic techniques, which may be automatic, semi-automatic, or manual. Off-line diagnostics are required only when a failure in the Mode S system cannot be isolated to a single LRU by the BIT/FIT features. Information concerning the Mode S system maintenance concept is presented under the following headings:

- a. On-Site Maintenance
- b. Integrated Diagnostics
- c. Central Maintenance Facility (CMF)
- d. Depot Level Maintenance
- e. Deferred Maintenance
- f. Preventive Maintenance

7.1.1.1 On-Site Maintenance. Figure 7-1 illustrates on-site repair actions. The integrated diagnostic techniques are used for isolation of faults to a single LRU. On-site repair consists of removal and replacement of the faulty LRUs. During maintenance, control of the system configuration may be transferred to the RMMS/MPS or to the on-site local terminal. The BIT/FIT feature provides failure detection capability and is capable of isolating the detected faults to the LRU within the Mode S system.

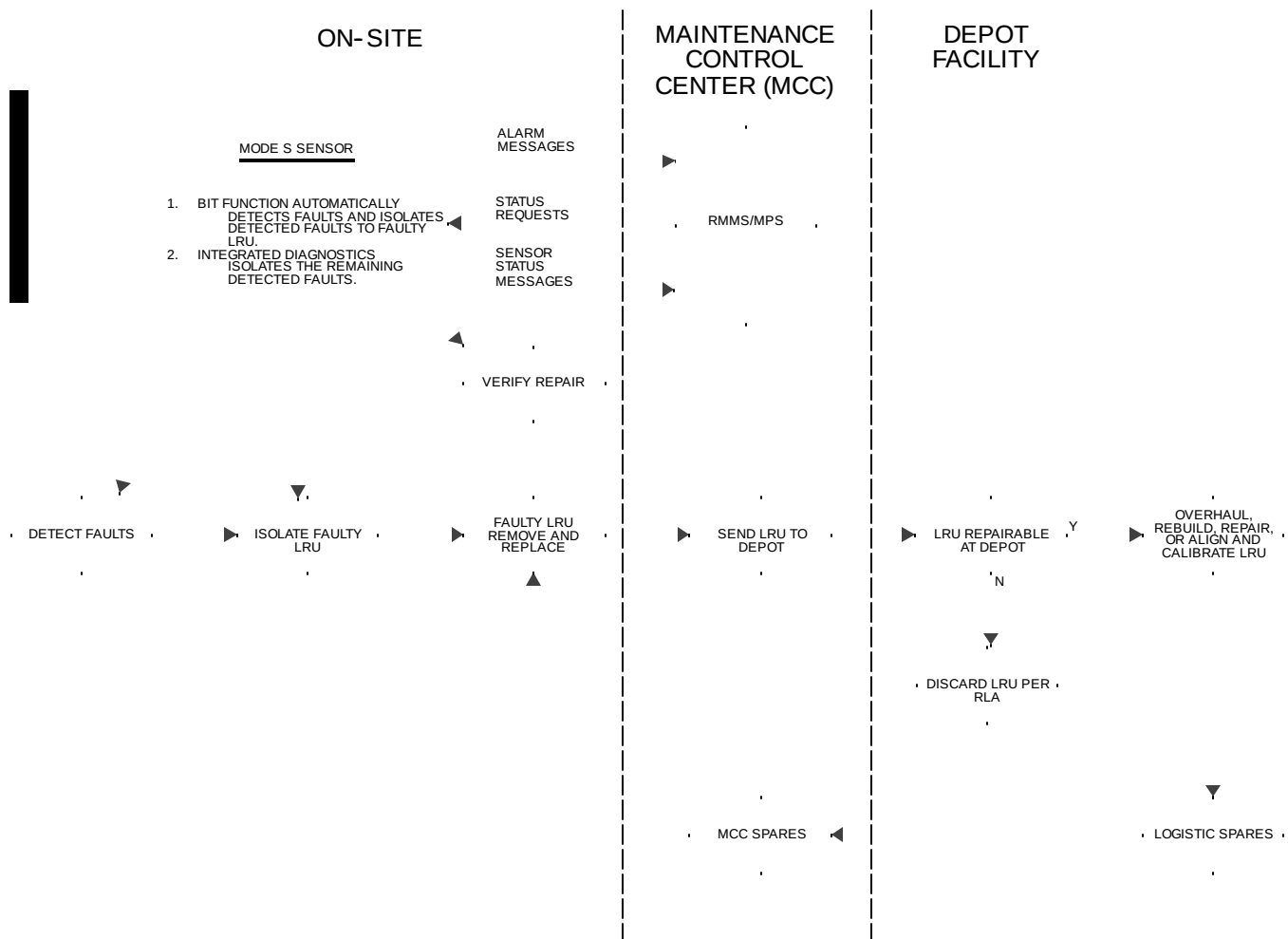


FIGURE 7-1. MODE S MAINTENANCE CONCEPT

7.1.1.1.1 The Built-In-Test (BIT)/Fault Isolation Test (FIT) feature will automatically detect and isolate failures to the faulty LRU in the Mode S system. The Mode S system status and the reference designation of the failed unit will be provided in the form of a message: (a) at the RMMS/MPS located at the Maintenance Control Center (MCC) (or currently at the MSIMC terminal) and/or (b) at the on-site local terminal via a keyboard request. Repair is accomplished by removal and replacement of the faulty LRU. Off-line diagnostics are required only when a failure in the Mode S system cannot be isolated by means of BIT. On-site corrective maintenance action requires one skilled technician. The following types of failures are some of the faults that will require dispatching of a technician to a site after all applicable tests have been performed: (1) Mode S system is down, (2) One channel is down, or (3) Single TOY clock or Azimuth Pulse Generator (APG) is down.

7.1.1.2 Integrated Diagnostics. On-site integrated diagnostics consist of Built-In-Test (BIT)/Fault Isolation Test (FIT) performance monitoring, which as a minimum, has the capability for detecting and isolating failures to a faulty line replaceable unit (LRU) in most cases. BIT/FIT will also be capable of detecting other faults to a group of LRUs. Performance monitoring is the software that periodically checks the performance and viability of various sensor hardware and software, produces reports on request, and generates alarms when fault conditions occur. The performance monitoring function coordinates and controls Built-In-Tests (BITs) for active and redundant components, and Fault Isolation Tests (FITs) for suspect components (LRUs).

7.1.1.3 Maintenance Control Center (MCC). The Mode S sensors will be monitored remotely by means of the RMMS/MPS at a Maintenance Control Center (MCC). When a failure occurs in the sensor equipment, an indication will be provided in the sensor status message. A maintenance technician will be alerted to perform necessary diagnostic tests, if needed, to isolate the failure. A technician will be dispatched from the MCC, if required (refer to paragraph 7.1.1.5 on Deferred Maintenance), to perform on-site corrective maintenance action. Faulty LRUs removed from the sensor sites will be taken to the MCC for disposition. Most LRUs will be forwarded to the assigned depot facility for repair. Certain LRUs are designated factory repairable and will be sent to the appropriate manufacturer or vendor for repair. Repaired LRUs will be stocked as spares at the MCC.

7.1.1.4 Depot Level Maintenance. Depot Maintenance involves repair of faulty LRUs and major overhaul, rebuild, repair, and refurbishment of Mode S equipment requiring more precise electronic test equipment or standards. Repaired LRUs will be sent to the appropriate CMF spares inventory.

7.1.1.5 Deferred Maintenance. Each Mode S sensor channel contains redundant elements which, when they fail, do not effect sensor performance and allow the sensor channels to continue to operate. Failures of these modules may be recorded and logged in memory. However, maintenance on these modules may be deferred until a channel fails. Failure of a channel requires switching to the redundant channel and immediate dispatching of a maintenance technician to the site. The technician will restore the failed channel by removing and replacing all failed modules and performing any preventive maintenance that is currently required. Deferred maintenance of failed redundant elements minimizes the frequency of visits to a site.

7.1.1.6 Preventive Maintenance. Scheduled or preventive maintenance tasks are required on the Mode S system, such as, adjustment, scheduled replacement of wearable items, and replacement of air filter elements. The maintenance technicians will coordinate these tasks with corrective maintenance tasks to minimize trips to the remote sites. Depending upon the availability of time, additional performance verification tests may be performed on a channel to verify the performance of analog parameters.

7.2 FAULT INDICATORS

If a fault is detected by the fault detection routines, BIT/FIT is automatically initiated to isolate the detected fault. When BIT/FIT has isolated the fault, an alarm message is sent to the RMMS/MPS which indicates the suspect unit that should be replaced. If the local terminal is connected to the DPS and in use with alarm reports enabled for display, the alarm reports may also be displayed on that terminal. The alarm reports sent to the local terminal are saved in an Alert Log maintained by the local terminal software. The Alert Log can be accessed for review from the local terminal main menu by pressing function key F1.

The reports to the RMMS are made each time a PRCSS channel is polled by the RMMS. (Messages are saved and queued in priority order between polls.) At each scheduled poll, messages are sent in the following order: hard alarms (red status conditions), soft alarms (yellow status error conditions that may lead to red status), restart messages (if a recovery or reinitialization has been performed), non-priority queue messages (responses and command error messages from application programs), and the site data reports. The data sent to the RMMS for site data reports is the same as that displayed on the local terminal as monitor reports (also called LU Data reports). The alarm messages report new failures, while the site data reports (or local terminal monitor reports) review the complete current status of the equipment.

Both alarm messages and LU Data (or monitor) reports identify faults in terms of logical units, and within each logical unit by data points. The form in which the monitor reports are displayed on the local terminal is shown in figure 7-1A. The LU Data Reports can be selected for viewing on the local terminal from the LU Data submenu. The LU Data submenu is accessed from the local terminal main menu by pressing function key F3. The desired reports can then be displayed and reviewed. The receipt of an alarm message increments the count of alert messages as displayed on the top status line of the local terminal main menu. The Alert log can then be accessed to obtain information about the report. Note the normal or “green” status is reported as “A”, failure or “red” status is “C”, and degraded or suspect “yellow” status is “E”. The form of an alarm report as maintained in the Alert Log on the local terminal is as follows:

```
10:52:15   RMS Alarm Msg   Ch B  LU22  2F  Blwr Unit 2  C1
```

This example of an alarm message indicates a failure (red status “C”), in PRCSS channel B, blower unit number 2 (logical unit 22, data point 2F). (The value field for this data point contains only a 0 for normal or a 1 for failed.) Where value fields contain pertinent additional information, a short definition of the value field contents is displayed. The contents of the value fields are further described in the fault analysis procedures. If more than one data point is displayed, the first data point LRU should be replaced and BIT/FIT initiated to verify the corrective action. If BIT/FIT finds the same fault, the second data point LRU should be replaced and BIT/FIT initiated. If BIT/FIT finds the same fault again, the third data point LRU should be replaced and BIT/FIT initiated. If BIT/FIT still finds a fault, manual corrective action is required (refer to the referenced Mode S system equipment instruction book indicated in Table 7-1). Tables 7-1 through 7-3 provide information concerning the Logical Unit (LU)/Data Point (DP), the LRU reference designation, and the applicable Mode S system equipment instruction book, which provides corrective maintenance instructions in Section 7 for that LRU. Data points are reportable to the RMMS/MPS and also can be obtained at the DPS local terminal.

Local Unit Name		DPS Channel	Logical Unit		Date Received	Time Received		Current Time
PRCSS Comm [Alert :000]		Ch A [SDR: 000]	24.DSP [Com: On] [Chan:A]		05-12-95 [Port:A]	09:20 [Print: Off]		09:28:05 [] 5.0
20	COMM LINE #1	A	Not Config		21	COMM LINE #2	A	Not Config
22	COMM LINE #3	A	Not Config		23	COMM LINE #4	A	Not Config
24	COMM LINE #5	A	Not Config		25	COMM LINE #6	A	Not Config
26	COMM LINE #7	E	OPNLK Fail		27	COMM LINE #8	E	OPNLK Fail
28	COMM LINE #9	E	OPNLK Fail		29	COMM LINE #10	E	OPNLK Fail
2A	COMM LINE #11	E	OPNLK Fail		2B	COMM LINE #12	E	OPNLK Fail
2C	COMM LINE #13	E	OPNLK Fail		2D	COMM LINE #14	E	OPNLK Fail
2E	COMM LINE #15	E	OPNLK Fail		2F	COMM LINE #16	A	Not Config
30	COMM LINE #17	A	Not Config		31	COMM LINE #18	A	Not Config
32	COMM LINE #19	A	Not Config		33	COMM LINE #20	A	Not Config
34	COMM LINE #21	A	Not Config		35	COMM LINE #22	A	Not Config
36	COMM LINE #23	A	Not Config		37	COMM LINE #24	A	Not Config
38	COMM LINE #25	A	Not Config		39	COMM LINE #26	A	Not Config
3A	COMM LINE #27	A	Not Config		3B	COMM LINE #28	A	Not Config
3C	COMM LINE #29	A	Not Config		3D	COMM LINE #30	A	Not Config
3E	COMM LINE #31	A	Not Config		3F	COMM LINE #32	A	Not Config
40	COMM LINE #33	A	Not Config		41	COMM LINE #34	A	Not Config
42	COMM LINE #35	A	Not Config		43	COMM LINE #36	A	Not Config
F1-Alert F2-SDR [Timer: F3-Decr. F4-Incr.] F5-Repeat F6-Beep F7-Print F8-Clear (PgUp/PgDn) (Home) (Ctrl/Alt more) (F10 Help) (Esc — Prev)								
Data Point No. (even)		Data Point Description		Status	Value Field			
Data Point No. (odd)		Data Point Description		Status	Value Field			

PRCSS COMMUNICATIONS REPORT (LU 24)

FIGURE 7-1A. TYPICAL LU DATA REPORT

7.2.1 Interrogator Monitor Report Fault Messages

Table 7-1 presents the Logical Unit and data points for the Interrogator units in Hexadecimal order, with associated LRU name, reference designator, and reference to the associated LRU reference designator, and the reference to the associated equipment instruction book. For additional information concerning the Interrogator logical units (LUs) and data points, refer to Section 3 of the Mode S Local Terminal Users Manual TI 6365.11. The applicable equipment instruction book contains corrective maintenance instructions. There are a few exceptions that require special instructions for some of the Interrogator data points that are listed in Table 7-1.

- For data points 24 and 25, prior to replacing the faulty LRU, a comparison should be made between the attenuator settings on the replacement LRU, prior to replacing the faulty LRU, and the settings on the Variable Site Parameter (VSP) card. If the settings do not agree, change the settings on the replacement LRU. Remove and replace the faulty LRU and initiate BIT/FIT.
- For data points 28, 29, 40, 41, and 57 through 5F, the receiver channels need to be aligned and the power detector offset VSPs may have to be changed after replacing any of these LRUs. Interfacing RF cables may be required that are matched sets and are semi-rigid (refer to the equipment instruction book before starting maintenance).

- c. For data points 2C and 2D, the entire TOY clock system is identified as suspect. Onsite troubleshooting is necessary to determine the specific faulty LRU. Therefore, a replacement LRU for each LRU of the TOY clock system should be brought to the site.

TABLE 7–1. INTERROGATOR HARDWARE MONITOR REPORT FAULT MESSAGES

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 26, 27, 28 (Interrogator A, B, C) (Display LUs for installed interrogators)</u>			
Data Point:			
20	Primary Modulator/Driver #1	*A8	TI 6365.6
21	Auxiliary Modulator/Driver #2	*A7	TI 6365.6
22	Primary Power Amplifier	*A16	TI 6365.6
23	Auxiliary Power Amplifier	*A13	TI 6365.6
24	Primary Output Level Control (#1)	*A11	TI 6365.6
25	Aux Output Level Control (#2)	*A10	TI 6365.6
26	Primary Capacitor Bank (#1)	*A15	TI 6365.6
27	Auxiliary Capacitor Bank (#2)	*A12	TI 6365.6
28	RF Receiver	*A20	TI 6365.7
29	IF Monopulse Receiver	*A5	TI 6365.7
2A	Video Processor/Monitor	*A3	TI 6365.7
2B	RF Test Target Generator	*A14	TI 6365.7
2C	TOY Clock #1	1A1	TI 6365.5
2D	TOY Clock #2	1A2	TI 6365.5
2E	Mode S Interface Board	*A4A109	TI 6365.4
2F	Mode S Message Decoder	*A4A110	TI 6365.4
30	ATCRBS Bracket Detector	*A4A103	TI 6365.4
31	ATCRBS Code Processor 1	*A4A104	TI 6365.4
32	ATCRBS Code Processor 2	*A4A105	TI 6365.4
33	ATCRBS Output FIFO	*A4A106	TI 6365.4

TABLE 7–1. INTERROGATOR HARDWARE MONITOR REPORT FAULT MESSAGES
(Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 26, 27, 28 (Interrogator A, B, C) (Display LUs for installed interrogators)</u>			
Data Point:			
34	Timing Generator	*A4A112	TI 6365.4
35	Timing Expansion	*A4A111	TI 6365.4
36	Modulation Control Unit	*A4A113	TI 6365.4
37	STC Generator	*A4A115	TI 6365.4
38	Independent ATCRBS	*A4A114	TI 6365.4
39	PMS Test Target Generator	*A4A107	TI 6365.4
3A	VSWR Monitor	*A4A116	TI 6365.4
3B	68020 CPU	*A4A122	TI 6365.4
3C	CSS I/O	*A4A117	TI 6365.4
3D	CPU Interface	*A4A118	TI 6365.4
3E	TOY/ARIES Interface	*A4A119	TI 6365.4
3F	Video Trigger Driver	1A3	TI 6365.5
40	Omni Diplexer	*A6	TI 6365.7
41	Monopulse Diplexer	*A19	TI 6365.7
42	5V Power Supply	*PS3	TI 6365.5
43	+15V Power Supply	*PS1	TI 6365.5
44	–15V Power Supply	*PS2	TI 6365.5
45	36V Power Supply	*PS4	TI 6365.5
46	52V Power Supply	*PS5	TI 6365.5
47	Blower Assembly	*A18	TI 6365.5
48	+5V Power Supply	1PS4	TI 6365.5
49	+24V Power Supply	1PS3	TI 6365.5

TABLE 7–1. INTERROGATOR HARDWARE MONITOR REPORT FAULT MESSAGES
(Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 26, 27, 28 (Interrogator A, B, C) (Display LUs for installed interrogators)</u>			
Data Point:			
4A	Bulk Power Supply	*PS6	TI 6365.5
4B	Cabinet Temperature Sensor 1	*RT1	TI 6365.5
4C	Cabinet Temperature Sensor 2	*RT2	TI 6365.5
4D	Cabinet Temperature Sensor 3	*RT3	TI 6365.5
4E	Cabinet Temperature Sensor 4	*RT4	TI 6365.5
4F	Cabinet Temperature Sensor 5	*RT5	TI 6365.5
50	Cabinet Temperature Sensor 6	*RT6	TI 6365.5
51	Cabinet Temperature Sensor 7	*A20A1RT7	TI 6365.5
52	Cabinet Temperature Sensor 8	*RT8	TI 6365.5
53	MSJB–1 Temperature Sensor 1	1RT1	TI 6365.5
54	MSJB–1 Temperature Sensor 2	1RT2	TI 6365.5
55	Azimuth Pulse Generator 1		
56	Azimuth Pulse Generator 2		
57	RF Transfer Switch No. 1 (Sigma)	*A9S1	TI 6365.5
58	RF Transfer Switch No. 2 (Delta)	*A9S2	TI 6365.5
59	RF Transfer Switch No. 3 (Omega)	*A9S3	TI 6365.5
5A	RF Transfer Switch No. 4 (Sigma)	*A14S1	TI 6365.5
5B	RF Transfer Switch No. 5 (Delta)	*A14S2	TI 6365.5
5C	RF Transfer Switch No. 6 (Omega)	*A14S3	TI 6365.5
5D	RF Transfer Switch No. 7 (Sigma)	*A13S1	TI 6365.5
5E	RF Transfer Switch No. 8 (Delta)	*A13S2	TI 6365.5
5F	RF Transfer Switch No. 9 (Omega)	*A13S3	TI 6365.5

TABLE 7–1. INTERROGATOR HARDWARE MONITOR REPORT FAULT MESSAGES
(Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 26, 27, 28 (Interrogator A, B, C) (Display LUs for installed interrogators)</u>			
Data Point:			
60	Local Oscillator	*A9	TI 6365.7
61	General Software Error	N/A	N/A
62	General Hardware Error	N/A	N/A
63	Primary Power Amplifier Blower	*B1	TI 6365.6
64	Primary Power Amplifier Blower Switch	*S2	TI 6365.6
65	General Software Configuration Error	N/A	N/A
Note: An * denotes 2, 3, or 8 (physical Interrogator A, B, or C).			

7.2.2 DPS Monitor Report Fault Messages

Table 7–2 lists the Logical Unit (LU IDs) and Data Points (DP number) applicable for each PRCSS channel of the Data Processing System (DPS). The LUs and DPs are listed in hexadecimal order with associated LRU name, reference designator, and reference to the DPS equipment instruction book TI 6365.3. For additional information concerning the DPS logical units (LUs) and data points (DPs), refer to Section 3 of the Mode S Local Terminal Users Manual TI 6365.11.

TABLE 7–2. DPS MONITOR REPORT FAULT MESSAGES

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 22 ENVIRONMENTAL</u>			
Data Point:			
20 thru 27	Contact Switches 1–8	B2/A2, Unit 6, B2–XA02–P2	TI 6365.3 TI 6365.3
28	+5 V Power Supply #1	B/PS1, B4–CB1 A2/CB1, A/FL1	TI 6365.3
29	+5 V Power Supply #2	B/PS2, B4–CB2 A2/CB1, A/FL1	TI 6365.3

TABLE 7-2. DPS MONITOR REPORT FAULT MESSAGES (Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 22 ENVIRONMENTAL</u>			
Data Point:			
2A	+5 V Power Supply #3	B/PS3, B4-CB3 A2/CB1, A/FL1	TI 6365.3
2B	SDS Power Supply	B6	TI 6364.3
2C, 2D	Not Used		
2E, 2F, 30	Blowers 1, 2, 3	B3, B4-CB4	TI 6365.3
31	Temp Sensor 1 (Inlet Temp.)	B3/A1, B2/A2, B2-XA02-P2	TI 6365.3
32	Temp Sensor 2 (Outlet Temp — Left)	A5, B2/A2, B2-XA02-P2	TI 6365.3
33	Temp Sensor 3 (Outlet Temp — Right)	A6, B2/A2 B2-XA02-P2	TI 6365.3
34	A/D Converter Status	B2/A2	TI 6365.3
<u>LU 23 RMS COMMUNICATIONS</u>			
Data Points:			
20	MPS Line Status	B2/A2, Unit 6, B2-XA02-P2	TI 6365.3
21	Local Terminal Line Status	B2/A2, B2-XA02-P2	TI 6365.3
22	Remote Terminal Line Status	B2/A2, B2-XA02-P2	TI 6365.3
23	Not Used		—
24	PSF Line Status	B2/A2, Unit 6, B2-XA02-P2	TI 6365.3
25	Magnetic Tape	B2/A1, B2-XA01-P2	TI 6365.3

TABLE 7-2. DPS MONITOR REPORT FAULT MESSAGES (Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 23 RMS COMMUNICATIONS</u>			
Data Points:			
26 thru 2B	Surv Line Status Lines 1 thru 6	B1/A2, Unit 6, B1/XA02-P2, B4/CB5, B6, 6A1/K1 thru K6	TI 6365.3
2C thru 31	Surv Line Status Lines 7 thru 12	B1/A3, Unit 11, B1/XA03-P2	TI 6365.3
32 thru 37	Surv Line Status Lines 13 thru 18	B1/A4, Unit 11, B1/XA04-P2	TI 6365.3
38 thru 3D	Surv Line Status Lines 19 thru 24	B1/A5, Unit 11, B1/XA05-P2	TI 6365.3
3E thru 43	Surv Line Status Lines 25 thru 30	B1/A6, Unit 11, B1/XA06-P2	TI 6365.3
44 thru 49	Surv Line Status Lines 31 thru 36	B1/A7, Unit 11, B1/XA07-P2	TI 6365.3
4A	RDR LAO PPA Line		
4B	RDR LAO PPB Line		
4C	RDR LAI PPA Line		
4D	RDR LAI PPB Line		
4E	RDR LAO Connect		
4F	RDR CA 1 Connect		
50	RDR Availability		
51	Chan to Chan Link Interface Status	B2/A2, B2/XA02-P2	TI 6365.3
<u>LU 24 PRCSS COMM LINES</u>			
Data Points:			
20 thru 25	Comm Line Status Lines 1 thru 6	B1/A13, Unit 6 B1-XA13-P2	TI 6365.3

TABLE 7-2. DPS MONITOR REPORT FAULT MESSAGES (Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
<u>LU 24 PRCSS COMM LINES</u>			
Data Points:			
26 thru 2B	Comm Line Status Lines 7 thru 12	B1/A12, Unit 10, B1-XA12-P2	TI 6365.3
2C thru 31	Comm Line Status Lines 13 thru 18	B1/A11, Unit 10, B1-XA11-P2	TI 6365.3
32 thru 37	Comm Line Status Lines 19 thru 24	B1/A10, Unit 10, B1-XA10-P2	TI 6365.3
38 thru 3D	Comm Line Status Lines 25 thru 30	B1/A9, Unit 10, B1-XA9-P2	TI 6365.3
3E thru 43	Comm Line Status Lines 31 thru 36	B1/A8, Unit 10, B1-XA8-P2	TI 6365.3
<u>LU 25 DPS HARDWARE</u>			

Data Points:

NOTE

If the LRU Ref. for LU25 designates replacement of an IOP card, the associated line adapter card types should be taken to the site for possible replacement. Likewise, if the LRU Ref. designates any line adapter; both the line adapter and its associated IOP should be taken to the site for replacement.

Configuration of MPU40 and CPU/IOP CCAs (marked with asterisks) may vary by site. Basic configuration is shown.

20	LARDR card (kit)	B1/A1	TI 6365.3
21	LASUR card (kit 1)	B1/A2	TI 6365.3
22	LASUR card (kit 2)	B1/A3	TI 6365.3
23	LASUR card (kit 3)	B1/A4	TI 6365.3
24	LASUR card (kit 4)	B1/A5	TI 6365.3
25	LASUR card (kit 5)	B1/A6	TI 6365.3
26	LASUR card (kit 6)	B1/A7	TI 6365.3
27	LACOM card (kit 5)	B1/A8	TI 6365.3
28	LACOM card (kit 4)	B1/A9	TI 6365.3

TABLE 7-2. DPS MONITOR REPORT FAULT MESSAGES (Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
LU 25	DPS HARDWARE		
Data Points:			
29	LACOM card (kit 3)	B1/A10	TI 6365.3
2A	LACOM card (kit 2)	B1/A11	TI 6365.3
2B	LACOM card (kit 1)	B1/A12	TI 6365.3
2C	LACOM card (basic)	B1/A13	TI 6365.3
2D	LASMT card	B2/A1	TI 6365.3
2E	LAMNT card	B2/A2	TI 6365.3
2F	MPU40 card (MIOP)*	B2/A3	TI 6365.3
30	CPU/IOP card	B2/A4	TI 6365.3
31	Spare Location	B2/A5	TI 6365.3
32	MPU40 card (CPU)*	B2/A6	TI 6365.3
33	Spare Location	B2/A7	TI 6365.3
34	MPU40 card (CPU)*	B2/A8	TI 6365.3
35	Spare Location	B2/A9	TI 6365.3
36	ITRI card	B2/A10	TI 6365.3
37	ITRI card	B2/A11	TI 6365.3
38	Spare Location	B2/A12	TI 6365.3
39	Spare Location	B2/A13	TI 6365.3
3A	GRAM card	B2/A14	TI 6365.3
3B	GRAM card	B2/A15	TI 6365.3
3C	NVMEM card	B2/A16	TI 6365.3
3D	Spare Location	B2/A17	TI 6365.3
3E	MPU40 (CPU)*	B2/A18	TI 6365.3
3F	Spare Location	B2/A19	TI 6365.3

TABLE 7-2. DPS MONITOR REPORT FAULT MESSAGES (Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
LU 25	DPS HARDWARE		
Data Points:			
40	MPU40 (CPU)*	B2/A20	TI 6365.3
41	Spare Location	B2/A21	TI 6365.3
42	Spare Location	B2/A22	TI 6365.3
43	CPU/IOP card (COMM IOP)*	B2/A23	TI 6365.3
44	Spare Location	B2/A24	TI 6365.3
45	Spare Location	B2/A25	TI 6365.3
46	Spare Location	B2/A26	TI 6365.3
47	Serial Bus	B2	TI 6365.3
48	Parallel Bus	B2	TI 6365.3
49	Mem Write Thru Link 0	B2/A14, B2, B2-XA14-P1	TI 6365.3
4A	Mem Write Thru Link 1	B2/A15, B2, B2-XA15-P1	TI 6365.3
4B	Inter Chan Bus 0	B2/A2, B2/XA02-P2	TI 6365.3
4C	Inter Chan Bus 1	B2/A2, B2/XA02-P2	TI 6365.3
4D	Unknown H/W Error		TI 6365.3
DISCOVERED FAULTS			
	IOP to LA Backplane Cable	B2-XA04-P1, B2-XA04-P3 B2-XA23-P1 B2-XZ23-P3	TI 6365.3

TABLE 7–2. DPS MONITOR REPORT FAULT MESSAGES (Continued)

Logical Unit/ Data Point Number (Hex)	Lowest Replaceable Unit (LRU)	LRU Ref. Designation (See Note)	Instruction Book Ref.
LU 25	DPS HARDWARE		
Data Points:			
	Indicator Lamps Fail	A2/DS1 thru DS4, A3/DS1 thru DS4, B4/K1, B2/A3, A3	TI 6365.3
	Audible Alarm Fail	A3 (A3/LS1)	TI 6365.3
	Circuit Breaker Open Indicator On	B4/CB1 thru CB4, B4/CB8	TI 6365.3
	Circuit Breaker Trips Open	B/PS1 thru PS3, B3, B6, B4/CB8 B4/CB1 thru CB4	TI 6365.3
	Fault Indicator On Power Supply	B4/CB1 thru CB3, B/PS1 thru PS3	TI 6365.3
	5 Volt Power Supply Out of Tolerance	B/PS1 thru PS3	TI 6365.3
	SDS Power Supply Out of Tolerance	B6	TI 6365.3
	Circuit Breaker Open Light & Alarm Fail	A3/DS1, A3 B4/CB1 thru CB5, B4/CB8	TI 6365.3

Note: For DPS–PRCSS Unit 4 or 5.

7.2.3 CPME Monitor Report Fault Messages

The following two paragraphs cover the CPME monitor report messages. For information concerning corrective maintenance procedures for CPME faults and the action taken to repair the CPME to the LRU, the user must refer to Section 6 of the Calibration Performance Monitor Equipment (CPME) TI 6365.8. For additional information concerning CPME logical units (LUs) and data points (DPs), refer to Section 4 of the Mode S Local Terminal Users Manual TI 6365.11.

7.2.3.1 The CPME summary conditions logical unit (LU2A) consists of data points presented in Table 7–3. The following information represents the site data reports (monitor reports) for logical unit LU2A. The CPME summary conditions monitor report indicates the overall condition of the CPMEs associated with the Sensor. This monitor report is distinguished from the CPME test conditions monitor report for LUs 2B through 2F (refer to Table 7–4) which shows the test conditions for each individual CPME associated with the Sensor (there can be from one to five CPMEs for each Sensor). Status and value pairs are transmitted for all data points in the LU in increasing numerical order of data point identification, starting with data point 20. The Status is important information and appears in the monitor report as an A (normal condition), C (red alarm), or E (yellow alarm). The Value field is always 0 for LU2A. The site data report presents summary information common to all CPMEs. For additional information concerning the CPME Summary Conditions Report (LU2A), refer to Section 3 of the Mode S Local Terminal Users Manual TI 6365.11.

NOTE

If it has been determined that CPME corrective maintenance has to be performed because an alarm condition (value C or E) appears in the status column of the monitor report for the CPME, the user must refer to Section 7 (Corrective Maintenance) of the Mode S Calibration Performance Monitor Equipment instruction book, TI 6365.8. To determine the corrective maintenance action to be taken, the user must refer to paragraphs 7.2.1 (CPME Test Set), 7.2.2 (CPME Test Set Tests), 7.3 (Onsite Corrective Maintenance), 7.3.1 (Onsite Fault Isolation), and 7.3.2 (LRU Replacement Procedures).

LU	0	MSG FUNCTION	STATUS	VALUE	STATUS	VALUE
STATUS	VALUE		STATUS	VALUE	STATUS	VALUE

Field	No. of Bits	Range of Values	Additional Information
LU Address	8	‘2A’ (Hex)	CPME 1 Summary Conditions
Message Function	8	“G”, “g” (ASCII)	“G” = Site Data Report “g” = Nonresponsive or Offline
Status	8	“A”, “C”, or “E” (ASCII)	“A” = Normal “C” = Red Alarm “E” = Soft Alarm (Yellow)
Value (DP = 20–48)	16	‘0’ (Hex)	Not Applicable, always ‘0’. (See Table 7–3.)

TABLE 7–3. CPME SUMMARY CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNIT 2A

Data Point Number (Hex)	Data Point Description	Value
20	Summary Front Face All-Call Report Missing Test Data	0
21	Summary Front Face All-Call Azimuth Test Data	0
22	Summary Front Face All-Call Range Test Data	0
23	Summary Front Face All-Call Capability Field Correct Test Data	0
24	Front Face All-Call Mode S ID Correct Test Data	0
25	Front Face Roll Call Report Missing Test Data	0
26	Front Face Roll Call Coarse Azimuth Test Data	0
27	Front Face Roll Call Range Test Data	0
28	Front Face Roll Call Altitude Test Data	0
29	Front Face Roll Call Code Test Data	0

TABLE 7–3. CPME SUMMARY CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNIT 2A (Continued)

Data Point Number (Hex)	Data Point Description	Value
2A	Front Face Roll Call Fine Azimuth Test Data	0
2B	Front Face ATCRBS Report Missing Test Data	0
2C	Front Face ATCRBS Azimuth Test Data	0
2D	Front Face ATCRBS Range Test Data	0
2E	Front Face ATCRBS Altitude Test Data	0
2F	Front Face ATCRBS Code Test Data	0
30	Ground Initiated Comm-B Data Link Missing Test Data	0
31	Ground Initiated Comm-B Data Link Error Test Data	0
32	Comm-A Comm-B Data Link Missing Test Data	0
33	Comm-A Comm-B B-bit Clear	0
34	Comm-A Comm-B Data Link Error Test Data	0
35	Comm-C Comm-D Data Link Missing Test Data	0
36	Comm-C Comm-D Dr Reset	0
37	Comm-C Comm-D Data Link Error Test Data	0
38	Back Face All-Call Report Missing Test Data	0
39	Back Face All-Call Azimuth Test Data	0
3A	Back Face All-Call Range Test Data	0
3B	Back Face All-Call Capability Field Correct Test Data	0
3C	Back Face All-Call Mode S ID Correct Test Data	0
3D	Back Face Roll Call Report Missing Test Data	0
3E	Back Face Roll Call Coarse Azimuth Test Data	0
3F	Back Face Roll Call Range Test Data	0
40	Back Face Roll Call Altitude Test Data	0
41	Back Face Roll Call Code Test Data	0
42	Back Face Roll Call Fine Azimuth Test Data	0
43	Back Face ATCRBS Report Missing Test Data	0
44	Back Face ATCRBS Azimuth Test Data	0
45	Back Face ATCRBS Range Test Data	0
46	Back Face ATCRBS Altitude Test Data	0
47	Back Face ATCRBS Code Test Data	0
48	CPME Status	0

7.2.3.2 The CPME 1 through 5 test conditions logical units (LU2B through LU2F) consists of data points presented in Table 7–4. The following information is applicable to the site data reports (monitor reports) for logical units LU2B through LU2F. The CPME test conditions monitor report shows the test values reported back by each CPME on the last interrogation. The number of these reports available will match the number of CPMEs associated with the Sensor. An attempt to produce a report for a non-existent CPME will cause the message “Selected report not available” to be displayed at the bottom of the local terminal’s screen. Status and value pairs are transmitted for all data points in the LU monitor report in increasing numerical order of data point identification, starting with data point 20. The status value reported in the CPME test conditions monitor report reflects alarm status and is interpreted as follows: (1) an A (green) status indicates a normal condition, (2) C (red) status indicates a hard alarm condition, and (3) E (yellow) status indicates a soft alarm condition. For additional information concerning the CPME (1–5) Test Conditions Reports (LU2B–LU2F), refer to Section 3 of the Mode S Local Terminal Users Manual TI 6365.11.

NOTE

If it has been determined that CPME corrective maintenance has to be performed because an alarm condition (value C or E) appears in the status column of the monitor report for the applicable CPME, the user must refer to Section 7 (Corrective Maintenance) of the Mode S Calibration Performance Monitor Equipment instruction book, TI 6365.8. To determine the corrective maintenance action to be taken, the user must refer to paragraphs 7.2.1 (CPME Test Set), 7.2.2 (CPME Test Set Tests), 7.3 (Onsite Corrective Maintenance), 7.3.1 (Onsite Fault Isolation), and 7.3.2 (LRU Replacement Procedures).

LU	0	MSG FUNCTION	STATUS	VALUE	STATUS	VALUE
STATUS	VALUE		STATUS	VALUE	STATUS	VALUE

Field	No. of Bits	Range of Values	Additional Information
LU Address	8	‘2B–2F’ (Hex)	2B = CPME 1 2C = CPME 2 2D = CPME 3 2E = CPME 4 2F = CPME 5
Message Function	8	“G”, “g” (ASCII)	“G” = Site Data Report “g” = Nonresponsive or Offline
Status	8	“A”, “C”, or “E”, (ASCII)	“A” = Normal (Green) “C” = Hard Alarm (Red) “E” = Soft Alarm (Yellow)
Value (DP20–49)	16	See Table 7–4	See Table 7–4

TABLE 7–4. CPME TEST CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNITS 2B THROUGH 2F

Data Point Number (Hex)	Data Point Description	Value	Range
20	Front Face All-Call Report Missing Test Data	0	NA
21	Front Face All-Call Azimuth Test Data	Measured Value (in Azimuth Units)	0–16383

TABLE 7-4. CPME TEST CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNITS 2B THROUGH 2F (Continued)

Data Point Number (Hex)	Data Point Description	Value	Range
22	Front Face All-Call Range Test Data	Measured Value (in Range Units)	-32768 to +32767 RUs
23	Front Face All-Call Capability Field Correct Test Data	0 = No extended capability report is available. 1 = Comm A/B extended capability report is available. 2 = Comm A/B/C and extended capability report is available. 3 = Comm A/B/C/D and extended capability report is available.	0-3
24	Front Face All-Code Mode S ID Correct Test Data	Measured Value, Mode S ID	$0-(2^{24}-1)$
25	Front Face Roll Call Report Missing Test Data	0	NA
26	Front Face Roll Call Coarse Azimuth Test Data	Measured Value (in Azimuth Units)	0-16383
27	Front Face Roll Call Range Test Data	Measured Value (in Range Units)	-32768 to +32767 RUs
28	Front Face Roll Call Altitude Test Data	Measured Value (in 100 ft. increments)	0-10 to +999
29	Front Face Roll Call Code Test Data	Measured Value ATCRBS ID Code	0-7777 (Octal)
2A	Front Face Roll Call Fine Azimuth Test Data	Measured Value (in Azimuth Units)	0-16383
2B	Front Face ATCRBS Report Missing Test Data	0	NA
2C	Front Face ATCRBS Azimuth Test Data	Measured Value (in Azimuth Units)	0-16383
2D	Front Face ATCRBS Range Test Data	Measured Value (in Range Units)	-32768 to +32767 RUs
2E	Front Face ATCRBS Altitude Test Data	Measured Value (in 100 ft. increments)	-10 to +999

TABLE 7–4. CPME TEST CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNITS 2B THROUGH 2F (Continued)

Data Point Number (Hex)	Data Point Description	Value	Range
2F	Front Face ATCRBS Code Test Data	Measured Value ATCRBS ID Code	0–7777 (Octal)
30	Ground Initiated Comm-B Data Link Missing Test Data	0	NA
31	Ground Initiated Comm-B Data Link Error Test Data	0	NA
32	Comm-A Comm-B Data Link Missing Test Data	0	NA
33	Comm-A Comm-B B-bit Clear	0	NA
34	Comm-A Comm-B Data Link Error Test Data	0	NA
35	Comm-C Comm-D Data Link Missing Test Data	0	NA
36	Comm-C Comm-D Dr Reset	0	NA
37	Comm-C Comm-D Data Link Error Test Data	0	NA
38	Back Face All-Call Report Missing Test Data	0	NA
39	Back Face All-Call Azimuth Test Data	Measured Value (in Azimuth Units)	0–16383
3A	Back Face All-Call Range Test Data	Measured Value (in Range Units)	–32768 to +32767 RUs
3B	Back Face All-Call Capability Field Correct Test Data	0 = No extended capability report is available. 1 = Comm A/B extended capability report is available. 2 = Comm A/B/C and extended capability report is available. 3 = Comm A/B/C/D and extended capability report is available.	0–3

TABLE 7-4. CPME TEST CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNITS 2B THROUGH 2F (Continued)

Data Point Number (Hex)	Data Point Description	Value	Range
3C	Back Face All-Call Mode S ID Correct Test Data	Measured Value Mode S ID	NA
3D	Back Face Roll Call Report Missing Test Data	0	NA
3E	Back Face Roll Call Coarse Azimuth Test Data	Measured Value (in Azimuth Units)	0-16383
3F	Back Face Roll Call Range Test Data	Measured Value (in Range Units)	-32768 to +32767 RUs
40	Back Face Roll Call Altitude Test Data	Measured Value (in 100 ft. increments)	-10 to +999
41	Back Face Roll Call Code Test Data	Measured Value ATCRBS ID Code	0-7777 (Octal)
42	Back Face Roll Call Fine Azimuth Test Data	Measured Value (in Azimuth Units)	0-16383
43	Back Face ATCRBS Report Missing Test Data	0	NA
44	Back Face ATCRBS Azimuth Test Data	Measured Value (in Azimuth Units)	0-16383
45	Back Face ATCRBS Range Test Data	Measured Value (in Range Units)	-32768 to +32767 RUs
46	Back Face ATCRBS Code Test Data	Measured Value ATCRBS ID Code	-10 to +999
47	Back Face ATCRBS Altitude Test Data	Measured Value (in 100 ft. increments)	0-7777 (Octal)
48	CPME Alert Status Missing	0	NA

TABLE 7-4. CPME TEST CONDITIONS MONITOR REPORT STATUS AND FAULT MESSAGES FOR LOGICAL UNITS 2B THROUGH 2F (Continued)

Data Point Number (Hex)	Data Point Description	Value	Range
49	CPME Alert Status	0 = No Fault Conditions. 1 = Miscellaneous Fault Conditions. 2 = Inside of enclosure is over temperature. 4 = Inside of enclosure is under temperature. 8 = Power Failure. 10 = Oscillator is out of phase lock.	0-10
4A	CPME ATCRBS Track	0 = Normal 1052 = ATCRBS Track Drop	
4B	CPME ModeS Track	0 = Normal 1062 = ModeS Track Drop	

Note: In least significant bit (LSB column; 1 AU = 1 Azimuth Unit, 1 RU = 1 Range Unit, and NA = Not Applicable.

7.3 FAULT ANALYSIS PROCEDURES

Information concerning the fault analysis procedures for the various Mode S system equipment such as the Interrogator, Data Processing System (DPS) and Calibration Performance Monitor Equipment (CPME) is presented in Section 7, paragraph 7.3 of the applicable Mode S system equipment instruction book as indicated in paragraph 7.4. In the Interrogator equipment instruction book, TI 6365.5, the fault analysis procedures provide methods and procedures for analyzing faults in the power distribution and environment control equipment. A signal flow diagram, the Interrogator fault isolation test (FIT) tree, wiring data, and parameter monitoring subjects are also discussed under the fault analysis procedures. Additional information for the Interrogator fault analysis procedures is provided in the Transmitter instruction book, TI 6365.6, the Receiver/RFTTG instruction book TI 6365.7, and Signal Processor Instruction book, TI 6365.4. In the DPS equipment instruction book TI 6365.3, the fault isolation consists of a table that associates DPS data points (that are used to report fault conditions) with LRU assemblies and, where necessary, also references supplemental maintenance procedures that isolate general failure indicators. In the CPME instruction book, TI 6365.8, the fault analysis procedures are presented in paragraph 7.3 (Onsite Corrective Maintenance) which is divided into onsite fault isolation to a defective LRU and replacement procedures for defective LRUs.

7.4 REMOVAL AND INSTALLATION PROCEDURES

Information concerning removal and installation procedures for the various Mode S system equipment such as the Interrogator, Data Processing System (DPS), and Calibration Performance Monitor Equipment (CPME) is presented in Section 7 of the applicable Mode S system equipment instruction book as indicated in the following list:

- a. Interrogator instruction book TI 6365.5
 1. Transmitter instruction book TI 6365.6
 2. Receiver/RFTTG instruction book TI 6365.7
 3. Signal Processor instruction book TI 6365.4
- b. Data Processing System (DPS) instruction book TI 6365.3
- c. Calibration Performance Monitor Equipment (CPME) instruction book TI 6365.8

7.5 CABLE WIRING DATA

The cable connection diagrams for the Mode S system configurations are described in the following list:

<u>Figure Number</u>	<u>Description</u>
7-2	MSJB-1, RF Transfer Switch, and Interrogator System Cabling Diagram for a Terminal Site
7-3	Interrogator and DPS System Cabling Diagram for a Terminal Site
7-4	MSJB-1, RF Transfer Switches, and Interrogator System Cabling Diagram for an Enroute Site
7-5	Interrogator to DPS System Cabling Diagram for an Enroute Site
7-6	DPS System Cable Configuration Diagram (6 Sheets)

The Mode S system cable configuration diagrams shown the cable connections between the MSJB-1, RF Transfer Switch(es), and DPS for a terminal site and an en route site (refer to Figure 7-2 through 7-5). The DPS system cable configuration diagram (Figure 7-6) shows the cable connections between the Data Processing System (DPS)-PRCSS cabinets (unit 4 and 5) via the primary Communications Junction Box (unit 6), the direct cable connections between DPS-PRCSS Channel A (unit 4) and DPS-PRCSS Channel B (unit 5), the cable connections from the DPS-PRCSS (Channel B and B) to the respective Channel A and B of the Interrogator (units 2 and 3), at a terminal site, and the cable connections from the DPS-PRCSS to the outside world via the primary Communications Junction Box (unit 6). Figure 7-6 also presents the cable connections from the DPS-PRCSS (units 4 and 5) to the Interrogator (units 2 and 8) at an en route site. The cable configuration diagrams also show the cable connections (when required) for the expansion Communications Junction Box (unit 10) and the expansion Surveillance Junction Box (unit 11) to the DPS-PRCSS cabinets (Channel A and B) and to the outside world. The cable designations (W####), the connector P## numbers, and the various junction box connector designations are shown on the cable configuration diagrams. Additional information concerning the functions of the various cable connectors located on the junction boxes, such as: (1) communication (COMM) lines, (2) building sensor (BLDG SNSR) lines, (3) Remote Maintenance Monitor System (RMT MAINT) lines, (4) Program Support Facility (PRGM SPRT) lines, (5) remote control (RMT CONT) lines, (6) local control (LCL CONT) lines, (7) primary radar (PRI RDR) lines, (8) surveillance (SR) lines, and (9) surveillance data selector control (SDS CONT) lines. Figures 7-7 through 7-10 show the cable connectors on the various junction boxes (units 6, 10, and 11) for the Mode S configurations. The input and output connector (J##) numbers, the function of that connector (such as: COMM A1, PRI RDR OUT B1), and the applicable channel (A or B) for that connector is shown on the various junction box configurations.

7.5.1 Mode S Interrogator, RF Transfer Switch, and MSJB-1 System Cable List

Information concerning Mode S Interrogators (units 2/3/8), RF transfer switches (units 9/13), and Mode S junction box 1 (MSJB-1, unit 1) system cable replacement is presented in the following list.

Mode S Interrogator, RF Transfer Switch, and MSJB-1 System Cable Information

<u>Figure Number</u>	<u>Description</u>
7-11	Interrogator and RF Transfer Switch Cable Interconnection Diagram for Terminal Site and Enroute Site
7-12	Interrogator Cable Assemblies W64-W66, W76-W78, and W100-W102
7-13	Interrogators (Units 2 and 3) — RF Transfer Switch (Unit 9) Harness Assembly W67 for a Terminal Site
7-14	Interrogators (Units 2 and 8) — RF Transfer Switches (Units 9 and 13) Harness Assembly W67 for an Enroute Site
7-15	Cable Pinout for Interrogator Cable Assemblies W68, W69, W73, and W74
7-16	Cable Pinout for Interrogator Cable Assemblies W70 and W71
7-17	Cable Pinout for Interrogator Cable Assemblies W72 and W75
7-18	Cable Pinout for Interrogator Cable Assemblies W79 and W80

7.5.2 Mode S DPS System Cable List

Information concerning Mode S DPS system cable replacement is presented in the following Mode S System Cable Lowest Replaceable Unit (LRU) List.

Mode S DPS System Cables LRU List

<u>Description</u>	<u>Part Number</u>	<u>Site Replaceable</u>	<u>Figure Number</u>
System Cable Type K1	3261 8399, 00-03	Yes	7-31
System Cable Type A1	3261 8423, 00-03	Yes	7-19
System Cable Type F1	3261 8472, 00-03	Yes	7-25
System Cable Type G1	3262 0007, 00-03	Yes	7-26
System Cable Type G2	3262 0015, 00-03	Yes	7-27
System Cable Type H1	3262 0023, 00-03	Yes	7-28
System Cable Type E1	3262 0049, 00-03	Yes	7-24
System Cable Type C1	3262 0064	Yes	7-22
System Cable Type B1	3262 0072	Yes	7-20
System Cable Type D1	3262 0080	Yes	7-23
System Cable Type J1	3262 0171	Yes	7-29
System Cable Type J2	3262 0189	Yes	7-30
System Cable Type B2	3588 0459	Yes	7-21

7.6 PACKING INSTRUCTIONS

Information concerning packing instructions (procedures) for the various Mode S system equipment (Interrogator, Data Processing System (DPS), and Calibration Performance Monitor Equipment-CPME) is presented in Section 7 of the applicable Mode S system equipment instruction book as indicated in paragraph 7.4.

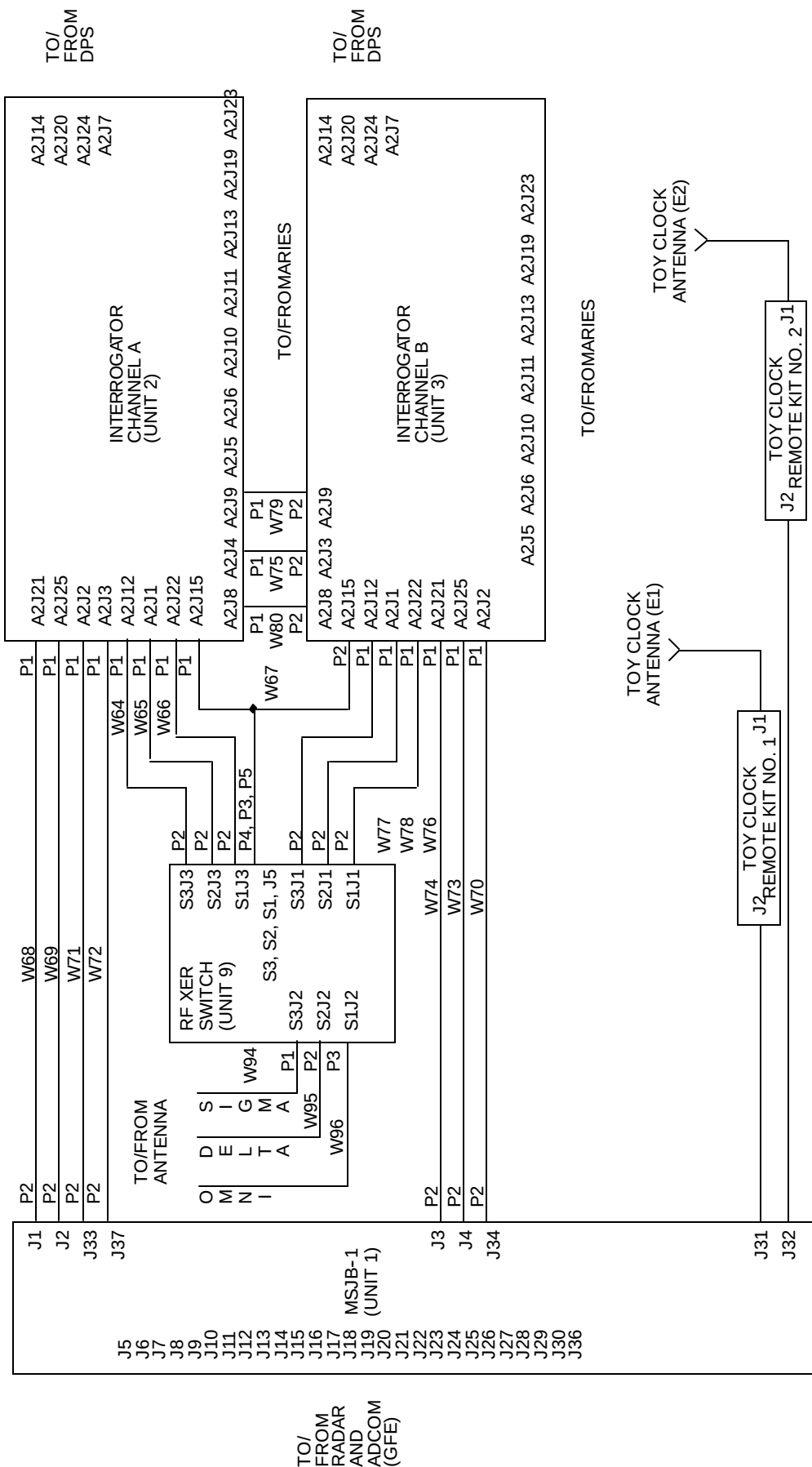


FIGURE 7-2. MSJB-1, RF TRANSFER SWITCH, AND INTERROGATOR SYSTEM CABLING DIAGRAM FOR A TERMINAL SITE

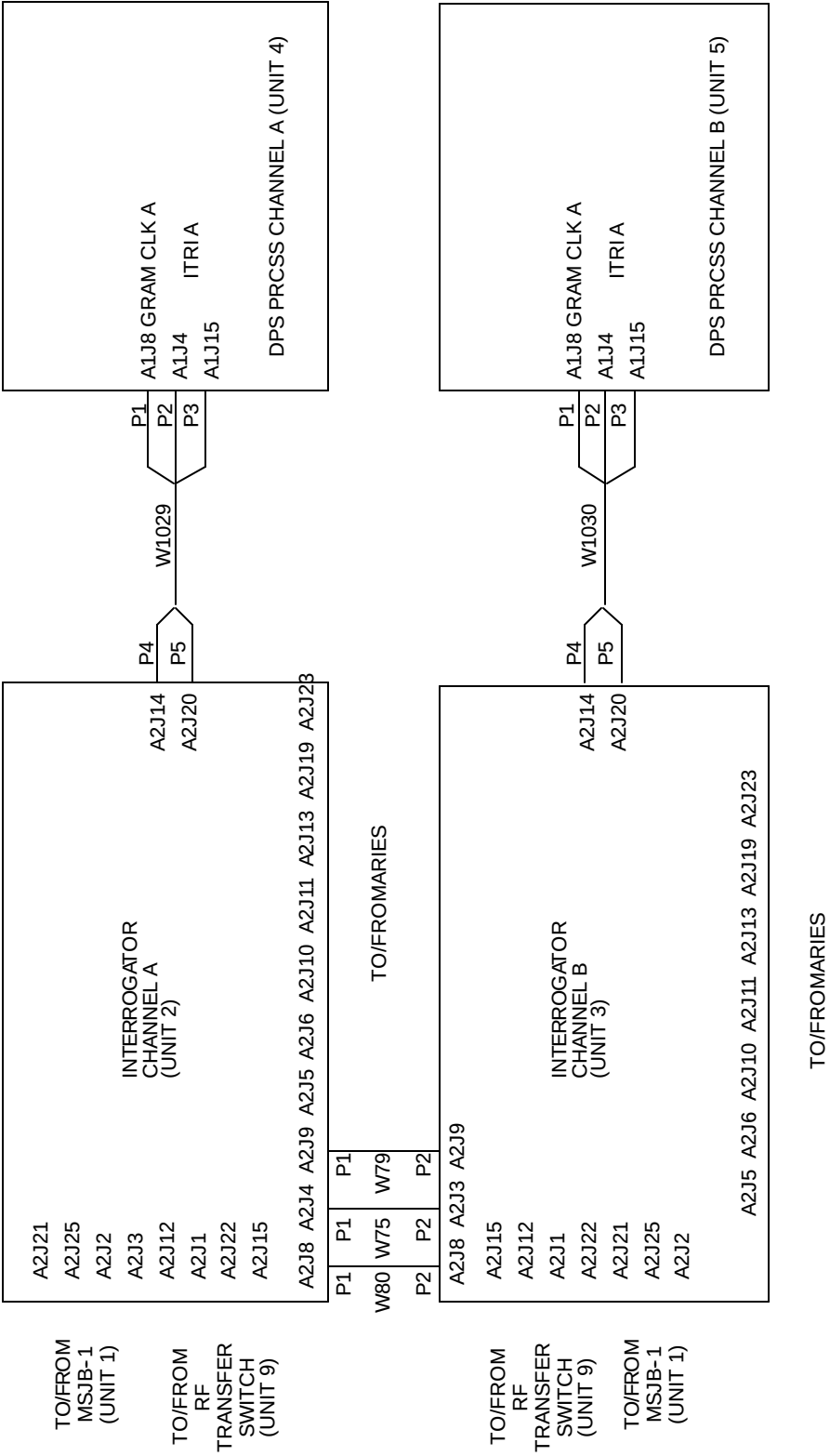


FIGURE 7-3. INTERROGATOR AND DPS SYSTEM CABLING DIAGRAM FOR A TERMINAL SITE

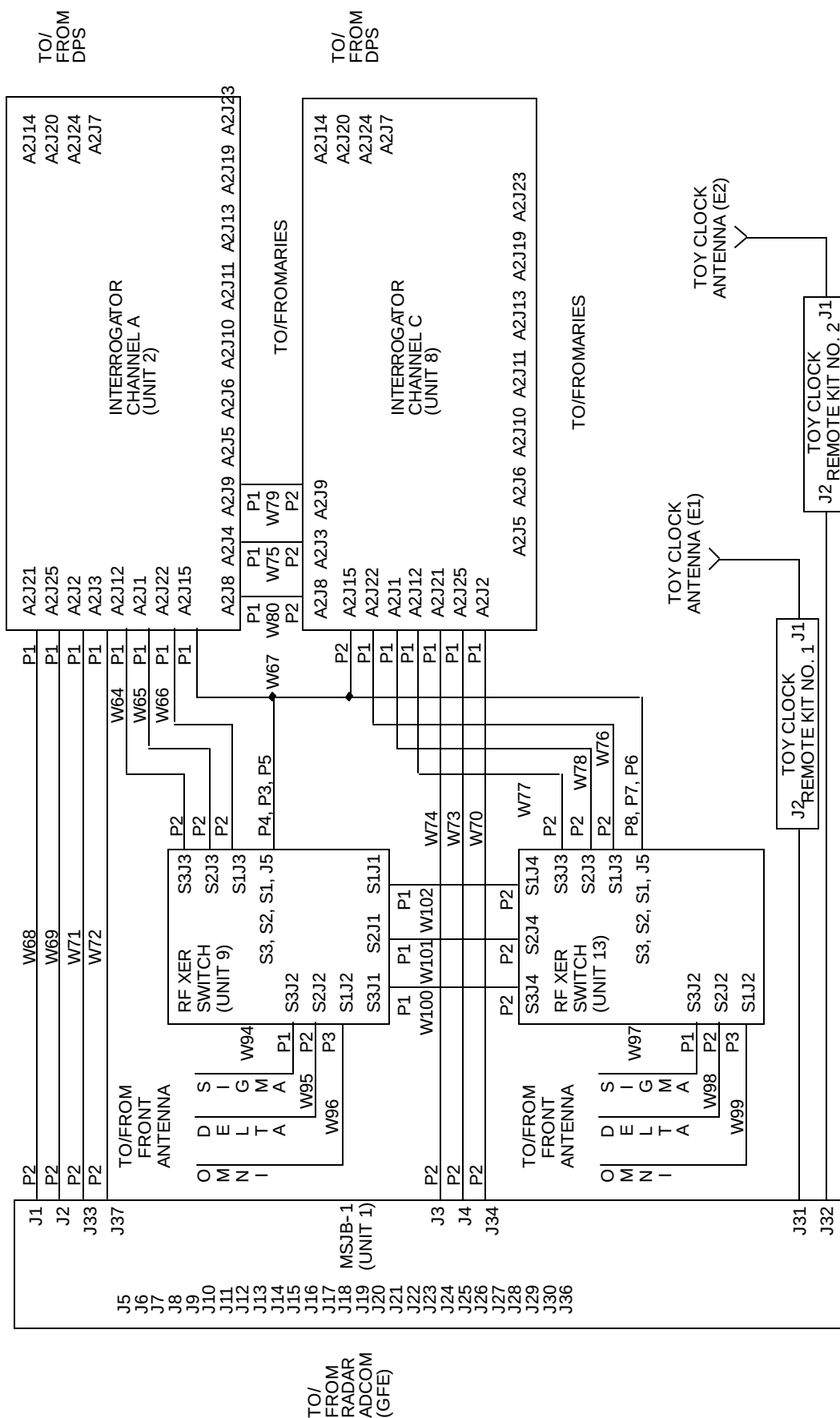


FIGURE 7-4. MSJB-1, RF TRANSFER SWITCHES, AND INTERGATOR SYSTEM CABLING DIAGRAM FOR AN EN ROUTE SITE

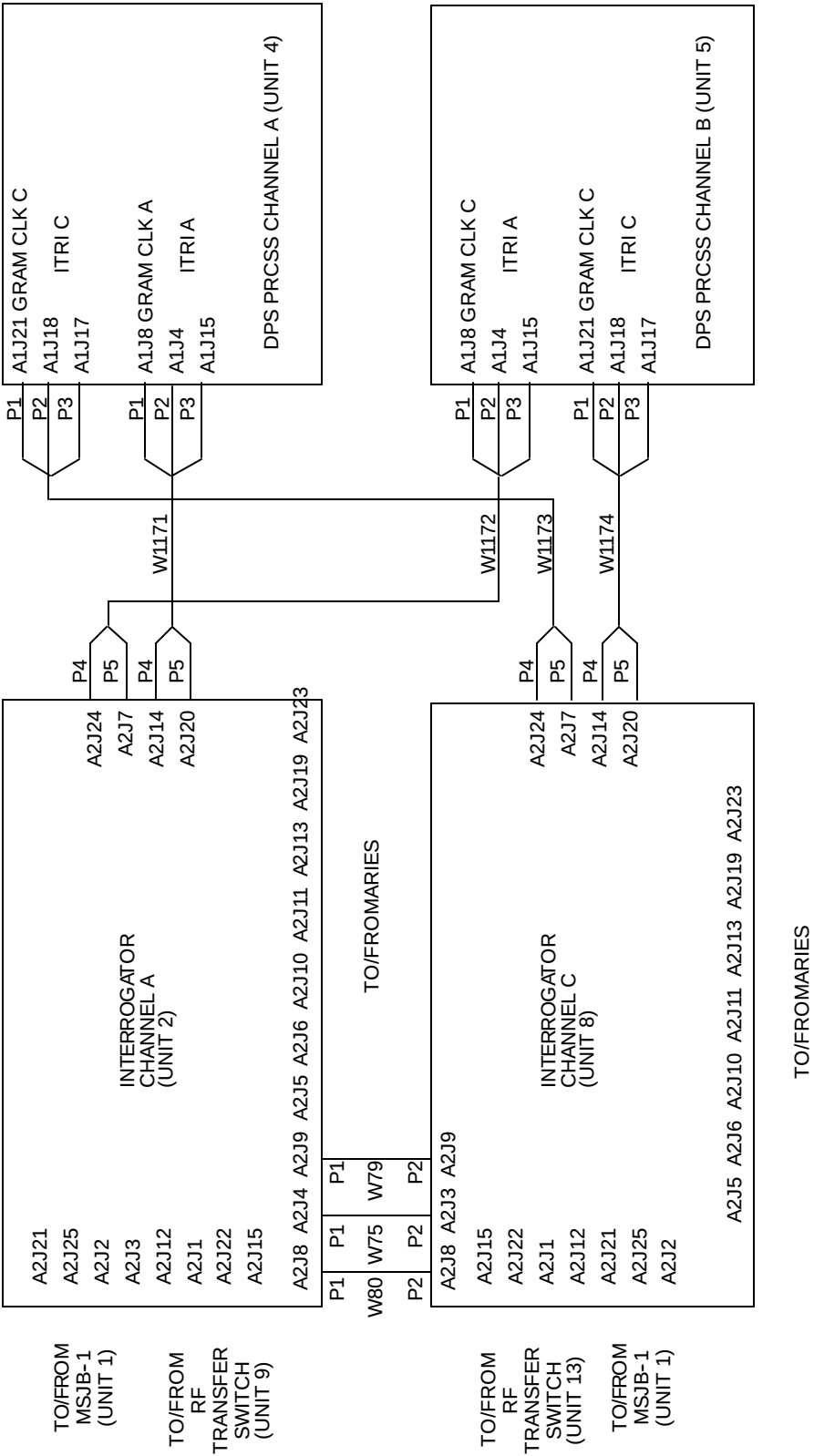


FIGURE 7-5. INTERROGATOR AND DPS SYSTEM CABLING DIAGRAM FOR AN EN ROUTE SITE

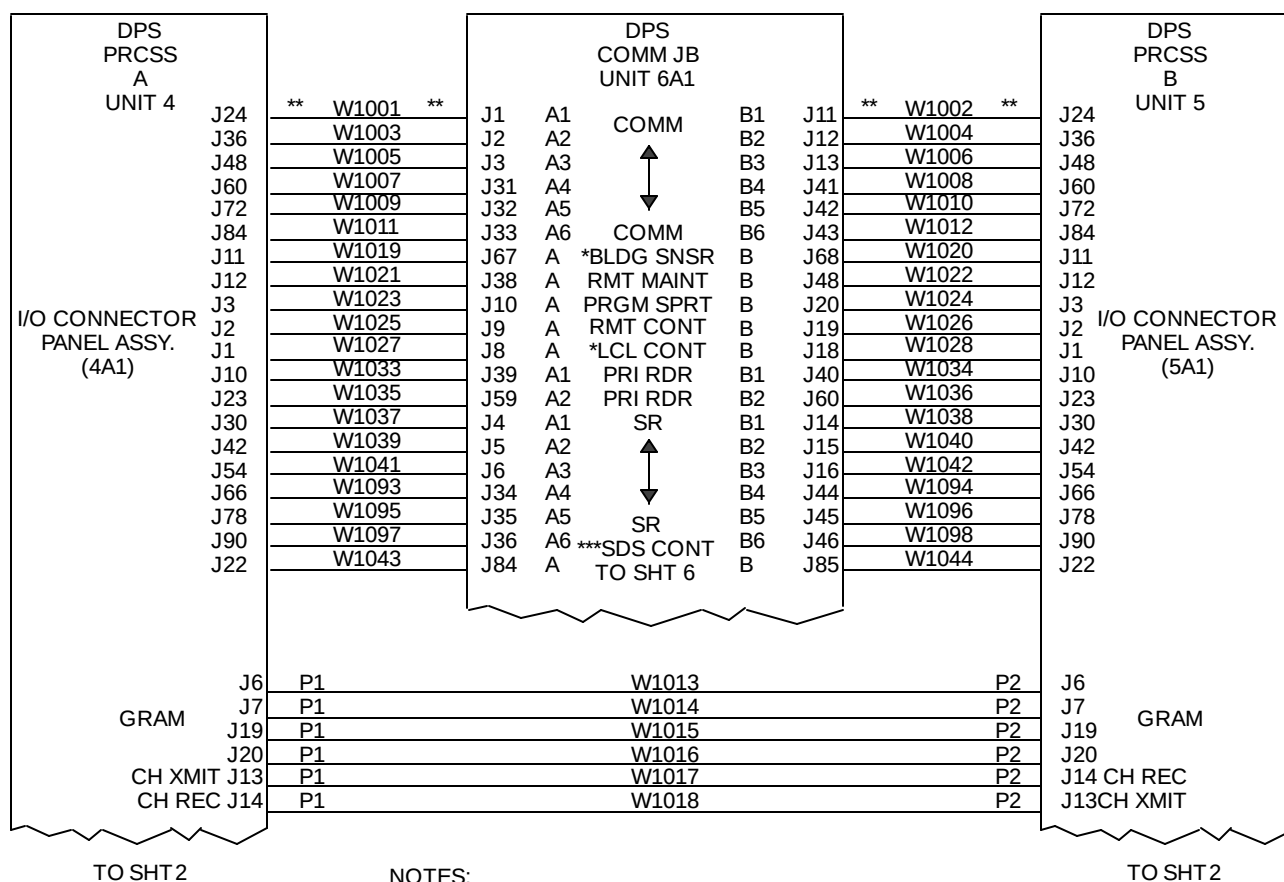


FIGURE 7-6. DPS SYSTEM CABLE CONFIGURATION DIAGRAM (SHEET 1)

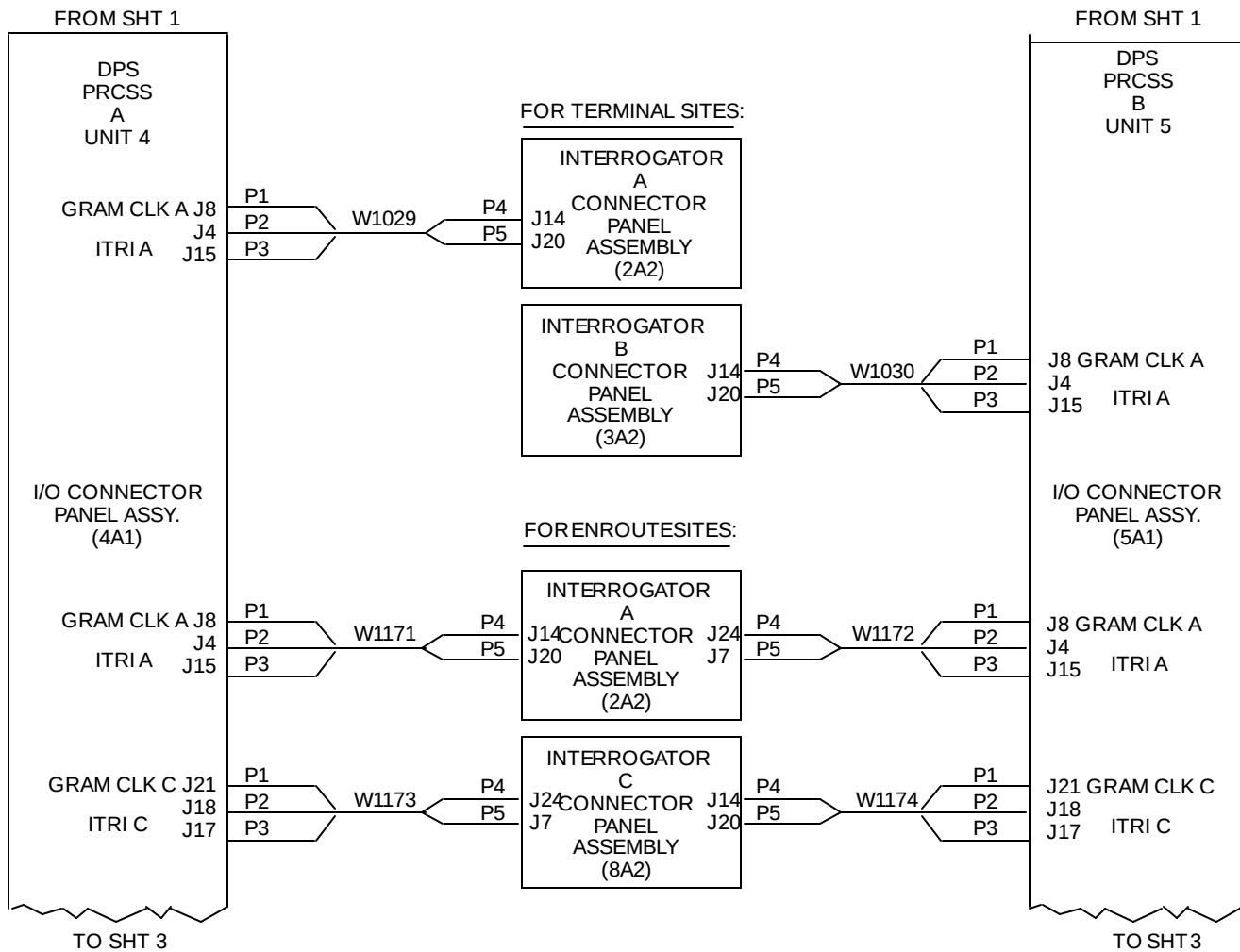
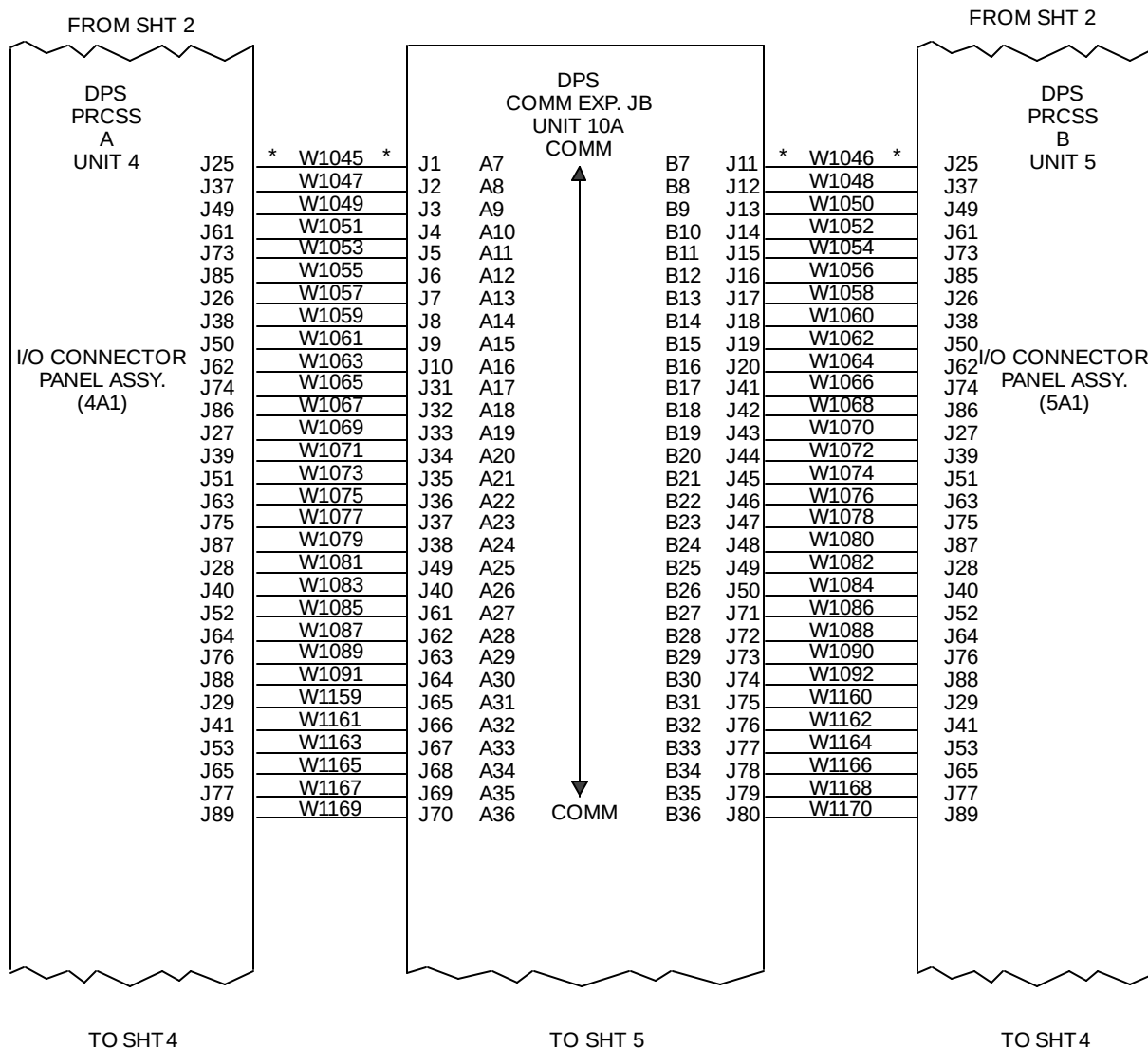


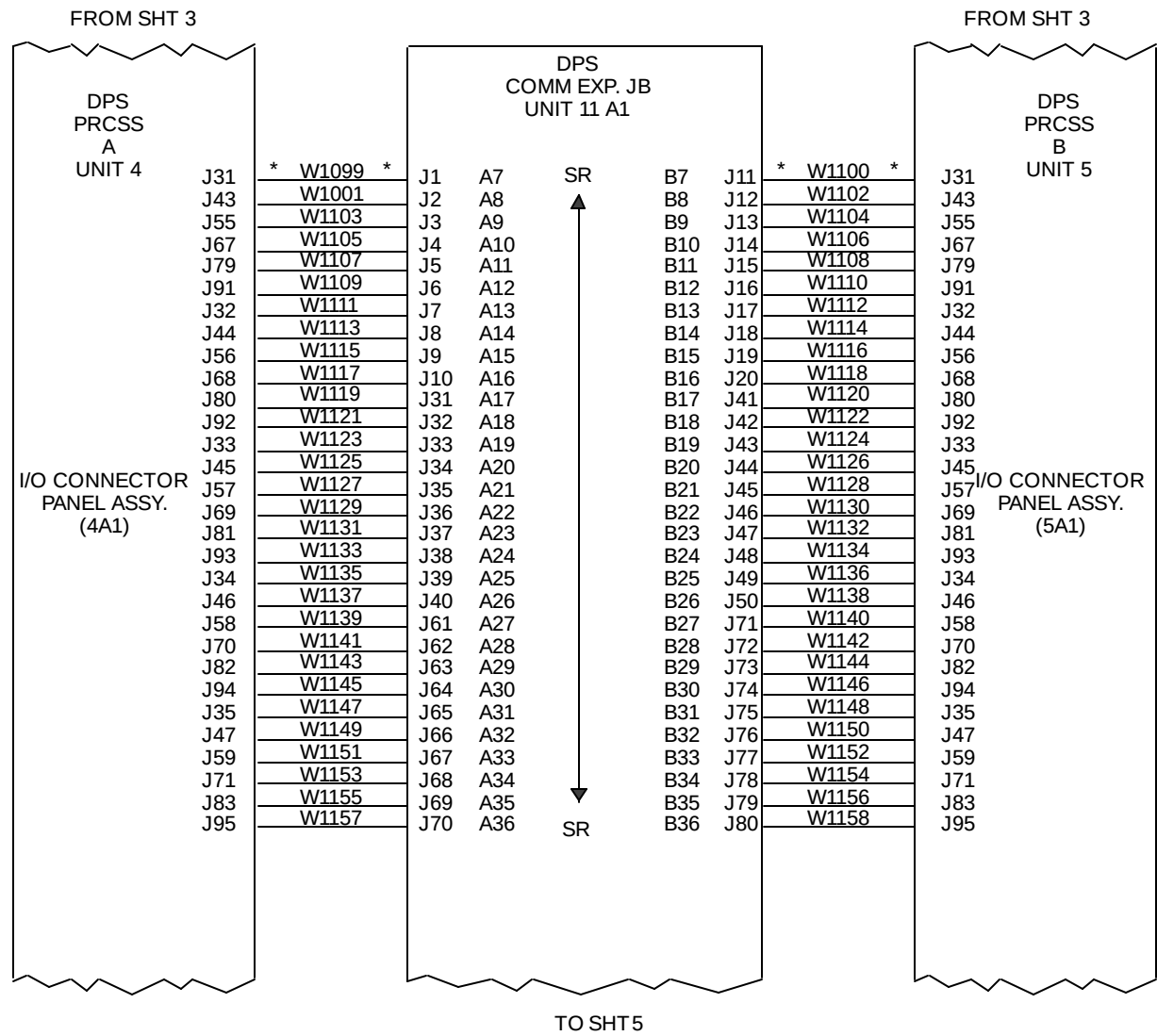
FIGURE 7-6. DPS SYSTEM CABLE CONFIGURATION DIAGRAM (SHEET 2)



NOTE:

1. * CABLE CONNECTORS TO PRCSS CABINETS ARE DESIGNATED P1.
CABLE CONNECTORS TO JUNCTION BOX ARE DESIGNATED P2.

FIGURE 7-6. DPS SYSTEM CABLE CONFIGURATION DIAGRAM (SHEET 3)



NOTE:
1. * CABLE CONNECTORS TO PRCSS CABINETS ARE DESIGNATED P1.
CABLE CONNECTORS TO JUNCTION BOX ARE DESIGNATED P2.

FIGURE 7-6. DPS SYSTEM CABLE CONFIGURATION DIAGRAM (SHEET 4)

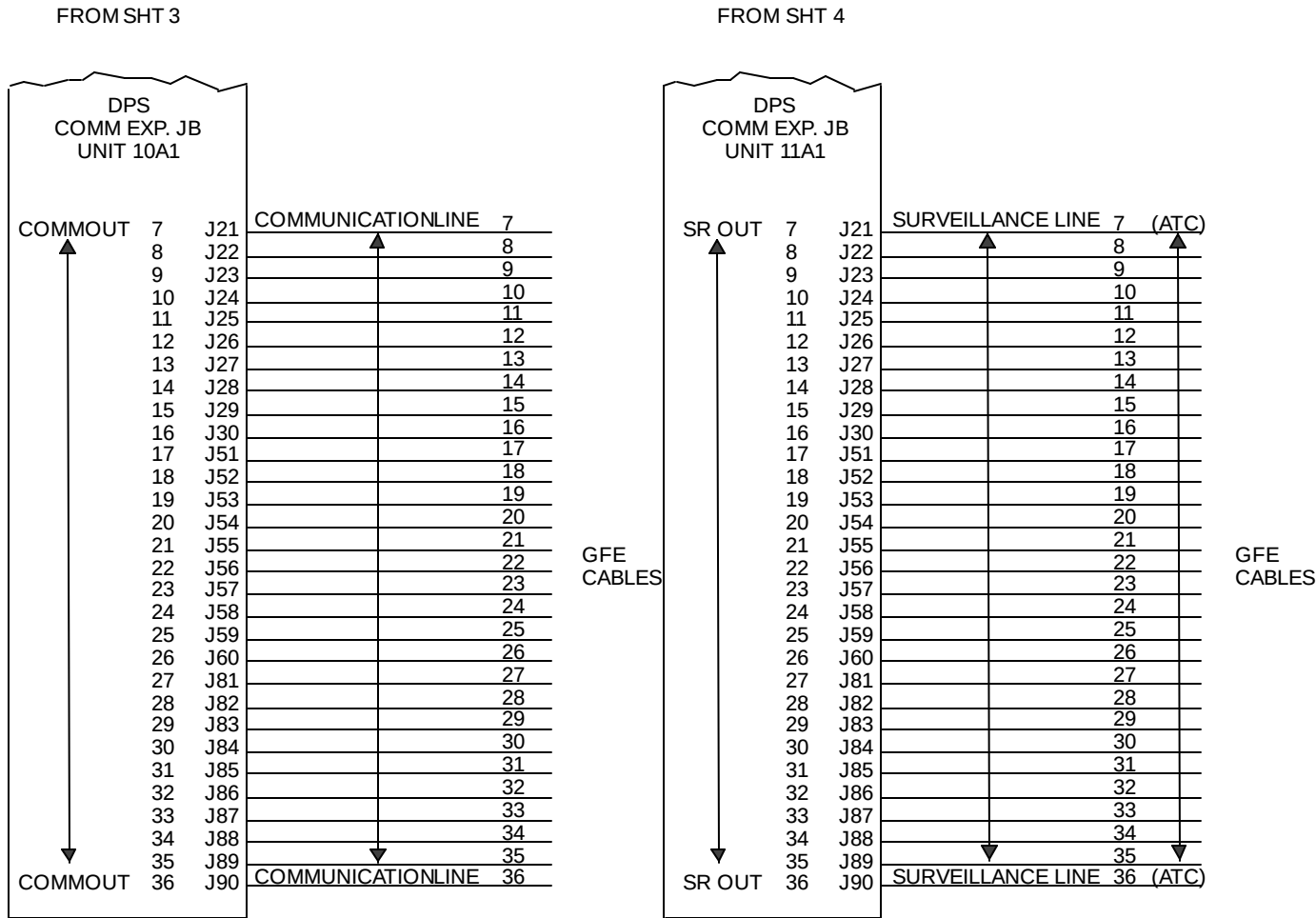
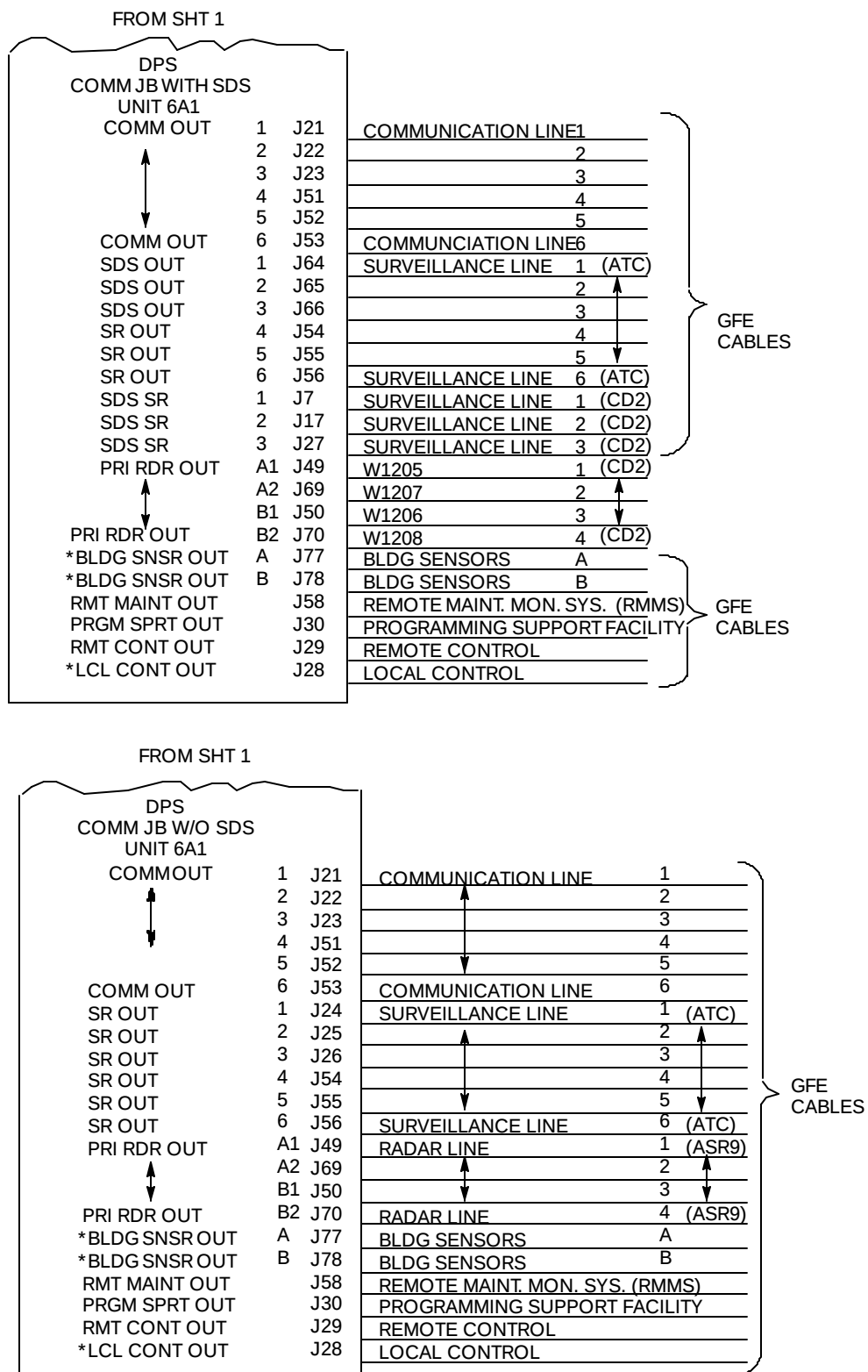


FIGURE 7-6. DPS SYSTEM CABLE CONFIGURATION DIAGRAM (SHEET 5)



NOTE : * INDICATES NOT USED IN THIS CONFIGURATION.

e7089_16_960530

FIGURE 7-6. DPS SYSTEM CABLE CONFIGURATION DIAGRAM (SHEET 6 — END)

COMM A1 J1	COMM A2 J2	COMM A3 J3	SR A1 J4	SR A2 J5	SR A3 J6		LCL CONT A J8	RMT CONT A J9	PRGM SPRT A J10
COMM B1 J11	COMM B2 J12	COMM B3 J13	SR B1 J14	SR B2 J15	SR B3 J16		LCL CONT B J18	RMT CONT B J19	PRSM SPRT B J20
COMM OUT 1 J21	COMM OUT 2 J22	COMM OUT 3 J23	SR OUT 1 J24	SR OUT 2 J25	SR OUT 3 J26		LCL CONT OUT J28	RMT CONT OUT J29	PRGM SPRTOUT J30
COMM A4 J31	COMM A5 J32	COMM A6 J33	SR A4 J34	SR A5 J35	SR A6 J36		RMT MAINT A J38	PRI RDR A1 J39	PRI RDR B1 J40
COMM B4 J41	COMM B5 J42	COMM B6 J43	SR B4 J44	SR B5 J45	SR B6 J46		RMT MAINT B J48	PRI RDR OUT A1 J49	PRI RDR OUT B1 J50
COMM OUT 4 J51	COMM OUT 5 J52	COMM OUT 6 J53	SR OUT 4 J54	SR OUT 5 J55	SR OUT 6 J56		RMT MAINT OUT J58	PRI RDR A2 J59	PRI RDR B2 J60
						BLDG SNSR A J67	BLDG SNSR 8 J68	PRI RDR OUT A2 J69	PRI ADR OUT B2 J70
						BLDG SNSR OUT A J77	BLDG SNSR OUT B J78		

COMMUNICATION JUNCTION BOX (UNIT 6) — BASIC CONFIGURATION

FIGURE 7–7. COMMUNICATION JUNCTION BOX (UNIT 6), BASIC CONFIGURATION

COMM A1 J1	COMM A2 J2	COMM A3 J3	SR A1 J4	SR A2 J5	SR A3 J6	SDS SR1 J7	LCL CONT A J8	RMT CONT A J9	PRGM SPRT A J10
COMM B1 J11	COMM B2 J12	COMM B3 J13	SR B1 J14	SR B2 J15	SR B3 J16	SDS SA2 J17	LCL CONT B J18	RMT CONT B J19	PRSM SPRT B J20
COMM OUT 1 J21	COMM OUT 2 J22	COMM OUT 3 J23				SDS SR3 J27	LCL CONT OUT J28	AMT CONT OUT J29	PRSM SPRTOUT J30
COMM A4 J31	COMM A5 J32	COMM A6 J33	SR A4 J34	SR A5 J35	SR A6 J36		RMT MAINT A J38	PRI ADR A1 J39	PRI ADR B1 J40
COMM B4 J41	COMM B5 J42	COMM B6 J43	SR B4 J44	SR B5 J45	SR B6 J46		RMT MAINT B J48	PRI RDR OUT A1 J49	PRI ADR OUT B1 J50
COMM OUT 4 J51	COMM OUT 5 J52	COMM OUT 6 J53	SR OUT 4 J54	SR OUT 5 J55	SR OUT 6 J58		RMT MAINT OUT J58	PRI ADR A2 J59	PRI ADR B2 J60
			SDS OUT 1 J64	SDS OUT 2 J65	SDS OUT 3 J66	BLDG SNSR A J67	BLDG SNSR 8 J68	PRI RDR OUT A2 J69	PRI ADR OUT B2 J70
						BLDG SNSROUT A J77	BLDG SNSROUT B J78		
			SDS CONT A J84	SDS CONT B J85					

COMMUNICATION JUNCTION BOX (UNIT 6) WITH SDS

FIGURE 7–8. COMMUNICATION JUNCTION BOX (UNIT 6) WITH SDS

COMM A7 J1	COMM A8 J2	COMM A9 J3	COMM A10 J4	COMM A11 J5	COMM A12 J6	COMM A13 J7	COMM A14 J8	COMM A15 J9	COMM A16 J10
COMM B7 J11	COMM B8 J12	COMM B9 J13	COMM B10 J14	COMM B11 J15	COMM B12 J16	COMM B13 J17	COMM B14 J18	COMM B15 J19	COMM B16 J20
COMM OUT 7 J21	COMM OUT 8 J22	COMM OUT 9 J23	COMM OUT 10 J24	COMM OUT 11 J25	COMM OUT 12 J26	COMM OUT 13 J27	COMM OUT 14 J28	COMM OUT 15 J29	COMM OUT 16 J30
COMM A17 J31	COMM A18 J32	COMM A19 J33	COMM A20 J34	COMM A21 J35	COMM A22 J36	COMM A23 J37	COMM A24 J38	COMM A25 J39	COMM A26 J40
COMM B17 J41	COMM B18 J42	COMM B19 J43	COMM B20 J44	COMM B21 J45	COMM B22 J46	COMM B23 J47	COMM B24 J48	COMM B25 J49	COMM B26 J50
COMM OUT 17 J51	COMM OUT 18 J52	COMM OUT 19 J53	COMM OUT 20 J54	COMM OUT 21 J55	COMM OUT 22 J56	COMM OUT 23 J57	COMM OUT 24 J58	COMM OUT 25 J59	COMM OUT 26 J60
COMM A27 J61	COMM A28 J62	COMM A29 J63	COMM A30 J64	COMM A31 J65	COMM A32 J66	COMM A33 J67	COMM A34 J68	COMM A35 J69	COMM A36 J70
COMM B27 J71	COMM B28 J72	COMM B29 J73	COMM B30 J74	COMM B31 J75	COMM B32 J76	COMM B33 J77	COMM B34 J78	COMM B35 J79	COMM B36 J80
COMM OUT 27 J81	COMM OUT 28 J82	COMM OUT 29 J83	COMM OUT 30 J84	COMM OUT 31 J85	COMM OUT 32 J86	COMM OUT 33 J87	COMM OUT 34 J88	COMM OUT 35 J89	COMM OUT 36 J90

COMMUNICATION EXPANSION JUNCTION BOX (UNIT 10)

FIGURE 7–9. COMMUNICATION EXPANSION JUNCTION BOX (UNIT 10)

SR A7 J1	SR A8 J2	SR A9 J3	SR A10 J4	SR A11 J5	SR A12 J6	SR A13 J7	SR A14 J8	SR A15 J9	SR A16 J10
SR B7 J11	SR B8 J12	SR B9 J13	SR B10 J14	SR B11 J15	SR B12 J16	SR B13 J17	SR B14 J18	SR B15 J19	SR B16 J20
SR OUT 7 J21	SR OUT 8 J22	SR OUT 9 J23	SR OUT 10 J24	SR OUT 11 J25	SR OUT 12 J26	SR OUT 13 J27	SR OUT 14 J28	SR OUT 15 J29	SR OUT 16 J30
SR A17 J31	SR A18 J32	SR A19 J33	SR A20 J34	SR A21 J35	SR A22 J36	SR A23 J37	SR A24 J38	SR A25 J39	SR A26 J40
SR B17 J41	SR B18 J42	SR B19 J43	SR B20 J44	SR B21 J45	SR B22 J46	SR B23 J47	SR B24 J48	SR B25 J49	SR B26 J50
SR OUT 17 J51	SR OUT 18 J52	SR OUT 19 J53	SR OUT 20 J54	SR OUT 21 J55	SR OUT 22 J56	SR OUT 23 J57	SR OUT 24 J58	SR OUT 25 J59	SR OUT 26 J60
SR A27 J61	SR A28 J62	SR A29 J63	SR A30 J64	SR A31 J65	SR A32 J66	SR A33 J67	SR A34 J68	SR A35 J69	SR A36 J70
SR B27 J71	SR B28 J72	SR B29 J73	SR B30 J74	SR B31 J75	SR B32 J76	SR B33 J77	SR B34 J78	SR B35 J79	SR B36 J80
SR OUT 27 J81	SR OUT 28 J82	SR OUT 29 J83	SR OUT 30 J84	SR OUT 31 J85	SR OUT 32 J86	SR OUT 33 J87	SR OUT 34 J88	SR OUT 35 J89	SR OUT 36 J90

SURVEILLANCE EXPANSION JUNCTION BOX (UNIT 10)

FIGURE 7–10. SURVEILLANCE EXPANSION JUNCTION BOX (UNIT 11)

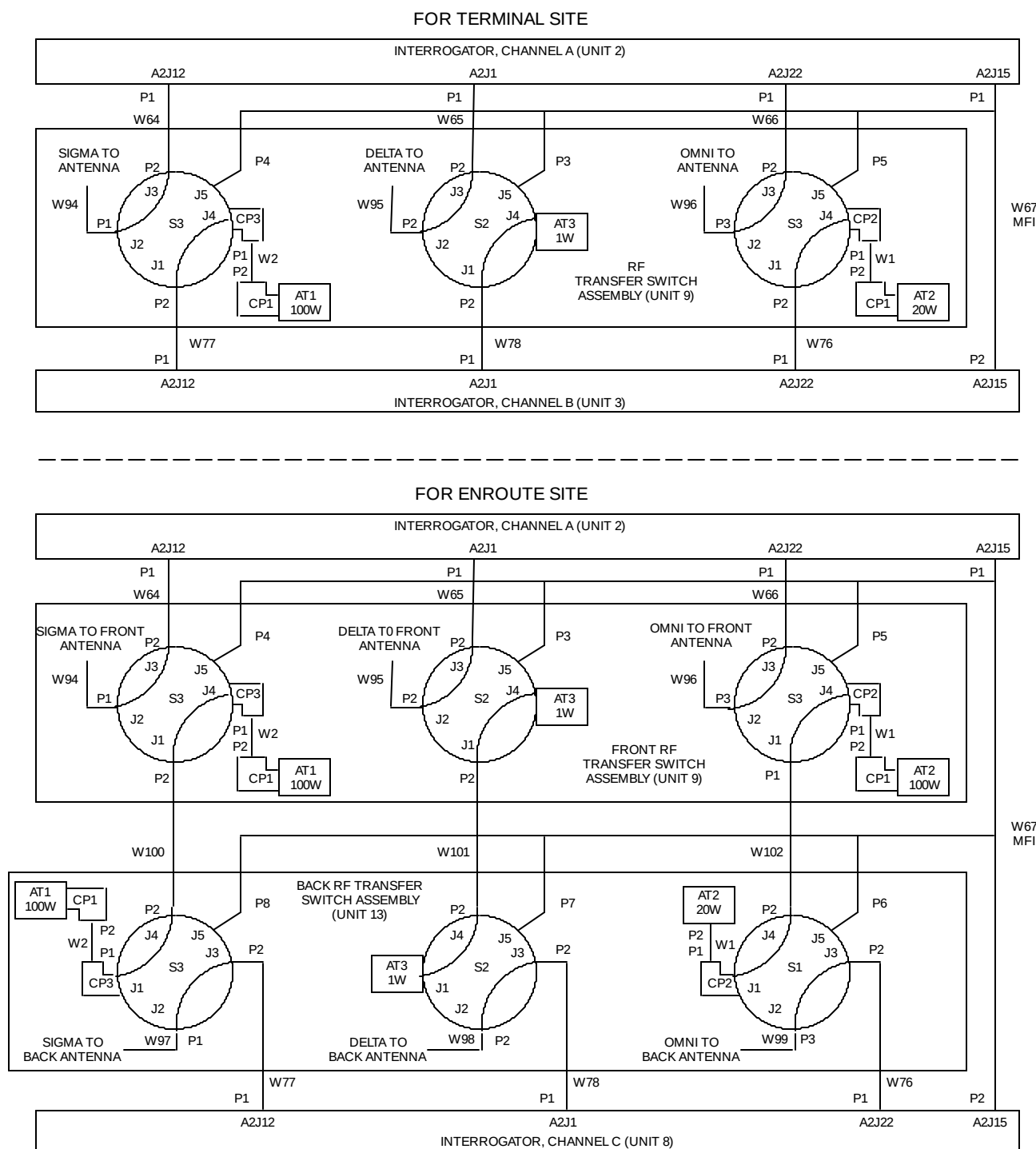
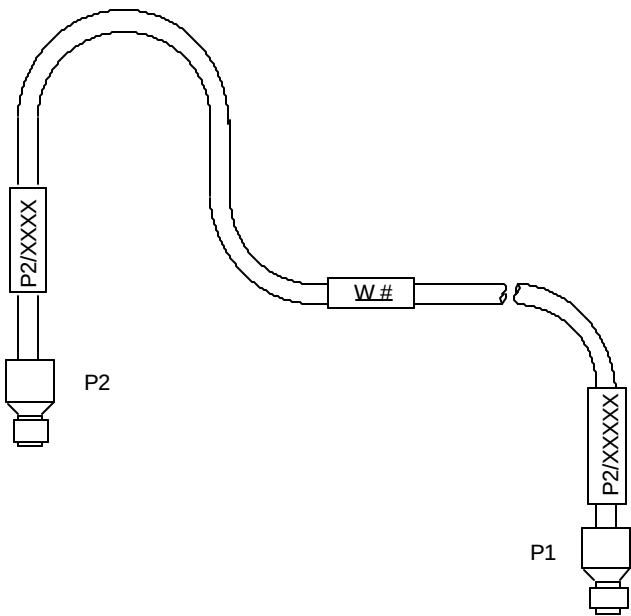


FIGURE 7-11. INTERROGATOR AND RF TRANSFER SWITCH CABLE INTERCONNECTION DIAGRAM FOR TERMINAL SITE AND EN ROUTE SITE



<u>COAXIAL CABLE NO.</u>	<u>PART NO.</u>	<u>FROM</u>	<u>TO</u>	<u>TERMINAL SITE</u>	<u>ENROUTE SITE</u>
W64	612J860H03	INTERR. A (UNIT 2)	RF TRANSFER	X	X
W65	(MATCHED PAIR)		SWITCH (UNIT 9)		
W66	612J861H01	INTERR. A (UNIT 2)	RF TRANSFER	X	X
	(ALT. 157C441G01)		SWITCH (UNIT 9)		
W76	612J862H01	INTERR. B (UNIT 3)	RF TRANSFER	X	—
	(ALT. 157C442G01)		SWITCH (UNIT 9)		
	612J861H01	INTERR. C (UNIT 8)	RF TRANSFER	—	X
	(ALT. 157441G01)		SWITCH (UNIT 13)		
W77	612J863H03	INTERR. B (UNIT 3)	RF TRANSFER	X	—
W78	(MATCHED PAIR)		SWITCH (UNIT 9)		
	612J860H03	INTERR. C (UNIT 8)	RF TRANSFER	—	X
	(MATCHED PAIR)		SWITCH (UNIT 13)		
W100	612J865H03	RF TRANSFER	RF TRANSFER	—	X
W101	(MATCHED PAIR)	SWITCH (UNIT 9)	SWITCH (UNIT 13)		
W102	612J866H01	RF TRANSFER	RF TRANSFER	—	X
	(ALT. 158C983G01)	SWITCH (UNIT 9)	SWITCH (UNIT 13)		

FIGURE 7–12. INTERROGATOR CABLE ASSEMBLIES W64–W66, W76–W78, AND W100–W102

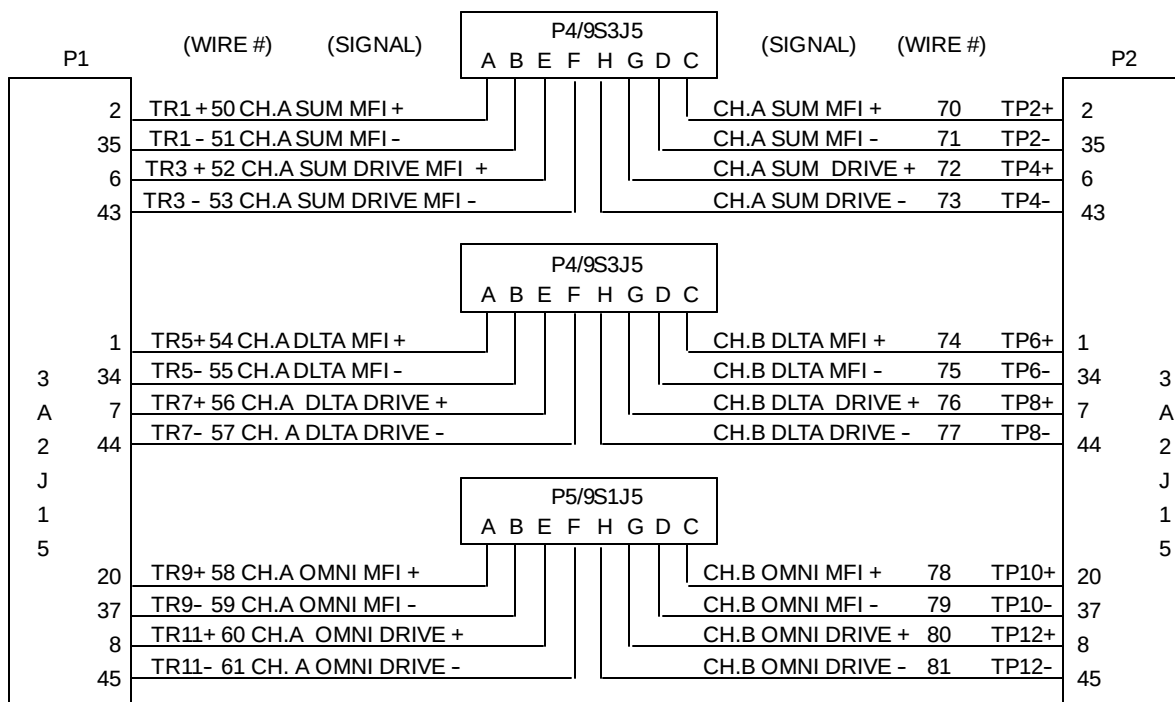


FIGURE 7-13. INTERROGATORS (UNIT 2 AND 3) RF TRANSFER SWITCH (UNIT 9) HARNESS ASSEMBLY W67 FOR A TERMINAL SITE

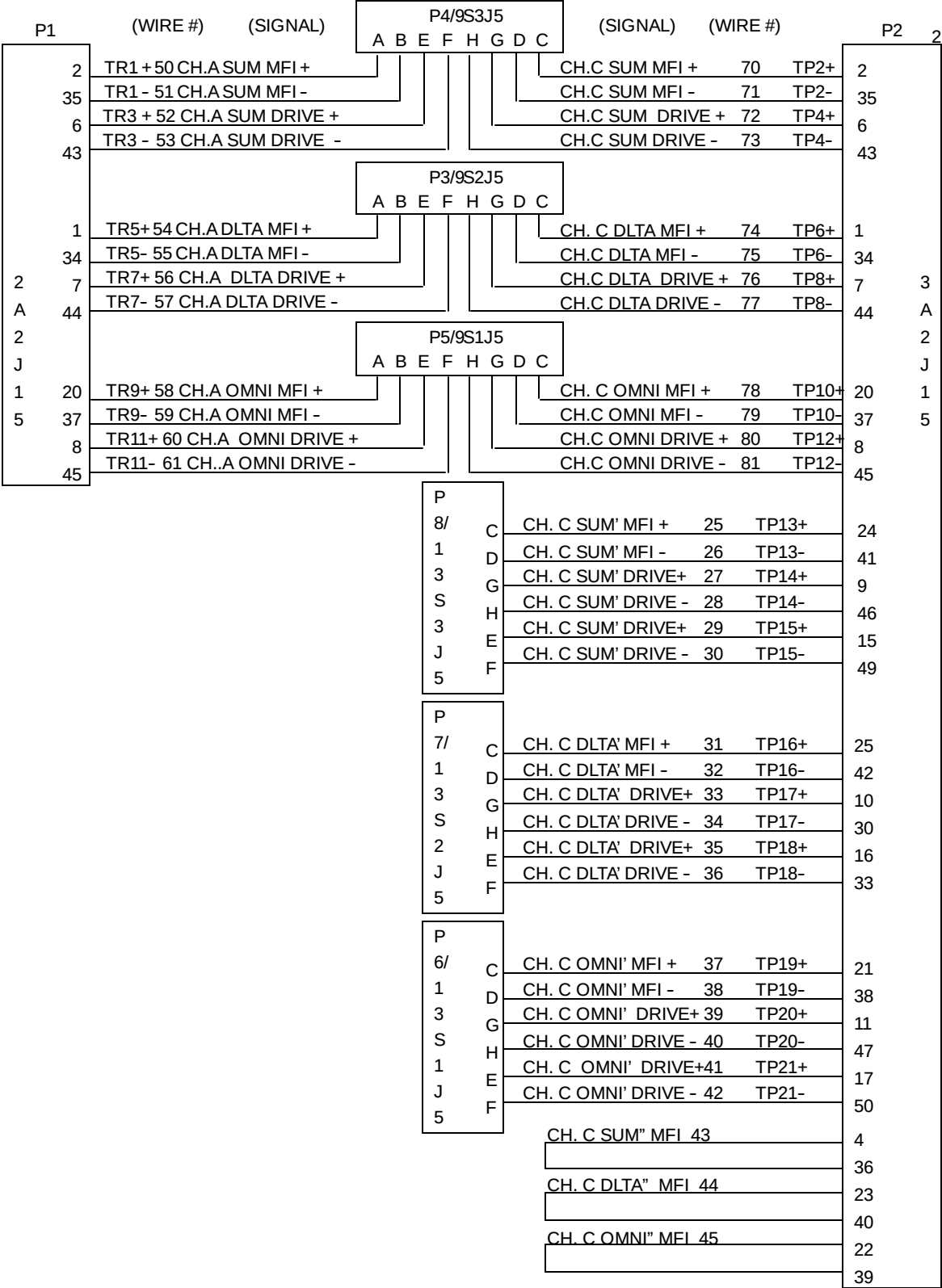
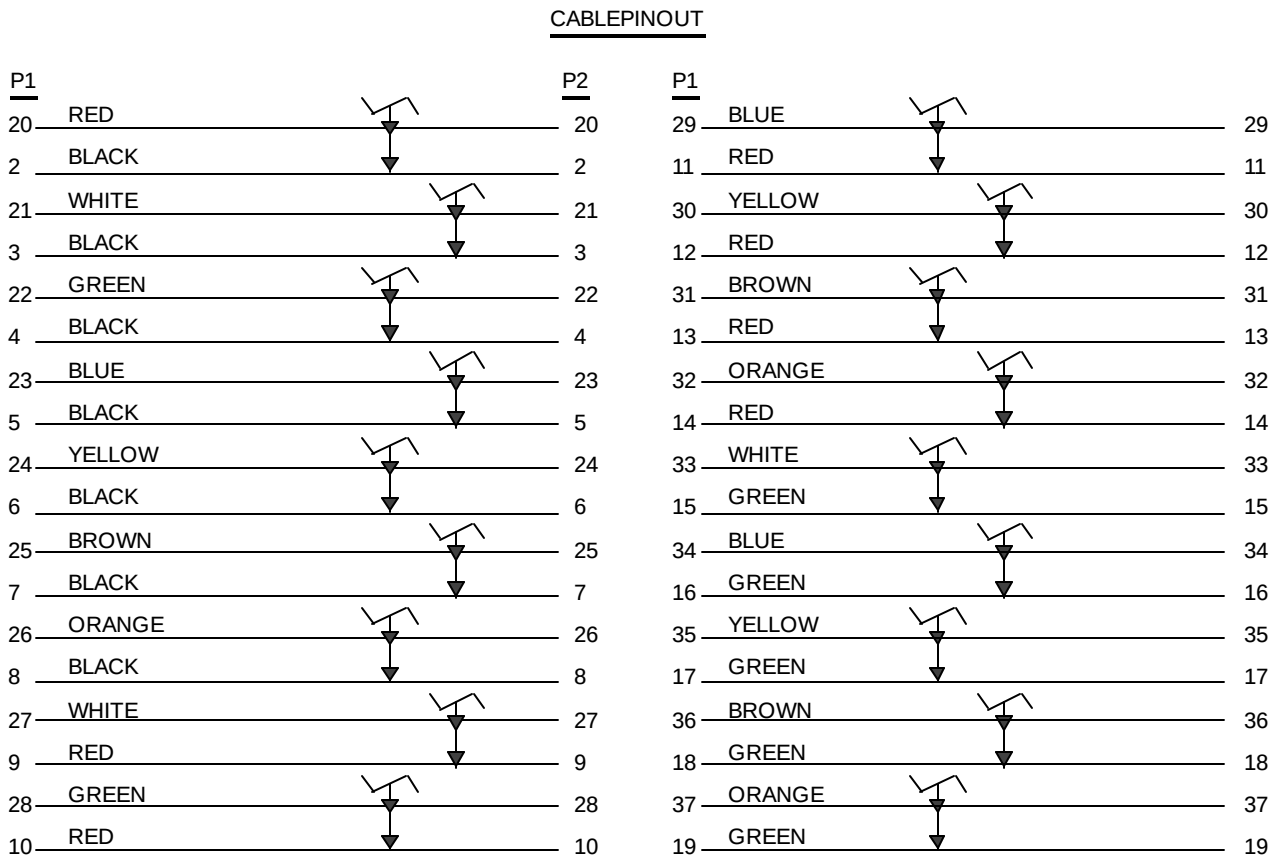
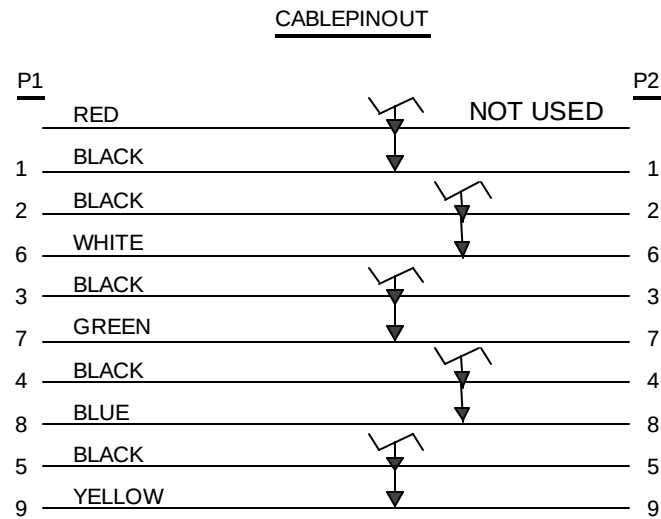


FIGURE 7-14. INTERROGATOR (UNITS 2 AND 8) RF TRANSFER SWITCHES (UNITS 9 AND 13) HARNESS ASSEMBLY W67 FOR AN EN ROUTE SITE



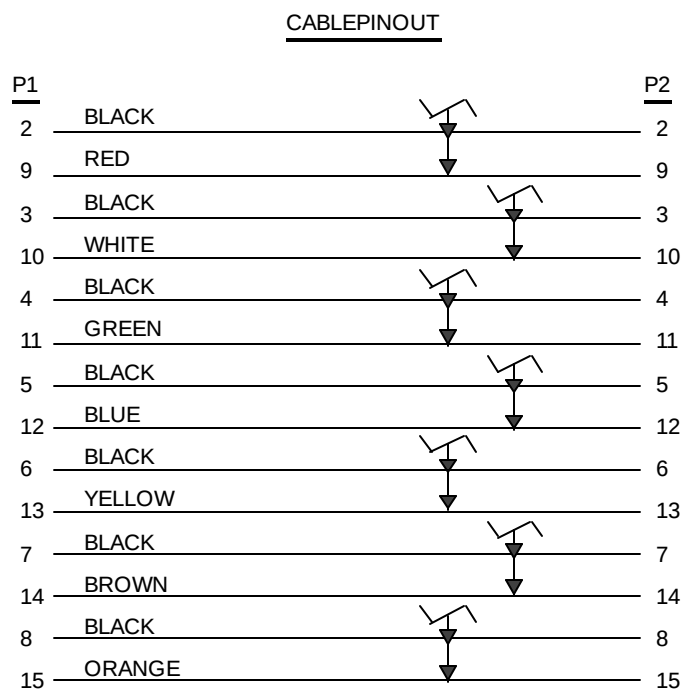
<u>CABLE NO.</u>	<u>PART NO.</u> <u>(USE)</u>	<u>FROM</u>	<u>TO</u>	<u>TERMINAL</u> <u>SITE</u>	<u>EN ROUTE</u> <u>SITE</u>
W68	158C502G01 (VID VFLT)	INTERR. A (UNIT 2)	MSJB-1 (UNIT 1)	X	X
W69	157C498G01 (VID TRIG)	INTERR. A (UNIT 2)	MSJB-1 (UNIT 1)	X	X
W73	157C499G01 (VID TRIG)	INTERR. B (UNIT 3) INTERR. C (UNIT 8)	MSJB-1 (UNIT 1) MSJB-1 (UNIT 1)	X —	— X
W74	157C500G01 (VID VFLT)	INTERR. B (UNIT 3) INTERR. C (UNIT 8)	MSJB-1 (UNIT 1) MSJB-1 (UNIT 1)	X —	— X

FIGURE 7-15. CABLE PINOUT FOR INTERROGATOR CABLE ASSEMBLIES W68, W69, W73, AND W74



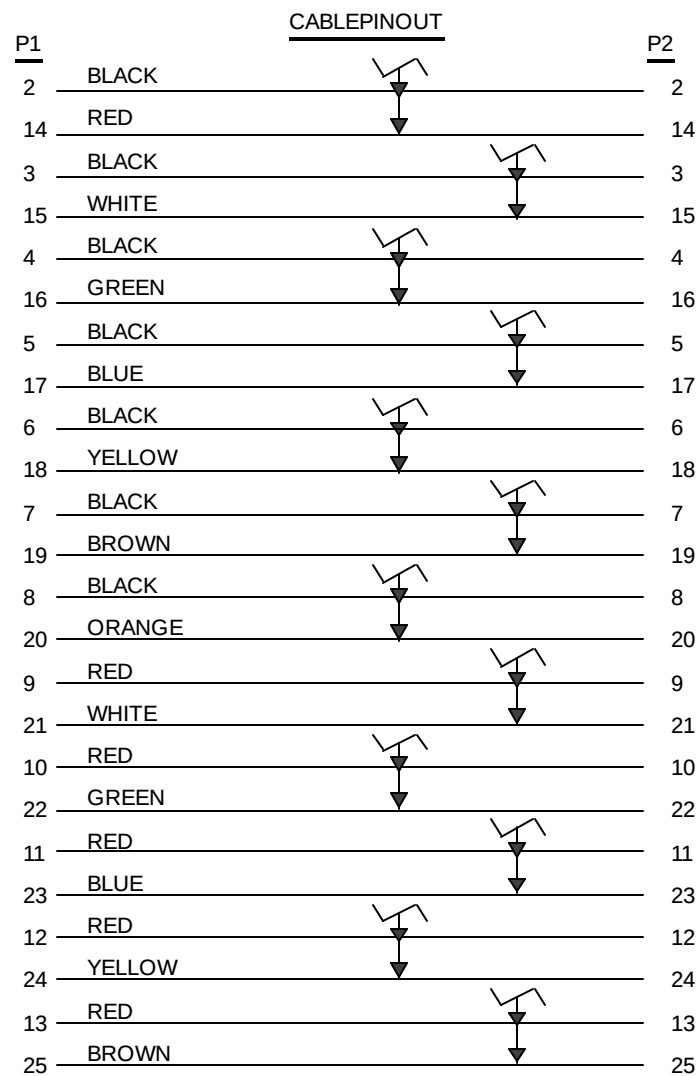
<u>CABLE NO.</u>	<u>PART NO.</u> <u>(USE)</u>	<u>FROM</u>	<u>TO</u>	<u>TERMINAL</u> <u>SITE</u>	<u>EN ROUTE</u> <u>SITE</u>
W70	157C497G01 (TOY CLK)	INTERR. B (UNIT 3)	MSJB-1 (UNIT 1)	X	—
		INTERR. C (UNIT 8)	MSJB-1 (UNIT 1)	—	X
W71	157C496G01 (TOY CLK)	INTERR. A (UNIT 2)	MSJB-1 (UNIT 1)	X	X

FIGURE 7-16. CABLE PINOUT FOR INTERROGATOR CABLE ASSEMBLIES W70 AND W71



<u>CABLE NO.</u>	<u>PART NO.</u> <u>(USE)</u>	<u>FROM</u>	<u>TO</u>	<u>TERMINAL</u> <u>SITE</u>	<u>EN ROUTE</u> <u>SITE</u>
W72	157C495G01 (APG)	INTERR. A (UNIT 2)	MSJB-1 (UNIT 1)	X	X
W75	157C494G01 (APG)	INTERR. A (UNIT 2)	INTERR. B (UNIT 3)	X	—
		INTERR. A (UNIT 2)	INTERR. C (UNIT 8)	—	X

FIGURE 7-17. CABLE PINOUT FOR INTERROGATOR CABLE ASSEMBLIES W72 AND W75



<u>CABLE NO.</u>	<u>PART NO.</u> <u>(USE)</u>	<u>FROM</u>	<u>TO</u>	<u>TERMINAL</u> <u>SITE</u>	<u>EN ROUTE</u> <u>SITE</u>
W79	158C530G01 (ABC)	INTERR. A (UNIT 2)	INTERR. B (UNIT 3)	X	—
	158C530G01 (MUC)	INTERR. A (UNIT 2)	INTERR. C (UNIT 8)	—	X
W80	158C530G01 (ABC)	INTERR. A (UNIT 2)	INTERR. B (UNIT 3)	X	—
	158C530G01 (MUC)	INTERR. A (UNIT 2)	INTERR. C (UNIT 8)	—	X

FIGURE 7–18. CABLE PINOUT FOR INTERROGATOR CABLE ASSEMBLIES W79 AND W80

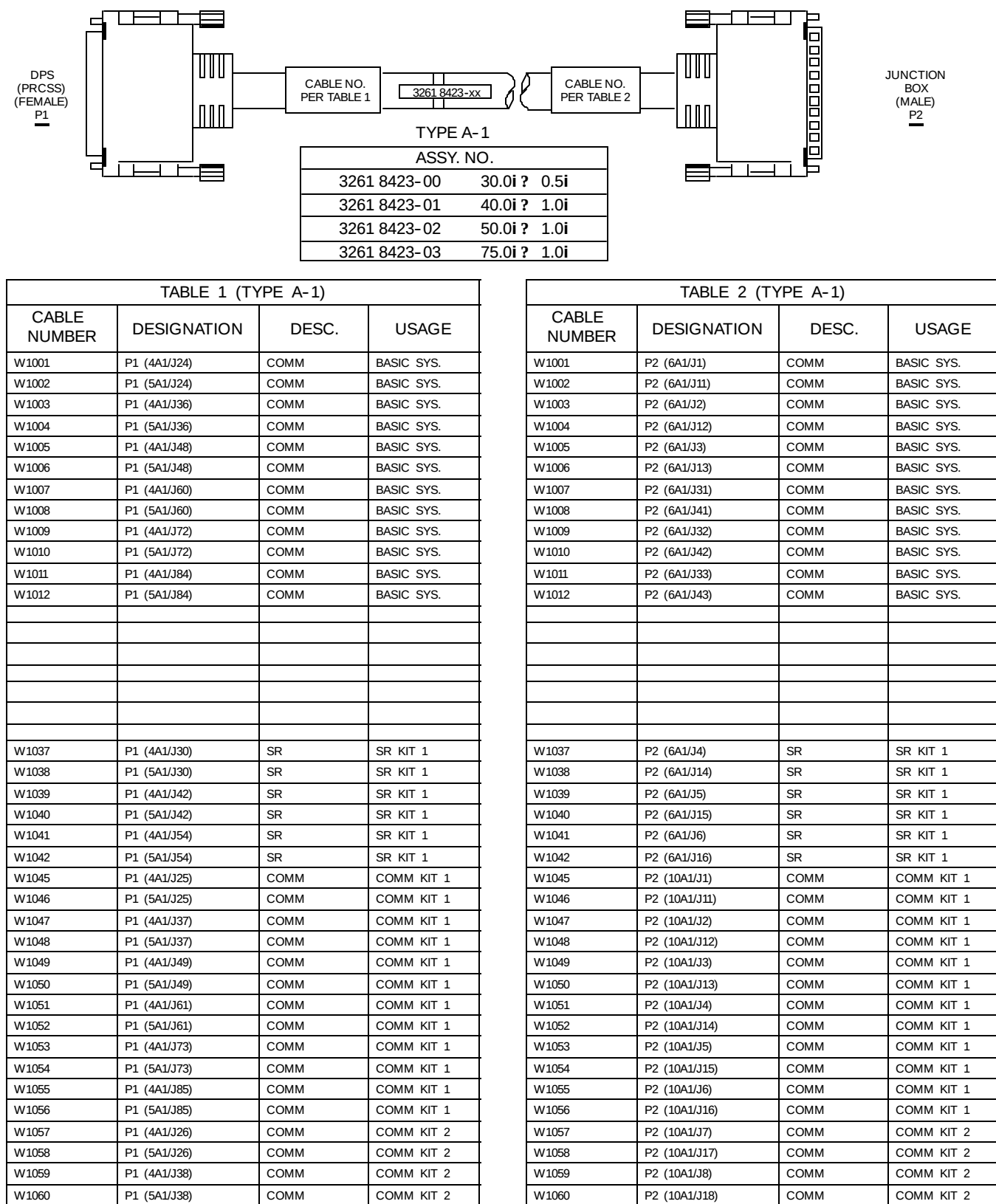


FIGURE 7–19. TYPE A–1 SYSTEM CABLE ASSEMBLY 3261 8423–XX CONNECTIONS (SHEET 1)

TABLE 1 (TYPE A-1)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1061	P1 (4A1/J50)	COMM	COMM KIT 2
W1062	P1 (5A1/J50)	COMM	COMM KIT 2
W1063	P1 (4A1/J62)	COMM	COMM KIT 2
W1064	P1 (5A1/J62)	COMM	COMM KIT 2
W1065	P1 (4A1/J74)	COMM	COMM KIT 2
W1066	P1 (5A1/J74)	COMM	COMM KIT 2
W1067	P1 (4A1/J86)	COMM	COMM KIT 2
W1068	P1 (5A1/J86)	COMM	COMM KIT 2
W1069	P1 (4A1/J27)	COMM	COMM KIT 3
W1070	P1 (5A1/J27)	COMM	COMM KIT 3
W1071	P1 (4A1/J39)	COMM	COMM KIT 3
W1072	P1 (5A1/J39)	COMM	COMM KIT 3
W1073	P1 (4A1/J51)	COMM	COMM KIT 3
W1074	P1 (5A1/J51)	COMM	COMM KIT 3
W1075	P1 (4A1/J63)	COMM	COMM KIT 3
W1076	P1 (5A1/J63)	COMM	COMM KIT 3
W1077	P1 (4A1/J75)	COMM	COMM KIT 3
W1078	P1 (5A1/J75)	COMM	COMM KIT 3
W1079	P1 (4A1/J87)	COMM	COMM KIT 3
W1080	P1 (5A1/J87)	COMM	COMM KIT 3
W1081	P1 (4A1/J28)	COMM	COMM KIT 4
W1082	P1 (5A1/J28)	COMM	COMM KIT 4
W1083	P1 (4A1/J40)	COMM	COMM KIT 4
W1084	P1 (5A1/J40)	COMM	COMM KIT 4
W1085	P1 (4A1/J52)	COMM	COMM KIT 4
W1086	P1 (5A1/J52)	COMM	COMM KIT 4
W1087	P1 (4A1/J64)	COMM	COMM KIT 4
W1088	P1 (5A1/J64)	COMM	COMM KIT 4
W1089	P1 (4A1/J76)	COMM	COMM KIT 4
W1090	P1 (5A1/J76)	COMM	COMM KIT 4
W1091	P1 (4A1/J88)	COMM	COMM KIT 4
W1092	P1 (5A1/J88)	COMM	COMM KIT 4
W1093	P1 (4A1/J66)	SR	SR KIT 1
W1094	P1 (5A1/J66)	SR	SR KIT 1
W1095	P1 (4A1/J78)	SR	SR KIT 1
W1096	P1 (5A1/J78)	SR	SR KIT 1
W1097	P1 (4A1/J90)	SR	SR KIT 1
W1098	P1 (5A1/J90)	SR	SR KIT 1
W1099	P1 (4A1/J31)	SR	SR KIT 2
W1100	P1 (5A1/J31)	SR	SR KIT 2
W1101	P1 (4A1/J43)	SR	SR KIT 2
W1102	P1 (5A1/J43)	SR	SR KIT 2
W1103	P1 (4A1/J55)	SR	SR KIT 2
W1104	P1 (5A1/J55)	SR	SR KIT 2
W1105	P1 (4A1/J67)	SR	SR KIT 2
W1106	P1 (5A1/J67)	SR	SR KIT 2
W1107	P1 (4A1/J79)	SR	SR KIT 2
W1108	P1 (5A1/J79)	SR	SR KIT 2
W1109	P1 (4A1/J91)	SR	SR KIT 2
W1110	P1 (5A1/J91)	SR	SR KIT 2
W1111	P1 (4A1/J32)	SR	SR KIT 3
W1112	P1 (5A1/J32)	SR	SR KIT 3
W1113	P1 (4A1/J44)	SR	SR KIT 3
W1114	P1 (5A1/J44)	SR	SR KIT 3
W1115	P1 (4A1/J56)	SR	SR KIT 3
W1116	P1 (5A1/J56)	SR	SR KIT 3

TABLE 2 (TYPE A-1)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1061	P2 (10A1/J9)	COMM	COMM KIT 2
W1062	P2 (10A1/J19)	COMM	COMM KIT 2
W1063	P2 (10A1/J10)	COMM	COMM KIT 2
W1064	P2 (10A1/J20)	COMM	COMM KIT 2
W1065	P2 (10A1/J31)	COMM	COMM KIT 2
W1066	P2 (10A1/J41)	COMM	COMM KIT 2
W1067	P1 (10A1/J32)	COMM	COMM KIT 2
W1068	P2 (10A1/J42)	COMM	COMM KIT 2
W1069	P2 (10A1/J33)	COMM	COMM KIT 3
W1070	P2 (10A1/J43)	COMM	COMM KIT 3
W1071	P2 (10A1/J34)	COMM	COMM KIT 3
W1072	P2 (10A1/J44)	COMM	COMM KIT 3
W1073	P2 (10A1/J35)	COMM	COMM KIT 3
W1074	P2 (10A1/J45)	COMM	COMM KIT 3
W1075	P2 (10A1/J36)	COMM	COMM KIT 3
W1076	P2 (10A1/J46)	COMM	COMM KIT 3
W1077	P2 (10A1/J37)	COMM	COMM KIT 3
W1078	P2 (10A1/J47)	COMM	COMM KIT 3
W1079	P2 (10A1/J38)	COMM	COMM KIT 3
W1080	P2 (10A1/J48)	COMM	COMM KIT 3
W1081	P2 (10A1/J39)	COMM	COMM KIT 4
W1082	P2 (10A1/J49)	COMM	COMM KIT 4
W1083	P2 (10A1/J40)	COMM	COMM KIT 4
W1084	P1 (10A1/J50)	COMM	COMM KIT 4
W1085	P2 (10A1/J61)	COMM	COMM KIT 4
W1086	P2 (10A1/J71)	COMM	COMM KIT 4
W1087	P2 (10A1/J62)	COMM	COMM KIT 4
W1088	P2 (10A1/J72)	COMM	COMM KIT 4
W1089	P2 (10A1/J63)	COMM	COMM KIT 4
W1090	P2 (10A1/J73)	COMM	COMM KIT 4
W1091	P2 (10A1/J64)	COMM	COMM KIT 4
W1092	P2 (10A1/J874)	COMM	COMM KIT 4
W1093	P2 (6A1/J34)	SR	SR KIT 1
W1094	P2 (6A1/J44)	SR	SR KIT 1
W1095	P2 (6A1/J35)	SR	SR KIT 1
W1096	P2 (6A1/J45)	SR	SR KIT 1
W1097	P2 (6A1/J36)	SR	SR KIT 1
W1098	P2 (6A1/J46)	SR	SR KIT 1
W1099	P2 (11A1/J1)	SR	SR KIT 2
W1100	P2 (11A1/J11)	SR	SR KIT 2
W1101	P2 (11A1/J2)	SR	SR KIT 2
W1102	P2 (11A1/J12)	SR	SR KIT 2
W1103	P2 (11A1/J3)	SR	SR KIT 2
W1104	P2 (11A1/J13)	SR	SR KIT 2
W1105	P2 (11A1/J4)	SR	SR KIT 2
W1106	P2 (11A1/J14)	SR	SR KIT 2
W1107	P2 (11A1/J5)	SR	SR KIT 2
W1108	P2 (11A1/J15)	SR	SR KIT 2
W1109	P2 (11A1/J6)	SR	SR KIT 2
W1110	P2 (11A1/J16)	SR	SR KIT 2
W1111	P2 (11A1/J7)	SR	SR KIT 3
W1112	P2 (11A1/J17)	SR	SR KIT 3
W1113	P2 (11A1/J8)	SR	SR KIT 3
W1114	P2 (11A1/J18)	SR	SR KIT 3
W1115	P2 (11A1/J9)	SR	SR KIT 3
W1116	P2 (11A1/J19)	SR	SR KIT 3

FIGURE 7–19. TYPE A–1 SYSTEM CABLE ASSEMBLY 3261 8423–XX CONNECTIONS (SHEET 2)

TABLE 1 (TYPE A-1)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1117	P1 (4A1/J68)	SR	SR KIT 3
W1118	P1 (5A1/J68)	SR	SR KIT 3
W1119	P1 (4A1/J80)	SR	SR KIT 3
W1120	P1 (5A1/J80)	SR	SR KIT 3
W1121	P1 (4A1/J92)	SR	SR KIT 3
W1122	P1 (5A1/J92)	SR	SR KIT 3
W1123	P1 (4A1/J33)	SR	SR KIT 4
W1124	P1 (5A1/J33)	SR	SR KIT 4
W1125	P1 (4A1/J45)	SR	SR KIT 4
W1126	P1 (5A1/J45)	SR	SR KIT 4
W1127	P1 (4A1/J57)	SR	SR KIT 4
W1128	P1 (5A1/J57)	SR	SR KIT 4
W1129	P1 (4A1/J69)	SR	SR KIT 4
W1130	P1 (5A1/J69)	SR	SR KIT 4
W1131	P1 (4A1/J81)	SR	SR KIT 4
W1132	P1 (5A1/J81)	SR	SR KIT 4
W1133	P1 (4A1/J93)	SR	SR KIT 4
W1134	P1 (5A1/J93)	SR	SR KIT 4
W1135	P1 (4A1/J34)	SR	SR KIT 5
W1136	P1 (5A1/J34)	SR	SR KIT 5
W1137	P1 (4A1/J46)	SR	SR KIT 5
W1138	P1 (5A1/J46)	SR	SR KIT 5
W1139	P1 (4A1/J58)	SR	SR KIT 5
W1140	P1 (5A1/J58)	SR	SR KIT 5
W1141	P1 (4A1/J70)	SR	SR KIT 5
W1142	P1 (5A1/J70)	SR	SR KIT 5
W1143	P1 (4A1/J82)	SR	SR KIT 5
W1144	P1 (5A1/J82)	SR	SR KIT 5
W1145	P1 (4A1/J94)	SR	SR KIT 5
W1146	P1 (5A1/J94)	SR	SR KIT 5
W1147	P1 (4A1/J35)	SR	SR KIT 6
W1148	P1 (5A1/J35)	SR	SR KIT 6
W1149	P1 (4A1/J47)	SR	SR KIT 6
W1150	P1 (5A1/J47)	SR	SR KIT 6
W1151	P1 (4A1/J59)	SR	SR KIT 6
W1152	P1 (5A1/J59)	SR	SR KIT 6
W1153	P1 (4A1/J71)	SR	SR KIT 6
W1154	P1 (5A1/J71)	SR	SR KIT 6
W1155	P1 (4A1/J83)	SR	SR KIT 6
W1156	P1 (5A1/J83)	SR	SR KIT 6
W1157	P1 (4A1/J95)	SR	SR KIT 6
W1158	P1 (5A1/J95)	SR	SR KIT 6
W1159	P1 (4A1/J29)	COMM	COMM KIT 5
W1160	P1 (5A1/J29)	COMM	COMM KIT 5
W1161	P1 (4A1/J41)	COMM	COMM KIT 5
W1162	P1 (5A1/J41)	COMM	COMM KIT 5
W1163	P1 (4A1/J53)	COMM	COMM KIT 5
W1164	P1 (5A1/J53)	COMM	COMM KIT 5
W1165	P1 (4A1/J65)	COMM	COMM KIT 5
W1166	P1 (5A1/J65)	COMM	COMM KIT 5
W1167	P1 (4A1/J77)	COMM	COMM KIT 5
W1168	P1 (5A1/J77)	COMM	COMM KIT 5
W1169	P1 (4A1/J89)	COMM	COMM KIT 5
W1170	P1 (5A1/J89)	COMM	COMM KIT 5

TABLE 2 (TYPE A-1)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1117	P2 (11A1/J10)	SR	SR KIT 3
W1118	P2 (11A1/J20)	SR	SR KIT 3
W1119	P2 (11A1/J31)	SR	SR KIT 3
W1120	P2 (11A1/J41)	SR	SR KIT 3
W1121	P2 (11A1/J32)	SR	SR KIT 3
W1122	P2 (11A1/J42)	SR	SR KIT 3
W1123	P2 (11A1/J33)	SR	SR KIT 4
W1124	P2 (11A1/J43)	SR	SR KIT 4
W1125	P2 (11A1/J34)	SR	SR KIT 4
W1126	P2 (11A1/J44)	SR	SR KIT 4
W1127	P2 (11A1/J35)	SR	SR KIT 4
W1128	P2 (11A1/J45)	SR	SR KIT 4
W1129	P2 (11A1/J36)	SR	SR KIT 4
W1130	P2 (11A1/J46)	SR	SR KIT 4
W1131	P2 (11A1/J37)	SR	SR KIT 4
W1132	P2 (11A1/J47)	SR	SR KIT 4
W1133	P2 (11A1/J38)	SR	SR KIT 4
W1134	P2 (11A1/J48)	SR	SR KIT 4
W1135	P2 (11A1/J39)	SR	SR KIT 5
W1136	P2 (11A1/J49)	SR	SR KIT 5
W1137	P2 (11A1/J40)	SR	SR KIT 5
W1138	P2 (11A1/J50)	SR	SR KIT 5
W1139	P2 (11A1/J61)	SR	SR KIT 5
W1140	P2 (11A1/J71)	SR	SR KIT 5
W1141	P2 (11A1/J62)	SR	SR KIT 5
W1142	P2 (11A1/J72)	SR	SR KIT 5
W1143	P2 (11A1/J63)	SR	SR KIT 5
W1144	P2 (11A1/J73)	SR	SR KIT 5
W1145	P2 (11A1/J64)	SR	SR KIT 5
W1146	P2 (11A1/J74)	SR	SR KIT 5
W1147	P2 (11A1/J65)	SR	SR KIT 6
W1148	P2 (11A1/J75)	SR	SR KIT 6
W1149	P2 (11A1/J66)	SR	SR KIT 6
W1150	P2 (11A1/J76)	SR	SR KIT 6
W1151	P2 (11A1/J67)	SR	SR KIT 6
W1152	P2 (11A1/J77)	SR	SR KIT 6
W1153	P2 (11A1/J68)	SR	SR KIT 6
W1154	P2 (11A1/J78)	SR	SR KIT 6
W1155	P2 (11A1/J69)	SR	SR KIT 6
W1156	P2 (11A1/J79)	SR	SR KIT 6
W1157	P2 (11A1/J70)	SR	SR KIT 6
W1158	P2 (11A1/J80)	SR	SR KIT 6
W1159	P2 (10A1/J65)	COMM	COMM KIT 5
W1160	P2 (10A1/J75)	COMM	COMM KIT 5
W1161	P2 (10A1/J66)	COMM	COMM KIT 5
W1162	P2 (10A1/J76)	COMM	COMM KIT 5
W1163	P2 (10A1/J67)	COMM	COMM KIT 5
W1164	P2 (10A1/J77)	COMM	COMM KIT 5
W1165	P2 (10A1/J68)	COMM	COMM KIT 5
W1166	P2 (10A1/J78)	COMM	COMM KIT 5
W1167	P2 (10A1/J69)	COMM	COMM KIT 5
W1168	P2 (10A1/J79)	COMM	COMM KIT 5
W1169	P2 (10A1/J70)	COMM	COMM KIT 5
W1170	P2 (10A1/J80)	COMM	COMM KIT 5

FIGURE 7–19. TYPE A–1 SYSTEM CABLE ASSEMBLY 3261 8423–XX CONNECTIONS (SHEET 3)

CABLE ASSEMBLY 3261 8423-XX (TYPE A-1) CONNECTORS AND WIRING TABLE



WIRING TABLE (TYPE A-1)			
WIRE COLOR	FROM	TO	PAIR NUMBER
	CONNECTOR A (P1)	CONNECTOR B (P2)	
RED	1	1	1
BLK	14	23	
DRAIN	JT1	JT2	
WHT	2	14	2
BLK	15	2	
DRAIN	JT1	JT2	
GRN	3	16	3
BLK	16	3	
DRAIN	JT1	JT2	
BLU	4	19	4
BLK	17	4	
DRAIN	JT1	JT2	
YEL	5	13	5
BLK	18	5	
DRAIN	JT1	JT2	
BRN	6	22	6
BLK	19	6	
DRAIN	JT1	JT2	
ORG	7	7	7
BLK	20	20	
DRAIN	JT1	JT2	
WHT	8	10	8
RED	21	8	
DRAIN	JT1	JT2	
GRN	9	9	9
RED	22	17	
DRAIN	JT1	JT2	
BLU	10	11	10
RED	23	24	
DRAIN	JT1	JT2	
YEL	11	12	11
RED	24	15	
DRAIN	JT1	JT2	
BRN	12	18	12
RED	25	25	
DRAIN	JT1	JT2	
GRY	13	21	SINGLE WIRE

FIGURE 7-19. TYPE A-1 SYSTEM CABLE ASSEMBLY 3261 8423-XX CONNECTIONS (SHEET 4 — END)

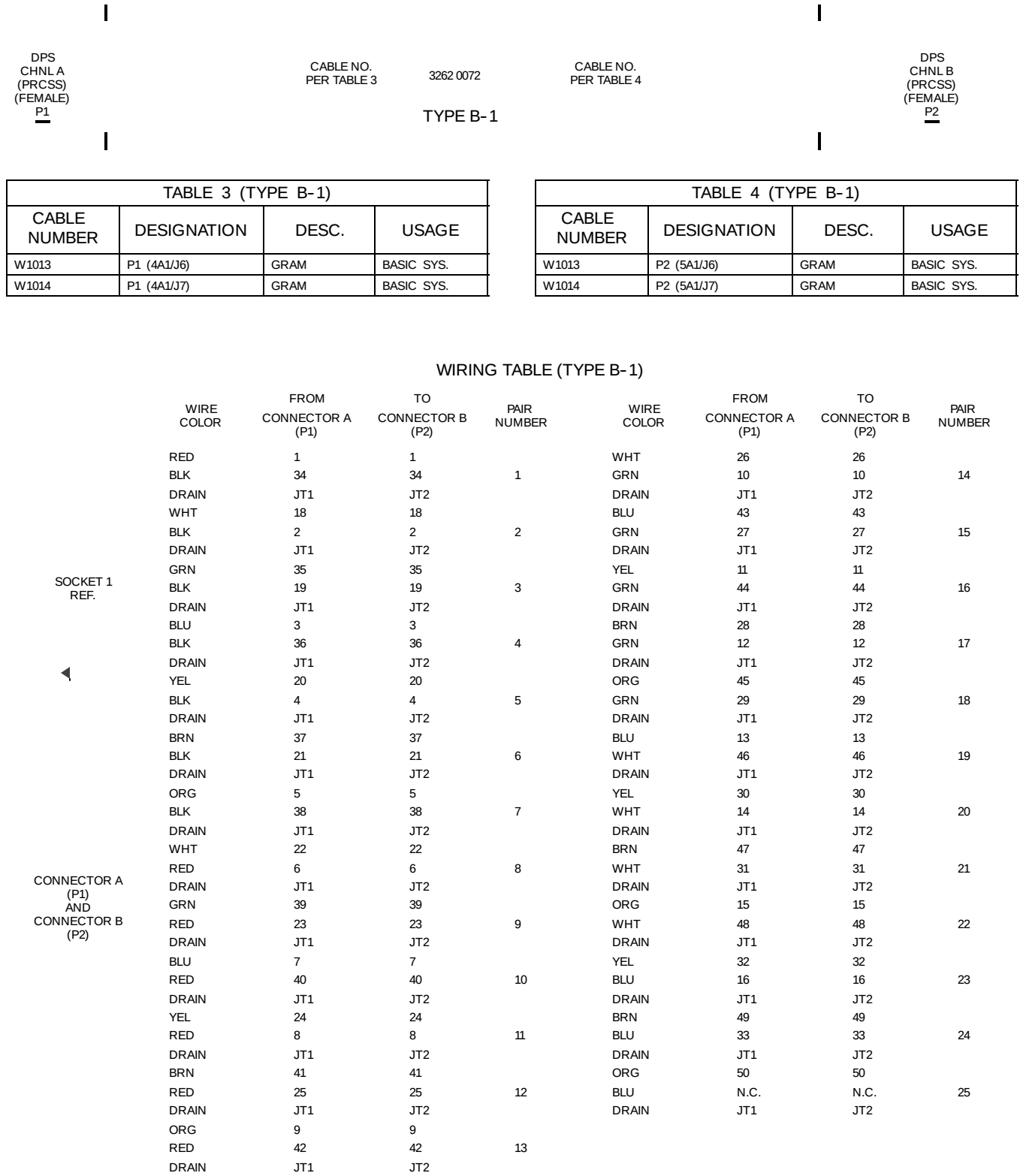


FIGURE 7-20. TYPE B-1 SYSTEM CABLE ASSEMBLY 3262 0072 CONNECTIONS

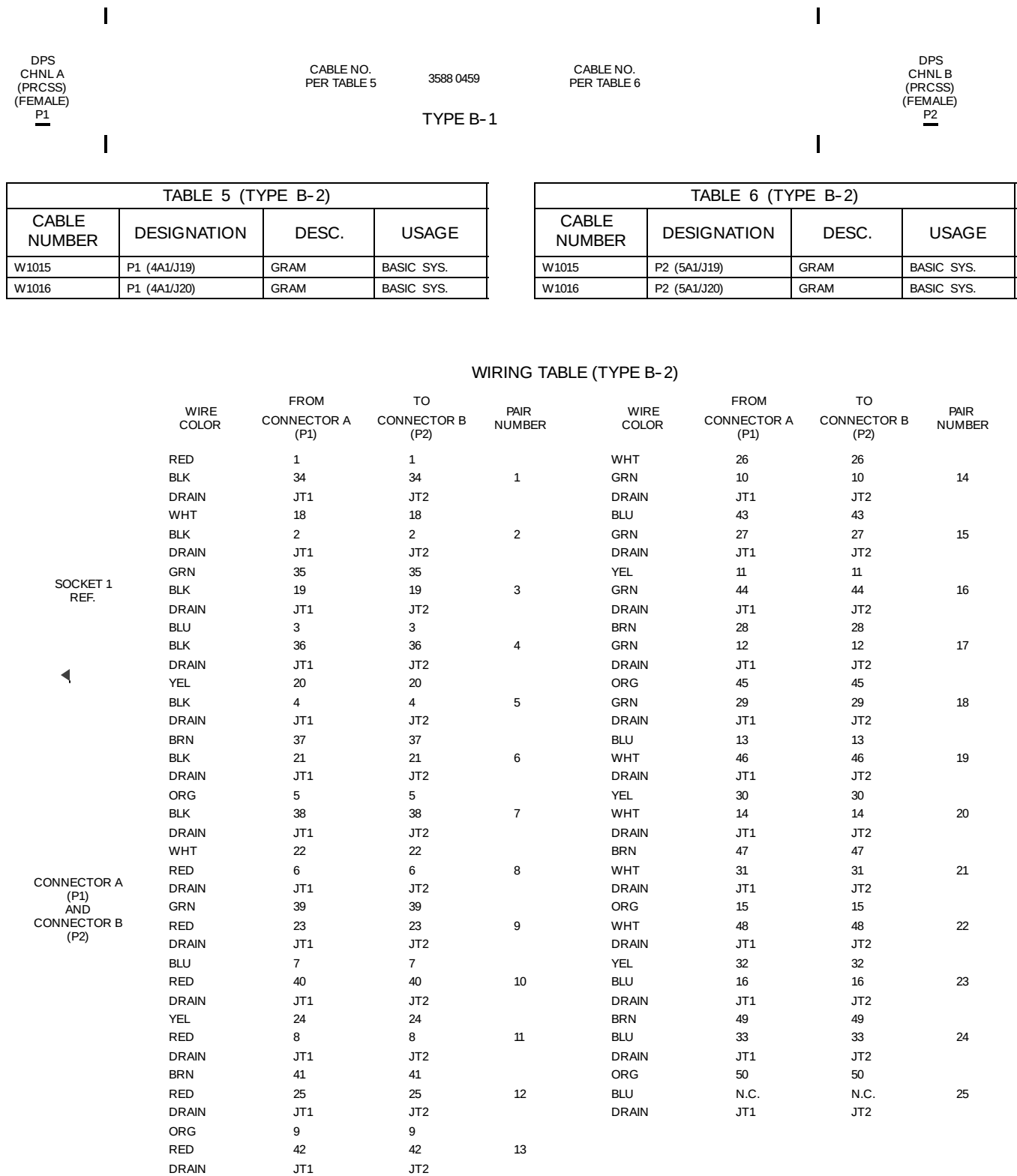


FIGURE 7-21. TYPE B-2 SYSTEM CABLE ASSEMBLY 3588 0459 CONNECTIONS

DPS
CHNL A
(PRCSS)
(MALE)
P1



CABLE NO.
PER TABLE 7

3262 0064

CABLE NO.
PER TABLE 8



DPS
CHNL B
(PRCSS)
(MALE)
P2

TYPE C-1

TABLE 7 (TYPE C-1)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1017	P1 (4A1/J13)	CH XMIT	BASIC SYS.
W1018	P1 (4A1/J14)	CH REC	BASIC SYS.

TABLE 8 (TYPE C-1)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1017	P2 (5A1/J14)	CH REC	BASIC SYS.
W1018	P2 (5A1/J13)	CH XMIT	BASIC SYS.

WIRING TABLE (TYPE C-1)

PIN 1 REF.	WIRE COLOR	FROM CONNECTOR A (P1)	TO CONNECTOR B (P2)	PAIR NUMBER
	RED	1	1	1
	BLK	6	6	
	DRAIN	JT1	JT2	2
	WHT	2	2	
	BLK	7	7	3
	DRAIN	JT1	JT2	
	GRN	3	3	4
	BLK	8	8	
	DRAIN	JT1	JT2	5
	BLU	4	4	
CONNECTOR A (P1) AND CONNECTOR B (P2)	BLK	N.C.	N.C.	4
	DRAIN	JT1	JT2	
	YEL	5	5	5
	BLK	9	9	
	DRAIN	JT1	JT2	

FIGURE 7-22. TYPE C-1 SYSTEM CABLE ASSEMBLY 3262 0064 CONNECTIONS

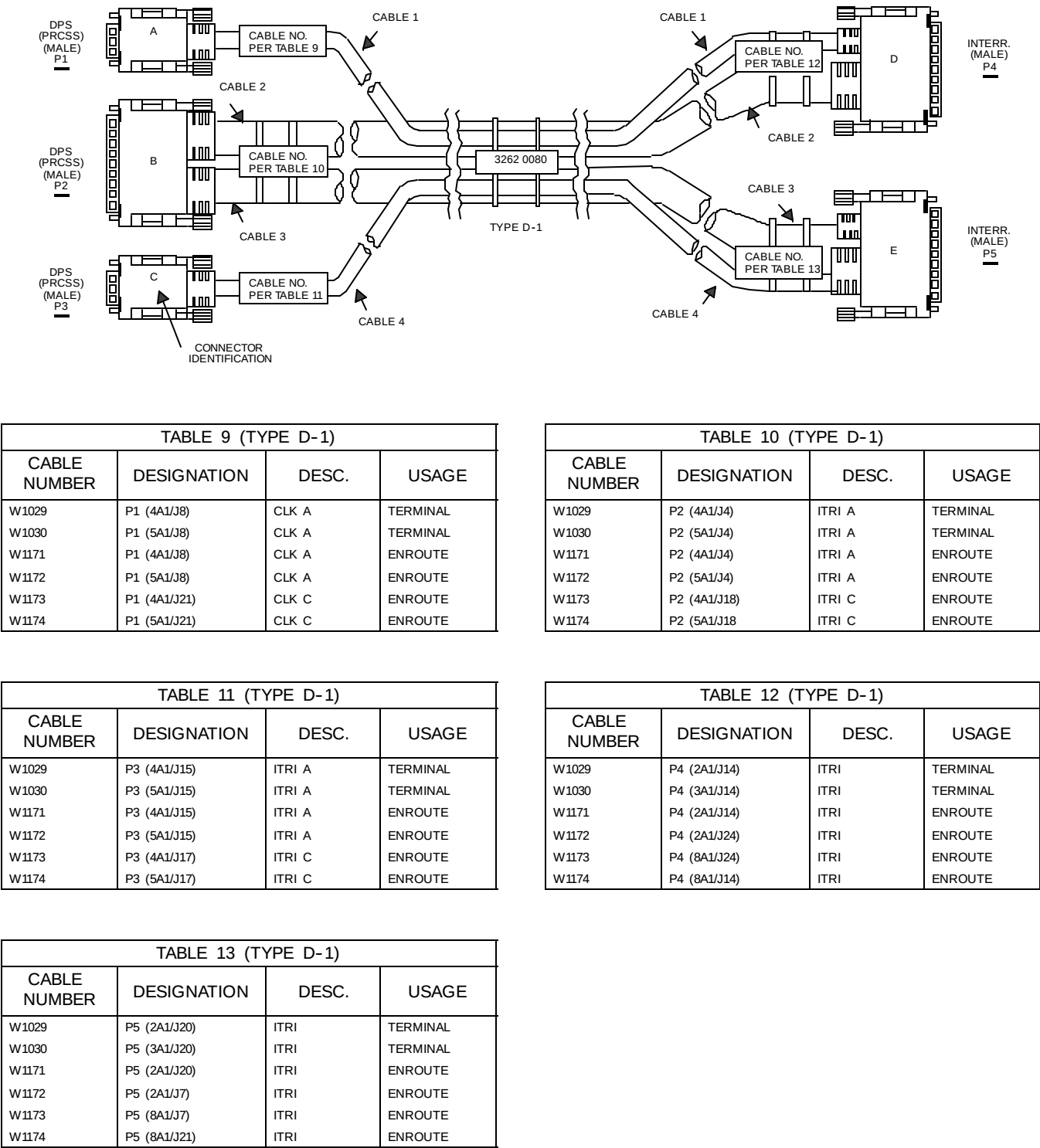


FIGURE 7-23. TYPE D-1 SYSTEM CABLE ASSEMBLY 3262 0080-01 CONNECTIONS (SHEET 1)

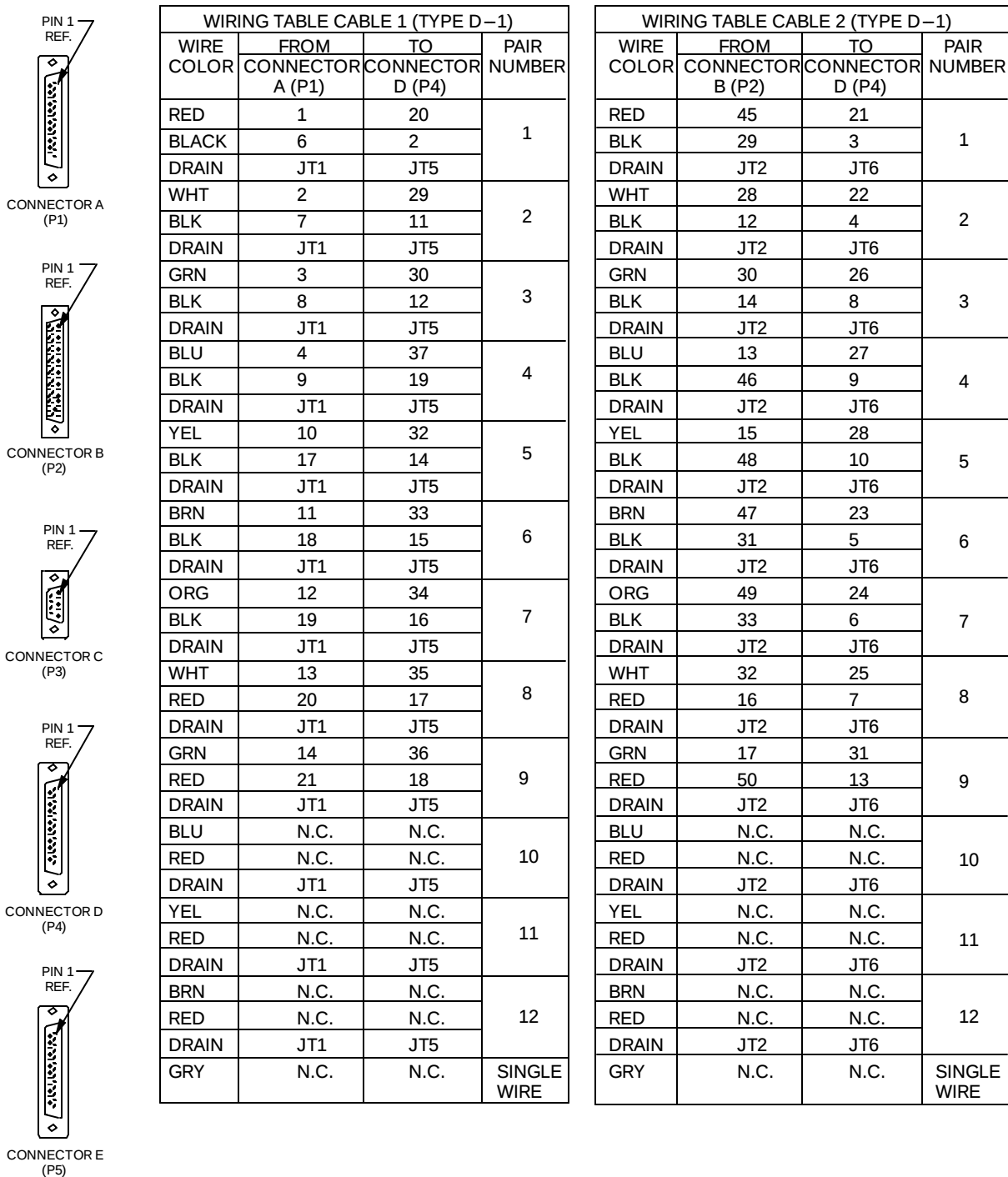


FIGURE 7-23. TYPE D-1 SYSTEM CABLE ASSEMBLY 3262 0080-01 CONNECTIONS (SHEET 2)

WIRING TABLE CABLE 3 (TYPE D-1)

WIRE COLOR	FROM CONNECTOR B (P2)	TO CONNECTOR E (P5)	PAIR NUMBER
RED	18	20	1
BLK	2	2	
DRAIN	JT3	JT7	2
WHT	1	21	
BLK	34	3	3
DRAIN	JT3	JT7	
GRN	3	22	4
BLK	36	4	
DRAIN	JT3	JT7	5
BLU	35	23	
BLK	19	5	6
DRAIN	JT3	JT7	
BRN	20	25	7
BLK	4	7	
DRAIN	JT3	JT7	8
ORN	22	26	
BLK	6	8	9
DRAIN	JT3	JT7	
WHT	5	27	10
RED	38	9	
DRAIN	JT3	JT7	11
GRN	7	28	
RED	40	10	12
DRAIN	JT3	JT7	
BLU	39	29	13
RED	23	11	
DRAIN	JT3	JT7	14
YEL	41	30	
RED	25	12	15
DRAIN	JT3	JT7	
BRN	24	31	16
RED	8	13	
DRAIN	JT3	JT7	17
ORN	26	32	
RED	10	14	18
DRAIN	JT	JT7	

WIRING TABLE CABLE 3 (TYPE D-1) (CONT.)

WIRE COLOR	FROM CONNECTOR B (P2)	TO CONNECTOR E (P5)	PAIR NUMBER
WHT	9	33	14
GRN	42	15	
DRAIN	JT3	JT7	15
BLU	11	34	
GRN	44	16	16
DRAIN	JT3	JT7	
YEL	43	35	17
GRN	27	17	
DRAIN	JT3	JT7	18
BRN	N.C.	N.C.	
GRN	N.C.	N.C.	19
DRAIN	JT3	JT7	
ORN	N.C.	N.C.	20
GRN	N.C.	N.C.	
DRAIN	JT3	JT7	21

WIRING TABLE CABLE 4 (TYPE D-1)

WIRE COLOR	FROM CONNECTOR C (P3)	TO CONNECTOR E (P5)	PAIR NUMBER
RED	1	36	1
BLK	6	18	
DRAIN	JT4	JT8	2
WHT	2	37	
BLK	7	19	3
DRAIN	JT4	JT8	
GRN	3	N.C.	4
BLK	8	N.C.	
DRAIN	JT4	JT8	5
BLU	4	N.C.	
BLK	9	N.C.	6
DRAIN	JT4	JT8	

FIGURE 7-23. TYPE D-1 SYSTEM CABLE ASSEMBLY 3262 0080-01 CONNECTIONS
(SHEET 3 — END)

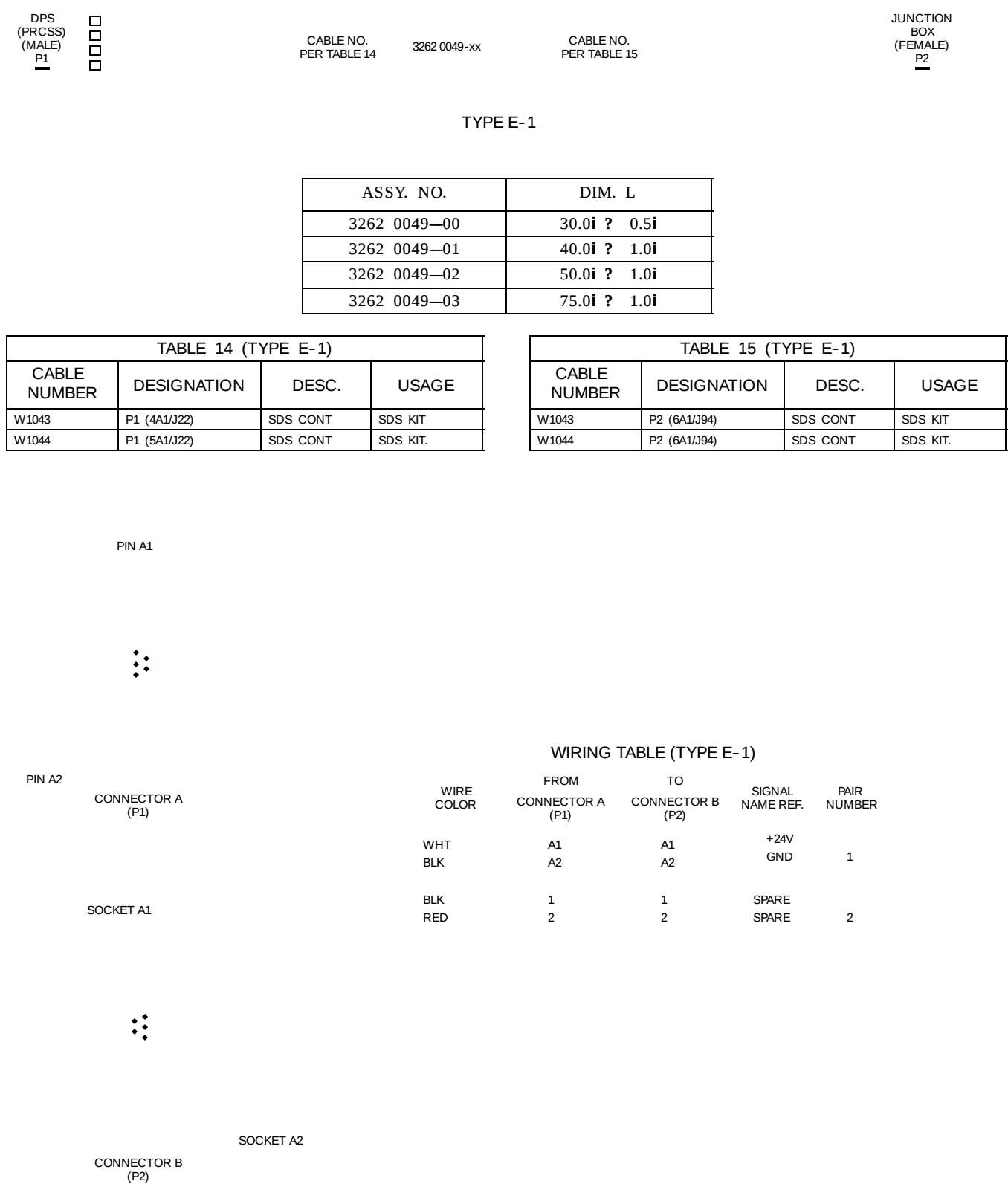
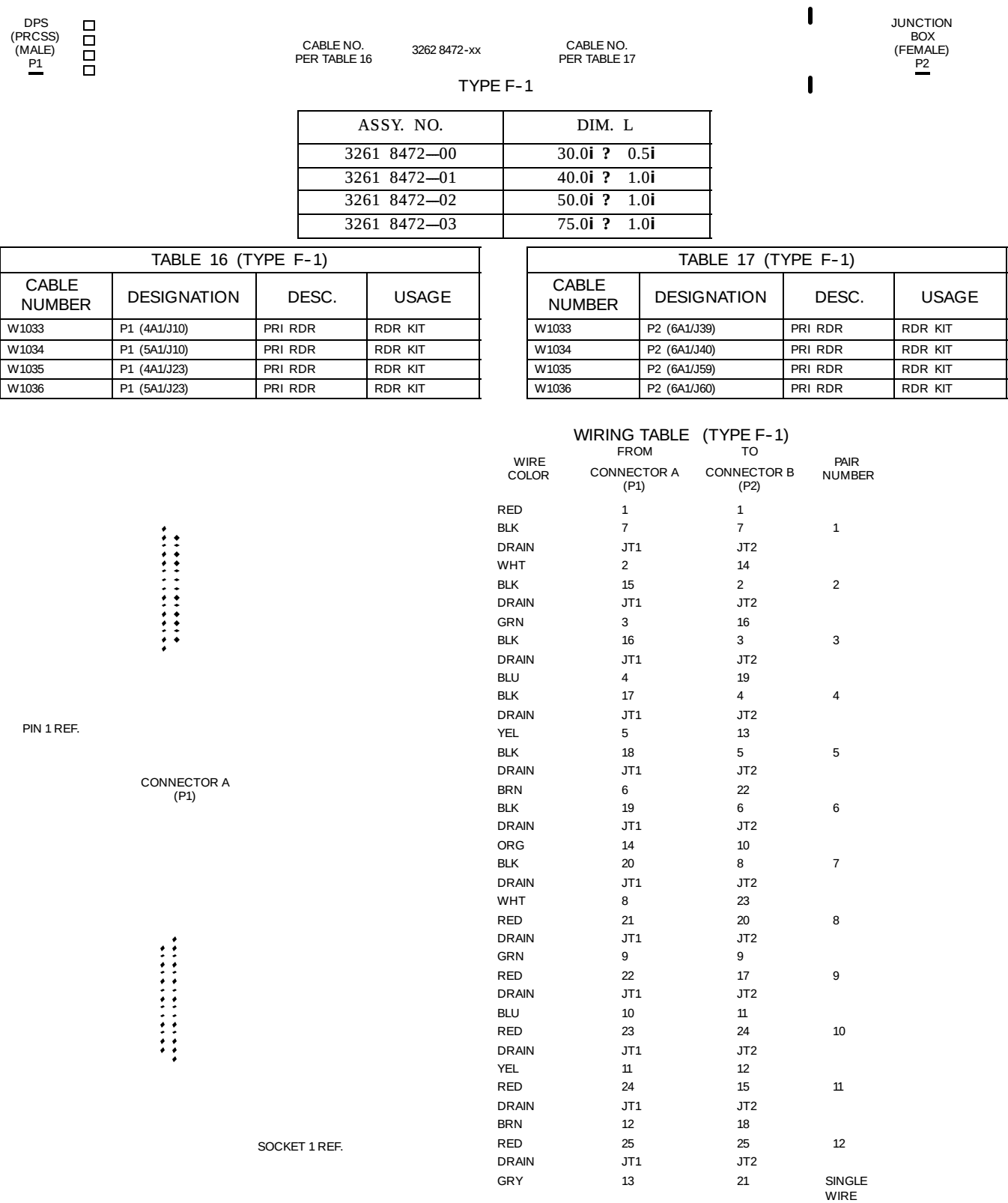


FIGURE 7–24. TYPE E–1 SYSTEM CABLE ASSEMBLY 3262 0049–XX CONNECTIONS



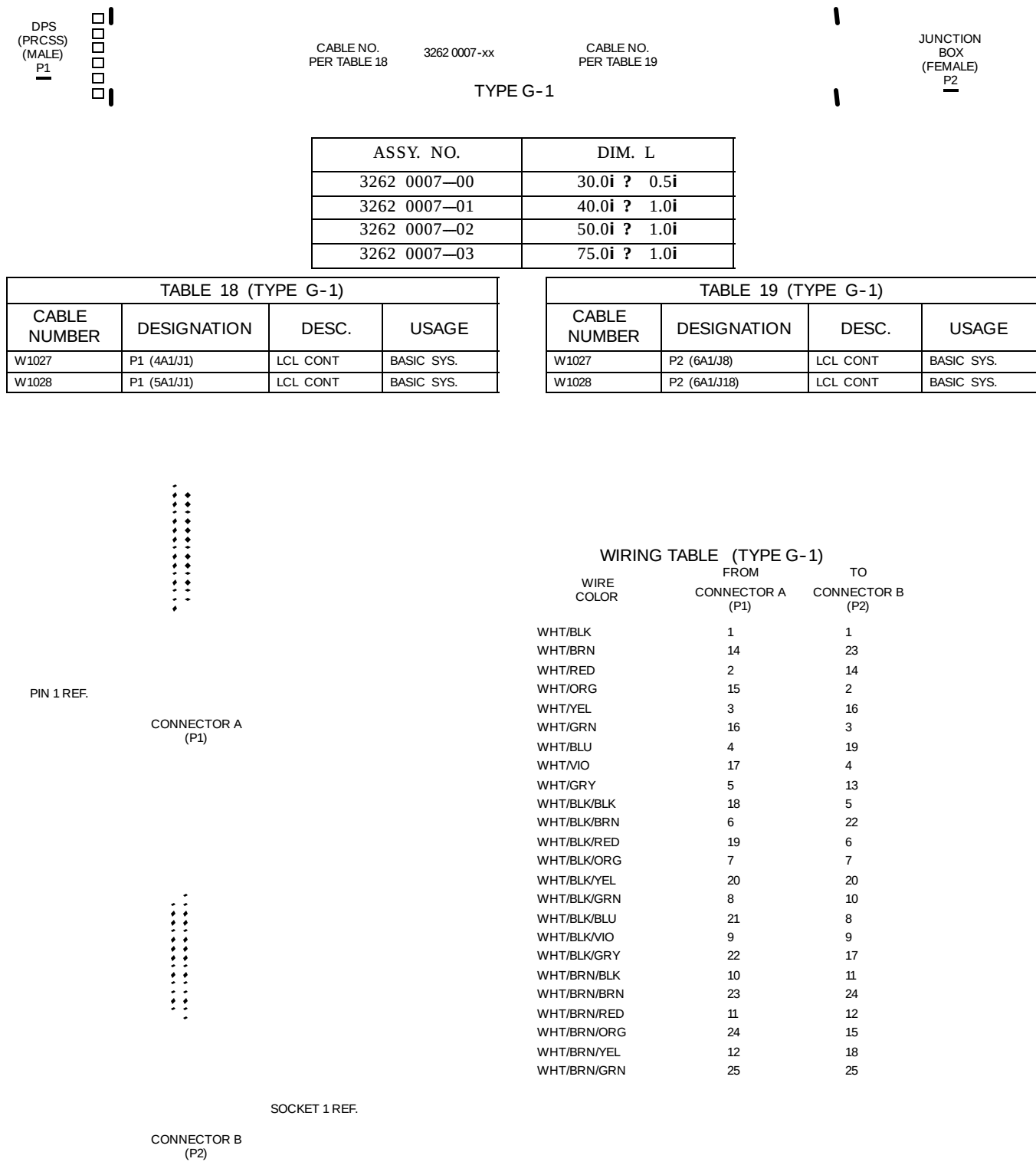


FIGURE 7-26. TYPE G-1 SYSTEM CABLE ASSEMBLY 3262 0007-XX CONNECTIONS

3262 0015-xx

JUNCTION
BOX
(FEMALE)
P2

TYPE G-2

ASSY. NO.	DIM. L
3262 0015—00	30.0i ? 0.5i
3262 0015—01	40.0i ? 1.0i
3262 0015—02	50.0i ? 1.0i
3262 0015—03	75.0i ? 1.0i

TABLE 20 (TYPE G-2)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1019	P1 (4A1/J11)	BLDG SNSR	BASIC SYS.
W1020	P1 (5A1/J11)	BLDG SNSR	BASIC SYS.

TABLE 21 (TYPE G-2)			
CABLE NUMBER	DESIGNATION	DESC.	USAGE
W1019	P2 (6A1/J67)	BLDG SNR	BASIC SYS.
W1020	P2 (6A1/J68)	BLDG SNR	BASIC SYS.

WIRING TABLE (TYPE G-2)

WIRE COLOR	FROM CONNECTOR A (P1)	TO CONNECTOR B (P2)	PAIR NUMBER
RED	1	1	1
BLK	14	23	
DRAIN	JT1	JT2	
WHT	2	14	2
BLK	15	2	
DRAIN	JT1	JT2	
GRN	3	16	3
BLK	16	3	
DRAIN	JT1	JT2	
BLU	4	19	4
BLK	17	4	
DRAIN	JT1	JT2	
YEL	5	13	5
BLK	18	5	
DRAIN	JT1	JT2	
BRN	6	22	6
BLK	19	6	
DRAIN	JT1	JT2	
ORG	7	7	7
BLK	20	20	
DRAIN	JT1	JT2	
WHT	8	10	8
RED	21	8	
DRAIN	JT1	JT2	
GRN	9	9	9
RED	22	17	
DRAIN	JT1	JT2	
BLU	10	11	10
RED	23	24	
DRAIN	JT1	JT2	
YEL	11	12	11
RED	24	15	
DRAIN	JT1	JT2	
BRN	12	18	12
RED	25	25	
DRAIN	JT1	JT2	
GRY	13	21	SINGLE WIRE

FIGURE 7-27. TYPE G-2 SYSTEM CABLE ASSEMBLY 3262 0015-XX CONNECTIONS

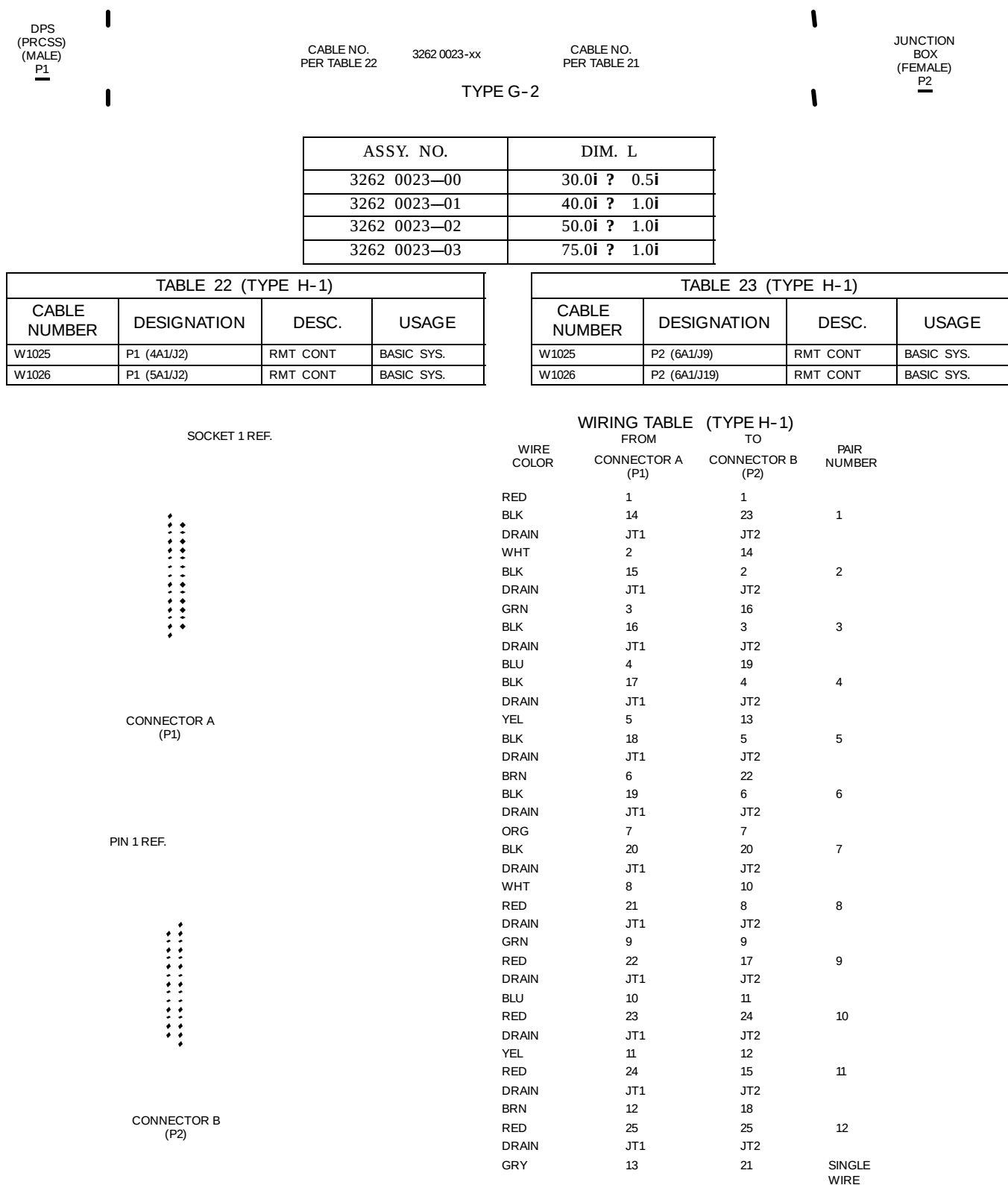
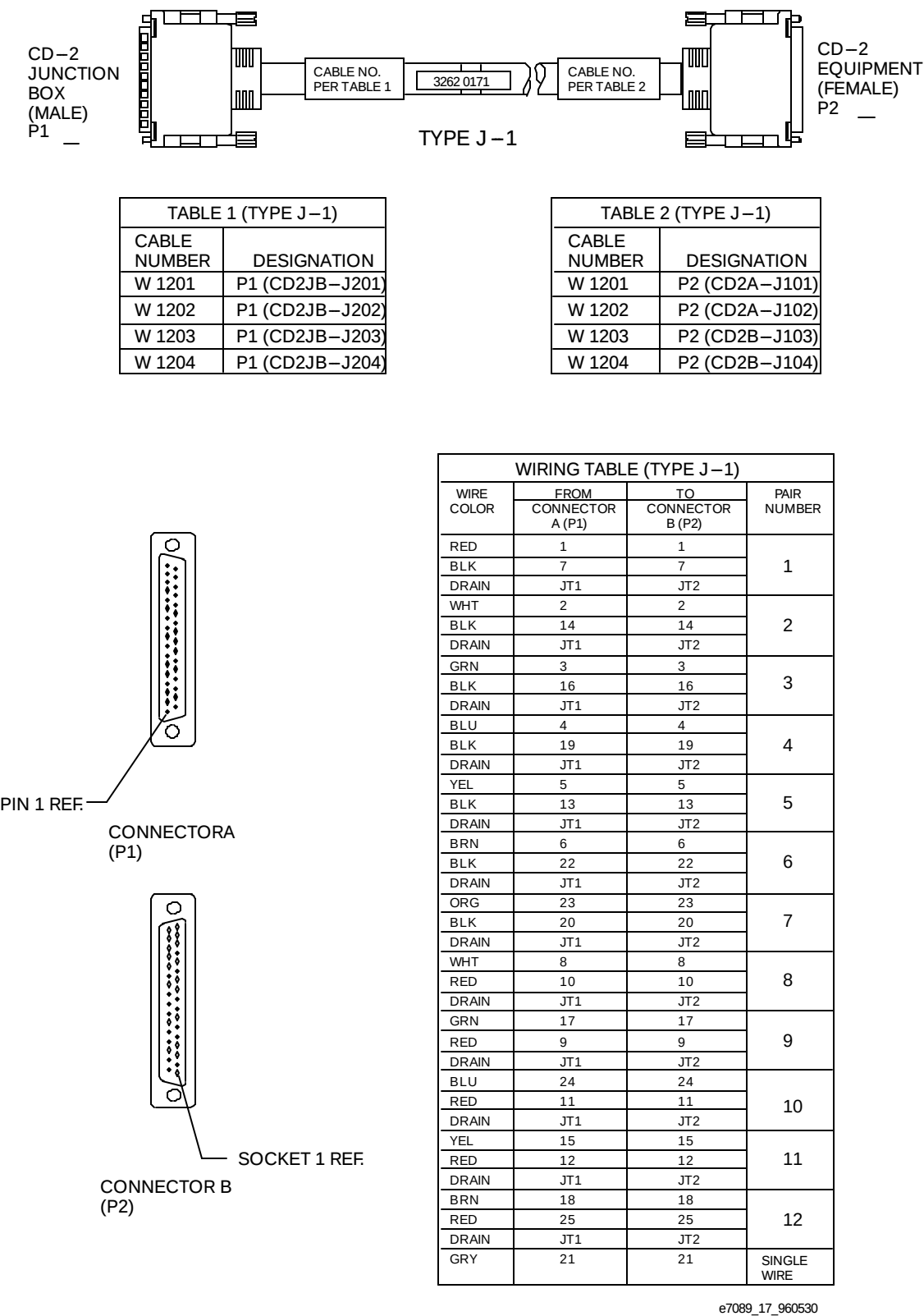


FIGURE 7-28. TYPE H-1 SYSTEM CABLE ASSEMBLY 3262 0023-XX CONNECTIONS



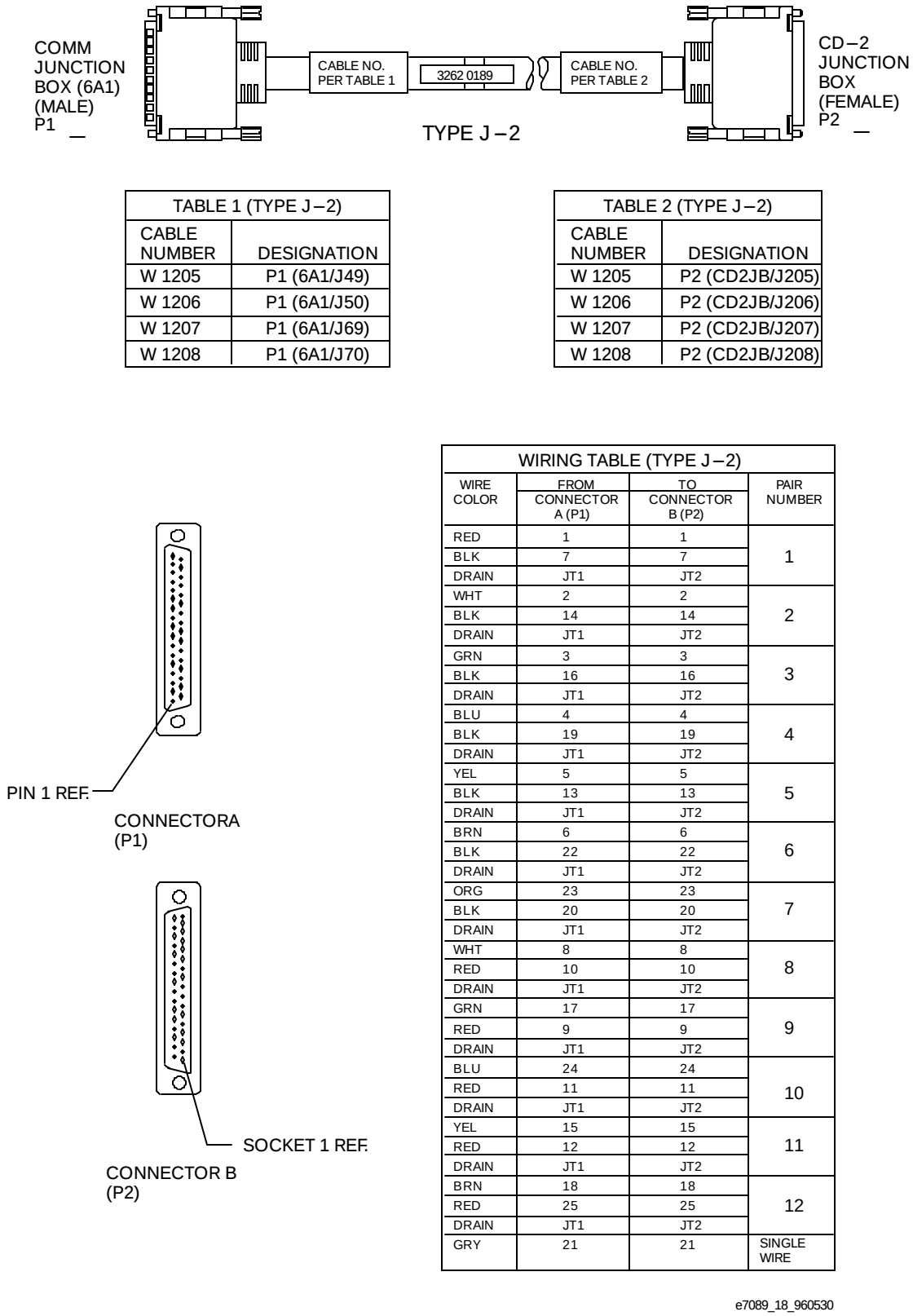
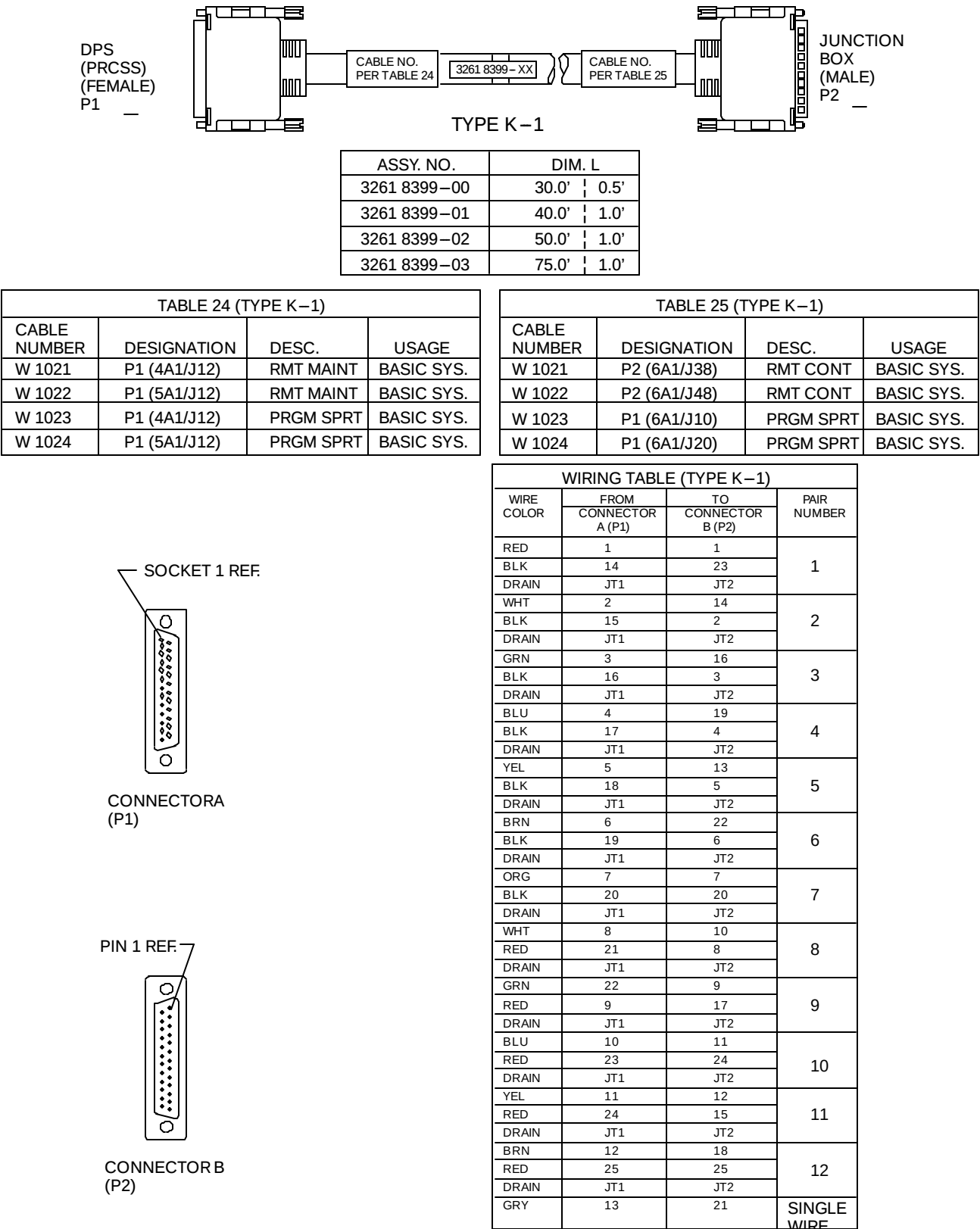


FIGURE 7-30. TYPE J-2 SYSTEM CABLE ASSEMBLY 3262 0189 CONNECTIONS



8.0 PARTS LIST

8.1 GENERAL

This parts list contains the replaceable parts of the MODE Select Beacon System Sensor as manufactured by the Joint Venture. The supporting illustrations are contained in Section 11 of this manual. The figure title of the parts list illustrations contains the area reference designation to aid in locating the parts listed in Section 8. The following subparagraphs explain each column entry in this Section.

8.1.1 Reference Designation

The reference designation of the assembly or part is listed in this column. Each assembly which has a separate figure in Section 11 will have a reference to that figure number directly below the reference designation.

8.1.2 Indent

A letter which indicates the relationship of each entry. Indent letter "A" would indicate the top unit and "B" parts or assemblies which form part of "A".

8.1.3 Name of Part/Description

This column contains the descriptive data required to identify the part. When the item name, description and part number are identical to a previously listed entry "Same as" is used in lieu of a description.

8.1.4 Mfgs. Code Number

This column contains the five-digit code assigned to the actual manufacturer of the part as shown in the Federal Supply Code for Manufacturers, Federal Cataloging Handbook H4-1. At the end of the parts listing, an explanation of the codes is given.

8.1.5 JAN/MIL Mfgs. Part No.

This column contains the Government standard part number, Government specification part number, or the Manufacturers Part Number(s) for each part. Whenever more than one part number is available, subsequent part numbers are listed beneath the first entry.

8.1.6 Notes

This column is left blank and is used for FAA field notes.

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
(Fig. 11-1)	A	DATA PROCESSING SYSTEM, FAATC Terminal Site FA 10203/1	08241	7103934 (DPS Equip. Option List)	
(Fig. 11-2)	A	DATA PROCESSING SYSTEM, FAATC Terminal Site FA 10203/2	08241	7103934 (DPS Equip. Option List)	
(Fig. 11-2)	A	DATA PROCESSING SYSTEM, FAATC Terminal Site FA 10203/3	08241	7103934 (DPS Equip. Option List)	
(Fig. 11-2)	A	DATA PROCESSING SYSTEM, Enroute Site less CD-2 FA 10203/4	08241	7103934 (DPS Equip. Option List)	
(Fig. 11-1)	A	DATA PROCESSING SYSTEM, Terminal Site w/ASR-9 FA 10203/5	08241	7103934 (DPS Equip. Option List)	
(Fig. 11-1)	A	DATA PROCESSING SYSTEM, Terminal Site less ASR-9 FA 10203/6	08241	7103934 (DPS Equip. Option List)	
1	B	JUNCTION BOX, FA 10208 (See TI 6365.5)	97942	1D23453G01	
2	B	INTERROGATOR CABINET FA 10202 (See TI 6365.5)	97942	612J422G01	
3 (Terminal Sites)	B	Same as 2			
4	B	PROCESSOR CHANNEL CABINET ASSEMBLY (See TI 6365.3)	08241	3594 1517	
5	B	Same as 4			

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
6	B	COMMUNICATION JUNCTION BOX ASSEMBLY FA 10213/1 (See TI 6365.3)	08241	3594 1715	
6	B	COMMUNICATION JUNCTION BOX ASSEMBLY with SDS FA 10213/2 (See TI 6365.3)	08241	2696 0161	
7		NOT USED			
8 (Enroute Sites)	B	Same as 2			
9	B	TRANSFER SWITCH ASSY (See TI 6135.7)	97942	1D23267G01	
10 (Sites 1 and 2 only)	B	COMMUNICATION JUNCTION BOX, Expansion FA10204 (See TI 6135.3)	08241	3594 1574	
11 (Sites 1 and 2 only)	B	SURVEILLANCE JUNCTION BOX, Expansion FA10214 (See TI 6135.3)	08241	3594 1582	
12		NOT USED			
13	B	Same as 9			
14 (For Future Expansion)	B	Same as 9			
A1, A2	B	TOY CLOCK ANTENNA, Remote Kit (See TI 6365.5)	97942	1D23490H02	
E1, E2	B	TOY CLOCK ANTENNA (See TI 6365.5)	97942	1D23490H04	
E3 thru E5 (Terminal Site only)	B	ARRESTOR, Electrical Surge	97942	1A20734H01	

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
4B1/A1, 5B1/A1 W1033 thru W1036	A	RADAR KIT: 30-foot long system cables	08241	3594 1525-00	
	A	RADAR KIT: 40-foot long system cables	08241	3594 1525-01	
	A	RADAR KIT: 50-foot long system cables	08241	3594 1525-02	
	A	RADAR KIT: 75-foot long system cables	08241	3594 1525-03	
	B	CIRCUIT CARD ASSEMBLY, LARDR	08241	3597 1522	
	B	CABLE ASSEMBLY	08241	3261 8472-00 thru 3621 8472-03	
	A	SURVEILLANCE KIT: 30-foot long system cables	08241	3594 1541-00	
	A	SURVEILLANCE KIT: 40-foot long system cables	08241	3594 1541-01	
	A	SURVEILLANCE KIT: 50-foot long system cables	08241	3594 1541-02	
	A	SURVEILLANCE KIT: 75-foot long system cables	08241	3594 1541-03	
	B	CIRCUIT CAD ASSEMBLY, LASUR	08241	3597 1563	
	B	CABLE ASSEMBLY	08241	3261 8423-00 thru 3261 8423-03	
4B1/A2 thru A7 5B1/A2 thru A7 (Dependent upon the number of kits)					
W1037 thru W1042, W1093 thru W1158 (Dependent upon the number of kits)					

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
	A	COMMUNICATIONS KIT: 30-foot long system cables	08241	3594 1558--00	
	A	COMMUNICATIONS KIT: 40-foot long system cables	08241	3594 1558--01	
	A	COMMUNICATIONS KIT: 50-foot long system cables	08241	3594 1558--02	
	A	COMMUNICATIONS KIT: 75-foot long system cables	08241	3594 1558--03	
4B1/A8 thru 4B1/A12 5B1/A8 thru 5B1/A12 (Dependent upon the number of kits)	B	CIRCUIT CARD ASSEMBLY, LACOM	08241	3597 1407	
W1045 thru W1092, W1159 thru W1170 (Dependent upon the number of kits)	B	CABLE ASSEMBLY	08241	3261 8423--00 thru 3261 8423--03	
	A	SDS KIT: 30-foot long system cables	08241	3594 1566--12	
	A	SDS KIT: 40-foot long system cables	08241	3594 1566--13	
	A	SDS KIT: 50-foot long system cables	08241	3594 1566--14	
	A	SDS KIT: 75-foot long system cables	08241	3594 1566--15	
4B4/CB8, 5B4/CB8	B	CIRCUIT BREAKER: 3-pole, 2 Amp, 120 Vac, 60 Hz	81349	M55629/8--024	
4B6	B	POWER SUPPLY ASSEMBLY	08241	3261 9942	

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1043, W1044 W1 thru W63	B	HARNESS ASSEMBLY, AC POWER	08241	3594 8025-01	
	B	HARNESS ASSEMBLY, POWER SUPPLY STATUS	08241	3594 8033	
	B	CABLE ASSEMBLY NOT USED AT SYSTEM LEVEL	08241	3262 0049-04 thru 3262 0049-07	
W64 (Fig. 11-3)	B	CABLE ASSEMBLY, Coax, 32.85 inches long	97942	612J860H01	
W64/P1	C	CONNECTOR, Plug, electrical	97942	1A20973H01	
			97942	1A20973H02	
W64/P2	C	CONNECTOR, Plug, electrical	97942	1A20971H01	
			97942	1A20971H02	
W65 (Fig. 11-3)	B	CABLE ASSEMBLY, Coax, 31.65 inches long	97942	612J860H02	
W65/P1	C	Same as W64/P1			
W65/P2	C	Same as W64/P2			
W66 (Fig. 11-3)	B	CABLE ASSEMBLY, Coax 23.75 inches long	97942	612J861H01	
W66/P1	C	Same as W64/P1			
W66/P2	C	Same as W64/P2			
W67 (Terminal Sites) (Fig. 11-4)	B	CABLE ASSEMBLY: 2 Conductors, 113 in. long	97942	1D24520G01	
W67/P1, P2	C	CONNECTOR, Plug, electrical	81349	M24308/2-5F	

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W67/P3 thru P5	C	CONNECTOR, Plug, electrical	96906	MS3476W12-10S	
W67 (Enroute Sites) (Fig. 11-4)	B	CABLE ASSEMBLY: 2 Conductors, 160 in. long	97942	1D24520G02	
W67/P1, P2	C	CONNECTOR, Plug, electrical	81349	M24308/2-5F	
W67/P3 thru P8	C	CONNECTOR, Plug, electrical	96906	MS3476W12-10S	
W68 (Fig. 11-5)	B	CABLE ASSEMBLY: 18 Twisted Pair Conductors, 360 in. long	97974	158C502G01	
W68/P1	C	CONNECTOR, Plug, electrical	81349	M24308/2-4F	
W68/P2	C	CONNECTOR, Plug, electrical	81349	M24308/4-4F	
W69 (Fig. 11-5)	B	CABLE ASSEMBLY, 18 Twisted Pair Conductors, 360 in. long	97942	157C498G01	
W69/P1	C	Same as W68/P1			
W69/P2	C	Same as W68/P2			
W70 (Fig. 11-5)	B	CABLE ASSEMBLY: 5 Twisted Pair Conductors, 360 in. long	97942	157C497G01	
W70/P1	C	CONNECTOR, Receptacle, electrical	81349	M24308/2-1F	
W70/P2	C	CONNECTOR, Receptacle, electrical	81349	M24308/4-1F	
W71 (Fig. 11-5)	B	CABLE ASSEMBLY: 5 Twisted Pair Conductors, 360 in. long	97942	157C496G01	
W71/P1	C	Same as W70/P1			
W71/P2	C	Same as W70/P2			

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W72 (Fig. 11-5)	B	CABLE ASSEMBLY: 7 Twisted Pair Conductors, 360 in. long	97942	157C495G01	
W72/P1	C	CONNECTOR, Plug, electrical	81349	M24308/2-2F	
W72/P2	C	CONNECTOR, Plug, electrical	81349	M24308/4-2F	
W73 (Fig. 11-5)	B	CABLE ASSEMBLY: 18 Twisted Pair Conductors, 360 in. long	97942	157C499G01	
W73/P1	C	Same as W68/P1			
W73/P2	C	Same as W68/P2			
W74 (Fig. 11-5)	B	CABLE ASSEMBLY: 18 Twisted Pair Conductors, 360 in. long	97942	157C500G01	
W74/P1	C	Same as W68/P1			
W74/P2	C	Same as W68/P2			
W75 (Fig. 11-5)	B	CABLE ASSEMBLY: 7 Twisted Pair Conductors, 360 in. long	97942	157C494G01	
W75/P1	C	Same as W72/P1			
W75/P2	C	Same as W72/P2			
W76 (Terminal Sites) (Fig. 11-3)	B	CABLE ASSEMBLY: Coax, 43.5 in. long	97942	612J862H01	
W76/P1	C	Same as W64/P1			
W76/P2	C	Same as W64/P2			

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W76 (Enroute Sites) (Fig. 11-3)	B	Same as W66			
W77 (Terminal Sites) (Fig. 11-3)	B	CABLE ASSEMBLY: Coax, 42.15 in. long	97942	612J863H01	
W77/P1	C	Same as W64/P1			
W77/P2	C	Same as W64/P2			
W77 (Enroute Sites) (Fig. 11-3)	B	Same as W64			
W78 (Terminal Sites) (Fig. 11-3)	B	CABLE ASSEMBLY: Coax, 42.15 in. long	97942	612J863H02	
W78/P1	C	Same as W64/P1			
W78/P2	C	Same as W64/P2			
W78 (Enroute Sites) (Fig. 11-3)	B	Same as W65			
W79 (Fig. 11-5)	B	CABLE ASSEMBLY: 2 Conductors, 96 in. long	97942	158C530G01	
W79/P1, P2	C	Connector, Plug, electrical	81349	M24308/4-3F	
W80 (Fig. 11-5)	B	Same as W79			
W81 thru W99		NOT USED AT SYSTEM LEVEL			
W100 (Fig. 11-3)	B	CABLE ASSEMBLY: Coax, 48 in. long	97942	612J865H01	
W100/P1, P2	C	Same as W64/P2			

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W101 (Fig. 11-3)	B	CABLE ASSEMBLY: Coax, 48 in. long	97942	612J865H02	
W101/P1, P2	C	Same as W64/P2			
W102 (Fig. 11-3)	B	CABLE ASSEMBLY: Coax, 48 in. long	97942	612J866H01	
W102/P1, P2	C	Same as W64/P2			
W103 thru W1000		NOT USED			
W1001 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 30 feet long	08241	3261 8423-00	
W1001 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 40 feet long	08241	3261 8423-01	
W1001 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 50 feet long	08241	3261 8423-02	
W1001 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 75 feet long	08241	3261 8423-03	
W1001/P1	C	CONNECTOR, Plug, electrical: 25 Removable socket contacts	08241 00779	2698 3452 207463-1	
	C	CONTACT, Electrical, Socket	81349	M39029/63-368	
W1001/P2	C	CONNECTOR, Plug, electrical: 25 Removable pin contacts	08241 00779	2698 3403 207464-2	
	C	CONTACT, Electrical, Pin	81349	M39029/64-369	

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1002 thru W1012 (Fig. 11-6)	B	Same as W1001			
W1013 (Fig. 11-7)	B	CABLE ASSEMBLY: 25 Twisted Pair Conductors, 9 feet long	08241	3262 0072	
W1013/P1, P2	C	CONNECTOR, Plug, electrical: 50 Removable socket contacts	08241 00779	2698 3478 205211-2	
	C	CONTACT, Electrical, Socket	81349	M39029/63-368	
W1014 (Fig. 11-7)	B	Same as W1013			
W1015 (Fig. 11-7)	B	CABLE ASSEMBLY: 25 Twisted Pair Conductors, 9 feet long	08241	3588 0459	
W1015/P1, P2	C	Same as W1013/P1			
	C	CONTACT, Electrical, Socket	81349	M39029/63-368	
W1016 (Fig. 11-7)	B	Same as W1015			
W1017 (Fig. 11-22)	B	CABLE ASSEMBLY: 5 Twisted Pair Conductors, 9 feet long	08241	3262 0064	
W1017/P1, P2	C	CONNECTOR, Plug, electrical: 9 Removable pin contacts	08241 00779	2698 3387 205204-4	
	C	CONTACT, Electrical, Pin	81349	M39029/64-369	
W1018 (Fig. 11-22)	B	Same as W1017			
W1019 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 30 feet long	08241	3262 0015-00	

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1019 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 40 feet long	08241	3262 0015-01	
W1019 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 50 feet long	08241	3262 0015-02	
W1019 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 75 feet long	08241	3262 0015-03	
W1019/P1	C	Same as W1001/P2			
W1019/P2	C	Same as W1001/P1			
W1020 (Fig. 11-9)	B	Same as W1019			
W1021 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 30 feet long	08241	3261 8399-00	
W1021 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 40 feet long	08241	3261 8399-01	
W1021 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 50 feet long	08241	3261 8399-02	
W1021 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 70 feet long	08241	3261 8399-03	
W1021/P1	C	Same as W1001/P1			

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1021/P2	C	Same as W1001/P2			
W1022 thru W1024 (Fig. 11-6)	B	Same as W1021			
W1025 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 30 feet long	08241	3262 0023-00	
W1025 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 40 feet long	08241	3262 0023-01	
W1025 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 50 feet long	08241	3262 0023-02	
W1025 (Fig. 11-6)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 75 feet long	08241	3262 0023-03	
W1025/P1	C	Same as W1001/P1			
W1025/P2	C	Same as W1001/P2			
W1026 (Fig. 11-6)	B	Same as W1025			
W1027 (Fig. 11-9)	B	CABLE ASSEMBLY: 24 Conductors, 30 feet long	08241	3262 0007-00	
W1027 (Fig. 11-9)	B	CABLE ASSEMBLY: 24 Conductors, 40 feet long	08241	3262 0007-01	
W1027 (Fig. 11-9)	B	CABLE ASSEMBLY: 24 Conductors, 50 feet long	08241	3262 0007-02	

PARTS LIST					
Ref. Desig.	Ident	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1027 (Fig. 11-9)	B	CABLE ASSEMBLY: 24 Conductors, 75 feet long	08241	3262 0007-03	
W1027/P1	C	Same as W1001/P2			
W1027/P2	C	Same as W1001/P1			
W1028 (Fig. 11-9)	B	Same as W1027			
W1029 (Fig. 11-8)	B	CABLE ASSEMBLY: 12, 12, 18 and 4 Twisted Pair Conductors, 15 feet long	08241	3262 0080-01	
W1029/P1	C	Same as W1017/P2			
W1029/P2	C	CONNECTOR, Plug, electrical: 50 Removable pin contacts	08241 00779	2698 3429 205212-3	
W1029/P3	C	CONTACT, Electrical, Pin	81349	M39029/64-369	
W1029/P4, P5	C	Same as W1017/P1			
	C	CONNECTOR, Plug, electrical: 37 Removable pin contacts	08241 00779	2698 3411 205210-3	
	C	CONTACT, Electrical, Pin	81349	M39029/64-369	
W1030 (Fig. 11-8)	B	Same as W1029			
W1031, W1032		NOT USED			
W1033 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 30 feet long (Part of Radar Kit)	08241	3261 8472-00	

PARTS LIST					
Ref. Design.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1033 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 40 feet long (Part of Radar Kit)	08241	3261 8472-01	
W1033 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 50 feet long (Part of Radar Kit)	08241	3261 8472-02	
W1033 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 75 feet long (Part of Radar Kit)	08241	3261 8472-03	
W1033/P1	C	Same as W1001/P2			
W1033/P2	C	Same as W1001/P1			
W1034 thru W1036 (Fig. 11-9)	B	Same as W1033			
W1037 thru W1042 (Fig. 11-6)	B	Same as W1001			
W1043 (Fig. 7-24)	B	CABLE ASSEMBLY: 4 Conductors, 30 feet long (Part of SDS Kit)	08241	3262 0049-00-04	
W1043 (Fig. 7-24)	B	CABLE ASSEMBLY: 4 Conductor, 40 feet long (Part of SDS Kit)	08241	3262 0049-01-05	
W1043 (Fig. 7-24)	B	CABLE ASSEMBLY: 4 Conductor, 50 feet long (Part of SDS Kit)	08241	3262 0049-02-06	

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1043 (Fig. 7-24)	B	CABLE ASSEMBLY: 4 Conductors, 75 feet long (Part of SDS Kit)	08241	3262 0049-03-07	
W1043/P1	C	CONNECTOR, Plug, electrical: 5 Removable pin contacts	14933	2698 3007	
	C	CONTACT, PWR, 10 amp	08241	2698 3015	
	C	CONTACT, Electrical, Pin	81349	M39029/64-369	
W1043/P2	C	CONNECTOR, Plug, electrical: 5 Removable socket contacts	14933	2698 7693	
	C	CONTACT, Electrical, Socket	81349	M39029/63-368	
	C	CONTACT, PWR, 10 amp, Receptacles	08241	2698 3718	
W1044 (Fig. 7-24)	B	Same as W1043			
W1045 thru W1170 (Fig. 11-6)	B	Same as W1001			
W1171 thru W1174	C	Same as W1029			
W1175 thru W1200		NOT USED			
W1201 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 30 feet long	08241	3262 0171-00	
W1201 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 40 feet long	08241	3262 0171-01	

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1201 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 50 feet long	08241	3262 0171-02	
W1201 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 75 feet long	08241	3262 0171-03	
W1201/P1	C	Same as W1001/P2			
W1201/P2	C	Same as W1001/P1			
W1202 thru W1204 (Fig. 11-9)	B	Same as W1201			
W1205 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 100 feet long	08241	3262 0189-00	
W1205 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 125 feet long	08241	3262 0189-01	
W1205 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 150 feet long	08241	3262 0189-02	
W1205 (Fig. 11-9)	B	CABLE ASSEMBLY: 12 Twisted Pair Conductors and one shielded, 175 feet long	08241	3262 0189-03	
W1205/P1	C	Same as W1001/P2			
W1205/P2	C	Same as W1001/P2			

PARTS LIST					
Ref. Desig.	Indent	Name of Part/Description	Mfrs. Code Number	JAN/MIL Mfrs. Part No.	Notes
W1206 thru W1208 (Fig. 11-9)	B	Same as W1205			
CPME	A	CALIBRATION PERFORMANCE MONITOR EQUIPMENT	55707	098544-0100	
Unit 1	B	BATTERY BOX	55707	083915-0001	
Unit 2	B	ELECTRONIC SUBSYSTEM, CPME	55707	098544-0150	
Unit 3	B	CPME ANTENNA	55707	447939-0001	

Manufacturer's Code Numbers

Code	Manufacturer's Name and Address	Code	Manufacturer's Name and Address
00779	AMP, Inc. 2800 Fulling Mill Middletown, PA 17105-3608	56501	Thomas and Betts Corp. 1001 Frontier Rd. Bridgewater, NJ 08807
08241	Lockheed Martin Tactical Systems, Inc. 590 Lancaster Avenue Frazer, PA 19355-1808	59964	TeleVideo Systems, Inc. 550 E. Brokaw Rd. P.O. Box 49048 San Jose, CA 95161-9048
14933	Defense Electronics Supply Center Dayton, OH 45444-5214	81349	Military Specifications
55707	Wilcox Electric Inc. A Sub of Northrop Corp. 1400 Chestnut Avenue Kansas City, MO 64127-2023	96906	Military Standards
		97942	Northrop Grumman Corp. Defense and Electronic Systems Ctr. Baltimore-Washington Airport P.O.Box 1897, (MS 984) Baltimore, MD 21203

9.0 INSTALLATION, INTEGRATION AND CHECKOUT

9.1 INTRODUCTION

The installation, integration, and checkout instructions for the Mode S sensor are contained in separate documentation referenced in the following paragraph.

9.2 INSTALLATION AND CHECKOUT

Installation, integration, and checkout documents are provided that describe how to install and check out the Mode S sensor at a site. Table 9–1 lists the document title and the corresponding installation document number. Additional references to other applicable installation documents and drawings are provided in Section 2 or 3 of the installation and checkout documents listed in Table 9–1.

TABLE 9–1. INSTALLATION AND CHECKOUT DOCUMENTATION

Document Title	Document Number
1. Site Installation Plan, (Sensor Equipment Installation) For Mode S Beacon System Sensor	
Sites 1 through 78	TM–PA–0092/500/XX through TM–PA–0092/577/XX
Sites 79 through 137	TM–PA–0092/000/XX through TM–PA–0092/058/XX
2. Site Preparation Report For Mode S Beacon System Sensor	
Sites 1 through 50	TM–PA–0018/250/XX through TM–PA–0018/299/XX
Sites 51 through 78	TM–PA–0092/600/XX through TM–PA–0092/627/XX
Sites 79 through 137	TM–PA–0092/628/XX through TM–PA–0092/687/XX
3. Site Test Plan (SITEAT) For Mode S Beacon System Sensor	
Sites 1 through 137	TM–PA–0018/350/XX
4. Site Test Procedure (SITEATP) For Mode S Beacon System Sensor	
Sites 1 through 50	TM–PA–0018/400/XX through TM–PA–0018/449/XX
Sites 51 through 78	TM–PA–0092/100/XX through TM–PA–0092/127/XX

TABLE 9–1. INSTALLATION AND CHECKOUT DOCUMENTATION (Continued)

Document Title	Document Number
Sites 79 through 129	TM–PA–0092/200/XX through TM–PA–0092/249/XX
Sites 130 through 137	TM–PA–0092/128/XX through TM–PA–0092/135/XX
5. Site As–Built Drawings	
Sites 1 through 78	TM–PA–0092/700/XX through TM–PA–0092/777/XX
Sites 79 through 129	TM–PA–0092/400/XX through TM–PA–0092/449/XX
Sites 130 through 137	TM–PA–0121/001/XX through TM–PA–0121/008/XX

10.0 COMPUTER SOFTWARE

10.1 INTRODUCTION

This section contains references to the software user, diagnostic, and maintenance documentation for the Mode S sensor.

10.2 PROGRAM HIERARCHY, DESCRIPTION AND LISTING INFORMATION

The Mode S program hierarchy, descriptions, and listings information are contained in the following documents:

1. Functional Requirements Document for Mode S Beacon System Sensor, TM-PA-0018/016/XX
2. Software Top Level Design Document for Interrogator Performance Monitor CPCI, TM-PA-0018/029/XX
3. Software Top Level Design Document for Reply-to-Reply Processor CPCI, TM-PA-0018/030/XX
4. As-Built Interrogator Performance Monitor CPCI, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0018/046/XX
5. As-Built Reply-to-Reply Processor CPCI, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0018/047/XX
6. Software Top Level Design Document for Data Processing Subsystem (DPS) and Support Software, for Mode S Beacon System Sensor, TM-PA-0028/028/XX
7. As-Built System Level Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0028/500/XX
8. Channel Management — As-Built, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/579/XX
9. As-Built Transaction Processing CPCI, Data Link CPC, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0028/502/XX
10. Surveillance Processing — As-Built, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/581/XX
11. Network Management — As-Built, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/582/XX
12. Remote Monitoring Subsystem — As-Built Monitoring, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/583/XX
13. Performance Monitoring — As-Built, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/584/XX
14. As-Built Data Extraction CPCI, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/586/XX
15. As-Built Data Reduction CPCI, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0028/509/XX

16. Support Software Applications — As-Built, Support Software, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/588/XX
17. PSF Support Software — As-Built, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/589/XX
18. As-Built System Performance Analysis Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0028/514/XX
19. As-Built Executive Programs CPCI, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/585/XX
20. I/O Firmware — As-Built, Software Detail Design Document for Mode S Beacon System Sensor, TM-PA-0092/590/XX

Note: XX represents the revision level

11.0 TROUBLESHOOTING SUPPORT AREA

11.1 INTRODUCTION

This section contains data referenced from the other sections of this book. The support data is presented in figure number order and is referenced from within each section. Table 11–1 lists the figures, by figure number and title.

TABLE 11–1. SUPPORT DATA FIGURES

<u>Figure</u>	<u>Title</u>
11–1	Mode S System Typical Terminal Site
11–2	Mode S System Typical Enroute Site
11–3	Interrogator Cable Assemblies, W64–W66, W76–W78, and W100–W102
11–4	Interrogator Cable Assembly, W67
11–5	Interrogator Cable Assemblies, W68–W75, W79, and W80
11–6	DPS Cable Assemblies, Processor to Junction Box (W1001–W1012, W1019–W1026, W1037–W1142, and W1045 thru W1170)
11–7	DPS Cable Assemblies, Processor to Processor (W1013–W1016)
11–8	DPS Cable Assemblies, Interrogator to Processor (W1029, W1030, and W1171–W1174)
11–9	DPS Cable Assembly, Processor to Junction Box (W1019, W1020, W1027, W1028, W1033–W1036, and W1201–W1208)
11–10	DPS System Cables Wiring and Signal Distribution Diagram (14 Sheets)
11–11	Mode S Power Distribution Information Diagram (2 Sheets)

NOTE

Figures 11–1 through 11–9 are referenced from the Parts List in Section 8.

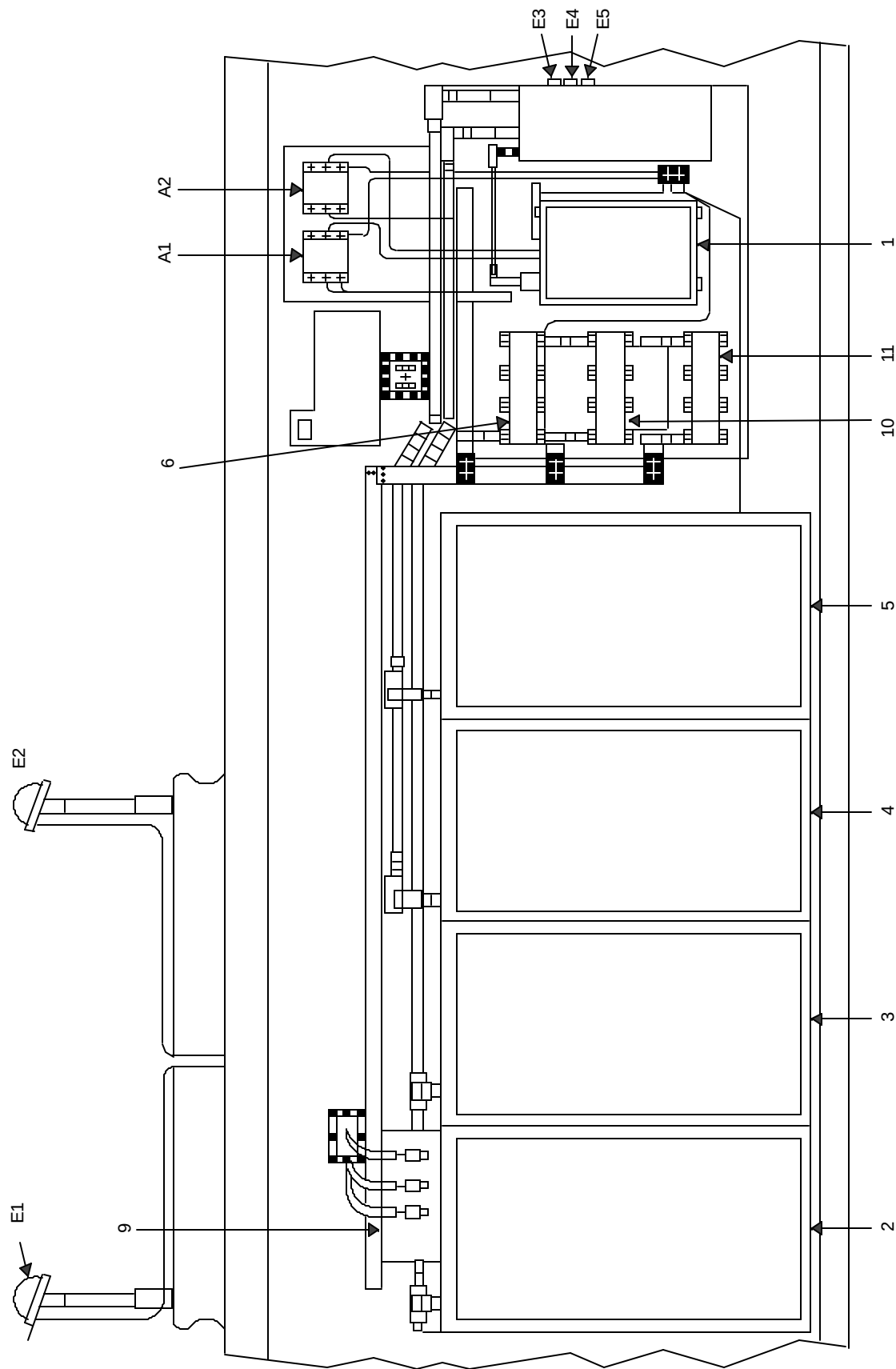


FIGURE 11-1. MODE S SYSTEM TYPICAL TERMINAL SITE

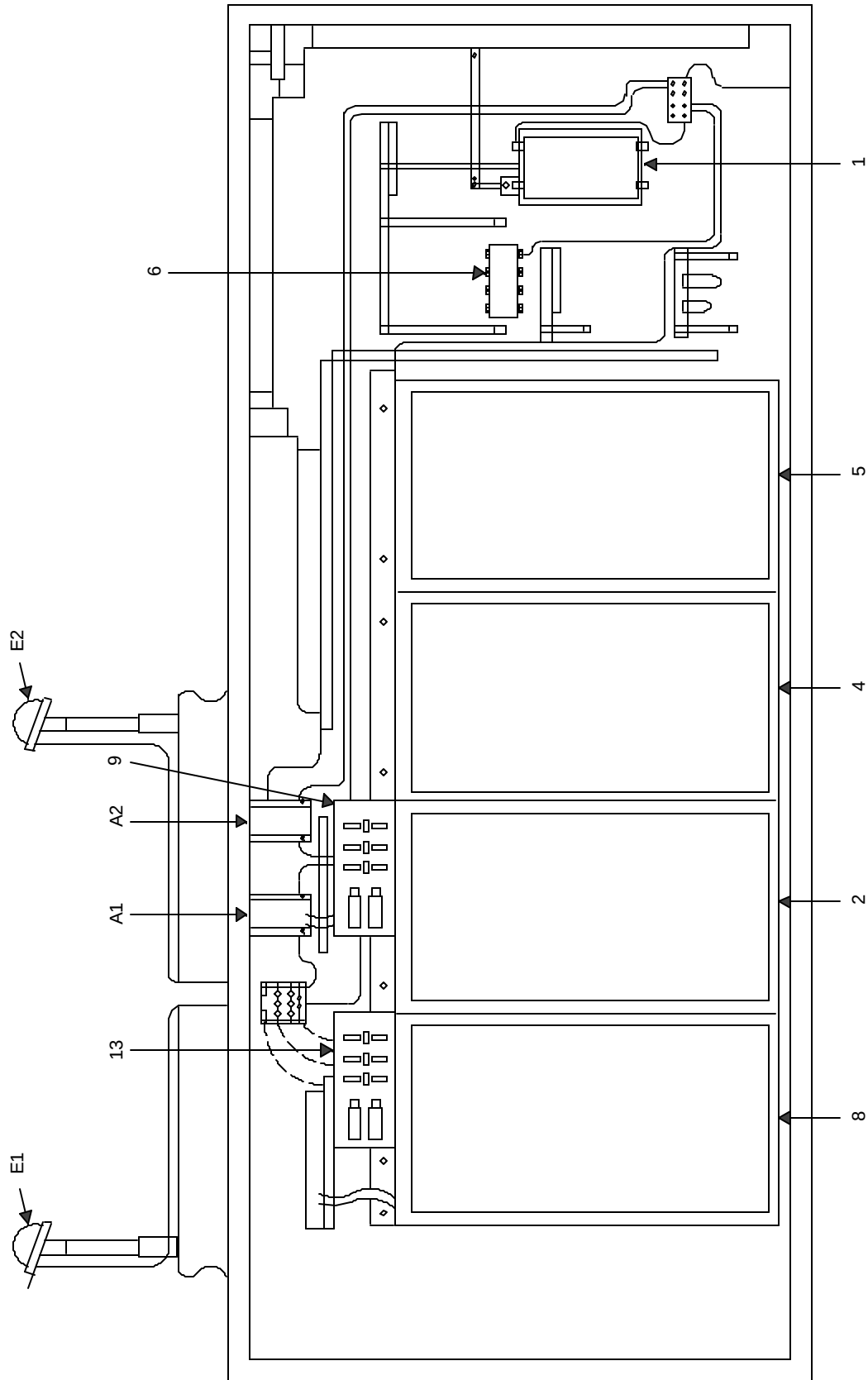


FIGURE 11-2. MODE S SYSTEM TYPICAL EN ROUTE SITE

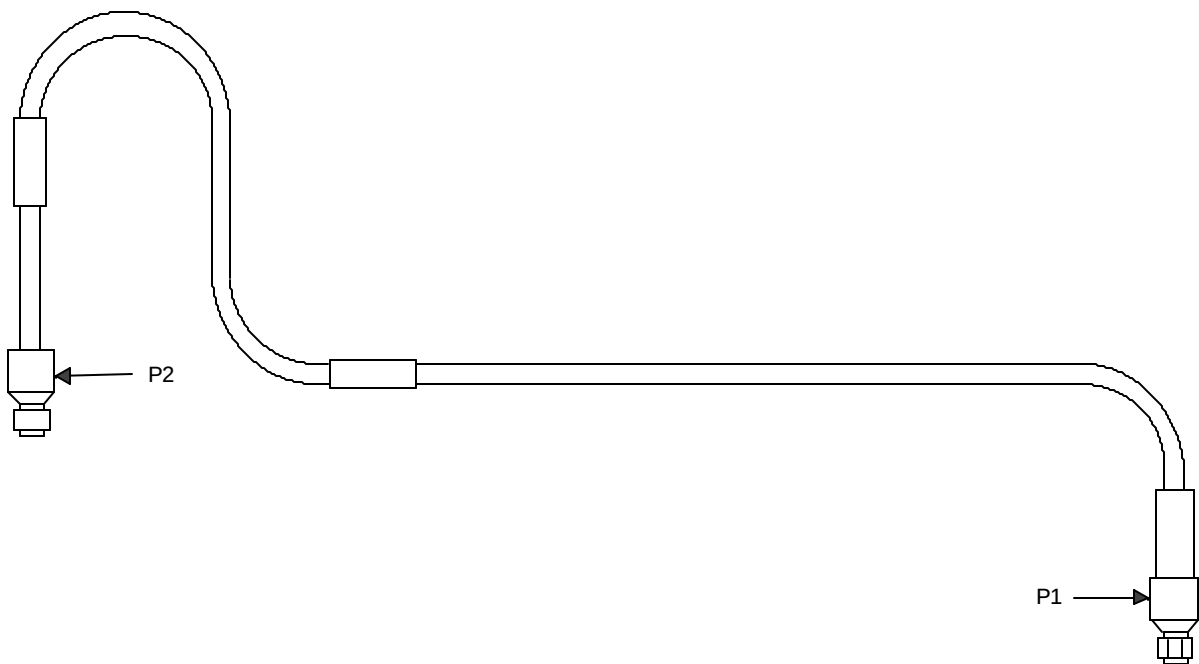


FIGURE 11-3. INTERROGATOR CABLE ASSEMBLIES W64-W66, W76-W78, AND W100-W102

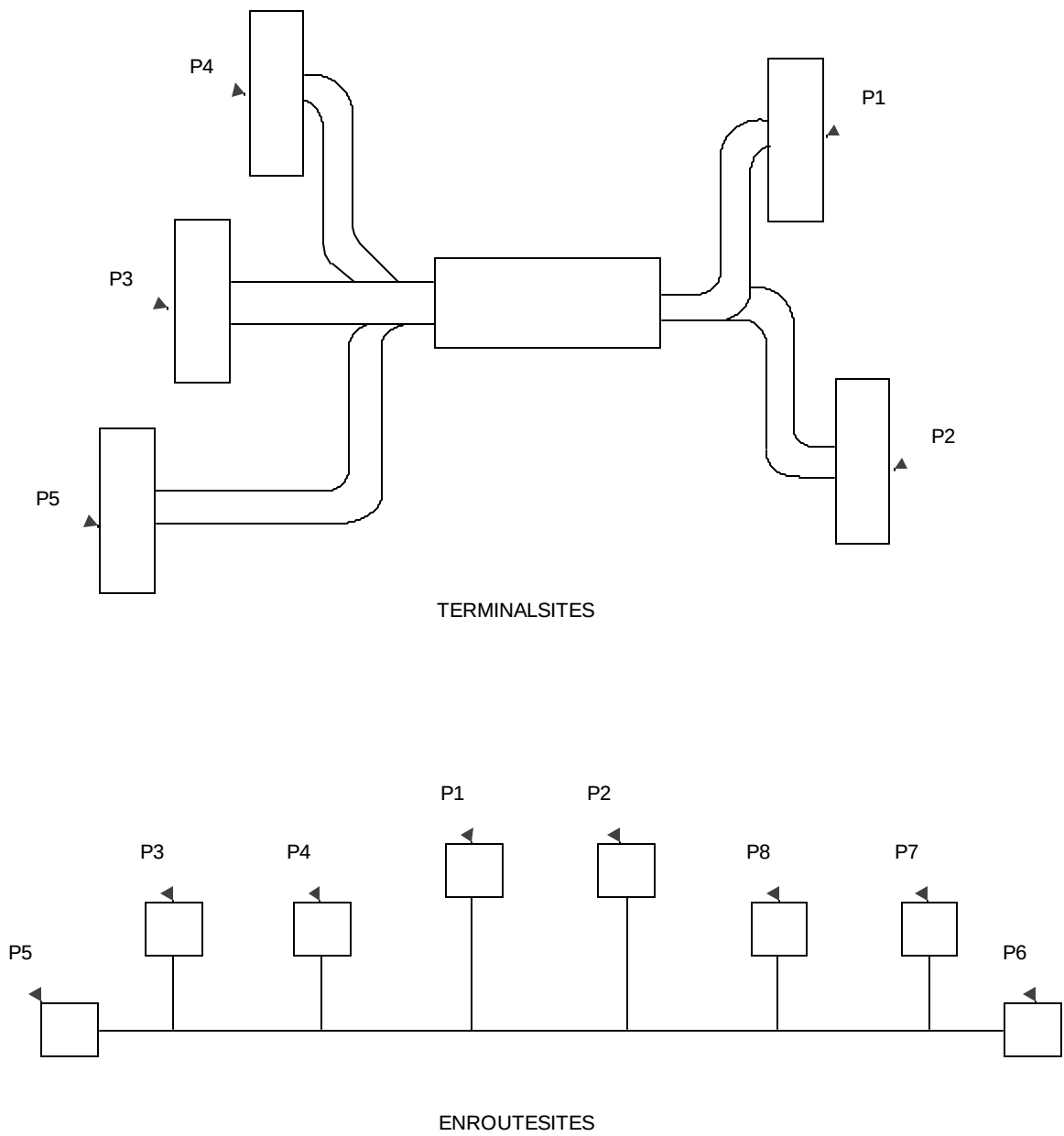


FIGURE 11-4. INTERROGATOR CABLE ASSEMBLY W67

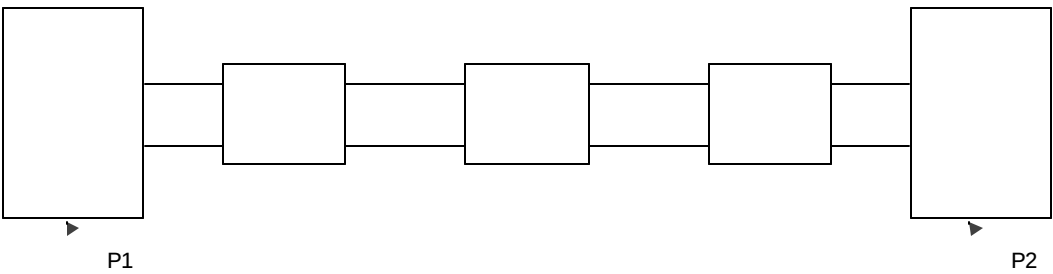


FIGURE 11-5. INTERROGATOR CABLE ASSEMBLIES W68-W75, W79, AND W80

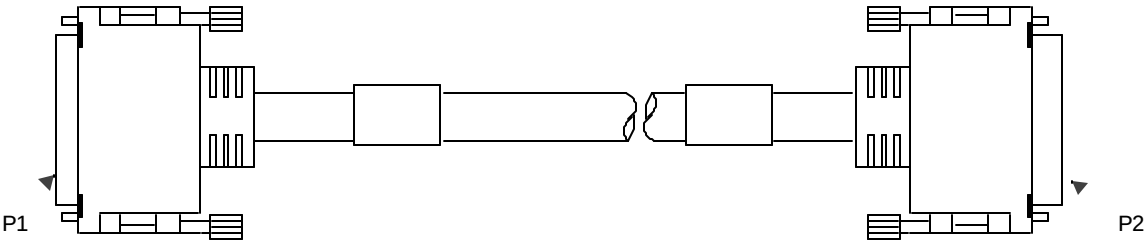


FIGURE 11-6. DPS CABLE ASSEMBLIES, PROCESSOR TO JUNCTION BOX (W1001-W1012, W1021-W1026, W1037-W1042 AND W1045-W1170)

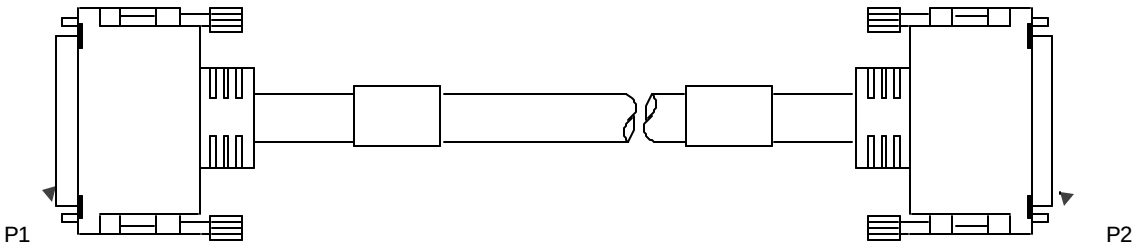


FIGURE 11-7. DPS CABLE ASSEMBLIES, PROCESSOR TO PROCESSOR (W1013-W1016)



FIGURE 11-8. DPS CABLE ASSEMBLIES, INTERROGATOR TO PROCESSOR (W1029, W1030, AND W1171-W1174)

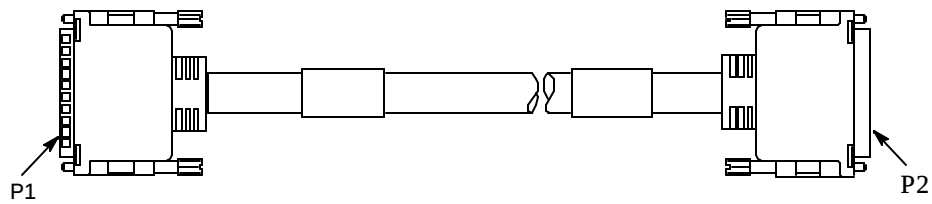


FIGURE 11-9. DPS CABLE ASSEMBLY, PROCESSOR TO JUNCTION BOX (W1019, W1020, W1027, W1028, W1033-W1036, AND W1201-W1208)

COMM(6)/COMM EXP(10)/SURV EXP(11)
JUNCTION BOX
P2

SPARE (GND)	1
SPARE (GND)	23
CHASSIS	
DRAIN	
CSD	0
CSD	1
CHASSIS	2
DRAIN	
CRD	0
CRD	1
CHASSIS	3
DRAIN	
CRS	0
CRS	1
CHASSIS	4
DRAIN	
CCS	0
CCS	1
CHASSIS	5
DRAIN	
CDM	0
CDM	1
CHASSIS	6
DRAIN	
CSG	0
CSG	7
SPARE (GND)	20
CHASSIS	
DRAIN	
CRR	0
CRR	1
CHASSIS	8
DRAIN	
CRT	0
CRT	1
CHASSIS	17
DRAIN	
CTT	0
CTT	1
CHASSIS	11
DRAIN	
CST	0
CST	1
CHASSIS	12
DRAIN	
CLL	0
CLL	1
CHASSIS	15
DRAIN	
CTM	0
CTM	1
CHASSIS	18
DRAIN	
CRL	0
CRL	1
CHASSIS	21

3261 8423-XX
SYSTEM CABLE TYPE A-1 BASIC
(COMMUNICATION AND SURVEILLANCE)

DPS
PROCESSOR (4/5)
P1

SPARE (GND)	1
SPARE (GND)	14
CHASSIS	
DRAIN	
CSD	0
CSD	1
CHASSIS	15
DRAIN	
CRD	0
CRD	1
CHASSIS	3
DRAIN	
CRS	0
CRS	1
CHASSIS	4
DRAIN	
CCS	0
CCS	1
CHASSIS	17
DRAIN	
CDM	0
CDM	1
CHASSIS	6
DRAIN	
CSG	0
CSG	7
SPARE (GND)	20
CHASSIS	
DRAIN	
CRR	0
CRR	1
CHASSIS	8
DRAIN	
CRT	0
CRT	1
CHASSIS	9
DRAIN	
CTT	0
CTT	1
CHASSIS	10
DRAIN	
CST	0
CST	1
CHASSIS	23
DRAIN	
CLL	0
CLL	1
CHASSIS	11
DRAIN	
CTM	0
CTM	1
CHASSIS	24
DRAIN	
CRL	0
CRL	1
CHASSIS	12
DRAIN	
CRL	1
CHASSIS	13

- NOTES:
- BOXES CONTAIN DIGIT 1 OR 2 AS A FUNCTION OF WHICH LINE ADAPTER PORT (PER PADDLE BOARD) THIS CABLE PLUGS INTO THE PROCESSOR (DPS).

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 1)

3261 8423-XX SYSTEM CABLE TYPE A-1 BASIC (COMMUNICATION AND SURVEILLANCE)

COMM/SURV CABLE LOCATIONS			COMM/SURV CABLE LOCATIONS			COMM/SURV CABLE LOCATIONS		
CABLE NO	FROM	TO	CABLE NO	FROM	TO	CABLE NO	FROM	TO
W1001	P1 (4A1-J24)	P2 (6A1-J1)	W1076	P1 (5A1-J63)	P2 (10A1-J46)	W1125	P1 (4A1-J45)	P2 (11A1-J34)
W1002	P1 (5A1-J24)	P2 (6A1-J11)	W1077	P1 (4A1-J75)	P2 (10A1-J37)	W1126	P1 (5A1-J45)	P2 (11A1-J44)
W1003	P1 (4A1-J36)	P2 (6A1-J2)	W1078	P1 (5A1-J75)	P2 (10A1-J47)	W1127	P1 (4A1-J57)	P2 (11A1-J35)
W1004	P1 (5A1-J36)	P2 (6A1-J12)	W1079	P1 (4A1-J87)	P2 (10A1-J38)	W1128	P1 (5A1-J57)	P2 (11A1-J45)
W1005	P1 (4A1-J48)	P2 (6A1-J3)	W1080	P1 (5A1-J87)	P2 (10A1-J48)	W1129	P1 (4A1-J69)	P2 (11A1-J36)
W1006	P1 (5A1-J48)	P2 (6A1-J13)	W1081	P1 (4A1-J28)	P2 (10A1-J39)	W1130	P1 (5A1-J69)	P2 (11A1-J46)
W1007	P1 (4A1-J60)	P2 (6A1-J31)	W1082	P1 (5A1-J28)	P2 (10A1-J49)	W1131	P1 (4A1-J81)	P2 (11A1-J37)
W1008	P1 (5A1-J60)	P2 (6A1-J41)	W1083	P1 (4A1-J40)	P2 (10A1-J40)	W1132	P1 (5A1-J81)	P2 (11A1-J47)
W1009	P1 (4A1-J72)	P2 (6A1-J32)	W1084	P1 (5A1-J40)	P2 (10A1-J50)	W1133	P1 (4A1-J93)	P2 (11A1-J38)
W1010	P1 (5A1-J72)	P2 (6A1-J42)	W1085	P1 (4A1-J52)	P2 (10A1-J61)	W1134	P1 (5A1-J93)	P2 (11A1-J48)
W1011	P1 (4A1-J84)	P2 (6A1-J33)	W1086	P1 (5A1-J52)	P2 (10A1-J71)	W1135	P1 (4A1-J34)	P2 (11A1-J39)
W1012	P1 (5A1-J84)	P2 (6A1-J43)	W1087	P1 (4A1-J64)	P2 (10A1-J62)	W1136	P1 (5A1-J34)	P2 (11A1-J49)
W1013	P1 (4A1-J30)	P2 (6A1-J4)	W1088	P1 (5A1-J64)	P2 (10A1-J72)	W1137	P1 (4A1-J46)	P2 (11A1-J40)
W1038	P1 (5A1-J30)	P2 (6A1-J14)	W1089	P1 (4A1-J76)	P2 (10A1-J63)	W1138	P1 (5A1-J46)	P2 (11A1-J50)
W1039	P1 (4A1-J42)	P2 (6A1-J5)	W1090	P1 (5A1-J76)	P2 (10A1-J73)	W1139	P1 (4A1-J58)	P2 (11A1-J61)
W1040	P1 (5A1-J42)	P2 (6A1-J15)	W1091	P1 (4A1-J88)	P2 (10A1-J64)	W1140	P1 (5A1-J58)	P2 (11A1-J71)
W1041	P1 (4A1-J54)	P2 (6A1-J6)	W1092	P1 (5A1-J88)	P2 (10A1-J74)	W1141	P1 (4A1-J70)	P2 (11A1-J62)
W1042	P1 (5A1-J54)	P2 (6A1-J16)	W1093	P1 (4A1-J66)	P2 (6A1-J34)	W1142	P1 (5A1-J70)	P2 (11A1-J72)
W1045	P1 (4A1-J25)	P2 (10A1-J1)	W1094	P1 (5A1-J66)	P2 (6A1-J44)	W1143	P1 (4A1-J82)	P2 (11A1-J63)
W1046	P1 (5A1-J25)	P2 (10A1-J11)	W1095	P1 (4A1-J78)	P2 (6A1-J35)	W1144	P1 (5A1-J82)	P2 (11A1-J73)
W1047	P1 (4A1-J37)	P2 (10A1-J2)	W1096	P1 (5A1-J78)	P2 (6A1-J45)	W1145	P1 (4A1-J94)	P2 (11A1-J64)
W1048	P1 (5A1-J37)	P2 (10A1-J12)	W1097	P1 (4A1-J90)	P2 (6A1-J36)	W1146	P1 (5A1-J94)	P2 (11A1-J74)
W1049	P1 (4A1-J49)	P2 (10A1-J3)	W1098	P1 (5A1-J90)	P2 (6A1-J46)	W1147	P1 (4A1-J35)	P2 (11A1-J65)
W1050	P1 (5A1-J49)	P2 (10A1-J13)	W1099	P1 (4A1-J31)	P2 (11A1-J1)	W1148	P1 (5A1-J35)	P2 (11A1-J75)
W1051	P1 (4A1-J61)	P2 (10A1-J4)	W1100	P1 (5A1-J31)	P2 (11A1-J11)	W1149	P1 (4A1-J47)	P2 (11A1-J66)
W1052	P1 (5A1-J61)	P2 (10A1-J14)	W1101	P1 (4A1-J43)	P2 (11A1-J2)	W1150	P1 (5A1-J47)	P2 (11A1-J76)
W1053	P1 (4A1-J73)	P2 (10A1-J5)	W1102	P1 (5A1-J43)	P2 (11A1-J12)	W1151	P1 (4A1-J59)	P2 (11A1-J67)
W1054	P1 (5A1-J73)	P2 (10A1-J15)	W1103	P1 (4A1-J55)	P2 (11A1-J3)	W1152	P1 (5A1-J59)	P2 (11A1-J77)
W1055	P1 (4A1-J85)	P2 (10A1-J6)	W1104	P1 (5A1-J55)	P2 (11A1-J13)	W1153	P1 (4A1-J71)	P2 (11A1-J68)
W1056	P1 (5A1-J85)	P2 (10A1-J16)	W1105	P1 (4A1-J67)	P2 (11A1-J4)	W1154	P1 (5A1-J71)	P2 (11A1-J78)
W1057	P1 (4A1-J26)	P2 (10A1-J7)	W1106	P1 (5A1-J67)	P2 (11A1-J14)	W1155	P1 (4A1-J83)	P2 (11A1-J69)
W1058	P1 (5A1-J26)	P2 (10A1-J17)	W1107	P1 (4A1-J79)	P2 (11A1-J5)	W1156	P1 (5A1-J83)	P2 (11A1-J79)
W1059	P1 (4A1-J38)	P2 (10A1-J8)	W1108	P1 (5A1-J79)	P2 (11A1-J15)	W1157	P1 (4A1-J95)	P2 (11A1-J70)
W1060	P1 (5A1-J38)	P2 (10A1-J18)	W1109	P1 (4A1-J91)	P2 (11A1-J6)	W1158	P1 (5A1-J95)	P2 (11A1-J80)
W1061	P1 (4A1-J50)	P2 (10A1-J9)	W1110	P1 (5A1-J91)	P2 (11A1-J16)	W1159	P1 (4A1-J29)	P2 (10A1-J65)
W1062	P1 (5A1-J50)	P2 (10A1-J19)	W1111	P1 (4A1-J32)	P2 (11A1-J7)	W1160	P1 (5A1-J29)	P2 (10A1-J75)
W1063	P1 (4A1-J62)	P2 (10A1-J10)	W1112	P1 (5A1-J32)	P2 (11A1-J17)	W1161	P1 (4A1-J41)	P2 (10A1-J66)
W1064	P1 (5A1-J62)	P2 (10A1-J20)	W1113	P1 (4A1-J44)	P2 (11A1-J8)	W1162	P1 (5A1-J41)	P2 (10A1-J76)
W1065	P1 (4A1-J74)	P2 (10A1-J31)	W1114	P1 (5A1-J44)	P2 (11A1-J18)	W1163	P1 (4A1-J53)	P2 (10A1-J67)
W1066	P1 (5A1-J74)	P2 (10A1-J41)	W1115	P1 (4A1-J56)	P2 (11A1-J9)	W1164	P1 (5A1-J53)	P2 (10A1-J77)
W1067	P1 (4A1-J86)	P2 (10A1-J32)	W1116	P1 (5A1-J56)	P2 (11A1-J19)	W1165	P1 (4A1-J65)	P2 (10A1-J68)
W1068	P1 (5A1-J86)	P2 (10A1-J42)	W1117	P1 (4A1-J68)	P2 (11A1-J10)	W1166	P1 (5A1-J65)	P2 (10A1-J78)
W1069	P1 (4A1-J27)	P2 (10A1-J33)	W1118	P1 (5A1-J68)	P2 (11A1-J20)	W1167	P1 (4A1-J77)	P2 (10A1-J69)
W1070	P1 (5A1-J27)	P2 (10A1-J43)	W1119	P1 (4A1-J80)	P2 (11A1-J31)	W1168	P1 (5A1-J77)	P2 (10A1-J79)
W1071	P1 (4A1-J39)	P2 (10A1-J34)	W1120	P1 (5A1-J80)	P2 (11A1-J41)	W1169	P1 (4A1-J89)	P2 (10A1-J70)
W1072	P1 (5A1-J39)	P2 (10A1-J44)	W1121	P1 (4A1-J92)	P2 (11A1-J32)	W1170	P1 (5A1-J89)	P2 (10A1-J80)
W1073	P1 (4A1-J51)	P2 (10A1-J35)	W1122	P1 (5A1-J92)	P2 (11A1-J42)			
W1074	P1 (5A1-J51)	P2 (10A1-J45)	W1123	P1 (4A2-J53)	P2 (11A1-J33)			
W1075	P1 (4A1-J63)	P2 (10A1-J36)	W1124	P1 (5A2-J33)	P2 (11A1-J43)			

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 2)

DPS
PROCESSOR (4/5)
P1

1	SPARE (GND)
14	SPARE
CHASSIS	DRAIN
2	MSDP --- 0
15	MSDP --- 1
CHASSIS	DRAIN
3	MRDP --- 0
16	MRDP --- 1
CHASSIS	DRAIN
4	MRSP --- 0
17	MRSP --- 1
CHASSIS	DRAIN
5	MCSP --- 0
18	MCSP --- 1
CHASSIS	DRAIN
6	MDMP --- 0
19	MDMP --- 1
CHASSIS	DRAIN
7	GND
20	SPARE
CHASSIS	DRAIN
8	MRRP --- 0
21	MRRP --- 1
CHASSIS	DRAIN
9	M RTP --- 0
22	M RTP --- 1
CHASSIS	DRAIN
10	MTTP --- 0
23	MTTP --- 1
CHASSIS	DRAIN
11	MSTP --- 0
24	MSTP --- 1
CHASSIS	DRAIN
12	MTRP --- 0
25	MICP --- 1
CHASSIS	DRAIN
13	MTRP --- 1

COMM
JUNCTION BOX (6)
P2

SPARE (GND)	1
SPARE	23
DRAIN	CHASSIS
MSDP --- 0	14
MSDP --- 1	2
DRAIN	CHASSIS
MRDP --- 0	16
MRDP --- 1	3
DRAIN	CHASSIS
MRSP --- 0	19
MRSP --- 1	4
DRAIN	CHASSIS
MCSP --- 0	13
MCSP --- 1	5
DRAIN	CHASSIS
MDMP --- 0	22
MDMP --- 1	6
DRAIN	CHASSIS
GND	7
SPARE	20
DRAIN	CHASSIS
MRRP --- 0	10
MRRP --- 1	8
DRAIN	CHASSIS
M RTP --- 0	9
M RTP --- 1	17
DRAIN	CHASSIS
MTTP --- 0	11
MTTP --- 1	24
DRAIN	CHASSIS
MSTP --- 0	12
MSTP --- 1	15
DRAIN	CHASSIS
MTRP --- 0	18
MICP --- 1	25
DRAIN	CHASSIS
MTRP --- 1	21

3261 8423-XX

SYSTEM CABLE TYPE A-1 BASIC
(PROGRAM SUPPORT FACILITY)

PROGRAM SUPPORT CABLE LOCATIONS

CABLE NUMBER	FROM	TO
W1023	P1 (4A1-J3)	P2(6A1-J10)
W1024	P1 (5A1-J3)	P2(6A1-J20)

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 3)

DPS
PROCESSOR (4/5)
P1

1	SPARE (GND)
14	SPARE
CHASSIS	
2	MSDM --- 0
15	MSDM --- 1
CHASSIS	
3	MRDM --- 0
16	MRDM --- 1
CHASSIS	
4	MRSRM --- 0
17	MRSRM --- 1
CHASSIS	
5	MCSM --- 0
18	MCSM --- 1
CHASSIS	
6	MDMM --- 0
19	MDMM --- 1
CHASSIS	
7	GND
20	SPARE
CHASSIS	
8	MRRM --- 0
21	MRRM --- 1
CHASSIS	
9	MRTM --- 0
22	MRTM --- 1
CHASSIS	
10	MTTM --- 0
23	MTTM --- 1
CHASSIS	
11	MSTM --- 0
24	MSTM --- 1
CHASSIS	
12	MLLM --- 1
25	MTMM --- 1
CHASSIS	
13	MRLM --- 1

COMM
JUNCTION BOX (6)
P2

SPARE (GND)	1
SPARE	23
CHASSIS	
DRAIN	
MSDM --- 0	14
MSDM --- 1	2
CHASSIS	
DRAIN	
MRDM --- 0	16
MRDM --- 1	3
CHASSIS	
DRAIN	
MRSRM --- 0	19
MRSRM --- 1	4
CHASSIS	
DRAIN	
MCSM --- 0	13
MCSM --- 1	5
CHASSIS	
DRAIN	
MDMM --- 0	22
MDMM --- 1	6
CHASSIS	
DRAIN	
GND	7
SPARE	20
CHASSIS	
DRAIN	
MRRM --- 0	10
MRRM --- 1	8
CHASSIS	
DRAIN	
MRTM --- 0	9
MRTM --- 1	17
CHASSIS	
DRAIN	
MTTM --- 0	11
MTTM --- 1	24
CHASSIS	
DRAIN	
MSTM --- 0	12
MSTM --- 1	15
CHASSIS	
DRAIN	
MLLM --- 1	18
MTMM --- 1	25
CHASSIS	
DRAIN	
MRLM --- 1	21

3261 8423-XX

SYSTEM CABLE TYPE A-1 BASIC.
(REMOTE MAINTENANCE MONITOR SYSTEM)

REMOTE MAINT CABLE LOCATIONS	
CABLE NUMBER	FROM
W1021	P1 (4A1-J12)
W1022	P1 (5A1-J12)
	TO
	P2(6A1-J38)
	P2(6A1-J48)

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 4)

DPS PROCESSOR A (4)	
P1	
1	CADRDAT 001
34	CADRDAT 000
CHASSIS	DRAIN
18	CADRDAT 011
2	CADRDAT 010
CHASSIS	DRAIN
35	CADRDAT 021
19	CADRDAT 020
CHASSIS	DRAIN
3	CADRDAT 031
36	CADRDAT 030
CHASSIS	DRAIN
20	CADRDAT 041
4	CADRDAT 040
CHASSIS	DRAIN
37	CADRDAT 051
21	CADRDAT 050
CHASSIS	DRAIN
5	CADRDAT 061
38	CADRDAT 060
CHASSIS	DRAIN
22	CADRDAT 071
6	CADRDAT 070
CHASSIS	DRAIN
39	CADRDAT P01
23	CADRDAT P00
CHASSIS	DRAIN
7	CADRDAT 081
40	CADRDAT 080
CHASSIS	DRAIN
24	CADRDAT 091
8	CADRDAT 090
CHASSIS	DRAIN
41	CADRDAT 101
25	CADRDAT 100
CHASSIS	DRAIN
9	CADRDAT 111
42	CADRDAT 110
CHASSIS	DRAIN
26	CADRDAT 121
10	CADRDAT 120
CHASSIS	DRAIN
43	CADRDAT 131
27	CADRDAT 130
CHASSIS	DRAIN
11	CADRDAT 141
44	CADRDAT 140
CHASSIS	DRAIN
28	CADRDAT 151
12	CADRDAT 150
CHASSIS	DRAIN
45	CADRDAT P11
29	CADRDAT P10
CHASSIS	DRAIN
13	CADRDAT 161
46	CADRDAT 160
CHASSIS	DRAIN
30	CADRDAT 171
14	CADRDAT 170
CHASSIS	DRAIN
47	SPARE
31	SPARE
CHASSIS	DRAIN
15	SPARE
48	SPARE
CHASSIS	DRAIN
32	SPARE
16	SPARE
CHASSIS	DRAIN
49	SPARE
33	SPARE
CHASSIS	DRAIN
50	SPARE
NC	NO CONNECTION
CHASSIS	DRAIN

DPS PROCESSOR B (%)	
P2	
1	CADRDAT 001
34	CADRDAT 000
CHASSIS	DRAIN
18	CADRDAT 011
2	CADRDAT 010
CHASSIS	DRAIN
35	CADRDAT 021
19	CADRDAT 020
CHASSIS	DRAIN
3	CADRDAT 031
36	CADRDAT 030
CHASSIS	DRAIN
20	CADRDAT 041
4	CADRDAT 040
CHASSIS	DRAIN
37	CADRDAT 051
21	CADRDAT 050
CHASSIS	DRAIN
5	CADRDAT 061
38	CADRDAT 060
CHASSIS	DRAIN
22	CADRDAT 071
6	CADRDAT 070
CHASSIS	DRAIN
39	CADRDAT P01
23	CADRDAT P00
CHASSIS	DRAIN
7	CADRDAT 081
40	CADRDAT 080
CHASSIS	DRAIN
24	CADRDAT 091
8	CADRDAT 090
CHASSIS	DRAIN
41	CADRDAT 101
25	CADRDAT 100
CHASSIS	DRAIN
9	CADRDAT 111
42	CADRDAT 110
CHASSIS	DRAIN
26	CADRDAT 121
10	CADRDAT 120
CHASSIS	DRAIN
43	CADRDAT 131
27	CADRDAT 130
CHASSIS	DRAIN
11	CADRDAT 141
44	CADRDAT 140
CHASSIS	DRAIN
28	CADRDAT 151
12	CADRDAT 150
CHASSIS	DRAIN
45	CADRDAT P11
29	CADRDAT P12
CHASSIS	DRAIN
13	CADRDAT 161
46	CADRDAT 160
CHASSIS	DRAIN
30	CADRDAT 171
14	CADRDAT 170
CHASSIS	DRAIN
47	SPARE
31	SPARE
CHASSIS	DRAIN
15	SPARE
48	SPARE
CHASSIS	DRAIN
32	SPARE
16	SPARE
CHASSIS	DRAIN
49	SPARE
33	SPARE
CHASSIS	DRAIN
50	SPARE
NC	NO CONNECTION
CHASSIS	DRAIN

3262 0072

SYSTEM CABLE TYPE B-1 BASIC (GRAM)

GRAM B-1 CABLE LOCATIONS

CABLE NUMBER	FROM	TO
W1013	P1 (4A1-J6)	P2 (5A1-J6)
W1014	P1 (4A1-J6)	P2 (5A1-J7)

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 5)

DPS PROCESSOR A (4)

P1

1	CADRDAT 181	1	CADRDAT 181
34	CHASSIS	34	CADRDAT 180
18	DRAIN	18	DRAIN
2	CADRDAT 191	2	CADRDAT 191
35	CHASSIS	35	DRAIN
19	CADRDAT 201	19	CADRDAT 201
3	CHASSIS	3	DRAIN
36	CADRDAT 211	36	CADRDAT 200
20	CADRDAT 210	20	DRAIN
4	CHASSIS	4	CADRDAT 211
37	CADRDAT 221	37	CADRDAT 210
21	CADRDAT 230	21	DRAIN
5	CHASSIS	5	CADRDAT 221
38	CADRDAT P21	38	CADRDAT 220
22	CHASSIS	22	DRAIN
6	CADRDAT 241	6	CADRDAT 231
39	CHASSIS	39	CADRDAT 230
23	CADRDAT 251	23	DRAIN
7	CHASSIS	7	CADRDAT 241
40	CADRDAT 261	40	CADRDAT 240
24	CHASSIS	24	DRAIN
8	CADRDAT 271	8	CADRDAT 251
41	CHASSIS	41	CADRDAT 250
25	CADRDAT 281	25	DRAIN
9	CHASSIS	9	CADRDAT 261
42	CADRDAT 291	42	CADRDAT 260
26	CHASSIS	26	DRAIN
10	CADRDAT 301	10	CADRDAT 271
43	CHASSIS	43	CADRDAT 270
27	CADRDAT 311	27	DRAIN
11	CHASSIS	11	CADRDAT 281
44	CADRDAT P31	44	CADRDAT 280
28	CHASSIS	28	DRAIN
12	CADRVALLID 1	12	CADRDAT 291
45	CHASSIS	45	CADRDAT 290
29	CDATREADY 1	29	DRAIN
13	CHASSIS	13	CADRDAT 301
46	CDATACCPT 1	46	CADRDAT 300
30	CHASSIS	30	DRAIN
14	GND (CARD)	14	CADRDAT 311
47	DRAIN	47	CADRDAT 310
31	SPARE	31	DRAIN
15	CHASSIS	15	CADRDAT P31
48	SPARE	48	CADRDAT P30
32	CHASSIS	32	DRAIN
16	SPARE	16	CADRVALLID 1
49	CHASSIS	49	CADRVALLID 0
33	CHASSIS	33	DRAIN
50	CHASSIS	50	CDATREADY 1
NC	NO CONNECTION	NC	CDATREADY 0
DRAIN	DRAIN	DRAIN	CDATACCPT 1

DPS PROCESSOR B (5)

P2

1	CADRDAT 181	1	CADRDAT 181
34	CHASSIS	34	CADRDAT 180
18	DRAIN	18	DRAIN
2	CADRDAT 191	2	CADRDAT 191
35	CHASSIS	35	DRAIN
19	CADRDAT 201	19	CADRDAT 201
3	CHASSIS	3	DRAIN
36	CADRDAT 211	36	CADRDAT 200
20	CADRDAT 210	20	DRAIN
4	CHASSIS	4	CADRDAT 211
37	CADRDAT 221	37	CADRDAT 210
21	CADRDAT 230	21	DRAIN
5	CHASSIS	5	CADRDAT 221
38	CADRDAT P21	38	CADRDAT 220
22	CHASSIS	22	DRAIN
6	CADRDAT 241	6	CADRDAT 231
39	CHASSIS	39	CADRDAT 230
23	CADRDAT 251	23	DRAIN
7	CHASSIS	7	CADRDAT 241
40	CADRDAT 261	40	CADRDAT 240
24	CHASSIS	24	DRAIN
8	CADRDAT 271	8	CADRDAT 251
41	CHASSIS	41	CADRDAT 250
25	CADRDAT 281	25	DRAIN
9	CHASSIS	9	CADRDAT 261
42	CADRDAT 291	42	CADRDAT 260
26	CHASSIS	26	DRAIN
10	CADRDAT 301	10	CADRDAT 271
43	CHASSIS	43	CADRDAT 270
27	CADRDAT 311	27	DRAIN
11	CHASSIS	11	CADRDAT 281
44	CADRDAT P31	44	CADRDAT 280
28	CHASSIS	28	DRAIN
12	CADRVALLID 1	12	CADRDAT 291
45	CHASSIS	45	CADRDAT 290
29	CDATREADY 1	29	DRAIN
13	CHASSIS	13	CADRDAT 301
46	CDATACCPT 1	46	CADRDAT 300
30	CHASSIS	30	DRAIN
14	GND (CARD)	14	CADRDAT 311
47	DRAIN	47	CADRDAT 310
31	SPARE	31	DRAIN
15	CHASSIS	15	CADRDAT P31
48	SPARE	48	CADRDAT P30
32	CHASSIS	32	DRAIN
16	SPARE	16	CADRVALLID 1
49	CHASSIS	49	CADRVALLID 0
33	CHASSIS	33	DRAIN
50	CHASSIS	50	CDATREADY 1
NC	NO CONNECTION	NC	CDATREADY 0
DRAIN	DRAIN	DRAIN	CDATACCPT 1

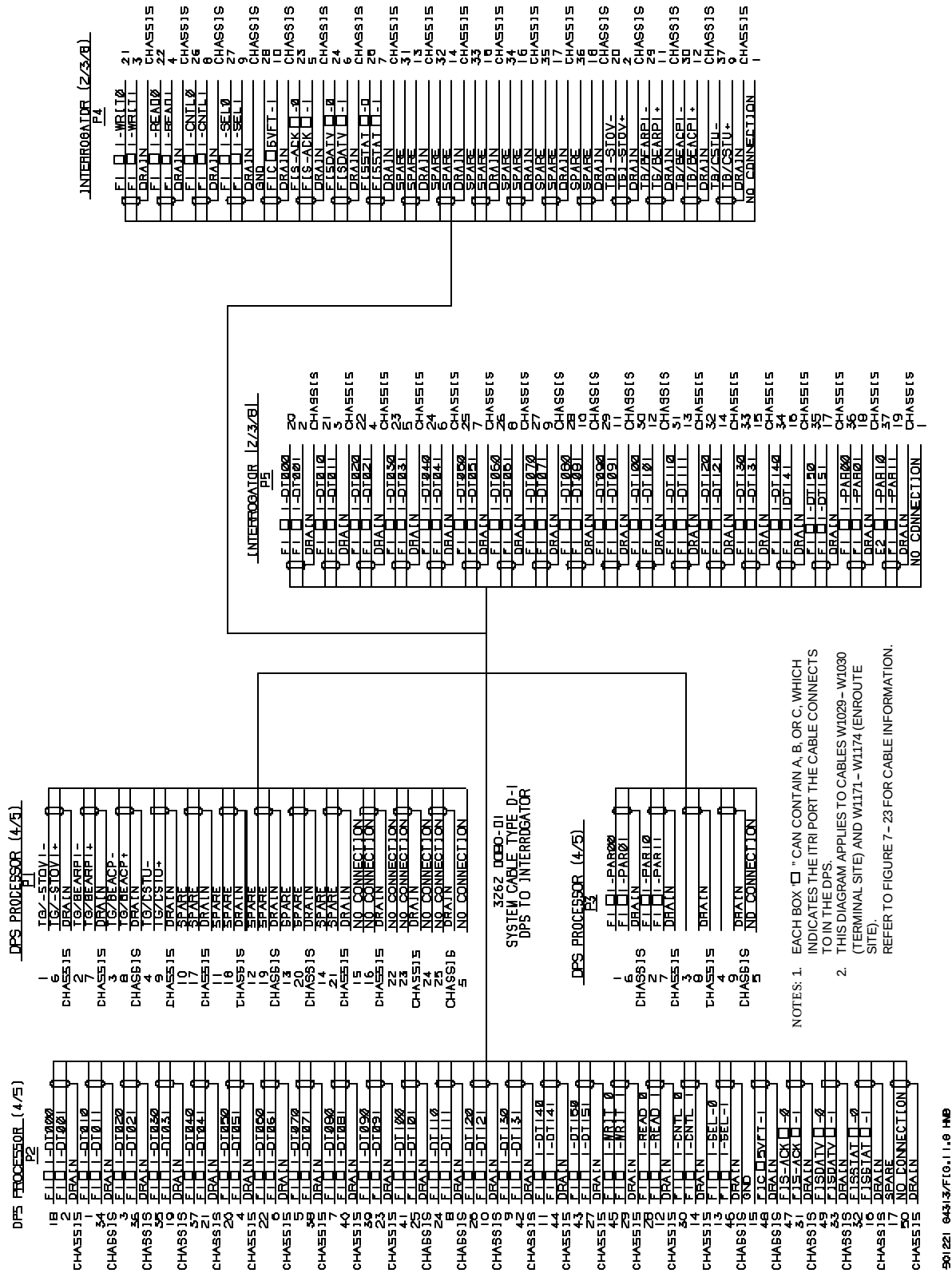
3388 0459

SYSTEM CABLE TYPE B-2 BASIC (GRAM)

CABLE NUMBER	FROM	TO
W1115	P1 (4A1-J19)	P2 (5A1-J19)
W1164	P1 (4A1-J20)	P2 (5A1-J20)

GRAM B-2 CABLE LOCATIONS

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 6)



DPS
PROCESSOR (4/5)
P1

1	SPARE (GND)	1
7	CSG	0
	DRAIN	
2	CSD	0
15	CSD	1
	DRAIN	
3	CRD	0
16	CRD	1
	DRAIN	
4	CRS	0
17	CRS	1
	DRAIN	
5	CCS	0
18	CCS	1
	DRAIN	
6	CDM	0
19	CDM	1
	DRAIN	
14	SPARE (GND)	
20	SPARE (GND)	
	DRAIN	
8	CRR	0
21	CRR	1
	DRAIN	
9	CRT	0
22	CRT	1
	DRAIN	
10	CTT	0
23	CTT	1
	DRAIN	
11	CST	0
24	CST	1
	DRAIN	
12	CLL	1
25	CTM	1
	DRAIN	
13	CLR	1

COMM
JUNCTION BOX (6)
P2

SPARE (GND)	1
CSG	0
DRAIN	
CSD	0
CSD	1
DRAIN	
CRD	0
CRD	1
DRAIN	
CRS	0
CRS	1
DRAIN	
CCS	0
CCS	1
DRAIN	
CDM	0
CDM	1
DRAIN	
SPARE (GND)	10
SPARE (GND)	8
DRAIN	
CRR	0
CRR	1
DRAIN	
CRT	0
CRT	1
DRAIN	
CTT	0
CTT	1
DRAIN	
CST	0
CST	1
DRAIN	
CLL	1
CTM	1
DRAIN	
CLR	1

3261 8472-XX
SYSTEM CABLE TYPE F-1 BASIC
(PRIMARY RADAR)

PRIMARY RADAR CABLE LOCATIONS

CABLE NUMBER	FROM	TO
W1033	P1 (4A1-J10)	P2(6A1-J39)
W1034	P1 (5A1-J10)	P2(6A1-J40)
W1035	P1 (4A1-J23)	P2(6A1-J59)
W1036	P1 (5A1-J23)	P2(6A1-J60)

- NOTES:
- BOXES CONTAIN DIGITS 1 OR 2 AS F FUNCTION OF WHICH LINE ADAPTER PORT (PER PADDLE BOARD) THIS CABLE PLUGS INTO THE PROCESSOR (DPS).

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 9)

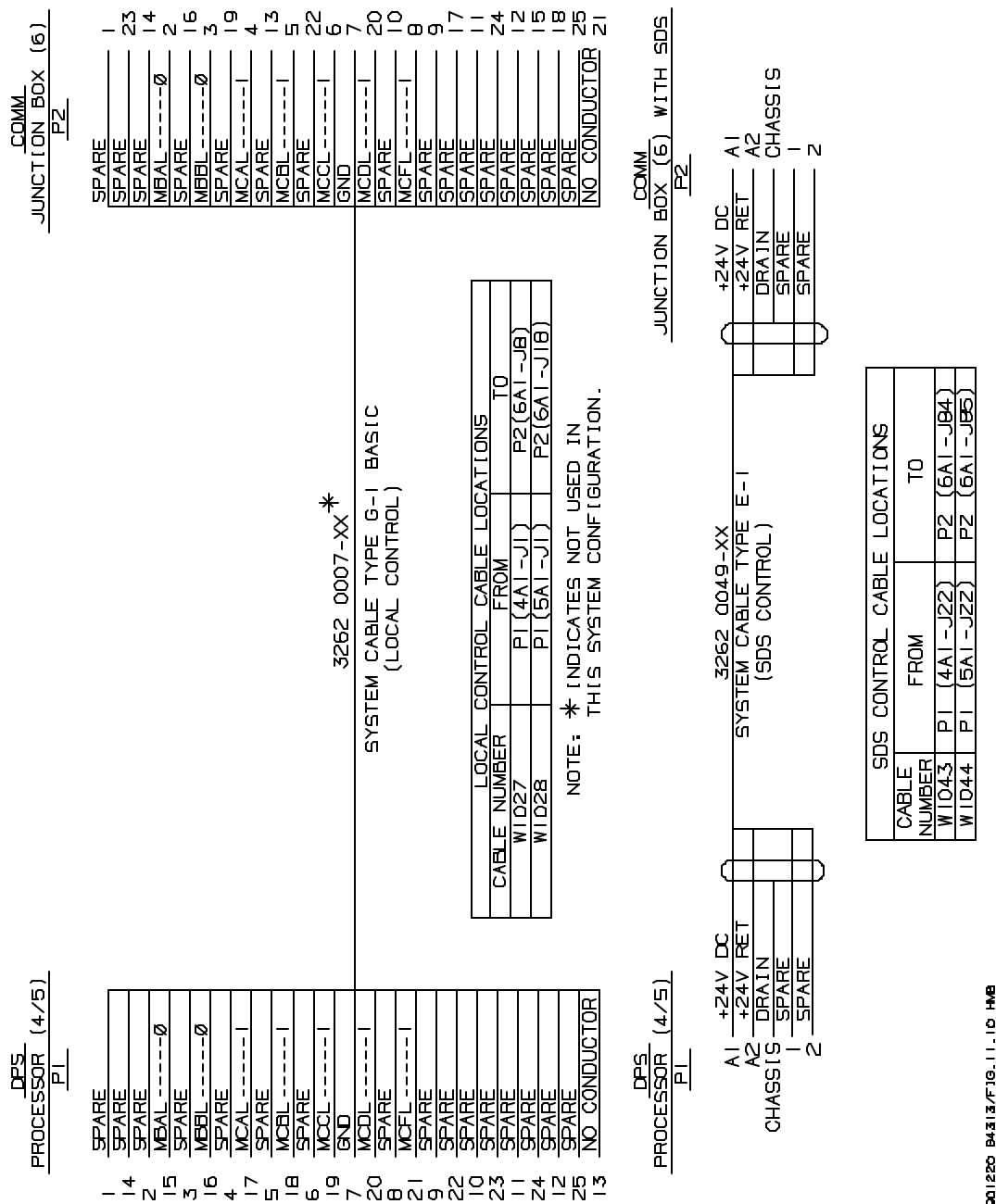


FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 10)

DPS
PROCESSOR (4/5)
P1

1	MCONT1	—	—	1
14	MCONTGND	—	0	
	DRAIN			
2	MCONTGND	—	0	
15	MCONT2	—	—	1
	DRAIN			
3	MCONT3	—	—	1
16	MCONTGND	—	0	
	DRAIN			
4	MCONTGND	—	1	
17	MCONT4	—	—	0
	DRAIN			
5	MCONT5	—	—	1
18	MCONTGND	—	0	
	DRAIN			
6	MCONT6	—	—	1
19	MCONTGND	—	0	
	DRAIN			
7	MCONT7	—	—	1
20	MCONTGND	—	0	
	DRAIN			
8	MCONT8	—	—	1
21	MCONTGND	—	0	
	DRAIN			
9	GND			
22	GND			
	DRAIN			
10	GND			
23	GND			
	DRAIN			
11	SPARE			
24	SPARE			
	DRAIN			
12	SPARE			
25	SPARE			
	DRAIN			
13	SPARE			

COMM
JUNCTION BOX (6)
P2

MCONT1	—	—	1	1
MCONTGND	—	0		23
DRAIN				CHASSIS
MCONTGND	—	0		14
MCONT2	—	—	1	2
DRAIN				CHASSIS
MCONT3	—	—	1	16
MCONTGND	—	0		3
DRAIN				CHASSIS
MCONTGND	—	1		19
MCONT4	—	—	0	4
DRAIN				CHASSIS
MCONT5	—	—	1	13
MCONTGND	—	0		5
DRAIN				CHASSIS
MCONT6	—	—	1	22
MCONTGND	—	0		6
DRAIN				CHASSIS
MCONT7	—	—	1	7
MCONTGND	—	0		20
DRAIN				CHASSIS
MCONT8	—	—	1	10
MCONTGND	—	0		8
DRAIN				CHASSIS
GND				9
GND				17
DRAIN				CHASSIS
GND				11
GND				24
DRAIN				CHASSIS
SPARE				12
SPARE				15
DRAIN				CHASSIS
SPARE				18
SPARE				25
DRAIN				CHASSIS
SPARE				21

3262 0015-XX *

SYSTEM CABLE TYPE G-2 BASIC
(BUILDING SENSOR)

BUILDING CABLE TYPE G-2 BASIC

CABLE NUMBER	FROM	TO
W1019	P1 (4A1-J11)	P2(6A1-J67)
W1020	P1 (5A1-J11)	P2(6A1-J68)

NOTE: * INDICATES NOT USED IN THIS
SYSTEM CONFIGURATION.

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 11)

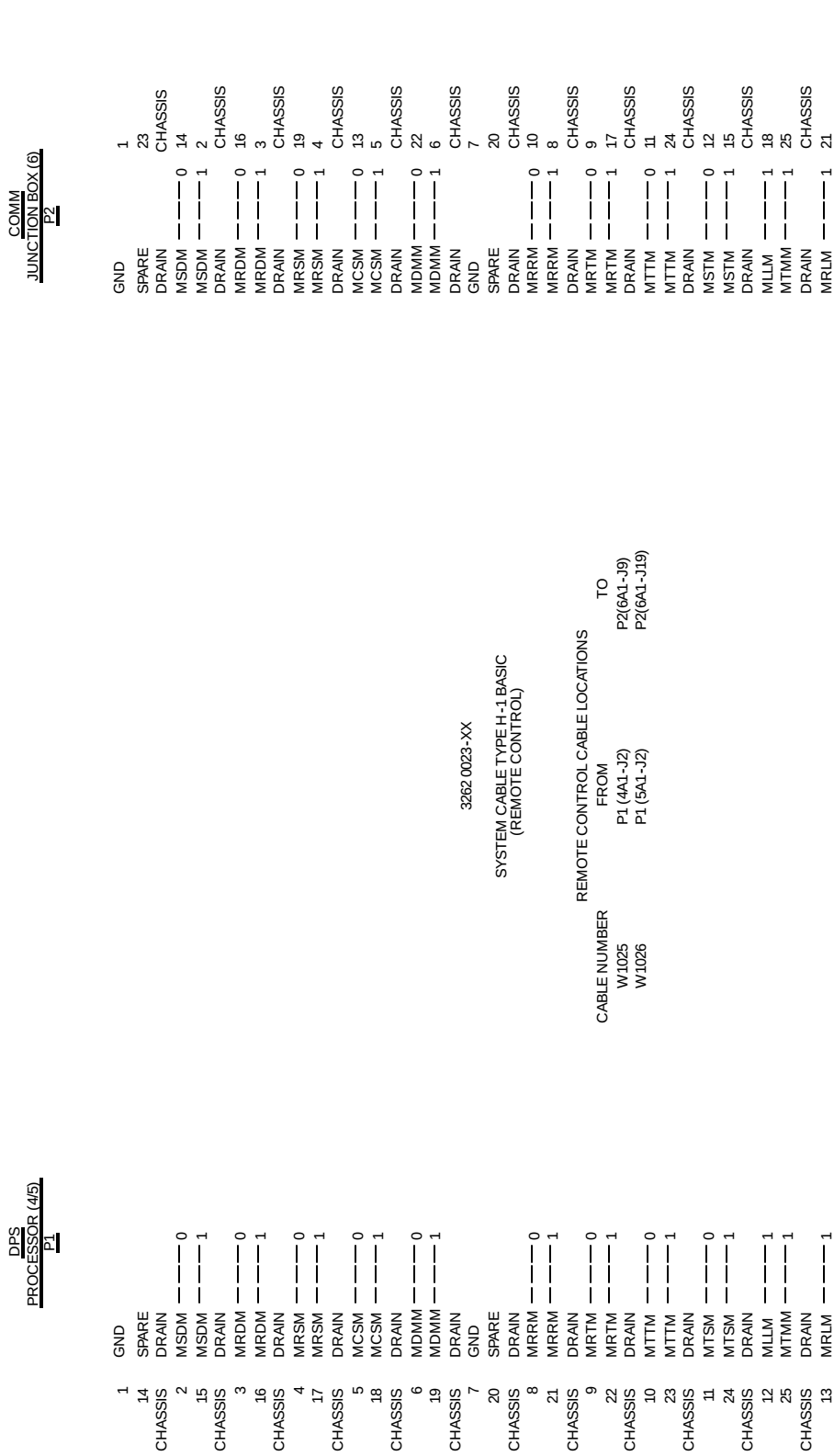


FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 12)

CD-2
JUNCTION BOX
P1

1	NO CONNECT
7	CSG ---- 0
CHASSIS	DRAIN
2	CSD ---- 1
14	CSD ---- 0
CHASSIS	DRAIN
3	CRD ---- 1
16	CRD ---- 0
CHASSIS	DRAIN
4	SPARE
19	SPARE
CHASSIS	DRAIN
5	SPARE
13	SPARE
CHASSIS	DRAIN
6	CDM ---- 1
22	CDM ---- 0
CHASSIS	DRAIN
23	CRR ---- 0
20	CRR ---- 1
CHASSIS	DRAIN
8	SPARE
10	SPARE
CHASSIS	DRAIN
17	CRT ---- 1
9	CRT ---- 0
CHASSIS	DRAIN
24	CTT ---- 1
11	CTT ---- 0
CHASSIS	DRAIN
15	SPARE
12	SPARE
CHASSIS	DRAIN
18	SPARE
25	SPARE
CHASSIS	DRAIN
21	SPARE

CD-2
AIB
P2

NO CONNECT	1
CSG ---- 0	7
DRAIN	CHASSIS
CSD ---- 1	2
CSD ---- 0	14
DRAIN	CHASSIS
CRD ---- 1	3
CRD ---- 0	16
DRAIN	CHASSIS
SPARE	4
SPARE	19
DRAIN	CHASSIS
SPARE	5
SPARE	13
DRAIN	CHASSIS
CDM ---- 1	6
CDM ---- 0	22
DRAIN	CHASSIS
CRR ---- 0	23
CRR ---- 1	20
DRAIN	CHASSIS
SPARE	8
SPARE	10
DRAIN	CHASSIS
CRT ---- 1	17
CRT ---- 0	9
DRAIN	CHASSIS
CTT ---- 1	24
CTT ---- 0	11
DRAIN	CHASSIS
SPARE	15
SPARE	12
DRAIN	CHASSIS
SPARE	18
SPARE	25
DRAIN	CHASSIS
SPARE	21

3262 0171-XX

SYSTEM CABLE TYPE J-1 BASIC
(CD-2 JUNCTION BOX TO CD-2)

CD-2 TO CD-2 JUNCTION BOX LOCATIONS		
CABLE NUMBER	FROM	TO
W1201	P1 (CD2JB-J201)	P2 (CD2A-J101)
W1203	P1 (CD2JB-J202)	P2 (CD2A-J102)
W1202	P1 (CD2JB-J203)	P2 (CD2B-J101)
W1204	P1 (CD2JB-J204)	P2 (CD2B-J102)

NOTE:

BOXES CONTAIN DIGIT 1 OR 2 AS A FUNCTION OF WHICH LINE ADAPTER PORT (PER PADDLE BOARD) IS USED IN THE PROCESSOR (DPS).

FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 13)

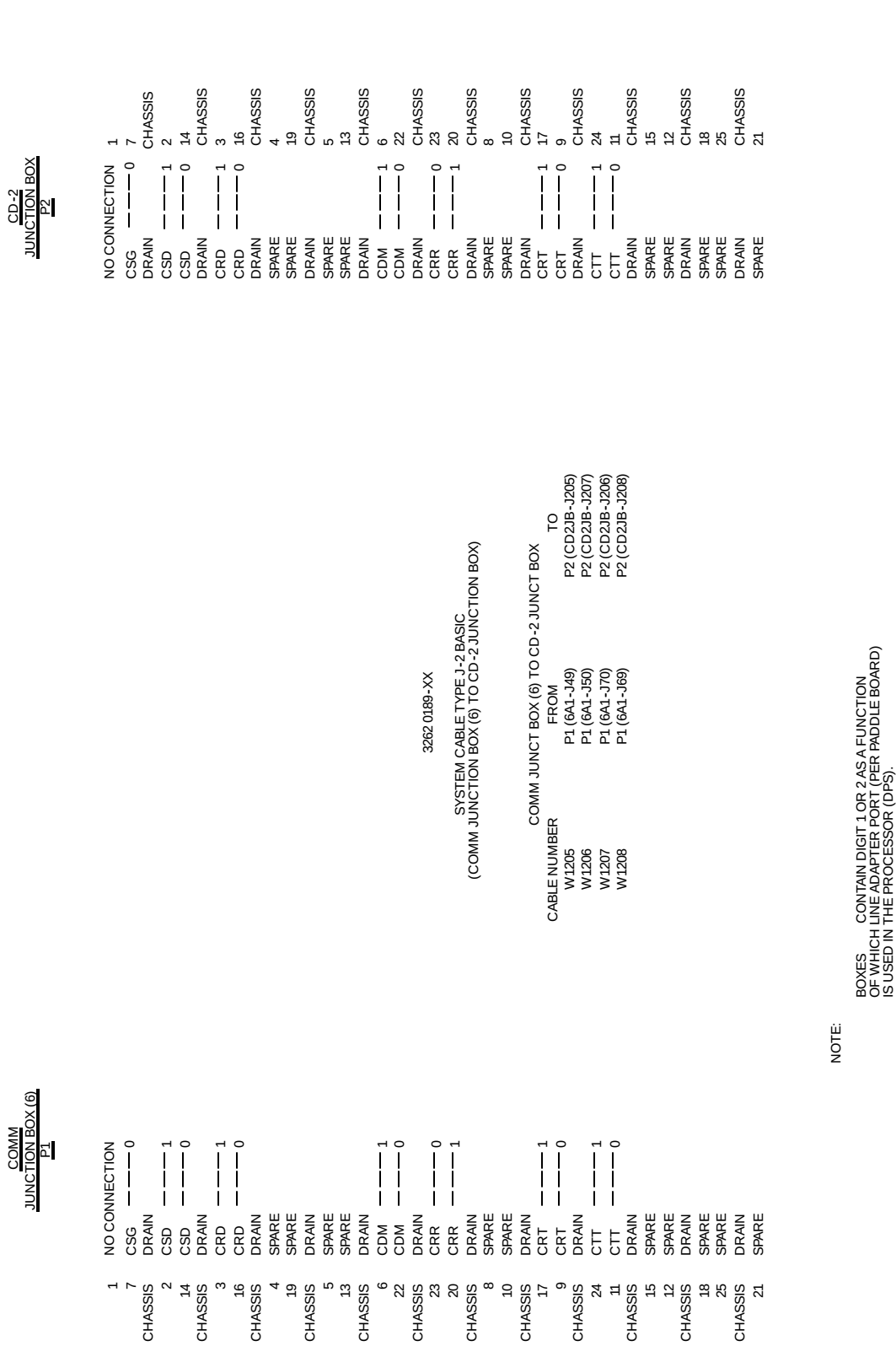


FIGURE 11-10. DPS SYSTEM CABLES WIRING AND SIGNAL DISTRIBUTION DIAGRAM (SHEET 14 — END)

MAIN POWER FROM
FAA PANEL BOARD
120/208 VAC,
3-PHASE, 4-WIRE,
60HZ

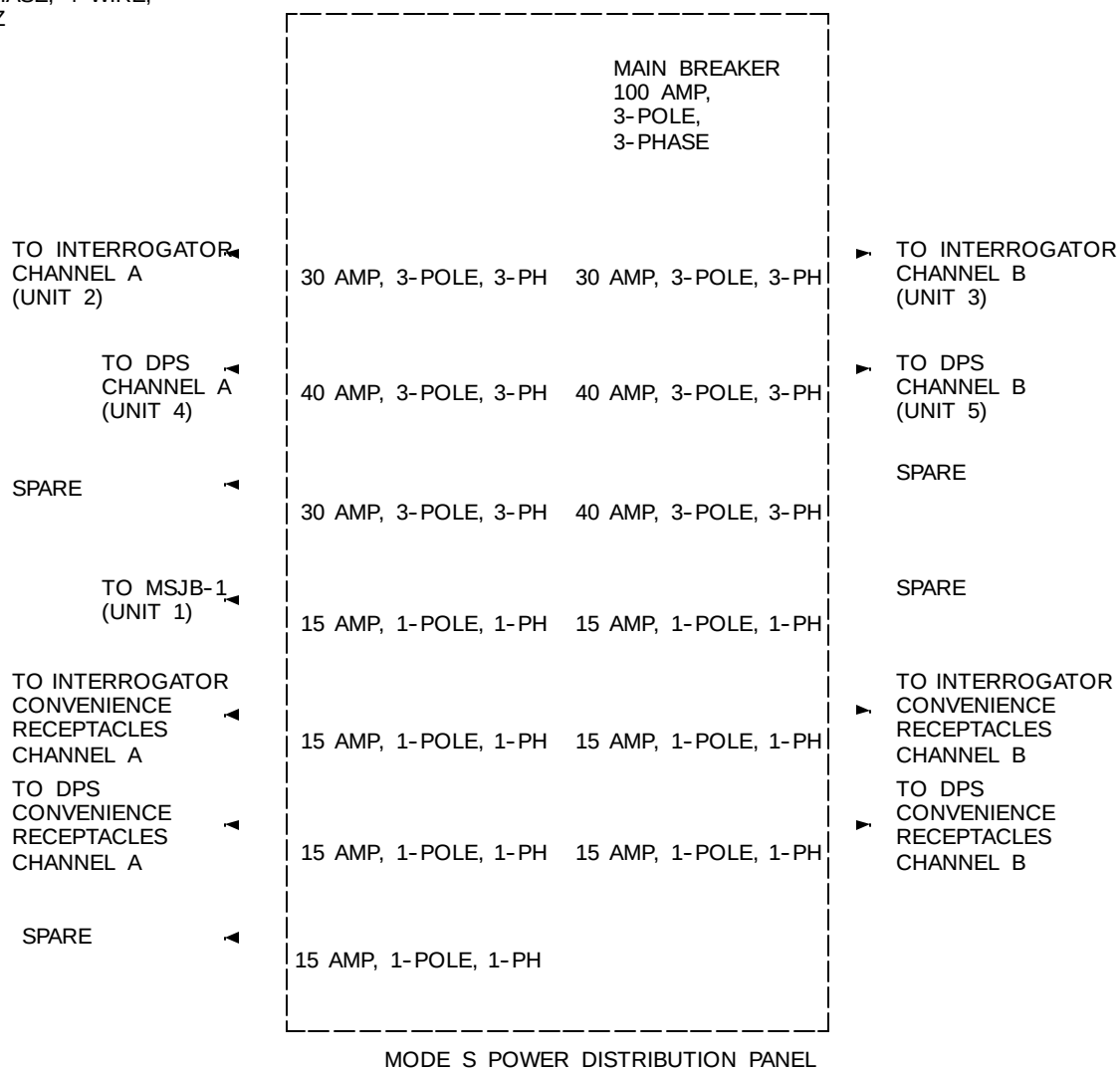
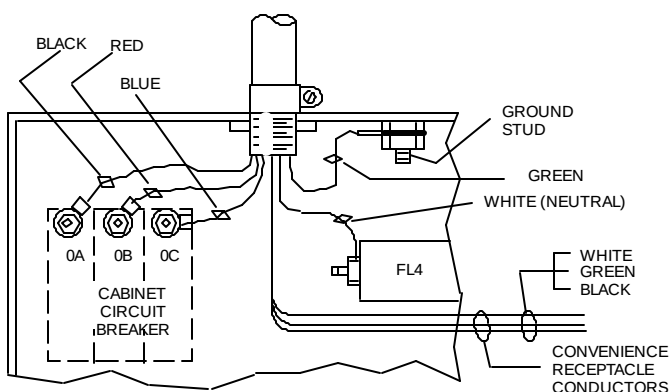


FIGURE 11-11. MODE S POWER DISTRIBUTION INFORMATION DIAGRAM (SHEET 1)

CONDUCTOR SCHEDULE				CONDUCTOR SCHEDULE		
BRANCH CIRCUIT	PHASES	CONDUCTOR COLOR	CONDUCTOR SIZE, AWG	CONDUIT RUN	BRANCH CIRCUITS	MAIN RUN CONDUIT DIA., IN.
DPS CHANNEL A (UNIT 4)	3	BLACK RED BLUE WHITE GREEN	6 6 6 6 10	A	DPS CHANNEL A DPS CHANNEL B DPS CONVENIENCE RECEPTACLES CHANNEL A DPS CONVENIENCE RECEPTACLES CHANNEL B	2
DPS CHANNEL B (UNIT 5)	3	BLACK RED BLUE WHITE GREEN	6 6 6 6 10	B	INTERR. CHANNEL A INTERR. CHANNEL B INTERR. CONVENIENCE RECEPTACLES CHAN. A INTERR. CONVENIENCE RECEPTACLES CHANN B/C	2
DPS CONVENIENCE RECEPTACLES CHANNEL A	1	BLACK WHITE GREEN	10 10 14	C	MSJB-1	1
DPS CONVENIENCE RECEPTACLES CHANNEL B	1	BLACK WHITE GREEN	10 10 14			
INTERROGATOR CHANNEL A (UNIT 2)	3	BLACK RED BLUE WHITE GREEN	8 8 8 8 10			
INTERROGATOR CHANNEL B/C (UNIT 3/8)	3	BLACK RED BLUE WHITE GREEN	8 8 8 8 10			
INTERROGATOR CONVENIENCE RECEPTABLES CHANNEL A	1	BLACK WHITE GREEN	10 10 14			
INTERROGATOR CONVENIENCE RECEPTABLES CHANNEL B/C	1	BLACK WHITE GREEN	10 10 14			
MSJB-1 (UNIT 1)	1	BLACK WHITE GREEN	10 10 14			

INPUT POWER TO INTERROGATOR
(TOP AREA OF INTERROGATOR CABINET)

INPUT POWER TO DPS MAIN CIRCUIT BREAKER ASSY.
AND FILTER BOX (TOP AREA OF PRCSS CABINET)



<u>3-PH INPUT</u>	<u>TO</u>	<u>WIRE COLOR</u>
01	A/TB1-A1	BLACK
02	A/TB1-A2	RED
03	A/TB1-A3	BLUE
NEUTRAL	A/TB1-A4	WHITE
GROUND	A/TB1-A5	GREEN
<u>1-PH INPUT</u>	<u>TO</u>	<u>WIRE COLOR</u>
01	A/TB2-A1	BLACK
NEUTRAL	A/TB2-A2	WHITE
GROUND	A/TB2-A3	GREEN

FIGURE 11-11. MODE S POWER DISTRIBUTION INFORMATION DIAGRAM
(SHEET 2 — END)

Appendix A GLOSSARY

This glossary provides standard definitions for commonly used acronyms and abbreviations in this Mode S book.

<u>Abbreviation</u>	<u>Definition</u>
AC	Alternating Current
ACK	Acknowledge
ACP	Azimuth Change Pulse
A/D	Analog-to-Digital
ADCOM	Air Defense Communications
ADJ	Adjust
ADRS	Address
ALU	Arithmetic and Logic Unit
AM	Amplitude Modulation
AMPL	Amplifier
APG	Azimuth Pulse Generator
ARIES	Aircraft Reply and Interference Environment Simulator
ARP	Azimuth Reference Pulse
ARSR	Air Route Surveillance Radar
ARTCC	Air Route Traffic Control Center
ASR	Airborne Surveillance Radar
ATC	Air Traffic Control
ATCRBS	Air Traffic Control Radar Beacon System
ATE	Automatic Test Equipment
ATTEN	Attenuator; Attenuation
AU	Azimuth Unit
AUX	Auxiliary
AZ	Azimuth
BCD	Binary Coded Decimal
BIT	Built-in-Test
BRKR	Breaker
BRKT	Bracket
C	Clear (Reset)
CCA	Circuit Card Assembly
CD	Common Digitizer
CJBX	Communications Junction Box
CKT	Circuit
CLK	Clock
CLR	Clear
CMD	Command
CMPT	Computer
CNTR or CTR	Counter
COMM	Communication

Appendix A GLOSSARY (Continued)

<u>Abbreviation</u>	<u>Definition</u>
CONT or CNTL	Control
CPCI	Computer Program Configuration Item
CPME	Calibration Performance Monitoring Equipment
CPU	Central Processing Unit
CRT	Cathode-Ray Tube
CSS	Computer Subsystem
CW	Continuous Wave
D/A	Digital-to-Analog
dB or db	Decibels
DC	Direct Current
DCDR	Decoder
DEC	Decimal
DEG	Degree
DET	Detector
DMA	Direct Memory Access
DPS	Data Processing System
DPSK	Differential Phase-Shift Keying
DRAM	Dynamic Random Access Memory
EEPROM	Electrically Erasable Programmable Read-Only Memory
EIB	Equipment Instruction Book
ELM	Extended Length Message
EMI	Electromagnetic Interference
ENBL	Enable
EOT	End of Test
EPROM	Erasable Programmable Read-Only Memory
ERR	Error
EXP	Expansion
FAA	Federal Aviation Administration
FD	Fault Detect
FE	Front End
FF	Flip-Flop
FIFO	First-In-First-Out
FIT	Fault Isolation Test
FWD	Forward
GEN	Generator
GND or GRD	Ground
GRAM	Global Random Access Memory
HEX	Hexidecimal
Hz	Hertz (Symbol for Cycles Per Second)
IATCRBS	Independent Air Traffic Control Radar Beacon System
IC	Integrated Circuit
ID	Identification
IF	Intermediate Frequency

Appendix A GLOSSARY (Continued)

<u>Abbreviation</u>	<u>Definition</u>
IFF	Identification Friend or Foe
I/O	Input/Output
INT or INTERR	Interrogator
IOP	Input/Output Processor
IPM	Interrogator Performance Monitor
ISLS	Interrogator Side-Lobe Suppression
ITRI	Interrogator Interface
J	Jack
JB	Junction Box
KCRT	Keyboard Cathode Ray Tube
LA	Line Adapter
LACOM	Communications Line Adapter
LAMNT	Maintenance Line Adapter
LARDR	Radar Line Adapter
LASMT	Magnetic Tape Line Adapter
LASUR	Surveillance Line Adapter
LED	Light-Emitting Diode
LO	Local Oscillator
LOG	Logarithm
LRU	Lowest Replaceable Unit
LSB	Least Significant Bit
LTI	Local Terminal Interface
LU	Logical Unit
MCU	Modulation Control Unit
MEM	Memory
MHz	Megahertz
MNPLS	Monopulse
Mode S	Mode Select Beacon System Sensor
MON	Monitor
MPS	Maintenance Processor Subsystem
MPU	Micro-Processor
MPU40	68040-based microprocessor unit circuit card
ms	Millisecond
MSB	Most Significant Bit
MSG	Message
MSJB	Mode S Junction Box
MUX	Multiplex, Multiplexer
NAS	National Airspace System
NEG	Negative
NM	North Mark
nmi	Nautical Miles
ns	Nanosecond
NVMEM	Nonvolatile Memory
OSC	Oscillator

Appendix A GLOSSARY (Continued)

<u>Abbreviation</u>	<u>Definition</u>
P	Plug
PAL	Programmable Array Logic
PAM	Pulse Amplitude Modulation
PARA	Paragraph
PCB	Printed Circuit Board
PMS	Performance Monitor(ing) System
PN	Part Number
POS	Positive
PPM	Pulse Position Modulation
PRCSS	(DPS) Processor Subsystem
PRF	Pulse Repetition Frequency
PRI	Primary
PROC	Processor
PROM	Programmable Read-Only Memory
PS	Power Supply
PSF	Program Support Facility
PWR	Power
RAM	Random Access Memory
RCV	Receive
RCVR	Receiver
REG	Register
RF	Radio Frequency
RFTTG	Radio Frequency Test Target Generator
RMS	Remote Maintenance Subsystem (DPS)
RMMS	Remote Maintenance Monitoring System
R/S	Radar/Surveillance
RU	Range Unit
RW	Read-Write
SAP	Site Adaptable Parameter
SDS	Surveillance Data Selector
SEC	Second
SIG PROC	Signal Processor
SLS	Side-Lobe Suppression
SNR	Signal-to-Noise Ratio
SNSR	Sensor
SPI	Special Position Identification
SR	Shift Register
SRAM	Static Random Access Memory
STAT	Status
SYNC	Synchronize, Synchronizer, Synchronous
SW	Switch
TACP	Test Azimuth Change Pulse
TARP	Test Azimuth Reference Pulse
TB	Terminal Board

Appendix A GLOSSARY (Continued)

<u>Abbreviation</u>	<u>Definition</u>
TCAS	Traffic Alert and Collision Avoidance System
TEMP	Temperature
TGT	Target
TOY	Time of Year
TRACAB	Tower Radar Approach Control Facility (Tower Cab)
TRACON	Tower Radar Approach Control Facility
TTG	Test Target Generator
V	Volts
VAC	Volts Alternating Current
VDC	Volts Direct Current
VID	Video
VSP	Variable Site Parameter
VSWR	Voltage Standing-Wave Ratio
XCVR	Transceiver
XFR	Transfer
XMT	Transmit
XMTR	Transmitter